SALIENT FEATURES OF SURAT METRO RAIL NETWORK (Phase-I)

1.0 ROUTE LENGTH

S.No.	Corridors	Underground (kms)	Elevated/At Grade (kms)	Total (kms)
01	Sarthana-Dream City	6.47 kms	15.14 kms	21.61 kms
02	Bhesan-Saroli	-	18.74 kms	18.74 kms
	Total	**6.47 kms**	**33.88 kms**	**40.35 kms**

2.0 NUMBER OF STATIONS

S.No.	Corridors	Underground	Elevated	Total
01	Sarthana-Dream City	6	14	20
02	Bhesan-Saroli	-	18	18
	Total	**06 nos**	**32 nos**	**38 nos**

3.0 TRACK GAUGE

Standard Gauge (SG): 1435 mm for both the corridors i.e.
(i) Sarthana –Dream City &
(ii) Bhesan – Saroli

4.0 TRAFFIC FORECAST

(i) Full Day Ridership Summary

Corridor	Direction	2021 - Without TOD	2026 - Without TOD	2026 - With TOD	2036 - Without TOD	2036 - With TOD	2046 - Without TOD	2046 - With TOD
Corridor 1	Sarthana-Dream City	82360	166550	208080	261070	369240	324500	453670
Corridor 1R	Sarthana-Dream City	83030	167960	205020	263410	355480	344110	464650
Corridor 2	Bheshan-Saroli	68610	102250	127150	221060	328380	272440	426310
Corridor 2R	Bheshan-saroli	65650	97130	123420	228770	342060	282040	441800
Total Ridership		**299650**	**533890**	**663670**	**974310**	**1395160**	**1223090**	**1786430**

(ii) Maximum Passengers: Peak Hour Peak Direction Traffic (PHPDT)

Corridor	2021_ Without TOD (Peak Hour)	2026_With out TOD (Peak Hour)	2026_With TOD (Peak Hour)	2036_With out TOD (Peak Hour)	2036_With TOD (Peak Hour)	2046_ Without TOD (Peak Hour)	2046_With TOD (Peak Hour)
Corridor 1b: Sarthana - Dream City	6,549	13,340	16,099	20,856	28,144	24,247	32,065
Corridor 1bR: Sarthana-Dream City	6,992	13,536	16,424	20,677	25,872	26,587	33,543
Corridor 2d: Bheshan-Saroli	3,918	5,698	7,466	12,573	19,944	15,448	25,849
Corridor 2dR: Bheshan-Saroli	3,693	5,236	6,951	12,098	19,902	15,175	25,165

5.0 SYSTEM DESIGN

(a) Station Dwell Time	**30 seconds**
(b) Train Composition	**3 cars**
(c) Average Speed	**33 Kmph**
(d) Speed (i) Max Design Speed (ii) Max Operating Speed	 **90 kmph** **80 kmph**

6.0 TRACTION POWER SUPPLY

a) Voltage	:	750 V DC
b) Current Collection	:	Third Rail Current Collection System
c) Power Supply source	:	i) Corridor-I (Sarthana to Dream City) - Grid Sub Stations of M/s. GETCO ii) Corridor-II (Bhesan to Saroli) -Grid Sub Station of M/s. Torrent Power
d) SCADA system	:	Provided

7.0 ROLLING STOCK

a) Coach Dimensions		
(i) Height	:	3.9m
(ii) Width	:	2.9m
(iii) Coach Body	:	Stainless Steel/Aluminium
b) Axle load (@ 8 persons per sqm of Standee area)	:	16 t
c) Seating arrangements	:	Longitudinal
d) Train Composition (for 3 car train)	:	DMC+TC+DMC
e) Coach Carrying Capacity		
(i) DMC	:	247 (Seating-43, Standing-204)
(ii) TC	:	270 (Seating -50, Standing - 220)
f) Train Carrying Capacity (3 car train)	:	764 passengers (Seating- 136, Standing-628)
g) Control System	:	Train Based Monitor & control system (TCMS/TIMS)

8.0 MAINTENANCE FACILITIES

Maintenance Depots (02 Nos.)	:	(i) Dream City (ii) Bhesan

9.0 SIGNALLING, TELECOMMUNICATION & TRAIN CONTROL

a) Type of Signalling : Continuous Automatic train control with cab signaling and ATO

b) Telecommunication : Integrated System with Fibre Optic cable, SCADA, Train Radio, PA system etc.

10.0 FARE COLLECTION : Automatic Fare Collection System with Combination of Contactless Smart Card for multiple journey and computerized Contactless Smart Token for single Journey.

11.0 CONSTRUCTION METHODOLOGY : Elevated Viaduct carried over pre-stressed concrete box shaped girder/U-Girder/Double U-Girder/I-girder with pile/open foundation and underground construction using TBM or Cut & cover or NATM techniques.

12.0 TOTAL ESTIMATED COST : (Dec '2018 prices) Rs 9,338 crores without taxes and duties. Taxes and duties are estimated to be Rs. 1322.78 crores. Total Cost: Rs 10,660.78 crores **=Rs 10661 Crores**

13.0 TOTAL ESTIMATED COMPLETION COST : Rs.12,020 Crores (i/c cost of land, PPP component, IDC for loan and all taxes and duties).

14.0 FINANCIAL INDICES

(a) ECONOMIC INTERNAL RATE OF RETURN (EIRR) : **17.20%**

(b) FINANCIAL INTERNAL RATE OF RETURN (FIRR) : i**) 6.90%** (with all Taxes & Duties) without additional PD, TOD & VCF income.

ii) **7.38%** (with all taxes & duties) with additional PD,TOD &VCF income

EXECUTIVE SUMMARY

0.1 BACKGROUND

0.1.1 Surat city, located on the western part of India in the State of Gujarat is also known as a ”Economic power house” of Gujarat. The city is located 284 Kms south of Gandhi Nagar, 265Kms of Ahmedabad and 289 Kms north of Mumbai. The economy of the entire city is based mainly on two industries, the textile industries of man made fibers/fabrics and the diamond cutting and polishing industry. Surat is one of the most dynamic cities of India having faster growth rate due to immigration from various parts of Gujarat and other States of India too. National Highway-8 (NH-8) passes within 16 Kms. of municipal boundaries of Surat and is located midway on the Ahmedabad-Mumbai route.

0.1.2 Surat is one of the oldest inhabited cities in the world and densely populated with an average 13680 persons/sq.km accommodating about 44.67 Lakhs people as per census 2011. Average decadal growth of population of Surat since 2001 to 2011 was about 55.29%. The census of India 2011 has revealed that one out of seven zones of Surat i.e. central zone has registered even negative growth. Surat city is spread over an area of 326.515 sq. kms. The Surat city is located at 21.18°N72.83°Ewith 21.195 Latitude and 72.8194 Longitude Surat’s high population growth rate, coupled with high economic growth rate has resulted in an ever increasing demand for transport creating excessive pressure on the existent transport system. With high growth in transport demand over the years, congestion on roads has been increasing due to phenomenal rise in private transport. Absence of an efficient full-fledged public transport system coupled with rapid growth in the use of personalized vehicles has led to high consumption of fossil fuel and increase in environmental pollution. Surat has also been selected as one of the hundred Indian cities to be developed as a “Smart City”. The existing network of public transport systems including dedicated BRTS needs to be strengthened further in order to cope-up with rising demand of transport system.

The inter-city traffic volumes in Surat necessitated a full-fledged integrated multi model mass rapid passenger system. In view of this, Government of Gujarat (GoG) vide their letter No. MEGA/MD/DMRC/IC dated 27.01.2016 requested Delhi Metro Rail Corporation (DMRC) to prepare a Detailed Project Report (DPR) for Metro Rail Project in Surat City.

The DPR for Surat Metro rail project had already been prepared and submitted to SMC and MEGA vide DMRC's letter no DMRC/GM/CS/04/Surat-DPR/2016/1150 dated 24th July'2017. Subsequently, in August' 2017 Ministry of Housing and Urban Affairs announced a new metro policy-2017 and returned the said proposal for incorporation of new provisions. The Commissioner, Surat Municipal Corporation(SMC) vide his e-mail dated 16th September'2017 requested DMRC to review and revise the detailed project report(DPR) in view of new metro rail policy. Thus the said DPR for Surat Metro Rail project has been revised incorporating new provisions such as comprehensive mobility plan(CMP), alternative analysis, last mile connectivity, transit oriented development(TOD), value capture finance(VCF), property development(PD), public private partnership(PPP), unified metropolitan transport authority(UTMA), options for central assistance etc.

The DPR has been updated upto December 2018 level.

0.1.3 The following two Metro Corridors have been identified for their inclusion in Surat Metro Rail Network (Phase-I).

Table 0.1: Surat Metro Corridors

S.No.	Corridor	Underground	Elevated/At Grade (Kms)	Total (Km)
01	**Phase-1** Corridor-1 **Sarthana to Dream City**	6.47 Kms	15.14 Kms	21.61 Kms
02	Corridor-2 **Bhesan to Saroli**	--	18.74 Kms	18.74 Kms
	Total	**6.47 Kms**	**33.88 Kms**	**40.35 ms**

0.1.4 Project Area

0.1.4.1 The Detailed Project Report (DPR) is prepared for the proposed 02 (two) Corridors extending in North, South, East, West and Central parts of Surat City, as such most parts of Surat are covered with these proposed corridors. The project comprises various important and prominent areas along the proposed corridors under Surat Metro Rail Network (Phase-I) as per the details given hereunder:-

0.1.4.2 The colonies and important/prominent areas covered along the two metro corridors are

i) Sarthana – Dream City

This corridor provides metro connectivity to Gadhpur Township, Sarthana Nature Park, Nana Varachhe village, Kalakunj Junction, Kapodra and Puna village, Labheshwar Chowk area, Big Diamond Market (Mini Bazar) of Surat, Retail Sari Market, Surat Railway Station, Muskati Hospital, Bhagal junction, Gandhi Bagh, Chowk Bazar, Majura Gate, Roopali Canal, Altha Treatment, Althan Gram, VIP road Women IIT, Bhimrad, Convention City and Dream City.

ii) Bhesan - Saroli

Prominent areas like Bhesan Treatment Plant, International Cricket Stadium, SMC Botanical Garden, Palanpur, LP Savani School, Performing Art Centre, TGB Circle, Aquarium, Badri Narayan Temple, Star Bazar, Tapi River, Athawa Chopati, Majura Gate, UdhanaDarwaja, Kamala Darwaja, Anjana Junction, Parvat Pataya, Magub Village, Bharat Cancer Hospital, Saroli etc. are connected through this route.

0.2 TRANSPORTDEMAND FORECAST

Metro Policy 2017 rightly proposes that the proposal for central assistance for identified Metro Rail project should be mooted by the State Governments on two counts- Firstly, because very nature of urban transport is such that it has intricate connect with other issues of urban development making it logical that only those who have overall perspective and feel of the city formulate the plan for urban transport fro that city and secondly also because, the “Urban Development” is a State Subject of the Constitution.

It is in this context that the DPR for Surat Metro is being submitted for central assistance is not only being submitted through state government for various reason but Government of Gujarat(GOG) apart from its own 50% contribution of

equity is also contributing towards the cost of the land and towards taxes and duties. More particularly the proposal for Surat Metro takes into account the overall urban development and urban transport plan of state government.

The policy further requires that metro rail development should follow a “systems approach” that is comprehensive approach to planning of urban land use and transport infrastructure.

The proposal is required to evaluate alternative modes for the most optimum mobility plan for the city. And as such the policy puts a mandatory pre-requisite for the city having a Comprehensive Mobility Plan (CMP) for planning metro rail in any city.

With the above in perspective the policy allows cities with two million and plus population tostart planning for mass transit systems including metro rail based on CMP. In this context, it is specified that Surat is a unique case where population has already exceeded 5 million mark and shall have approximately 07 million population by the year 2021.

Also Surat being the “Economic Power House” of Gujarat, has substantial daily floating population from other cities. Additionally international tourist footfall is also at Surat.

As per requirement of the city the DPR of Surat Metro Rail Project has been prepared by DMRC which have sufficient expertise to formulate DPR for other cities in the country. DPR of Surat Metro has been prepared keeping in view the growing need of the city in consultation with GoG and Central Government.

In the year 2017, a Comprehensive Mobility Plan (CMP) for Surat city has been got completed by Surat Municipal Corporation (SMC) through Centre of Excellence in Urban Transport (CEPT), Ahmedabad.

Gist of Comprehensive Mobility Plan (CMP) – 2017 of Surat City related to MRTS Project as provided by Centre of Excellence in Urban Transport,(CEPT), Ahmedabad, the agency engaged by Surat Municipal Corporation (SMC) to update the CMP, has been included in this DPR which mainly contains the following :-

Demand Analysis for Alternative CMP Surat Metro Corridors
Background
Approach for analysis of alternative corridors
Demand Assessment
Corridor Characteristics
Capacity of existing transport system

Summary
Observations
Conclusions

0.2.1 The estimation of traffic demand on the proposed metro corridors under Surat Metro Rail Project (Phase-I) has been carried out by "The Centre of Excellence in Urban Transport, CEPT University, Ahmedabad who were entrusted this assignment by Surat Municipal Corporation (SMC) vide their letter No. CE.Spl.Cell/Out/339 dated 31.08.2016. The important highlights are summarized below:

(a) The following development area is considered as the study area :

The first step of the process was to define the study area for modelling purposes. Surat is the major urban area in the region, with strong linkages to urban centers in and around Gujarat. The Dedicated Freight Corridor (DFC) passing through the region along with the Delhi-Mumbai Industrial corridor (DMIC) is expected to trigger huge investment and growth in the near future.

The Development Plan of 2035 has identified growth nodes in Olpad and Kamrej with residential areas to be developed in these areas. The recent Development Plan 2035 Surat Urban Development Authority has increased its area to include all the above mentioned developments from 715 Sq. km to 1351sq.km. Hence for this study, the Development Plan area is considered as the study area.

(i) Surat Urban Development Authority (SUDA)

SUDA was constituted in 1978. It now has a spatial extent measuring 1351 sq.km. Its jurisdiction lies both right and left banks of the Tapi river, which includes the area of Surat Municipal Corporation (SMC), two INAs, 15 urban areas and 182 villages.

(ii) Surat Municipal Corporation (SMC)

The area within the Surat Municipal Corporation limits consists of:

The traditional city centre, known as the 'Old city' with relatively high-density development, large concentration of commercial activities and narrow streets.

The southern and eastern part of the city is recognized as the industrial zone of the city, with large and small industries and low income residential areas.

Presence of textile market along certain areas of the Ring road.

(b) Growth Potential of Surat

The total population of the study area was 50.87 Lakh in 2011 with an annual growth rate of 4.3 % (2001-2011). It is estimated that in year 2016 there are 59.29 people residing in the study area. The future population is considered based on Surat Development Plan 2035. Based on growth trend estimated, population for year 2046 is projected to be around 1.25 crore and population for year 2026 and 2036 has been taken from the DP 2035.

0.2.2 MODEL DEVELOPMENT

A standard four stage modelling process has been followed to project travel demand and metro ridership in future. An overview of the modelling approach is presented below. This public transport model has been developed using EMME 4.0.8 software. For the analysis of travel pattern four trip purposes - Home based Work, Education, Others, Non-home based, has been taken into account.

0.2.2.1 Modelling framework

A detailed four stage modelling has been carried out for the Public Transport Operations Plan and Comprehensive Mobility Plan being prepared for Surat. The study area defined has been taken as the modelling area and has been subdivided into 515 internal TAZs and 6 external TAZs. A public transport model consisting of major road network in the study area along with the existing public transport network of City bus, BRTS was prepared and calibrated for the base year 2016. The same model has been used for the metro demand modelling exercise.

0.2.2.2 Model Period and Horizon Years

The base year has been taken as 2016. A 30 year forecast period has been taken with 2021 being the start year of the project and 2046 being the horizon year. Intermediate years for forecasts have been taken as 2026 and 2036.The model period for assignment is one hour morning peak. Appropriate expansion factors have been used for computing average daily ridership.

0.2.2.3 Scenario Specification Framework

The following scenarios are developed for this study:

Table 0.2Scenario Development

Future scenarios	Network	Demand
Gradual Growth Development	Existing network +Proposed Metro & other PT improvement	As per DP Population and Employment distribution
Rapid Growth Development		Intensification along proposed metro alignments

0.2.2.4 Model Software

For the four stages modelling process, MS-Excel and SPSS has been used for Trip Generation and Trip Distribution purposes, while EMME has been used for Trip Distribution and Assignment Process. Mode split has been carried out based on "Willingness to Shift Survey", conducted as part of this study.

0.2.2.5 Data Collection

This study has based its information from several studies undertaken in the recent past, in addition to some primary surveys and secondary surveys undertaken.

0.2.3 TRAFFIC DATA

0.2.3.1 Full Day Metro Ridership

Full Day Ridership Summary for both the corridors is given in Table as hereunder:-

Table 0.3: Full Day Ridership Summary

Corridor	Direction	2021 - Without TOD	2026 - Without TOD	2026 - With TOD	2036 - Without TOD	2036 - With TOD	2046 - Without TOD	2046 - With TOD
Corridor 1	Sarthana-Dream City	82360	166550	208080	261070	369240	324500	453670
Corridor 1R	Sarthana-Dream City	83030	167960	205020	263410	355480	344110	464650
Corridor 2	Bheshan-Saroli	68610	102250	127150	221060	328380	272440	426310
Corridor 2R	Bheshan-saroli	65650	97130	123420	228770	342060	282040	441800
Total Ridership		299650	533890	663670	974310	1395160	1223090	1786430

0.2.3.2 Maximum Passenger Peak Hour Per Direction (PHPDT)

Maximum PHPDT for both the corridors is given in Table as hereunder:-

Table:0.4 :Maximum PHPDT(2021, 2026, 2036, 2046)

Sr. No.	Corridor	2021_ Without TOD (Peak Hour)	2026_Without TOD (Peak Hour)	2026_With TOD (Peak Hour)	2036_Without TOD (Peak Hour)	2036_With TOD (Peak Hour)	2046_Without TOD (Peak Hour)	2046_With TOD (Peak Hour)
1	Corridor 1b: Sarthana -Dream City	6,549	13,340	16,099	20,856	28,144	24,247	32,065
2	Corridor 1bR: Sarthana-Dream City	6,992	13,536	16,424	20,677	25,872	26,587	33,543
3	Corridor 2d: Bheshan-Saroli	3,918	5,698	7,466	12,573	19,944	15,448	25,849
4	Corridor 2dR: Bheshan-Saroli	3,693	5,236	6,951	12,098	19,902	15,175	25,165

0.3 SYSTEM SELECTION

Metro Rail Policy (2017) and the Appraisal Guidelines for Metro Rail Project Proposals (September, 2017) issued by Government of India Ministry of Housing and Urban Affairs (MOHUA) mandates compulsory incorporation of Alternatives Analysis in the Detailed Project Report of Metro Rail for submitting the same to the central government while seeking their approval.

More importantly, DPR for Surat Metro has been prepared to be in line with the study “Comprehensive Transport Forecast Study” done by Centre of Excellence in Urban Transport (CEPT), Ahmedabad, who were commissioned by Surat Municipal Corporation (SMC) in the year 2016 vide their letter No. CE.Spl.Cell/Out/339 dated 31-08-2016 for undertaking demand assessment for alternative metro corridor proposals whose final report was submitted to State Government in April, 2017, in terms of content was analogous to a Comprehensive Mobility Plan for Surat City and the study had proposed corridors for different modes of urban transport based on a comprehensive Alternative Analysis In this connection it is worthwhile to upfront examine, as to how the content of the study match up to the content requirement of Alternative Analysis as mandated by the new Appraisal Framework

The following section provides a clear picture of whether CEPT University, Ahmedabad analysis has done adequate Alternative Analysis or not before factoring Surat Metro Rail Project in its report.

Further, the CEPT's study report recommends the following corridors as potential metro network:-

1) Dream City – Kamraj (28.90 kms)
2) Bhesan – Umbhel (26.30 kms)
3) Majura Gate – Karamala (15.80 kms)

Based on the demand analysis, it can be concluded that all of three corridors may be considered for metro. However, in the 1st phase following two metro corridors/corridor-sections are recommended:-

Dream City – Sarthana
Bhesan – Saroli

For Phase-II, Majura Gate - Ved can be considered. The extension to Karamala based on road network developments and demand build can be considered after demand assessment. Similarly, extensions from Sarthana – Kamrej/Vav and Saroli-Umbhel can be taken up in subsequent phases.

0.3.1 The following systems are mainly available for Urban Mass Transit:

i) Heavy Metro System with a capacity of 90,000 passengers per hour per direction of traffic (PHPDT). The capital city Delhi has gone for a Heavy Metro System.

ii) Medium Capacity Metro with PHPDT of 30,000 to 45,000. Cities like Bangalore, Chennai, Kolkata, etc. Have gone for Medium Capacity Metro.

iii) Light Metro with PHPDT of upto 30,000. Trivandrum and Kozhikode cities are planned for Light Metro Systems.

iv) **Light Rail Transit System**: Modern trams-Street Cars running on Rails at grade or elevated with sharp curves of 24m radius. These are extremely popular and operating in large number of European countries. Generally the stations are spaced at 500m to 1 km and have high acceleration and deceleration characteristics. In most of the countries, they are operating at-grade with prioritized signalling at road inter-section.

v) **Monorail System**

Monorail trains operate on grade separated dedicated corridors with sharp curves of up to 50m radius. This is a rubber tyred based rolling stock, electrically propelled on concrete beams known as guide-ways. The system is extremely suitable in narrow corridors as it requires minimum right of way on existing roads

and permits light and air and is more environmental friendly. This is prevalent in several countries for traffic densities of over 30,000 PHPDT.

vi) **Other Rail Based Systems:** A number of options are available but have not been introduced in India. Some of these are very briefly mentioned below:

(a) Maglev

This is an advanced Rail based transit system in which Magnetic Levitation is used to raise the vehicles above the rail surface. Rail wheel interaction is thus avoided and very high speeds are attainable. Maglev Levitation can either be due to attractive force or due to repulsive forces.

(b) Linear Induction Motor (LIM) Train System

This is also an advanced Rail based transit system in which propulsion is through a Linear Induction Motor whose stator is spread along the track. The rotor is a magnetic material provided in the under frame of train. In this technology, the attractive force is not transmitted through rail-wheel interaction and therefore there is no limitation on account of adhesion. This technology is most appropriate for turnouts, as the height of the tunnel can be reduced to lower height of cars.

(c) Bus Rapid Transit System (BRT)

This system involves operation of buses on a dedicated corridor (except of traffic integration) at a high frequency to achieve PHPDT. For providing a very high transport capacity say 20,000 PHPDT, about 200 buses shall be required per hour *i.e.,* at headway of 20 seconds. Such a high PHPDT can be achieved by providing two lanes of traffic in each direction and elimination of traffic intersection on the route.

(d) Automated Guide way Transit System

The term is used for systems other than conventional rail based system on grade separated guide ways. The system can be rail based or rubber tire based but fully automated guided systems with driver less operation.

0.3.2 SELECTION OF MODE

Selection of a particular mode for any pre-determined traffic corridor depends mainly on demand level of a corridor Right of Way (ROW) on the road and the capacity of the mode. The demand forecast is estimated considering the traffic

growth for about 30 years. Other considerations in mode choice are location of building lines, possibility of increasing ROW. Cost of some mode may vary depending upon the location in view of engineering constraints. Therefore final choice of mode to be adopted for a particular corridor is based on techno economic considerations. As regards the location of a particular mode like at-grade, elevated and underground, depends upon the ROW. If ROW is 20 m or more, elevated alignment is preferred over underground as the cost of underground alignment is 2- 2½ times of elevated alignment.

In view of above, it is recommended to adopt a stable, tested and reliable Metro technology i.e. **Medium Capacity Metro System**.

0.4 GEOMETRIC DESIGNING PARAMETERS AND ALIGNMENT DESCRIPTION

0.4.1 General

Geometrical standards are adopted for horizontal and vertical alignments, route description, etc. The proposed corridors of Surat Metro Rail Project (Phase-I) will consist of Standard Gauge (SG) lines, For underground corridors, track centers are governed by the dimension of island platform and for elevated (side platform), it is governed by sharpest curvature on the alignment.

The geometrical design norms are based on international practices adopted for similar metro systems with standard gauge on the assumption that the maximum permissible speed on the section is limited to 80kmph. Planning for any higher speed is not desirable as the average inter-station distance is about 1.00km and trains will not be able to achieve higher speed.

The elevated tracks will be carried on box-shaped/U-shaped/Double U-shaped elevated decking supported by single circular piers, generally spaced at 25-m centres and located on the median of the road to extent possible. The horizontal alignment and vertical alignment are, therefore, dictated to a large extent by the geometry of the road and ground levels followed by the alignment.

The underground tracks will be carried in separate tunnels to be drilled by Tunnel Boring Machine. Stations will, however, be constructed by cut and cover method or NATM technique. Two stations are planned to be stacked stations.

0.4.2 Geometric Design Parameters

The design parameters related to the Metro system described herewith have been worked out based on a detailed evaluation, experience and internationally

accepted practices. Various alternatives were considered for most of these parameters but the best-suited ones have been adopted for the system as a whole.

0.4.3 Horizontal Alignment

As far as possible, the alignment follows the existing roads. This leads to introduction of horizontal curves. On consideration of desirable maximum cant of 110 mm and cant deficiency of 85 mm on Metro tracks, the safe speed on curves of radii of 300 m or more is 80 km/h. On elevated sections minimum radius of 120m has been used at one location having speed potential upto 40 km/h. However in underground section desirable minimum radius of curve shall be 300 m for ease of working of Tunnel Boring Machine (TBM).

0.4.3.1 Horizontal Curves

Table 0.5 : Horizontal Curves

Description	Underground Section	Elevated Section
Desirable Minimum radius	300 m	200 m
Absolute minimum radius	200 m (only c/c)	120 m
Minimum curve radius at stations	1000 m	1000 m
Maximum permissible cant (Ca)	125 mm	125 mm
Maximum desirable cant	110 mm	110 mm
Maximum cant deficiency (Cd)	85 mm	85 mm

0.4.3.2Transition Curves

It is necessary to provide transition curves at both ends of the circular curves for smooth riding on the curves and to counter act centrifugal force. Due to change in gradients at various locations in the corridor, it is necessary to provide frequent vertical curves along the alignment. In case of ballast less track, it is desirable that the vertical curves and transition curves of horizontal curves do not overlap. These constraints may lead to reduced lengths of transition curves at certain locations. The transition curves have certain minimum parameters:

Length of Transitions of Horizontal curves (m)

Minimum : 0.44 times actual cant or cant deficiency (in mm), whichever is higher.

Desirable : 0.72 times actual cant or cant deficiency, (in mm), whichever is higher.

Overlap between transition curves and vertical curves not allowed.

Minimum straight between two Transition curves (in case of reverse curves): either 25 m or Nil.

Minimum straight between two Transition curves (in case of same flexure curves): either 25 m or both curves should be converted in to the compound curve by introducing single transition between the two circulars.

Minimum curve length between two transition curves: 25 m

0.4.3.3Vertical Alignment and Track Centre

(a) Elevated Sections

The viaducts carrying the tracks will have a vertical clearance of minimum 5.5 m above road level. For meeting this requirement with the 'Box' shaped pre-stressed concrete girders, the rail level will be about 9.8 m above the road level with U-girder, rail level may come to about 9.5 m above ground level. However, at stations which are located above central median, the rail level will be 13.5 m above the road level with concourse at mezzanine. These levels will, however, vary marginally depending upon where the stations are located.

(b) Underground sections

Rail level at midsection in tunneling portion shall be kept at least 12.0 m below the ground level. At stations, the desirable depth of rail below ground level is 13.5 m, so that station concourse can be located above the platforms. However, this rail level may be further lowered if alignment passing under built-up areas.

Track center in underground sections are follows:

Track center in underground sections are follows: Sections where stations are to be constructed by cut & cover and running section by TBM to accommodate 13 m wide platform	16.05 m (for lesser width of platform, track center to be reduced)
Sections where stations are to be constructed by NATM and running section by TBM to facilitate Construction of stations	22.00 m

(c) Gradients

Normally the stations shall be on level stretch. In exceptional cases, station may be on a grade of 0.1 % between stations, generally the grades

may not be steeper than 3.0 %. However, where existing road gradients are steeper than 2% or for Switch Over Ramps gradient up to 4% (compensated) can be provided in short stretches on the main line.

(d) Vertical Curves

Vertical curves are to be provided when change in gradient exceeds 0.4%. However, it is recommended to provide vertical curves at every change ofgradient.

(e) Radius of vertical curves:

On main line		
(i) Desirable	:	2500 m
(ii) Absolute minimum	:	1500 m
Other Locations	:	1500 m
Minimum length of vertical curve	:	20 m

0.4.4ROUTE ALIGNMENT

02 (two) metro corridors identified for their inclusion in Surat Metro Rail Network (Phase-I) are as follows:-

Table 0.7: Route Alignment

S.No.	Corridors	Length (Km)
01	Sarthana-Dream City	21.61 Kms
02	Bhesan-Saroli	18.74 Kms
	Total	**40.35 Kms**

0.5 CIVIL ENGINEERING

0.5.1 Underground construction

For the underground section running under the road, cut and cover method of the underground construction can be employed for the construction of the underground sections. However keeping in view the inconvenience to the traffic movement, it is proposed to tunnel through by using Tunnel Boring Machine (TBM) or New Austrian Tunneling Method (NATM) in the overburden soil mass. Tunnel excavation for a major length of underground section is expected to be carried out by Tunnel Boring Machines. There are few stretches along the underground alignment where Cut & Cover method has been considered for construction after Switch Over Ramp (SOR). Tunnel boring machines (TBMs)

capable of drilling in soft soil with a minimum finished internal diameter of 5.6 m. can be successfully employed for boring tunnels through this stratum. The tunnels are proposed with a minimum overburden of 6.0m.

0.5.2 Underground Stations

All the 06 underground stations have been proposed to be constructed by cut and cover with top-down or bottom up methods depending upon geological conditions and land/space availabilities. The diaphragm walls for such station constructions would be 80 to 120 cm. thick and will function as a permanent side wall of the station. It is, therefore, necessary to construct the diaphragm walls absolutely watertight and with the required concrete strength as has been done in other phases of Delhi Metro station construction. By resorting to top-down method the surface could be restored quickly and further excavations and construction of the station will not hamper the surface activity.

0.5.3 Cut and Cover Method of Construction of Underground Stations

Cut and Cover mainly consists of following steps:

1. Diversion of utilities
2. Construction of support walls
3. Excavation between the support walls along with the installation of struts between the two walls to keep them in position.
4. Construction of tunnel/structure and removal of temporary struts.
5. Back filling and restoration of the surfaces

0.5.4 Utility Diversion:

It is suggested that all utilities falling within excavation area are diverted away in advance to avoid damage to such utilities during the excavation/ construction phase. The cross utilities, however has to be kept supported. It is suggested that pressure water pipelines crossing the proposed cut area are provided with valves on both sides of the cut so that the cut area can be isolated in case of any leakage to the pipeline to avoid flooding of the cut/damage to the works.

0.5.5 Elevated Section - Choice of Superstructure

The choice of superstructure has to be made keeping in view the ease of constructability and the maximum standardization of the formwork for a wide span ranges.

The segmental construction has been chosen mainly due to the following advantages:

- Segmental construction is an efficient and economical method for a large range of span lengths and types of structures. Structures with sharp curves and variable super elevation can be easily accommodated.

- Segmental construction permits a reduction of construction time as segments may be manufactured while substructure work proceeds and assembled rapidly thereafter.
- Segmental construction protects the environment as only space required for foundation and sub-station is required at site. The superstructure is manufactured at a place away from busy areas and placement of superstructure is done with the system erected from piers at heights.

- Segments are easy to stack in the casting yard/stacking yard in more than one layer, thereby saving in requirement of space.

- It is easier to transport smaller segments by road trailers on city roads.

- It is easy to incorporate last minute changes in span configuration if the site situation so warrants.

- Interference to the traffic during construction is significantly reduced.

- Segmental construction contributes towards aesthetically pleasing structures and good finishes.

- The overall labour requirement is less than that for conventional methods.
- Better quality control is possible in the casting yard
.
- During construction, the technique shows an exceptionally high record of safety.

0.5.6 Types of Superstructures for Elevated Section

(a) Pre-cast segmental box girder using external unbounded tendon
(b) Pre-cast segmental U-Channel Superstructure with internal pre-stressing.
(c) Double U-girder with internal pre-stressing
(d) I-girder with internal pre-stressing (in transition spans)

0.5.7 Construction Methodology

For the elevated sections it is recommended to have pre-cast segmental construction for super structure for the viaduct. For stations also the superstructure is generally of pre-cast members. The pre-cast construction will have following advantages:-

- Reduction in construction period due to concurrent working for substructure and superstructure.
- For segmental, pre-cast element (of generally 3.0m length), transportation from construction depot to site is easy and economical.
- Minimum inconvenience is caused to the public utilizing the road as the superstructure launching is carried out through launching girder requiring narrow width of the road.
- As the pre-cast elements are cast on production line in a construction depot, very good quality can be ensured.
- The method is environment friendly as no concreting work is carried at site for the superstructure.

0.5.8 Pre-cast Construction

For viaducts segmental pre-cast construction requires a casting yard. The construction depot will have facilities for casting beds, curing and stacking area, batching plant with storage facilities for aggregates and cement, site testing laboratories, reinforcement steel yard and fabrication yard etc. An area of about 3.0 ha to 3.5 ha is required for each construction depot.

For casting of segments both long line and short line method can be adopted. However the long line method is more suitable for spans curved in plan while short line method is good for straight spans. A high degree of accuracy is required for setting out the curves on long line method for which pre calculation of offsets is necessary. Match casting of segments is required in either method. The cast segments are cured on the bed as well as in stacking
yard. Ends of the segments are to be made rough through sand blasting so that gluing of segments can be effective.

The cast segment will be transported on trailers and launched in position through launching girders.

0.5.8.1 Launching Scheme

Launching girder is specially designed for launching of segments. The suggested launching scheme is designed in such a way that initially the launching girder is erected on pier head at one end of the work. The segments are lifted in sequence and when the lifting is over, they are dry matched while hanging from the launching girder. After dry matching, the Segments are glued with epoxy and pre-stressed from one end. The girder is lowered on the temporary / permanent bearings after pre-stressing. The launching girder then moves over the launched span to next span and the sequences continue.

0.5.9 Construction of Stations

It is proposed to construct the elevated stations with elevated concourse over the road at most of the locations to minimize land acquisition. Because of different entity than viaduct, a separate structural configuration is required to be proposed, although this may necessitate the break in the launching operations at each station location

In the cross section of station structure, viaduct column will be located on the median/footpath and other supporting columns/piers shall be provided on other sides of the road within ROW.

In some the cases which suffer space constraint, the cross section is supported by a single viaduct column and cantilever arms to eliminate the columns in the right of way.

0.5.10Grade of Concrete

It is proposed to carry out construction work with design mix concrete through computerized automatic Batching Plants with following grade of concrete for various members as per design requirement/durability considerations.

i)	Piles	-	M -35
ii)	Pile cap and open foundation	-	M -35
iii)	Piers	-	M -40
iv)	All precast element for viaduct and station	-	M -45
v)	Cantilever piers and portals	-	M -45/M -60
vi)	Other miscellaneous structure	-	M -30

For all the main structures, permeability test on concrete sample is recommended to ensure impermeable concrete.

0.5.10.1 TUNNELLING

0.5.10.1.1 TBM

A tunnel boring machine (TBM), also known as a "mole", is a machine used to excavate tunnels with a circular cross section through a variety of soil and rock strata.

Tunnel boring machines are used as an alternative to drilling and blasting methods in rock and conventional "hand mining" in soil. TBMs have the advantages of limiting the disturbance to the surrounding ground and producing a smooth tunnel wall. This significantly reduces the cost of lining the tunnel, and makes them suitable to use in heavily urbanized areas. This is because tunnelling with TBMs is much more efficient and results in shortened completion times.

0.5.10.1.2 NATM

The New Austrian tunnelling method (NATM), also known as sequential excavation method (SEM), is a method of modern tunnel design and construction. The name NATM was intended to distinguish it from the old Austrian tunnelling approach. The fundamental difference between this new method of tunnelling, as opposed to earlier methods, comes from the economic advantages made available by taking advantage of the inherent geological strength available in the surrounding rock mass to stabilize the tunnel.

0.5.10.1.3 Cross Passage

Cross-passages are reinforced concrete structures built in between either two tunnels (as in case of twin tunnel systems) or a tunnel and the ground surface. They are provided to serve two primary purposes namely, emergency escape and maintenance work. As a standard practice, during construction of a tunnel system, depending upon length of the tunnel and soil characteristics, a pre-calculated number of cross-passages are planned.

0.5.10.1.4 Precast Tunnel Segments

Tunnels play a key role in assisting the movement of people, goods, and special services. The functionality of tunnels depends on the structural and durability performance of its **lining system.** Tunnel lining systems act as lines of defense against large overburden loads and complex geotechnical surrounding conditions. The use of **precast concrete tunnel segments** has been escalating due to its efficient and economical installation process compared to that of normal cast in-situ lining practice. Generally, precast concrete tunnel segments

are designed for 120 years of service life. Normally, each ring consists of 05 segments and 01 key stone. Each segment weighs about 3 tons.

0.5.11 Geo-technical Investigations

0.5.11.1 Methodology of Investigation

The investigation has been planned to obtain the subsurface stratification in the proposed project area and collect soil samples for laboratory testing to determine the engineering properties such as shear strength, along with basic engineering classification of the subsurface stratum to arrive at the foundation design parameters.

For Geotechnical investigation work, boring / drilling rig has been installed at the specified borehole locations. Stability of rig was ensured by making the ground level. Boring has been advanced by shell and auger method in soil and sampling carried out at regular interval in the bore hole.

The rig deployed was suitable for and had arrangement for boring, conducting Standard Penetration Test (SPT), collection of Undisturbed Soil Sample (UDS) and Disturbed Soil Samples (DS).

0.5.11.2 Standard Penetration Tests (SPT)

Standard Penetration Tests (SPT) has been conducted at different depths in these bore holes. SPT split spoon sampler of standard dimensions has been driven into the soil from the borehole bottom using 63.5 kg hammer falling from 75 cm height. The SPT weight was mechanically lifted to the specified height and allowed to fall freely on the anvil with the use of cathead winch with one to one and half turn of the drum. The efficiency of the SPT blows is expected to be 65 to 75% in this system, and can be directly used in most of analyses involving “N” values.

Blow counts for the penetration of every 15 cm were recorded and the N is reported as the blow counts for 30 cm penetration of the sampler leaving the first 15 cm penetration as seating driver.

When the number of blows exceeded 50 to penetrate the first or second 15 cm length of the sampler, the SPT N is regarded as more than 100 as described in IS 2131 – 1981. The test is terminated in such case and a record of penetration of the sampler under 50 blows is made. SPT refusal is recorded when there is no penetration of the sampler at any stage and also when a rebound of the sounding

system is recorded. These tests were conducted at close intervals of 1.5m at shallow depths so that a continuous SPT N profile is available.

0.5.11.3 Disturbed Sampling in Boreholes:

Disturbed soil samples collected in the SPT sampler have been preserved in polythene covers and transported to the laboratory. One more polythene cover was provided to prevent the loss of moisture during the transit period.

0.5.11.4 Undisturbed Sampling in Boreholes

Undisturbed samples were collected using 100mm dia and 600mm long MS tubes as per IS 2132:1981 provided with sampler head with ball check arrangement. These samples are expected to be “good” for conducing direct shear tests and for determining filed density and void ratio.

The laboratory testing was done following the testing procedures given in the relevant parts of IS-2720. No rock has been encountered near both the corridors of Surat Metro under Phase I

0.5.11.5 Laboratory Investigation

The following laboratory tests were conducted on the selected samples recovered from the test boreholes:

(A) Tests conducted on Soil Samples

(a) Particle size analysis:

(i) Sieve analysis

(ii) Hydrometer analysis

(b) Consistency tests

(i) Liquid limit

(ii) Plastic limit

(c) Bulk & Dry density

(d) Water content

(e) Specific gravity

(f) Direct shear test

(g) Tri-axial test

All the above laboratory tests were carried out as per relevant Indian Standards. All the samples were identified and classified as per IS: 1498-1970.

0.5.11.6 General Geology & Related Characteristics

The sites investigated are located in Surat City for both the corridors proposed under Surat Metro Rail network (Phase-I). The general features are described below:

Introduction

Surat is a port city situated on the banks of the Tapi river. Damming of the Tapi caused the original port facilities to close; the nearest port is now in the Magdalla and Hazira area of Surat Metropolitan Region. The city is located at 21°10′N 72°50′E& has a total geographical area of 4109.74 sq. km. The district is situated in the southern part of the state. It is bounded by the Arabian Sea on the west, Bharuch in the North, the Valsad district in the south, Dangs district in the south-east, and the Tapi district of the in the east. The district has been divided into nine talukas.

Geology

Major geological formations exposed in the district are Quaternary alluvium, Tertiary limestone and sandstones and Deccan Trap basalt.

Geographically, the district belongs to the western coastlands of the Deccan peninsula. The main Sahyadri scarp is a little outside the limits of the district towards east, but it gives the district its orientation, landscape features and drainage pattern. Distinct zones, viz., hilly areas, piedmont slopes, alluvial plains and coastal plains.

Drainage

The Tapi is the major river which passes through the central parts of the district and flows towards the west. The river is perennial in nature. It originates in Madhya Pradesh near Betul and has about 62225 sq. Km. of catchment area. The average width of the river upstream of Kathor bridge in about 500m.Downstream of the bridge the average being about 700m.

Soil Type

The basaltic lava flows are covered by black clayey to loamy soil. It is in general ranges in thickness up to one meter. The colour of the soil turns brown due to high iron content at places. In the piedmont slope area the soil is shallow to moderately deep, moderate to severely eroded and non calcareous in nature.

The texture is silt clay loam to clay loams. The clay content varies from 30 to 60%. The water holding capacity of the soil is moderate. In the midland and flood plain areas, the soil is deep to very deep, light greyish to yellowish brown in colour. The texture is fine clay loam to sandy loam. The clay content varies from 25 to 60%. In coastal region the soil is deep to very deep dark grey to black colour. The texture is clay loam to silty loam. The area is affected by tide as well as leaching of salts from up land forming saline alkali soils. The content of clay in this is high and permeability is low (mud flats).

Hydrometeorology

The district has the meteorological station in the district town Surat, therefore, the climatologically data of Surat IMD station is discussed here. There are many rain gauge stations being monitored by different state government agencies. The Water Resources Investigation Circle (WRI) under the department of Narmada Water Resources, Govt. of Gujarat, monitors most of the rain gauges stations and also collects and compiles the rainfall data collected by different agencies.

0.5.12 Land

In order to minimise land acquisitions and to provide good accessibility form either directions, the metro alignments are located mostly along the center of the roads, which lie on the corridor. But, at some locations the geometrics of the roads especially at road turnings may not match with geometric parameters required for metro rail systems. In such cases, either the alignment will be off the road or some properties abutting the road would get affected. Further, some land is required for various purposes as detailed below:

0.5.12.1 Land Requirement for following Major Components

MRTS Structure (including Route Alignment), Station Building, Platforms, Entry/Exit Structures, Traffic Integration Facilities, Depots, etc.
Receiving/Traction Sub-stations
Radio Towers
Temporary Construction Depots and work sites.
Staff quarters, office complex and operation control centre(OCC)

0.5.12.2 Land for Underground stretches

No land at surface is required permanently for underground section, except for small areas for entry/exit structures, traffic integration, chilling plant and ventilation shafts at stations. These will be located either on footpath edge or in front marginal open setback of the building along the road.

0.5.12.3 Land required for elevated stretches

For elevated section, single pier supporting the viaduct will be located on the middle of road so that the existing roads remain in use as usual. Accordingly, necessary permission for using such right-of-way will have to be obtained from the concerned authorities. Elevated station is generally proposed with elevated concourse so that land is required only for locating the entry/exit structures. Traffic integration facilities are provided wherever the same are required and, but no land is proposed for acquisition.

The normal viaduct structure of elevated Metro is about 10 m (edge to edge) wide. Ideally the required right of way is 10m. However, for reasons of safety a clean marginal distance / setback of about 5 m is necessary from either edge of the viaduct (or 10 m on both sides of the centre line) wherein no structures are to be located. Also, it ensures road access and working space all along the viaduct for working of emergency equipments and fire brigade. In stretches, where the elevated alignment has to be located away from road, a strip of 20-m width is proposed for acquisition.

In view of the constraints on space on ground floor, it is proposed to provide the concourse area exactly below the Station Building at mezzanine level. All the stations in elevated stretch including terminal station are planned with side platforms. Normally, the ideal width required for station building in is 28.0m. The staircase giving access to concourse area from ground will be located at the edge of footpaths or in front marginal open setback of the buildings in the as far as possible in the open space. Nevertheless it is not possible to find open space at all the locations therefore acquisition of certain private structures is inevitable.

0.5.12.4 Land for Switch-over Ramps

Switch-over ramps are required for transition from the underground to elevated section or *vice versa*. The ramp covers a stretch at ground for the whole width of structure for two tracks (about 10.5m including the protection works). The length of ramp above ground depends on the existing ground slope and the gradient provided on Metro alignment (normally 3% to 4%). Thus the ramp is to be located in an area where sufficient road width is available or in an open area. On this corridor, three such ramps are provided on the both the corridors.

0.5.12.5 Land for Traction and Receiving Substation and Radio Towers

Two RSS are proposed to be located for Sarthana-Dream City Corridor (i) near Nature Park (4000 Sq.m) and (ii) Kapodra Station (3000 Sq.m)and one RSS having area as (4000 Sq.m) near Magub Station for Bhesan to Saroli Corridor. Hence, an area of 11000 Sq.m has been earmarked for RSS. No additional land proposed for locating radio towers. These will be accommodated in the land already acquired. Land required for RSS will be as under:-

Table 0.7 : Details of Land for RSS

S. No.	Name of Corridor	Area (m^2)	Ownership
1.	Sarthana-Dream City	7000	Government
2.	Bhesan – Saroli	4000	Government
	Total	**11000**	

0.5.12.6 Land Requirement for Stations & Running section

As indicated earlier, the ROW of the roads along which the alignment is planned is sufficiently wide and hence no land is required for acquisition as long as the alignment is straight and in the centre of the road. However, at curved portions, the alignment could not be kept in the centre of the road and land acquisition at such locations is inevitable in spite of introduction of sharper curves.

To the extent possible the Entry and Exit points of stations (underground and elevated) were planned on the foot paths. But, for locating other station facilities such as chiller plants, ventilation shafts, underground water tanks, generator set room etc., land acquisition is proposed.

0.5.12.7 Summary of Land Requirement

Abstract of land requirements for different components of this corridor is given in following Table .However, the land requirement is summarized below:

a) Govt. Land permanently required for stations, Depot,Ramp and running section:**52.65ha.**

b) Private Residential and Commercial Required for stations, Ramp and Running section**:1.66 ha.**

c) **Total land required for the metro project**: **54.31 ha.**

d) Govt. Land for Property development(PD)

Corridor-I (Sarthana to Dream City) : 2.98 ha
Corridor-II(Bhesan to Saroli) :1.72 ha
Development of Stations on PPP mode : 1.07 ha
Total : 5.77 ha

Table 0.8 - Summary of Permanent Land Requirement

(All figures in Sq.m)

S.No.	Description	Sarthana to Dream City		Bhesan to Saroli	
		Govt.	Pvt.	Govt.	Pvt.
1.	Station	15,283.70	6215.10	8245	10351.90
2.	Running Section	3492.82		1301.94	
3.	Ramp	11160.00			
4.	Depot	240900		169500	
5.	Staff Quarter Office Complex and OCC	25000			
6.	RSS	7000		4000	
7.	Mid Shaft	2500			
8	Parking	19145.7	0	18948.70	0
	Total	324482.22	6215.10	201995.64	10351.90Sq.m

0.6 STATION PLANNING

0.6.1 General

The proposed Metro Rail at Surat has two corridors covering an approximated distance of about 40.35 Kilometer. The Corridor-I is partially Underground and partially Elevated having 20 stations and Corridor-II is entirely Elevated having 18 stations thus a total of 38 stations are proposed in entire Phase-I. Almost all the stations are on the road stations Kadarsha-Ni-Nal station is a part of development plan and is designed on the model of TOD (Transit Oriented Development) station having Property Development around the station.

Corridor-I run from Sarthana, Nature Park, Varchha Chopati Garden, Shri Swaminarayan Mandir Kalakunj, Kapodara, Labheshwar Chowk, Central Warehouse, Surat Railway Station, Maskati Hospital, Gandhi Baug, Kadarsha Ni Nal, Majura Gate, Rupali Canal, Althan Tenament, Althan Gam, VIP Road, Women ITI, Bhimrad Convention Centre and Dream City.

The corridor-I covers a total distance of 21.61 Km from dead end to dead end of which about 6.47 km is underground excluding Ramps on either side and balance corridor is elevated. It has 20 stations out of which 6 stations are underground and 14 stations are elevated.

Corridor II runs from Bheshan, Botanical Garden, Ugat Vaarigruh, Palanpur Road, L.P. Swani School, Performing Art Centre, Adajan Gam, Aquarium, Badri Narayan Temple, Athwa Chopati, Majura Gate, Udhana Darwaja, Kamela Darwaja, Anjana Farm, Model Town, Magob, Bharat Cancer hospital and Saroli.

It covers a total distance of about 18.74 Kilometer from Dead End to End of Track and the entire corridor is elevated stations having 18 stations.

The Stations have been located so as to serve passenger requirements and to enable convenient integration with other modes of transport. Efforts have been made to propose station locations at a uniform inter-station distance as feasible. Average inter-station distance is ~1 Kilometer, though it varies from 0.5867 Km to 1.8028 km due to land-use and topographic reasons.

The Minimum Inter-station distance in corridor-I between Central Warehouse Station and Surat Railway Stationis 586.70 meters and in corridor-II 590.2 meters is between L.P. Swani School and Performing Art Centre. The Maximum distance between stations on corridor-I is between Maskati Hospital and Gandhi Baug and 1802.80 meters and in corridor-II between Bharat Cancer Hospital and Saroli 1442.40 meters.

0.6.2 Rail Levels and Alignment

The rail levels in the stations are decided to provide sufficient clearance (Head room) for the traffic on the road. In the Elevated section, Rail Level is generally about 13 meter above Ground in order to maintain a clearance of 5.50 meter between the Road and the Station Structure. The alignment is planned generally in middle of the road in order to keep the land acquisition to minimum, and a two-level station design has been proposed. Entry/exit structures to the proposed stations and traffic integration areas have been planned in the open spaces available on the road sides.

0.6.3 Platforms

The stations have been planned with side platforms to avoid the viaduct structure from flaring in and out at stations, which obstructs the road traffic below. Care

has been taken to locate stations on straight alignment. However, in some stations, site constraints have become the deciding criteria and a curve of 1000m radius has been introduced in platform.

0.6.4 Planning and Design Criteria for Stations

1. The stations can be divided into public and non-public areas (those areas where access is restricted). The public areas can be further subdivided into paid and unpaid areas.
2. The platform level has adequate assembly space for passengers for both normal operating conditions and a recognized abnormal scenario.

3. The platform level at elevated stations is determined by a critical clearance of 5.5m under the concourse above the road intersection, allowing 3.3m for the concourse height, about 0.8m for concourse floor and 1.8 m for structure of tracks above the concourse. Further, the platforms are 1.09-m above the tracks. This would make the rail level in an elevated situation at least 13 meters above ground.

4. In the underground stations, platform level is determined by a critical clearance of 2.0m - 2.50-m above the station box, which would be 13.7 high. Allowing about 80 cm for the box structure, ~70 cm for rails /supporting structure and 1.09 m for rail to platform ht, would make the platforms in an underground situation at least 13.5-m below ground.

5. In the case of Up and Down Line Platforms Decked above each other the Rail level of First Platform is about -11.5 m and that of the Second Platform is about -20.5m below the ground.

6. The concourse contains automatic fare collection system in a manner that divides the concourse in two distinct areas. The “Unpaid Area” is where passengers gain access to the system, obtain travel information and purchase tickets. On passing through the ticket gates, the passenger enters the “Paid Area”, which includes access to the platforms.

7. The arrangement of the concourse is assessed on a station-by-station basis and is determined by site constraints and passenger access requirements. However, it is planned in such a way that maximum surveillance can be achieved by the ticket hall supervisor over ticket machines, automatic fare collection (AFC) gates, stairs and escalators. Ticket machines and AFC gates are positioned to minimize cross flows of passengers and provide adequate circulation space.

8. Sufficient space for queuing and passenger flow has been allowed at the ticketing gates.

9. Station entrances are located with particular reference to passenger catchment points and physical site constraints allowing for 30 meter right-of-way in order to provide a minimum of 3 lane road under the station builidng on either side of the median.

10. Office accommodation, operational areas and plant room space is required in the non-public areas at each station. The functions of such areas are given below in Table 5.5.1

11. The DG set, Bore Well, Pump House and Underground Water Tanks would be located generally in one area on ground within the Entry / Exit structures.

12. The system is being designed to maximize its attraction to potential passengers and the following criteria have been observed:

    Minimum distance of travel to and from the platform and between platforms for transfer between lines.

    Adequate capacity for passenger movements.

    Convenience, including good signage relating to circulation and orientation.

    Safety and security, including a high level of protection against accidents.

13. Following requirements have been taken into account:

    Minimum capital cost is incurred consistent with maximizing passenger attraction.

    Minimum operating costs are incurred consistent with maintaining efficiency and the safety of passengers.

    Flexibility of operation including the ability to adapt to different traffic conditions changes in fare collection methods and provision for the continuity of operation during any extended maintenance, repair period, etc.

    Provision of good visibility of platforms, fare collection zones and other areas, thus aiding the supervision of operations and monitoring of efficiency and safety.

    Provision of display of passenger information and advertising.

14. The numbers and sizes of staircases/escalators are determined by checking the capacity against AM and PM peak flow rates for both normal and emergency conditions such as delayed train service, fire etc.

15. I n order to transfer passengers efficiently from street to platforms and vice versa, station planning has been based on established principles of pedestrian flow and arranged to minimize unnecessary walking distances and cross-flows between incoming and outgoing passengers.

16. Passenger handling facilities comprise of stairs/escalators, lifts and ticket gates required to process the peak traffic from street to platform and vice-versa (these facilities must also enable evacuation of the station under emergency conditions, within a set safe time limit).

0.7TRAIN OPERATION PLAN

0.7.1 Operation Philosophy

The under lying operation philosophy is to make the MRT System more attractive and economical, the main features being:

- Selecting the most optimum frequency of Train services to meet sectional capacity requirement during peak hours on most of the sections.
- Economical & optimum rain service frequency not only during peak period, but also during off-peak period.
- A short train consists of 3 coaches.
- Multi-tasking of train operation and maintenance staff.

0.7.2Train Operation Plan

Based on the projected PHPDT demand, Train operation plan with train carrying capacity @ 6 persons per square meter of standee area for the Surat Metro Corridor-I: Dream City-Sarthana' &'Corridor-II: Bhesan-Sroli' for the year 2021,2026,2036and 2046 are given below:

1. <u>Corridor-I:DreamCity–Sarthana</u>

Train Operation Plan for Corridor-I: Dream City-Sarthana has been planned in such a way that there is Single loop of train operation end to end.

i) Year 2021:

Train operation in 'Dream City to Sarthana' Loop is at 6.5min headway with3-Car train. This results in following train operation:

'Dream City to Sarthana' Section (Refer Attachment I/A1)

- 6.5 min Effective Headway with 3-car train.
- Available Peak Hour Peak Direction Capacity of 7052 @ 6 persons per square meter of standee area.
- Available Peak Hour Peak Direction Capacity of 8972 @ 8 persons per square meter of standee are a under dense loading conditions.
- The maximum PHPDT demand of 6992 is in the Section between Maskati Hospital to Surat Railway Station and demand in the remaining sections is in the range of 6609 to 119 only. The planned capacity of 7052 (8972 under dense loading) is more than the PHPDT demand.

Traffic demand and train capacity for this corridor is the year 2021 is tabulated and represented on a chart enclosed as Attachment I/A1.

ii) Year2026:

Train operation in 'Dream City to Sarthana' Loop is at 3.5 min head way with3-Car train. This results in following train operation:

'Dream City to Sarthana' Section (Refer Attachment I/B1)

3.5min Effective Headway with 3-car train.
Available Peak Hour Peak Direction Capacity of 13097 @ 6 persons per square meter of standee area
Available Peak Hour Peak Direction Capacity of 16663 @ 8 persons per square meter of standee area under dense loading conditions.
The maximum PHPDT demand of 13536 is in the Section between Surat Railway Station to Central Ware House and demand in the remaining sections is in the range of 13495 to 615 only. The planned capacity of 13097 (16663 under dense loading) is slightly less than the PHPDT demand in only two (zero, with dense loading capacity) sections out of nineteen sections.

Traffic demand and train capacity for this corridor is the year 2026 is tabulated and represented on a chart enclosed as Attachment I/B1.

iii) Year2036:

Train operation in 'Dream City to Sarthana' Loop is at 2.25 min headway with3-Car train. This results in following train operation:

'Dream City to Sarthana'Section(Refer Attachment I/C1)

- 2.25min Effective Headway with 3-car train.
- Available Peak Hour Peak Direction Capacity of 20373 @ 6 persons per square meter of standee area.
- Available Peak Hour Peak Direction Capacity of 25920 @ 8 persons per square meter of Standee area under dense loading conditions.
- The maximum PHPDT demand of 20856 is in the Section between Chowk Bazar to Maskati Hospital and demand in the remaining sections is in the range of 20677 to 2150 only. The planned capacity of 20373 (25920 under dense loading) is slightly less than the PHPDT demand in only two (zero, with dense loading capacity) sections out of nineteen sections.

Traffic demand and train capacity for this corridor is the year 2036 is tabulated and represented on a chart enclosed as Attachment I/C1.

iv) Year2046:

Train operation in 'Dream City to Sarthana' Loop is at 1.75 min headway with3-Car train. This results in following train operation:

'Dream City to Sarthana' Section (Refer Attachment I/D1)

- 1.75 min Effective Headway with 3-car train.
- Available Peak Hour Peak Direction Capacity of 26194 @ 6 persons per square meter of standee area.
- Available Peak Hour Peak Direction Capacity of 33326 @ 8 persons per square meter of standee area under dense loading conditions.
- The maximum PHPDT demand of 26587 is in the Section between Surat Railway Station to Central Ware House and demand in the remaining sections is in the range of 25236 to 3095 only. The planned capacity of26194 (33326 under dense loading) is slightly less than the PHPDT

demand in only one (zero, with dense loading capacity) section out of nineteen sections.

- Traffic demand and train capacity for this corridor is the year 2046 is tabulated and represented on a chart enclosed as Attachment I/D1.

2. Corridor-II: Bhesan– Saroli

Train Operation Plan for Corridor-II: Bhesan- Saroli has been planned in such a way that there is Single loop of train operation end to end.

i) Year2021:

Train operation in 'Bhesan to Saroli' Loop is at10min headway with 3-Car train. This results in following train operation:

'Bhesan to Saroli' Section (Refer Attachment I/A2)

- 10 min Effective Headway with 3-car train.
- Available Peak Hour Peak Direction Capacity of 4584 @ 6 persons per square meter of standee area.
- Available Peak Hour Peak Direction Capacity of 5832 @ 8 persons per square meter of standee area under dense loading conditions.
- The maximum PHPDT demand of 3918 is in the Section between Udhana Darwaja to Kamela Darwaja and demand in the remaining sections is in the range of 3848 to 245 only. The planned capacity of 4584 (5832 under dense loading) is more than the PHPDT demand.

Traffic demand and train capacity for this corridor is the year 2021 is tabulated and represented on a chart enclosed as Attachment I/A2.

ii) Year2026:

Train operation in 'Bhesan to Saroli' Loop is at 8 min headway with 3-Car train. This results in following train operation:

'Bhesan to Saroli' Section (Refer Attachment I/B2)

- 8 min Effective Headway with 3-car train.
- Available Peak Hour Peak Direction Capacity of 5730 @ 6 persons per square meter of standee area.
- Available Peak Hour Peak Direction Capacity of 7290 @ 8 persons per square meter of standee area under dense loading conditions.

- The maximum PHPDT demand of 5698 is in the Section between Udhana Darwaja to Kamela Darwaja and demand in the remaining sections is in the range of 5666 to 629 only. The planned capacity of 5730 (7290 under dense loading)is more than the PHPDT demand.

- Traffic demand and train capacity for this corridor is the year 2026 is tabulated and represented on a chart enclosed as Attachment I/B2.

iii) Year2036:

Train operation in 'Bhesan to Saroli' Loop is at 3.75 min headway with 3-Car train. This results in following train operation:

'Bhesan to Saroli' Section (Refer Attachment I/C2)

- 3.75min Effective Headway with 3-car train.
- Available Peak Hour Peak Direction Capacity of 12224 @ 6 persons per square meter of standee area.
- Available Peak Hour Peak Direction Capacity of 15552 @ 8 persons per square meter of standee area under dense loading conditions.
- The maximum PHPDT demand of 12573 is in the Section between Badri Narayan Temple to Athwa Chaupati and demand in the remaining sections is in the range of 12098 to 2649 only. The planned capacity of 12224 (15552 under dense loading) is slightly less than the PHPDT demand in only one (zero, with dense loading capacity) section out of seventeen sections.

Traffic demand and train capacity for this corridor is the year 2036 is tabulated and represented on a chart enclosed as Attachment I/C2.

v) Year 2046

Train operation in 'Bhesan to Saroli' Loop is at 3 min headway with3-Car train. This results in following train operation:

'Bhesan to Saroli' Section(Refer Attachment I/D2)

- 3 min Effective Headway with 3-car train.
- Available Peak Hour Peak Direction Capacity of 15280 @ 6 persons per square meter of standee area.
- Available Peak Hour Peak Direction Capacity of 19440 @ 8 persons per square meter of standee area under dense loading conditions.

- The maximum PHPDT demand of 15448 is in the Section between Badri Narayan Temple to Athwa Chaupati and demand in the remaining sections is in the range of 15175 to 3487only.The planned capacity of 15280 (19440 under dense loading) is slightly less than the PHPDT demand in only one (zero, with dense loading capacity) section out of seventeen sections.

Traffic demand and train capacity for this corridor is the year 2046 is tabulated and represented on a chart enclosed as Attachment I/D2.

The above Train Operation Plan is based on calculations on the basis of available traffic data. In case of any mismatch in the capacity provided and the actual traffic, the capacity can be moderated suitably by adjusting the Headway.

The PHPDT capacity provided on the two corridors in different years of operations tabulated below:

Table: 0.9 :Capacity Provided for Corridor-I: DreamCity– Sarthana

Sections	Year	Headway (min)	No.of Rakes	Rake Consist	No. of Coaches	Max. PHPDT Demand	PHPDT Capacity Available
Dream City to Sarthana	2021	6.5	15	3-car	45	6992	7052 (8972*)
Dream City to Sarthana	2026	3.5	26	3-car	78	13536	13097 (16663*)
Dream City to Sarthana	2036	2.25	41	3-car	123	20856	20373 (25920*)
Dream City to Sarthana	2046	1.75	5 2	3-car	156	26587	26194 (33326*)

Table: 0.10:Capacity Provided for Corridor-II: Bhesan-Saroli

Sections	Year	Headway (min)	No. of Rakes	Rake Consist	No. of Coaches	Max. PHPDT Demand	PHPDT Capacity Available
Bhesan to Saroli	2021	10	9	3-car	27	3918	4584 (5832*)
Bhesan to Saroli	2026	8	11	3-car	33	5698	5730 (7290*)
Bhesan to Saroli	2036	3.75	21	3-car	63	12573	12224 (15552*)
Bhesan to Saroli	2046	3	26	3-car	78	15448	15280 (19440*)

***** @ 8persons per square meter of standee area.

0.8TRAIN MAINTENANCE DEPOTS

0.8.1 Functions of Depot-cum-Workshop

It is proposed to establish one depot- cum- workshop for each Corridor with following functions:

(i) Major overhaul so fall the trains.
(ii) All minor schedules and repairs.
(iii) Lifting for replacement of heavy equipment and testing thereafter.
(iv)Repair of heavy equipment's.

0.8.2 Depot Planning

The Depot planning is based on following assumptions:

a. Enough space should be available for establishment of a Depot- Cum- workshop.

b. All inspection lines, workshop lines, stab lingines are designed to accommodate two train sets of 3- Car each and space earmarked for future provision.

c. All Stabling lines are designed o accommodate two trains of3- Car each.

d. Allstablinglinesareplannedintheproposeddepot-cum-workshopassuming adequate space availability. In case of space constraints, if any, stabling Facilities may need to be created atterminalstationsorelsewheretocatertotherequiredstabilityfacilities.

e. Incase of space constraint for depot storeyed Stabling lines canal so be planned.

In broad terms, based on the planned Rolling Stock requirements, this chapter covers conceptual design on following aspects and will work as a guide for detailed design later:

- Lay out of Stabling-shed, Inspection shed, minor repairs and heavy repair overhauling workshop and cleaning of Rolling Stock.
- Operational and functional safety requirements.
- Ancillary buildings for other maintenance facilities.
- Electrical & Mechanical Services, power supply and distribution system.
- Water Supplies, Drainage &Sewerage.

0.8.3 Maintenance Philosophy

Monitoring of the performance of all key Rolling Stock equipment by suitable advanced condition monitoring techniques available. The concepts to evolve the need based maintenance regime, which can be suitably configured in the form of schedules like daily check, “A” checks, “B” type checks, “IOH” and “POH”.

Labour intensive procedures are kept to the minimum. Automation with state of the art machinery to ensure quality with reliability.

Increase in the periodic maintenance intervals with predictive maintenance based on condition monitoring.

Multi skilling of the Maintenance staff to ensure quality and productivity in their performance.

Periodic review of maintenance practices to update replacement cycle of critical components based on experience.

Energy conservation is given due attention

0.8.4MaintenanceSchedule

The following maintenance schedule has been envisaged for conceptual design of depot assuming approx 264 kms. Running per train per day, taking in consideration the passenger load of 2021, 2026, 2036 & 2046 respectively.

Table 0.11 Maintenance Schedule

Type of Schedule	Interval	Work Content	Locations
Daily	Daily	Check on the train condition and function at every daily service completion. Interval cleaning/mopping off loran dwalls with Vacuum cleaner.	Stabling Lines
"A" Service Check	5,000 Km (approx.15 days)	Detailed in section and testing of sub- systems, under frame, replacement/Topping up of oils &lubricants.	Inspection Bays
"B" Service Check	15,000Km (approx.45 days)	Detailed Inspection of 'A' type tasks plus Itemsatmultiplesof15,000Km ('B' type tasks)	Inspection Bays
Intermediate Overhaul (IOH)	420,000Km, (3and half Years approx.) whichever is earlier	Check and testing of all sub-assemblies (Electrical +Mechanical).Overhaul of pneumatic valves, Compressor. Condition based maintenance of sub- systems to bring them to original condition. Replacement of parts and rectification, trial run.	Workshop
Periodical Overhaul (POH)	840,000Km, (7Years approx.) whichever is earlier	Dismantling of all sub-assemblies, bogies suspension system, traction motor, gear, control equipment, air-conditioning unit's Etc .Overhauling to bring the mto original condition. Checking repair and replacement as necessary. Inspection and trial.	Workshop
Heavy Repairs	-	Changing of heavy item such as bogies, traction motor, wheel sets/axles, gear cases& axle boxes etc.	Workshop

The above Schedule may need slight revision based on the actual earned kilometers per train and the specific maintenance requirements of Rolling Stock finally procured.

0.9 POWER REQUIREMENT

Electricity is required for operation of Metro system for running of trains, station services (e.g. lighting, lifts, escalators, signaling & telecom, fire fighting etc) and workshops, depots & other maintenance infrastructure within premises of metro system. The power requirements of a metro system are determined by peak-hour demands of power for traction and auxiliary applications. Broad estimation of auxiliary and traction power demand is made based on the following requirements:-

(i) Specific energy consumption of rolling stock 60 kWh/1000 GTKM in case of 750V dc Traction at Pantograph level is considered as per MOUD guideline vide letter No. 14011/9/2014-UT.II-Part I, dated 21.04.2017.

(ii) Elevated/at –grade station load – initially 250 kW, which will increase to 300 kW in the year 2046.

(iii) Underground station load – initially 2000 kW, which will increase to 2200 kW in the year 2046.

(iv) Depot & Metro Bhawan auxiliary load - initially 2000 kW (each), which will increase to 2200 KW in the year 2046.

Keeping in view of the train operation plan and demand of auxiliary and traction power, power requirements projected for the year 2021, 2026, 2036 and 2046 are summarized in table below:-

Table 0.12 Power Demand Estimation (MVA)

Corridor	Load	Year			
		2021	2026	2036	2046
Sarthana to Dream City	Traction	6.16	10.83	16.16	20.44
	Auxiliary	24.09	24.76	25.51	26.93
Corridor-1 20 Stations (21.61 km)	**Sub-total**	**30.25**	**35.59**	**41.67**	**47.37**
Bhesan to Saroli	Traction	3.96	5.00	9.15	11.16
	Auxiliary	8.03	8.31	8.71	9.39
Corridor-2 18 Stations (18.74 km)	**Sub-total**	**11.99**	**13.31**	**17.86**	**20.55**

0.9.1 Need for High Reliability of Power Supply

The proposed Surat metro system is being designed to handle about 26,587 passengers per direction during peak hours when trains are expected to run at 1.75 minutes intervals. Incidences of any power interruption, apart from affecting train running, will cause congestion at stations. Interruption of power at night is likely to cause alarm and increased risk to traveling public. Lack of illumination at stations, non-visibility of appropriate signages, disruption of operation of lifts and escalators is likely to cause confusion, anxiety and ire in commuters, whose tolerance level are low on account of stress. Effect on signal and communication may affect train operation and passenger safety as well. Therefore, uninterrupted power supply is mandatory for efficient metro operations.

To ensure reliability of power supply, it is essential that both the sources of Supply and connected transmission & distribution networks are reliable and have adequate redundancies built in. Therefore, it is desirable to obtain power supply at high grid voltage of 220 kV, 132 kV or 66 kV from stable grid sub-stations and further transmission & distribution is done by the Metro Authority itself.

0.9.2 Sources of Power Supply

The high voltage power supply network of Surat city was studied in brief. The city has 220 kV and 66 kV network to cater to various types of demand in vicinity of the proposed corridors. A meeting was held on dated 05.04.2017 & 06.04.2017

with M/s Torrent Power, GETCO & Surat Municipal Corporation official, and various sub-stations sites had been inspected to finalize the Input Power Supply sources & Supply Voltage.

Keeping in view the reliability requirements, Four Receiving Sub-stations (RSS) are proposed to be set up for corridor – 1 (2 RSS) and corridor – 2 (2 RSS). This is an economical solution without compromising reliability. Based on the discussions in meeting with M/s Torrent Power, M/s GETCO & Surat Municipal Corporation, it is proposed to avail power supply for traction as well as auxiliary services from the following grid sub-stations at 66 kV voltage through cable feeders

0.9.3 750VDCThird Rail Current Collection System

For the 750V dc Third Rail Current Collection System, Bottom current collection with the use of composite Aluminum steel third rail on main lines is envisaged from reliability and safety considerations. Low carbon steel third rail available indigenously is proposed for the depot because of reduced current requirements.

0.10 VENTILATION AND AIR-CONDITIONING SYSTEM

0.10.1 INTRODUCTION:

This chapter covers the Ventilation and Air-conditioning (VAC) system requirements for the underground sections of the proposed Surat Metro. It includes the following:

- Station Air-conditioning System
- Ventilation System for station plant rooms
- Station Smoke Management System
- Tunnel Ventilation System

0.10.2 Need for Ventilation and Air Conditioning

The underground stations of the Metro Corridor are built in a confined space. A large number of passengers occupy concourse and the platforms, especially at the peak hours. The platform and concourse areas have a limited access from outside and do not have natural ventilation, it is therefore, essential to provide forced ventilation and air-conditioning in the stations and inside the tunnel for the purpose of:

- Supplying fresh air for the physiological needs of passengers and the authority's staff;

- Removing body heat, obnoxious odours and harmful gases like carbon dioxide exhaled during breathing;
- Preventing concentration of moisture generated by body sweat and seepage of water in the sub-way;
- Removing large quantity of heat dissipated by the train equipment like traction motors, braking units, compressors mounted below the under-frame, lights and fans inside the coaches, A/c units etc.;
- Removing vapour and fumes from the battery and heat emitted by light fittings, water coolers, Escalators, Fare Gates etc. working in the stations;
- Removing heat from air conditioning plant and sub-station and other equipment, if provided inside the underground station.

This large quantity of heat generated in M.R.T. underground stations cannot be extracted by simple ventilation. It is, therefore, essential to provide mechanical cooling in order to remove the heat to the maximum possible extent. As the passengers stay in the stations only for short periods, a fair degree of comfort conditions, just short of discomfort are considered adequate.

0.10.3 Design parameters for VAC system

Based on the above, the following VAC system design parameters are assumed in the present report.

(1) Outside ambient conditions:

This is based upon Indian Weather Data 2017 issued by ISHRAE recommended design conditions of Surat for 1% criteria, as under

1% Criteria

Summer :	36.4 DB, 22.9 WB
Monsoon:	31.4 DB, 27.7 WB

For Surat Metro Underground Corridor it is suggested to use 1% criteria, which is defined as the conditions, when the DB or WB temperatures are likely to exceed for only 1% of the total time.

(2) Inside design conditions:

Platform/Concourse areas - 27 deg. C at 55 % RH

(3) Tunnel design conditions

Normal conditions – Max. DB 40 deg. C

Congested conditions -- Max. DB 45 deg. C

(4) Minimum fresh air - 10 % or 18 cmh / person (In station public areas).

0.10.4 Space Requirement for VAC System

The station air conditioning equipment plant rooms are normally located at each end of the concourse for the two level stations. The approximate area for air handling equipment room would be 400m^2 to 500m^2 at each end of the station. There shall be supply shafts and exhaust shafts of about 8 m^2 to 10m^2 each at each end of the stations. The space requirements for Labheshwar Chowk and Maskati Hospital U/G stations will require special considerations as the platform for UP and DN Line are at different level i.e. One-Over other.

0.11 SIGNALLING AND TRAIN CONTROL SYSTEM

0.11.1 Signalling

The Signalling shall provide the highest security level to ensure that the operational activities are developed following strict safety requirements. At the same time, it shall meet the requirements for efficient train operations and high quality of service.

The proposed signalling system design for metro line corridor is as under:

Continuous Automatic Train Control System (CATC)
Unattended Train Operation (UTO) System
Automatic Train Operation (ATO)System
Radio based Automatic Train Control (ATC) System
Automatic Train Protection (ATP) System
On board Equipment
Cab Signalling
Fall-Back Block System
Interlocking device
Track side Radio equipment
Track Vacancy Detection System
Electric Point Machine
Track side Signals
Centralized Traffic Control System
Power Supply of Signalling
Cable for Signalling
Half Height Integrated Platform Gate (PG)
Display of CCTV images from Train to OCC
Onboard Radio Antennas, Large Video Screen, MMIs, etc

0.11.2 Overview of Signalling System

It is expected to carry large number of passengers by maintaining shorter spacing between trains requiring a very high level of safety enforcement and reliability. At the same time heavy investment in infrastructure and Rolling stock necessitates optimization of its capacity to provide the best services to the people.

The requirements of the Surat Metro Corridor planned to be achieved by adopting following basic principles of signaling System: -

The Train Control and Monitoring shall be ensured from Centralized Traffic control System located at Operation Control Centre (OCC). OCC equipment shall be connected to station equipment room through optical fiber network.

The CBTC (Communication based Train Control) based system shall be provided in main line & depot (except workshop area) for train operation & primary mode of detection. Secondary detection shall be through Axle Counter.

Computer Based Interlocking System shall be designed on failsafe philosophy. In case of failure of any equipment, the equipment shall fail on safe side or more restrictive state. In such case the signalling System shall authorized movement of train in normal and degraded operations.

Track side equipment shall be connected through Electronic Interlocking (to Station Equipment Room) by secure links to ensure safe movement of train.

Provide high level of safety with trains running at shorter headways ensuring continuous safe train separation.

Eliminate accidents due to driver passing Signal at Danger by continuous speed monitoring and automatic application of brake in case of disregard of signal / warning by the driver.

Provide safety and enforce speed limit on the sections having permanent and temporary speed restrictions.

Improve capacity with safer and smoother operations. Driver will have continuous display of Target Speed in his cab enabling him to optimize the speed potential of the track section. It provides signal / speed status in the cab even in bad weather.

Increased productivity of rolling stock by increasing line capacity and train speeds, and enabling train to arrive at its destination sooner. Hence more trips will be possible with the same number of rolling stock.

Improve maintenance of Signaling and Telecommunication equipment by monitoring System status of trackside and train borne equipment and enabling preventive maintenance.

Signaling& Train Control System on the line shall be designed to meetthe required headway during peak hours.

For monitoring inside train saloon, signaling system shall provide radio transmission media to transfer live streams to OCC controller on large video screen & MMI.

To avoid any accident at platform, Integrated Passenger Gate shall be provided, which will be a barrier between the track and platform accessible to passengers. Signaling and Rolling Stock interfaces shall be provided for Passenger Gate System.

0.12 TELECOMMUNICATION AND AUTOMATIC FARE COLLECTION SYSTEMS

0.12.1Introduction

The Telecommunication system acts as the communication backbone for Signaling systems and other systems such as SCADA, AFC etc and provides Telecommunication services to meet operational and administrative requirements of the metro network.

0.12.2 Overview

The Telecommunication facilities proposed are helpful in meeting the requirements for operation of trains:

1. Supplementing the Signalling system for efficient train operation.
2. Exchange of managerial information
3. Crisis management during emergencies
4. Passenger information system

The proposed Telecom system will cater to the following requirements:

Radio System

Backbone network using Optical Fiber Cable (OFC)

Ethernet & WAN Network.

Station to Station dedicated communication

Telephone System with Telephone Exchanges, Telephones and their Recording

Centralized Recording System (CDRS)

Centralized Clock System

Closed Circuit Television (CCTV) System

Passenger Information & Display System within the station &trains and from Central Control to each station, Integrated Passenger Announcement System

Train Traffic Control, Maintenance Control, Emergency Control, Assistance to Train Traffic Control.

Data Channels for Signalling, SCADA, Automatic Fare Collection

Power Supply of Telecommunications, and

Cables for Telecommunications etc.

0.12.3 Automatic Fare Collection System

Metro System handles large number of passengers. Ticket issue and fare collection play a vital role in the efficient and proper operation of the system. To achieve this objective, ticketing system shall be simple, easy to use / operate and maintain, easy on accounting facilities, capable of issuing single / multiple journey tickets, amendable for quick fare changes and require overall less manpower. In view of the above computer based automatic fare collection system is proposed**.** Seamless ticketing is now being thought of for Surat Metro Rail.

Automatic Fare Collection system is recommended to be adopted as this will enable the commuters to travel hassle free by different modes of transport viz. Metro, suburban trains, buses, water transport (whenever introduced) and even taxies without purchasing multiple tickets for each mode separately.

Automatic fare collection systems have the following advantages:

1. Less number of staff required.
2. Less possibility of leakages of revenue due to 100% ticket check by control gates.
3. Recycling of ticket fraudulently by staff avoided.
4. Efficient and easy to operate.
5. System is amenable for quick fare changes.
6. Management information reports generation is easy.

7. System has multi operator capabilities. Same Smart Card can be used for other applications also.
8. AFC systems are the world wide accepted systems for Metro environment.

The proposed AFC system shall be of Contactless Smart Token / Card type. For multiple journeys, the stored value smart card shall be utilized and for the single journey, the smart media shall be as utilized as contactless smart token. The equipments for the same shall be provided at each station counter / booking offices and at convenient locations and will be connected to a local area network with a computer in the Station Master's room. Equipment and installation cost of Contactless Smart Card / Token based AFC system is similar to magnetic ticket based AFC system, but Contactless system proves cheaper due to reduced maintenance, less wear and tear and less prone to dusty environment.

It is proposed, the smart NCMC (National Common Mobility card) standard model for implementation of AFC system in Surat Metro. The AFC system as per the guidelines issued by Govt of India shall enable seamless travel by different metros and other transport systems across the city besides retail shopping and purchases.

The AFC system shall support the EMV (Europay, MasterCard, and Visa) and RuPay based open loop ticketing following the NCMC standard model for interoperability with other operators by use of non-proprietary standard so that the interface is scalable to other networks (transit operator/ retail outlets/parking/Toll etc) in Surat. The AFC equipments shall support EMV, RuPay, QR, NFC (Near field communication) based ticketing, integration of clearing house, smart card host system of Financial Institutions and integration of mobile application with AFC system.

0.12.4 Gate

Retractable Flap Type/Paddle Type Control Gates are proposed which offer high throughput, require less maintenance and are latest in modern systems internationally. All these gates will have a functionality of Auto Top on smart cards in case balance goes below the threshold value (as per choice / business rule).

The gate should also capable to NFC enabled Mobile Tickets or any latest type of Ticket media at the time of procurement/installation. The AFC system shall provide access control solutions, offering both access control devised and hardware which can be tailored to accept any ticket media readily available in market (Barcode, QR code, NFC etc).

0.12.4.1 Gate Function

a) Gate arrays shall be the normal-means of controlling entry to and exit from the paid areas. Control shall be by means of actuating a physical barrier on recognition of a valid ticket or card by the gate. The barrier may be a bi-parting leaves, centre flaps, end flaps or other configuration however the use of tripod or turnstile type gates is not acceptable. The gate shall be capable of operating either in normally open or normally closed mode.

b) Where required, barriers shall be provided to separate paid and unpaid areas of the concourse. The barriers shall meet local public safety requirements and be aesthetically merged with station engineering.

0.12.4.2 Features

Power Failure - In the event of a total power failure to the gates, the gates shall open to allow unrestricted user access. All latch gates shall automatically unlatch where electric locks are installed.

Concourse Emergency Mode - All AFC gates shall open whenever the Concourse Operating Mode is in emergency. An Emergency Push Button independent of the SC shall be provided in each Excess Fare Office.

Ergonomics- The engineering of the gate arrays should be such that the passenger uses reader placed on the right hand side while passing through the gate. The display and Contact less Smart Card (CSC) reader associated with each gate shall be grouped such that they bias the passenger towards the aisle through which the passenger should pass.

0.12.3Types of Gates

(a) **Passenger Entry Gate**:- The Passenger Entry Gate shall control the entry of passengers into the paid area by validating the fare media.

(b) **Passenger Exit Gate**: - The Passenger Exit Gate shall control the exit from the paid area by validating the fare media.

(c) **Passenger Reversible Gate**:- The Passenger Reversible Gate shall combine the features of the Entry and Exit gates. It shall be capable of being switched by the Station Computer from entry mode to exit mode and vice-versa depending on the operational requirements of passenger flow. Reversible Gates shall also function automatically,

based on the side from where the Passenger approaches first.

(d) Staff / Emergency Gate: - Normally situated adjacent to the Excess fare Office and kept open during emergency situations.

0.12.4 Spacing

Spacing for passenger gates shall be based generally on the following dimensional criteria:

a) Gate Centre spacing: - Standard gates 880mm
b) Aisle width: - Standard gates 465 - 580mm

0.12.5 Ticket Vending Machine (TVM) & Self-Service Ticketing Kiosks

The TVM should provide the convenience for the passengers to procure ticket on their own, without the need to queue at the ticket sale counter.

At all stations, Passenger Operated Ticket Vending Machines (Automatic Ticket Vending Machines) are proposed. The TVM's will provide convenience to passengers to avoid standing in queues at ticket booths and provide them international standard service. This will be used for

1. Dispensing Smart Tokens for single journey
2. Add Value in Smart card by paying money using Bank Notes or through Credit Card /Debit card /pre-Paid card.
3. Return the remaining money through Bank Notes and Coins (Min 2 types).

0.12.5.1 Functions

a) Enable passengers to purchase tickets for journey.

b) The touch-point including the screen interface should be customizable in terms of the text, graphics and video. It should be able to support the promotion of any preferred products.

c) The machines shall accept payment in the form of bank notes, coins and credit / debit cards and shall interact with the passengers via a touch screen display and receipt printer.

d) A reject button shall be provided to enable a passenger to abort a transaction before a token issue cycle has commenced.

e) The bank note reader shall accept notes inserted in any orientation

(any way up or round) and change shall be provided via a combination of note and coin re-circulating mechanism, which minimises the number of times the station staffs need to replenish the machines with change.

0.12.6 Ticket Reader/Add Value Machines

These machines will be used to know the Card/Token balance and can also be used as Add value device in case payment for Card top up is made through alternate Internet based channel like net banking, Credit/Debit card (Payment gateway) etc.

0.12.7 Recharge Card Terminal Machine (RCTM)

RCTM will be used to recharge the Card using Credit Card /Debit card /Pre Paid card as well as bank Note

0.13 DISABLED FRIENDLY FEATURES

0.13.1 Introduction

The objective of making this chapter is to create a user-friendly mass transport system in India which can ensure accessibility to persons with disabilities, people travelling with small children or are carrying luggage, as well as people with temporary mobility problems (e.g. a leg in plaster) and the elderly persons.

The design standards for universal access to Public Transport Infrastructure including related facilities and services, information, etc. would benefit people using public transport.

The access standards given here are extracted from Indian Roads Congress Code, IRC 103: 2012, Guidelines for Pedestrian Facilities; Model Building Bye-Laws, 2011 and National Building Code, 2005. Central Public Works Department's (CPWD) "Space Standards for Barrier Free Built Environment for Disabled and Elderly Persons", 1998 and 2013 edition (under revision by MoUD), and international best practices / standards. Further, it has also been attempted to provide guidelines/ standards for alighting and boarding area, approach to station, car parking area, drop-off and pick-up areas, taxi/auto rickshaw stand, bus stand/stop, footpath (sidewalk), kerb ramp, road intersection, median/pedestrian refuge, traffic signals, subway and foot over bridge etc. to achieve a seamless development around metro stations.

0.13.2 Content

1. **Rail Transport**

2. **Metro Rail Station**
   - Way finding
   - Signage
   - Automated Kiosks
   - Public Dealing Counters
   - Audio-visual Displays
   - Public Telephones
   - Rest Areas/Seating
   - Tactile Paving - Guiding & Warning
   - Doors
   - Steps & Stairs
   - Handrails
   - Ramps
   - Lifts/Elevators
   - Platform/Stair Lift
   - General and Accessible toilets
   - Drinking Water Units
   - Visual Contrasts
   - Emergency Egress/Evacuation

3. **Street Design**
   - Footpath (Sidewalk)
   - Kerb Ramp
   - Road Intersection
   - Median/Pedestrian Refuge
   - Traffic Signals
   - Subway and Foot Over Bridge

Alighting and Boarding Area Approach
- Car Park
- Drop-off and Pick-up Areas
- Taxi/Auto Rickshaw Stand
- Bus Stand/Stop

0.13.3 METRO RAILWAY STATIONS

1. LEVEL APPROACH

Approach route should not have level differences. If the station is not on the same level as the walkway or pathway, it should a ramp.

Walkway surfaces should be non-slip.

Approach walkway should have tactile pavements for persons with visual impairments.

2. STATION ENTRANCES AND EXITS

These should have a minimum width of 1800mm and is level or ramped.

3. RESERVATION AND INFORMATION COUNTERS

Should have clear floor space of at least 900 mm x 1200 mm in front of the counters;

There should be at least one low counter at a height of 750 mm to 800 mm from the floor with clear knee space of 750 mm high by 900 mm wide by 480 mm deep.

At least one of the counters should have an induction loop unit to aid people with hearing impairments; and

The counters should have pictographic maps indicating all the services offered at the counter and at least one of the counter staff should be sign language literate.

4. TOILET FACILITIES

There should be at least one unisex accessible toilet

- Ticket Gates

At least one of the ticket gates should:

Be minimum 900 mm wide to allow a wheelchair user through; and
Have a continuous line of guiding paver for people with visual impairments.

5. PLATFORMS

The Platforms should:

Have a row of warning paver installed 600mm before the track edge (photo 6);

Have non-slip and level flooring;
Have seating areas for people with ambulatory disabilities;
Be well illuminated lux level 35 to 40;
There should be no gap or difference in level between the train entry door and the platform.
All platforms should inter-connect by means of an accessible routes or lifts; and provide accessible level entrance to the train coach.

6. WAY FINDING

Way finding references should be available at decision points.
Colour can be used to identify routes and provide assistance in locating doors, walls and hazards. Proper colour contrast between different elements greatly improves visibility for all users and is critical for persons with low vision. For example, colour contrasting of door frames can assist in locating doors, and likewise floors should be contrasted with walls. In addition, furniture should contrast with walls and floors so as not to create an obstacle.
Structural elements such as columns should be colour contrasted or brightly marked so as to be visible to those who may have a visual disability.
Generally, patterns on flooring should be avoided or else should be minimal and small to avoid visual confusion.
In addition to identifying hazards or warnings, tactile floor surfaces can also be used to inform that there is a change in area (e.g. leaving a corridor and entering a boarding area).
Tactile systems should be consistent throughout the building. For example, terminals should not have carpeting in some boarding areas and tile in others as this may create confusion for those who rely on tactile surfaces to guide them to their destination.
Good lighting assists those with a visual disability to see better and allows people who have a hearing impairment to lip read easier. However, care should be taken to properly direct lighting and to use matte finishes on floors, walls and signage, so as not to create glare which may create difficulties for all travelers.
Blinds can be used to adjust lighting levels in areas where the natural lighting changes significantly throughout the day.

7. SIGNAGE

Signs must be clear, concise, and consistent. All travelers need clear information about the purpose and layout of terminals to maintain a sense

of direction and independent use of all facilities. Using internationally and nationally established symbols and pictograms with clear lettering and Braille ensures universal accessibility cutting across regional/cultural and language barriers. A cohesive information and signage system can provide visual (e.g. signs, notice boards), audible (e.g. public address and security systems, induction loops, telephones, and infrared devices), and/ or tactile information (e.g. signs with embossed lettering or Braille).

8. SIGN DESIGN SPECIFICATIONS

The sign should be in a prominent position.
The face of the sign should be well-illuminated by natural or artificial light.
Letters should be simple such as Arial, Helvetica medium, and san serif or similar and numbers should be Arabic.
The colour of the text should be in a colour that contrasts with the sign board.
The sign board should also contrast with the wall on which it is mounted.
The surface of the sign should not be reflective.
Some signs such as those adjacent to or on a toilet door may be embossed so that they can be read by touch.
Illuminated signs should not use red text on a dark background.
Signs should be supplemented by Braille where possible.

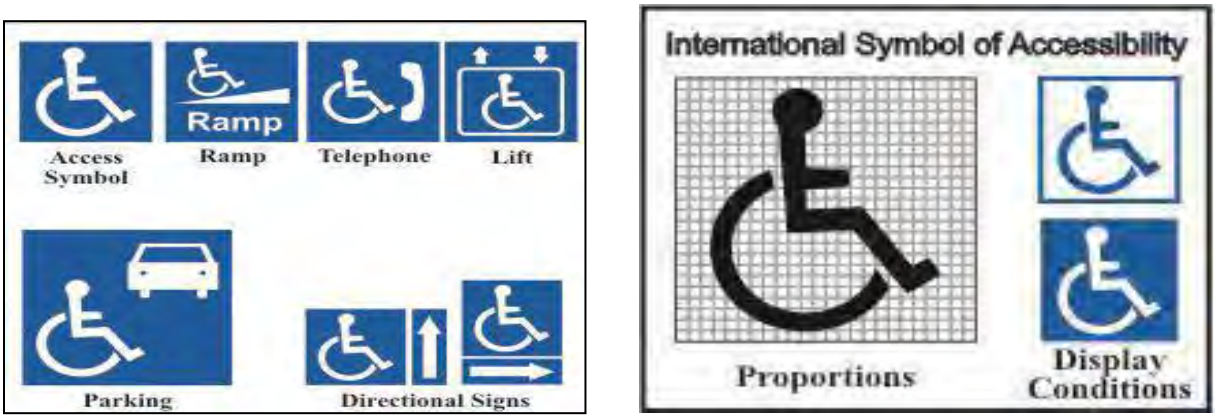

Fig.0.1 Way finding signage & International Symbol of Accessibility

0.14 ENVIRONMENTAL AND SOCIAL IMPACT ASSESSMENTS

0.14.1 Objective and Scope of the Study

The objective of the Environment and Social Impact Assessment study is to facilitate the Surat Municipal Corporation(SMC) evaluate the environmental impacts of its proposed activity. SMC proposes to apply for loan to seek financial support from multilateral funding agencies. Thus, the objective of the study is to conduct Environmental Impact Assessment as per requirement of multilateral

funding agencies. The scope of EIA includes the impacts resulting from pre-construction, during construction and operation phases of the proposed metro alignment in Surat. In addition, it is proposed to establish environmental baseline and safeguard measures for protection of environment for sustainable development during project cycles.

0.14.2 Approach and Methodology

The SMC has considered different alternative corridors. The underlying principles for evaluation for each corridor, without affecting the overall usefulness of the corridor, are minimum private land acquisition, least disturbance to properties, minimal disturbance to ecology/biodiversity. In the analysis of alternatives, a comparison of scenario with and without the project has also been made. The final alternative was fixed based on Technical Feasibility, Socio-economic acceptability, and Environmental sustainability for Metro Corridors. The environmental study is carried out for the alignment proposed by SMC. The approach is to follow the sequence of steps adopted in an EIA study. The basic concept is to ascertain the existing baseline conditions and assess the impacts as a result of construction and operation of the project. The changes likely to occur in different components of the environment viz. physical, biological / ecological, environmental and socio-economic etc. have been studied, analyzed and quantified, wherever possible. The identification of parameters for data generation and impact assessment are important. The analysis of assessment depends upon the reliable data generated/ available on environmental attributed. This study has documented the baseline data for various parameters of physical, ecological and environmental pollution (air, water and noise). The impacts are assessed for various phases of project cycle namely:

Impacts due to project location,
Impacts due to project design,
Impacts due to project construction, and
Impacts due to project operation.

The impacts are categorized as negative and positive. The cost of management and monitoring programs were estimated and budgeted for.

0.14.3 Environmental Scoping

Baseline environmental status in and around the proposed project depicts the existing environmental conditions of the location. Baseline data was collected for various/environmental attributes so as to compute the impacts that are likely to arise due to proposed project.

The scope of the present study includes detailed characterization of following environmental components, which are most likely to be influenced by the proposed project:

Land Environment
Water Quality (Surface + Ground water)
Meteorological conditions
Ambient Air Quality
Noise Levels
Biodiversity
Socio Economic studies.

The information presented in this chapter has been acquired from various sources. Data on land environment has been collected and compiled from various reports and field surveys. The data on water, air, noise quality, and biodiversity were collected through field studies, sampling in January and March 2017. Climatological data was collected from India meteorological Department. Efforts have been made to compile the available data from literature, books, maps and reports. The methodology adopted for data collection is highlighted wherever necessary. Environmental Attributes and Frequency of Baseline Survey is presented in Table given below:-

Table 0.13: Environmental Attributes and Frequency of Monitoring

S. No		Attribute	Parameter	No. of Samples	Source
	LAND ENVIRONMENT				
1		Geology	Geological Status	---	Literature review
2		Seismology	Seismic Hazard	---	Literature review
	WATER ENVIRONMENT				
3		Ground Water	Physical, Chemical and Biological parameters	7	Sampling locations
4.		Surface Water	Physical, Chemical and Biological parameters	3	Sampling locations
	AIR, NOISE AND METEOROLOGY				
4		Ambient Air Quality	PM_{10}, SO_2, NO_x	36	Sampling/Monitoring locations
5		Noise	Noise levels in dB (A) Leq, Lmax, Lmin, L_{10}, L_{50}, L_{90}	36	Sampling/Monitoring locations
6.		Soil	Physico-chemical parameters	7	Sampling Locations
7		Vibration	PPV mm/s, V_{rms} mm/s, VdB_{rms}(wrt $2.54x10^{-5}$ mm/s)	18	Sampling/ Monitoring Locations
	SOCIO-ECONOMIC				

S. No		Attribute	Parameter	No. of Samples	Source
8		Socio-economic aspects	Socio-economic profile	Once	Field Studies, Literature review.
	ECOLOGY				
9		Trees	Number	Once	Filed Studies

0.14.4 Socio-Economic Impact

0.14.4.1 Socio-Economic Impact on PAPs

The metro alignment runs between Bhesan and Saroli and Sarthana and Dream City Land is mainly required for viaduct, construction of stations and allied services, construction of Depot including laying of stabling lines, workshops, washing lines, administrative buildings and Water, Sewage and Effluent treatment systems in addition to storage facilities. Additionally, land is also required for RSS. There is no residential unit affected by the land acquisition for the proposed metro corridors. A total of 18 commercial establishments are getting affected due to the project due to viaduct and stations. The commercial establishments are having 24 employees. Among the shops being affected 9 are located at SMC Shopping Complex, One at SMC Hospital Station on Corridor-1 between Sarthana- Dream City. Simultaneously there are 8 shops affected due to the sharp curve at Textile Market 451 near Millenium Textile Market in basement and three floors between Torrent Power and Anjana Farm stations on Bhesan- Saroli corridor of Surat Metro.

0.14.5 Positive Environmental Impacts

Based on project particulars and existing environmental conditions, potential impacts that are likely to result from the proposed Surat metro corridors development have been identified and wherever possible these have been quantified. This section deals with the positive impacts of the project. The introduction of the corridor will also yield benefits from non-tangible parameters such as savings due to equivalent reduction in road construction and maintenance, vehicle operating costs, less atmospheric air pollution and socio-economic benefits of travel time, better accessibility, better comfort and quality of life. However, all benefits cannot be evaluated in financial terms due to non-availability of universally accepted norms. The parameters such as economic growth, improvement in quality of life, reduction in public health problems due to reduction in pollution, etc have not been quantified.

Various positive impacts have been listed under the following headings:

Employment Opportunities;
Enhancement of Economy;
Mobility, Safety and reduced accidents;
Traffic Congestion Reduction;
Reduced Fuel Consumption;
Reduced Air Pollution;
Reduction in Number of Buses/ Auto rickshaws

0.14.6 Impacts due to project locations

During this phase, those impacts, which are likely to take place due to the layout of the project, have been assessed. These impacts are:

- Project Affected People (PAPs)
- Change of Land use;
- Loss of trees/forest;
- Utility/Drainage Problems,
- Socio-economic impacts;
- Impact on Historical and Cultural Monuments;

0.14.7 Impacts due to project construction

Although environmental hazards related to construction works are mostly of temporary nature. Appropriate measures should be included in the work plan and budgeted for. The most likely negative impacts related to the construction works are:

- Top Soil erosion, pollution and health risk at construction site,
- Traffic diversion and risk to existing building,
- Excavated soil disposal problems,
- Dust Generation,
- Increased water demand,
- Impact due to Supply of Construction Material,
- Disposal of Construction and Demolition Waste,
- Impacts due to batching plant and casting yard,
- Noise Pollution,

0.15SECURITY MEASURES FOR A METRO SYSTEM

0.15.1 Introduction

Metro is emerging as the most favoured mode of urban transportation system. The inherent characteristics of metro system make it an ideal target for terrorists and miscreants. Metro systems are typically open and dynamic systems which carry thousands of commuters. Moreover the high cost of infrastructure, its economic impotence, being the life line of city high news value, fear & panic and man casual ties poses greater threat to its security. Security is a relatively new challenge in the context of public transport. It addresses problems caused intentionally. Security differs from safety which addresses problems caused accidentally. Security problems or threats are caused by people whose actions aim to undermine or disturb the public transport system and/or to harm passengers or staff. These threats range from daily operational security problems such as disorder, vandalism and assault to the terrorist threat.

0.15.2 Necessity of Security

It is well known that public transportation is increasingly important for urban areas to prosper in the face of challenges such as reducing congestion and pollution. Therefore, security plays an important role in helping public transport system to become the mode of choice. Therefore, excellence in security is a pre-requisite for Metro system for increasing its market share. Metro railway administration must ensure that security model must keep pace rapid expansion of the metro and changing security scenario.

0.15.3Three Pillars of Security

Security means protection of physical. Human and intellectual assets either from criminal interference, removal of destruction by terrorists or criminals or incidental to technological failures or natural hazardous events. There are three important pillars of security as mentioned under:

i. The human factor (ii) Procedures and (iii)Technology

Staff engaging with the passengers create a sense of re-assurance which cannot fully be achieved by technology. For human factor to be more effective staff has to be qualified, trained, well equipped and motivated. They should be strained, drilled and tested. The security risk assessment is the first step for understanding the needs and prioritizing resources. The organization of security should be clear and consistent. Security incidents, especially major

ones, often happen without warning. Emergency and contingency plans must be developed communicated and drilled in advance.

There are number of technologies which can be used to enhance security e.g. surveillance systems. The objectives of the security systems are to differ i.e., making planning or execution of on attack too difficult, detect the planned evidence before it occurs deny the access after in plan of attack has been made and to mitigate i.e. lessen the impact severity as the attack by appropriate digits.

0.16 DISASTER MANAGEMENT MEASURES

0.16.1"Disaster is a crisis that results in massive damage to life and property, uproots the physical and psychological fabric of the affected communities and outstrips the capacity of the local community to cope with the situation." Disasters are those situations which cause acute distress to passengers, employees and outsiders and may even be caused by external factors. As per the disaster management act, 2005 *"disaster" means a catastrophe, mishap, calamity or grave occurrence in any area, arising from natural or manmade causes, or by accident or negligence which results in substantial loss of life or human suffering or damage to, and destruction of, property, or damage to, or degradation of, environment, and is of such a nature or magnitude as to be beyond the coping capacity of the community of the affected area".* As per World Health Organization (WHO):

"Any occurrence that causes damage, economic disruption, loss of human life and deterioration of health and services on a scale sufficient to warrant an extra ordinary response from outside the affected community or area."

A disaster is a tragic event, be it natural or manmade, which brings sudden and immense agony to humanity and disrupts normal life. It causes large scale human suffering due to loss of life, loss of livelihood, damages to property and persons and also brings untold hardships. It may also cause destruction to infrastructure, buildings, communication channels essential services, etc.

0.16.2 Need for Disaster Management Measures

The effect of any disaster spread over in operational area of Surat Metro is likely to be substantial as Surat Metro is expected to deal with thousands of passengers daily in underground tunnels, viaducts and stations. Disaster brings about sudden and immense misery to humanity and disrupts normal human life

in its established social and economic patterns. It has the potential to cause large scale human suffering due to loss of life, loss of livelihood, damage to property, injury and hardship. It may also cause destruction or damage to infrastructure, buildings and communication channels of Metro. Therefore there is an urgent need to provide for an efficient disaster management plan.

0.16.3 Objectives:

The main objectives of this Disaster Management Measures are as follows:

Save life and alleviate suffering.

Provide help to stranded passengers and arrange their prompt evacuation.

Instill a sense of security amongst all concerned by providing accurate information.

Protect Metro Rail property.

Expedite restoration of train operation.

Lay down the actions required to be taken by staff in the event of a disaster in Surat Metro in order to ensure handling of crisis situation in coordinated manner.

To ensure that all officials who are responsible to deal with the situation are thoroughly conversant with their duties and responsibilities in advance. It is important that these officials and workers are adequately trained in anticipation to avoid any kind of confusion and chaos at the time of the actual situation and to enable them to discharge their responsibilities with alertness and promptness.

0.17 MULTI MODAL TRAFFIC INTEGRATION AT METRO STATIONS

The proposed metro in Surat has two corridors totaling 40.35kms, consisting of 38 stations. Its two corridors–Sarthana-Dream City and Bhesan to Saroli traverses across the city in the east-west and south west-north east direction respectively. Along its path, the corridors cover many important origins and destinations. However, its reach will not be enough to reach all origins and destinations in the city. Fortunately, Surat has planned for a multi-modal system, where the proposed metro will be complemented by the BRTS and Surat city bus services currently under implementation. The need is now to ensure that people have safe, comfortable and secure access to the above modes as well as seamless integration facilities.

0.17.1 Last Mile Connectivity

Last mile connectivity actually impacts on overall ridership and usage of any mass transit system. It is essential to provide or improve last mile connectivity and link them for ease of access for users. Even if street infrastructure that comprises of last mile connectivity falls outside Metro or BRT's jurisdiction and control, they remain critical components of an effective public transportation. The easier it is to access the Metro or BRT, the more likely people are to use it. It has been seen that access beyond a distance of 400-500m is not comfortable enough to attract people onto the public transport system. The unavailability of this type of service is one of the main constraints to the use of public transport in urban areas.

Last Mile Connectivity refers to the provision of travel service from home or workplace to the nearest public transportation mode. A trip is considered as the entire journey between origin and destination. Commuters may utilize and combine different modes of transport for the entire trip. Metro or BRTS may cater to a majority of this kind of trip, but commuters always need to complete the access and egress part on their own.

0.17.2 Interchange Stations

Interchange stations permit riders from particular mode of transport to move and board same or different mode in order to reach their desired destination. Interchanges can be of several types, depending on available modes of transport. Transfer may happen between the same modes or among different available public transport. Interchanges bring public transport together. For example metro station can have interchanges with metro on particular corridor connecting metro station of different corridor or, Metro stations may have interchange with BRTS, city services and even rail. Interchange helps to integrate various means of transport system under a common node.

Interchanges increases flexibility of movement and gives commuter option to avail multiple public transports at a common station. BRTS is the high capacity public transport system while metro and rail are considered to have even higher capacity than BRTS.

0.18 TRANSIT ORIENTED DEVELOPMENT(TOD) PLAN AND VALUE CAPTURE FINANCE(VCF)

New Metro Rail Policy 2017 emphasizes that "Transit Oriented Development (TOD)" with proposed intermodal integration, universal accessibility, adequate

walkways and pathways for Non-Motorized Transport (NMT), stations for public bike sharing, commensurate parking lots for cycles and personal vehicles, as well as adequate arrangement for receiving and dispatch of feeder buses at all metro stations. The commitment by the State Government to adhere the guidelines issued by the central government w.r.t. TOD and adoption of VCF framework should be an integral part of the project proposal. The commitment should inter alia include commitment of transfer of the financial benefits accruing in the influence zone of the metro alignment on account of the TOD policies and VCF framework directly to the Special Purpose Vehicle (SPV)/agency implementing the metro rail project. The project report should specify the proposed quantum of such benefits being transferred to the project. This requirement would form a mandatory part of all metro rail project proposals.

Commercial/property development at stations and on other urban land has been used as a key instrument for maximizing revenues in metro rail/ railway systems in cities around the world. Notable examples are Hong Kong and Tokyo metro rail.

0.18.1 OBJECTIVES OF TOD

TOD integrates land use and transport planning to develop compact growth centers within the influence zone of 500-800 m on either side of the transit stations i.e. areas within walking distance, to achieve the following objectives:

To promote the use of public transport by developing high density zones in the influence area, which would increase the share of transit and walk trips made by the residents/ workers to meet the daily needs and also result in reduction in pollution and congestion in the influence area.

To provide all the basic needs of work/ job, shopping, public amenities, entertainment in the influence zone with mixed land-use development which would reduce the need for travel.

To establish a dense road network within the development area for safe and easy movement and connectivity of NMT and pedestrians between various uses as well as to transit stations.

To achieve reduction in the private vehicle ownership, traffic and associated parking demand.

To develop inclusive habitat in the influence area so that the people dependent on public transport can live in the livable communities within the walkable distance of transit stations.

To integrate the Economically Weaker Sections (EWS) and affordable housing in the influence zone by allocating a prescribed proportion of built-up area for them in the total housing supply.

To provide all kinds of recreational/entertainment/ open spaces, required for a good quality of life in the influence area.

To ensure development of safe society with special attention to safety of women, children, senior citizen and differently abled by making necessary amendments to the building bye laws.

To prevent urban sprawl by accommodating the growing population in a compact area with access to the transit corridor, which would also consolidate investments and bring down the infrastructure cost for development.

0.18.2 Value Capture Methods

A comparative study on land based financing tools being used in India and the world shows a large number of diverse VCF tools being used. The main types of VCF methods are given below.

Land value tax–is considered the most ideal Value Capture tool which apart from capturing any value increment, helps stabilize property prices, discourage speculative investments and is considered to be most efficient among all Value Capture methods. Maharashtra and Tamil Nadu, through State laws, have expanded the scope of this mechanism to cover urban land also. Globally, land value tax is widely used in Denmark, Australia, and New Zealand.

Fees for changing land use (agricultural to non-agricultural)–land revenue codes provide for procedures to obtain permission for conversion of land use from agricultural to non-agricultural use.

Betterment levy–one-time upfront charge on the land value gain caused by public infrastructure investment. This occurs in two forms – revenue source for improvement schemes and for specific projects. In India, the Mumbai Metropolitan Regional Development Authority (MMRDA) Act, 1974 provides for levying betterment charges for specific projects. The Hyderabad Municipal Corporation Act, 1955 originally provided for the levy of betterment charges to meet the costs of internal infrastructure and services in the case of development projects. In the late nineties, the Government of Andhra Pradesh amended the

Act to enhance the scope of such levy to include external betterment. Under this concept, the municipal authority is empowered to collect external betterment charges at the time of according approval to layouts or sub-divisions of plot or issue of building permit for the laying of trunk water lines, development of freeways/major roads, percent of the land value gain attributable to public investment.

Development charges(Impact fees) are area-based and link the development charge to the market value of land by carrying out periodic revisions. This is the most widely used land based fiscal tool in States. States like Andhra Pradesh, Gujarat, Maharashtra, Tamil Nadu and Madhya Pradesh levy Impact Fee and collect it upfront while granting development permissions. Impact fee is widely used to fund infrastructure in the United States. The Government of Andhra Pradesh in the late nineties also permitted Hyderabad Municipal Corporation to levy Impact Fees to mitigate the impacts of construction of commercial buildings, which lead to increase in traffic and necessitate decongestion measures. This is meant to address citywide problems emanating from high-density commercial development and is expected to be utilized for the Capital Improvement and Decongestion Plan. This includes works such as road widening, link roads, slip roads, parallel roads, junction improvements including traffic signals, flyovers, rail over-bridges, rail under-bridges, modern lighting on major roads, development of major storm water drains, riverfront and parks and for Geographic Information System (GIS) applications.

Transfer of Development Rights (TDRs)–used for trading development rights. Maharashtra, Karnataka and Gujarat have enabling laws for using TDRs for developing open spaces, promoting affordable housing, etc. In New York City, TDRs are given for preservation of the property owners for loss in revenue on their properties.

Premium on relaxation of rules or additional FSI/FAR–widely used in States such as Maharashtra, Karnataka, Gujarat, Tamil Nadu, etc. to allow for additional development rights beyond the permissible limits in the State Town Planning Laws and Regulations. Sale of additional Floor Area Ratio (FAR) is an important Value Capture tool in Brazil and France. The French Land-use Policy restricts the land owners building right to a low baseline FAR and additional FAR has to be purchased.

Vacant Land Tax (VLT)–is applicable on those landowners who have not yet initiated construction on their lands. In Andhra Pradesh, the Greater Hyderabad Municipal Corporation (GHMC) imposes a tax of 0.5% of the registration value

of the land if not used exclusively for agriculture purpose or is vacant without a building.

Tax Increment Financing (TIF)–is one of the most popular Value Capture tools in many developed countries, especially the United States. In TIF, the incremental revenues from future increases in property tax or a surcharge on the existing property tax rate is ring-fenced for a defined period to finance some new investment in the designated area. Tax Increment Financing tools are especially useful to finance new investments in existing habitations. Some of the Smart City Proposals have planned for TIF in their area-based developments (ABD).

Land Acquisition and Development –acquiring and developing land could be adopted as a useful Value Capture method to mobilize resources. In Hyderabad, impact fees are levied on all new developments within a one-kilometer wide growth corridor on both sides of the Outer Ring Road (ORR). Another innovative „Road widening Scheme" is being implemented in Hyderabad in which the Municipal Corporation gives additional FAR and relaxes zoning for property owners who give land free of cost for road widening.

Land pooling System (LPS)–is a form of land procurement where all land parcels in an area are pooled, converted into a layout, infrastructure developed, and a share of the land, in proportion to original ownership, returned as reconstituted parcels. In India, States such as Gujarat and Haryana have used land assembly programs where the owners agree to exchange their barren lands for infrastructure-serviced smaller plots. Gujarat has used these tools to guide the development of Ahmedabad city and its surrounding infrastructure. The State of Andhra Pradesh has used LPS to get land for Amravati, its new Capital City. Such LPS are also a common feature in countries like Japan and Germany.

0.18.3 Potential of TOD And VCF In Surat Metro Corridors

In this connection it is clarified that the total cost of Surat Metro Rail Project (Phase-I) works out to be Rs. 9338 crores plus applicable taxes totaling to Rs 10,661crores (inclusive of all taxes).

Once the contribution from the private sector for the proposed PPP portion amounting to Rs 396 crores (without taxes & duties) is excluded, the total cost of the project stands at Rs.8942crore plus applicable taxes.

Thus against this total project cost of Rs. 9338crore(plus applicable taxes), direct contributions from Transit Oriented Development (TOD) is expected to be Rs.396 crores and as this amount is expected to directly flow to Surat Metro Project through the mechanism decided by Central Govt. and Government of Gujarat(GoG), it is only the balance amount of Rs. 8942crores(plus applicable taxes) which is proposed to be funded by Central and State Governments by equity and other sources including Debt.

The Revenue of Surat Metro mainly consists of fare box collection, other property developments, advertisements, parking's, transit oriented developments (TOD), Value Capture Finance(VCF) etc.

Approximate revenue expected to be generated due to TOD along the Surat metro corridors of Phase-I, shall be amounting to Rs. 1343 crores and details of calculations have been forwarded by Surat Municipal Corporation (SMC) through their letter No. CE SP Cell/out/1199 dated 22-12-2017 Further, it has been stated that 60% of total revenue from sale of FSI could be transferred towards MRTS project and considered for DPR purpose too. The prevailing rate of FSI chargeable is 40% of the Jantri rate. Regarding VCF, an amount equivalent to Rs. 2738 crores worked out from 5% cess on property tax and stamp duty charges.

0.19 COST ESTIMATES

Project Cost estimates for the Surat Metro Rail Project (Phase-I) network has been prepared covering civil, electrical, signalling and telecommunication works, rolling stock, environmental protection, rehabilitation, considering 750V DC third rail traction at December' 2018 price level, both for Capital and Operation & Maintenance costs.

While preparing cost estimates, various items have generally been grouped under three major heads on the basis of:-

(i) Route km. Length of alignment

(ii) No. of units of that item and

(iii) Item being an independent entity.

All items related with alignment, whether in underground or elevated or at-grade construction, permanent way OHE, signaling and telecommunication, have been estimated on rate per route km/km basis. Route km. cost for underground alignment construction, excludes station lengths. Station lengths (190m) have to be done by cut and cover in general and by tunneling under

compelling exceptional circumstances. The rates adopted for underground stations include cost of civil structures and architectural finishes. Similarly, cost of elevated and at grade stations includes civil work for station structures, architectural finishes, platform roofing, etc. Provisions for electrical and mechanical works, air conditioning, lifts, escalators, etc. have been worked out separately. These rates do not include cost of permanent way, O.H.E., power supply, signaling and telecommunication, automatic fare collection (AFC) installations, for which separate provisions have been made in the cost estimates. Similarly, for other items like Rolling stock, Traction & Power, VAC, etc. costs have been summed up separately. In remaining items, viz. land, utility diversions, rehabilitation, etc. the costs have been assessed on the basis of each item taken as an independent entity.

Initially cost estimate for Surat Metro (Phase-I) had been formulated adopting the Unit rates of various components of Ahmedabad Metro project. Escalation charges had been applied @5% per year to bring these costs to Current price level, wherever necessary in terms of MoUD's letter No.F.No.K-14011/58/2013-MRTS-1 (Vol.I) dated 27.10.2016.Taxes & Duties such as Custom Duty, CGST,SGST and IDST etc wherever applicable, had been worked out on the basis of prevailing rates and included in the cost estimates separately.

Now, on the request of MEGA Cost estimate has been revised, adopting the cost components in line with Ahmedabad Metro project (Ph-II) and Delhi Metro Phase-IV in accordance of MoUHA letter no. K-14011/5/2017-UT-IIdated 23.01.2019 forwarded by MEGA vide their letter No. GMRCL/DPR Revision/JAN-19 dated 25-01-2019.

In accordance of MoHUA letter No. K-14011/2018-MRTS-IIIdated 15.02.2019 forwarded by MEGA through their e-mail dated 16.02.2019 contingencies have been removed as land component and cost estimate has been revised accordingly.

The overall Capital Cost for the Surat Metro rail network under Phase-I at December' 2018 price level works out to **Rs.10,661Crores** including applicable Taxes & Duties of **1322.78 Crores** as tabulated hereunder:

Table 0.14 –Corridor-wise Details of Capital Cost

Sr. No.	Name of the corridor	Capital Cost (Rs. Crore)	Taxes & Duties (Rs. Crore)	Total (Rs. Crore)
1.	Sarthana-Dream City	5810	849.30	6659.30
2.	Bhesan–Saroli	3528	473.48	4001.48
	Total	**9338**	**1322.78**	**10660.78 =10,661**

0.19.1 PPP Model for Unbundled Components of Surat Metro

As communicated by Surat Municipal Corporation (SMC) vide their letter no. CE Spl Cell/Out/1199 Dated 22-12-2017, 04 nos stations as mentioned below will be considered on PPP model through a developer to be engaged by metro SPV but with VGF/Land support to the PPP agencies from SMC:-

Corridor-1 (Sarthana to Dream City)

(i) Rupali Canal
(ii) Althan tenement

Corridor-2 (Bhesan to Saroli)

(i) Aquarium(Star Bazar)
(ii) Athwa Chaupati

However, the stations proposed above on PPP model may be changed depending upon financial feasibility.

In addition to above, it has also been conveyed that two nos elevated metro stations (including one for future requirement) inside Dream City area shall be implemented by Metro SPV with the VGF/Land support to the PPP agencies from DREAM Ltd.

For both the corridors of Surat Metro Project(Phase-I) three unbundled components namely (1)Lifts,(2) Escalators and (3) AFC gates are proposed under PPP model, in addition to civil cost of 05 nos elevated stations mentioned above. Details of cost under PPP model is given hereunder:-

Table 0.14 :Cost of proposed Components to be considered on PPP model (incrores)

S.No	Description	Estimated Cost*
(A)	**Corridor-1: Sarthana to Dream City**	
01	Station-Rupali Canal (civil cost)	26.00
02	Station-Althan Tenament (civil cost)	26.00
03	Station-Dream City (civil cost)	26.00
04	Lifts & Escalators (20 X 3.50)	70.00
05	Automatic Fare Collection(AFC) Gates (20 X 3.50)	70.00
	Sub-Total (a)	**218.00**
(B)	**Corridor-2: Bhesan to Saroli**	
01	Station-Aqurium (civil cost)	26.00
02	Station- Athwa Chaupati (civil cost)	26.00
03	Lifts & Escalators (18 X 3.50)	63.00
04	Automatic Fare Collection(AFC) Gates (18 X 3.50)	63.00
	Sub-Total(b)	**178.00**
	Grand Total (a)+(b)	**396.00**

*plus charges on account of applicable taxes, contingencies, GC etc.

0.19.2 Integration with Surat Railway Station MMTH project

In a series of meetings taken place at Surat in the office of The Commissioner, Surat Municipal Corporation(SMC), it had been intimated that Surat Railway station MMTH project shall be integrated with underground metro station, city bus/BRTS, interstate buses(GSRTC) and Indian Railways. Further, in view of Surat Railway station MMTH master plan, it has been decided to provide an underground tunnel road for vehicular traffic in the portion of MMTH along Lambe Hanuman road, which shall be above the level of underground metro corridor and UG metro station with a sufficient cushion of 2 to 3 mtr from ground level. The estimated cost of proposed tunnel road shall be Rs 109.13 crores as intimated by Surat Municipal Corporation (SMC) vide their e-mail dated 2nd May' 2018.The cost of construction of this underground road shall be borne by SMC/GoG As such, the cost has not been added in the cost of Surat Metro Rail Project(Phase-I) for working out of financial indices such as FIRR,EIRR etc. Executing agency for tunnel road will be decided by Metro SPV in consultation with SMC.

0.20 FINANCING OPTIONS, FARE STRUCTURE AND FINANCIAL VIABILITY

0.20.1 Introduction

The Surat Metro Rail Project consists of two corridors (From Sarthna to Dream City and Bhesan to Saroli) are proposed to be constructed with an estimated cost of Rs.10661.00 Crore with land cost and all taxes. The route length of the

metro system and estimated cost at December-2018 price level without and with all taxes are placed in table given below as under:

Table 0.15 Cost Details

Corridor no.	Name of Corridor	Distance (KMs)	Estimated cost with Land cost and without all taxes (Rs/Crore)	Estimated cost with land cost and all taxes (Rs/Crore)
1	Sarthana to Dream City	21.61	5810.00	6659.30
2	Bhesan to Saroli	18.74	3528.00	4001.48
Total		40.350	9338.00	10660.78 = 10661 cr

The estimated cost at December-2018 price level includes an amount of Rs.14.06 Crore plus applicable taxes as one-time charges towards security i.e. cost of weapons, barricades and hand held and door detector machine etc. However, the recurring cost towards salary and allowances of security personal have not taken in to account in the FIRR calculation since providing required security at metro stations shall be the responsibility of state police department.

0.20.2 Investment Cost

For the purpose of calculating the Financial Internal Rate of Return (FIRR), the completion cost with all taxes except land and contingencies has been calculated by taking escalation factor @5.00% per annum. It has been assumed that the Government of Gujarat will provide the land worth Rs.1234 crore either free of cost or it shall provide Interest Free Subordinate Debt in lieu thereof. With the implementation of GST from 1st July 2017, the Excise Duty, Central Sales Tax/State VAT and Service Tax have been subsumed into Central Goods and Service Tax (CGST), State Goods and Service Tax (SGST), Integrated Goods and Service Tax (IGST). The effective CD rate under project import scheme, post GST, works out to 24.49% {Basic CD @ 5%, IGST (CGST & SGST) @ 18% and cess} on the imported portions, CGST and SGST @ 6% each and IGST @ 12% on indigenously manufactured items. The Interest Free Subordinate Debt is repayable in 5 equal installments after repayment of Multilateral/Overseas Development Assistance Loan.

It is assumed that the construction work will start on 01.04.2019 and is expected to be completed on 31.03.2024 with Revenue Opening Date (ROD) as 01.04.2024 for both the corridors. The total completion costs duly escalated and shown in the table 0.16 have been taken as the initial investment as below:-

Table 0.16 Year –wise Investment

Figures in Rs. Crore

Financial Year	Estimated Cost including cost of land and all taxes & duties at December 2018 Price Level	Completion Cost including cost of land cost and all taxes & duties
2019-20	1,722	1,757
2020-21	2,439	2,595
2021-22	2,797	3,106
2022-23	2,382	2,751
2023-24	1,320	1,629
Total	10,660	11,838

0.20.3 Fare Structure

The fare structure for the FY 2024-25 as per the proposed fare slabs is shown in the table given below:-

Table 0.17 Fare Structure in 2023-24

Distance in kms.	Fare (Rs)
0-2	10
2-4	20
4-6	30
6-12	40
12-18	50
>18	60

Considering the increase in the Consumer Price Index (CPI) and input costs of operation since then, the fare structure has been escalated by using @14.00% once in every two years.

0.20.4 Financial Internal Rate of Return (FIRR)

The Financial Internal Rate of Return (FIRR) obtained for 30 years life cycle business model including construction period without additional PD and TOD & VCF Income is **6.90%** and **with** additional PD and TOD & VCF Income is **7.38%.**

0.20.5 Financing Options

The objective of funding metro systems is not necessarily enabling the availability of funds for construction but coupled with the objective of financial closure are other concerns, which are of no less importance: -

Ensuring low project cost

Ensuring debt funds at low rates of interest

Creating self sustainable system in the long run by

- Low infrastructure maintenance costs
- Longer life span
- Setting fares which minimise dependence on subsidies

Recovering returns from both direct and indirect beneficiaries

Rail based mass transit systems are characterised by heavy capital investments coupled with long gestation period leading to low financial rates of return although the economic benefits to the society are immense. Such systems generate externalities, which do not get captured in monetary terms and, therefore, do not flow back to the system. However, experience all over the world reveals that both construction and operations of metro are highly subsidised. Government involvement in the funding of metro systems is a foregone conclusion. Singapore had a 100% capital contribution from the government, Hong Kong 78% for the first three lines and 66% for the later 2 lines. The Phase-I, Phase-II as well as Phase-III of Delhi MRTS project, Chennai and Bengaluru metros are also funded with a mixture of equity and debt (ODA) by GOI & concerned State Governments.

0.20.6 Alternative Models of Financing

The financing option shall depend upon selection of the dedicated agency created to implement the project. The prominent models are: -

1 Special Purpose Vehicle under the State Government Control (Delhi Metro Rail Corporation (DMRC) /Bangalore Metro Rail Corporation (BMRC)/Chennai Metro Rail Corporation/Mumbai Metro Rail Corporation

2 Design-Build-Finance-Operate-Transfer (DBFOT), and

3 Public Private Partnership (PPP).

0.20.6.1 SPV Model: -The corridor is a standalone one and therefore forming a separate SPV may be in the name of Surat Metro Rail Corporation may be desirable MOUD vide letter no. F.No. K-14011/03/2017-UT-V-Part(1) dated 6th July 2017 has proposed for sharing of overall taxes post Goods and Service Tax (GST) regime in the ratio of 1:2.

Surat Municipal Corporation (SMC) vide email dated 02.05.2018 stated that metro station at Surat Railway Station shall be integrated with Multi Model Traffic Hub (MMTH) project and one underground tunnel road shall also be provided above the metro station for vehicular traffic along Lambe Hanuman road. The estimated cost of Rs. 109.13 crore will be borne by SMC for above Multi Model Traffic Hub integration.

0.20.6.2 ODA Loan from JICA/ADB/AFD:

JICA had provided loan assistance to Phase-I, Phase-II and ongoing Phase-III of Delhi MRTS projects to the extent of 60%, 54.47% & 51.32% respectively and also provided loan assistance to Chennai Metro, Mumbai Metro and MEGA. The Department of Economic Affairs, MOF, GOI vide letter no. F.1/9/2014-Japan.I dated 12th July 2018 has issued new set of guidelines on Operational Rules of the Japan-India Special ODA (Official Development Assistance) for Metro and Railway Projects in India. As per the discussions made with JICA officials, JICA may extend only the modified step loan as per above said guidelines for the new projects in India at an interest rate of 0.20% per annum. The tenure of the loan is 40 years with 12 years moratorium period. JICA may fund the project to the extent of 85% of the cost of the project excluding the cost of the land, cost of Rehabilitation and Resettlement and taxes and duties.In case JICA not agreeing to provide loan for the Surat Metro project or provide less loan assistance than the projected, the difference shall have to be sourced either from Asian Development bank (ADB) or Agency Franchise de Development (AFD) or European Investment Bank (EIB) or Asian Infrastructure Investment Bank (AIIB) by GOI or from Domestic Financial Institutions against guarantee from GOI.

0.20.6.3 Loan from Domestic Financial Institutions:

Domestic Financial Institutions may have to be taped in the case of any shortfall in securing the targeted loan amount from bilateral/multilaterals institutions. In such a situation Government of India may have to provide sovereign guarantee as is being done to obtain the loan from JICA for Delhi

Metro, ADB for Jaipur Metro, AFD for Kochi Metro and EIB for Lucknow Metro. The new SPV shall raise the required loan from IIFCL and or such other Domestic Financial Institutions. In order to keep the debt servicing sustainable so as to keep the Debt-Service Coverage Ratio (DSCR) above 1.2 at all points of time, interest subsidy between the domestic borrowing rate of interest and Multilateral/Overseas Development Loan rate of interest shall be provided to SPV.

0.20.6.4 Private Participation under SPV:

As already stated in above para, the private participation either for complete provisioning of metro rail or for some unbundled components will form an essential requirement for all metro rail project proposals seeking central financial assistance as per new Metro Policy 2017 issued by Ministry of Housing and Urban Affairs (MH&UA).The cost of unbundled activities of Rs.452 crore involving civil construction of 5 stations, AFC and Lifts & Escalators of all the corridors may be funded by engaging concessionaire in line with Kochi Metro, Nagpur Metro and Noida Metro for a period of 10 years.

0.20.6.5 DBFOT Model:

In this model, the private firm will be responsible for financing, designing, building, operating and maintaining of the entire project. The contribution of Government of Gujarat will be limited to cost of land only. Such a project become eligible for Viability Gap Funding (VGF) upto 20% from the Central Government provided the state government also contribute same or more amount towards the project. The metro being a social sector project not much private parties are available to bid for such a project. Besides quite expectedly the private operator may demand assured rate of return in the range of 16% to 18% or a comfort of guaranteed ridership.

0.20.6.6 PPP Model: - In this model, Government funds the fixed infrastructure such as land and basic civil structures, and a private investor funds all the systems such as rolling stock, signalling, power supply, traction, track, fare collection, E&M works etc including station architectural finishes. An example of this is Delhi Metro Airport line. Under this arrangement the government's investment will be about 57% of the cost of the Project and the PPP operator funds the remaining 43%. Under this model the concessionaire, operates and maintains the system to the required and agreed service and safety

levels. All the revenues will accrue to the Operator and at the end of the concession period the project is handed over to the owner. Ridership risks are taken by the operator or shared by the operator and owner. The PPP operator pays a specified amount every year to the Govt. out of his revenues. It could be that he may need a viability gap funding (VGF) even. The VGF (positive or negative) will be known only after competitive bidding.

0.21 ECONOMIC APPRIASAL

Economic benefits are social and environmental benefits which are quantified and then converted into money cost and discounted against the cost of construction and maintenance for deriving Economic Internal Rate of Return (EIRR). When actual revenue earned from fare collection, advertisement and property development are discounted against construction and maintenance cost, interest (to be paid) and depreciation cost, Financial Internal rate of Return (FIRR) is obtained. Therefore, EIRR is viewed from socio-economic angle while FIRR is an indicator of financial profitability and viability of any project.

0.21.1 After generating the cost and benefit stream table, economic performance indicators are derived and are presented in **table 0.18**. Project period is 2019-2048, With reference to completion cost of capital with tax, **EIRR** is found to be **17.20 %** and **B/C ratio** as **4.77** and with 12% discount, EIRR is **4.65%**and B/C ratio is **1.63**. NPV without discount is Rs **149337**Cr. And with 12% discount rate, NPV is Rs.**8287Cr**. which shows that the project is economically viable.

Table 0.18

Economic Indicator Values (on 2048-49)

	WITHOUT DISCOUNT	WITH DISCOUNT (12%)
Cumulative cost (Cr.)	39561	13127.87
Cumulative benefit(Cr.)	188898	21414.43
Benefit Cost Ratio	4.77	1.63
NPV(Cr.)	149337	8287
EIRR	**17.20%**	**4.65%**

0.22 IMPLEMENTATION PLAN

0.22.1 Way Forward for Implementing Surat Metro Project (Phase-I)

On receipt of the Detailed Project Report, following action will be required for implementing the Surat Metro (Phase-I):-

Approval to the Detailed Project Report to be taken from Gujarat State Government (Cabinet approval).

The DPR to be forwarded to the Ministry of Urban Development(GOI),Planning Commission and Finance Ministry with the request for approving the Metro project and for financial participation.

Signing of an MOU between Gujarat State Government and Government of India giving all details of the Joint Venture bringing out the financial involvement of each party, liability for the loans raised, the administrative control in the SPV, policy in regard to fare structure, operational subsidy, if any, etc.

A dedicated Special Purpose Vehicle (SPV) shall need to be set up, as Metro Company headquartered in other city namely Ahmedabad/ Gandhinagar may not be in position to effectively implement Surat Metro Project and its subsequent Operation & Maintenance. Organization depending upon the methodology of implementation need to be developed. In case project is done by appointing General Consultants, sufficient number of Engineers in SPV organization to be taken on deputation/market. In case the work is intended to be done on deposit terms from the organization like DMRC, SPV's organization may be kept lean and thin.

The Metro Railways (Amendment) Act-2009 can readily be made use of for implementation of Surat Metro by declaring Surat City as Metropolitan Area in terms of clause- c of section 243 P of Constitution.

Request to GOI for a notification for making the Metro Railways (Amendment) Act 2009 applicable to Surat Metro.

In view of new Metro Rail Policy 2017, the State Government should formulate the funding plan for executing this project and get the same approved by the Government of India. The loan portion of the funding will have to be tied up by State Government in consultation with the Government of India.

The Government should freeze all developments along the corridors suggested. For any constructions near the proposed alignment a system of No Objection Certificate should be introduced as per provisions of Metro Railway Act so that in fructuous expenditure at a later stage is avoided.

The Metro Railways (Amendment) Act-2009 can readily be made use of for implementation of Surat Metro by declaring Surat City as Metropolitan Area.

In view of new metro policy Unified Metropolitan Transport Authority (UMTA) should be formed within one year of Submission of Proposal of any metro project to GOI.

0.22.2 Implementation on Delhi Metro/Chennai Metro Model

Special Purpose Vehicle (SPV) shall have to take action for appointment of General Consultants for project management including preparation of tender documents. Till the General Consultants are in position, SPV should appoint an interim Consultant for all preliminary and enabling jobs such as land acquisition, detailed design of civil structures, utility diversions, etc.

The proposed date of commissioning of the both corridor with suggested dates of important milestones is given in Table as follows:

Table 0. 19:Implementation Schedule through DMRC model Phase I

S. No.	Item of Work	Completion Period
1	Submission of Final DPR to State Govt.	D
2	Approval of DPR by State Government	D + 0.5 month
3	Submission of DPR for Approval of Ministry of Urban Development (MoUD).	D + 1 month
4	Appoint interim Consultant for preliminary works	D + 3 months
5	Approval of Central Government	D+4 months
6	Arrangement of Funding Source/Agency	D +6 months
7	Appoint General Consultant	D +12 months
8	Tendering, Execution of works and Procurement of equipments, coaches and installations	D +57 months
9	Testing and Commissioning	D +60 moths
10	Revenue Operation	D +60 months

0.23 CONCLUSIONS AND RECOMMENDATIONS

The project is essentially a social necessity as it aims to improve the overall health of the city by way of reducing congestion on roads coupled with to reduced consumption of fossil fuels thereby reducing environmental pollution. This is evident from the fact the EIRR of the project is of the order of **17.20%** indicating that the project is very well justified in terms of social benefits that will accrue to the Society/Surat City. Therefore, implementation of this project is inescapable.Since the Gujarat Government is providing requisite land for PD and being social sector project, it is advisable to take up the job on DMRC model. Accordingly, the corridors are recommended for implementation.

As a way forward the metro network needs to be expanded in due course as the city population is already more than 50 lakhs and further city is growing in rapid manner. Therefore, the further study may be taken up when the execution of Phase-I is going on so that the DPR prepared and approved can be obtained before completion of Phase-I and the same project team can continue to work for Phase-II.

0.24 MISCELLANEOUS

0.24.1 In reference to MoHUA letter No.K-14011/3/2018-MRTS dated 15.02.2019 forwarded by MEGA vide email dated 16.02.2019, the contingencies have been removed on land component, cost estimate &funding pattern revised accordingly. Revised Cost Estimate and funding pattern had already been sent to MEGA and SMC vide letter No. DMRC/GM(CS)/04/Surat-DPR/2016/341 dated: 18.02.2019 **(Annexure- 0.1)**

0.24.2 Government of India (GoI) has already approved the proposal for “Surat Metro Rail Project” vide their letter No. K-14011/03/2018-MRTS-III dated 09.03.2019 as communicated by MEGA through their e-mail dated 13.03.2019. Copy of approval letter dated 09.03.2019 is enclosed herewith as **Annexure- 0.2.**

(Annexure- 0.1)

CIN No. L74899DL 1995G01066150
Tel. : 23417910/12
फैक्स Fax : 23417921

दिल्ली मेट्रो रेल कॉर्पोरेशन लि०
DELHI METRO RAIL CORPORATION LTD.
(भारत सरकार एंव दिल्ली सरकार का संयुक्त उपक्रम)
(A JOINT VENTURE OF GOVERNMENT OF INDIA AND GOVT. OF DELHI)

DMRC/GM(CS)/04/Surat-DPR/2016/341

18-02-2019

Sh Amit Gupta
CGM/Corporate Planning
MEGA
Block No.1 Ist Floor Karam Yogi Bhawan Sector 10A,
Gandhi Nagar-382010

Dear Sir,

Subject :- DPR for Surat Metro Rail Project (Phase-I)
- Revision of Cost and funding pattern thereof

Ref :- (i)MOUHA letter no K 14011/3/2018-MRTS-III dated 15-02-2019
(ii)MEGA's e-mail dated 16-02-2019

In reference to above, cost estimate for Surat Metro Rail Project (Phase I) has been revised removing contingencies on land cost component and also revised the funding pattern accordingly. The revised cost estimate and revised funding pattern are enclosed herewith.

2)This is for further appropriate necessary action in the matter at your end please.

Thanking you.

Encls:-As above

Yours Faithfully,

(T.Gupta)
Chief Engineer/CS-I

CC :-
The City Engineer, Surat Municipal Corporation(SMC), Mahanagar Seva Bhawan Muglisara,SURAT-395003

(मेट्रो भवन, फायर ब्रिगेड लैन, बाराखम्बा रोड, नई दिल्ली-110001)
Metro Bhawan, Fire Brigade Lane, Barakhamba Road, New Delhi-110001

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(Annexure- 0.2)

K-14011/03/2018-MRTS-III
Government of India
Ministry of Housing and Urban Affairs
(MRTS-III)

Nirman Bhawan, New Delhi
Dated the 09th March, 2019

To,

The Chief Secretary,
Government of Gujarat
1st block, 5th Floor
Sachivalaya,Gandhinagar
Gujarat

Subject: Surat Metro Rail Project.

Sir,

With reference to your letter No. MGA/122016/SF/97/DH dated 12.9.2018 on the above mentioned subject, I am directed to inform that Government of India (GoI) has approved the proposal for "Surat Metro Rail Project" which comprises two corridors of a total length 40.35 Km. (a) Sarthana to Dream City for a length of 21.61 km and (b) Bhesan to Saroli for a length of 18.74 km, at a total completion cost of the project is Rs. 12,020.32 crore. (Rupees Twelve Thousand Twenty Crore and Thirty Two Lakh only).

2. The funding pattern for the above project is as under:

S.No.	Source	Amount (Rs. in Crores)	% Contribution
1	Equity by GoI	1487.53	15.70%
2	Equity by GoG	1487.53	15.70%
3	SD for Central Taxes by GOG (50%)	407.71	4.30%
4	SD for Central Taxes by GOI (50%)	407.71	4.30%
5	Loan from bilateral/ multilateral agencies	5484.25	57.35%%
6	Upfront VCF and proceeds from Monetization	250.00	2.65%
7	**Total cost excluding Land, R&R and PPP component**	9474.73	**100.00%**
8	Subordinate Debt for Land including R&R cost by GoG	1234.00	
9	SD for State Taxes by GOG	677.59	
10	**Sub Total**	11386.32	
11	PPP component	452.00	
12	**Sub Total**	11838.32	
13	IDC for Loan to be borne by GoG	182.00	
14	**Total completion cost**	12020.32	

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3. A detailed sanction letter along with terms and conditions of the sanction will be issued in due course.

Yours faithfully,

(V.S. Pandey)
Director (MRTS)
Tel. 011-23061916

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CHAPTER -1

INTRODUCTION

1.1 BACKGROUND

Surat city, located on the western part of India in the State of Gujarat is also known as a ”Economic power house” of Gujarat. The city is located 284 Kms south of Gandhi Nagar, 265 kms of Ahemdabad and 289 Kms north of Mumbai. The economy of the entire city is based mainly on two industries, the textile industries of man made fibres /fabrics and the diamond cutting and polishing industry. Surat is one of the most dynamic cities of India having faster growth rate due to immigration from various parts of Gujarat and as well as from the other States of India. National Highway-8 (NH-8) passes within 16 Kms. of municipal boundaries of Surat and is located midway on the Ahmedabad-Mumbai route.

Surat is one of the oldest inhabited cities in the world and densely populated with an average 13680 persons/sq.km accommodating about 44.67 Lakhs people as per census 2011. Average decadal growth of population of Surat since 2001 to 2011 was about 55.29%. The census of India 2011 has revealed that one out of seven zones of Surat i.e. central zone has registered even negative growth. Surat city is spread over an area of 326.515 sq. kms. The Surat city is located at 21.18°N72.83°E with 21.195 Latitude and 72.8194 Longitude Surat’s high population growth rate, coupled with high economic growth rate has resulted in an ever increasing demand for transport creating excessive pressure on the existent transport system. With high growth in transport demand over the years, congestion on roads has been increasing due to phenomenal rise in private transport.

Absence of an efficient full-fledged public transport system coupled with rapid growth in the use of personalized vehicles has led to high consumption of fossil fuel and increase in the level of environmental pollution. Surat has also been selected as one of the hundred Indian cities to be developed as a “Smart City”. The existing network of public transport systems including dedicated BRTS needs to be strengthened further in order to cope-up with rising demand of transport system.

The inter-city traffic volumes in Surat necessitated a full-fledged integrated multi model mass rapid passenger system. In view of this, Government of Gujarat (GoG) vide their letter No. MEGA/MD/DMRC/IC dated 27.01.2016 requested Delhi Metro Rail Corporation (DMRC) to prepare a Detailed Project Report (DPR) for Metro Rail Project in Surat City.

The DPR for Surat Metro rail project had already been prepared and submitted to SMC and MEGA vide DMRC's letter no DMRC/GM/CS/04/Surat-DPR/2016/1150 dated 24th July'2017. Subsequently, in the month of August' 2017 Ministry of Housing and Urban Affairs announced a new metro policy-2017 and returned the said proposal for incorporation of new provisions. The Commissioner, Surat Municipal Corporation (SMC) vide his e-mail dated 16th September'2017 requested DMRC to review and revise the detailed project report(DPR) in view of new metro rail policy. Thus the said DPR for Surat Metro Rail project has been revised incorporating new provisions such as comprehensive mobility plan(CMP), alternative analysis, last mile connectivity, transit oriented development(TOD), value capture finance (VCF), property development(PD), public private partnership(PPP), unified metropolitan transport authority (UTMA), options for central assistance etc.

The DPR has been updated upto December 2018 level.

1.2 DEMOGRAPHIC PROFILE OF SURAT CITY

Surat - a vibrant city of Gujarat has witnessed phenomenal growth in population and employment and trend is expected to continue in future. Availability of various employment opportunities attracts immigration not only from different parts of Gujarat but also from other States of India too.

Surat city is one of the fast growing regions in India, it comprises of 07 Zones, 29 Wards, 116 Councils and decadal growth rate of 55.29% (as per 2011 census) and spreads over the area of 326.515 Sq. Km. Surat City is projected to have population as 69.36 Lakhs in the year 2021.The population of Surat City (comprising of Surat Municipal Corporation, Out Growth area & Urban Agglomeration) has increased from 0.49 million in 1971 to 4.59 million in the year 2011 (Table 1.1). However, decadal growth rate for population for SMC including outgrowth and urban Agglomeration has declined from 85.20% during 1971-81 to 63.30% in 2001-2011.

Table 1.1: POPULATION GROWTH IN SURAT CITY DURING 1971-2011
(Comprising of areas under Surat Municipal Corporation, Out Growth area & Urban Agglomeration)

S.NO.	CENSUS	POPULATION	GROWTH RATE (%)
01	1971	4,92,700	71.1% *
02	1981	9,12,600	85.20%
03	1991	15,19,000	66.40%
04	2001	28,11,614	85.10%
05	2011	45,91,246	63.30%

*Population in 1961 was 2,88,000, source: SMC

1.2.1 Total population in areas falling under the jurisdiction of Surat Municipal Corporation, Out Growth Area & Urban Agglomeration as per Census-2011 has been tabulated in Table 1.2.

Table 1.2 : TOTAL PUPUOATION IN SURAT AS PER CENSUS-2011

S.No.	Description	Male	Female	Total
01	Surat Municipal Corporation	25,43,145	19,23,681	**44,66,826**
02	Surat City-Out Growth Area	22,658	12,126	34,784
03	Surat City-Urban Agglomeration	53,570	36,066	89,636
	Total	**26,19,373**	**19,71,873**	**45,91,246**

Source: SMC

1.2.2 ZONE-WISE POPULATION, AREA, DENSITY & GROWTH RATE (2001-2011)

Zone-wise population, area, density and growth rate in Surat City for the year 2001-2011 has been tabulated in Table 1.3.

Table 1.3: ZONE-WISE POPULATION, AREA, DENSITY & GROWTH RATE (2001-2011)

Sl.No.	Zone	Area (Sq.Kms)	Population		Density per SQ Km.	Decade Growth 2001-2011(%age)
			2001 Census	2011 Census		
01	Central	8.18	4,13,641	4,08,760	49,971	-1.18
02	South West	111.912	2,42,466	3,47,447	3,105	43.30
03	South	61.764	4,07,980	6,95,028	11,253	70.36

04	South East	19.492	3,97,257	7,48,304	38,390	88.37
05	East	37.525	7,11,516	11,37,138	30,303	59.82
06	North	36.363	4,16,370	7,05,163	19,392	69.36
07	West	51.279	2,87,144	4,24,986	8,288	48.00
	Total	**326.515**	**28,76,374**	**44,66,826**	**13,680**	**55.29**

Source: SMC

1.2.3 MID YEAR POPULATION ESTIMATES

Mid year populations in between year 2011 to 2021 has been estimated adopting the average annual growth rate as 4.5% considering Decennial Population Growth Rate between 2001-2011 of Surat City as 55.29% and tabulated in Table 1.4.

Table 1.4: MID-YEAR POPULATION

Year	Decennial Population Growth Rate Between 2001-2011 of Surat City 55.29% i.e. average Annual Growth Rate is 4.5%	
	Per 100 Persons	Total Estimated Population
2011	100	44,66,826
2012	104.5	46,67,833
2013	109.2	48,77,774
2014	114.11	50,97,095
2015	119.25	53,26,690
2016	124.61	55,66,112
2017	130.22	58,16,701
2018	136.08	60,78,457
2019	142.21	63,52,273
2020	148.6	66,37,703
2021	155.29	69,36,534

Source :SMC

1.3 PRESENT SCENARIO OF TRANSPORT IN SURAT CITY

In the absence of adequate full fledged public transport system, the use of individual modes such as cycles, 2- wheelers, cars etc. and un-organized IPT has increased in Surat City. Cycling and walking constitute about 45% of the total trips and this feature needs to be integrated within the upcoming public transport initiatives. As per the records available, the total road length as on the year ending on 31-03-2016 is 3575 kms out of which 624 Kms is un-surfaced. The total vehicles registered in Surat as per the figures indicated by Regional transport Office (RTO), Surat are tabulated in Table 1.5:-

Table 1.5 : Registered Vehicles in Surat

SN	Type of Vehicle	01/04/2013 To 31/01/2014	01/04/2014 To 31/01/2015	Remarks
01	Motor Cycle	103492	138291	
02	Auto Rickshaw	1385	2914	
03	Jeep	447	463	
04	Private Auto Rick.	17	16	
05	Motor Car	20325	29391	
06	Taxicab	153	217	
07	Maxicab	63	16	
08	Stage Carriage	9	58	
09	Passenger Bus	107	35	
10	School Bus	86	91	
11	Private Service Veh.	26	8	
12	Police Van	1	11	
13	Goods Truck	767	1044	
14	Tanker	27	27	
15	Tempo	0	0	
16	Other Light Veh.	3692	3753	
17	Tran. Trailor	101	35	
18	Non Tras. Trailor	0	0	
19	Tractor	424	601	
20	Ambulance	23	16	
21	Others	239	174	
	Total	**131384**	**177161**	**Yearly Growth rate@34.84%**

*Source: Surat RTO

According to Comprehensive Mobility Plan (CMP), 2008 of Surat City, it has been estimated that internal trip rates in Surat stands elevated from 0.75 in the year 2005 to 0.8, 0.9 and 1.0 by the year 2011, 2021 and 2031 respectively. However, no change is foreseen in external trips. Trip rates and total trips have been, summarized in Table 1.6.

Table 1.6: EXISTING AND FUTURE TRAVEL DEMAND (Excluding Walk Trips)

Year	Population	Passenger Trips		Passenger Trips		Total
		Internal	External	Internal	External	
2005	3,607,596	0.75	0.5	2,705,697	1,803,798	4,509,495
2011	4,499,006	0.80	0.5	3,599,205	2,249,503	5,848,708
2021	6,500,003	0.90	0.5	5,850,003	3,250,002	9,100,005
2031	8,500,000	1.00	0.5	8,500,000	4,250,000	12,750,000

Source: CMP-2008 of Surat City

CMP-2008 also recommends LRTS to be in place in addition to dedicated BRTS corridors/routes in Surat City based on the studies of Gujarat Infrastructure Development Board (GIDB) and further indicates that the proposed LRTS would replace BRTS on some parts of the network, wherever found essential and justified. Presently CMP-2008 of Surat City is under up-datation and likely to be completed by the end of year 2017

1.4 EARLIER STUDIES FOR THE SURAT CITY

Under-mentioned Studies for Surat City have been conducted earlier for assessing the traffic conditions of the city and measures to be taken:-

In 1992, CRRI had produced Comprehensive Traffic and Transportation Study with SMC.

Development Plan for Surat prepared by SUDA in 1998, approved in 2004.

City Corporate Plan – 2001 (Plan initiated by SMC and prepared by CEPT)

Surat Vision – 2020 (Undertaken by SMC and assisted by CEPT)

City Development Plan – 2005 (initiative under JnNURM by SMC with technical support from CEPT)

CTTS – 2006 Draft Final Report (Study initiated by SMC and prepared by CRRI)

Integrated Public Transport Study – 2006 Draft (assignment by GIDB, prepared by CES)

Sustainable Transport System for Surat – 2007 (CEPT Study)

Comprehensive Mobility Plan – 2008 (initiated by SMC, technical support by CEPT)

Bus Rapid Transit System (BRTS) Plan – 2008 (initiated by SMC, technical support by CEPT)

Bus Rapid Transit System (BRTS) Phase 2 & 3 – 2012 (initiated by SMC, technical support by CEPT)

Service Level Benchmarks in Urban Transport for Surat City -2012

1.5 After conducting extensive route surveys and studies in Surat City and also considering technical feasibilities, DMRC had worked out various options of probable routes/corridors on the basis of recommendations and suggestions of Centre for Environmental Planning and Technology (CEPT)' University, Ahmedabad who is conducting the traffic and transportation surveys/studies, and presented before the concerned authorities of Surat Municipal Corporation (SMC) on various occasions. Details are envisaged in Table 1.7 as Annexure-'A'. Subsequently, on the suggestions of SMC & CEPT, following three corridors were shortlisted and presented before High Powered Committee (HPC) in a meeting held on 13.01.2017 at Ahmedabad under the Chairmanship of Hon'ble Dy. Chief Minister, Govt. of Gujarat:-

1) **Sarthana**-Nana Varachha – Railway Station- Chowk- Majuragate – Exhibition Centre-**Dream City**
2) **Bhesan**-L.P. Savani-Majuragate – MithiKhadi-**Saroli**
3) **Majuragate**-Bhagal-Katorgem**-Ved**

HPC considered all the options and after their detailed deliberations, approved the following 02 (two) metro routes/corridors to be included under Phase-I of Surat Metro Rail Project, which has been communicated through Minutes of Meeting (MoM) sent vide letter No. GUDC/METRO/HPC/2017/182 dated 23.01.2017:-

1. Corridor-I : **Sarthana to Dream City**
(via Nana Varachha, Railway Station, Chowk, Majuragate, Exhibition Centre)

2. Corridor-II : **Bhesan to Saroli**
(via L.P. Savani, Majuragate, MithiKhadi)

Details of above mentioned corridors are given in Table 1.8.

Table 1.8

MRTS CORRIDORS IN SURAT CITY (Ph-I)
(Proposed by DMRC and in- principle approved by Government of Gujarat)

S.No.	Alignment	Length (in Kms)	Remarks
01	**Phase-1** Corridor-1 **Sarthana to Dream City** (via Nana Varchha, Railway Station, Maskati Hospital, Gandhi Baug, Majura Gate, Roopali Canal, AlthanTenament, VIP road, Surat Women ITI, Convention Centre)	21.61 Kms	
02	Corridor-2 **Bhesan to Saroli** (via Ugat, Botanical Garden,UgatVaarigruh, LP Savani School, AdajanGam,Aquerium, Badri Narayana Temple, AthwaChaupati, Majura Gate, UdhanaDarwaja,KamelaDarwaja, Model Town,Magub ,Bharat Cancer Hospital)	18.74 Kms	
	Total	**40.35 Kms**	

1.6 STRUCTURE OF DETAILED PROJECT REPORT (DPR)

This report includes the chapters as under:-

1) Chapter-1 : Introduction
2) Chapter-2 : Travel Demand Forecast
3) Chapter 3 : System Selection
4) Chapter 4 : Geometric Designing Parameters and Alignment Description
5) Chapter 5 : Civil Engineering
6) Chapter 6 : Station Planning
7) Chapter 7 : Train Operation Plan & Rolling Stock
8) Chapter 8 : Train Maintenance Depots
9) Chapter 9 : Power Supply and System Parameters
10) Chapter 10 : Ventilation & Air Conditioning System
11) Chapter 11 : Signalling And Train Control System
12) Chapter 12 : Telecommunication & Automatic Fare Collection
13) Chapter 13 : Disabled Friendly Features
14) Chapter 14 : System Environment & Social Impact Assessment
15) Chapter 15 : Security Measures For a Metro System
16) Chapter 16 : Disaster Management Measures
17) Chapter 17 : Multi Modal traffic Integration at Metro Stations

18) Chapter 18 : Transit Oriented Development(TOD) Plan And Value Capture Finance(VCF)
19) Chapter 19 : Cost Estimates
20) Chapter 20 : Financing Options, Fare Structure And Financial Viability
21) Chapter 21 : Economic Appraisal
22) Chapter 22 : Implementation Plan
23) Chapter 23 : Conclusion And Recommendations

Annexure-'A'

Table 1.7

VARIOUS OPTIONS OF METRO CORRIDORS IN SURAT CITY, SUGGESTED BY DMRC

S. N	Options	Corridors	Details	Length (Kms)
01	Option-1	Corridor-I	**Gadhpur Road Jn. to Dream City** (via Varachha Road, Lambe Hanuman Road, Railway Station, Raj Marg, Dumas Road, Vesu Main Road)	24.5 Kms
		Corridor-2a	**Bhesan Road to Civil Hospital** (via L.P.Savani , Adajan Road, HaziraMarg, Chaupati Ring Road, Navsari Road)	11.8 Kms
			Total	**36.30 Kms**
02	Option-2	Corridor-I	**Gadhpur Road Jn. to Dream City** (via Varachha Road, Lambe Hanuman Road, Railway Station, Raj Marg, Dumas Road, Vesu Main Road)	24.5 Kms
		Corridor-2ab1	**Bhesan Road to KamathMahadev** on Surat-Navasari Road (via Ugat, LP Savani Road, Adajan Road, Majura Gate, Kharwarnagar, Swami NarayanaMandir)	17.00 Kms
			Total :	**41.50 Kms**
03	Option-3	Corridor-1	**Godhpur Road Jn. to Dream City** (via Varachha Road, Lambe, Hanuman Road, Railway Station, Raj Marg, Dumas Road, Vesu Main Road)	24.50 Kms
		Corridor-2ab2	**Bhesan Road to Anjana,** near Canal Road (via Ugat, LP Savani, Adajan, Majuragate, KamelaDarwaza)	14.00 Kms
			Total :	**38.5 Kms**
04	Option-4	Corridor-1	**Gadhpur Road Jn. to Dream City** (via Varachha Road, Lambe, Hanuman Road, Railway Station, Raj Marg, Dumas Road, Vesu Main Road)	24.50 Kms
		Corridor-2ab1	**Bhesan Road to KamathMahadev** on Surat-Navasari Road (via Ugat, LP Savani Road, Adajan Road, Majura Gate, Kharwarnagar, Swami NarayanaMandir)	17.00 Kms
		Corridor-3	**Civil Hospital to Shri Swami NarayanaGurukul** (via Kshetrapal Dada Marg, GopiTalao, Kotsafil Road, KaskiwadBadhal Road, Saiyadpura Main Road, Ved Road)	8.50 Kms
			Total:	**50.00 Kms**
05	Option-5	Corridor-I	**Gadhpur Road Jn. to Dream City** (via Varachha Road, Lambe, Hanuman Road, Railway Station, Raj Marg, Dumas Road, Vesu Main Road)	24.5 Kms
		Corridor-2ab2	**Bhesan Road to Anjana,** near Canal Road (via Ugat, LP Savani, Adajan, Majuragate, KamelaDarwaza)	14.00 Kms
		Corridor-3	**Civil Hospital to Shri Swami Narayan Gurukul** (via Kshetrapal Dada Marg, GopiTalao, Kotsafil Road, KaskiwadBadhal Road, Saiyadpura Main Road, Ved Road)	8.50 Kms
			Total:	**47.00 Kms**
06	Option- 6	Corridor-1	**Gadhpur Road Jn. to Dream City** (via Varachha Road, Lambe, Hanuman Road, Railway Station, Raj Marg, Dumas Road, Vesu Main Road)	24.50 Kms
		Corridor-2a	**Bhesan Road to Civil Hospital** (via L.P.Savani , Adajan Road, HaziraMarg, Chaupati Ring Road, Navsari Road)	11.8 Kms
		Corridor-3	**Civil Hospital to ShriSwaminarayanGurukul** (via Kshetrapal Dada Marg, GopiTalao, Kotsafil Road, KaskiwadBadhal Road, Saiyadpura Main Road, Ved Road)	8.50 Kms

			Total	**44.80 Kms**
07	Option-7	Corridor-I	**Gadhpur Road Jn. to Dream City** (via Varachha Road, Lambe, Hanuman Road, Railway Station, Raj Marg, Dumas Road, Vesu Main Road)	24.50 Kms
		Corridor-2ab3	**Bhesan Road to Saroli on Bardoli Road** (via Ugat, L.P. Savani, Adajan, Majuragate,KamelaDarwaza, ParvatPatiya)	18.50 Kms
			Total:	**43.00 Kms**
08	Option-8	Corridor-I	**Gadhpur Road Jn. to Dream City** (via Varachha Road, Lambe, Hanuman Road, Railway Station, Raj Marg, Dumas Road, Vesu Main Road)	24.50 Kms
		Corridor-2ab3	**Bhesan Road to Saroli on Bardoli Road** (via Ugat, L.P. Savani, Adajan, Majuragate,KamelaDarwaza, ParvatPatiya)	18.50 Kms
		Corridor-3	**Civil Hospital to ShriSwaminarayanGurukul** (via Kshetrapal Dada Marg, GopiTalao, Kotsafil Road, KaskiwadBadhal Road, Saiyadpura Main Road, Ved Road)	8.50 Kms
			Total	**51.50 kms**
09	**Option-9**	**Corridor-1M**	**Chharra to Dream City** **(**via Nana Varchha, Railway Station, Chowk, Majura Gate, Bhatar Char Rasta, Sarsana, Khajod char rasta**)**	**22.40 Kms**
		Corridor-2ab3	**Bhesan Road to Saroli on Bardoli Road** (via Ugat, L.P. Savani, Adajan, Majuragate,KamelaDarwaza, ParvatPatiya)	18.50 Kms
			Total:	**40.90 Kms**
10	Option-10	Corridor-1M	**Chharra to Dream City** (via Nana Varchha, Railway Station, Chowk, Majura Gate, Bhatar Char Rasta, Sarsana, Khajod char rasta)	22.40 Kms
		Corridor-2ab3	**Bhesan Road to Saroli on Bardoli Road** (via Ugat, L.P. Savani, Adajan, Majuragate,KamelaDarwaza, ParvatPatiya)	18.50 Kms
		Corridor-3	**Civil Hospital to ShriSwaminarayanGurukul** (via Kshetrapal Dada Marg, GopiTalao, Kotsafil Road, KaskiwadBadhal Road, Saiyadpura Main Road, Ved Road)	8.50 Kms
			Total	**49.40 Kms**
11	Option-11	Corridor-IM1	**Charra to Dream City** (via Nana Varchha, Railway Station, Chowk, Majura Gate, Bhatar Char Rasta, L.P. Savani, Mangalam heights)	23.00 Kms
		Corridor-2ab3	**Bhesan Road to Saroli on Bardoli Road** (via Ugat, L.P. Savani, Adajan, Majuragate,KamelaDarwaza, ParvatPatiya)	18.50 Kms
			Total:	**41.50 Kms**
	Option-12	Corridor-1M1	**Charra to Dream City** (via Nana Varchha, Railway Station, Chowk, Majura Gate, Bhatar Char Rasta, L.P. Savani, Mangalam heights)	23.00 Kms
		Corridor-2ab3	**Bhesan Road to Saroli on Bardoli Road** (via Ugat, L.P. Savani, Adajan, Majuragate,KamelaDarwaza, ParvatPatiya)	18.50 Kms
		Corridor-3	**Civil Hospital to ShriSwaminarayanGurukul** (via Kshetrapal Dada Marg, GopiTalao, Kotsafil Road, KaskiwadBadhal Road, Saiyadpura Main Road, Ved Road)	8.50 Kms
			Total:	**50.00 Kms**

CHAPTER-2

Travel Demand Forecast

2.1 INTRODUCTION

Metro Policy 2017 rightly proposes that the proposal for central assistance for identified Metro Rail project should be mooted by the State Governments on two counts- Firstly, because very nature of urban transport is such that it has intricate connect with other issues of urban development making it logical that only those who have overall perspective and feel of the city formulate the plan for urban transport fro that city and secondly also because, the "Urban Development" is a State Subject of the Constitution.

It is in this context that the DPR for Surat Metro is being submitted for central assistance is not only being submitted through State Government for various reason but Government of Gujarat (GOG) apart from its own 50% contribution of equity is also contributing towards the cost of the land and towards taxes and duties. More particularly the proposal for Surat Metro takes into account the overall urban development and urban transport plan of State Government.

The policy further requires that metro rail development should follow a "systems approach" that is comprehensive approach to planning of urban land use and transport infrastructure.

The proposal is required to evaluate alternative modes for the most optimum mobility plan for the city. And as such the policy puts a mandatory pre-requisite for the city having a Comprehensive Mobility Plan (CMP) for planning metro rail in any city.

With the above in perspective the policy allows cities with two million and plus population to start planning for mass transit systems including metro rail based on CMP. In this context, it is specified that Surat is a unique case where population has already exceeded 5 million mark and shall have approximately 07 million population by the year 2021.

Also Surat being the "Economic Power House" of Gujarat, has substantial daily floating population from other cities. Additionally international tourist footfall is also at Surat.

As per requirement of the city the DPR of Surat Metro Rail Project has been prepared by DMRC which have sufficient expertise to formulate DPR for

other cities in the country. DPR of Surat Metro has been prepared keeping in view the growing need of the city in consultation with GoG and Central Government.

Such proposal of Surat Metro is consistent with the vision of city as articulated in Comprehensive Mobility Plan (CMP). Also it is clarified that due to unique position of Surat in the country both central and state governments have been proactively involved in planning, approval, construction and operations of Surat Metro Rail.

2.2 DOES SURAT HAVE A MOBILITY PLAN

In the year 2017, a Comprehensive Mobility Plan (CMP) for Surat city has been got completed by Surat Municipal Corporation (SMC) through Centre of Excellence in Urban Transport (CEPT), Ahmedabad. which mainly contains the following:-

(1)Chapter 1:- Introduction

Overview
Project Background
Surat past Efforts Towards Mobility Management
Scope of Study
Study Approach
Methodology
Introduction to Surat & Surrounding Regions
Surat
SUDA
Outside SUDA
Plan Period
Study Area Delineation
Traffic Analysis Zone
Data Collection
Report Structure and Chapter Scheme

(2) Chapter 2:- Growth Trends

Physical Growth in Surat
Land Use
Land use in Study Area
Demographic Trends
Population Distribution
Socio Economic Characteristics
Family Size and Composition
Gender Status and Household Size
Education
Income Distribution
Economy Surat
Workers in Surat
Employment Generating Activities
Employment Estimates Surat

Summary

(3) Chapter 3: Existing Transportation Systems

Vehicular Growth in Surat
Road Network
Regional Network
Urban Road Network
Road widths
Bridges, Flyovers, ROB & RUB
Road Network Pattern and Completeness
Network Regional Transport-Flows and Terminals/Nodes
Railway
GSRTC Network
Public Transport
City Bus Service
BRTS-sitilink
Public Transport Coverage
Intermediate Public Transport(IPT) Modes
Non-Motorised Transport(NMT)
Existing NMT Infrastructure in Surat
Parking
System Performance
Traffic Volume
Network Speeds
Accidents
Air Quality
Benchmarking Urban Transport in Surat
Summary

(4.) Chapter 4 People Movement in Surat

Household Surveys
Household Income and Vehicle ownership
Expenditure on transportation
Travel Characteristics
Trip Rate
Mode Choice
Trip Purpose and Mode Share
Trip Length frequency distribution
Willingness to shift
Summary

(5) Chapter 5 Urban Freight Movement in Surat

Introduction
Defining Urban Freight
Institutional Responsibilities
Urban freight Vehicles and Time Zones
Primary Data Base
External Freight Movement
Commodity Movement
Goods Vehicular Flows
Truck Flows-Major Commodities

Goods Vehicular Flows-Vehicle Load
Vehicular Composition
Urban Freight Generators
Internal Goods Movement
Manufacturing Activity Cluters
Wholesale Trade Activity Concentrations
Textiles Industries in Urban Freight
Textile- Supply Chain Analysis
Externalities to Urban Freight
Urban Freight and Road Safety
Urban Freight and Environment
Land Use Conflicts
Congestions
Industrial Areas and Parking Management
Summary

(6) Chapter 6: Issues and Challenges in Surat

Managing Growth
Inadequacies in Road Network
Congestion
Safety
Lack of Public Transport
Uncontrolled Parking
Environment
Summary

(7) Chapter 7: Growth Dynamics in Surat

Growth Direction
Proposed Future Development
Population Projections
Employment Projections
Network Development
Travel Demand Model Development
Modelling Framework
Model Software
Four Stage Modelling Process
Future year Auto(private vehicles)assignment and analysis
Future year for Integrated Public Transport assignment and analysis
Evaluating the Scenario

(8) Chapter 8: Mobility Vision for City

Stakeholder Consultation
Online Survey Results
Vision Statements
Goals

(9) Chapter 9: Mobility Improvement Measures

Proposal for Integrated Land use and Urban Transport
Proposal for Road network Development Plan
Principles for CMP Road Network Development
CMP Network Development

Future Urban Road Network
Proposed ROWs of Road network
Proposed River Bridges, RUB/ROB
Road alignment along DFC corridor
Costing Road Proposals
Proposal for Integrated Transport System
Regional Bus Service-GSRTC
Road Transit Corridor
City Bus Service
Depots and Terminals
Proposal for Interchanges
Fare Integration
Intelligent Transport Management System(ITMS)
Water Transport for Surat
Transit Oriented Development
TOD in Surat
Infrastructure cost for TOD
Value Capture and Self-Financing for TOZ
Proposal for NMT Facility Improvement Plan
NMT Approach in India
Alternate Approach to Planning NMT
Planning Process and Strategy
Network selection for Bicycle Lanes in Surat
Network selection for Footpaths
NMT Street Design Principles
Supporting Initiatives
Mobility Management Measures
Parking
Accident Management
Accident Monitoring Cell
Junction Improvement
Setting up speed limits for the city
Design Improvements
Urban Freight Proposals
Fiscal Measures

(10) Chapter 10: Implementation Program

Prioritization of projects
Identification of Funding Agency
Implementing Agencies
Existing Institutional Arrangement-Urban Transport
Existing Institutional Framework in the study area
Proposed Institutional Arrangement-Urban Transport
MATA
Surat Municipal Corporation(SMC) as the main Co-ordinating agency

Gist of Comprehensive Mobility Plan (CMP) – 2017 of Surat City related to MRTS Project as provided by Centre of Excellence in Urban Transport,(CEPT), Ahmedabad, the agency engaged by Surat Municipal Corporation (SMC) to

update the CMP, has been included in this DPR which mainly contains the following:-

Demand Analysis for Alternative CMP Surat Metro Corridors
Background
Approach for analysis of alternative corridors
Demand Assessment
Corridor Characteristics
Capacity of existing transport system
Summary
Observations
Conclusions

2.2.1 Demand Analysis for Various Surat Metro Corridors(CMP)

2.2.1.1 Background

A Comprehensive Mobility Plan for the city of Surat has been prepared for the year 2046 which analyses the existing travel demand pattern and city structure and proposes road network and tentative rapid transit network for the future year of 2046.

The Comprehensive Mobility Plan aims to retain the compact structure of the city of Surat and proposes intensification of development along the rapid transit corridors. The report recommends the following corridors as potential metro network:

1. Dream city –Kamrej (28.9 km)
2. Bhesan–Umbhel (26.3 km)
3. Majuragate to Karamala (15.8 km)

This report presents demand analysis and network characteristics, in order to assess suitability of metro for these three corridors.

2.2.1.2 Approach for analysis of alternative corridors

For assessing suitability of metro mode for the proposed rapid transit corridors, following factors have been considered:

1. Demand assessment along the corridor
2. Characteristics of the corridor
3. Capacity of existing transport system and its ability to handle future demand

2.2.1.3 Demand Assessment

An assessment of potential ridership on the three CMP metro corridors for the future years of 2026 and 2046 has been undertaken. An analysis of future PT ridership is important to assess suitability of a metro or another rapid transit mode.

The demand outputs interims of peak passenger flows (PHPDT) for the year 2026 and horizon year 2046 has been presented in the table 2.1. It can be seen that Dream City to Kamrej has the highest passenger flow of 16000 PHPDT in 2026 going up to 46000 PHPDT in 2046. The other two corridors - Bhesan – Umbhel and Majuragate to Karamala has a demand of around 24000 PHPDT and 19500 PHPDT in 2046 respectively.

2.2.1.4 Corridor Characteristics

A description of the corridors in terms of the road widths and adjacent land uses has been presented below:

i. RTC 1: Dream City to Kamrej/ Vav

Dream city to Kamrej/ Vav corridor is about 29 km long cutting across the city centre and also connecting two far ends of the city. This corridor can be divided into three parts, i.e. Dream City to Majura Gate, Majura Gate to Surat railway station, Railway station to Sarthana and Sarthana to Kamrej/ Vav based on the adjacent land use characteristics.

The section from proposed Dream city to Majura Gate is about 8km, of which Majura Gate to Althan is fairly densely developed with mixed uses-industries, educational institutions, commercial and lower and middle income residential land uses. The section from Althan to Dream City is currently developing with residential and commercial land uses coming up in this area. Recently, SGCCI has developed a huge Exhibition Centre on this corridor towards proposed Dream City, which is likely to be a major attraction node in near future. Dream city is the one of the prestigious projects of Surat for which master planning is completed and the construction is underway. Envisaged as the diamond hub of India, it is expected to generate huge employment in future. The existing Row of entire corridor connecting from Dream City approach to Majura Gate is ranging from 18-24 m which is proposed to be widened up to 30 m as per the CMP.

Majura Gate to Railway station corridor passes through the city centre area which is very densely built up. The development consists of commercial, retail local markets, government buildings etc. The inner city is the major employment and attraction area in the city. The existing Right of Way (RoW) from Majura Gate to Railway station is ranging from 15 to 24 m which is not ideal for any kind of at-grade transit system, but being the major node of the city, it is important to connect this area with rapid transit system. The area around Railway station is also a major node. GSRTC and the existing city bus terminals are also located in close proximity to the station, owing to which this node is already been planned and developed as the multimodal transit hub.

There is already an operational BRT from Railway station to Kamrej via Sarthana and Valak. The section from Railway station to Sarthana is highly dense with mostly commercial and retail markets. It also provides direct access to Hirabaug and surrounding area which is the existing diamond hub of the city. There are many power loom clusters situated along the corridor which operate 24 x 7. The small segment from Railway station to Hirabaug is having a Row of 36m currently, which is proposed to be widened to 45m in the CMP.

The section from Sarthana to Kamrej is currently developing; Kamrej is already connected to the railway station directly through BRT. There is an alternative node -Vav, 3 kms south of Kamrej on the national highway, which can also be considered for connecting through a rapid transit corridor in future. The section from Sarthana to Vav via Pisodarais proposed to be 45m RoW in CMP while Sarthana to Kamrej is already having BRT system operational with 60m RoW.

ii. RTC 2: Bhesan to Umbhel

Bhesan to Umbhel corridor of around 28km length passing through city center via Adajan, Majura Gate, Kamela Darwaja and Saroli. This corridor can be divided into three parts; i.e. Bhesan to Adajan, Adajan to Saroli and Saroli to Umbhel.

Bhesan to Adajan is the main street of the western part of the city with existing RoW ranging from 24m to 36m. A very small segment of around 3 km from Bhesan to L P Savani has 24m RoW which is proposed to be widened to

30m in CMP proposal. Roads with 30m and above are appropriate for development of any kind of rapid transit system.

Bhesan to Adajan is the section with predominant upper and middle income group residential development. Along the corridor, mixed land use with retail shops and market on the road abutting properties can be seen. Many schools, hospitals and recreational activities are also located along this corridor. This corridor provides direct connectivity with city center area.

The corridor from Adajan to Saroli via Athwa gate, Majura gate, Kamela Darwaja, is predominantly commercial. There are many government offices, colleges and major hospitals in the section from Adajan to Majura Gate. Civil Hospital and B D Mahavir Heart Hospital are the two big hospitals located along this section. Majura Gate and Udhana are major destinations because of textile and ancillary industries lying on this corridor. Udhana junction is on the BRT corridor connecting Sachin. Udhna to Kamela Darwaja is a high employment zone because of location of textile markets and industries along this section. Kamela Darwaja to Parvat Patiya through Mithi khadi is again predominantly lined with textile markets and industries on both side. From Parvat Patiya to Saroli, a lot of new textile markets are coming up on both sides.

The existing RoW on Ring Road is 60m;KamelaDarwaja to Canal corridor, it is 24m and again from Canal corridor to Saroli, it is 60m. There is a proposal of widening from 24m to 30m the section from Kamela Darwaja to Canal corridor in CMP.

As per initial discussions with SMC, there is also a proposal of station for High Speed Rail (HSR) near Umbhel, owing to which major external demand is expected at this node in future. Saroli to Umbhel corridor via Chedcha and Ladvi is about 8km long. It passes through the vacant area currently but there is a proposal of continuous road from Saroli to Umbhel with 45m of RoW. In future, this corridor would provide connectivity to HSR station and vacant land along this corridor may have an opportunity to be developed as high density commercial and residential land use.

iii. RTC 3: Majura Gate to Karmala

The corridor from Majura Gate to Karmala via Ved has a length of about 15 km. Currently the city does not have direct connection across the river, and there is no development on the other side of the river. There is a proposal of connection across the river from Ved to Karmala with 45m of RoW that will also attract development to happen across the river and will also provide direct connection to city center from other side of the river.

The Majura Gate to Ved via Chowk and Katargam has length of about 8km, with varying RoW from 18m to 36m. A lot of government offices, local and informal markets are located along this corridor. The Katargam area has existing textile industries on one side and the other side of the corridor has predominantly residential land use with middle and upper middle income group of societies. Ved is an existing gamtal having predominantly residential development. There are also many educational institutes are located along the corridor.

2.2.1.5 Capacity of existing transport system

An analysis of vehicular traffic along the corridor with and without the rapid transit corridor has been undertaken, in order to ascertain the importance of the transit corridor. This would help assess the need for the project - what will be the status if the alternative is not implemented. Two indicators have been looked at:

1. Congestion relief because of metro: % corridor congested with VCR more than 0.9 in with metro and without metro scenario.
2. Travel time benefits: Travel time along the corridor and time savings in with metro and without metro scenario.

The results of this analysis has been presented in the summary table below.

2.2.1.6 Summary

Table 2.1 : Summary of Analysis

Rapid Transit Corridors (RTC)	Approx. Length (km)	Passenger demand on the corridor (PHPDT) (with Project scenario) – 2026	Passenger demand on the corridor (PHPDT) (with Project scenario) - 2046	Existing Predominant Land use characteristics along the corridor	Proposed Road widths (m) in CMP	Congestion relief (2046)		Pvt. Veh. Time in Min. (with metro)	Pvt. Veh. Time in Min. (without metro)	Vehicular travel time saving (with versus without metro)
						With metro - % of Corridor congested	Without metro - % of Corridor congested			
1. Dream City to Kamrej/ Vav	29	16000	34000			29%	43%	78.24	88.01	11%
A. Dream city to Majura Gate	8	7300	19500	Residential and commercial	Existing is 18 - 24 and proposed to be widened to 30m in CMP	55%	69%	23.0	22.9	0%
B. Majura Gate to Railway Station	4	16000	33500	Commercial and historically developed local markets	Existing is 15 - 24 m	61%	76%	18.74	22.53	17%
C. Railway Stn to Sarthana	9	15000	34000	Diamond industries, Textile looms & Industries and commercial	Existing is 36 - 60 and proposed to be widened to 45 - 60 m in CMP	15%	38%	19.53	24.21	19%
D. Sarthana to Kamrej/ Vav	7	4000	11000	Under development phase, opportunity to develop as TOD	Existing Sarthana to Kamrej - 60 m Propsoed from Sarthana to Vav - 45m	0%	2%	16.97	18.37	8%
2. Bhesan to Umbhelv	28	6500	24000			28%	42%	81.1	88.9	9%

Rapid Transit Corridors (RTC)	Approx. Length (km)	Passenger demand on the corridor (PHPDT) (with Project scenario) – 2026	Passenger demand on the corridor (PHPDT) (with Project scenario) - 2046	Existing Predominant Land use characteristics along the corridor	Proposed Road widths (m) in CMP	Congestion relief (2046)		Pvt. Veh. Time in Min. (with metro)	Pvt. Veh. Time in Min. (without metro)	Vehicular travel time saving (with versus without metro)
						With metro - % of Corridor congested	Without metro - % of Corridor congested			
A. Bhesan to Adajan	7	5800	16500	Residential and mixed land use with recreational activities	Existing 24 - 36 m and proposed to be widened to 30 - 36 m in CMP	12%	18%	15.6	17.5	11%
B. Adajan to Saroli	11	6500	24000	Commercial, Textile markets & industries and public buildings	Existing 36 - 60 m and proposed to be widened to 60 m in CMP throughout	69%	88%	43.5	48.6	10%
C. Saroli to Umbhel	8	3000	10000	Mostly vacant	45 m RoW proposed in CMP	0%	15%	22.0	22.9	4%
3. Majura Gate to Karmala	15	4400	19500			25%	44%	46.1	59.6	23%
A. Majura Gate to Ved	7	4400	19500	Residential and Katargamtextile power looms	Existing 18 - 36 m Proposed to 36 m (except section in old city)	48%	64%	26.2	34.3	24%
B. Ved to Karmala	9	4300	19200	Mostly vacant	Existing - No connection across river 45 m proposed in CMP with river bridge connection	0%	22%	19.9	25.4	22%

2.2.1.7 Observations

Rapid Transit Corridor 1: Dream City to Vav

Rapid Transit Corridor 1 section from Majura Gate to Sarthana section has the maximum PHPDT about 34000 in year 2046 which is considerably high. In the year 2026 also, the demand is 15000 PHPDT. In order to accommodate passenger flow of this magnitude, a metro system is more appropriate. The section from Dream City to Majura Gate is not fully developed as yet and has the proposed diamond hub 'Dream City' coming up in the future. The demand along this section is therefore expected to increase from 7300 PHPDT in 2026 to 19500 PHPDT in 2046. The rapid transit corridor in this section would support the proposed Dream City development and hence can be considered for metro along with Majura Gate to Sarthana section.

The demand build up from Sarthana-Kamrej/Vav is comparatively slower as this section is currently vacant and the development of the area is expected to happen post 2026. This section can therefore considered for metro in Ph- II with and without project analysis shows that this corridor if planned as a metro, would reduce the congested network length to 30% in comparison to 44% in the 'Without metro' scenario. The travel time saving along this corridor would be around 10% in with metro scenario.

Rapid Transit Corridor 2: Bhesan to Umbhel

Bhesan –Umbhel rapid transit corridor connects western part of the city to the city centre and commercial area coming up along Saroli corridor. As presented in the summary above, the section from Bhesan to Saroli (i.e. A & B) has16500-24000 PHPDT in 2046 according to demand assessment. In the year 2026, the passenger demand is around 5800-6500 PHPDT. Saroli-Umbhel section demand in 2026 is 3000 PHPDT increasing to 10000 PHPDT. However, as and when the High Speed Rail corridor and station comes up, external passenger demand would also get added to this corridor. With and without project scenario shows an overall 10% benefit in terms of private vehicular travel time along the corridor.

Since the passenger flow from Bhesan- Saroli section is more than 15000 PHPDT in 2046, it is proposed to consider metro as the mode for Bhesan-Saroli section. Extension of this corridor to Umbhel may be considered in Phase II based on HSR project implementation and demand analysis.

Rapid Transit Corridor 3: Majura Gate to Karmala

This corridor has a passenger demand of around 19000 PHPDT in year 2046. Majura Gate to Ved section is about 7km, which is predominantly residential in the Ved area and has textile industries and power looms towards Katargam. Looking at the demand, this corridor can be recommended for metro system, however owing to road width constraints in the Majura Gate to Katargam section, underground metro may have to be considered. The section from Ved to Karmala is completely vacant and currently there is no connection across the river. This road network has been proposed with 45 m RoW in CMP which will also attract development in future. This proposed growth towards north of the city may take time to build up. The 2026 demand for both the sections is not very high (around 4400 PHPDT). This corridor may hence be considered in Phase II. In terms of travel time, significant savings to the tune of 25% is observed.

2.2.1.8 Conclusions

Based on the demand analysis, it can be concluded that all three corridors may be considered for metro. However, in the first phase following corridor /corridor sections are recommended:

Dream City- Sarthana

Bhesan-Saroli

For Phase II, Majura Gate- Ved can be considered. The extension to Karamala based on road network development and demand build can be considered after demand assessment. Similarly, extensions from Sarthana-Kamrej/Vav and Saroli-Umbhel can be taken up in the subsequent phase.

2.3 Travel Demand Assessment and Forecasting

2.3.1 Study Background

A metro system is being proposed for the city of Surat, for which Delhi Metro Rail Corporation (DMRC) is preparing a Detailed Project Report. The Centre of Excellence in Urban Transport, CEPT University was commissioned by Surat Municipal Corporation in the year 2016 vide their letter no. CE-Spl. Cell/Out/339 dated 31.08.2016 for undertaking demand assessment for alternative metro corridor proposals. The objective of this study was to carry out demand forecasting for metro corridors which could aid in preparing the Detailed Project Report. CEPT University Ahmedabad has forwarded final report on Traffic & Transportation assessment for Surat Metro Rail Project in two volumes (Vol. I & Vol. II) vide their letter No. CEPT/SM/2017/04/01 dated 24-04-2017. The same shall be sent separately.

As per the Comprehensive Mobility Plan and Alternative Analysis, three corridors - Dream City to Sarthana, Bhesan to Saroli, Majura Gate to Ved are being proposed for metro (Refer Map 2-1).Alternative alignments of the corridors have been studied and evaluated using a demand modeling exercise to arrive at the most suitable one based on the ridership levels.

2.4 Past Studies

Several transport studies have been carried out till date for Surat which served as a useful base to understand past trends and the city's context:

In 1992, CRRI had produced Comprehensive Traffic and Transportation Study with SMC.

Development Plan for Surat prepared by SUDA in 1998, approved in 2004.

City Corporate Plan – 2001 (Plan initiated by SMC and prepared by CEPT)

Surat Vision – 2020 (Undertaken by SMC and assisted by CEPT)

City Development Plan – 2005 (initiative under JnNURM by SMC with technical support from CEPT)

CTTS – 2006 Draft Final Report (Study initiated by SMC and prepared by CRRI)

Integrated Public Transport Study – 2006 Draft (assignment by GIDB, prepared by CES)

Sustainable Transport System for Surat – 2007 (CEPT Study)

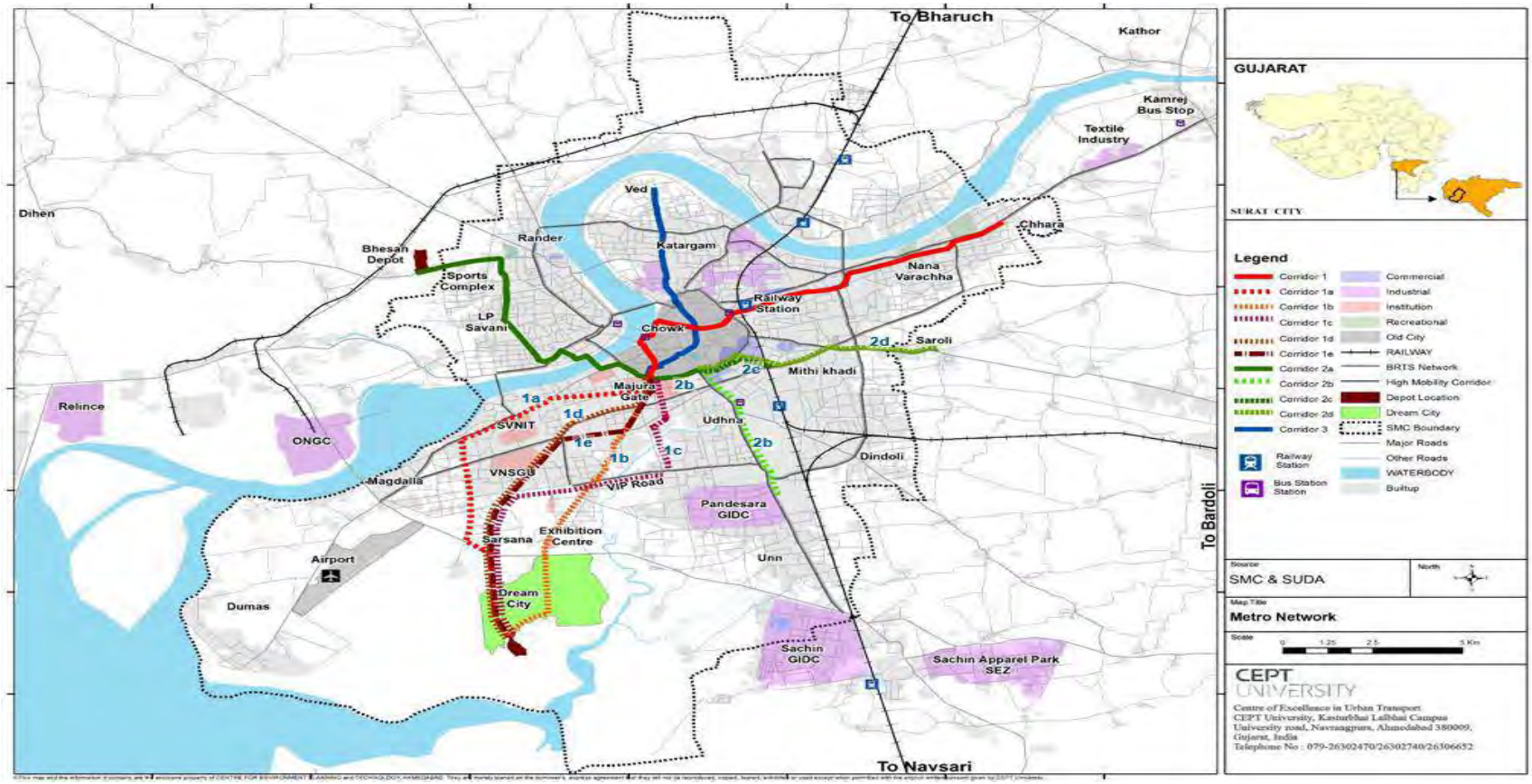
Comprehensive Mobility Plan – 2008 (initiated by SMC, technical support by CEPT)

Bus Rapid Transit System (BRTS) Plan – 2008 (initiated by SMC, technical support by CEPT)

Bus Rapid Transit System (BRTS) Phase 2 & 3 – 2012 (initiated by SMC, technical support by CEPT)

Service Level Benchmarks in Urban Transport for Surat City -2012

Comprehensive Mobility Plan for Surat City - 2017



Map 2-1: Metro Corridors Options

Source: CoE-UT, CEPT University

2.5 Study Area Delineation

The first step of the process was to define the study area for modelling purposes. Surat is the major urban area in the region, with strong linkages to urban centers in and around Gujarat. The Dedicated Freight Corridor (DFC) passing through the region along with the Delhi-Mumbai Industrial corridor (DMIC) is expected to trigger huge investment and growth in the near future.

The Development Plan of 2035 has identified growth nodes in Olpad and Kamrej with residential areas to be developed in these areas. The recent Development Plan 2035 Surat Urban Development Authority has increased its area to include all the above mentioned developments from 715 Sq. km to 1351sq.km. Hence for this study, the Development Plan area is considered as the study area.

Surat Urban Development Authority (SUDA)

SUDA was constituted in 1978. It now has a spatial extent measuring 1351 sq.km. Its jurisdiction lies both right and left banks of the Tapi river, which includes the area of Surat Municipal Corporation (SMC), two INAs, 15 urban areas and 182 villages.

Surat Municipal Corporation (SMC)

The area within the Surat Municipal Corporation limits consists of:

- The traditional city centre, known as the 'Old city' with relatively high-density development, large concentration of commercial activities and narrow streets.
- The southern and eastern part of the city is recognized as the industrial zone of the city, with large and small industries and low income residential areas.
- Presence of textile market along certain areas of the Ring road.

2.5.1 Growth Potential of Surat

The total population of the study area was 50.87 Lakh in 2011 with an annual growth rate of 4.3 % (2001-2011). It is estimated that in year 2016 there are 59.29 people residing in the study area. The future population is considered based on Surat Development Plan 2035. Based on growth trend estimated, population for year 2046 is projected to be around 1.25crore and population for year 2026 and 2036 has been taken from the DP 2035.

Table 2-2: Population Projections

Year	Population	CAGR
2011	50,81,856	-
2016	59,29,821	3.13%
2021	73,00,000	3.69%
2026	85,50,000	3.53%
2031	98,00,000	3.34%
2036	110,00,000	3.14%
2041	118,50,000	2.86%
2046	125,00,000	2.60%

The following presents the proposed developments which are likely to have an impact on spatial growth of Surat city:

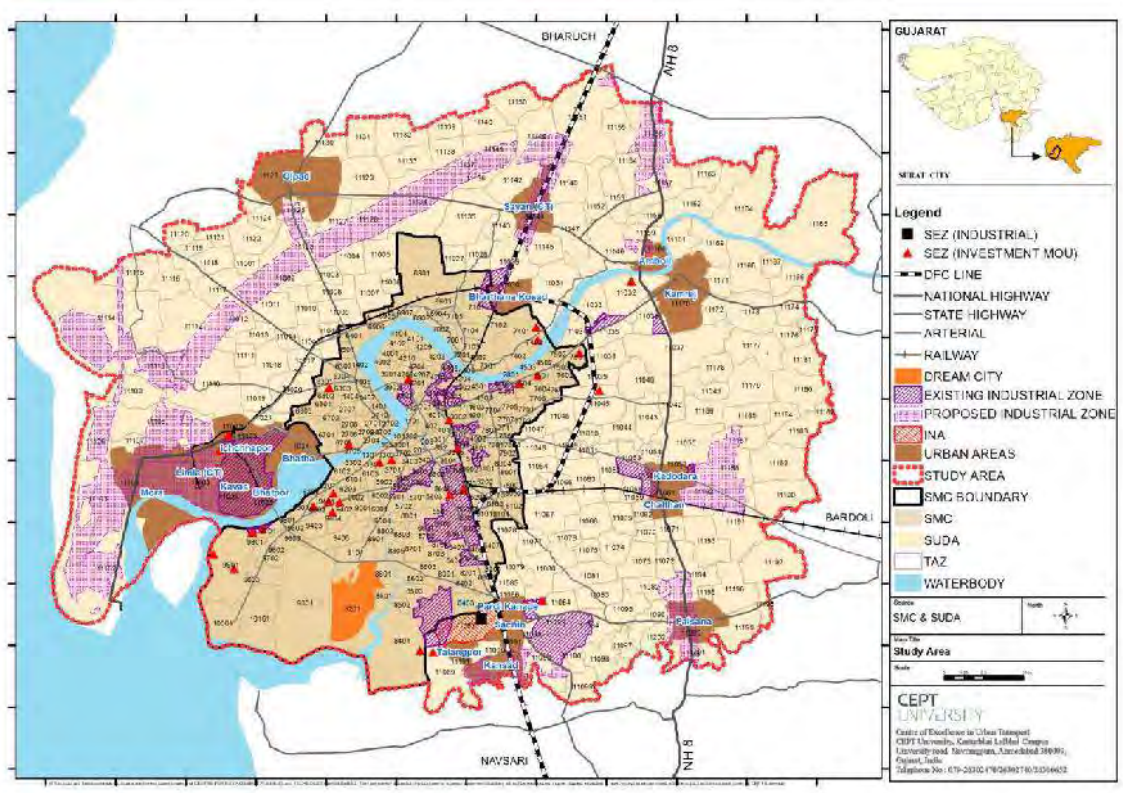
The Dedicated Freight corridor (DFC) passing through the region along with the Delhi-Mumbai Industrial corridor (DMIC) is expected to trigger huge investment and growth in the near future. Surat industrial growth region zone has also been identified as a part of the recent Development Plan 2035. As a part of the proposal, an industrial belt is identified towards the west of SMC connecting industrial location in Hazira to the DFC corridor near Gothan. Along with this, an industrial corridor along the broad gauge railway line near Gothan- Kosad is also proposed.

SUDA had declared industrial zone in Pandesara GIDC, Udhna GIDC, Katargam GIDC, and Navagam GIDC. SEZ were also proposed in Sachin GIDC (SURSEZ Diamond & Gem Dev Corporation) and Vanj GIDC (Surat Apparel Park Gujarat Industrial Dev. Corporation). Apart from these, GIDC has also developed Hazira Special Investment Region.

The Proposed Draft Development Plan 2035 aims to formulate an efficient and inclusive development of SUDA region. Some salient features are as follows:

- Population is expected to reach 110 Lakh by the year 2035. Hence land use categorization has been done with a view to accommodate an additional 60 Lakh population.
- 9 major growth nodes to accommodate development have been identified, namely; Olpad, Kathor, Kamrej, Kadodara, Palsana, Dream City, Hazira and Tena.
- Increase in percentage of urbanized land from 289 sq.km in the year 2015 to 787 sq.km.
- Zone dedicated as Gamtal in Development Plan 2004 has been kept intact. A village extension zone of about 200-300m, from the boundary of the Gamtals has been proposed in order to ensure natural growth of villages. If population is less than 5000, the Village Extension Zone should be 200m from the Gamtal boundary, otherwise 300m.

- Total residential area proposed is 358 sq km, which accounts for about 48% of the total urbanized area proposed.
- Existing hazardous industries within Surat city area, measuring about 4.3 sq km, are proposed to be shifted to the newly added Pinjarat village. Total area proposed as industrial zone is 136 sq km.
- In order to link Hazira Industrial Zone with the rest of the country, there is a need to establish strong dedicated transportation and communication network. In the light of this, an industrial corridor along broad gauge railway line near Gothan - Kosad has been proposed. This corridor would link Hazira to the existing Delhi - Mumbai railway line and is about 30kms long.
- The region would be developed through TP Scheme, under GTPUD Act by forming a special purpose vehicle exclusively for the purpose.
- A 'Dream City' is being planned to provide a global trading platform to the diamond traders. The Dream City will develop as a smart city which would help in developing skills in the diamond sector. The zone is proposed in part of Khajod village.
- New commercial zone has been proposed in Chaltan village, near NH-8, which has a potential to develop as a commercial centre for APMC. Total commercial zone proposed in Development Plan 2035 is 12 sq km, accounting for about 1.50% of total urbanized area.
- A 90m wide Outer Ring Road (ORR) has already been proposed for Surat city. A new Regional Ring Road of 120m width and 116km length in SUDA area, acting as a gateway for traffic towards Mumbai and Ahmedabad is proposed. Another 60m wide ring road is proposed between the two ring roads to accommodate the proposed residential zone. All three ring roads are connected to the growth nodes and other major city centres through a network of major radials of 90m, 60m, and 45mwidth.
- Apart from these, there is additional road network proposed for SUDA. The total road length proposed in Development Plan 2035 is more than 2200 km, out of which total major roads proposed have a total length of 917km.
- Other proposals include Dream City project in Khajod area, development of multimodal transportation hub at Surat Railway Station etc.



Map 2-2: Study area-Existing Administrative boundaries and identified Growth Centers

Source: CoE-UT, CEPT University

The study area defined thus is an amalgam of:

Table 2-3: Study area Details

Study Region	Area (sq. km)	Comprises of
SMC	326.52	89 wards
OLD SUDA	715	SMC area, 11 urban area {Bharthana Kosad (CT), Bhatha (OG), Bhatpor (OG), Chalthan (CT), Ichchhapor (CT), Kadodara (CT) ,Kansad(M), Kavas(OG), PardiKanade(CT) , Sachin(CT), Talangpor (OG) and Varelli (CT)} ,1 INA (Sachin INA)86 villages
NEW SUDA (Since 2015)	1351	SMC area, Old SUDA, 4 urban areas (Amboli (CT), Limla (CT), Mora (CT), Sayan (CT)), 1 INA (Hazira) and 96 villages
Total	**1351**	**SMC area, 2 INA, 15 urban areas and 182 villages**

Source: CoE-UT, CEPT University

2.6 Zoning

The study area is subdivided into smaller areas known as Traffic Analysis Zones (TAZs). TAZs are delineated taking into account various factors like administrative boundaries, physical barriers such as water bodies, railway lines, highways and homogenous land uses.

The following the criteria were used while delineating the TAZ boundaries.

Road network

Existing and planned transportation facilities; and

Boundary compatibility

Physical geography (physical barrier such as river, canals, railway lines, etc.)

Census/Administrative boundaries: Entities using travel demand models often need to analyze travel patterns relevant to certain predefined political geographies, such as wards, cities, districts etc.

- Socioeconomic data (existing and future)
- Homogeneous land uses, where feasible

In SMC, 89 wards were further split to form TAZs. Based on the above criteria, around 313 TAZs were identified. Similar exercise was also carried for the SUDA area and in most of the cases the concept of one village as one TAZ is adopted, in some cases the bigger villages were split into two or more TAZs. TAZs inside the study area, or model boundaries, is defined as internal zones. Those zones outside the study area along the model boundaries are defined as external zones. The area, outside the study area is aggregated into larger zones along the various directions of travel, which are known as external zones. Six external zones have been identified as per the direction of incoming/outgoing traffic as below:

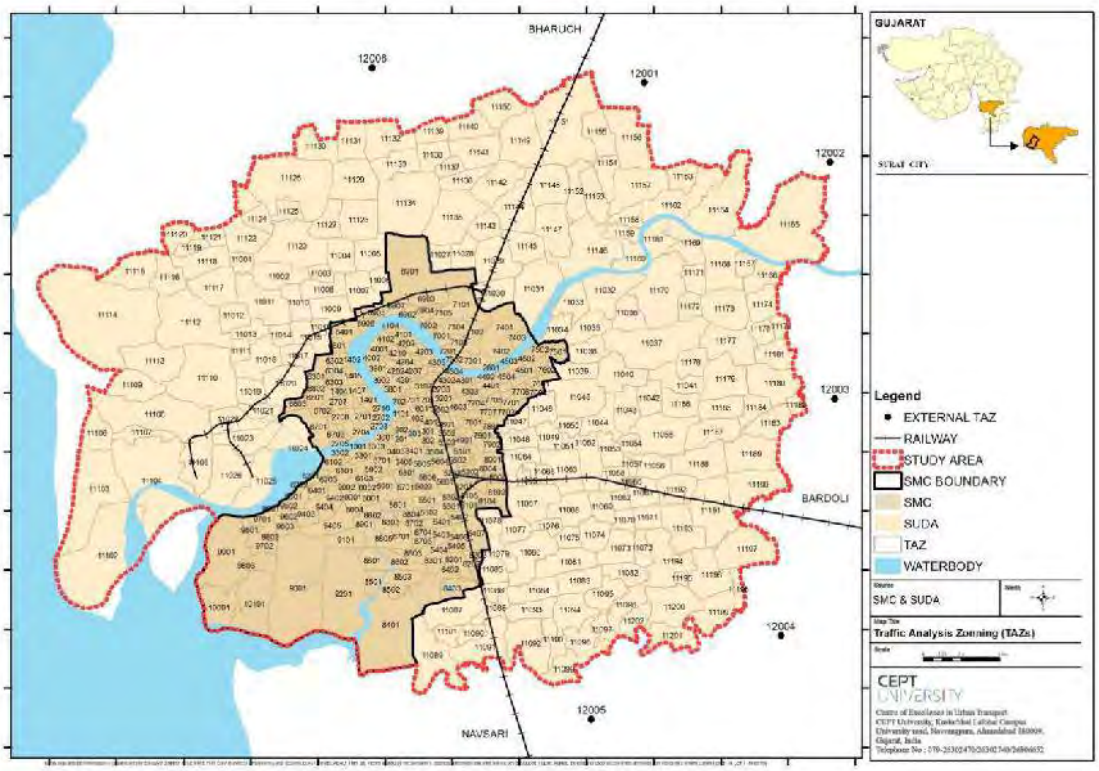
Table 2-4: Traffic direction from external TAZs

Sr. No.	From Direction	Code
1	To / From Mangrol	12001
2	To / From Umbarpada	12002
3	To / From Mandvi	12003
4	To / From Bardoli	12004
5	To / From Navsari	12005
6	To / From Ankleshwar	12006

Thus, a total of 521 zones which includes 515 internal and 6 external zones has been use in the transport model.

Table 2-5: TAZ distribution in study area

Area	No. of TAZ
SMC	313
SUDA (old)	102
SUDA (new)	100
Total Study area (Internal)	515
Outside study are (External)	6
Total No of TAZ	**521**



Map 2-3: Study area zoning

2.7 Model Development

A standard four stage modelling process has been followed to project travel demand and metro ridership in future. An overview of the modelling approach is presented below. This public transport model has been developed using EMME 4.0.8 software. For the analysis of travel pattern four trip purposes - Home based Work, Education, Others, Non-home based, has been taken into account.

2.7.1 Modelling framework

A detailed four stage modelling has been carried out for the Public Transport Operations Plan and Comprehensive Mobility Plan being prepared for Surat. The study area defined has been taken as the modelling area and has been

subdivided into 515 internal TAZs and 6 external TAZs. A public transport model consisting of major road network in the study area along with the existing public transport network of City bus, BRTS was prepared and calibrated for the base year 2016. The same model has been used for the metro demand modelling exercise.

2.7.2 Model Period and Horizon Years

The base year has been taken as 2016. A 30 year forecast period has been taken with 2021 being the start year of the project and 2046 being the horizon year. Intermediate years for forecasts have been taken as 2026 and 2036.The model period for assignment is one hour morning peak. Appropriate expansion factors have been used for computing average daily ridership.

2.7.3 Scenario Specification Framework

The following scenarios are developed for this study:

Table 2-6: Scenario Development

Future scenarios	Network	Demand
Gradual Growth Scenario	Existing network + Proposed Metro & other PT improvement	As per DP Population and Employment distribution
Rapid Growth Scenario		Intensification along proposed metro alignments

2.7.4 Model Software

For the four stages modelling process, MS-Excel and SPSS has been used for Trip Generation and Trip Distribution purposes, while EMME has been used for Trip Distribution and Assignment Process. Mode split has been carried out based on “Willingness to Shift Survey”, conducted as part of this study.

2.7.5 Data collection

This study has based its information from several studies undertaken in the recent past, in addition to some primary surveys and secondary surveys undertaken.

Following sets of data collected from primary surveys have been used for this study:

2.7.5.1 Primary Data

1. Household Survey Data

There are estimated 14,14,626 households in the study area in the year 2016. For estimating the base year demand, the study has used the available household survey data carried out as part of Public Transport Operations Plan for Surat. An additional 3000 households were surveyed as part of this study taking the total household samples to be around 15777 (1.12%) households for the city of Surat.

2. Terminal Surveys

Terminal surveys were carried out to estimate the external public transport demand at major transport nodes of GSRTC and railway station. In the year 2016, passenger counts and origin-destination surveys of passengers were carried out at five such nodes - Surat railway station, Surat Central Bus Station, Udhna Bus depot, Kamrej Bus Station and Adajan Bus Station.

3. Cordon Point/ Classified Volume Counts & Road Side Interview Surveys

These surveys were carried out to estimate the external private and goods vehicles demand for the study area. Classified Volume Counts and Road Side Interviews were carried out at 7 cordon points for 24 hours for all modes in the year 2016. A total of 22,000 vehicles were interviewed through Road Side Interview method for all 7 cordon points. Further, for an existing situation analysis, additional CVC counts were carried out. Some CVC counts were available from the previous study of Benchmarking for the city of Surat in 2015-16 and ongoing study on PT Operations Plan.

4. Speed and Delay surveys

Travel speed surveys were carried out on major corridors of the city as part of the Benchmarking study of Surat in the year 2015. Additional surveys in 2016 were carried out on some major roads which were not covered in the past studies. This information was critical for calibration of the traffic assignment model for the base year.

5. Screen Line Surveys

Classified Volume Count Surveys at screen line locations were carried out to calibrate the transport model. Screen line surveys in the year 2016 were carried out at 27 screen line locations in the study area for 16 hours.

6. Occupancy Surveys

Occupancy surveys are carried out to calibrate the transport model. Occupancy surveys in the year 2016 were carried out at 27 screen line locations in the morning and evening peak.

7. Activity Surveys (Employment Data)

For estimating the employment in the base year, property tax data of SMC was used. There are 76 property tax wards in SMC, having a total of 3.62 Lakhs non-residential properties (2016). A primary survey of around 3000 non-residential properties across 76 property tax wards was carried out as a part of this study to compute trip generation rates for different types of non-residential properties. This was subsequently used for generating employment estimates in the study area.

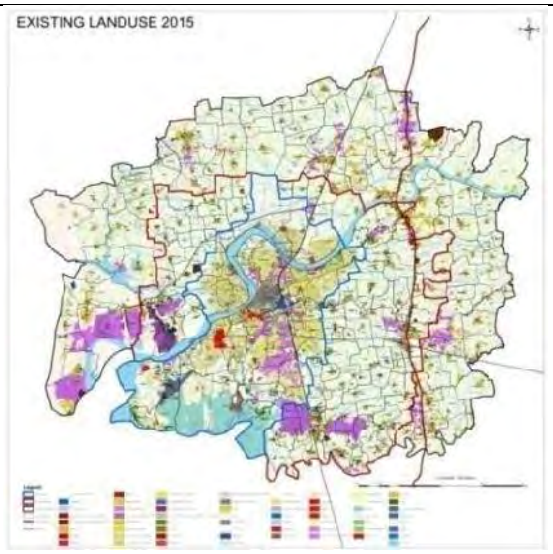
8. Willingness to Pay and Shift survey

Willingness to Pay and Shift survey has been carried out in 2016 at activity ends (non-residential property ends) to assess the likely shift to integrated public transport system of metro, BRTS and city bus under different conditions. A total of 1000 samples have been carried out.

2.7.5.1 Secondary Data

2.7.5.1.1 Existing Land use

Four Development Plans have been prepared till date for the city of Surat- 1978, 1995, 2004 and the recent plan of 2016.These were reviewed to analyse proposed land use scenarios. Additional surveys have also been carried out along the proposed metro corridor to have a detailed mapping of land use.



Map 2-4: Existing Land Use Plan – 2015

Map 2-5: Proposed Development Plan - 2035

Source: Draft Development Plan 2035

Table 2-7:Land Use Classification

No	Type of Zone	SUDA Area in 2014	% of Urbanized land	SUDA Area in 2015	% of Urbanized land
1	Residential	102.49	51.9	136.94	47.43
2	Commercial	4.66	2.3	5.71	1.97
3	Industrial	38.89	19.71	74.61	25.84
4	Educational Public Purpose	13.6	6.89	17.52	6.06
5	Recreation garden and open space	2.34	1.18	2.46	0.85
6	Transport & Communication	35.31	17.89	51.48	17.83
7	Agriculture	517.71		1062.81	
8	Non-Urbanized Area				
	Total	715		1351.53	

Source: Surat Development Plan (2004, 2035)

The existing land use distribution (for the year 2015) shows that there has been a substantial increase in the area under residential use, followed by the area under industrial use. The development is occurring in the south, south-west and towards the north, north-eastern side of SUDA, with planned and unplanned growth. The land use trend of Surat shows that increase in industries have led to growth in residential areas. It is observed that the walled city is a densely populated area, with a huge concentration of commercial activity. The commercial land use has increased from 466 hectares to 571 hectares in a span of just one year; thereby depicting the importance of Surat as an important trade centre.

The speed with which industries have grown in SUDA has been so tremendous that now there is no land available for industrial growth. The industrial zones dedicated in DP 2004 have already been occupied to its full capacity. Some hazardous and obnoxious industries are located within residential areas of Surat city. This can negatively affect the health, environment and overall livability of the people. These polluting industries occupy an area of about 4.3 sq km. There is a proposal in DP 2035 to shift these industries and locate them outside of the residential areas.

2.7.5.1.2 Details of Existing and Proposed Road Networks

For preparing a base model, network coding of the study area for both base and future year is important. The existing road network has been taken for developing the base year road network for the study area. For the proposed road network, Draft Development Plan of SUDA 2035 has been referred.

2.7.5.1.3 Details of Public Transport Services

In order to develop an understanding of base year public transit services, detailed service information from Surat Siti link Limited, Rainbow Tours and Travels Private Limited and GSRTC for regional services in the study area were collected. Data on operational routes, location of stops, fare structure, fleet sizes, service frequency, journey times, vehicle capacity etc. were collected apart from details of current ridership levels. This information was used for developing a base year public transport model in EMME.

2.8 Existing Network and Travel Characteristics

The city of Surat is well connected by road, rail and air transport. Several National and State Highways pass through the city; as a result traffic coming through the city is very high. The city's transportation system is predominantly road based. Dependence on individual modes and intermediate public transport is high due to the absence of sufficient public transport system. Though roads are generally wide, network design and development is not complete.

This chapter provides an overview of the existing road transportation system outlines the travel characteristics and presents the demand forecasting carried out for estimating ridership on metro.

2.8.1 Road Network: Pattern, Length and Distribution

The street network in Surat can be classified as ring-radial, with grid iron pattern at local level. There are eight radials and one complete ring (inner ring road). Majority of the vehicular load is taken by the inner ring road which

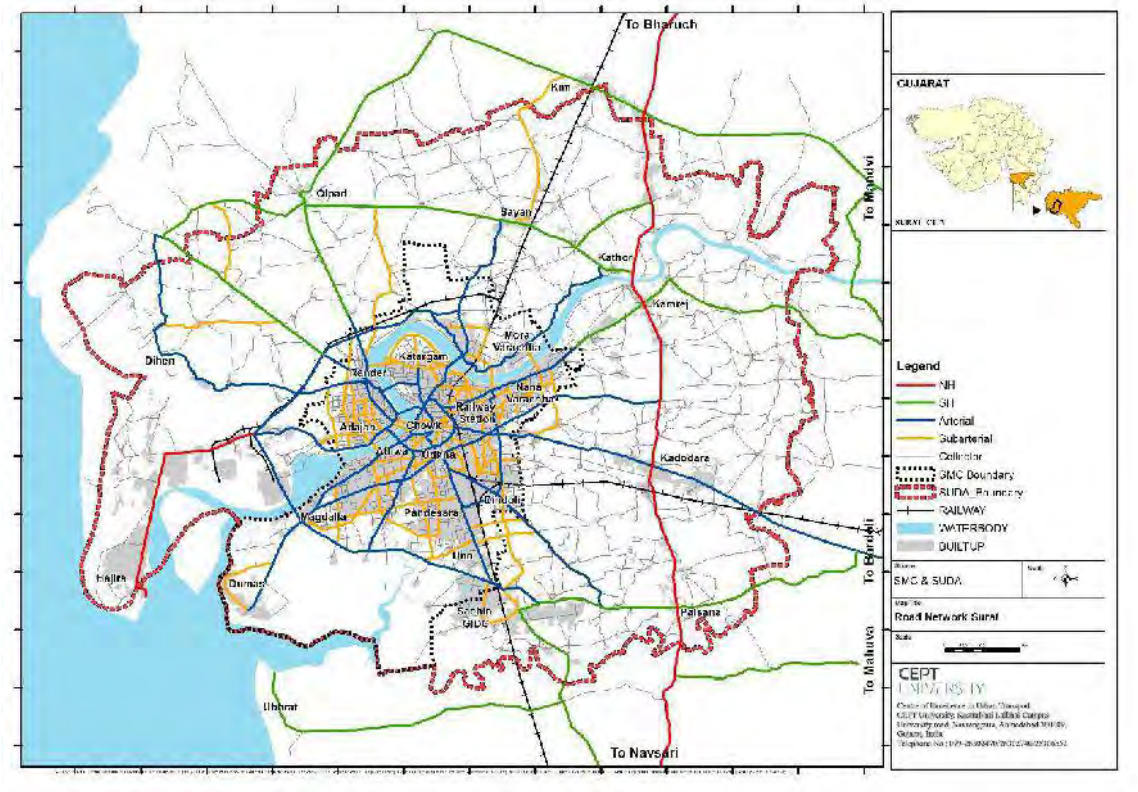
connects to all important roads. All roads in the walled city are congested due to high volumes of traffic and constrained widths varying from 12m to 18m. Within the developed area, road density is high but roads are narrower. The existing road network length of Surat is around 2230 kms (2016).

Other than the National Highway Authority which maintains the National Highways, the State Roads and Building Department along with the two urban local bodies; SMC and SUDA, are responsible for developing, operating and maintain road infrastructure.

Table 2-8: Total road network length in study area in year 2016

Jurisdictional Area	Road length (km)	% of road
SMC	1005	45 %
SUDA	1225	55%
Total Study Area	2230	100%

Source:CoE-UT, CEPT University



Map 2-6: Existing road network in study area

Source: CoE-UT, CEPT University

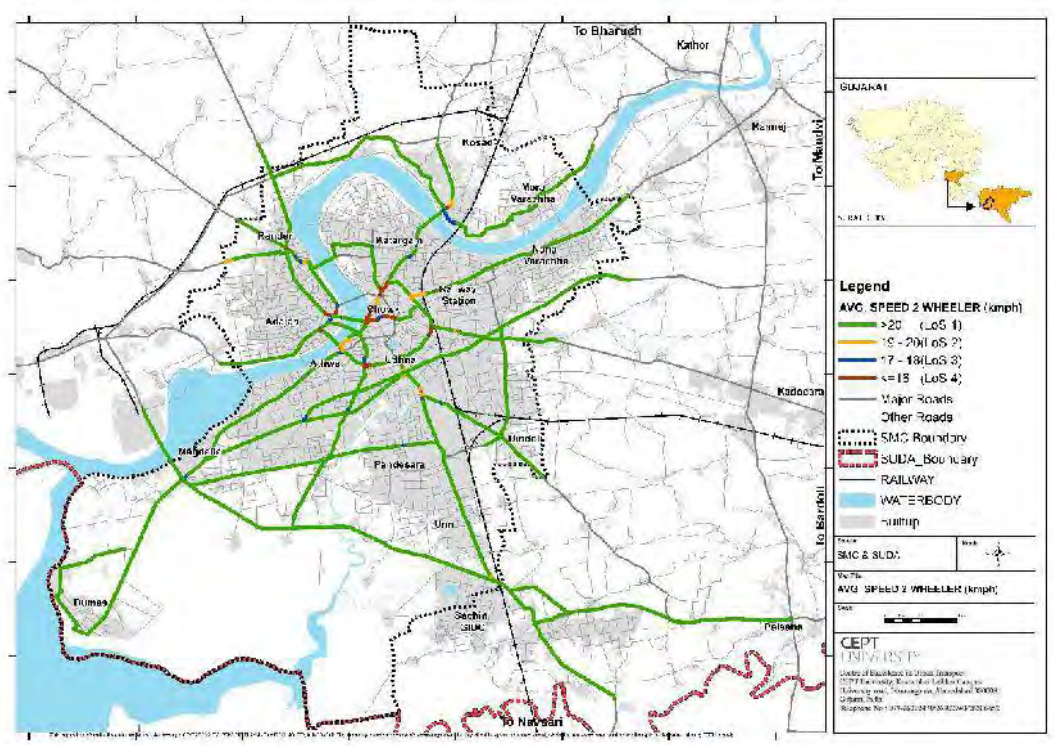
2.8.2 Details of Existing and Proposed Road Networks

For preparing a base model on Emme, network coding of the study area for both base and future year is important. The existing road network has been taken for developing the base year road network for the study area. For the proposed road network, draft Development Plan of SUDA 2035 has been referred.

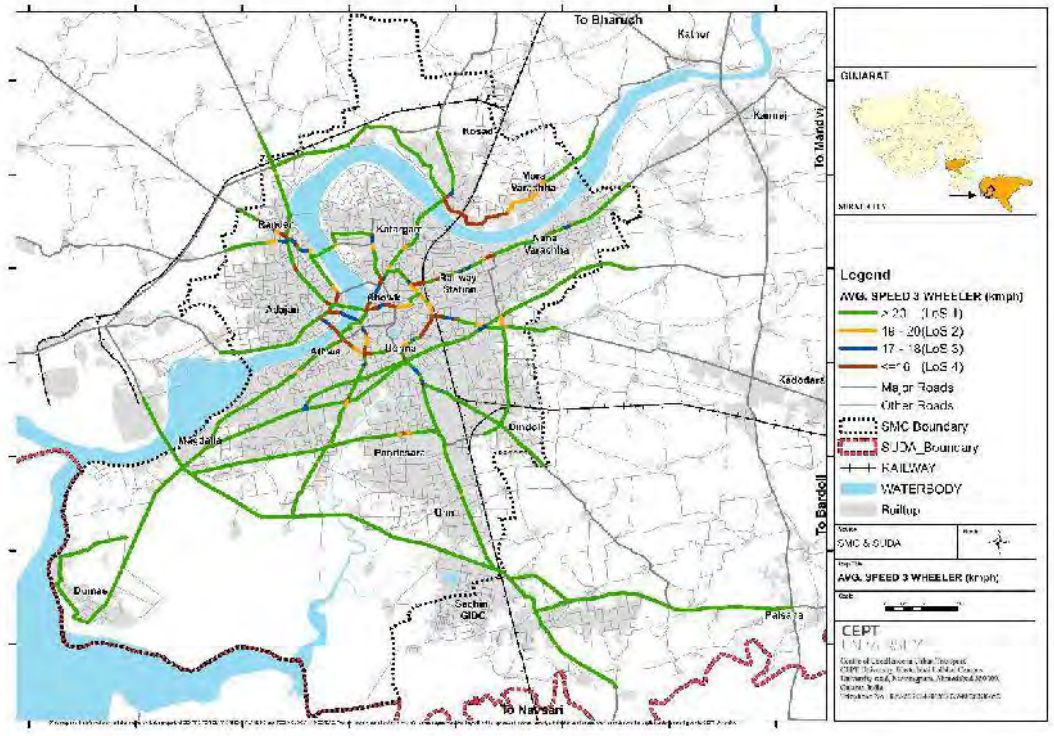
2.8.3 Travel Speeds

To ascertain travel speeds across the network, speed and delay surveys were also carried out. The Map below shows travel speeds across the network. The travel speeds along most of the corridors are more than 25 kmph except old city area.

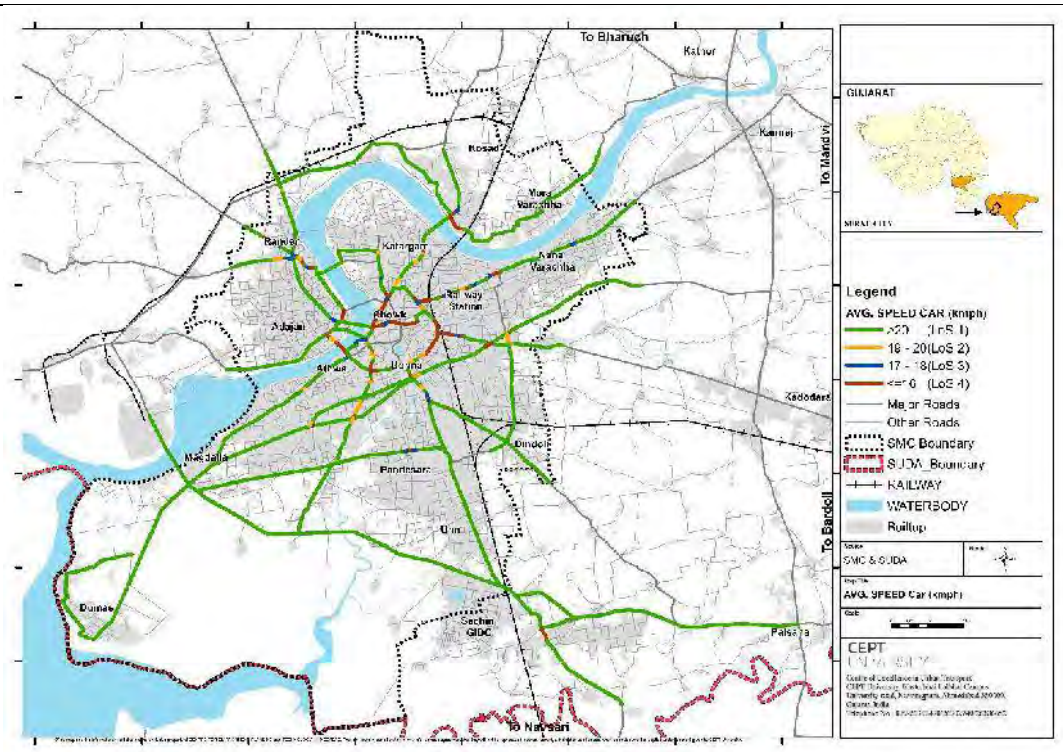
In Surat city the speed profile of the strategic network has high average speed on NH and SH and lower travel speeds on MDR and internal roads. The average speed of 2Wh is 30 kmph, followed by 4Wh and 3Wh i.e. 29 kmph and 22 kmph respectively. Though public transport share in Surat is very low, the average speed of public transport is 22 kmph, i.e. BRTS and city bus together.



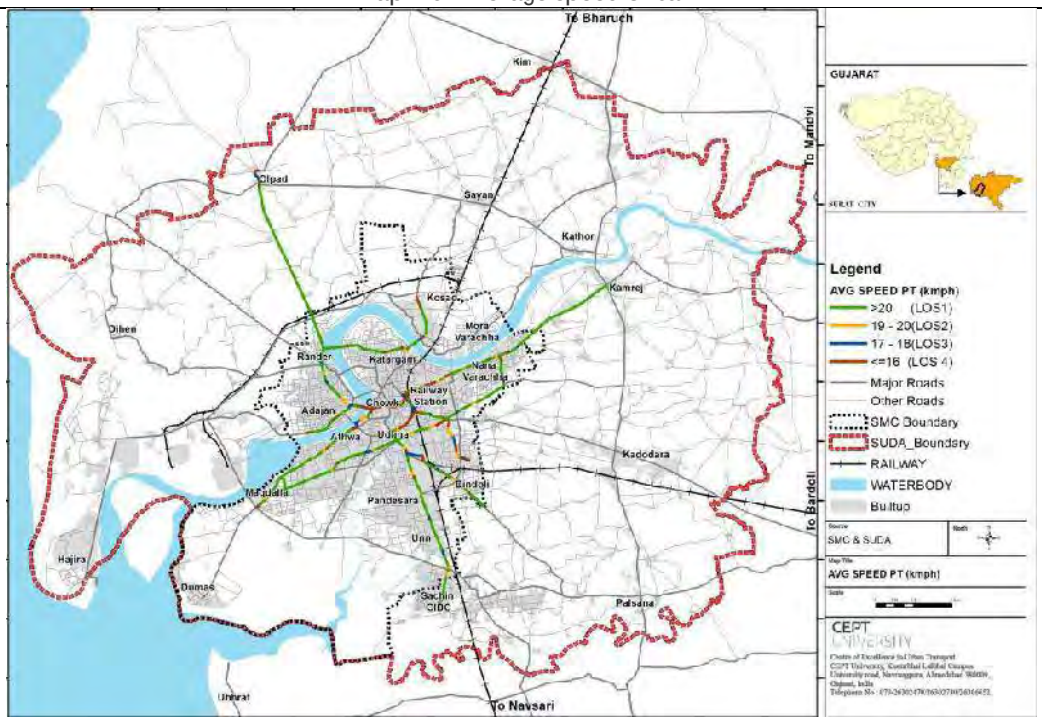
Map 2-7: Average speed of 2 wheeler



Map 2-8: Average speed of 3 wheeler



Map 2-9: Average speed of car



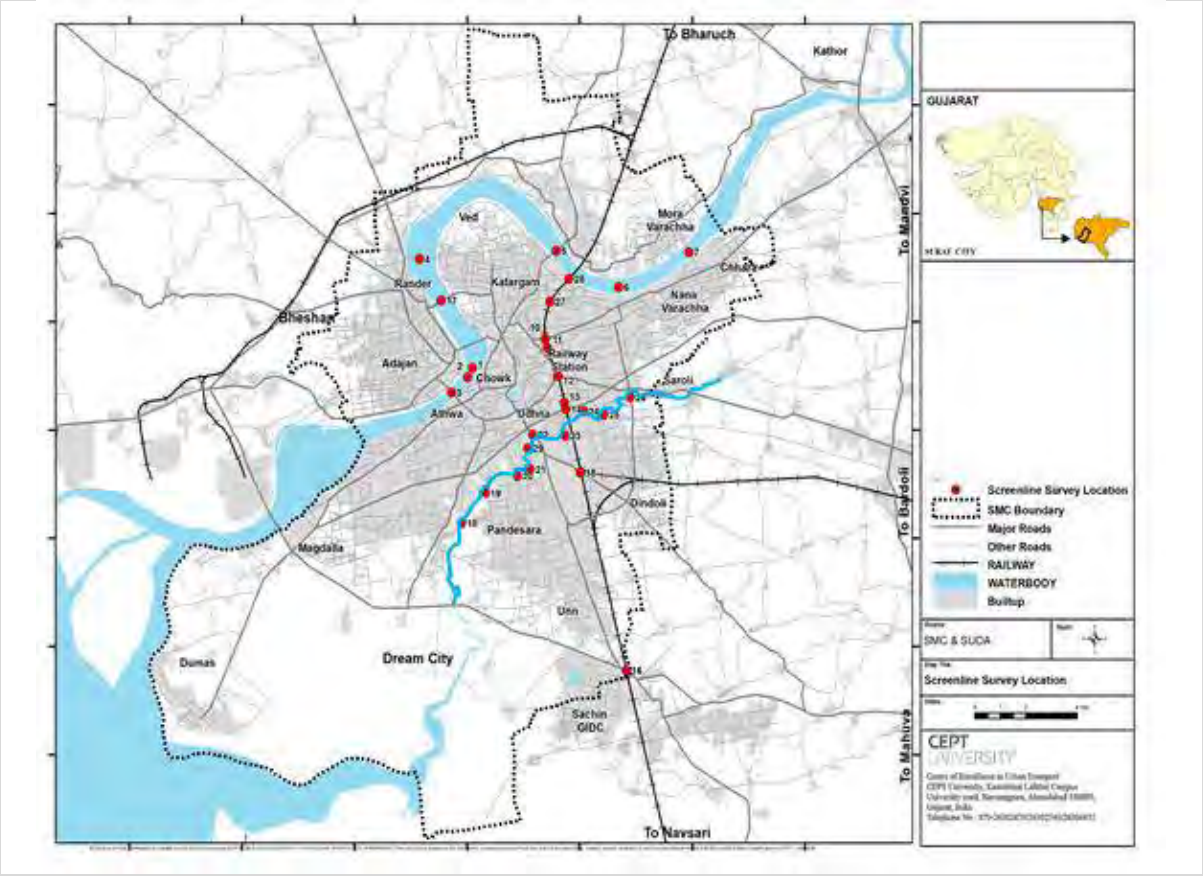
Map 2-10: Average speed of public transport

Source: CoE-UT, CEPT University

2.8.4 Temporal traffic variation

Classified Volume Count Surveys have been carried out on major screen lines i.e. Tapi river, railway line and khadi. Screen lines are physical barriers like railway line, river passing through the city dividing them into parts. The river bridges, rail over-bridges or rail underpasses and Khadi bridges have been taken as the screen line locations. The main objectives with which CVC surveys has been carried out are to analyze temporal traffic pattern in the city and to validate the private vehicular assignment model.

There were 27 screen line locations where classified volume count surveys were conducted for 16 hours (7:00 am to 11:00 pm). The surveys were carried out by installing video cameras at the screen line locations. This was followed by manual classified traffic counts from the captured videos. The traffic surveys were carried out in the months of August and September 2016 and while doing so Government holidays and rainy days were avoided for surveys.



Map 2-11: Screen line locations for CVC

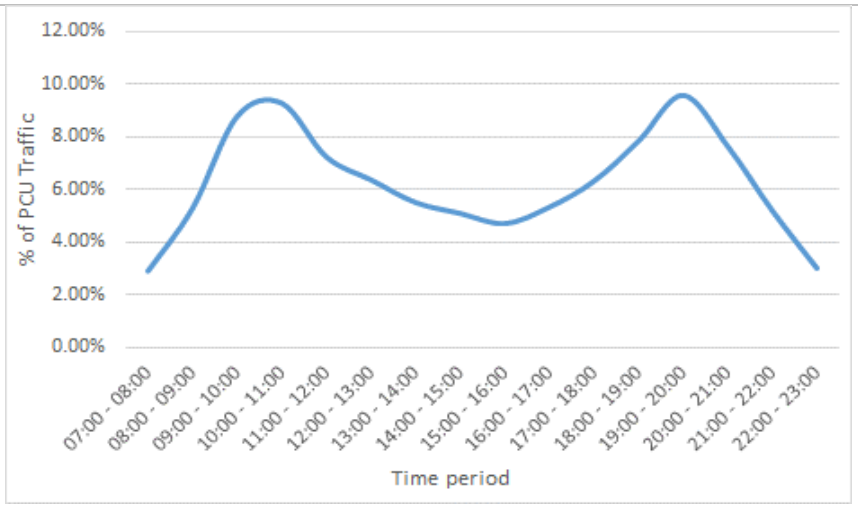
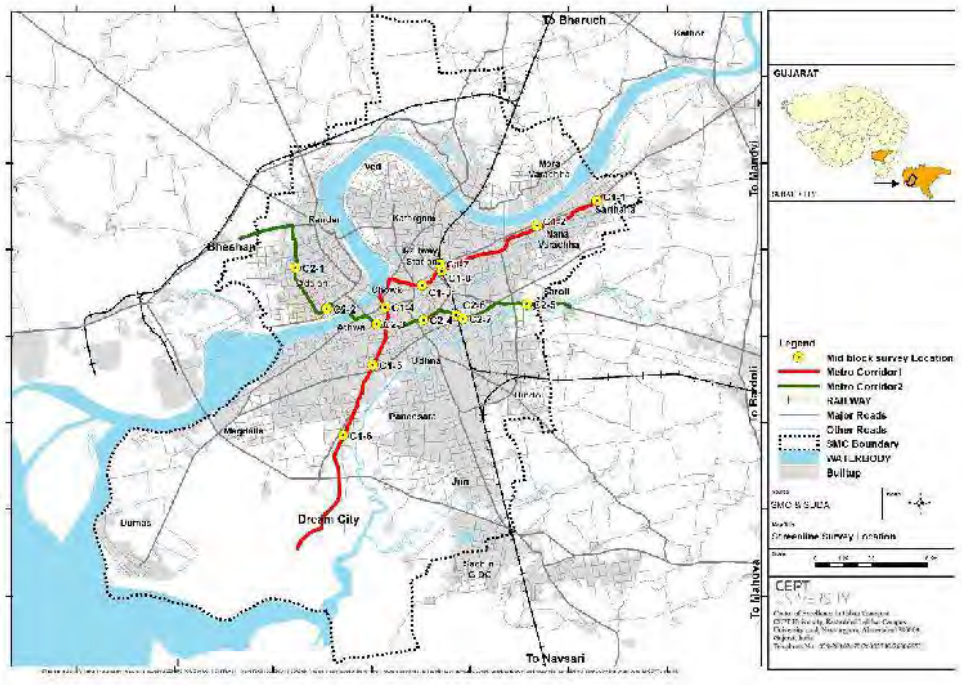


Figure 2-1: Temporal traffic distribution

The graph above represents comprehensive traffic pattern of 27 locations. The morning peak hour is 9:30 to 10:30 am which contributes nearly 10% of the day traffic, whereas evening peak hour is 7:00 to 8:00 pm.

2.8.5 CVC (Mid-block) on Metro corridors

This survey was conducted in the month of February 2017 on normal weekday for duration of 16 hours. This survey was carried out to understand the traffic flow on the proposed Metro corridors. The survey was conduction for 15 locations; 11 locations on Corridor-1 i.e. Dream City to Sarthana and 8 locations on Corridor-2 i.e. Bheshan to Saroli for one day (16 hours).



Map 2.12:Midblock CVC along final metro corridors

Table: 2.9: Traffic Volume in PCUs (16 hours) along proposed metro corridors

Location No.	Location Name	Direction	No. of Vehicles							PCUs
			2Wh	3Wh	4Wh	Bus	BRT	Others	Total	Total
Corridor1 : Dream City to Sarthana										
C1-1	Sarthana Nature Park	Towards Kamrej	40133	6754	11519	1058	76	1549	**61089**	**56513**
		From Kamrej	49388	8265	13673	886	80	2172	**74464**	**67351**
C1-2	Shree Swami AtmanandSaraswati Institute of Technology	Towards Sarthana Nature Park	117427	13071	19144	916	104	1739	**152401**	**129062**
		From Sarthana Nature Park	132631	14975	17673	850	80	1815	**168024**	**141554**
C1-3	Mahidhapura Police Station	Towards Surat Railway Station	26370	7264	962	331	0	138	**35065**	**30924**
		From Surat Railway Station	31575	9771	2948	346	0	373	**45013**	**40453**
C1-4	MachhiwadKhari Road	Towards Majura Gate	18684	4469	3015	41	0	355	**26564**	**23498**
		From Majura Gate	13763	4278	1922	43	0	244	**20250**	**18770**
C1-5	Central Bank of India - Bhatar Road	Towards Bhatar Road	18407	7085	2405	102	0	201	**28200**	**25699**
		From Bhatar Road	20304	8313	3324	61	0	225	**32227**	**29409**
C1-6	Aashirwad Enclave	Towards Citilite, Parle Point	11053	2269	5506	55	0	368	**19251**	**17556**
		From Citilite, Parle Point	10845	2145	5266	55	0	341	**18652**	**16936**
C1-7	Khand Bazar Underpass	Towards Varaccha	69647	12225	9632	462	0	1173	**93139**	**80833**
		From Varaccha	48210	11661	6592	267	0	682	**67412**	**59287**
C1-8	Lambe Hanuman Underpass	To Delhi Gate	36403	9599	6179	613	3	837	**53634**	**48470**
		From Delhi Gate	13111	9712	990	367	0	318	**24498**	**24277**

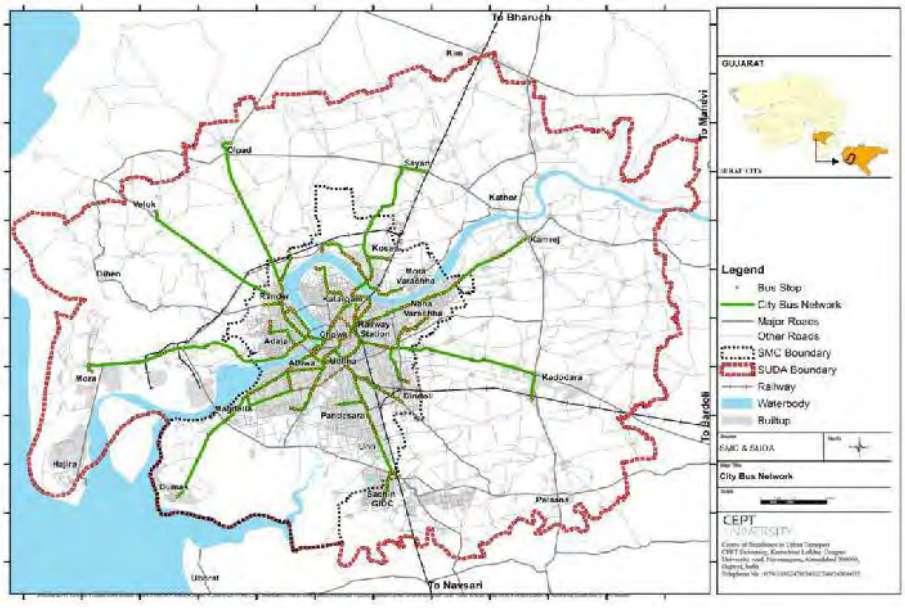
Location No.	Location Name	Direction	No. of Vehicles							PCUs
			2Wh	3Wh	4Wh	Bus	BRT	Others	Total	Total
Corridor2: Bheshan to Saroli										
C2-1	VidhyaKunSankul	Towards Rajpoint, Pal	8616	3690	1665	153	0	153	**14277**	**13464**
		From Rajpoint, Pal	8860	3472	1632	183	0	195	**14342**	**13325**
C2-2	AdajanGaam Circle	Towards HaziraRoad	28801	6121	11437	311	89	617	**47376**	**42856**
		From Hazira Road	26262	6530	13880	359	88	606	**47725**	**43759**
C2-3	Majura Gate	Towards Majura Gate	77152	12953	15558	453	0	1416	**107532**	**93545**
		From Majura Gate	86336	16855	20525	490	0	2156	**126362**	**111121**
C2-4	MaanDarwaja	Towards UdhnaDarwaja	66262	25715	21855	704	0	2619	**117155**	**111436**
		From UdhnaDarwaja	72927	28429	29443	700	0	3433	**134932**	**130190**
C2-5	Magob	Towards Sahara Darwaja	26337	6861	9278	791	0	4921	**48188**	**52918**
		From Sahara Darwaja	24722	7022	10572	665	0	5256	**48237**	**53020**
C2-6	Anjana Railway Crossing	Towards KamelaDharwaja	25616	12750	1491	6	0	1897	**41760**	**45686**
		From KamelaDharwaja	24505	11667	1549	0	0	1970	**39691**	**43646**
C2-7	Surat Municipal Corporation Sankalith Ward Office - Limbayat Zone	Towards Kharwarnagar	41438	3683	9101	209	70	3199	**57700**	**54063**
		From Kharwarnagar	42795	3216	9403	231	49	3292	**58986**	**54736**

2.8.6 Existing Public Transport network

Till the year 2004, Gujarat State Road Transport Corporation (GSRTC) operated the urban transport service in the city of Surat. Over the years, the routes served by City bus service have increased, but fleet availability has reduced drastically and since May 2007, all urban services have been withdrawn. A new system under the overall supervision of SMC with private participation has been operational since August 2007.In order to develop an understanding of base year public transit services, detailed service information from Surat Siti link Limited, Rainbow Tours and Travels Private Limited and GSRTC for regional services in the study area were collected. Data on operational routes, location of stops, fare structure, fleet sizes, service frequency, journey times, vehicle capacity etc. were collected apart from details of current ridership levels. This information was used for developing a base year public transport model in EMME.

2.8.6.1 Surat City Bus Services (CBS)

There are 29 routes plying within the study area with approximately90 buses on road. System has scheduled to operate each route at 15 min. frequency. Today, the sole system is carrying average about 45000 riders per day which share only about 0.8% of total motorized demand. Recently, Surat city has planned a network of 547 Km for city bus services integrating with BRTS. 1000 buses are planned to be procured to ply on city bus routes. 575 buses are already ordered and 300 buses are tendered.

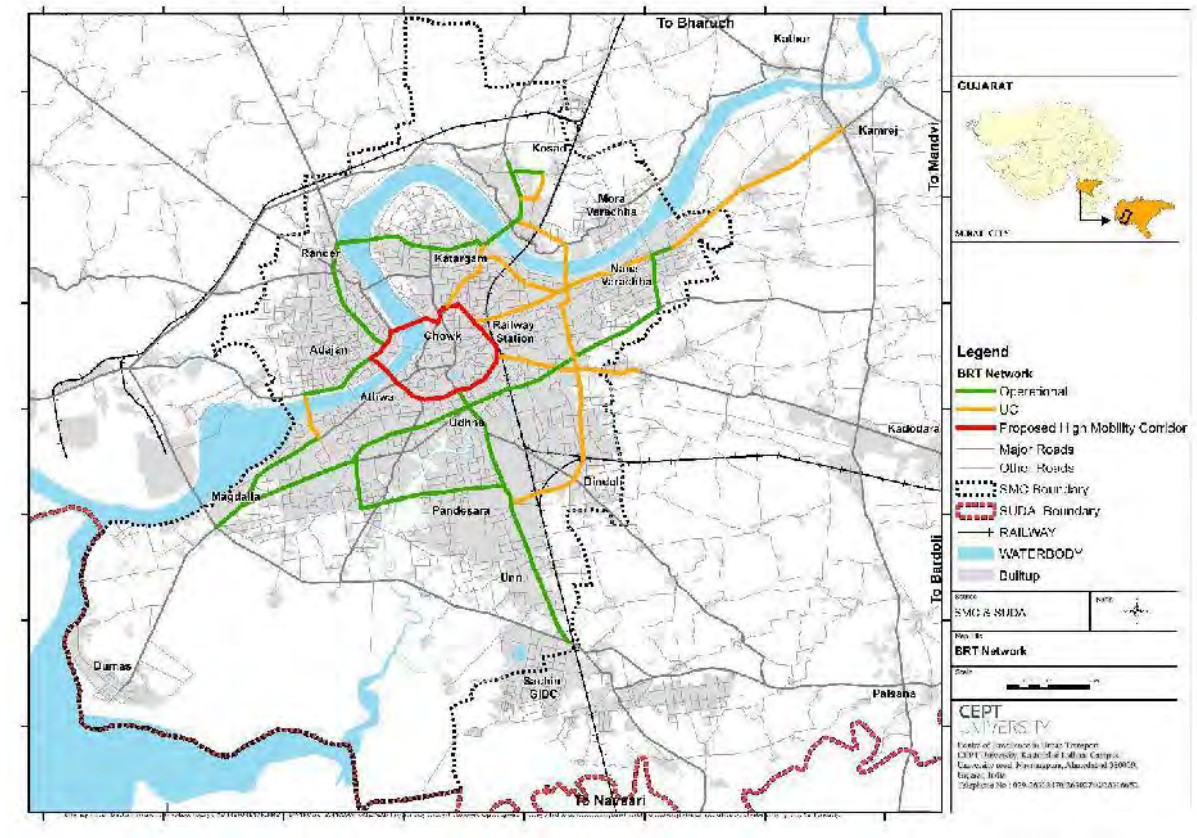


Map 2-13: Existing city bus routes

Source: CoE-UT, CEPT University

2.8.6.2 Bus Rapid Transit System – Siti link

Siti link has inaugurated BRTS in 2014 with the 30km of network as Phase I. Currently 59km (October 2016) of network is in operation along with 5 BRTS routes which carries about 45000 passengers in a day. Siti link has a plan of total of 101km network and it is under implementation currently, connecting major transit nodes, residential and commercial hubs, recreational and public areas of the city, with provisions for NMT (pedestrian pathways and cycle tracks). The average speed of BRTS buses is 24 kmph which is quite high as compared to existing city bus services and auto rickshaws.



Map 2-14: Existing and proposed BRTS

Source: CoE-UT, CEPT University

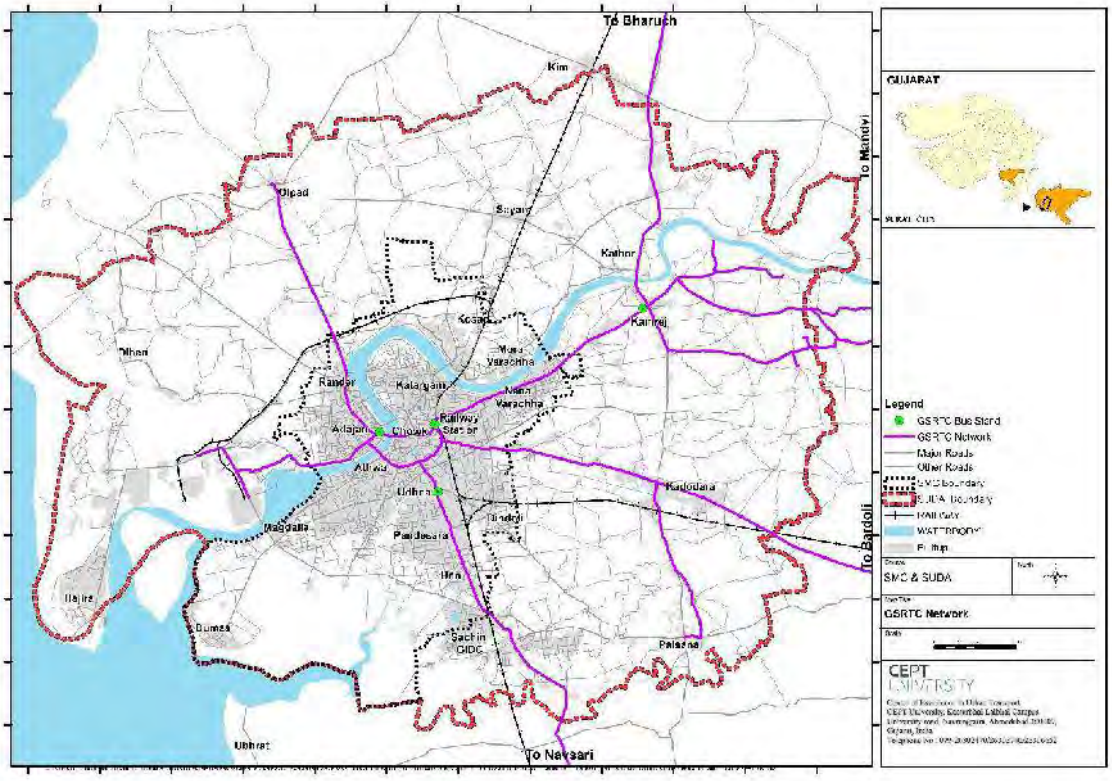
2.8.6.3 GSRTC

The Gujarat State Road Transport Corporation (GSRTC) operates regional bus transport passenger services both within Gujarat and to neighbouring states. These services provides from Surat to suburban areas/villages within SUDA area such as Olpad, Kamrej, Choryasi, Limba, Bhatpor, Kamrej, Kholeshwar, Dhatva, Jat Bharthana and Asta and other cities outside Gujarat such as Nasik, Auragabad, Malegaoun, Mumbai, Pune, Shirdi, etc.

The major depots and pick up points within the city are from Udhana Gam depot, Adajan Patiya depot and Surat Railway station (CBS). Total 398 routes are plying under five different categories form Surat city; they are

1. Linear Regional Bus route service -11 routes
2. Linear Sub Urban Bus routes service - 12 routes
3. Regional Urban routes (within Gujarat) service - 304 routes
4. Regional Urban routes (out of Gujarat) service - 66 routes
5. Volvo Bus routes - 5 routes

For modelling purpose, all these routes have been merged together according its direction finally 12 GSRTC routes have been coded in to the model with combined frequency. The map below shows the routes alignment which has been coded in to the model.



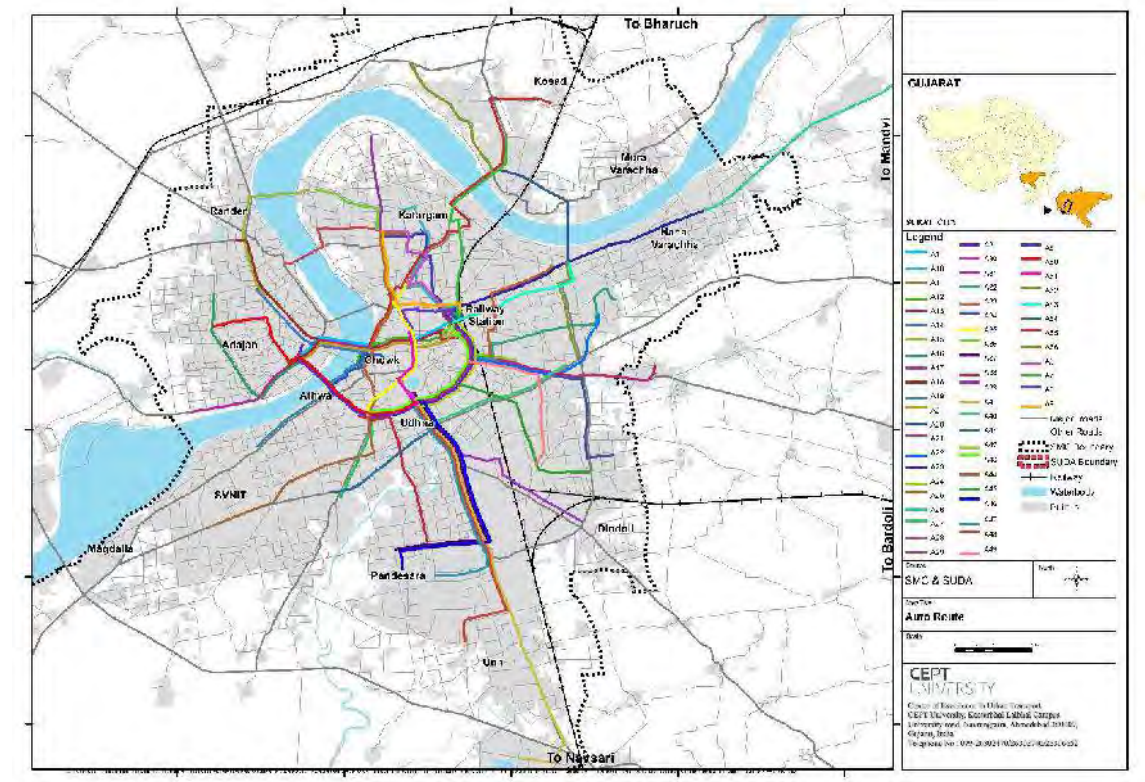
Map 2-15: GSTRC Network and major terminal

Source: CoE-UT, CEPT University

2.8.6.4 Intermediate Public Transportation:

The city of Surat is known for auto rickshaw transportation city since years. Currently there are about 38000 registered autos in the city after year of 2006, seems quite high ownership of autos. Around 10 new autos get added to the city every day.

An absence of an efficient bus service has resulted in transfer of trips from buses to illegally operating autos on a point to point service. Around 8.6 lakh trips are being undertaken as per 2016 household surveys on these auto rickshaws. There are about 52 shared auto rickshaws routes identified as part of the study. The autos have unorganized operations and suffer from problems of overloading and poor vehicle quality. They also are a major cause of increased congestion and pollution levels in the city. Most of the autos run as shared autos along the major road networks; also they have better frequency and coverage in comparison to bus network.



Map 2-16:Predominant shared auto rickshaws routes

Source: CoE-UT, CEPT University

2.8.7 Socio-economic Characteristics

Household survey data analysis gives an appreciation of the demographic, socio-economic and travel characteristics of residents of the study area. It is important to understand the travel needs of the people, their tendency to travel, preferences for travel modes, ability to pay for travel and their desires of travel within the area. The Socio-economic characteristics of the city can be understood from the following sections:

2.8.7.1 Gender status and Household size

The social development characteristics have been assessed with the help of indicators including sex ratio and Household size. As per Census 2011, the

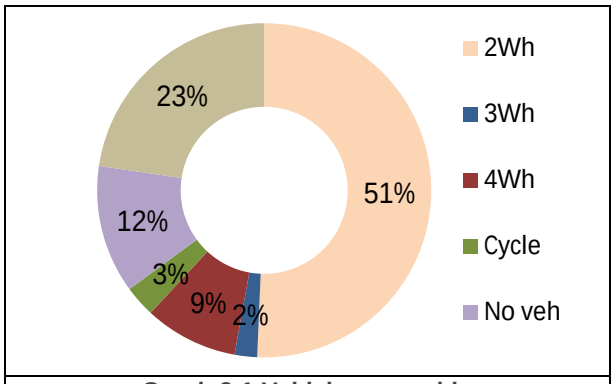
Sex ratio (females per thousand males) for SUDA area stands at 758 .The average family size in the total SUDA region is 4.6.

2.8.7.2 Workforce Participation Rate (WPR)

Work Force Participation Rate (WPR) is defined as a ratio of total number of workers in an area to the total population of that area. A low ratio indicates high dependency of population on few jobs. At the same time, areas with high employment potential and having migrant workers show unusually high ratio due to number of workers living in the area with their families outside the area. Surat being the major hub of diamond and textile industries has 40 % WPR within SMC and 41% WPR within SUDA. The overall study area WPR average is 41%.

2.8.7.3 Household Income and Average Vehicle Ownership

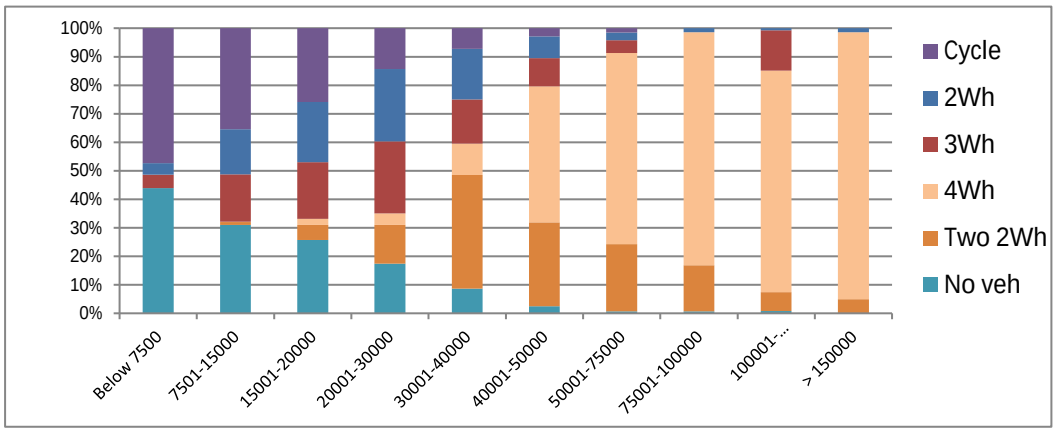
The city of Surat is known for one of the highest income city of India. As per 2016 Household surveys, Surat has averge household income of Rs. 31300 per month within SMC and Rs. 26150 only for SUDA region. Average HH income for study area together is Rs. 29100 per month.



Graph 2-1:Vehicle ownership

Source: Household surveys, CoE-UT, CEPT

No. of motorized vehicles per house hold is the vehicle ownership. Around 85% of householdsare having motorized vehicles. Surat is called as two wheeler city. As per HH survey, each HH owns atleast one two wheeler, where as cycle & car ownership is quite less. There is about 273 two wheelers &23 four wheelers per 1000 population.



Graph 2-2:Household income & vehicle ownership

Source: Household surveys, CoE-UT, CEPT

The above figure shows vehicle ownership by different income groups. As income increases, share of 4Wh ownership is increases and share of 2Wh ownership is decreases.

2.8.8 Travel Characteristics

Travel characteristics, are defined in terms of trip rate, mode choice, geographical distribution etc.These indicators are the guiding principles in determining the system needs and its growth. Socio-economic characteristics such as income, age, sex, occupation, vehicle ownership, etc., are related to travel characteristics of individual households.

2.8.8.1 Trip rate

The per capita trip rate for the study area (all modes) is observed to be 1.60. The same excluding walk is 0.96. The per capita motorized trip rate is 0.93 and the public transport trip rate is 0.02 which is significantly low. Intermediate Para transit trip rate is 0.12 which is higher than public transport rate due to influence of auto rickshaws in the city.

Table:2.10: Trip characteristics of study area

Particulars	Study Area
Year of Household Survey	2016
Households surveyed	15777
Trip-Rate Total	1.59
Trip rate – Motorised	0.93
By excluding walk trips (<0.5km) which are generally short in length, the estimated PCTR is	0.96
Public transport trip rate	0.01

Source: Household surveys, CoE-UT, CEPT University

2.8.8.2 Mode share

Two wheeler trips are a dominant share in the mode choice, accounting for 62% share in motorized mode. Public transport mode share is quite low as it is serviced only by BRTS and City bus service. The share of walk and cycle trips is about 43% , which is quite high, indicating compact and mixed land use development of the city. South eastern part of the city has lower and middle income group housing character which has high NMT & PT share whereas the south western part of the city has middle and higher income group housing, leading to more motorized (majorly four wheeler) trips share. Auto rickshaw is the most prominent form of public transport and has a motorised share of 17%.

Table: 2.11: Modal share - 2016

Particulars	Study area
Walk	41%
Bicycle	2%
Two Wheelers	36%
Car	2%
Auto Rickshaw	10%
Bus	0.56%
BRTS	0.36%
Others (School bus, Staff bus, ST bus, Rail)	9%
Total	**100%**

Source: Household surveys, CoE-UT, CEPT

2.8.8.3 Trip Purpose and mode share

About 28% of work trips are made by two wheelers, whereas only 0.52% trips are made by PT. Work trips made by three wheelers contributes to 4.6%. For education purpose, about 13% of trips are made by walk and cycle.

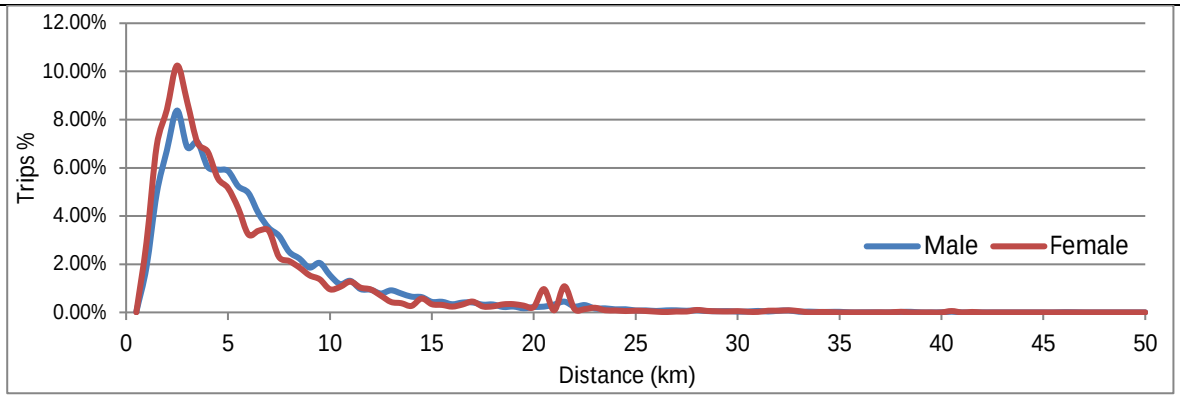
Table 2.12: Trip purpose share

Mode	Walk	Bicycle	2 wheeler	4 wheeler	3 wheeler	PT	Others	Total
Work	6.46%	1.09%	28.42%	1.94%	4.62%	0.52%	0.73%	43.78%
Education	11.68%	0.82%	4.40%	0.06%	4.69%	0.51%	7.95%	30.10%
Others	22.01%	0.03%	2.68%	0.13%	1.19%	0.04%	0.03%	26.12%
Total	**40.15%**	**1.94%**	**35.50%**	**2.12%**	**10.49%**	**1.07%**	**8.72%**	**100.00%**

Source: Household Survey, CoE-UT, CEPT University

2.8.8.4 Trip length frequency distribution

The following figure shows the trip length frequency distribution for all trips in the study area. It can be seen that more than 80% of the trips are of less than 7 Km length. It can be observed that females travel more for shorter trip length when compared to male.



Graph 2-3:Trip length frequency distribution in study area

Source: Household Survey, CoE-UT, CEPT University

The average trip length of the study area is 5.01 Km. Average trip length by modes is shown in the table below. It can be seen that the average trip length by public transport is around 10.3 Km, followed by four wheelers which have a trip lengths of 7.9 Km.

Table:2.13: Mode wise average trip length

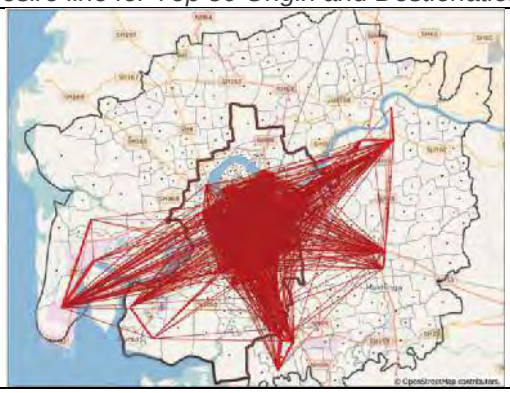
Modes	Average Trip length (km)
Two Wheeler	5.8
Four Wheeler	7.9
Three Wheeler	6.1
Public Transport	10.3
Walk	2.88
Bicycle	4.1

Source: Household Survey, CoE-UT, CEPT University

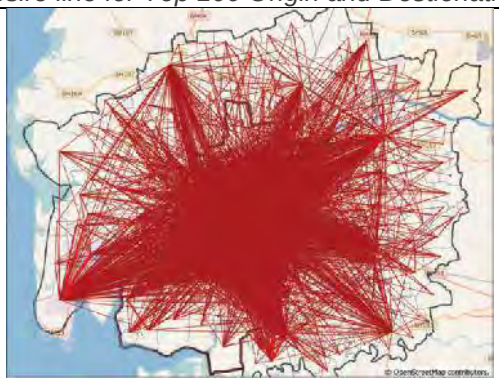
The figures below shows major traffic is along the eastern and western part of the cities, central and southern part for top 50 origin and destination (OD) pairs. The major attractor points for top 50 OD pairs include Hazira, Sachin GIDC, Kamrej, Varaccha area, Pandesara GIDC and some areas of the old city. Areas like Kamrej and Hazira show high intensity of trips in top 100 OD pairs. Additional areas like Kadodara and Mota Varacha also add up in this list.

Desire line for Top 50 Origin and Destionation

Desire line for Top 100 Origin and Destionation



Desire line for Top 200 Origin and Destionation



Desire line for all Origin and Destionation

Figure 2-2: Travel pattern –desire line diagram

2.9 Growth Dynamics

Surat is one of the fastest growing cities in India; it is ranked seventh in the country in terms of the population size and is the second most populous cities in Gujarat (Census of India). The study area includes Surat Municipal Corporation, 2 Industrial Notified Areas, 15 urban areas and 182 villages. The total population of the study area was 50.87 lakh in 2011 with an annual growth rate of 4.3% from 2001 to2011. It is estimated that in year 2016, there are 59.29 inhabitants in the study area.

The city has experienced rapid growth in population during the past four decades. The decadal growth rates between 2001 and 2011 have been one of the highest in the country and the highest in the state of Gujarat. High growth rate experienced by the city over four successive decades has been a major feature in the city's growth, necessitating prompt responses in supply of infrastructure.

The table below places Surat along with other urban agglomerations in the state of Gujarat. It clearly shows that though the decadal growth rate of Surat has reduced since the last decade, it still records the highest amongst the other urban agglomerations.

Table: 2.14: Surat U.A. comparison with other U.A. in Gujarat

City	Population 2001	Growth Rate (1991-2001)	Sex-ratio 2001	Population 2011	Growth Rate (2001-2011)	Sex-ratio 2011
Ahmedabad - U.A.	4519278	36.44%	885	6,361,084	40.75%	899
Surat U.A.	2811464	85.09%	760	4591246	63.30%	753
Vadodara U.A.	1492398	32.44%	905	1,670,806	11.95%	921
Rajkot U.A.	1002160	53.12%	906	1,286,678	28.39%	908
Source: Census of India						

This chapter deals with the growth in the study area in terms of population and employment which forms the basis on which productions and attractions for demand analysis has been computed. The total population of the study area is estimated to be 59.29 lakhs with a compound annual growth rate of 3.1% between year 2011 and 2016. The population of the study area is projected to be around 73 lakhs in 2021.

2.9.1 Existing Population Distribution for year 2016

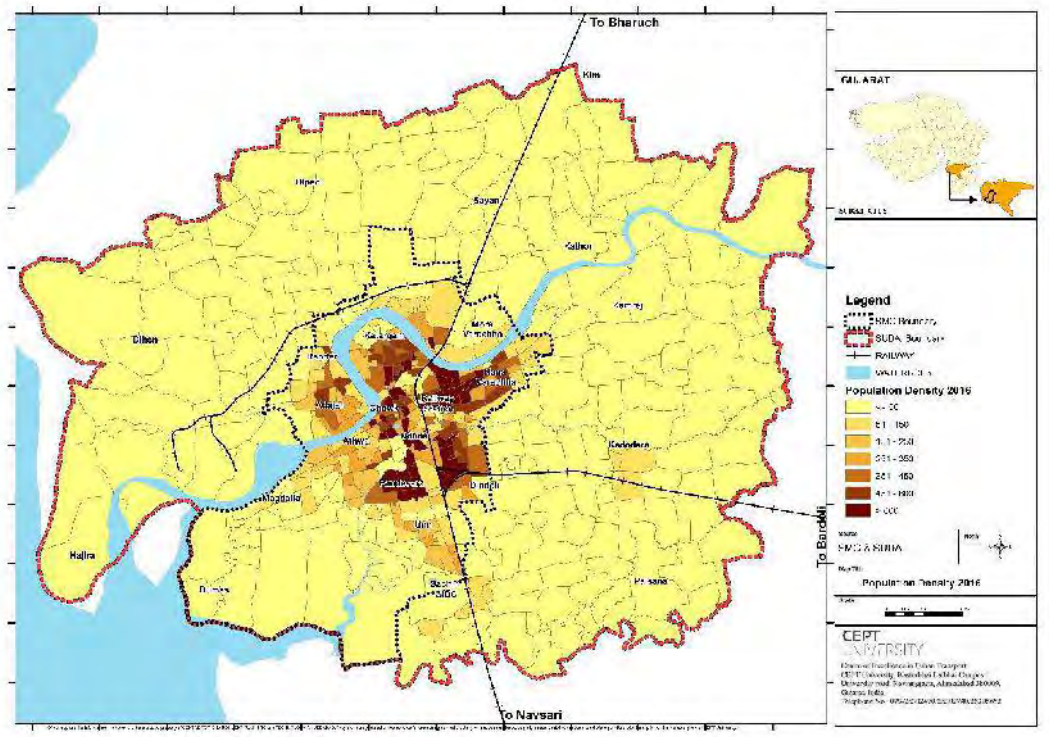
The estimated population of 59.29 lakhs for year 2016 was distributed at TAZ level based on built up and existing land use characteristics in the TAZ.

The distributed population in different parts of SMC and SUDA are shown in Table 2-14 and Map 2-14 below:

Table:2.15: SUDA area Growth

Areas	Area (Ha)	Population 2001	Population 2011	Population 2016	Density 2001 (pph)	Density 2011 (pph)	Density 2016 (pph)
SMC	**32652**	**2876374**	**4466826**	**5213426**	**88**	**137**	**160**
Central Zone	978	413641	408760	487259	423	418	498
East Zone	3643	707768	1134822	1290382	194	311	354
North Zone	3293	428573	717016	845206	130	218	257
South East Zone	2154	371341	722400	835806	172	335	388
South West Zone	11052	247473	351879	419166	22	32	38
South Zone	6516	420434	706963	834406	65	108	128
West Zone	5015	287144	424986	501200	57	85	100
SUDA	**102224**	**475636**	**615030**	**716395**	**5**	**6**	**7**
Municipality	709	17384	28327	30901	25	40	44
Census Towns	2645	56927	104176	129764	22	39	49
Out Growth	2504	14711	26488	28895	6	11	12
Industrial Notified Areas	653	3293	2680	1930	5	4	3
Village	95712	383321	453359	524905	4	5	5
Grand Total	**134876**	**3352010**	**5081856**	**5929821**	**25**	**38**	**44**

Source: Census of India



Map 2-17: Population Density - 2016

Source: CoE-UT, CEPT University

2.9.2 Future Population Projection and Distribution

The Draft Development Plan (DP) prepared for year 2035 has already projected population for year 2021, 2031 and 2035. As the study area is the same as in the Development Plan, population of study area for the intermediate years of 2021, 2031 and 2036 have been taken from the DP. The total population of the study area in the horizon year 2046 is projected to be 125 lakhs by considering slightly decline in growth rate after 2 decades. The CAGR of the region considered is around 2.60% from 2011 to 2046.

To estimate employment in the study area, a sample survey from non-residential properties were conducted and data from industrial department and the chamber of commerce. IEM and EM data was also used for the same. As from the employment rate, the base year employment is estimated to be 29.28 lakhs in the study area while it projected to 62.50 lakhs for the horizon year by looking at existing rate and various future proposals in the study area.

The following table shows population and employment projections of the study area.

Table:2.16: Population and Employment Projections

Year	Population	CAGR	Employment	CAGR
2011	50,81,856	-	-	-
2016	59,29,821	3.13%	29,28,559	-
2021	73,00,000	3.69%	36,06,200	4.25%
2026	85,50,000	3.53%	42,32,250	3.75%
2031	98,00,000	3.34%	48,60,800	3.44%
2036	110,00,000	3.14%	54,69,200	3.17%
2041	118,50,000	2.86%	59,10,780	2.85%
2046	125,00,000	2.60%	62,50,000	2.56%

Source: Census of India, SMC, SUDA, CoE-UT

2.9.3 Future growth alternatives for study area

There are two growth scenarios which have been considered for this study:

1. Gradual Growth: This is the Business as Usual scenario for development which is based on the proposed land use scenario of the Development Plan.
2. Rapid Growth: This is the Transit Oriented Development (TOD) Scenario. This considers intensification of development along proposed metro corridors through high density residential and commercial developments.

2.9.3.1 Gradual Growth: Population Distribution

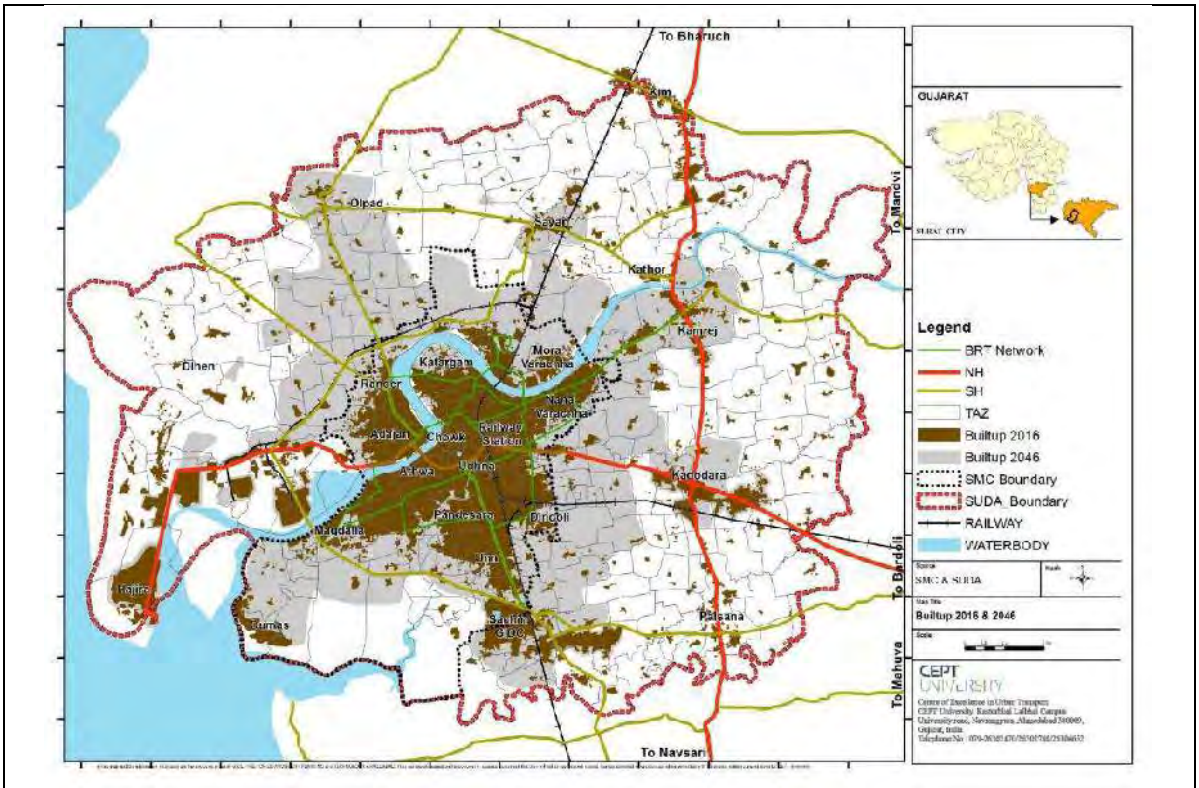
The total population of the study area in the horizon year 2046 is projected to 125 lakhs. Distribution of population plays the vital role in development, and densification plays a significant role in transit ridership. It is assumed that the growth of each TAZs would be based on trend based growth direction. That

may take a longer time period to realize the benefits of shift towards public transport modes i.e. market forces will dictate where growth will happen with effect of development plan prepared by SUDA for year 2035. Table 2-15 shows the population distribution for 'Without TOD' Scenario.

Table: 2.17: Population Distribution for Gradual Growth scenario

Areas	Area (Ha)	Population				Density (pph)			
		2021	2026	2036	2046	2021	2026	2036	2046
SMC	**32652**	**7300000**	**6090761**	**7383204**	**7867870**	**195**	**202**	**226**	**241**
Central Zone	978	467482	424978	388161	365620	478	434	397	374
East Zone	3643	1521699	1447039	1427876	1416143	418	397	392	389
North Zone	3293	1072426	1088735	1184176	1242610	326	331	360	377
South East Zone	2154	966211	908077	878952	861120	449	422	408	400
South West Zone	11052	627090	869063	1292271	1551378	57	79	117	140
South Zone	6516	1024334	1043945	1141481	1201196	157	160	175	184
West Zone	5015	672552	809745	1070287	1229802	134	161	213	245
SUDA	**102224**	**948206**	**1958417**	**1616796**	**4632131**	**9**	**19**	**35**	**45**
Census Towns	2645	155246	205062	293889	348273	59	78	111	132
Industrial Notified Areas	653	1677	1707	1863	1958	3	3	3	3
Municipality	709	37757	46600	63029	73087	53	66	89	103
Out Growth	2504	35306	45064	62781	73629	14	18	25	29
Village	95712	718220	1659984	3195235	4135184	8	17	33	43
Grand Total	**134876**	**7300000**	**8500000**	**11000000**	**12500000**	**54**	**63**	**82**	**93**

Source: CoE-UT, CEPT University

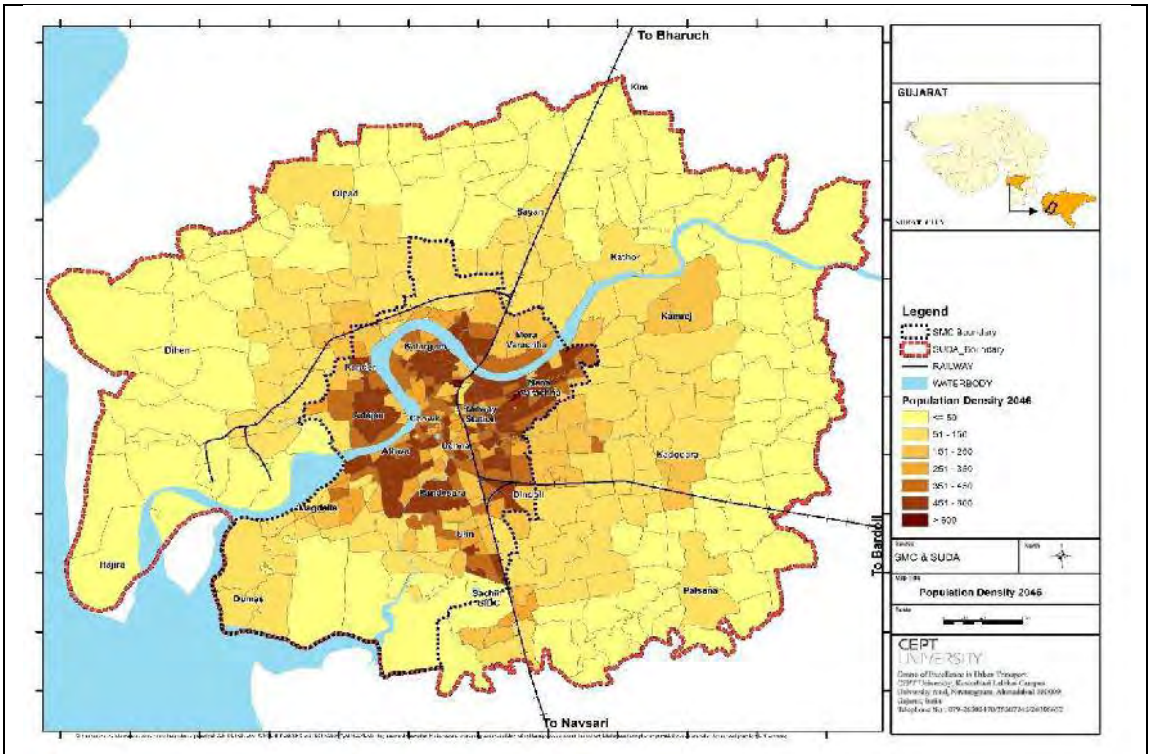


Map 2-18: Existing and 2046 ProposedDeveloped(without TOD) Area

For population distribution, Development Plan prepared by SUDA for year 2035 has been considered. The distribution carried out has been as per Development Plan, potential growth areas, economic development centres, industrial areas and taking into account high flood risk areas. A ratio method was used for population distribution and density checks were applied to arrive at holding capacity. Excess population, if any was redistributed in the study area again using ratio method but excluding the zones with excess population. A map above defined the existing developed are and how the growth of future development will happen in Gradual Growth scenario.

The developed area of study area in base year 2016 is 231sq. km. and net population density is 255 pph. With the distribution it is expected that developed area in year 2046 will grow to 625sq.km and net density is expected to reduce to 200 pph from 255 pph.

The distribution suggests that population will spread beyond the existing SMC area and around 65% of the 2046 population will reside in current SMC area, 10% in the SMC adjoining area and rest in SUDA area. Growth in the SUDA area is expected to be in the new growth centers envisaged as a part of the Development Plan along with the other industrial proposals getting realized.

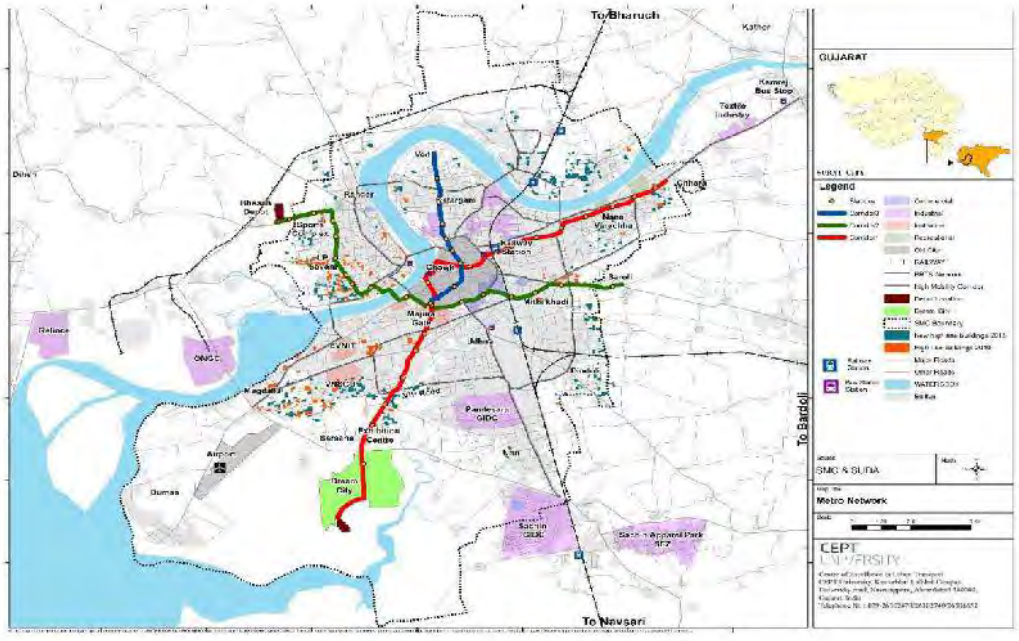


Map 2-19:Future 2046 Population Density – Without TOD

2.9.3 .2 Rapid Growth Scenario: Population Distribution

This scenario envisages intensification of development along metro corridor facilitated through policy incentives of the government. Map 2-20 shows the

proposed metro corridors. However, as most of the area through which the metro passes is intensely developed, scope for future development in predominantly in the peripheral areas as can be seen in the map below.



Map 2-20: Employment Activities for Metro Option

Source: CoE-UT, CEPT University

Thus, for development along metro corridors, higher density is proposed in TAZs along the metro corridors, specifically in the peripheral areas as shown in Map 2-20. This was done taking into consideration growth potential along corridor, existing and proposed land use and existing density levels.

Average range of projected population densities of TAZs along metro corridors have been presented in the table below:

Table 2-18: Population densities along metro corridors (With TOD Scenarios)

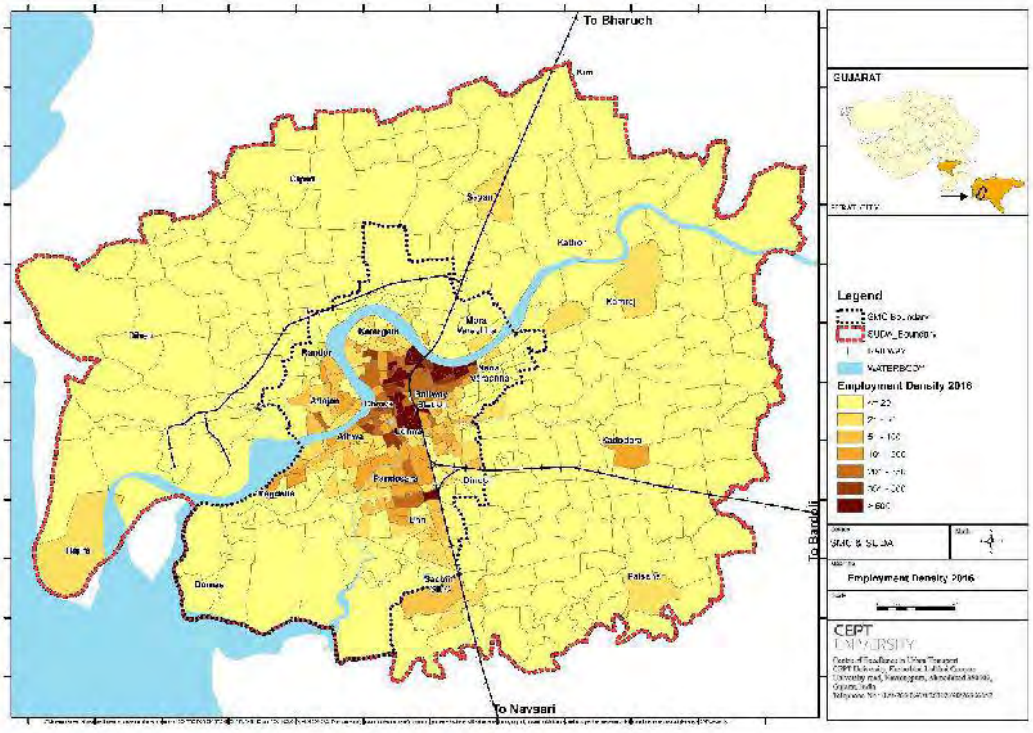
Corridor and Segment	Population			
	2021	2026	2036	2046
Corridor 1				
Dream City to Majuragate	135 - 166	187 - 215	280 - 294	317 – 342
Majuragate to Sarthana	534	501	442	416
Corridor 2				
Bhesan to Majuragate	255	295	359	398
Majuragate to Saroli	314	335	369	389
Corridor 3				
Majuragate to Ved	456	445	427	424

It can be seen in the table above that population densities are declining along two corridors - Majuragate to Sarthana and Majuragate to Ved. This is due to commercialization in this area as a result of which textile markets, diamond industries and retail markets are now located here. Over a period of time, this

corridor is expected to be further intensified by commercial land use leading to a further decline in the population densities.

2.9.3.3 Existing Employment Distribution

Base year employment was estimated for SMC and SUDA area separately. In case of SMC area, the property tax data on non-residential properties and a sample survey was conducted to estimate the per unit area employment generation by the different property types a sample size of 0.85% was taken and the same expanded to the rest of the area. In case of SUDA area data was collected from the industries department and the chamber of commerce on the number of establishments and the employment there off. IEM and EM data was also used to estimate the overall employment of the base year. The same was crosschecked with the home base work attraction numbers generated by the extensive house hold survey conducted across the study area.



Map 2-21:Employment density 2016

Source: CoE-UT, CEPT University

The table below represents the estimated employment for the study area:

Table 2-19: Employment estimation

Area	Estimated Employment 2016	Percentage of total employment	Employment rate
SMC	2501653	85%	48%
SUDA	426906	15%	60%
Total Employment	2928559	100%	49%

Source: Primary Survey, 2016

2.9.3.4 Employment Distribution: Gradual Growth Scenario

In case of 2021 the existing distribution of employment is assumed with 85% in current SMC area and 15% in SUDA. Owing to the industrial growth proposed in the 2035 Development Plan of SUDA the proportion is changed accordingly for year 2026, 2036 and 2046 by considering following points:

- Existing and Proposed land use
- land availability for development
- Future growth nodes as per DP
- Future corridors
- Industrial Entrepreneurs' Memorandum

Unlike population distribution for Gradual Growth scenario, there will not be any difference at TAZs level for employment densities across various metro network options.

Table 2-20: Projected Employment SMC and SUDA

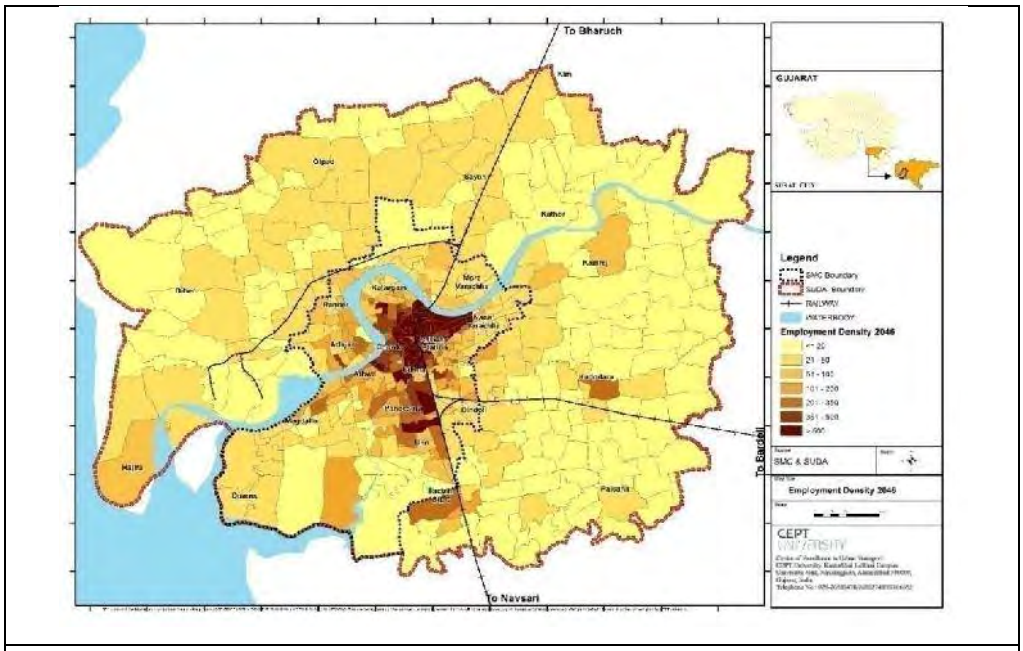
Year	Employment Rate	SMC	SUDA	Total	Current SMC area	Current SUDA area
2016	49.39%	25,01,653	4,26,907	29,28,559	85%	15%
2021	49.40%	30,80,513	5,25,687	36,06,200	85%	15%
2026	49.50%	32,88,162	9,44,088	42,32,250	78%	22%
2036	49.72%	39,60,298	15,08,902	54,69,200	72%	28%
2046	50%	43,84,570	18,65,430	62,50,000	70%	30%
Source: CoE-UT, CEPT University						

2.9.3.5 Employment Distribution: Rapid Growth

The following map shows employment density distribution for 'Gradual Growth' and 'Rapid Growth' scenarios along metro corridors for the year 2046. The density map of both scenarios for year 2026, 2036 has been provided in the annexure. The comparison shows higher density along corridor in TOD scenario wherever land is available for development.

Table 2.21: Employment densities along metro corridors (With TOD Scenarios)

Corridor and Segment	Employment Density			
	2021	2026	2036	2046
Corridor 1				
Dream City to Majuragate	29 - 53	68 - 90	120 - 146	141 - 181
Majuragate to Sarthana	317	374	441	484
Corridor 2				
Bhesan to Majuragate	53	73	97	112
Majuragte to Saroli	202	236	275	298
Corridor 3				
Majurgate to Ved	202	237	277	304



Map 2-22:Employment density - 2046 – Without TOD

2.10 Conceptual development of alternative Metro options

A standard four stage modelling process has been followed to project travel demand and metro ridership in future. This public transport model has been developed using EMME 4.2 software. An overview of the modelling approach is presented below followed at different stages.

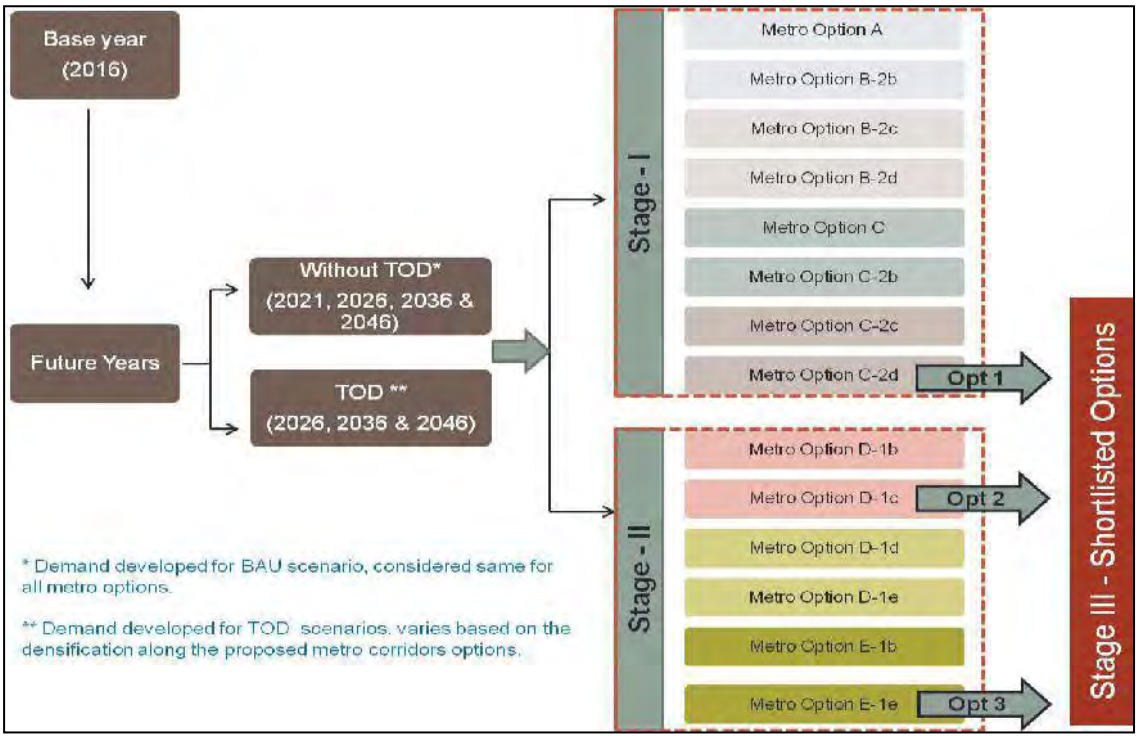


Figure 2.3: Conceptual framework for metro network options

2.10.1 Modelling framework

A travel demand model has been developed as part of Operations Plan for the city of Surat. The same model is carried forward for demand assessment for metro rail system for the city. A public transport model has been developed for 1-hour peak period. The major road network in the study area along with the existing public transport network of City bus, BRTS and proposed networks of metro has been taken into account for developing future scenarios. For the analysis of travel pattern four trip purposes - Home based Work, Education, Others, Non-home based, has been taken into account.

This modelling exercise has been carried out for 2016(Base Year) and future year of 2021. The following table shows the PT network taken for different years:

Table 2.22: PT Network Development Inputs

Year	Public Transport Network
2016	197km city bus network, 29 routes 58.78 km of BRTS network, 5 routes 153km of GSRTC network within study area, 12 GSRTC routes coded (directionally routes are combined by adding frequency) 161km of Shared Auto rickshaw network, 52 routes
2021, 2026, 2036 & 2046	Proposed city bus network of 450 km, 48routes 112 km of BRTS network, 12 routes 12 GSRTC routes coded (directionally routes are combined by adding frequency) Various Alternatives for metro option

For each of these years, demand scenarios of Gradual Growth and Rapid Growth have been considered for modelling. The data inputs used for modelling are as below:

Table 2-23: Data Inputs for Modelling Process

Type of data	Details
Demand Data	Existing demand – Household Surveys, RSI at Cordon points Future demand - Population, Employment data (nonresidential survey) existing and proposed, Student enrollment existing and proposed, Willingness to shift survey
Supply Data	Road network – Physical attributes, Network speeds, Delays at junctions
	Public Transport network – stops, corridors, routes, service details
Model calibration data	Travel speeds and travel time on major network (2015-16) Classified Volume Counts, Vehicle Occupancy Surveys

2.10.2 Four stage Modelling Process

As part of the modelling process, trip generation and distribution models have been developed as part of Public Transport Operations Plan project. These have been used for this project also.

1. Trip Generation

Trip generation model estimates number of trip productions and trip attractions to each TAZ. Trip productions are estimated based on household socio economic trip making characteristics. Trip attractions are estimated from employment & student enrolment of each TAZ.

Trip Production Model: Trip production model without walk trip for study area are present in following table, except Non Home Based trips which are marginal (0.001 % of all trips) and their estimation may not create significant impact in trip generation model. It shows strong relation between Home Based Work with resident workers and Home Based Education with resident student in each TAZ.

Table 2-24:Production Model

Dependent Variable (Productions)	Independent Variable	R^2	t	Equation
HBW	Resident Workers	0.99	211.4	2.156x
HBE	Resident Students	0.99	216.4	1.967x
HBO	Population	0.81	40.27	0.441x
NHB	Population	0.22	3.42	0.0025x

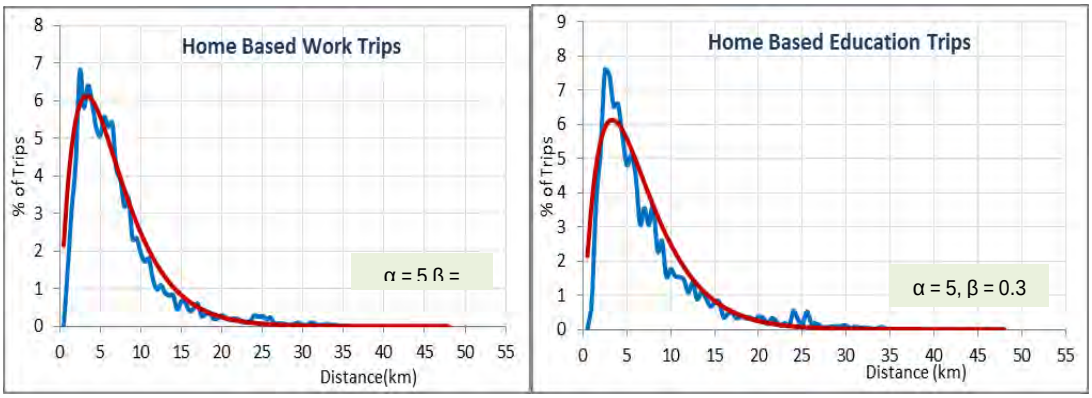
Trip Attraction Model: The model develops for trip attraction variables like employment and student enrolment in each TAZ. Table shows R^2 of Home Based Work and Home Based Education are between 0.70 - 0.80, which seems to be positive for trip attraction model.

Table 2-25:Attraction Model

Dependent Variable (Attraction)	Independent Variable	R2	t	Equation
HBW	Employment	0.82	34.81	0.9936x
HBE	Students Enrolment	0.67	16.33	3.4004x
HBO	Employment	0.63	17.23	0.7189x
NHB	Employment	0.49	5.74	0.0126x

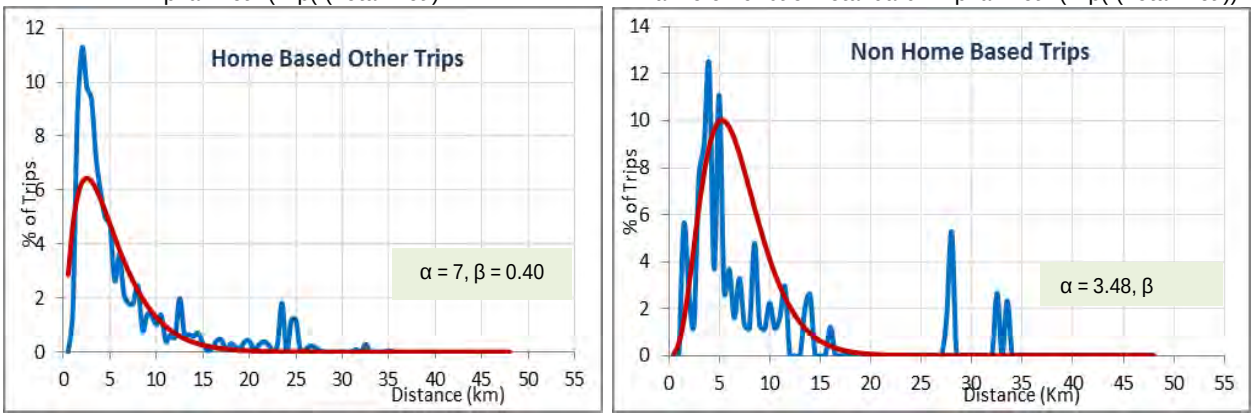
2. Trip Distribution Model

Trip distribution model works on gravity model which is based on assumption that trip interaction between TAZs is directly proportional to relative attractiveness of each zone, while inversely proportional to spatial separation between the TAZs. This model was developed for each of the four purposes - Home Based Work, Home Based Education, Home Based Other and Non Home Based trips. The following figure shows the modelled and actual Trip Length Frequency Distribution (TLFD) for Home Based Work and Home Based Education trips. Combined Tanner's function was used for trip distribution.



82.5 % Coincidence RatioTanners' Function- Standard: Alpha*Dist.*(Exp(-(Beta*Dist.)

79.08 % Coincidence Ratio
Tanners' Function- Standard: Alpha*Dist.*(Exp(-(Beta*Dist.))



64.5 % Coincidence Ratio
Tanners' Function- Standard: Alpha*Dist.*(Exp(-Beta*Dist.))

47.5 % Coincidence Ratio
Tanner's Function (Alternative 1): Dist.^Alpha*(Exp(Beta*-Dist.))

Graph 2-4:Trip Length Frequency Distribution – Different Purpose wise

Source: Comprehensive Mobility Plan for Surat, CEPT Research and Development Foundation

3. Mode split

A detailed 'Willingness to Shift' survey has been conducted and analyzed through mode shift modelling framework. The attributes taken for analysis are travel time and travel cost, through which utility equations have been generated through logit model. Table2-24 below shows the estimated PT share for future years. In 2021, the start year of the metro PT share remains the same. However, in subsequent years, Rapid Growth scenario, there are

differences in population and employment distribution in comparison to the BAU scenario, which also results in mode share differentials.

Table 2.26:Modelled Public transport share for year 2021, 2026, 2036 and 2046

Year	Growth Scenarios	
	Gradual Growth	Rapid Growth
2021	10.6%	-
2026	15.0%	17.0%
2036	25.6%	29.1%
2046	26.8%	32.5% - 34.5%

Value of Time (VoT) become the critical parameter for private vehicle users to take the decision to shift on public transport system from their own vehicles. Below is the VoT for different modes have been analysed from the surveys for the base year.

Table 2.27: Value of Time (VoT) of private vehicle users – Base year

Mode	VoT (Rs./min.)
2 Wh	**1.72**
3 Wh Shared	**1.10**
3 Wh Private	**1.22**
4 Wh.	**3.10**

4. Transit Assignment

Table 2.28:Public Transport Demand Matrix

PT Demand	Internal	External
Internal	7,59,606 **(93%)**	26,933 **(3%)**
External	23,718 **(3%)**	3,098 **(0.38%)**

Transit assignment is the final stage of the four stage modeling process. Base transit demand (internal & external) is about 8.13 lakhs per day. One hour peak hour demand i.e. 10% of the full demand is assigned on the model considering PT and IPT mode. The external demand is derived from regional terminal surveys where external PT demand was captured. The share of the PT (City bus, BRTS and GSRTC within study area) trips is 1.75% in motorized trips with 0.92 lakhs demand (Internal to internal within study area) whereas IPT(auto rickshaw) is about 16% in motorized trips with 6.66 lakh demand. External to external demand is about 0.03 lakhs which was captured as part of terminal surveys but does not make much significant in overall scenario.

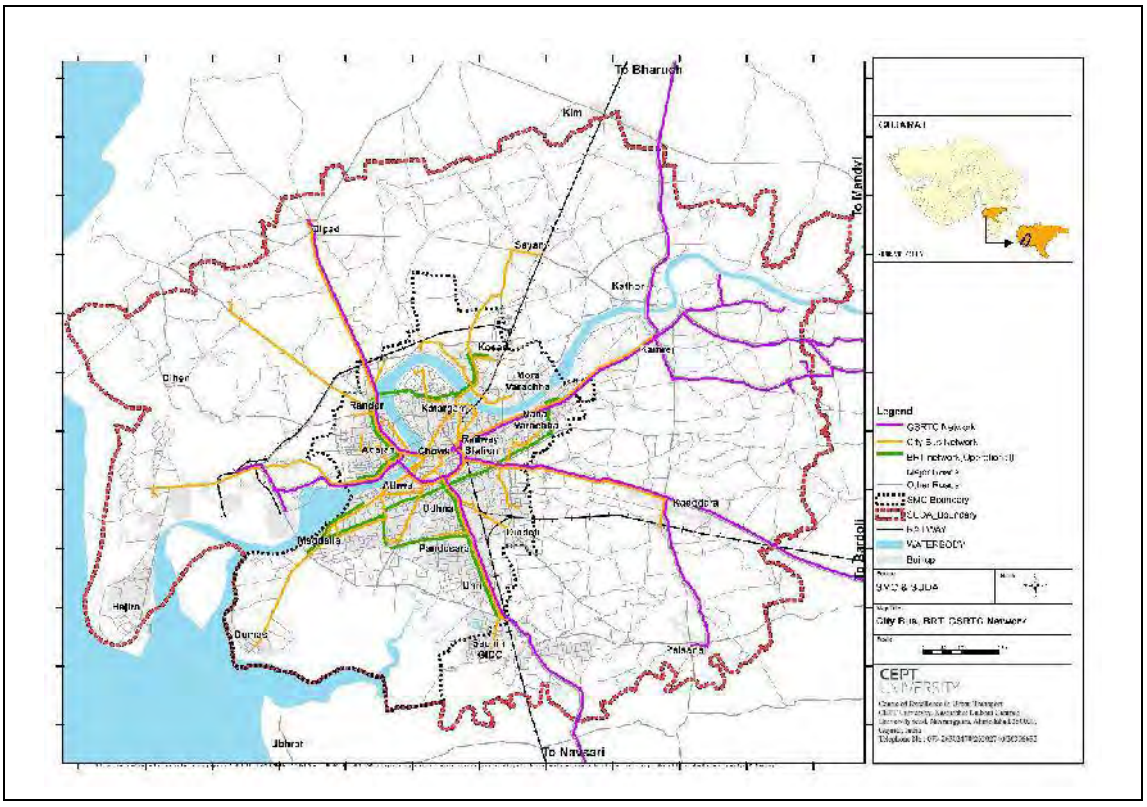
2.10.3 Base PT Model Development

A base year network has been developed in GIS which has been ARC GIS and the modelling process has been carried out in travel demand modeling software i.e. EMME 4.2.

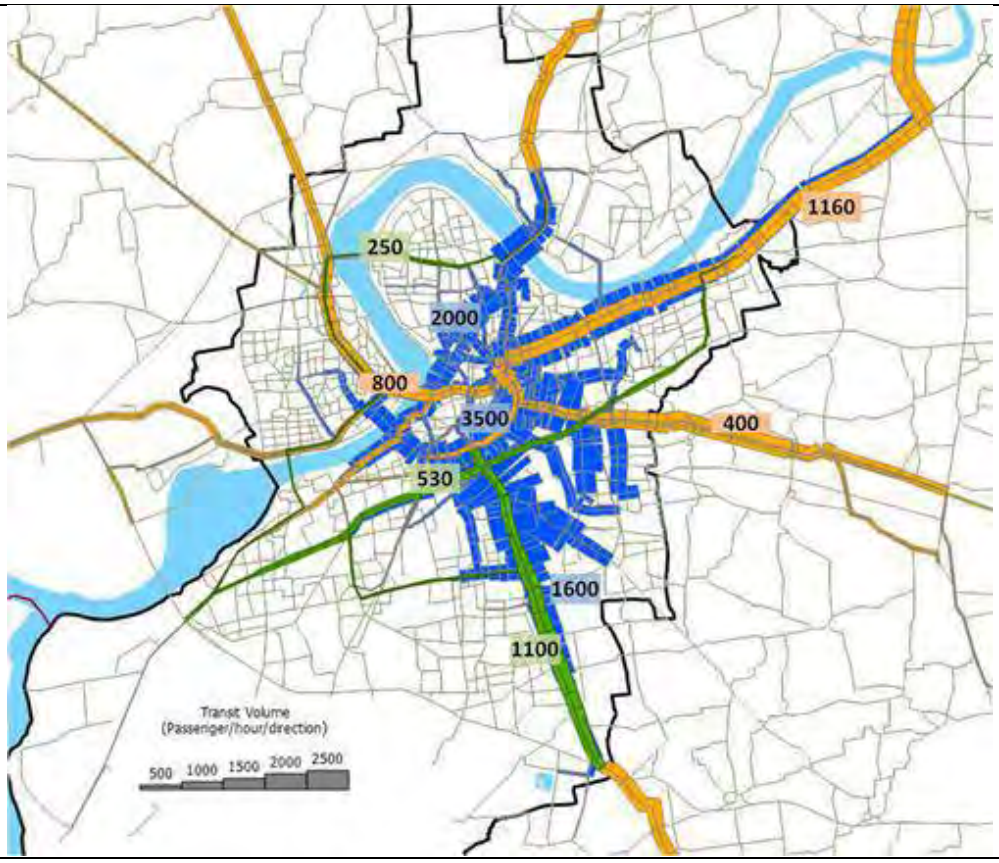
In September 2016 base year, when surveys and model development was carried out, Operational BRTS network was 58.78 km and 29 city bus routes were in operation. All together there are about 398 regional, sub urban and local GSRTC routes were plying or passing through Surat. For modelling purpose, 12 corridors/ directional routes have been identified and combined frequency were coded in the model for simplicity, as it mostly take care of external PT demand. Use of auto rickshaw is higher than city bus services, therefore base year transit model have been developed with PT services as well with IPT services to perceive the transit flow on the network. Since the model has been developed for the peak hour, only peak hour frequency has been taken for each route.

As discussed in the previous section, household surveys and RSI survey were carried out in within the study area and at cordon location respectively in 2016, which have been used for developing the base year demand matrix. For capturing the external public transport demand, passenger surveys at major bus terminal were also carried out.

A total of 8.13 lakh public transport trips were assigned on PT and IPT mode as trips with auto rickshaw is about 16% of motorized trip as compared with 1.75% of PT trips.



Map 2.23 Base year Transit (BRTS + GSRT TC Comprehensive Network) Network



Map 2-24 :Base year – Transit flows

2.10.4 Future Model Development

As discussed in Section 2.1, three corridors have been proposed for metro:

1. Dream City to Sarthana
2. Bhesan to Saroli
3. Majura Gate to Ved

For corridors 1 and 2, several alternative alignments have been considered to arrive on an alignment with good catchment area and ridership levels. These alternative alignments are presented in the table below:

Table2.29: Metro Corridor alternatives

Corridor	Alternative alignments	Details	Length (km)	No. of Stations
1 (Sarthana to Dream City)	1a	**Sarthana to Dream City** (viaSarthana Nature Park, Nana Varchha, Railway Station, Chowk, Majuragate, Parlepoint, Athwa, Vesu)	26.64	22
	1b	**Sarthana to Dream City** (via Sarthana Nature Park, Nana Varchha, Railway Station, Chowk, Majuragate, Bhatar Char rasta, Sarsana-Exhibition Centre, Khajod char rasta)	22.68	24
	1c	**Sarthana to Dream City**	26.18	21

Corridor	Alternative alignments	Details	Length (km)	No. of Stations
		(via Sarthana Nature Park, Nana Varchha, Railway Station, Chowk, Majuragate, Bamroli Road, VIP Road, Mangalam heights)		
	1d	**Sarthana to Dream City** (via Sarthana Nature Park, Nana Varchha, Railway Station, Chowk, Majuragate, Swami Vivekanand garden, AnuvratDwar, VNSGU, Mangalam heights)	24.46	21
	1e	**Sarthana to Dream City** (via Sarthana Nature Park, Nana Varchha, Railway Station, Chowk, Majuragate, Bhatar Char rasta, AnuvratDwar, VNSGU, Mangalam heights)	24.69	22
2 (Bhesan to Saroli)	2a	**Bhesan to New Civil Hospital** (via Ugat, L.P. Savani, Adajan, Majuragate)	11.45	11
	2b	**Bhesan to KamnathMahadev** (via Ugat, L.P. Savani, Adajan, Majuragate, Kharwarnagar, Swaminarayanmandir)	16.78	15
	2c	**Bhesan to MithiKhadi Road** (via Ugat, L.P. Savani, Adajan, Majuragate, KamelaDarwaza)	13.62	13
	2d	**Bhesan to Saroli** (via Ugat, L.P. Savani, Adajan, Majuragate, KamelaDarwaza, ParvatPatiya)	18.08	16
3	3	**Majuragate to Ved**(via Bhagal, Katargam)	8.06	8

2.10.4.1 Demand Scenarios

The demand forecasting exercise for alternative metro options were carried out and developed for two growth strategies these are:-

1. Gradual Growth Scenario and;
2. Rapid Growth Scenario

2.10.4.1 Network Scenarios

These scenarios with proposed metro corridors were evaluated in three stages as describe below to arrive at appropriate metro corridor option for the study area.

Table2.30: Evaluation stages of Metro Option

Stages	Network Scenarios	Corridor considered	Total Length of Network (km)

Stages	Network Scenarios	Corridor considered	Total Length of Network (km)
Stage-I (8 network scenarios)	A	Corridor 1a + Corridor 2a	38.09
	B-2b	Corridor 1a + Corridor 2b	43.42
	B-2c	Corridor 1a + Corridor 2c	40.26
	B -2d	Corridor 1a+ Corridor 2d	44.72
	C	Corridor 1a + Corridor 2a + Corridor 3	46.15
	C-2b	Corridor 1a + Corridor 2b + Corridor 3	51.48
	C-2c	Corridor 1a + Corridor 2c + Corridor 3	48.32
	C-2d	Corridor 1a + Corridor 2d+ Corridor 3	52.76
Stage-II (6network scenarios)	D-1b	Corridor 1b + Corridor 2d	40.76
	D-1c	Corridor 1c + Corridor 2d	44.26
	D-1d	Corridor 1d + Corridor 2d	44.54
	D-1e	Corridor 1e + Corridor 2d	44.77
	E-1b	Corridor 1b + Corridor 2d + Corridor 3	48.82
	E-1e	Corridor 1e + Corridor 2d+ Corridor 3	50.83
Stage –III (3 Selected network scenarios from stage I & II)	C-2d	Corridor 1a + Corridor 2d+ Corridor 3	52.76
	E-1b	Corridor 1b + Corridor 2d + Corridor 3	48.82
	E-1e	Corridor 1e + Corridor 2d+ Corridor 3	50.83
Final Selected Network	**D-1b**	**Corridor 1b + Corridor 2d**	**39.4**

2.10.5 Scenario evaluation

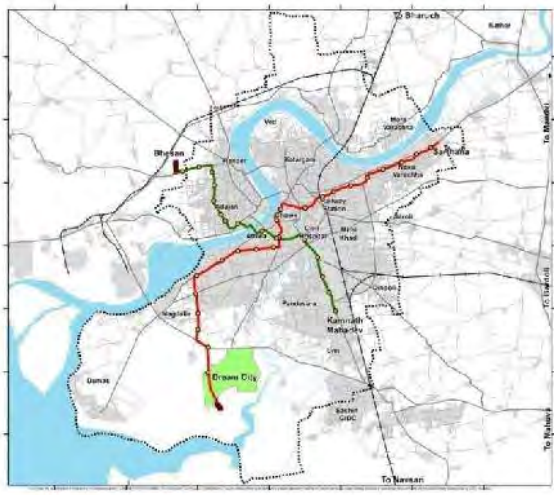
2.10.5.1 Stage 1

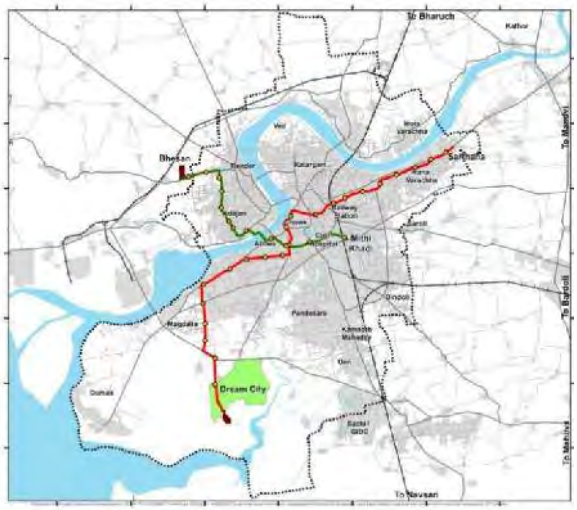
As can be seen in table above, a total of 8 metro network options were modeled, considering 4 alternative alignments of Corridor 2 (2a, 2b, 2c & 2d). These were discussed with Surat Municipal Corporation on 2nd Dec 2016. Of these, Corridor 2d (Bhesan to Saroli) alignment was attracting maximum ridership on metro and hence was found most suitable.

Map 2-25: Stage 1 – 8 Metro Network Option

Option A

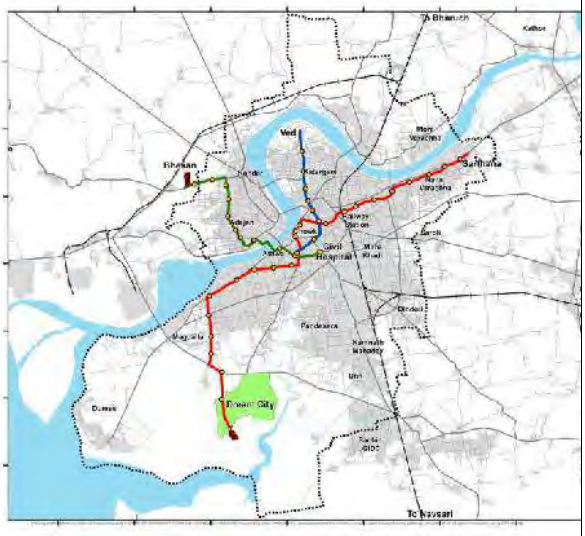
Option B-2b



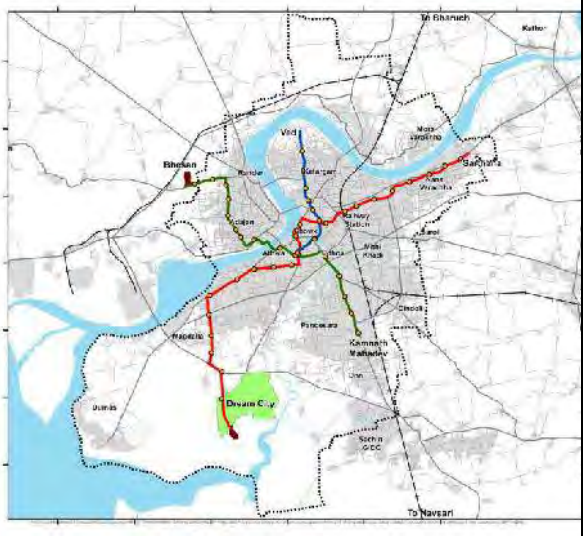


Option B-2c

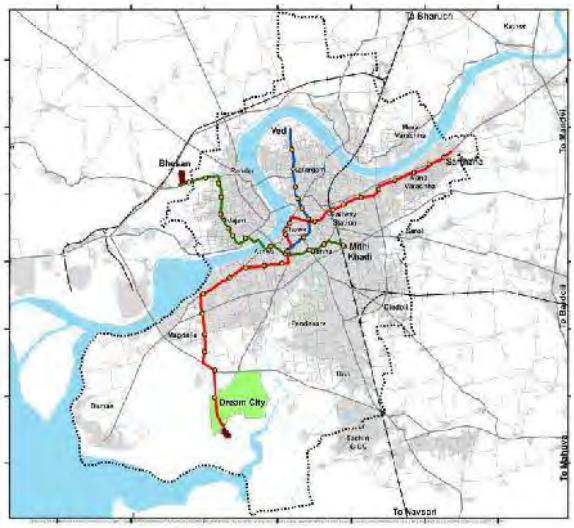
Option B-2d



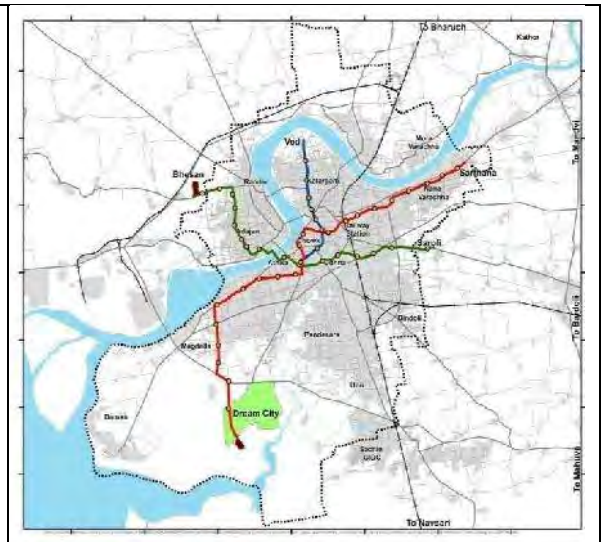
Option C



Option C-2b



Option C-2c



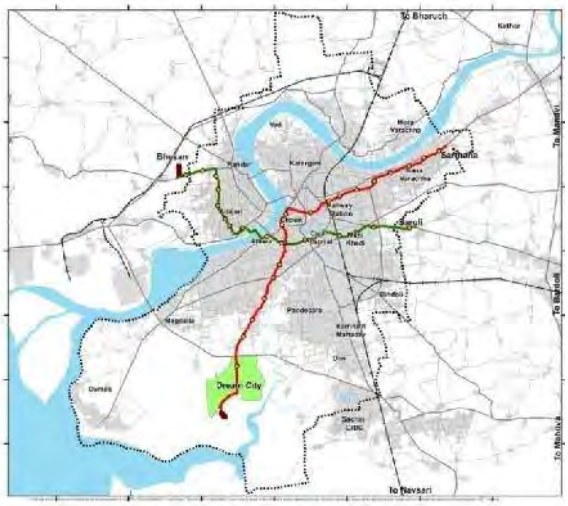
Option C-2d

20.10.5.2 Stage 2

Stage 1 analysis also showed ridership on Corridor 1 segment from Dream City-Majura Gate to be low in comparison to other segments. Therefore, testing of alternative alignments for this segment was suggested by the SMC.

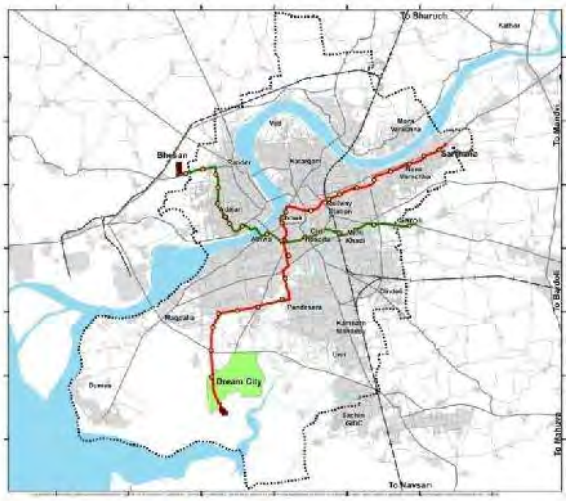
As shown in the table above, a total of 6 network options were modeled, for analyzing 5 alternative alignments of corridor 1 (1a, 1b, 1c, 1d, 1e). These were discussed with SMC and DMRC on 6th Dec and 17th Dec 2016. Corridor 1b seemed to be performing better in terms of ridership. It was however suggested to finalise ridership numbers for corridor three alignment alternatives – 1a, 1b, 1e.

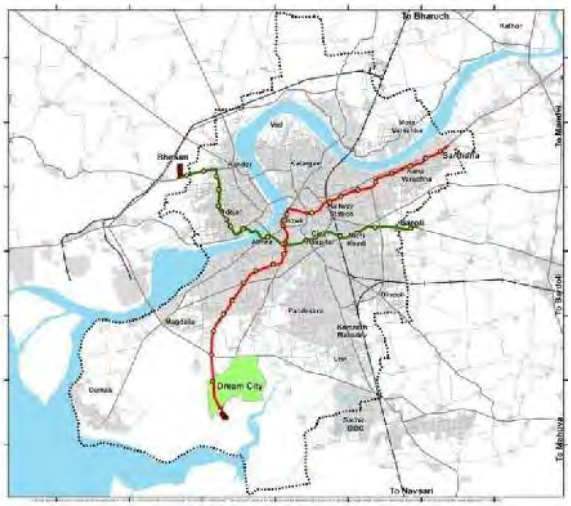
Map 2-26 : Stage 2–6Metro Network Options



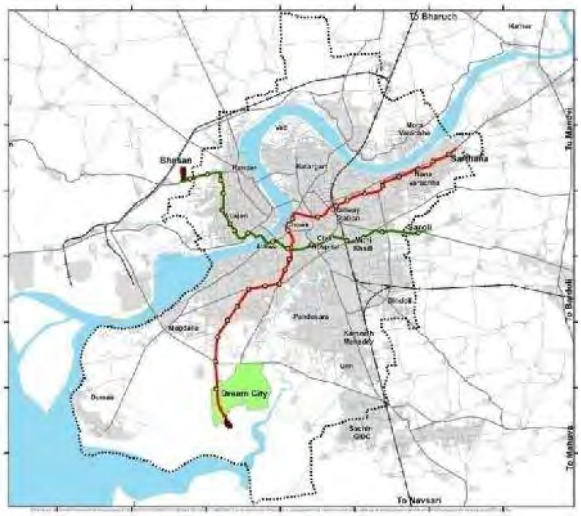
Option D-1b

Option D-1c

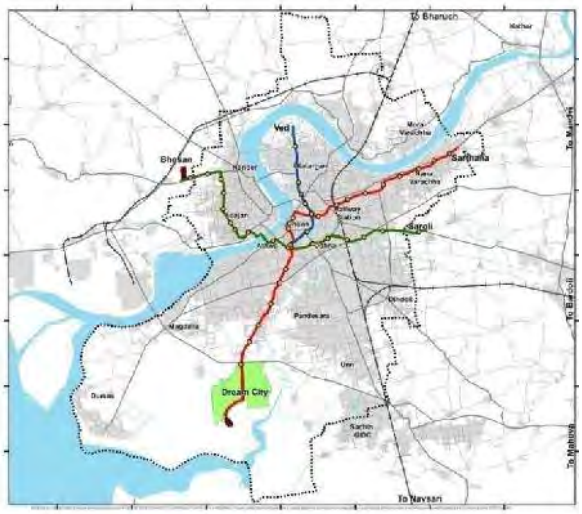




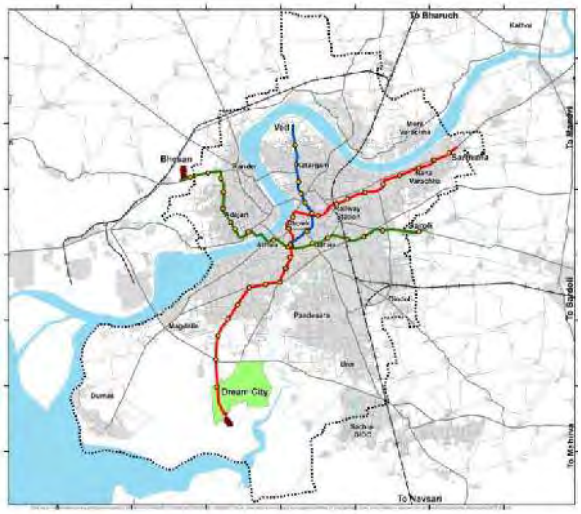
Option D-1d



Option D-1e



Option E-1b

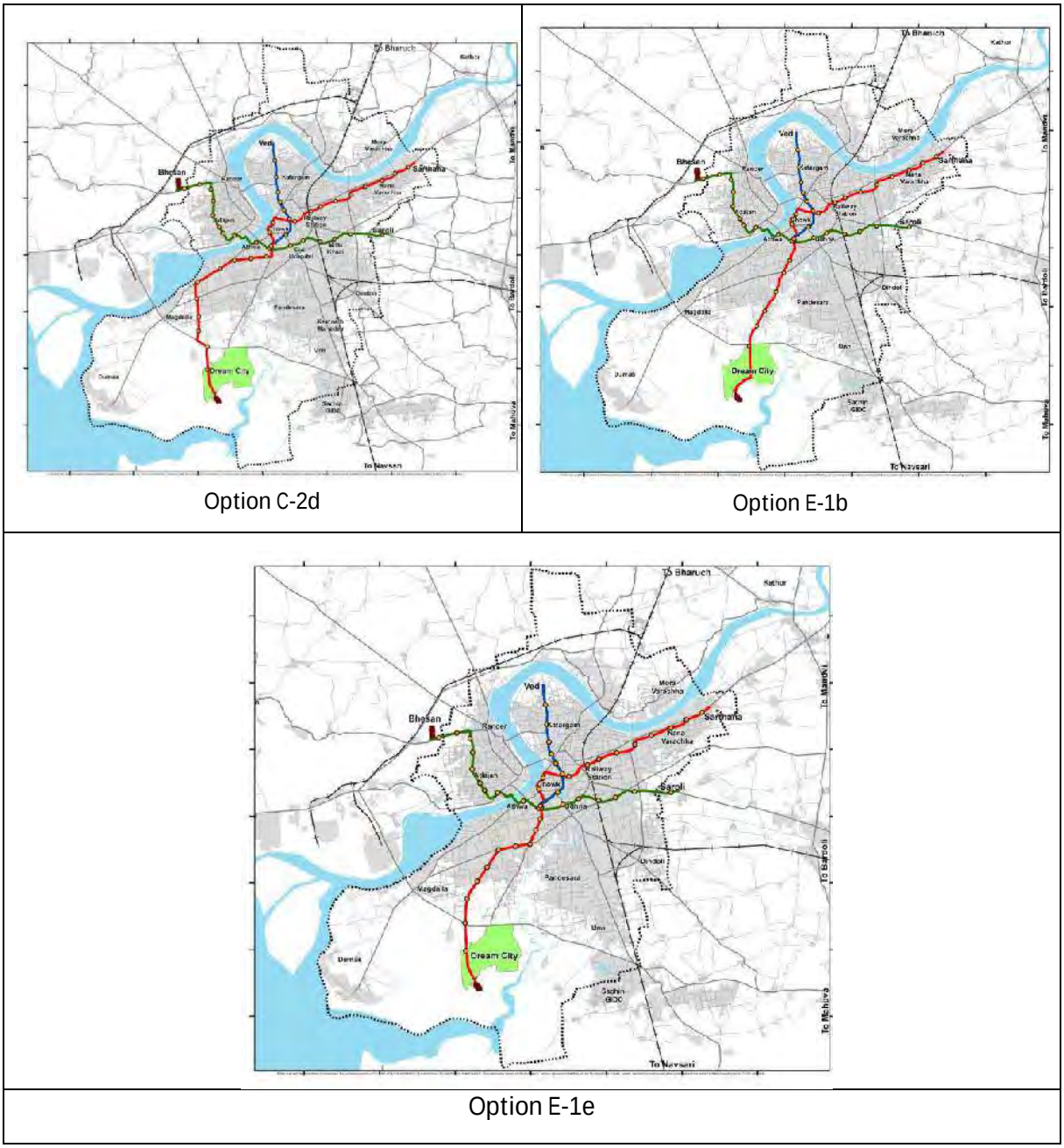


Option E-1e

2.10.5.3 Stage 3

For 3 alternative alignment scenarios of Corridor 1 (1a, 1b & 1e), demand forecasts for 2021 and 2046 were carried out. The results for the three network options are shown as model outputs in this report.

Map 2-27: Stage 3– Alternative Metro Network Options



Option C-2d

Option E-1b

Option E-1e

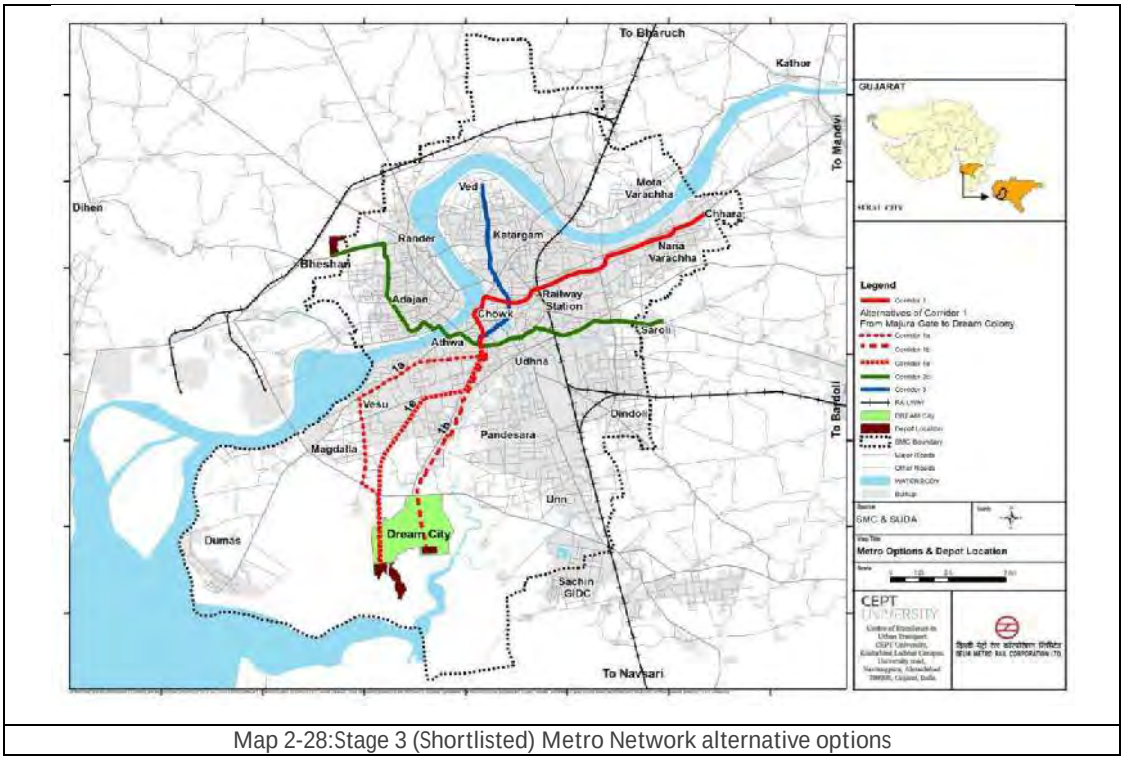
2.10.6 Model outputs for Stage 3 Metro network options

The following table shows details of the 3 network scenarios modelled in Stage

Table 2-31:Metro Network options details

Metro Network Option	Corridors Description	Network Length (km)
C-2d	**Corridor 1a:Sarthana to Dream City** (via Sarthana Nature Park, Nana Varchha, Railway Station, Chowk, Majuragate, Parlepoint, Athwa, Vesu) **Corridor 2d:Bhesan to Saroli** (via Ugat, L.P. Savani, Adajan, Majuragate, KamelaDarwaza, ParvatPatiya) **Corridor 3: Majuragate to Ved**(via Bhagal, Katargam)	52.76
E-1b	**Corridor 1b:Sarthana to Dream City** (via Sarthana Nature Park, Nana Varchha, Railway Station, Chowk, Majuragate, Bhatar Char rasta, Sarsana- Exhibition Centre, Khajod char rasta) **Corridor 2d:Bhesan to Saroli** (via Ugat, L.P. Savani, Adajan, Majuragate, KamelaDarwaza, Parvat Patiya) **Corridor 3: Majuragate to Ved** (via Bhagal, Katargam)	48.82
E-1e	**Corridor 1e:Sarthana to Dream City** (via Sarthana Nature Park, Nana Varchha, Railway Station, Chowk, Majuragate, Bhatar Char rasta, Anuvrat Dwar, VNSGU, Mangalam heights) **Corridor 2d: Bhesan to Saroli** (via Ugat, L.P. Savani, Adajan, Majuragate, Kamela Darwaza, Parvat Patiya) **Corridor 3: Majuragate to Ved** (via Bhagal, Katargam)	50.83

The following map shows the three metro corridors and alternative alignments of Corridor 1 segment (Dream City to Majura Gate)



Map 2-28:Stage 3 (Shortlisted) Metro Network alternative options

2.10.7 Results summary

Demand modelling for the network options Option C-2d, E-1b and E-1e were undertaken for both - With TOD and Without TOD demand scenarios. The following tables present the ridership summary in terms of metro boarding's and boarding's per km for the years 2021and 2046.

Table 2-32:Scenarios without TOD – Year 2021

Option	Mode	Boarding's	Transfer rate	% Boarding's	Metro Network Length (km)	Metro Boarding's per km
C-2d	Bus	5,69,850	**1.66**	46%	52.76	**5,375**
	BRT	3,83,440		31%		
	Metro	2,83,580		23%		
	Total	**12,36,870**		**100%**		
E-1b	Bus	5,57,630	**1.67**	45%	48.82	**6,493**
	BRT	3,71,080		30%		
	Metro	3,17,010		25%		
	Total	**12,45,720**		**100%**		
E-1e	Bus	5,63,320	**1.66**	45%	50.83	**5,930**
	BRT	3,74,370		30%		
	Metro	3,01,420		24%		
	Total	**12,39,110**		**100%**		

Table 2.33:Scenarios without TOD – Year 2046

Option	Mode	Boardings	Transfer rate	% Boardings	Metro Network Length (km)	Metro Boardings per km
C-2d	Bus	27,97,680	**1.76**	45.5%	52.76	24,078
	BRT	20,75,660		33.8%		
	Metro	12,70,350		20.7%		
	Total	**61,43,690**		**100%**		
E-1b	Bus	27,56,500	**1.77**	45%	48.82	27,578
	BRT	20,57,400		33%		
	Metro	13,46,340		22%		
	Total	**61,60,240**		**100%**		
E-1e	Bus	27,75,620	**1.77**	45%	50.83	26,546
	BRT	20,40,690		33%		
	Metro	13,49,350		22%		
	Total	**61,65,660**		**100%**		

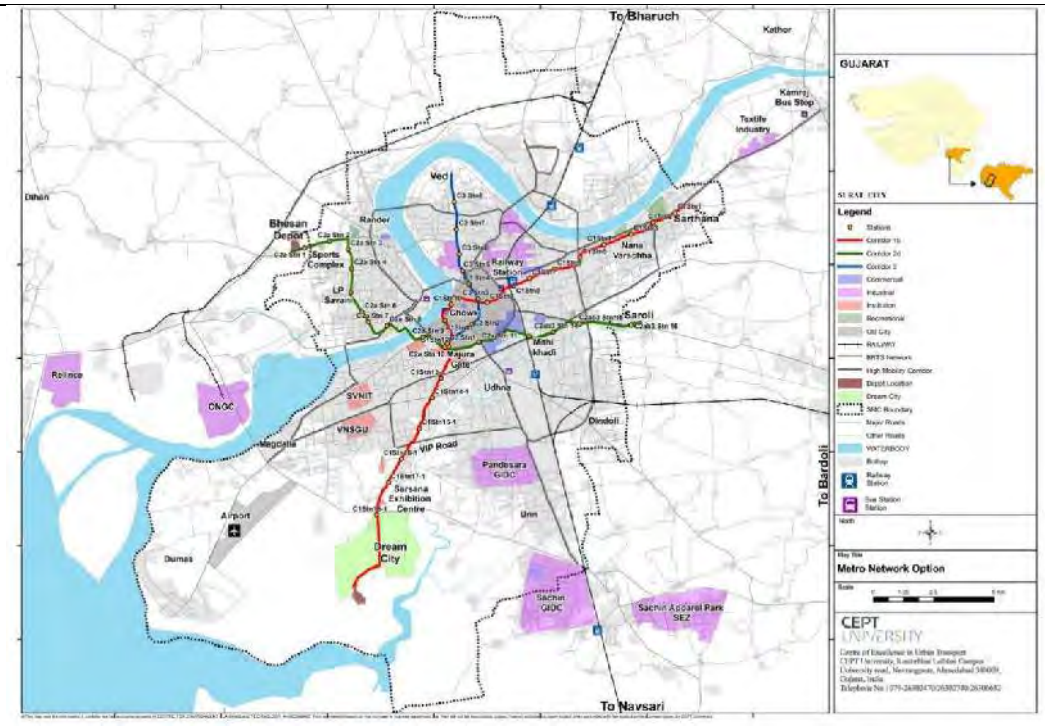
Table 2-34:Scenarios with TOD – Year 2046

Option	Mode	Boardings	Transfer rate	% Boardings	Metro Network Length (km)	Metro Boardings per km
C-2d	Bus	33,77,510	**1.83**	44.3%	52.76	32,654
	BRT	25,31,580		33.2%		
	Metro	17,22,810		22.6%		
	Total	**76,31,900**		**100%**		
E-1b	Bus	35,97,590	**1.85**	44%	48.82	43,167
	BRT	25,34,330		31%		
	Metro	21,07,410		26%		
	Total	**82,39,330**		**100%**		
E-1e	Bus	36,69,830	**1.86**	44%	50.83	40,600
	BRT	25,58,630		31%		
	Metro	20,63,680		25%		
	Total	**82,92,140**		**100%**		

2.10.8 Conclusion

A detailed evaluation of alternative metro corridor alignments helped in identifying most suitable option based on ridership and connectivity. The metro option E-1b seems to be most efficient option, and comprises of:

i. **Corridor 1b: Sarthana to Dream City** (via Sarthana Nature Park, Nana Varchha, Railway Station, Chowk, Majuragate, Bhatar Char rasta, Sarsana-Exhibition Centre, Khajod Charrasta)
ii. **Corridor 2d: Bhesan to New Civil Hospital** (via Ugat, L.P. Savani, Adajan, Majuragate, KamelaDarwaza, Parvat Patiya)
iii. **Corridor 3: Majuragate to Ved** (via Bhagal, Katargam)



Map 2-29:Metro Network Option E-1b

These results were presented to the High Power Committee on 13thJanuary 2017 and it was decided to initially go ahead with two corridors in Phase I, i.e. Corridor 1 (Sarthana to Dream City) and Corridor 2 (Bhesan to Saroli). Corridor 3 (Majuragate to Ved) could be taken up in subsequent phase.

As decided by the High Power Committee, based on the initial ridership numbers, transport model has been prepared to estimate the ridership **only** for selected two corridors as below:-

i) Corridor 1:Sarthana to Dream City:21.61 Kms(via Nature Park,VarachhaChopatiGarden,Surat Railway Station, Chowk Bazar, Majuragate, AlthanGam, ExhibitionCentre, Khajod Char rasta)

ii) Corridor 2 : Bhesan to Saroli : 18.74 Kms(via Ugat Vaarigruh, LP SavaniSchool, AdajanGam, Majuragate, Kamela Darwaza, Magob)

2.11 Auto (Private Vehicle) Assignment

Surat city has about 85% trips made by private modes as per the 2016 household survey undertaken. It is known that increased motorisation leads to traffic congestion and increase in travel time. A strategic private travel demand model was also developed for the study area to assess the congestion levels in the study area in future years. This analysis has looked at private vehicle km (VKT) and private vehicle hours traveled (VHT) savings.

The Surat Development Plan 2035 has proposed the strategic road network for the study area considering the future developments and land use. The same has been adopted for the future year transport network coding. Similarly, in terms of demand, both without and with metro demand scenarios were considered for assigning on the future DP network. The model outputs of two scenarios, i.e. Without metro and With metro have been compared and analysed. The outputs have been presented below;

Table 2-35 Auto model outputs (Full day)

Year	Without Metro			
	PT Share	Demand (PCU Demand)	PCU Km	PCU Hrs.
2016	1.84%	3024090	22704150	781870
2021	10.60%	3496080	55751870	1939010
2026	15.00%	4144480	70394110	2585250
2036	25.65%	4812010	85407810	3217900
2046	26.85%	5398470	100748550	4131470
Year	**With Metro**			
	PT Share	**Demand (PCU Demand)**	**PCU Km**	**PCU Hrs.**
2016	1.80%	3024090	22704150	781870
2021	10.60%	3496080	55751870	1939010
2026	17.00%	4039970	67012090	2374700
2036	29.11%	4534210	76753680	2649570
2046	34.60%	4769730	83575470	2762090
Year	**Savings**			
	PT Share	**Demand (PCU Demand)**	**PCU Km**	**PCU Hrs.**
2016	-	-	0.00%	0.00%
2021	-	-	0.00%	0.00%
2026	-	-	4.80%	8.14%
2036	-	-	10.13%	17.66%
2046	-	-	17.05%	33.15%

For the horizon year of 2046, 33% travel time and 17% vehicle km savings have been observed in with metro scenario compared to Without metro scenario from the model outputs.

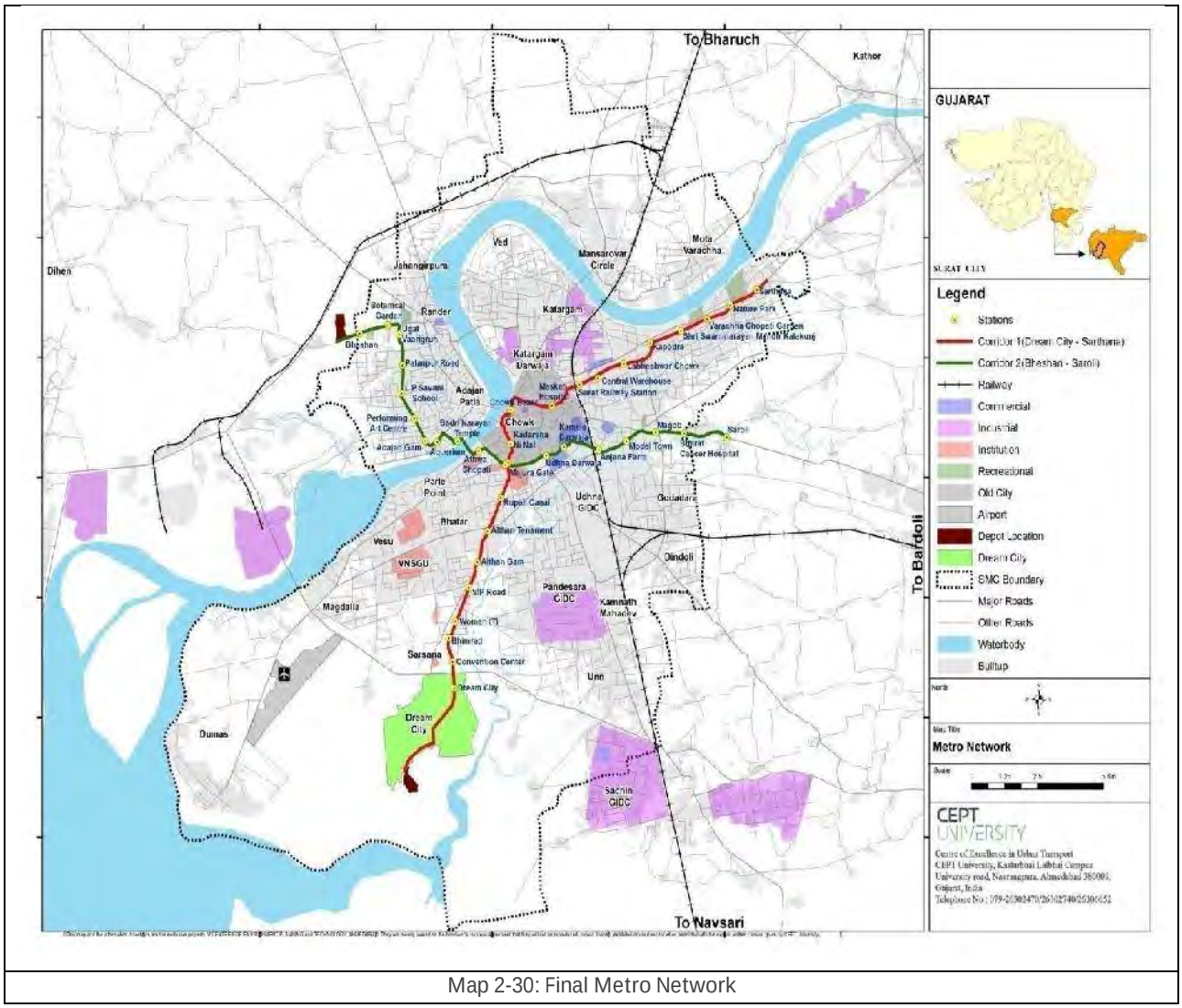
2.11.1 Model Outputs for the Selected Network

Based on the inputs from High Power Committee, a detailed field assessment of Corridor 1 and 2 was carried out by DMRC/SMC to finalise the station locations. The following table and map presents the details of the stations for both the corridors:

Table 2-36: Stations Location Chainage

Sr.No	CORRIDOR-I : SARTHANA TO DREAM CITY(Length: 21.61kms)		CORRIDOR-II: BHESHAN TO SAROLI (Length: 18.74 km)	
	Station Name	Chainage(m)	Station Name	Chainage(m)
	Dead End	(-) 450	Dead End	(-) 949.630
1	Sarthana	0.0	Bhesan	0.0
2	Nature Park	1238.1	Botanical Garden	950.60
3	VarachhaChopati Garden	2108.5	Ugat Vaarigruh	1617.5
4	Shri Swaminarayan Mandir Kalakunj	3196.6	Palanpur Road	2738.9
5	Kapodra	4237.0	L.P. Savani school	3831.60
6	Labheshwar Chowk	5506.60	Performing Art Centre	4421.80
7	Central Warehouse	6668.9	Adajan Gam	5155.00
8	Surat Railway Station	7255.6	Aquerium	5770.10
9	Maskati Hospital	8487.0	Badri Narayan Temple	6891.50
10	Chowk Bazar	10289.8	AhthwaChopati	7863.50
11	Kadarsha Ni Nal	11721.7	Majura Gate	9019.20
12	Majura Gate	12313.8	Udhana Darwaja	10646.20
13	Roopali Canal	13611.5	Kamela Darwaja	11247.20
14	AlthanTenament	14696.2	Anjana Farm	12593.90
15	Althan Gam	15843.50	Model Town	13636.30
16	VIP Road	16699.1	Magob Gam	14763.10
17	Surat Women ITI	17808.40	Bharat Cancer Hospital	15899.40
18	Bhimrad	18659.5	Saroli	17341.80
19	Convention Centre	19829.40	DEAD END	17791.90
20	Dream City	20710.3		18741.53
	DEAD END	21160.3		
		21610.3		

Source: DMRC



Map 2-30: Final Metro Network

2.12 Results Summary

2.12.1.1 Boarding Summary and PHPDT

The summary model outputs in terms of boarding's for PT modes are presented in the tables below.

Table 2-37: Mode wise Boarding Summary(2021, 2026, 2036, 2046)

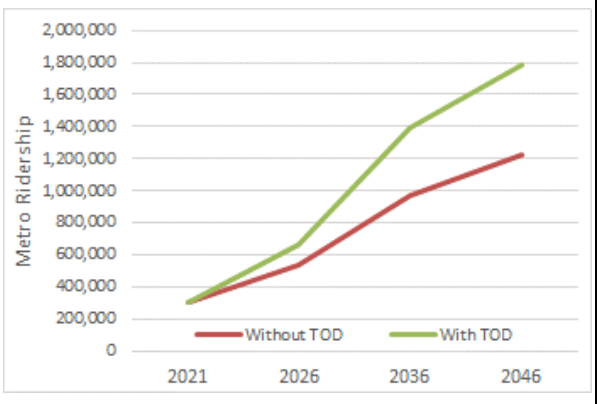
Scenarios	Mode	Boarding's	Transfer rate	% Boarding's
2021 Without TOD	Bus	516,360	1.71	41%
	BRT	457,300		36%
	Metro	299,650		24%
	Total	**1,273,310**		100%
2026 Without TOD	Bus	853,980	1.64	39%
	BRT	793,170		36%
	Metro	533,890		24%

Scenarios	Mode	Boarding's	Transfer rate	% Boarding's
	Total	**2,181,040**		100%
2026 With TOD	Bus	869,130	1.65	37%
	BRT	825,650		35%
	Metro	663,670		28%
	Total	**2,358,450**		100%
2036 Without TOD	Bus	2,129,440	1.74	43%
	BRT	1,791,850		37%
	Metro	974,310		20%
	Total	**4,895,600**		100%
2036 With TOD	Bus	2,502,620	1.81	42%
	BRT	1,994,720		34%
	Metro	1,395,160		24%
	Total	**5,892,500**		100%
2046 Without TOD	Bus	2,744,630	1.78	44%
	BRT	2,226,070		36%
	Metro	1,223,090		20%
	Total	**6,193,790**		100%
2046 With TOD	Bus	3,623,590	1.85	44%
	BRT	2,812,810		34%
	Metro	1,786,430		22%
	Total	**8,222,830**		100%

It can be seen that by the horizon year of 2046, around 12.2 lakh boardings is expected on metro in 'Without TOD' scenario and 17.8 lakh boardings in 'With TOD' scenario. Overall metro boarding summary has been presented in the table below:

Table 2.38 : Overall Metro boardings summary (2021, 2026, 2036, 2046)

Sr. No.	Scenarios	Metro Boardings (Full Day)
1	2021 Without TOD	299,650
2	2026 Without TOD	533,890
3	2026 With TOD	663,670
4	2036 Without TOD	974,310
5	2036 With TOD	1,395,160
6	2046 Without TOD	1,223,090
7	2046 With TOD	1,786,430



For maximum PHPDT for future years, same has been presented below:

Link load for maximum PHPDT for future years, has been presented below:

Table 2.39: Maximum PHPDT(2021, 2026, 2036, 2046)

Sr. No.	Corridor	2021_Without TOD (Peak Hour)	2026_Without TOD (Peak Hour)	2026_With TOD (Peak Hour)	2036_Without TOD (Peak Hour)	2036_With TOD (Peak Hour)	2046_Without TOD (Peak Hour)	2046_With TOD (Peak Hour)
1	Corridor 1b: Sarthana - Dream City	6,549	13,340	16,099	20,856	28,144	24,247	32,065
2	Corridor 1bR: Sarthana- Dream City	6,992	13,536	16,424	20,677	25,872	26,587	33,543
3	Corridor 2d: Bheshan- Saroli	3,918	5,698	7,466	12,573	19,944	15,448	25,849
4	Corridor 2dR: Bheshan- Saroli	3,693	5,236	6,951	12,098	19,902	15,175	25,165

Note: 1bR and 2dR are the name of reverse direction.

2.12.1.2 : Metro – Metro Transfers

There is only one metro-metro interchange at Majuragate because of the two metro corridors. Below is the transfer passengers details for the year 2021 and 2046.

Table2.40: Peak Hour Transfer passenger details (metro line – metro line)

Year 2021					
Line	**MRT1: Dream city to Sarthana**	**MRT1R: Sarthana to Dream city**	**MRT2: Bheshan to Saroli**	**MRT2R: Saroli to Bheshan**	**Total**
MRT1: Dream city to Sarthana	-	-	75	25	99
MRT1R: Sarthana to Dream city	-	-	750	394	1144
MRT2: Bheshan to Saroli	330	23	-	-	353
MRT2R: Saroli to Bheshan	603	100	-	-	704
Total	933	123	824	419	2300
Year 2046 - Without TOD					
Line	**MRT1: Dream city to Sarthana**	**MRT1R: Sarthana to Dream city**	**MRT2: Bheshan to Saroli**	**MRT2R: Saroli to Bheshan**	**Total**
MRT1: Dream city to Sarthana	-	-	364	227	591
MRT1R: Sarthana to Dream city	-	-	983	1953	2937
MRT2: Bheshan to Saroli	1805	316	-	-	2122
MRT2R: Saroli to Bheshan	830	537	-	-	1367
Total	2635	853	1347	2181	7016
Year 2046 - With TOD					
Line	**MRT1: Dream city to Sarthana**	**MRT1R: Sarthana to Dream city**	**MRT2: Bheshan to Saroli**	**MRT2R: Saroli to Bheshan**	**Total**
MRT1: Dream city to Sarthana	-	-	1322	654	1976
MRT1R: Sarthana to Dream city	-	-	886	2238	3124
MRT2: Bheshan to Saroli	1985	775	-	-	2760
MRT2R: Saroli to Bheshan	796	1640	-	-	2436
Total	2781	2416	2208	2892	10296

2.12.1.3 Trip and link load for corridor finalized

Directional link loads including boarding and alighting are furnished for Corridor-I (Sarthana-Dream City) in the Table 2.41 & for Corridor-II (Bhesan-Saroli) in Table 2.42:-

Table 2.41: Corridor 1: Sarthana to Dream City (Peak hour)

Station Name	2021_Without TOD (Peak Hour)			2026_Without TOD (Peak Hour)			2026_With TOD (Peak Hour)			2036_Without TOD (Peak Hour)			2036_With TOD (Peak Hour)			2046_Without TOD (Peak Hour)			2046_With TOD (Peak Hour)		
	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load
Dream City	93	-	93	538	-	538	1476	-	1476	1746	-	1746	5113	-	5113	2384	-	2384	7555	-	7555
Convention Center	27	-	119	41	-	579	347	0	1823	126	-	1872	1446	-	6558	174	-	2559	2251	-	9806
Bhimrad	26	0	145	39	0	618	148	0	1971	114	2	1984	698	6	7250	155	4	2710	1037		10831
Woman ITI	24	0	169	37	0	655	141	0	2112	106	2	2087	667	6	7910	135	4	2841	1011	11	11832
VIP Road	1485	5	1649	2201	49	2807	2711	98	4725	3593	224	5456	4299	477	11732	4145	389	6598	4517	909	15439
Althan Gam	509	1	2158	751	5	3553	1291	11	6004	1408	20	6844	2400	43	14088	1683	29	8251	2566	59	17946
AlthanTenament	488	1	2645	903	10	4447	839	17	6826	1260	3	8101	1797	8	15877	1326	3	9574	2169	12	20102
Rupali Canal	302	2	2945	497	26	4918	723	29	7520	1021	159	8963	1345	149	17073	1533	204	10903	2993	136	22960
Majura Gate - 1	1442	104	4283	2774	177	7515	3402	435	10486	3680	478	12165	4516	1660	19929	4342	646	14599	4640	2218	25381
Kadarsha Ni Nal	383	7	4659	884	19	8380	959	36	11409	1406	47	13524	1416	132	21212	1690	51	16238	1656	192	26845
Chawk Bazar	1981	221	6420	4441	296	12525	4819	636	15591	7106	569	20062	8061	1640	27633	7397	1105	22530	7190	3549	30486
Maskati Hospital	343	213	6549	778	268	13035	934	426	16099	1058	264	20856	1160	649	28144	1968	251	24247	2246	667	32065
Surat Railway Station	686	1085	6150	1840	1534	13340	1884	1901	16082	1900	3193	19562	2027	5525	24646	2540	3529	23258	2239	6035	28269
Central Warehouse	182	649	5683	378	943	12775	464	1162	15383	632	1250	18945	804	1894	23556	920	1375	22803	1045	1916	27398
LabheshwarChowk	131	966	4847	277	1270	11782	322	1704	14001	396	1207	18134	479	1888	22147	413	1623	21594	459	2396	25461
Kapodra	108	2057	2898	213	3890	8105	271	4849	9423	436	5590	12980	547	7116	15578	1176	4621	18149	1301	6312	20451
ShriSwaminarayanMandirKalakunj	26	910	2013	63	3383	4785	78	3071	6430	117	3294	9804	152	3124	12606	466	3022	15593	492	2753	18190
VarchaChopati Garden	0	407	1606	0	975	3810	0	1367	5063	0	1356	8448	0	1720	10886	1	1449	14145	0	1826	16364
Nature Park	-	939	667	-	1937	1873	-	2692	2371	-	3941	4507	-	5654	5232	-	7225	6920	-	9281	7083
Sarthana	-	667	-	-	1873	-	-	2371	-	-	4507	-	-	5232	-	-	6920	-	-	7083	-
Total Boardings	8236			16655			20809			26105			36927			32448			45367		
Sarthana	561	-	561	1719	-	1719	2134	-	2134	4331	-	4331	4918	-	4918	6976	-	6976	6817	-	6817
Nature Park	681	0	1242	1736	0	3455	2372	0	4506	3547	0	7879	4621	-	9539	4168	0	11145	4651	-	11467
VarchaChopatiGarden	442	0	1684	1095	0	4550	1455	0	5961	1669	0	9548	2007	0	11546	2057	0	13201	2406	0	13873
ShriSwaminarayanMandirKalakunj	954	12	2626	3546	36	8060	3214	43	9133	3955	63	13440	3683	91	15138	5754	119	18837	5978	171	19680
Kapodra	1683	29	4280	3122	94	11089	3818	111	12840	4874	160	18153	6323	154	21308	5201	357	23681	7068	190	26558
LabheshwarChowk	1367	53	5594	1831	150	12770	2439	170	15108	1815	226	19743	2612	240	23680	1907	351	25236	2770	298	29030
Central Warehouse	685	62	6217	916	149	13536	1074	183	16000	1078	143	20677	1709	153	25237	1620	269	26587	2871	141	31760

Station Name	2021_Without TOD (Peak Hour)			2026_Without TOD (Peak Hour)			2026_With TOD (Peak Hour)			2036_Without TOD (Peak Hour)			2036_With TOD (Peak Hour)			2046_Without TOD (Peak Hour)			2046_With TOD (Peak Hour)		
	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load
Surat Railway Station	1439	663	6992	1888	1929	13495	2311	1886	16424	2411	3308	19781	3737	3101	25872	2983	4807	24763	5220	3438	33543
Maskati Hospital	59	442	6609	71	1056	12510	137	1057	15505	114	1168	18727	328	1094	25107	193	1038	23917	640	861	33323
Chawk Bazar	275	2023	4860	479	4118	8871	831	4423	11912	1281	6804	13204	2853	7212	20748	1696	8700	16913	4193	9332	28184
Kadarsha Ni Nal	20	287	4593	51	697	8225	70	731	11251	95	1165	12134	191	1136	19803	118	1354	15678	306	1401	27089
Majura Gate - 1	129	1621	3101	250	3061	5414	513	3716	8049	671	3428	9377	2040	4236	17608	944	5159	11462	2690	5691	24088
Rupali Canal	1	376	2726	6	712	4708	18	918	7149	93	879	8591	89	1100	16597	140	985	10616	175	1219	23043
AlthanTenament	1	450	2277	19	793	3935	23	860	6311	52	1150	7493	81	1395	15283	82	642	10056	123	986	22180
Althan Gam	1	543	1734	4	842	3097	8	1387	4932	17	1562	5948	30	2542	12770	25	1926	8155	28	3174	19034
VIP Road	5	1567	173	63	2375	785	85	2868	2148	332	3634	2646	312	5235	7847	540	4903	3792	518	7556	11995
Woman ITI	0	25	148	0	40	745	0	148	2001	2	110	2538	6	674	7180	4	151	3646	11	1069	10937
Bhimrad	0	63	85	0	90	656	0	218	1783	2	264	2277	6	957	6229	4	378	3272	-	1495	9442
Convention Center	-	27	58	-	41	615	-	349	1435	-	127	2150	-	1463	4766	-	177	3095	-	2285	7157
Dream City	-	58	-	-	615	-	-	1435	-	-	2150	-	-	4766	-	-	3095	-	-	7157	-
Total Boardings	8303			16796			20502			26339			35546			34412			46465		
Grand Total: Corridor 1 Boardings (Peak hour)	16539			33451			41311			52444			72473			66860			91832		

Table 2.42: Trip and Link Load for Bhesan – Saroli Corridor-2 (Peak hour)

Station Name	2021_Without TOD (Peak Hour)			2026_Without TOD (Peak Hour)			2026_With TOD (Peak Hour)			2036_Without TOD (Peak Hour)			2036_With TOD (Peak Hour)			2046_Without TOD (Peak Hour)			2046_With TOD (Peak Hour)		
	Pass. Boarding	Pass. Alighting	Link load	Pass Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load
Bheshan	57	-	57	185	-	185	334	-	334	907	-	907	2242	-	2242	1286	-	1286	3361	-	3361
Botanical Garden	187	-	245	444	0	629	686	0	1020	1742	-	2649	3471	0	5714	2201	0	3487	4468	0	7828
UgatVaarigruh	492	0	737	861	0	1490	1134	-	2154	1501	-	4150	2276	0	7990	2396	-	5883	3669	0	11497
Palanpur Road	326	0	1063	481	0	1970	649	0	2803	1109	0	5258	1642	0	9632	1122	0	7005	1732	0	13229
L.P.Savani School	540	3	1600	828	3	2795	1148	3	3948	1568	5	6821	2382	6	12008	1586	6	8585	2592	7	15814
Performing Art Centre	272	6	1866	438	6	3226	616	5	4558	1464	5	8281	2374	7	14374	2135	6	10714	3359	9	19165
Adajan Gam	292	7	2151	517	6	3737	773	8	5324	1045	17	9309	1499	28	15846	1279	23	11970	1922	42	21045
Aquerium	424	29	2545	657	48	4346	800	53	6071	1373	137	10544	1860	225	17481	1592	166	13396	2198	261	22981
Badri Narayan Temple	278	29	2794	661	62	4945	770	55	6786	1222	234	11532	1213	265	18430	1034	317	14112	1080	176	23884
AthwaChopati	334	18	3111	465	31	5379	546	40	7293	1143	102	12573	1687	173	19944	1468	132	15448	2121	241	25764
Majura Gate - 2	1048	409	3750	1308	1021	5666	1568	1394	7466	1257	1866	11963	2317	2870	19391	2074	2426	15096	3419	3334	25849
UdhnaDarwaja	575	562	3763	812	896	5581	927	1245	7148	1715	2243	11435	2317	3930	17778	1948	3015	14029	2888	5296	23441
KamelaDarwaja	570	415	3918	751	634	5698	747	763	7133	1473	979	11929	1612	1492	17897	1647	833	14842	1793	1535	23699
Anjana Farm	209	279	3848	240	523	5415	284	707	6710	440	793	11576	559	1246	17210	488	949	14381	644	1498	22846
Model Town	280	389	3739	275	657	5033	241	881	6071	218	1218	10576	281	1779	15712	655	1246	13790	1002	2051	21797
Magob	977	1384	3332	1302	2173	4161	1492	2395	5168	3931	4498	10009	5106	5127	15691	4333	4796	13327	6384	5727	22454
Bharat Cancer Hospital	-	67	3265	-	148	4013	-	229	4939	-	404	9605	-	905	14786	-	666	12661	-	1394	21060
Saroli	-	3265	-	-	4013	-	-	4939	-	-	9605	-	-	14786	-	-	12661	-	-	21060	-
Total Boardings	**6861**			**10225**			**12715**			**22108**			**32838**			**27244**			**42632**		
Saroli	3160	-	3160	3876	-	3876	4813	-	4813	11013	-	11013	16277	-	16277	14584	-	14584	23498	-	23498
Bharat Cancer Hospital	61	1	3221	138	2	4012	226	4	5036	428	11	11430	953	27	17203	591	0	15175	1462	-	24960

Station Name	2021_Without TOD (Peak Hour)			2026_Without TOD (Peak Hour)			2026_With TOD (Peak Hour)			2036_Without TOD (Peak Hour)			2036_With TOD (Peak Hour)			2046_Without TOD (Peak Hour)			2046_With TOD (Peak Hour)		
	Pass. Boarding	Pass. Alighting	Link load	Pass Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load	Pass. Boarding	Pass. Alighting	Link load
Magob	1318	846	3693	1993	1271	4734	2387	1448	5974	4351	4091	11690	5865	5193	17874	4178	5496	13857	5584	7554	22991
Model Town	209	412	3490	375	388	4721	532	375	6131	437	832	11295	658	1155	17377	599	1009	13447	879	1672	22197
Anjana Farm	300	202	3587	519	220	5019	681	260	6551	916	433	11778	1434	554	18258	970	505	13912	1521	673	23045
KamelaDarwaja	518	448	3657	780	563	5236	976	577	6951	1349	1130	11998	2299	1273	19283	1292	1185	14020	2550	1393	24203
UdhnaDarwaja	435	656	3436	659	862	5034	841	1000	6793	1889	1788	12098	3037	2418	19902	2388	2051	14356	4076	3113	25165
Majura Gate - 2	480	1020	2897	1184	1236	4982	1675	1530	6937	1736	2049	11785	2754	3518	19139	2535	2301	14590	3400	3983	24582
AthwaChopati	31	170	2757	62	277	4768	88	325	6700	231	734	11283	364	965	18538	300	909	13980	440	1324	23698
Badri Narayan Temple	13	335	2435	59	555	4272	46	738	6009	297	923	10657	4	1929	16612	447	979	13448	322	1495	22525
Aquerium	28	344	2119	57	520	3808	58	653	5415	198	969	9886	495	1352	15756	279	1112	12615	408	1702	21231
Adajan Gam	9	271	1856	8	477	3340	14	718	4711	26	778	9132	52	1091	14717	32	936	11711	22	1399	19854
Performing Art Centre	1	235	1622	1	410	2931	2	588	4124	3	1125	8010	6	1568	13156	4	1401	10314	9	2003	17860
L.P.Savani School	2	487	1136	2	743	2189	2	1078	3048	5	1292	6724	7	1957	11206	6	1495	8825	8	2494	15374
Palanpur Road	0	423	713	0	640	1549	0	845	2204	0	1292	5432	0	1839	9368	0	1407	7418	0	2078	13296
UgatVaarigruh	0	436	277	0	870	679	0	1142	1062	-	2728	2704	-	3961	5407	-	3556	3863	-	5345	7951
Botanical Garden	-	190	87	-	418	260	-	603	458	-	1406	1298	-	2519	2888	-	1887	1976	-	3491	4460
Bheshan	-	87	-	-	260	-	-	458	-	-	1298	-	-	2888	-	-	1976	-	-	4460	-
Total Boardings	**6565**			**9713**			**12341**			**22879**			**34205**			**28205**			**44179**		
Grand Total: Corridor 2Boardings (Peak hour)	**13426**			**19938**			**25056**			**44987**			**67043**			**55449**			**86811**		

2.12.1.4 Trip length distribution for both the final corridors is given as under in Table: 2.43 as hereunder:-

Table 2.43: Trip Length Frequency Distribution – Metro Passengers

Distance Range (km)	2021 Without TOD			2026 Without TOD			2026 With TOD			2036 Without TOD			2036 With TOD			2046 Without TOD			2046 With TOD		
	C1	C2	C1+C2	C1	C2	C1+C2	C1	C2	C1+C2	C1	C2	C1+C2	C1	C2	C1+C2	C1	C2	C1+C2	C1	C2	C1+C2
0-2	1.6%	0.0%	0.9%	1.9%	0.0%	1.2%	1.4%	0.0%	0.9%	1.5%	0.0%	0.8%	1.1%	0.1%	0.6%	2.5%	0.0%	1.4%	1.1%	0.0%	0.6%
2-4	11.6%	30.7%	18.8%	10.3%	29.9%	15.9%	8.7%	26.0%	13.5%	11.0%	29.0%	18.4%	9.6%	24.8%	15.8%	11.0%	28.7%	18.1%	10.2%	24.0%	16.0%
4-6	18.2%	14.7%	14.4%	18.0%	15.4%	14.9%	15.3%	14.6%	12.9%	12.6%	11.3%	11.1%	10.4%	10.1%	9.3%	13.4%	10.3%	11.0%	10.5%	9.2%	9.0%
6-9	31.0%	29.2%	26.5%	31.1%	26.2%	26.5%	30.5%	27.2%	26.2%	29.5%	25.4%	25.4%	27.4%	26.0%	23.7%	27.3%	25.7%	24.0%	25.2%	25.4%	22.2%
9-12	24.9%	18.7%	23.2%	23.2%	19.6%	22.5%	25.1%	20.6%	24.3%	25.9%	20.5%	23.6%	26.8%	22.0%	24.9%	25.4%	19.7%	22.8%	26.4%	21.9%	24.4%
12-15	10.1%	5.7%	11.8%	11.9%	7.0%	13.8%	13.7%	9.3%	15.6%	12.9%	9.7%	12.8%	15.2%	12.1%	15.4%	12.6%	10.4%	12.8%	15.4%	13.0%	15.6%
15-18	2.4%	1.1%	3.6%	3.4%	1.9%	4.4%	4.7%	2.3%	5.4%	5.9%	3.9%	6.6%	7.5%	5.0%	8.3%	6.9%	5.1%	8.2%	8.7%	6.5%	9.9%
>18	0.1%	0.0%	0.8%	0.2%	0.0%	0.8%	0.7%	0.0%	1.1%	0.8%	0.0%	1.3%	2.0%	0.0%	2.0%	0.9%	0.0%	1.6%	2.4%	0.0%	2.2%
Total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**
C1: Corridor 1 – Sarthana to Dream City																					
C2: Corridor 2 – Bheshan to Saroli																					
C1+C2: Corridor 1 + Corridor 2																					

2.12.1.5 Summary of hourly & day's boarding for both the corridors(Sarthana-Dream City &Bhesan- Saroli) is given in Table 2.44 and 2.45 as hereunder:-

Table 2.44: Peak Hour Ridership Summary

Corridor	Direction	2021 - Without TOD	2026 - Without TOD	2026 - With TOD	2036 - Without TOD	2036 - With TOD	2046 - Without TOD	2046 - With TOD
Corridor 1	Sarthana-Dream City	8236	16655	20808	26107	36924	32450	45367
Corridor 1R	Sarthana-Dream City	8303	16796	20502	26341	35548	34411	46465
Corridor 2	Bheshan-Saroli	6861	10225	12715	22106	32838	27244	42631
Corridor 2R	Bheshan-Saroli	6565	9713	12342	22877	34206	28204	44180
Total Ridership		29965	53389	66367	97431	139516	122309	178643

Table 2.45: Full Day ridership summary

Corridor	Direction	2021 - Without TOD	2026 - Without TOD	2026 - With TOD	2036 - Without TOD	2036 - With TOD	2046 - Without TOD	2046 - With TOD
Corridor 1	Sarthana-Dream City	82360	166550	208080	261070	369240	324500	453670
Corridor 1R	Sarthana-Dream City	83030	167960	205020	263410	355480	344110	464650
Corridor 2	Bheshan-Saroli	68610	102250	127150	221060	328380	272440	426310
Corridor 2R	Bheshan-Saroli	65650	97130	123420	228770	342060	282040	441800
Total Ridership		299650	533890	663670	974310	1395160	1223090	1786430

CHAPTER – 3

SYSTEM SELECTION

3.0 INTRODUCTION

Metro Rail Policy (2017) and the Appraisal Guidelines for Metro Rail Project Proposals (September, 2017) issued by Government of India Ministry of Housing and Urban Affairs (MOHUA) mandates compulsory incorporation of Alternatives Analysis in the Detailed Project Report of Metro Rail for submitting the same to the central government while seeking their approval.

More importantly, DPR for Surat Metro has been prepared to be in line with the study "Comprehensive Transport Forecast Study done by Centre of Excellence in Urban Transport (CEPT), Ahmedabad, who were commissioned by Surat Municipal Corporation (SMC) in the year 2016 vide their letter No. CE.Spl.Cell/Out/339 dated 31-08-2016 for undertaking demand assessment for alternative metro corridor proposals whose final report was submitted to State Government in April, 2017, in terms of content was analogous to a Comprehensive Mobility Plan for Surat City and the study had proposed corridors for different modes of urban transport based on a comprehensive Alternative Analysis In this connection it is worthwhile to upfront examine, as to how the content of the study match up to the content requirement of Alternative Analysis as mandated by the new Appraisal Framework.

The following section provides a clear picture of whether CEPT University, Ahmedabad analysis has done adequate Alternative Analysis or not before factoring Surat Metro Rail Project in its report.

3.1 COMPARISON OF ALTERNATIVE ANALYSIS FRAME WORK IN CEPT's STUDY AND AS REQUIRED FOR IN THE NEW APPRAISAL GUIDELINES OF GOI

The Metro Rail Policy (2017) outlines five broad categories of Mass Rapid Transit Systems – Busways and Bus Rapid Transit Systems, Light Rail Transit (LRT),

Metro Rail and Regional Rail. The comprehensive study done by CEPT University, Ahmedabad has after due analysis suggested as given hereunder:-

3.1.1 Bus Rapid Transit System (BRTS)

Currently city is operating a total of 102 km BRT network in addition to proposed 10 km of high mobility corridor. Based on the alternative analysis, additional 94 km of potential BRT corridor has been identified.

BRT system with the dedicated corridor is proposed as a complementary service to the Metro system. This is to avoid competition among the two systems. From railway station to Sarthana corridor, the passenger demand is very high. Therefore to avoid overcrowding on the system and to provide easy commute, along with existing BRT network, metro corridor has also been proposed to accommodate the future demand.

3.1.2 Metro Corridors

The following three corridors are proposed as potential metro corridors:

1. Dream city – Kamrej (28.9 km)
2. Bhesan – Umbhel (26.3 km)
3. Majuragate to Karamala (15.8 km)

A total of 71 km of metro network has been proposed. Tentative distance between two stations would be around 1km.

3.1.3 City Bus Services

City bus services are the primary mode of travel and also work as a feeder service to the rapid transit system. In future, city bus would have to be planned around 900km network covering all the major roads which are above and equal to 18m RoW. The concept of integrated services is proposed to be adopted by allowing the city bus services to use the BRT corridor for short sections in order to provide seamless transfer opportunities for passengers.

As per the future projections of passenger ridership of 36 lakhs/day on bus in the year 2046, a total fleet of around 5000 buses would be required (assuming 700 passengers /bus/day).

Presently, route rationalization activities have been undertaken for the city bus services. However, as the city grows, the city structure also undergoes transformation and needs to be updated regularly. It is proposed to undertake a route rationalization and operations planning exercise every five years, so that the public transport system responds to the travel demand pattern of the city residents.

3.2 Model Output for Public Transport assignment

Based on the observed traffic flows on the rapid transit corridors, following have been screened as possible rapid transit corridors. The screening has been done on the basis of flows more than 4000 passengers in the peak hour per direction.

Table 3.1: Proposed rapid transit corridors (2046)

S.N	Corridor Name	Via	Km
1.	Majuragate to Karamala	Ved	15.80
2.	Dream City – Kamrej/Vav	Sarthana – Exhibition Centre, Bhatar Char rasta and Sarsana, Majura Gate, Rly Station and Sarthana	29
3.	Bhesan – Umbhel	Adajan, Majura gate, Kamela Darwaja and Saroli	26.30
4.	Adajan to Hazira	Pal	14
5.	Sherdi to Unn (half ring)	Kapodara, Saroli, Devadh, Saniya Kande	30
6.	Jahangirpura to Olpad	Talad	11
7.	Kosad to Sayan	Gothan	9
8.	Punagam to Valthan	Kosmada	11
9.	Unn to Khajod	-	6.3
10.	Dindoli to Bhatiya	-	10.2
11.	Kamrej to Vav	2.3	
	Total length of Rapid Transit Corridor (Tentative)		**165.00**

It can be seen that around 165 kms of rapid transit corridors are proposed for the future year 2046. Further based on the ridership and passenger flow, following three corridors may be considered as metro, while the other eight corridors for BRTS system.

Majuragate to Karamala
Dream city - Kamrej
Bhesan - Umbhel

Thus, in the year 2046, the CMP scenario proposes an integrated public transport system consist of around 71 km of metro; 196 km of BRT and 889 km of city bus network.

3.2.1 Proposed Multimodal Affordable Transport Authority (MATA)

Recently Gujarat Government has announced the setting up of a Multimodal Affordable Transport Authority (MATA) to develop a holistic and integrated multi-modal systems approach so as to move towards an energy efficient and sustainable future. This authority will work for integrated transport management of sky, land & water routes. This authority will be setup at the state level and will ensure the integration of all modes of transport at the state level.

3.2.2 The alternative Analysis undertaken in CEPT's study has to be juxtaposed with the Alternative Analysis done in DPR of Surat Metro and the same has to be further seen in the context of PHPDT along the selected corridors which exclude other modes of public transport along the corridors.

In this connection it is again clarified upfront that the Mobility Planning of Surat City is the responsibility of Government of Gujarat who has assigned this job of comprehensive mobility study to CEPT University, Ahmedabad.

The comparison of Alternative Analysis proposed in CEPT's study/DPR with the requirement of the Appraisal of Guidelines is provided as under:

Table 3.2: Comparison of Alternative Analysis

S. No.	Provisions of Appraisal Guidelines Framework	Covered in CEPT's Report/DPR	Chapters Paragraphs Covered
1	Objective of Alternative Analysis .All transportation objectives covered All project impacts evaluated Stakeholders opinion considered Select locally preferred objective	Yes	Chapter one of CEPT's report itself succinctly puts all these objectives upfront.
2	Stage I Development, Qualitative and Quantitative evaluation of screening criteria- Mobility Effects Conceptual Engineering Effects Financial and social effects Environmental and Social Effects Cost Effectiveness and Affordability	Yes	Described in various chapters below
3	Evaluation Parameter for Mobility Effect-Travel Demand Forecasting	Yes	Chapter 3 of CEPT's report deals with existing Transport System, Vehicular Growth, Intermediate Public Transport (IPT) modes, Non-Motorised Transport (NMT) Benchmarking Urban Transport in Surat etc. Chapter-7 of CEPT's report deals with Growth Dynamics in Surat Employment Projections, Travel demand model, Development etc. Chapter 10 of CEPT's report deals with priority of projects,

			identification of funding agency, multi-model affordable transport authority etc.
4	Evaluation Criteria Conceptual Engineering-parameters-geotechnical, civil structures, station planning (bus/rail), utilities, rights of ways, other parameters parking, intermodal connectivity etc	Yes	Chapter-9 of CEPT's report deals with mobility improvement measures, road network development plan, CMP network development, ROWs./RUB/ROB, road alignment along DFC Corridor, Urban Freight proposals, TOD, NMT etc.
5	Evaluation Parameter- Social and Environmental Assessment	Yes	The Environment and Social impact (EIA & SIA) analysis has been got done for Surat Metro Rail Project and included in DPR.
6.	Evaluation Parameter- Cost Effectiveness and Affordability	Yes	Chapter 9 of CEPT's report deals with costing of roads, flyovers, ROB/RUB etc. Chapter 19 of DPR deals with working out cost estimates of metro corridors, cost reduction measures, standardization, indigenization and inducing competition etc.
7.	Evaluation Parameter- Financial and Economic	Yes	Financial Internal Rate of Return (FIRR) and Economic Internal Rate of Return (EIRR) have been got worked out and included in DPR.
8.	Stage III: Alternative Evaluation Identification of alternatives that are most likely to meet Purpose and need identified for the Project	Yes	CEPT's study indicates/recommends the implementation of Integrated Public Transport System by the year 2046, as hereunder:-

	Avoid or minimize environmental and community impacts Allow for effective and feasible project phasing and constructability Provide a cost effective transportation investment The evaluation of alternatives should include a no build alternative (without project)		Metro : 71 Kms BRTS : 196 Kms City Bus Service: 889 Kms

3.3 JUSTIFICATION OF SURAT METRO BASED ON ALTERNATIVE ANALYSIS

It is instructive to note that Metro Rail Projects in Surat City has been proposed /undertaken after conducting in-depth planning and dithering. In this connection it is to be noted that CEPT's study had done for both scenarios i.e. Gradual Growth Scenario and Rapid Growth Scenario. CEPT's analysis had substantively focused on expansion of BRTS network and City Bus network in Surat, it has recommended the integrated public transport system in the year 2046, consisting of 71 Km of metro, 196 Km of BRTS and 889 Km of city bus network.

Further, the CEPT's study report recommends the following corridors as potential metro network:-

1) Dream City – Kamraj (28.90 kms)
2) Bhesan – Umbhel (26.30 kms)
3) Majura Gate – Karamala (15.80 kms)

Based on the demand analysis, it can be concluded that all of three corridors may be considered for metro. However, in the 1st phase following two metro corridors/corridor-sections are recommended:-

Dream City – Sarthana
Bhesan – Saroli

For Phase-II, Majura Gate - Ved can be considered. The extension to Karamala based on road network developments and demand build can be considered after

demand assessment. Similarly, extensions from Sarthana – Kamrej/Vav and Saroli-Umbhel can be taken up in subsequent phases.

3.4 OPTIONS FOR PUBLIC TRANSPORT SYSTEM

Over and above the above mentioned Alternative Analysis that emerges from the CEPT's study and which broadly follows the requirement of the New Appraisal Guidelines, in this chapter of the DPR of Surat Metro, following detailed Alternative Analysis has been done for which the following Mass Transit Modes have been considered :

i) Heavy Metro System with a capacity of 90,000 passengers per hour per direction of traffic (PHPDT). The capital city Delhi has gone for a Heavy Metro System.

ii) Medium Capacity Metro with PHPDT of 30,000 to 45,000. Cities like Bangalore, Chennai, Kolkata, etc. have gone for Medium Capacity Metro.

iii) Light Metro with PHPDT of upto 30,000. Trivandrum and Kozhikode cities are planned for Light Metro Systems.

iv) Light Rail Transit Systems: Modern trams-Street Cars running on Rails at grade or elevated with sharp curves of 24m radius. These are extremely popular and operating in large number of European countries. Generally the stations are spaced at 500m to 1 km and have high acceleration and deceleration characteristics. In most of the countries, they are operating at-grade with prioritized signaling at road inter-section.

v) Monorail System:

Monorail trains operate on grade separated dedicated corridors with sharp curves of up to 50m radius. This is a rubber tyred based rolling stock, electrically propelled on concrete beams known as guide-ways. The system is extremely suitable in narrow corridors as it requires minimum right of way on existing roads and permits light and air and is more environmental friendly. This is prevalent in several countries for traffic densities of over 20,000 PHPDT.

vi) Other Rail Based Systems: A number of options are available but have not been introduced in India. Some of these are very briefly mentioned below:

a) Maglev

This is an advanced Rail based transit system in which Magnetic Levitation is used to raise the vehicles above the rail surface. Rail wheel interaction is thus avoided and very high speeds are attainable. Maglev Levitation can either be due to attractive force or due to repulsive forces.

b) Linear Induction Motor (LIM) Train

This is also an advanced Rail based transit system in which propulsion is through a Linear Induction Motor whose stator is spread along the track. The rotor is a magnetic material provided in the under frame of train. In the technology the tractive force is not transmitted through rail-wheel interaction, and so there is no limitation on account of adhesion. This technology is most appropriate for turnouts, as the height of the tunnel can be reduced to lower height of cars.

c) Bus Rapid Transit System(BRT)

This system involves operation of buses on a dedicated corridor (except of traffic integration) at a high frequency to achieve PHPDT. For providing a very high transport capacity say 20,000 PHPDT, about 200 buses shall be required per hour *i.e.,* at headway of 20 seconds. Such a high PHPDT can be achieved by providing two lanes of traffic in each direction and elimination of traffic intersection on the route.

d) Automated Guide way Transit System

The term is used for systems other than conventional rail based system on grade separated guide ways. The system can be rail based or rubber tire based but fully automated guided systems with driver less operation.

3.5 Salient features of various Mass Transit Systems

The salient features of the various Transit Systems are summarized as under:-

Table 3.3: Various Transit Systems

Parameter	Heavy Metro	Medium Metro	Light Metro	LRT	Electric Bus
Axle Load	17 T	16T	11T to 12 T	10 T	10 T
PHPDT range	More than 45000	More than 25000 and upto 45000	Upto 25000	Upto 10000	Upto 8000
Number of cars	6 cars or more	4 to 6 cars	3 to 4 cars	3 to 9 units/cars train	Electric Bus
Car dimensions	2.9 m/3.2 m wide 22 m long	2.9m/3.2m wide 22 m long	2.7m/2.9m wide 18 m long	2.5 to 2.65 Length 7-8 M and unit length 21m, 33 m,43m	Length 24m Width 2.5M
Car capacity	300 persons per car	300 persons per car	200 persons per car	160 per Tram	150 per bus
Viaduct particulars	Box Girder/ U Girder/ double u Girder	Box Girder/ U Girder/ double U Girder	Box Girder/ U Girder/ double U Girder	Box Girder/ U Girder	Box Girder/ U Girder
Length of platform required	185m to 210m	140m	90m	50m	50m
CAPEX/km	Rs 250 Crore/Km	225 crores/Km	175 crore/km	120 crore/km	80 crore/km
O&M/km	Depends on No. of Passengers carried/day and number of train trips				
Deck width of viaduct	9 m to 10 m	8.5m to 9.0 m	7.0 to 8.5 m	7.0 to 8.5 m	7.0 to 8.5 m
Distance between pillars	25m to 37m spans	25m to 37m spans	25m to 37m spans	25m to 37m spans	25m to 37m spans
Speed (Max)	85 KMPH	85 KMPH	85 KMPH	50 Kmph	50 Kmph

3.6 CAPACITY OF VARIOUS MODES (as per the recommendations of Working Group on Urban Transport for 12th Five Year Plan)

In their report on **Urban Transport for 12th Five Year Plan**, the Working Group has set the guidelines for the choice of different modes is as follows:

Table 3.5: Guidelines for Choice of Different Modes

SYSTEM	PHPDT IN 2021	POPULATION IN 2011	AVG. TRIP LENGTH
Metro Rail #	>=15000 for at least 5km continuous length	More than 20 Lakhs	More than 7 Km
LRT primarily at grade	=<10,000	More than 10 Lakhs	More than 7 Km
Monorail @@	=<10,000	More than 20 Lakhs	About 5-6 Km
Bus Rapid Transit System	>=4,000 and upto 20000	More than 10 Lakhs	>5 Km
Organised City Bus Service as per urban bus specifications		>1 lac, >50,000 in case of hilly towns	>2 to 3 Km

 # for having Metro Rail, the city should have a ridership of at least 1 million on organized public transport (any mode)
@@ Monorail is desirable only as a feeder system or where the narrow roads are flanked on either side by high rise buildings. In monorail while the cost of construction, operation and maintenance is almost the same as elevated metro rail, the carrying capacity is much lesser.

3.7 SELECTION OF MODE

Selection of a particular mode for any pre-determined traffic corridor depends mainly on demand level of a corridor Right of Way (ROW) on the road and the capacity of the mode. The demand forecast is estimated considering the traffic growth for about 30 years. Other considerations in mode choice are location of building lines, possibility of increasing ROW. Cost of some mode may vary depending upon the location in view of engineering constraints. Therefore final choice of mode to be adopted for a particular corridor is based on techno economic considerations. As regards the location of a particular mode like at-grade, elevated and underground, depends upon the ROW. If ROW is 20 m or more, elevated alignment is preferred over underground as the cost of underground alignment is 2- 2½ times of elevated alignment.

3.8 DAILY RIDERSHIP ON CORRIDOR

Two corridors have been identified in Surat Metro Rail Project (Phase-I) as per details given underneath:

Corridor-I : Sarthana to Dream City
Corridor-II : Bhesan to Saroli

TRAFFIC DATA
(As forwarded by Surat Municipal Corporation/CEPT)

a) Full Day Metro Boarding

Table 3.6: Full Day Metro Boarding

Scenario No.	Scenarios	Metro Boardings (Full Day)
01	2021 Without TOD	2,99,650
02	2026 Without TOD	533,890
03	2036 Without TOD	9,74,310
04	2046 Without TOD	12,23,090

b) Full Day Boarding (Different Modes)

Table 3.7: Full Day Boarding

Scenarios	Mode	Boardings	Transfer rate	%Boardings
2021 Without TOD	Bus	5,16,360	1.71	41%
	BRT	4,57,300		36%
	Metro	2,99,650		24%
	Total	12,73,310		100%
2026 Without TOD	Bus	8,53,980	1.64	39%
	BRT	7,93,170		36%
	Metro	5,33,890		24%
	Total	21,81,040		100%
2036 Without TOD	Bus	21,29,440	1.74	43%
	BRT	17,91,850		37%
	Metro	9,74,310		20%
	Total	**48,95,600**		**100%**
2046 Without TOD	Bus	27,44,630	1.78	44%
	BRT	22,26,070		36%
	Metro	12,23,090		20%
	Total	**61,93,790**		**100%**

c) Maximum Peak Hour Peak Direction Traffic (PHPDT)

Table 3.8: Peak Hour Peak Direction Traffic
(For the year 2021,2026,2036 & 2046)

Sr.No.	Corridor	2021 Without TOD (Peak Hour)	2026 Without TOD (Peak Hour)	2036 Without TOD (Peak Hour)	2046 Without TOD (Peak Hour)
	Corridor-I				
1.	Corridor 1b: Dream City-Sarthana (Up Direction)	6,549	13340	20,856	24,247
2.	Corridor 1bR: Sarthana-Dream City (Down Direction)	6,992	13,536	20,677	26,587
	Corridor-II				
3.	Corridor 2d: Bhesan-Saroli (Up Direction)	3,918	5698	12,573	15,448
4.	Corridor 2dR: Saroli-Bhesan (Down Direction)	3,693	5236	12,098	15,175

3.9 MODE SELECTION FOR SURAT METRO CORRIDORS

Road-based systems can optimally carry up to a maximum of 8,000 PHPDT. With an aim of reduction in road traffic and with the PHPDT of more than 8000 assumed on the above corridors, there can be two options namely 1) Mono Rail and 2) Light Capacity Metro. Light Metro system caters to a PHPDT upto 30,000 and for PHPDT in between 30,000 to 45,000 medium capacity metro will cater adequately. Therefore, in view of PHPDT figures and other traffic data explained in Para 3.4 above, it is hereby recommended to adopt Medium capacity metro system, keeping in view the system being provided to be suitable for future 75 years

Moreover, if metro system is in place in Surat City, Gujarat, the daily and peak hourly traffic on various sections of road is expected to be reduced considerably. It may be seen that with the proposed metro corridors in Surat City, Gujarat, the road traffic will be reduced not only on the road along the corridor but also in the surrounding road network in its influence area.

Hence, keeping in view the above points, it is recommended to adopt an stable, tested and reliable metro technology i.e. **Medium Capacity Metro System** in Surat City, Gujarat.

CHAPTER - 4

GEOMETRIC DESIGNING PARAMETERS AND ALIGNMENT DESCRIPTION

4.1 GENERAL

This chapter deals with geometrical standards adopted for horizontal and vertical alignments, route description, etc. The proposed corridors of Surat Metro Rail Project (Phase-I) will consist of Standard Gauge (SG) lines, For underground corridors, track centers are governed by the dimension of island platform and for elevated (side platform), it is governed by sharpest curvature on the alignment.

The geometrical design norms are based on international practices adopted for similar metro systems with standard gauge on the assumption that the maximum permissible speed on the section is limited to 80kmph. Planning for any higher speed is not desirable as the average inter-station distance is about 1.00km and trains will not be able to achieve higher speed.

The elevated tracks will be carried on box-shaped/U-shaped/Double U-shaped elevated decking supported by single circular piers, generally spaced at 25-m centres and located on the median of the road to extent possible. The horizontal alignment and vertical alignment are, therefore, dictated to a large extent by the geometry of the road and ground levels followed by the alignment.

The underground tracks will be carried in separate tunnels to be drilled by Tunnel Boring Machine. Stations will, however, be constructed by cut and cover method or NATM technique. Two stations are planned to be stacked stations.

4.2 GEOMETRIC DESIGN PARAMETERS

The design parameters related to the Metro system described herewith have been worked out based on a detailed evaluation, experience and internationally accepted practices. Various alternatives were considered for most of these parameters but the best-suited ones have been adopted for the system as a whole.

4.2.1 Horizontal Alignment

As far as possible, the alignment follows the existing roads. This leads to introduction of horizontal curves. On consideration of desirable maximum cant of 110 mm and cant deficiency of 85 mm on Metro tracks, the safe speed on curves of radii of 300 m or more is 80 km/h. On elevated sections minimum radius of 120m has been used at one location having speed potential upto 40 km/h. However in underground section desirable minimum radius of curve shall be 300 m for ease of working of Tunnel Boring Machine (TBM). For maximum permissible speed on curve with various radii, Table 4.2may be referred.

4.2.2 Horizontal Curves

Table 4.1

Description	Underground Section	Elevated Section
Desirable Minimum radius	300 m	200 m
Absolute minimum radius	200 m (only c/c)	120 m
Minimum curve radius at stations	1000 m	1000 m
Maximum permissible cant (Ca)	125 mm	125 mm
Maximum desirable cant	110 mm	110 mm
Maximum cant deficiency (Cd)	85 mm	85 mm

4.2.3 Transition Curves

It is necessary to provide transition curves at both ends of the circular curves for smooth riding on the curves and to counter act centrifugal force. Due to change in gradients at various locations in the corridor, it is necessary to provide frequent vertical curves along the alignment. In case of ballast less track, it is desirable that the vertical curves and transition curves of horizontal curves do not overlap. These constraints may lead to reduced lengths of

Transition curves at certain locations. The transition curves have certain minimum parameters:

Length of Transitions of Horizontal curves (m)

Minimum : 0.44 times actual cant or cant deficiency (in mm), whichever is higher.

Desirable : 0.72 times actual cant or cant deficiency, (in mm), whichever is higher.

Overlap between transition curves and vertical curves not allowed.

Minimum straight between two Transition curves (in case of reverse curves): either 25 m or Nil.

Minimum straight between two Transition curves (in case of same flexure curves): either 25 m or both curves should be converted in to the compound curve by introducing single transition between the two circulars.

Minimum curve length between two transition curves: 25 m

4.2.4 Vertical Alignment and Track Centre

(a) Elevated Sections

The viaducts carrying the tracks will have a vertical clearance of minimum 5.5 m above road level. For meeting this requirement with the 'Box' shaped pre-stressed concrete girders, the rail level will be about 9.8 m above the road level, with U-girder rail level may come to about 9.5 m above ground level However, at stations which are located above central median, the rail level will be 13.5 m above the road level with concourse at mezzanine. These levels will, however, vary marginally depending upon where the stations are located.

(b) Underground sections

Rail level at midsection in tunneling portion shall be kept at least 12.0 m below the ground level. At stations, the desirable depth of rail below ground level is 13.5 m, so that station concourse can be located above the platforms. However, this rail level may be further lowered if alignment passing under built-up areas.

Track center in underground sections are follows:

Track center in underground sections are follows: Sections where stations are to be constructed by cut & cover and running section by TBM to Accommodate 13 m wide platform	16.05 m (for lesser width of platform, track center to be reduced)
Sections where stations are to be constructed by NATM and running section by TBM to facilitate Construction of stations	22.00 m

(c) Gradients

Normally the stations shall be on level stretch. In exceptional cases, station may be on a grade of 0.1 %. Between stations, generally the grades may not be steeper than 2.5 %. However, where existing road gradients are steeper than 2% or for Switch Over Ramps gradient up to 4% (compensated) can be provided in short stretches on the main line.

(d) Vertical Curves

Vertical curves are to be provided when change in gradient exceeds 0.4%. However, it is recommended to provide vertical curves at every change of gradient.

(e) Radius of vertical curves:

On main line		
(i) Desirable	:	2500 m
(ii) Absolute minimum	:	1500 m
Other Locations	:	1500 m
Minimum length of vertical curve	:	20 m

4.2.5 Design Speed

The maximum sectional speed will be 80 km/h. However, the applied cant, and length of transition will be decided in relation to normal speeds at various locations, as determined by simulation studies of alignment, vertical profile and station locations. Computerized train simulation studies need to be conducted with proposed gradients at the time of detailed design stage. This is with the objective of keeping down the wear on rails on curves to the minimum.

Table 4.2 -Cant, Permitted Speed & Minimum Transition Length for Curves

RADIUS	CANT	MAXIMUM PERMISSIBLE SPEED	MINIMUM DISTANCE BETWEEN ADJACENT TRACKS	
			UNDERGROUND	ELEVATED AND AT-GRADE
meters	mm	kmph	mm	Mm
3000	15	80	3500	3650
2800	15	80	3500	3650
2400	20	80	3500	3650
2000	20	80	3500	3650
1600	25	80	3500	3650

RADIUS	CANT	MAXIMUM PERMISSIBLE SPEED	MINIMUM DISTANCE BETWEEN ADJACENT TRACKS	
			UNDERGROUND	ELEVATED AND AT-GRADE
meters	mm	kmph	mm	Mm
1500	30	80	3500	3650
1200	35	80	3500	3650
1000	45	80	3500	3700
800	55	80	3550	3700
600	70	80	3550	3750
500	85	80	3600	3750
450	95	80	3600	3800
400	105	80	3650	3800
350	110	75	3650	3800
300	110	70	3700	3850
200	110	55	3800	3950
150*	110	45	4000	4050
150*	0	30	4000	4050
120*	110	40	4000	4150
120*	0	25	4000	4150

*The curves of 120 and 150 meters radii are used in depot and depot connections.

Notes: (a) The track spacing is without any column/structure between two tracks and is with equal cant for both outer and inner tracks.

(b) Track spacing shown is not applicable to stations which should be calculated depending on specific requirement.

(c) Figures for any intermediate radius of curvature may be obtained by interpolating between two adjacent radii. For higher radii, values may be extrapolated.

4.2.6 Station Locations

Stations have been located so as to serve major passenger destinations and to enable convenient integration with other modes of transport. However effort has also been made to propose station locations, such that inter station distances are as uniform as possible. The average spacing of stations is close to 1.00 km.

4.3 TRACK STRUCTURE

Track on Metro Systems is subjected to intensive usage with very little time for day-to-day maintenance. Thus it is imperative that the track structure selected for Metro Systems should be long lasting and should require minimum or no maintenance and at the same time, ensure highest level of

safety, reliability and comfort, with minimum noise and vibrations. The track structure has been proposed keeping the above philosophy in view. Two types of track structures are proposed for the corridors under Surat Metro Rail Project (Phase-I). The normal ballasted track in Depot (except inside the Workshops, inspection lines and washing plant lines). The ballast less track is recommended on Viaducts and inside tunnels as the regular cleaning and replacement of ballast at such locations will not be possible.

For the depots, ballasted track is recommended as ballast less track on formation is not suitable due to settlement of formations. Ballast less track in depot is required inside the workshop, on inspection lines and washing plant lines.

From considerations of maintainability, riding comfort and also to contain vibrations and noise levels, the complete track is proposed to be joint-less and for this purpose even the turnouts will have to be incorporated in LWR/CWR.

The track will be laid with 1 in 20 canted rails and the wheel profile of Rolling Stock should be compatible with the rail cant and rail profile.

4.4 RAIL SECTION

Keeping in view the proposed axle load and the practices followed abroad, it is proposed to adopt UIC-60 (60 kg. /m) rail section. Since on main lines, sharp curves and steep gradients would be present, the grade of rail on main lines should be 1080 Head Hardened as per IRS-T-12-2009. As these rails are not manufactured in India at present, these are to be imported. For the Depot lines, the rails of grade 880 are recommended, which are available indigenously.

4.5 BALLASTLESS TRACK ON MAIN LINES

On the viaducts, it is proposed to adopt plinth type ballastless track structure with RCC derailment guards integrated with the plinths. Further, it is proposed to adopt fastening system complying to performance criteria laid down by Indian Railways on ballastless track structures, with a base-plate spacing of 60 cm. on viaducts. In the underground sections, similar track structure with a base plate spacing of 70 cm is proposed on slab after 1st stage concrete.

4.6 BALLASTLESS/BALLASTED TRACK IN DEPOT

The ballast less track in Depot may be of the following types:

Supported on steel pedestal for inspection lines.

Embedded rail type inside the Workshop.

Plinth type for Washing line.

Track is to be laid on PRC sleepers with sleeper spacing of 65 cm. All the rails are to be converted into rail panels by doing flash butt/ Thermit welding.

4.7 TURNOUTS

All turn-outs/crossovers on the main lines and other running lines shall be as under:

Table 4.3– Turn-Outs

S. No.	Description	Turn out Type
01	Main Line	1 in 9
02	Depot/Yard Lines	1 in 7

4.8 BUFFER STOPS

On main lines and Depot lines, friction buffer stops with mechanical impact absorption (non-hydraulic type) will be provided. In elevated portion, the spans on which friction buffer stops are to be installed will be designed for an additional longitudinal force, which is likely to be transmitted in case of Rolling Stock hits, the friction Buffer Stops.

4.9 RAIL STRUCTURE INTERACTION

For continuing LWR/CWR on Viaducts, the elevated structures will be adequately designed for the additional longitudinal forces likely to be transmitted as a result of Rail-Structure interaction. Rail structure interaction study will determine the need and locations of Rail Expansion Joints (REJ) required to be provided.

4.10 ROUTE ALIGNMENT

02 (Two) - Corridors identified for implementation in Phase-I of Surat Metro Rail Project network are as follows:-

Table 4.4

S. No.	Corridors	Total (km)
i)	Sarthana to Dream City	21.61 Kms.
ii)	Bhesan to Saroli	18.74 Kms.
	Total	**40.35 Kms.**

Both the above mentioned Corridors have been shown in Fig. 4.1

4.11 CORRIDOR–I : SARTHANA TO DREAM CITY

The features of **Sarthana to Dream City** along with the details of route alignment have been described below:-

4.11.1 References

(a) Chainages

Chainage at Sarthana end (proposed Sarthana Metro Station) has been reckoned as '0' and it increases towards Dream City

(b) Coordinates

Coordinates system adopted for topographical survey is WGS 84 for Northing and Easting. However ground elevations are with respect to GTS bench mark of survey of India.

(c) Directions

Direction from Sarthana to Dream City has been named as 'Down line' and 'up line' is converse of it.

4.11.2 Description of the Route Alignment of the Corridor

(Corridor-I :Sarthana to Dream City)

This corridor runs from North-Eastern parts of Surat to Central to Southern parts of Surat. The corridor originates from Sarthana near to Gadhpur Township on Varachha Road (Surat-Kamraj Road) where it gets integrated with Poshottam Bus Stop of BRTS and moves further to take left turn towards Nature Park by the side of existing Flyover before arriving at Varachha Chopati Garden and then at Shri Swami Narayan Mandir Kalakunj and alignment then ramps down to be underground with null point at chainage 3805mand then runs towards Kapodhora. Thereafter, it runs towards Labheshwar Chowk and then Central Warehouse on Lambe Hanuman Road before integrating at Surat Railway Station (Indian Railways) and existing Surat Bus Terminal of Gujarat State Road Transport Corporation (GSRTC). The alignment then crosses the railway lines atchainage 7420.50m by underground construction method (TBM) beneath underpass as proposed due to integration with Surat railway station Development MMTH project The alignment further continues towards Chowk Bazar via Maskati Hospital on GandhiBaug Road.After Chowk Bazar Station, the alignment ramps out at

chainage 10270m in Gandhi Baug Area to be elevated and runs towards Kadarsha Ni Nal on Dayalji Ashram Margand then it gets integrated at Majura Gate with Majura Gate Metro Station of Corridor-2 : Bhesan to Saroli and then takes turn towards southern parts of Surat and reach Roopali Canal Station on GurunanakMarg (Bhatar Road) and continues further towards AlthanTenament and Althan Gam area on Shri Satya Sai Marg (Althan Road) before reaching on VIP road. Thereafter, the corridor runs towards Women ITI and Bhimrad Station on Bhimrad- Althan Canal Road and continues further towards Convention Centre on Althan- Sarsana Road and crosses ring road before terminating at Dream City on Dream City Road. One Train Maintenance Depot has been proposed on right hand side of road in Dream City. The Sarthana-Dream City Corridor has been shown in Fig. 4.2.

4.11.2.1 A list of Stations along with chainages, inter- station distance, type of station etc. to be provided on Corridor: I (Sarthana to Dream City) is given in Table 4.4(A) as follows:-

Table 4.4(A): List of Stations onSarthana- Dream City Corridor

CORRIDOR-I (SARTHANA TO DREAM CITY) OF SURAT METRO								
S.No	**Station Name**	**Chainage(m)**	**Inter Distance Between Two Stations.**	**Rail Level**	**Ground Level**	**Depth/ Hight**	**U/G / ELEVATED**	**Remark's**
0	**DEAD END**	(-) 450						
1	**SARTHANA**	0.0	450	72.800	58.936	13.864	ELEVATED	SIDE PLATEFORM
2	**NATURE PARK**	1238.1	1238.1	72.500	58.722	13.778	ELEVATED	SIDE PLATEFORM
3	**VARCHHA CHOPATI GARDEN**	2108.5	870.4	72.400	58.755	13.645	ELEVATED	SIDE PLATEFORM
4	**SHRI SWAMINARAYAN MANDIR KALAKUNJ**	3196.6	1088.1	71.700	58.067	13.633	ELEVATED	SIDE PLATEFORM
5	**KAPODRA**	4237.0	1040.4	44.000	58.059	-14.059	U/G	ILAND
6	**LABHESHWAR CHOWK**	5506.6	1269.6	45.200	56.678	-11.478		Side Plateformfor UP Line & Down Line at Different Level i.e One -Over Other
				36.200	56.654	-20.454	U/G **(For Bottom Line Track)**	
7	**CENTRAL WAREHOUSE**	6668.9	1162.3	41.600	55.390	-13.790	U/G	
8	**SURAT RAILWAY STATION**	7255.6	586.7	40.900	54.717	-13.817	U/G	ILAND
9	**MASKATI HOSPITAL**	8487.0	1231.4	43.300	54.376	-11.076	U/G **(For Top Line Track)**	Side Plateformfor UP Line & Down Line at Different Level i.e One -Over Other
				34.000	54.379	-20.379	U/G **(For Bottom Line Track)**	
10	**CHOWK BAZAR**	10289.8	1802.8	42.600	54.206	-11.606	U/G	ILAND
11	**KADARSHA NI NAL**	11721.7	1431.9	67.600	52.818	14.782	ELEVATED	SIDE PLATEFORM

12	**MAJURA GATE**	12313.8	592.1	75.500	51.574	23.926	ELEVATED	SIDE PLATEFORM
13	**ROOPALI CANAL**	13611.5	1297.7	67.300	53.625	13.675	ELEVATED	SIDE PLATEFORM
14	**ALTHAN TENAMENT**	14696.2	1084.7	65.300	51.692	13.608	ELEVATED	SIDE PLATEFORM
15	**ALTHAN GAM**	15843.5	1147.3	64.600	50.867	13.733	ELEVATED	SIDE PLATEFORM
16	**VIP ROAD**	16699.1	855.6	64.400	50.844	13.556	ELEVATED	SIDE PLATEFORM
17	**SURAT WOMEN ITI**	17808.4	1109.3	63.700	50.044	13.656	ELEVATED	SIDE PLATEFORM
18	**BHIMRAD**	18659.5	851.1	62.500	49.000	13.500	ELEVATED	SIDE PLATEFORM
19	**CONVENTIO N CENTRE**	19829.4	1169.9	62.800	49.300	13.500	ELEVATED	SIDE PLATEFORM
20	**DREAM CITY**	20710.3	880.9	62.500	49.000	13.500	ELEVATED	SIDE PLATEFORM
	DEAD END	21160.3	450.0					

4.11.3 Switch Over Ramp (SOR)

Vertical alignment of this corridor changes from underground to elevated or *vise versa* at two locations, as described below:-

Table 4.5 -Details of Switch over Ramps

S. No.	Chainage (m)		Length (m)	Remarks
	From	to		
1	3560	4050	490m	After crossing Shri Swami Narayan Mandir Kalakunj Station, sufficient space is available on road to accommodate ramp structure.
2	10450	10490	440m	In Gandhi Baug area near Chowk Bazar, sufficient space is available to accommodate ramp structure.

4.11.4 Critical Locations

Few locations along the alignment are critical from alignment point of view. These locations are:

Crossing of Railway Lines (near Surat Railway Station) at deeper depth beneath underpass road as proposed due to integration with Surat railway station Development MMTH project.

Crossing of existing Flyover and Surat Metro's Corridor-2 (Bhesan to Saroli) at Majura Gate

Underground Alignment near to LabeshwarChowk Station and Maskoti Hospital Station (one above other)

(a) **Crossing of Railway Lines (near Surat Railway Station).** Due to integration with proposed Surat railway station MMTH project, the existing underpass has been planned at -5m level and shall have to be accommodated above the proposed Metro Station (Surat Railway Station) at chainage 7255.60. Therefore, metro station needed to be

further deepened to accommodate the said underpass above its top slab, with appropriate overburden etc. Therefore, the arrangement showing the tunnels and station box etc. has been shown in fig.

Fig 4:3 Crossing of Railway Line (Near Surat Railway Station)

(b) Crossing of existing Flyover and Surat Metro's Line-2 at Majuragate

This Corridor crosses existing Flyover and Surat Metro's Corridor-2 (Bhesan –Saroli) at Majuragatenear chainage 9175m. The alignment shall have to be made higher to cross over the above mentioned structures safely by providing suitable vertical gradient. Therefore, the arrangement indicating the said crossing is shown in Figure 4.4 hereunder:-

Fig.4.4: Crossing of Flyover and Metro Line-2 at Majura Gate

(c) Underground Alignment near to Labheshwar Chowk Station and Moskati Hospital Station (one above other)

Due to space constraints and heavy congestion on road areas, two underground stations namely Labheshwar Chowk and Moskati Hospital have been planned on side platform pattern for UP line and Down line at different levels i.e. one-over other. To accommodate these two level stations up line and down line tunnels have been planned one above other by tapering the inter tunnel distance on both sides of the stations. Therefore, the said arrangement indicating providing tunnels one over other is shown in Figure 4.5 hereunder:-

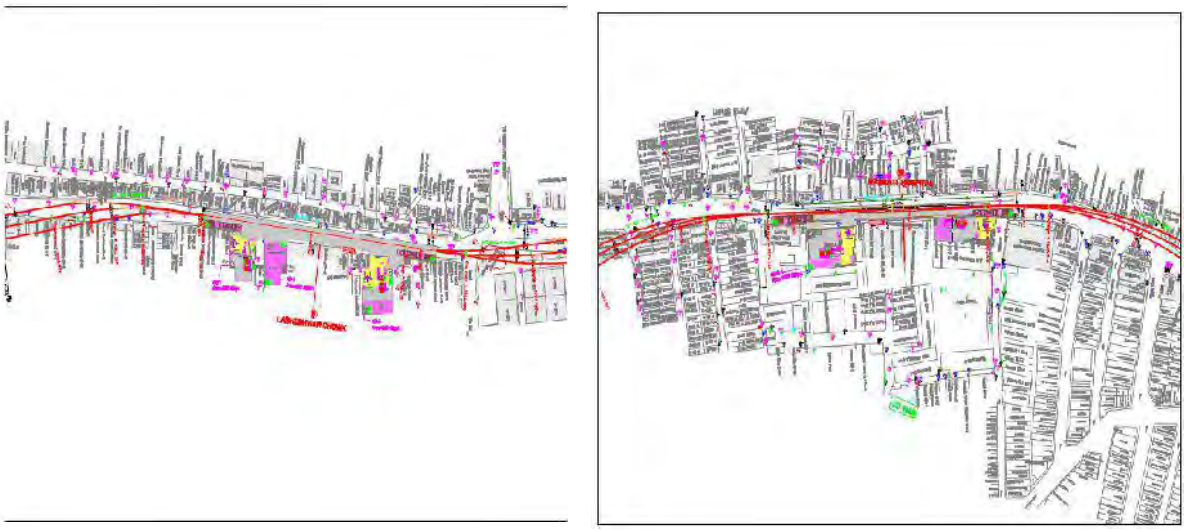


Fig. 4.5 : Tapering of UP line & Down line Tunnels near Labheshwar Chowk Station and Maskati Station

4.11.5 Archeological Survey of India (ASI) protected monuments along the corridor

(a) Dargah Khwaja Dana Sahab Rouza

Alignment passes by the side of DargahKhwaja Dana SahabRouza at the distance of 400m from the Metro Corridor-I.

(b) Old English Tomb

Alignment passes by the side of Old English Tomb and is situated approximately 1450 m away from Metro's Corridor-I.

(c) Tomb of Khwaja Safar Sulemani

Alignment passes by the side of "Tomb of Khwaja Safar Sulemani" which is situated approximately 400m away from Metro's Corridor-I.

(d) Old Dutch and Armenian Tomb and Cemetery

Old Dutch and Armenian Tomb and Cemetery is situated approximately 1100 m from Metro's Corridor-I.

4.11.6 Alignment Underground Verses Elevated

Construction cost of underground metro line is almost two to three times of that elevated one. Therefore efforts have made to keep alignment elevated as far as it is technically possible. However there are certain stretches along the corridor which are underground due to various reasons. The underground stretches are as under:-

Table 4.6 -Details of underground sections

S. No	Chainage(m)		Length(m)
	From	To	
1.	3805.00	10270.00	6465m
		Total	6465m

From chainage 3805.00m to chainage10270m, section has been envisaged as Underground section due to inadequate space available on road in the areas like Kapodra, Labheshwar Chowk, Central Warehouse, Surat Railway Station, Muskati Hospital and Gandhi Baug and Row is less than 20m through which alignment is passing and planning elevated metro alignment on such narrow road will involve demolition of large number of structures. It is also required to provide metro stations at interval of approximately 1.00 Km, in this stretch to serve inhabitants, which further requires more width and resulting in more demolitions.

4.11.7 Stations integrating with different modes of Transport

While deciding station locations efforts have been made to make integration of metro system with other mode of transport such as existing BRTS and railways as far as possible. The details of stations which will have integration with other mode of transport are given in the Table 4.7.

Table 4.7 -Details of Integrating Stations

S. No.	Station	Remarks
1.	Sarthana	Integration with existing Poshottam Bus stop of BRTS
2.	Surat Railway Station	Integration with proposed MMTH project comprising of railway station, GSRTC, BRT/City bus, IPT etc
3.	Majura Gate	Integration with proposed metro station on the Surat Metro'sCorridor-2 (Bhesan-Saroli).
4.	VIP road	Integration with existing Althan Bus Stop of BRTS
5.	Gandhi Baug	Integration with city bus depot
6.	Rupali Canal	Integration with BRT bus station

4.11.8 Depot

Land for depot location for this line has been identified in Dream City area. It is about 2.0 km away from Ring Road/NH. Depot connectivity has been planned from Dream City Station.

4.11.9 Other Main Features of the Corridor

a) This corridor provides metro connectivity to Gadhpur Township, Sarthana Nature Park, NanaVarachhe village, Kalakunj Junction, Kapodra and Punavillage, Labheshwar Chowk area, Big Diamond Market (Mini Bazar) of Surat, Retail Sari Market, Surat Railway Station, Muskati Hospital,Bhagaljunction, Gandhi Bagh, Chowk Bazar, MajuraGate, RoopaliCanal,Althan Treatment, Althan Gram, VIP road Women IIT, Bhimrad, Convention City and Dream City.

b) This Corridor is integrated with Corridor-2 (Bhesan-Saroli) of Surat Metro at Majura Gate. This integration provides metro connectivity to eastern parts of Surat city.

c) Corridor is integrated with Surat Railway Station. This integration provides metro connectivity to commuters of neighbouring cities.

d) About 32% length of the corridor is underground.

e) Total 20 stations have been proposed on this corridor; out of these 14 stations are elevated and remaining 06stations are underground.

4.11.10 Technical Features

(a) Route Length

The total route length from dead end to dead end is 21.61 Km as per the break-up details given hereunder:

Underground

i) TBM (i/c station length)	:	6.005km
ii) Ramp	:	0.465 km
Elevated	:	15.14 km

Total	:	21.61 kms

(b) Horizontal Curves

There are total 62 curves on the alignment of this corridor. Sharpest radius is 122.515m which has been used at twolocation in the elevated section whereas largest radius is 20002.52m which has been used at one location. Total length of alignment on curves is nearly 45.93 (including transition length of curves). Details of curves are given in the Table 4.8A & 4.8B below:-

Table 4.8(A) – Abstract of Horizontal Curves

Abstract of Horizontal Curves CORRIDOR-I (SARTHANA TO DREAM CITY) OF SURAT METRO				
S. No.	Radius (m)	Nos. Occurrences	Curved Length With TL (m)	% w. r. t. total curved length
1	>120m - 510m	35	6644.44	66.95%
2	>510m - 1010m	17	2454.58	24.73%
3	>1010m - 2010m	5	407.637	4.11%
4	>2010m - 5010m	2	190.47	1.92%
5	>5010m	3	227.441	2.29%
	Total	62	9924.6	100.00%

Table 4.8 (B): Statement of Horizontal Curves

CORRIDOR-I (SARTHANA TO DREAM CITY) OF SURAT METRO (Horizontal Curve Details)								
Curve No.	Hand of Arc	Radius (m)	Arc Length (m)	Transition Length (m)		Included Angle	Tangent (m)	Staright Length (m)
				L1	L2	D M S		301.759
1	Right	1002.515	68.749	25	25	03 55 44.939	34.388	94.079
2	Left	1002.515	27.221	25	25	01 33 20.614	13.611	66.619
3	Right	1102.515	27.071	25	25	01 24 24.637	13.536	28.547
4	Right	402.515	45.676	55	32.918	06 30 06.455	22.863	0
5	Right	1002.515	100.74	32.918	25	05 45 26.923	50.412	64.057
6	Left	1002.515	33.338	25	25	01 54 19.296	16.671	360.977
7	Left	202.515	64.856	55	55	18 20 56.753	32.708	148.941
8	Right	202.515	79.149	55	25	22 23 34.758	40.086	141.546
9	Right	2002.515	42.301	25	25	01 12 37.090	21.151	78.946
10	Left	2012.515	25.234	25	25	00 43 06.230	12.617	29.911
11	Left	1802.515	27.723	25	25	00 52 52.349	13.862	185.818
12	Right	2112.515	25.236	45	45	00 41 04.016	12.618	63.745
13	Left	602.515	85.512	25	25	08 07 54.268	42.828	59.078
14	Left	1002.515	37.78	50	50	02 09 33.228	18.892	0
15	Right	302.515	37.205	50	50	07 02 47.507	18.626	25.514
16	Left	482.515	28.447	25	25	03 22 40.579	14.228	0
17	Right	1002.515	76.067	25	25	04 20 50.653	38.052	831.577
18	Right	1008.12	53.401	25	25	03 02 06.093	26.707	86.94
19	Left	1008.12	50.578	55	55	02 52 28.374	25.294	142.171
20	Left	308.12	140.983	55	55	26 12 58.083	71.748	303.896
21	Right	308.12	126.075	55	55	23 26 38.155	63.932	148.846
22	Right	308.12	27.156	55	55	05 02 58.745	13.587	182.808
23	Left	508.12	58.749	55	55	06 37 28.229	29.407	125.001
24	Left	308.12	94.978	55	55	17 39 41.119	47.869	146.934

25	Right	308.12	161.27	25	25	29 59 19.070	82.528	94.266
26	Left	1008.12	39.64	25	25	02 15 10.567	19.823	42.436
27	Left	1008.12	193.619	25	25	11 00 15.131	97.108	35.407
28	Right	1008.12	36.791	55	55	02 05 27.660	18.398	347.798
29	Left	308.12	80.482	55	55	14 57 57.030	40.471	528.285
30	Right	308.12	105.561	55	55	19 37 45.992	53.303	189.747
31	Right	308.12	181.229	55	55	33 42 00.407	93.321	664.177
32	Left	308.12	537.872	55	55	100 01 07.761	367.326	248.262
33	Right	178.12	159.056	55	55	51 09 48.783	85.271	34.515
34	Left	136.515	211.422	40	40	88 44 04.721	133.533	44.798
35	Right	280.515	53.008	40	40	10 49 37.281	26.583	0
36	Left	302.515	27.162	40	40	05 08 39.866	13.59	173.518
37	Right	1002.515	305.812	25	55.007	17 28 40.036	154.103	0
38	Right	292.515	159.629	55.007	45	31 16 01.232	81.856	0
39	Left	202.515	34.697	45	45	09 48 59.097	17.391	168.448
40	Left	502.515	101.526	55	55	11 34 33.010	50.937	42.116
41	Left	1002.515	40.526	25	25	02 18 58.165	20.266	45.285
42	Right	132.515	55.951	50	50	24 11 29.185	28.398	25.629
43	Left	502.515	39.829	50	50	04 32 28.361	19.925	477.837
44	Right	15002.52	48.152	20	20	00 11 02.030	24.076	704.554
45	Right	1502.515	33.943	25	25	01 17 39.744	16.972	95.912
46	Left	1502.515	26.599	25	25	01 00 51.542	13.3	84.444
47	Right	1002.515	171.318	25	25	09 47 28.302	85.868	123.623
48	Left	212.515	62.119	55	55	16 44 52.428	31.283	75.526
49	Right	1002.515	33.819	25	25	01 55 58.238	16.911	115.148
50	Right	252.515	36.238	55	55	08 13 20.585	18.15	295.226
51	Left	162.515	28.136	55	55	09 55 10.718	14.103	0
52	Right	182.515	25.082	55	55	07 52 25.645	12.561	127.704
53	Left	15002.52	37.495	20	20	00 08 35.501	18.747	1051.908
54	Left	20002.52	31.794	15	15	00 05 27.855	15.897	697.134
55	Right	1002.515	41.747	25	25	02 23 09.333	20.877	105.153
56	Left	202.515	45.851	55	55	12 58 20.332	23.024	170.934
57	Left	252.515	48.732	55	55	11 03 26.436	24.442	220.578
58	Right	132.515	34.894	55	55	15 05 14.218	17.549	42.156
59	Right	132.515	30.739	55	55	13 17 26.968	15.439	33.257
60	Left	152.515	83.181	55	55	31 14 56.369	42.653	1584.553
61	Right	122.515	54.785	55	55	25 37 15.974	27.858	53.303
62	Left	122.515	54.785	55	55	25 37 15.968	27.858	2250.559

(c) Gradient

Existing ground profile of corridor is rising from Sarthana towards Nature Park in between chainage510 to 720.00 and 720.00 to 1080.00, @0.238% and 0.139% respectively. Thereafter, the terrain is almost flat upto chainage

1395. Then, there is a fall @-1.357% upto chainage 1689.80. After crossing Varachha Chopati Garden there is a continuous rise upto chainage 2713.5@ 0.60%. After crossing Shri Swami Narayan Mandir Kalakunj Station, there is a fall @-3.36% in between chainage 3310.0 to 4134.30. Thereafter, after crossing Labeshwar Chowk Station, there is a further fall @-3.486% in between chainage 5670.90 to 5975.0. At Surat Railway Station location the terrain is almost flat. Thereafter, a rise @2.042% is observed in between chainage 8040 to 8397.40 followed by a fall @-1.778% in between chainage 8650.0 to 8920 and @-2.100% in between chainage 8920 to chainage 9420.00 before attaining a rise of 1.316% in between chainage 9420to chainage 9800.00 and @ 2.429% upto chainage 10195.30. Further, there shall be a rise before reaching Kadarsha Ni Nal Station at the rate of 3.247% upto chainage 11155.4 and 0.029% upto chainage 11,600.

Change of grade takes place at 64 locations along this corridor. Flattest grade is level which has been provided at the stations. Level gradient has been provided at 21(Twenty one) locations and about 28.22% length of corridor is on level gradient. Steepest gradient on the route is 3.486% which has been provided in between chainage 5640.9 to 5975.00. While designing the vertical alignment efforts have been made to place stations on higher altitude than the running section to get benefit of gravitational force for the acceleration and retardation of the trains. A statement showing details of gradients provide along the corridor is given in the following Table 4.9A & 4.9B.

Table 4.9A: Abstract of Gradients

Abstract of Gradients (CORRIDOR-I (SARTHANA TO DREAM CITY) OF SURAT METRO)				
S. NO.	Description	Nos. Occurrences	Length (m)	% w. r. t. total Alignment length
1	Level	21	6097.74	28.22%
2	> 0% to = 1%	16	5365.07	24.83%
3	> 1% to = 2%	18	6132.72	28.38%
4	> 2% to = 3%	5	1900.54	8.79%
5	> 3% to = 4%	4	2114.23	9.78%
	Total	64	21610	100.00%

Table 4.9 (B) : Statement of Gradients

CORRIDOR-I (SARTHANA TO DREAM CITY) OF SURAT METRO (Vertical Curve Details) (As per centre Line)							
S. No.	Chainage		Length	Rail Level		Gradient	Remarks
	From	To		From	To		
1	-450.0	154.3	604.3	72.8	72.8	0.000%	Level
2	154.3	510.0	355.7	72.8	71.5	-0.365%	Fall
3	510.0	720.0	210.0	71.5	72	0.238%	Rise
4	720.0	1080.0	360.0	72	72.5	0.139%	Rise
5	1080.0	1395.0	315.0	72.5	72.5	0.000%	Level
6	1395.0	1689.8	294.8	72.5	68.5	-1.357%	Fall
7	1689.8	1958.9	269.1	68.5	72.4	1.449%	Rise
8	1958.9	2280.0	321.1	72.4	72.4	0.000%	Level
9	2280.0	2713.5	433.5	72.4	75	0.600%	Rise
10	2713.5	2871.8	158.3	75	75	0.000%	Level
11	2871.8	3100.0	228.2	75	71.7	-1.446%	Fall
12	3100.0	3310.0	210.0	71.7	71.7	0.000%	Level
13	3310.0	4134.3	824.3	71.7	44	-3.360%	Fall
14	4134.3	4415.0	280.7	44	44	0.000%	Level
15	4415.0	4720.0	305.0	44	37	-2.295%	Fall
16	4720.0	5010.0	290.0	37	37.7	0.241%	Rise
17	5010.0	5230.0	220.0	37.7	45.2	3.409%	Rise
18	5230.0	5670.9	440.9	45.2	45.2	0.000%	Level
19	5670.9	5975.0	304.1	45.2	34.6	-3.486%	Fall
20	5975.0	6560.0	585.0	34.6	41.6	1.197%	Rise
21	6560.0	6790.0	230.0	41.6	41.6	0.000%	Level
22	6790.0	7140.0	350.0	41.6	40.9	-0.200%	Fall
23	7140.0	7360.0	220.0	40.9	40.9	0.000%	Level
24	7360.0	7780.0	420.0	40.9	34.5	-1.524%	Fall
25	7780.0	8040.0	260.0	34.5	36	0.577%	Rise
26	8040.0	8397.4	357.4	36	43.3	2.042%	Rise
27	8397.4	8650.0	252.6	43.3	43.3	0.000%	Level
28	8650.0	8920.0	270.0	43.3	38.5	-1.778%	Fall
29	8920.0	9420.0	500.0	38.5	28	-2.100%	Fall
30	9420.0	9800.0	380.0	28	33	1.316%	Rise
31	9800.0	10195.3	395.3	33	42.6	2.429%	Rise
32	10195.3	10389.5	194.2	42.6	42.6	0.000%	Level
33	10389.5	11155.4	765.8	42.6	67.47	3.247%	Rise
34	11155.4	11600.0	444.6	67.47	67.6	0.029%	Rise
35	11600.0	11812.5	212.5	67.6	67.6	0.000%	Level

36	11812.5	12155.3	342.8	67.6	75.5	2.305%	Rise
37	12155.3	12480.0	324.7	75.5	75.5	0.000%	Level
38	12480.0	13280.0	800.0	75.5	63.6	-1.488%	Fall
39	13280.0	13510.0	230.0	63.6	67.3	1.609%	Rise
40	13510.0	13710.0	200.0	67.3	67.3	0.000%	Level
41	13710.0	14030.0	320.0	67.3	62.4	-1.531%	Fall
42	14030.0	14320.0	290.0	62.4	62	-0.138%	Fall
43	14320.0	14585.7	265.7	62	65.3	1.242%	Rise
44	14585.7	14805.0	219.3	65.3	65.3	0.000%	Level
45	14805.0	15140.0	335.0	65.3	61.7	-1.075%	Fall
46	15140.0	15480.0	340.0	61.7	61.4	-0.088%	Fall
47	15480.0	15740.0	260.0	61.4	64.6	1.231%	Rise
48	15740.0	16010.5	270.5	64.6	64.6	0.000%	Level
49	16010.5	16310.0	299.5	64.6	60.4	-1.402%	Fall
50	16310.0	16600.0	290.0	60.4	64.4	1.379%	Rise
51	16600.0	16800.0	200.0	64.4	64.4	0.000%	Level
52	16800.0	17100.0	300.0	64.4	59.2	-1.733%	Fall
53	17100.0	17380.0	280.0	59.2	60.9	0.607%	Rise
54	17380.0	17700.0	320.0	60.9	63.7	0.875%	Rise
55	17700.0	17920.0	220.0	63.7	63.7	0.000%	Level
56	17920.0	18200.0	280.0	63.7	59.4	-1.536%	Fall
57	18200.0	18505.5	305.5	59.4	62.5	1.015%	Rise
58	18505.5	18833.8	328.2	62.5	62.5	0.000%	Level
59	18833.8	19300.0	466.2	62.5	59	-0.751%	Fall
60	19300.0	19725.0	425.0	59	62.8	0.894%	Rise
61	19725.0	19940.0	215.0	62.8	62.8	0.000%	Level
62	19940.0	20220.0	280.0	62.8	60	-1.000%	Fall
63	20220.0	20480.0	260.0	60	62.5	0.962%	Rise
64	20480.0	21160.3	680.3	62.5	62.5	0.000%	Level

(d) Two Nos. underground Stations namely (i) Labheshwar Chowk Station and (ii) Maskati Hospital Station shall be constructed as two level stations i.e. Side platform for up line and Down line at different levels (one-over other). Therefore, one tunnel shall run beneath the other in order to retrieve in station box at different levels. The vertical Curve details for upline and bottom line in between chainage 4100 to 10500 have been shown in Table : 4.9© and 4.9(d).

Table: 4.9(c)
(As per UP Line Track CH: 4100 to 10500)

S. No.	Chainage		Length	Rail Level		Gradient	Remarks
	From	To		From	To		
1	4129.7	4415.0	285.3	44	44	0.000%	Level
2	4415.0	4720.0	305.0	44	37	-2.295%	Fall
3	4720.0	5010.0	290.0	37	36	-0.345%	Fall
4	5010.0	5402.0	392.0	36	45.2	2.347%	Rise
5	5402.0	5636.0	234.0	45.2	45.2	0.000%	Level
6	5636.0	5994.3	358.3	45.2	34.35	-3.028%	Fall
7	5994.3	6560.0	565.7	34.35	41.6	1.282%	Rise
8	6560.0	6790.0	230.0	41.6	41.6	0.000%	Level
9	6790.0	7140.0	350.0	41.6	40.9	-0.200%	Fall
10	7140.0	7360.0	220.0	40.9	40.9	0.000%	Level
11	7360.0	7780.0	420.0	40.9	34.5	-1.524%	Fall
12	7780.0	8040.0	260.0	34.5	31.5	-1.154%	Fall
13	8040.0	8390.0	350.0	31.5	43.3	3.371%	Rise
14	8390.0	8664.0	274.0	43.3	43.3	0.000%	Level
15	8664.0	8949.0	285.0	43.3	32.9	-3.649%	Fall
16	8949.0	9420.0	471.0	32.9	28	-1.040%	Fall
17	9420.0	9800.0	380.0	28	33	1.316%	Rise
18	9800.0	10180.0	380.0	33	42.6	2.526%	Rise
19	10180.0	10385.0	205.0	42.6	42.6	0.000%	Level
20	10385.0	11155.4	770.4	42.6	67.47	3.228%	Rise
21	11155.4	11600.0	444.6	67.47	67.47	0.000%	Fall

Table: 4.9(d)
(As per Bottom Line Track CH: 4100 to 10500)

CORRIDOR-I (SARTHANA TO DREAM CITY) OF SURAT METRO (Vertical Curve Details)
(As per Bottom Line Track CH: 4100 to 10500)

S. No.	Chainage		Length	Rail Level		Gradient	Remarks
	From	To		From	To		
2	4133.0	4415.0	282.0	44	44	0.000%	Level
3	4415.0	4720.0	305.0	44	37	-2.295%	Fall
4	4720.0	5160.0	440.0	37	39	0.455%	Rise
5	5160.0	5380.0	220.0	39	36.2	-1.273%	Fall
6	5380.0	5636.0	256.0	36.2	36.2	0.000%	Level
7	5636.0	5960.0	324.0	36.2	38.5	0.710%	Rise
8	5960.0	6560.0	600.0	38.5	41.6	0.517%	Rise

9	6560.0	6790.0	230.0	41.6	41.6	0.000%	Level
10	6790.0	7140.0	350.0	41.6	40.9	-0.200%	Fall
11	7140.0	7360.0	220.0	40.9	40.9	0.000%	Level
12	7360.0	7780.0	420.0	40.9	34.5	-1.524%	Fall
13	7780.0	8120.0	340.0	34.5	36.8	0.676%	Rise
14	8120.0	8300.0	180.0	36.8	34	-1.556%	Fall
15	8300.0	8680.0	380.0	34	34	0.000%	Level
16	8680.0	8930.0	250.0	34	40.1	2.440%	Rise
17	8930.0	9420.0	490.0	40.1	28	-2.469%	Fall
18	9420.0	9800.0	380.0	28	33	1.316%	Rise
19	9800.0	10209.3	409.3	33	42.6	2.346%	Rise
20	10209.3	10399.7	190.5	42.6	42.6	0.000%	Level
21	10399.7	11155.4	755.6	42.6	67.47	3.291%	Rise
22	11155.4	11600.0	444.6	67.47	67.6	0.029%	Rise

4.12 Corridor - 2 : Bhesan to Saroli

4.12.1 References

(a) Chainages

Chainage starting from Bhesan Station as 0.00 increases towards Saroli.

(b) Co-ordinates

Coordinates system adopted for topographical survey is WGS 84 for Northing and Easting. However ground elevations are with respect to GTS bench mark of Survey of India.

(c) Directions

Direction from Bhesan to Saroli has been named as ‘Up line’ and ‘Down line’ is converse of it.

4.12.2 Description of the Route Alignment of the Corridor

(Corridor 2: Bhesan to Saroli)

This Corridor runs from North-Western parts of Surat to Central area and finally terminate in South-Eastern parts of Surat. The entire corridor is elevated corridor. It originates at Bhesan on Rander-Bhesan Road and connected with the proposed Train Maintenance Depot at Bhesan. From Bhesan it runs towards Botanical Garden and Ugat Vaarigruh on Ugat road

and reaches Palanpur Road. Then the alignment runs on LP Savani Road and reaches LP Savani School Station and Performing Art Centre near Star Bazar Mall. Alignment then takes turns on Old Adanjan Road and reaches Adjan Gam, Aquarium and Badri Narayan Temple before crossing Tapi River in between chainage 7143 m to 7500 m. After crossing Tapi River, the alignment reaches at Athwa Chopati Centre and continues further on Surat-Dumas Road and reaches Majura Gate where it integrates with the proposed station on Surat Metro's Corridor-I (Sarthana-Dream City) near chainage 9175 m and runs parallel to the existing Flyover and reaches Udhana Darwaja and Kamela Darwaja on ring road (Jaya Prakash Narayan Marg). The alignment then switch over on Mithi Khadi road and reaches Anjana Farm and Model Town before following canal road to arrive at Magub, Bharat Cancer Hospital and finally terminates at Saroli on Bardoli Road. The Bhesan-Saroli Corridor has been shown in Fig.4.6.

4.12.2.1 A list of stations along with chainages, inter-station distances, type of stations etc. proposed to be provided on Corridor:2 (Bhesan to Saroli) is given in Table: 4.10(A) as follows:-

Table: 4.10(A)
List of Stations on Bhesan- Saroli Corridor

CORRIDOR-II BHESHAN TO SAROLI OF SURAT METRO								
S.No	**Station Name**	**Chainage(m)**	**Inter Distance Between Two Stations.**	**Rail Level**	**Ground Level**	**Depth/ Hight**	**U/G / ELEVATED**	**Remark's**
0	**DEAD END**	(-) 949.630						
1	**BHESHAN**	0.0	949.63	65.900	52.362	13.538	ELEVATED	SIDE PLATEFORM
2	**BOTANICAL GARDEN**	950.6	950.6	65.800	52.015	13.785	ELEVATED	SIDE PLATEFORM
3	**UGAT VAARIGRUH**	1617.5	666.9	65.400	51.682	13.718	ELEVATED	SIDE PLATEFORM
4	**PALANPUR ROAD**	2738.9	1121.4	65.000	51.482	13.518	ELEVATED	SIDE PLATEFORM
5	**L.P. SAVANI SCHOOL**	3831.6	1092.7	65.100	51.449	13.651	ELEVATED	SIDE PLATEFORM
6	**PERFORMING ART CENTRE**	4421.8	590.2	65.500	51.604	13.896	ELEVATED	SIDE PLATEFORM
7	**ADAJAN GAM**	5155.0	733.2	65.300	51.247	14.053	ELEVATED	SIDE PLATEFORM
8	**AQUARIUM**	5770.1	615.1	63.600	50.009	13.591	ELEVATED	SIDE PLATEFORM
9	**BADRI NARAYAN TEMPLE**	6891.5	1121.4	64.400	50.714	13.686	ELEVATED	SIDE PLATEFORM
10	**AHTHWA CHAUPATI**	7863.5	972.0	66.300	52.529	13.771	ELEVATED	SIDE PLATEFORM
11	**MAJURA GATE**	9019.2	1155.7	67.100	53.431	13.669	ELEVATED	SIDE PLATEFORM
12	**UDHANA DARWAJA**	10646.2	1627.0	67.700	53.804	13.896	ELEVATED	SIDE PLATEFORM
13	**KAMELA DARWAJA**	11247.2	601.0	68.000	54.535	13.465	ELEVATED	SIDE PLATEFORM
14	**ANJANA FARM**	12593.9	1346.7	68.000	54.135	13.865	ELEVATED	SIDE PLATEFORM
15	**MODAL TOWN**	13636.3	1042.4	68.500	54.935	13.565	ELEVATED	SIDE PLATEFORM

16	**MAGOB GAM**	14763.1	1126.8	76.500	55.692	20.808	ELEVATED	SIDE PLATEFORM
17	**BHARAT CANCER HOSPITAL**	15899.4	1136.3	68.300	54.914	13.386	ELEVATED	SIDE PLATEFORM
18	**SAROLI**	17341.8	1442.4	71.500	57.310	14.190	ELEVATED	SIDE PLATEFORM
	DEAD END	17791.9	450.1					

4.12.3 Main features of the corridor

Main features of the corridor are as detailed below:

(a) This corridor provides connectivity to North Western, Central and South Eastern parts of Surat.

(b) The route provides integration with Surat Metro's proposed line (Sarthana-Dream City) at Majura Gate enabling the passengers to go in almost all directions of Surat.

(c) Prominent areas like Bhesan Treatment Plant, International Cricket Stadium, SMC Botanical Garden, Palanpur, LP Savani School, Performing Art Centre, TGB Circle, Aquerium, Badri Narayan Temple, Star Bazar, Tapi River, Athawa Chopati, Majura Gate, Udhana Darwaja, Kamala Darwaja, Anjana Junction, Parvat Pataya, Magub Village, Bharat Cancer Hospital, Saroli etc. are connected through this route.

4.12.4 Interchange Stations

An interchange provides a great utility and flexibility for the system as a whole and it decreases the time required for travel within the city. The proposed corridor has one interchange stations namely Majura Gate.

4.12.5 Depot

Land for depot locations for this corridor has been identified in Bhesan Area. It is situated on Rander-Bhesan Road and is approximately 1.0 km away from proposed Bhesan Station-Depot – Connectivity has been planned from Bhesan Station.

4.12.6 Critical Locations

(a) Sharp S-Curve in between Ugat Vaarigruh and Palanpur Road

From alignment point of view providing a sharp curve to follow road with two right angle turnings within a short length enabling alignment to reach at LP Savani Road is a critical location. Providing of sharp S-Curve has been shown in Figure 4.7 hereunder:

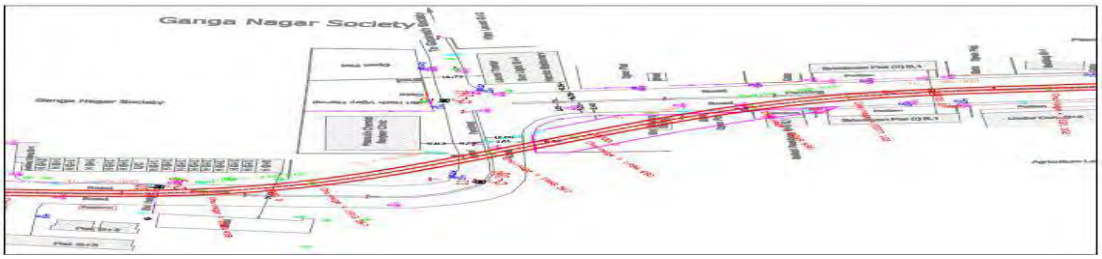


Fig. 4.7: Sharp S-curve in between UgatVaarigruh and Palanpur Road

(b) Crossing Tapiriver : AfterBadri Narayan Temple on old Adajan Road, the alignment crosses Tapi River at Chainage 7143 m to Chainage 7500 m before reaching AthwaChopatiStation on Surat-Dumas Road. Crossing of Tapi River has been shown in Fig.4.8 hereunder:-

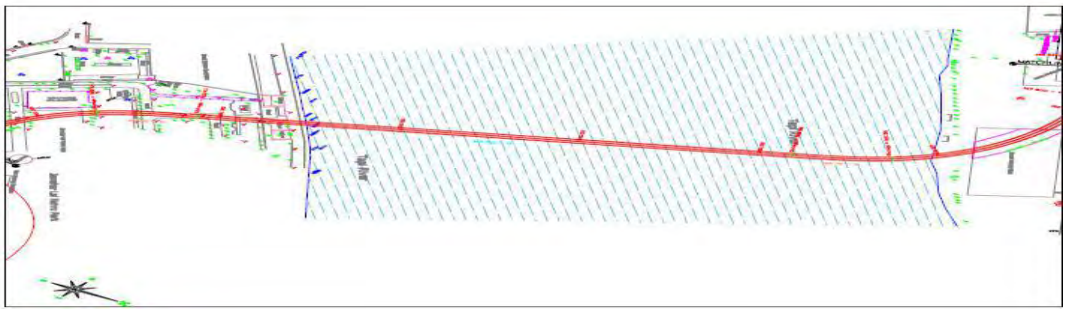


Fig.4.8: Crossing Tapi River

(c) Crossing of Railway Lines of Indian Railways at Surat –Mumbai Route.

This alignment crosses Railway Line (Indian Railways) onSurat to Mumbai Route) at Chainage 12,400m. One underpass also exists at this location. Crossing of Railway Line has been shown in Figure 4.9 as here under:-

Fig. 4.9: Crossing of Railway Lines on Surat – Mumbai Route

4.12.7 Stations integrating with different modes of Transport

While deciding station locations efforts have been made to make integration of metro system with other mode of transport such as proposed metro system, railways, BRTS etc. as far as possible. The details of stations which will have integration with other mode of Transport are given in Table 4.10(b).

Table 4.10 (b) – Details of Integrating Stations

S.No.	Name of Station	Remarks
01	Palanpore Road	Integration with city bus station
02	LPSwani Road	Integrating with city bus station
03	Aquarium	Integrating with BRT station
04	Majura Gate	Metro intersection junction
05	Udhna Darwaja	Integration with BRT station
06	Model Township	Integration with BRT station
07	Magob	Integration with BRT station

4.12.8 Technical Features

(a) Route Length

The total route length from Bhesan to Saroli Station is 18.74 km. Whole alignment is elevated alignment.

(b) Horizontal Curves

There are total 66 curves on the alignment of this corridor. Sharpest radius is 122.515 m, whereas largest radius is 8002.515m. Details of curves are given in the Table 4.11 (A) & 4.11 (B) below:-

Table 4.11(A)- Abstract of Horizontal Curves

Abstract of Horizontal Curves (CORRIDOR-II BHESHAN TO SAROLI) OF SURAT METRO)				
S. No.	Radius (m)	Nos. Occurrences	Curved Length With TL (m)	% w. r. t. total curved length
1	>120m - 510m	33	5978.59	59.07%
2	>510m - 1010m	15	2055.98	20.31%
3	>1010m - 2010m	8	842.807	8.33%
4	>2010m - 5010m	7	809.236	8.00%
5	>5010m	3	434.694	4.29%
	Total	66	10121	100.00%

Table 4.11 (B) : Statement of Horizontal Curves

CORRIDOR-II BHESHAN TO SAROLI) OF SURAT METRO (Horizontal Curve Details)								
Curve No.	Hand of Arc	Radius (m)	Arc Length (m)	Transition Length (m)		Included Angle	Tangent (m)	Staright Length (m)
				L1	L2	D M S		230.652
1	Left	1802.515	29.198	25	25	00 55 41.121	14.599	238.734
2	Left	1002.515	63.98	25	25	03 39 23.682	32.001	96.264
3	Right	1002.515	94.771	25	25	05 24 58.958	47.421	386.012
4	Right	2002.515	249.076	25	25	07 07 35.588	124.699	280.82
5	Left	2602.515	25.193	25	25	00 33 16.710	12.597	52.064
6	Right	1002.515	33.306	25	25	01 54 12.546	16.654	91.126
7	Right	122.515	195.485	25	25	91 25 16.574	125.592	28.771
8	Left	1002.515	126.943	25	25	07 15 18.280	63.557	94.539
9	Left	142.515	30.127	55	55	12 06 44.100	15.12	26.123
10	Right	182.515	27.413	55	55	08 36 19.642	13.732	55.62
11	Right	502.515	33.868	55	55	03 51 41.567	16.94	33.826
12	Right	602.515	27.026	40	40	02 34 12.154	13.515	0
13	Left	602.515	26.629	40	40	02 31 56.092	13.317	110.413
14	Left	1002.515	93.961	25	25	05 22 12.140	47.015	36.237
15	Right	362.515	27.082	55	55	04 16 49.088	13.547	0
16	Left	1802.515	26.419	25	25	00 50 23.194	13.21	35.127
17	Right	1702.515	27.517	25	25	00 55 33.764	13.759	25.147
18	Left	3002.515	30.109	20	20	00 34 28.430	15.055	157.167
19	Right	3002.515	27.686	25	25	00 31 41.925	13.843	149.103
20	Left	122.515	34.623	55	55	16 11 30.701	17.428	61.141
21	Right	1002.515	58.376	25	25	03 20 10.779	29.196	74.784
22	Left	5502.515	26.11	25	25	00 16 18.760	13.055	395.272
23	Right	8002.515	251.751	25	25	01 48 08.873	125.886	293.497
24	Left	8002.515	26.833	15	15	00 11 31.629	13.417	109.803
25	Right	502.515	26.994	40	40	03 04 39.953	13.5	58.123
26	Left	142.515	188.317	55	55	75 42 35.560	110.765	140.643
27	Left	182.515	57.143	50	50	17 56 18.253	28.807	82.87
28	Right	122.515	192.485	35	35	90 01 05.088	122.554	0
29	Left	192.515	27.308	35	35	08 07 38.576	13.677	0
30	Right	192.515	71.051	35	35	21 08 45.601	35.934	0
31	Left	122.515	129.834	55	55	60 43 06.531	71.762	141.699
32	Right	122.515	148.87	55	55	69 37 15.089	85.183	331.046
33	Left	122.515	129.027	55	55	60 20 27.752	71.221	240.287
34	Right	122.515	73.051	55	55	34 09 47.661	37.648	128.673
35	Left	1102.515	31.236	25	25	01 37 23.848	15.619	0
36	Right	802.515	82.323	45	45	05 52 38.979	41.198	102.593
37	Left	202.515	33.244	55	55	09 24 19.799	16.66	276.094
38	Left	302.515	90.792	55	55	17 11 44.939	45.74	333.477

39	Right	1602.515	27.255	25	25	00 58 28.025	13.628	54.089
40	Left	552.515	156.547	50	50	16 14 02.154	78.801	0
41	Right	452.515	32.233	55	55	04 04 52.488	16.123	0
42	Left	452.515	51.049	55	55	06 27 49.031	25.552	0
43	Right	252.515	40.529	55	55	09 11 45.734	20.308	54.613
44	Right	1202.515	26.743	25	25	01 16 27.153	13.372	0
45	Left	1002.515	42.231	25	25	02 24 48.961	21.119	72.6
46	Left	172.515	41.434	55	55	13 45 40.413	20.817	99.918
47	Right	302.515	26.313	50	50	04 59 01.236	13.165	173.056
48	Left	152.515	25.976	50	50	09 45 29.941	13.019	0
49	Right	122.515	121.522	50	50	56 49 53.111	66.287	0
50	Left	202.515	37.092	50	50	10 29 38.476	18.598	0
51	Right	232.515	30.861	55	55	07 36 16.768	15.453	273.455
52	Right	2502.515	25.357	25	25	00 34 50.029	12.679	105.825
53	Left	2502.515	36.541	25	25	00 50 11.827	18.271	166.814
54	Left	122.515	31.81	55	55	14 52 34.187	15.995	248.973
55	Left	1002.515	67.536	25	25	03 51 35.332	33.781	0
56	Right	1002.515	84.181	25	25	04 48 39.963	42.115	57.788
57	Left	502.515	100.045	45	45	11 24 24.848	50.188	95.22
58	Right	1502.515	25.363	25	25	00 58 01.874	12.682	397.478
59	Right	152.515	70.373	55	55	26 26 14.639	35.825	192.46
60	Left	302.515	42.428	55	55	08 02 08.601	21.249	400.026
61	Right	602.515	61.81	55	55	05 52 39.861	30.932	265.448
62	Right	2502.515	36.84	25	25	00 50 36.505	18.421	520.357
63	Right	802.515	36.359	45	45	02 35 45.000	18.182	63.54
64	Left	322.515	146.48	55	55	26 01 21.427	74.526	106.097
65	Right	502.515	323.732	50	50	36 54 40.526	167.707	148.645
66	Left	3002.515	287.51	25	25	05 29 11.142	143.865	226.026

(c) Gradient

Existing ground profile of the corridor is falling @-2.056% in between chainage 220 to chainage 400 after taking off at Bhesan Station and then rising @ 0.115% in between chainage 400 to 660 and @ 1.833% in between chainage 660 to 840 before reaching Botanical Garden Station. After crossing Botanical Garden there is a rise @ 1.837% in between chainage 1280.0 to 1481.4 and thereafter there is flat terrain upto chainage 1725.0. Before reaching Palanpur Road Station there is a rise @ 0.922% in between chainage 2198.5 to 2600. Level gradient has been provided at 19 (Nineteen) locations and about 30.36% length of Corridor is on level gradient steepest gradient on the route is 2.519% which has been provided to between chainage 14870 to 15140.

Change of grade takes place at 62 locations along this corridor. Flattest grade is level which has been provided at the stations. While designing the vertical alignment efforts have been made to place stations on higher altitude than the

running section to get benefit of gravitational force for the acceleration and retardation of the trains. A statement showing details of gradients provide along the corridor is given in the following Table 4.12(A) & 4.12 (B).

Table 4.12(A)
Abstract of Gradients
Corridor-II Bhesan to Saroli of Surat Metro

S. NO.	Description	Nos. Occurrences	Length (m)	% w. r. t. total Alignment length
1	Level	19	5689.46	30.36%
2	> 0% to = 1%	20	7347.05	39.20%
3	> 1% to = 2%	16	4077.54	21.76%
4	> 2% to = 3%	7	1627.43	8.68%
5	> 3% to = 4%	0	0	0.00%
	Total	62	18741	100.00%

Table 4.12(B)
Statement of Gradients

CORRIDOR-II BHESHAN TO SAROLI) OF SURAT METRO (Vertical Curve Details)							
S. No.	Chainage		Length	Rail Level		Gradient	Remarks
	From	To		From	To		
1	-949.6	-680.0	269.6	61.9	61.9	0.000%	Level
2	-680.0	-340.0	340.0	61.9	62.2	0.088%	Rise
3	-340.0	-100.0	240.0	62.2	65.9	1.542%	Rise
4	-100.0	220.0	320.0	65.9	65.9	0.000%	Level
5	220.0	400.0	180.0	65.9	62.2	-2.056%	Fall
6	400.0	660.0	260.0	62.2	62.5	0.115%	Rise
7	660.0	840.0	180.0	62.5	65.8	1.833%	Rise
8	840.0	1088.8	248.8	65.8	65.8	0.000%	Fall
9	1088.8	1280.0	191.2	65.8	61.7	-2.144%	Rise
10	1280.0	1481.4	201.4	61.7	65.4	1.837%	Rise
11	1481.4	1725.0	243.6	65.4	65.4	0.000%	Level
12	1725.0	2198.5	473.5	65.4	61.3	-0.866%	Fall
13	2198.5	2600.0	401.5	61.3	65	0.922%	Rise
14	2600.0	2877.1	277.1	65	65	0.000%	Level
15	2877.1	3194.9	317.8	65	61.1	-1.227%	Fall
16	3194.9	3495.0	300.1	61.1	60.8	-0.100%	Fall
17	3495.0	3686.2	191.2	60.8	65.1	2.249%	Rise
18	3686.2	3960.0	273.8	65.1	65.1	0.000%	Level
19	3960.0	4130.0	170.0	65.1	62	-1.824%	Fall
20	4130.0	4320.0	190.0	62	65.5	1.842%	Rise
21	4320.0	4530.0	210.0	65.5	65.5	0.000%	Level
22	4530.0	4820.0	290.0	65.5	61.6	-1.345%	Fall

23	4820.0	5040.0	220.0	61.6	65.3	1.682%	Rise
24	5040.0	5289.6	249.6	65.3	65.3	0.000%	Level
25	5289.6	5610.0	320.4	65.3	63.6	-0.531%	Fall
26	5610.0	5920.0	310.0	63.6	63.6	0.000%	Level
27	5920.0	6220.0	300.0	63.6	61	-0.867%	Fall
28	6220.0	6520.0	300.0	61	60.5	-0.167%	Fall
29	6520.0	6740.0	220.0	60.5	64.4	1.773%	Rise
30	6740.0	7060.0	320.0	64.4	64.4	0.000%	Level
31	7060.0	7300.0	240.0	64.4	61.5	-1.208%	Fall
32	7300.0	7520.0	220.0	61.5	64.4	1.318%	Rise
33	7520.0	7700.0	180.0	64.4	66.3	1.056%	Rise
34	7700.0	7965.0	265.0	66.3	66.3	0.000%	Level
35	7965.0	8140.0	175.0	66.3	62.7	-2.057%	Fall
36	8140.0	8510.0	370.0	62.7	62.3	-0.108%	Fall
37	8510.0	8900.0	390.0	62.3	67.1	1.231%	Rise
38	8900.0	9210.0	310.0	67.1	67.1	0.000%	Level
39	9210.0	9691.9	481.9	67.1	66.6	-0.104%	Fall
40	9691.9	9960.0	268.1	66.6	64.4	-0.821%	Fall
41	9960.0	10520.0	560.0	64.4	67.7	0.589%	Rise
42	10520.0	10750.0	230.0	67.7	67.7	0.000%	Level
43	10750.0	11105.7	355.7	67.7	68.1	0.112%	Rise
44	11105.7	11404.9	299.2	68.1	68.1	0.000%	Level
45	11404.9	11757.9	353.1	68.1	62	-1.728%	Fall
46	11757.9	12100.0	342.1	62	61.3	-0.205%	Fall
47	12100.0	12465.2	365.2	61.3	68	1.834%	Rise
48	12465.2	12746.2	281.0	68	68	0.000%	Level
49	12746.2	13120.0	373.8	68	66	-0.535%	Fall
50	13120.0	13490.0	370.0	66	68.5	0.676%	Rise
51	13490.0	13780.0	290.0	68.5	68.5	0.000%	Level
52	13780.0	14220.0	440.0	68.5	68.2	-0.068%	Fall
53	14220.0	14580.0	360.0	68.2	76.5	2.306%	Rise
54	14580.0	14870.0	290.0	76.5	76.5	0.000%	Level
55	14870.0	15140.0	270.0	76.5	69.7	-2.519%	Fall
56	15140.0	15440.0	300.0	69.7	65.4	-1.433%	Fall
57	15440.0	15800.0	360.0	65.4	68.3	0.806%	Rise
58	15800.0	16010.0	210.0	68.3	68.3	0.000%	Level
59	16010.0	16380.0	370.0	68.3	64.9	-0.919%	Fall
60	16380.0	16740.0	360.0	64.9	65.7	0.222%	Rise
61	16740.0	17000.0	260.0	65.7	71.5	2.231%	Rise
62	17000.0	17791.9	791.9	71.5	71.5	0.000%	Level

CHAPTER - 5

CIVIL ENGINEERING

5.1 GENERAL

This chapter deals with civil underground and elevated structure, Geotechnical investigation, construction methods, land requirements, Utility services and Traffic diversion during construction etc.

5.2 CIVIL STRUCTURES

5.2.1 Underground Construction

For the underground section running under the road, cut and cover method of the underground construction can be employed for the construction of the underground sections. However keeping in view the inconvenience to the traffic movement, it is proposed to tunnel through by using Tunnel Boring Machine (TBM) or New Austrian Tunneling Method (NATM) in the overburden soil mass. Tunnel excavation for a major length of underground section is expected to be carried out by Tunnel Boring Machines. There are few stretches along the underground alignment where Cut & Cover method has been considered for construction after Switch Over Ramp (SOR). Tunnel boring machines (TBMs) capable of drilling in soft soil with a minimum finished internal diameter of 5.6 m. can be successfully employed for boring tunnels through this stratum. The tunnels are proposed with a minimum overburden of 6.0m.

5.2.2 Underground Stations

All the 06 underground stations have been proposed to be constructed by cut and cover with top-down or bottom up methods depending upon geological conditions and land/space availibilities. The diaphragm walls for such station constructions would be 80 to 120 cm. thick and will function as a permanent side wall of the station. It is, therefore, necessary to construct the diaphragm walls absolutely watertight and with the required concrete strength as has been done in other phases of Delhi Metro station construction. By resorting to top-down method the surface could be restored quickly and further excavations and construction of the station will not hamper the surface activity.

5.2.3 Cut and Cover Method of Construction of Underground Stations

Cut and Cover mainly consists of following steps:

1. Diversion of utilities
2. Construction of support walls
3. Excavation between the support walls along with the installation of struts between the two walls to keep them in position.
4. Construction of tunnel/structure and removal of temporary struts.
5. Back filling and restoration of the surface

5.2.4 Utility Diversion:

It is suggested that all utilities falling within excavation area are diverted away in advance to avoid damage to such utilities during the excavation/ construction phase. The cross utilities, however has to be kept supported. It is suggested that pressure water pipelines crossing the proposed cut area are provided with valves on both sides of the cut so that the cut area can be isolated in case of any leakage to the pipeline to avoid flooding of the cut/damage to the works.

5.2.5 Support Walls:

Most commonly used support wall is RCC Diaphragm Wall. The advantage of diaphragm wall is that the same can be used as part of permanent structure. The modern techniques are now available where water-stop can be inserted at the joints of two diaphragm wall panels to avoid seepage through the joints. It is also now possible to ensure the verticality of the diaphragm wall panels to avoid any infringement problem later on. Typically the diaphragm wall of 80 cm to 1 meter thickness is sufficient to do the cut and cover construction. The various advantages of diaphragm wall are as follows.

(a) It is rigid type of support system and therefore ensures the maximum safety against settlement to the adjacent structures.

(b) Can be used as part of the permanent structure and, therefore, considered economical.

(c) With diaphragm wall it is possible to construct an underground structure by top down method. In this method top slab is cast once the excavation is reached to the top slab level with rigid connections to the diaphragm wall which can be achieved by leaving couplers in the diaphragm wall

reinforcement at appropriate level. This top slab then acts as strut between the two support walls and gives much more rigidity and safety to the construction. Excavation thereafter can be completed. This also helps in restoration of the surface faster without waiting for full structure to be completed.

The other support walls which can be used depending on the site conditions are as follows:

(a) Sheet Piles

'Z'/'U' sheet piles can be used as temporary support wall. This can be advantageous where it is possible to re-use the sheet pile again and again and therefore, economy can be achieved However the main concern remains, driving of sheet piles causes vibrations/noise to the adjacent buildings. This may sometimes lead to damage to the building and most of the time causes inconvenience to the occupants of the building. Situation becomes more critical if sensitive buildings are adjacent to the alignment like hospitals, schools, laboratories, etc. Silent pile driving equipments however are now available and can be used where such problems are anticipated.

(b) Retaining Casing Piles

This is suitable for situation where the cut and cover is to be done in partly soil and partly rock. The top soil retaining structure can be done with the help of Casing pile which is then grouted with cement slurry. This is considered suitable in case of shallow level, non-uniform, uneven nature of rock head surface which render the construction of sheet piles/diaphragm wall impracticable. These are suitable up to 7-meter depth. The common diameter used for such casing pile is 2.00-2.50 m dia.

(c) Soldier Piles and Lagging

Steel piles (H Section or I section) are driven into the ground at suitable interval (normally 1-1.5 m) centre-to-centre depending on the section and depth of excavation. The gap between two piles is covered with suitable lagging of timber planks/shot-creting /steel sheets/GI sheets during the process of excavation.

(d) Secant Piles

Secant Piles are cast-in-situ bored piles constructed contiguously to each other

so that it forms a rigid continuous wall. This is considered an alternative to diaphragm wall where due to soil conditions it is not advisable to construct diaphragm wall from the consideration of settlement during the trenching operation. 800 to 1000 mm dia piles are commonly used. Two alternate soft piles are driven and cast in such a way that the new pile partly cuts into earlier constructed piles. This new pile is constructed with suitable reinforcement. With this, alternate soft and hard pile is constructed. This has got all the advantages of diaphragm wall. However, this wall cannot be used as part of permanent structure and permanent structure has to be constructed in- side of this temporary wall.

5.2.6 Anchors

As an alternative to the struts, soil/rock anchors can be used to keep these support walls in position. This gives additional advantage as clear space is available between two support walls and progress of excavation & construction is much faster as compared to the case where large number of struts is provided which create hindrance to the movement of equipments and material & thus affects the progress adversely.

The combination of all the type of retaining walls, struts/anchors may be necessary for the project to suit the particular site. Based on the above broad principle, the support walls system for cut and cover shall be chosen for particular locations.

5.2.7 Elevated Section - Choice of Superstructure

The choice of superstructure has to be made keeping in view the ease of constructability and the maximum standardization of the formwork for a wide span ranges.

The segmental construction has been chosen mainly due to the following advantages:

- Segmental construction is an efficient and economical method for a large range of span lengths and types of structures. Structures with sharp curves and variable super elevation can be easily accommodated.

- Segmental construction permits a reduction of construction time as segments may be manufactured while substructure work proceeds and assembled rapidly thereafter.

- Segmental construction protects the environment as only space required for foundation and sub-station is required at site. The superstructure is manufactured at a place away from busy areas and placement of superstructure is done with the system erected from piers at heights.

- Segments are easy to stack in the casting yard/stacking yard in more than one layer, thereby saving in requirement of space.

- It is easier to transport smaller segments by road trailers on city roads.

- It is easy to incorporate last minute changes in span configuration if the site situation so warrants.
- Interference to the traffic during construction is significantly reduced.
- Segmental construction contributes towards aesthetically pleasing structures and good finishes.
- The overall labour requirement is less than that for conventional methods.
- Better quality control is possible in the casting yard.
- During construction, the technique shows an exceptionally high record of safety.

5.2.8 Types of Superstructures for Elevated Section

(a) Pre-cast segmental box girder using external unbounded tendon

(b) Pre-cast segmental U-Channel Superstructure with internal pre-stressing.

(c) Double U-girder with internal pre-stressing
(d) I-girder with internal pre-stressing (in transition spans)

Comparative advantages/disadvantages of the above types are given below:

(a) Pre-cast Segmental Box Girder using External Unbounded Tendon.

This essentially consists of precast segmental construction with external pre-stressing and dry joints and is by far most preferred technique in fast track projects. In such construction the pre-stressing is placed outside the structural concrete (inside the box section) and protected with high density polyethylene tubes, which are grouted with special wax or cement. The match cast joints at the interface of two segments are provided with shear keys as in traditional

segmental construction. However, epoxy is dispensed with because water tight seal at the segment joints is not required because tendons are laid externally & protected by special wax or cement.

The main advantages of dry-jointed externally pre-stressed precast segmental construction can be summarized as follows:-

- Simplification of all post-tensioning operations, especially installation of tendons.
- Reduction in structural concrete thickness as no space is occupied by the tendons inside the concrete.
- Good corrosion protection due to tendons in polyethylene ducts, the grout inspection is easier and leaks, if any, can be identified during the grouting process.
- Simplified segment casting. There is no concern about alignment of tendons.
- Increased speed of construction.
- The elimination of the epoxy from the match-cast joints reduces costs and increases speed of construction further.
- Replacement of tendons in case of distress is possible and can be done in a safe and convenient manner.
- Facility for inspection and monitoring of tendons during the entire service life of the structure.

Precast Segmental Box Girder using internal tendon is also use.

(b) Precast Segmental U-Channel Superstructure with Internal Pre-stressing.

The single U type of viaduct structure is also a precast segmental construction with internal pre-stressing and requires gluing and temporary pre-stressing of segments. The match cast joints at the interface of two segments are also provided with shear keys. The main advantages for this type of structural configuration of superstructure are:

1. Built in sound barrier.

2. Built in cable support and system function.

3. Possibility to lower the longitudinal profile by approximately 1m compared to conventional design.

4. Built in structural elements capable to maintain the trains on the bridge in case of derailment (a standard barrier design allow this)

5. Built in maintenance and evacuation path on either side of the track.

(c) Double U-girder with internal pre-stressing

Girders of various spans(19m,22m, 25m and 30m) are cast in casting yard, pre-stressed internally. These girders are transported to site in trailors and launched in position by using double cranes of suitable capacity one on either end. Great advantage of these girders is launching being done in the night without disturbing the normal traffic. It will have better quality control due to the fact that all the girders are shop manufactured.

(d) I-girders with internal pre-stressing

In transition spans and other locations, where found necessary due to ir-regular widths and spans, I-girders with internal pre-stressing may be provided in order to have continuous structure.

Twin U-girders are normally economical as compared to segmental U-girders and box girders.

In view of above, Twin U-girders is recommended for adoption.

5.3 CONSTRUCTION METHODOLOGY

For the elevated sections it is recommended to have pre-cast segmental construction for super structure for the viaduct. For stations also the superstructure is generally of pre-cast members. The pre-cast construction will have following advantages:-

- Reduction in construction period due to concurrent working for substructure and superstructure.

- For segmental, pre-cast element (of generally 3.0m length), transportation from construction depot to site is easy and economical.

- Minimum inconvenience is caused to the public utilizing the road as the superstructure launching is carried out through launching girder requiring narrow width of the road.

- As the pre-cast elements are cast on production line in a construction depot, very good quality can be ensured.

- The method is environment friendly as no concreting work is carried at site for the superstructure.

5.4 PRE-CAST CONSTRUCTION

5.4.1 Casting of Segments

For viaducts segmental pre-cast construction requires a casting yard. The construction depot will have facilities for casting beds, curing and stacking area, batching plant with storage facilities for aggregates and cement, site testing laboratories, reinforcement steel yard and fabrication yard etc. An area of about 3.0 ha to 3.5 ha is required for each construction depot.

For casting of segments both long line and short line method can be adopted. However the long line method is more suitable for spans curved in plan while short line method is good for straight spans. A high degree of accuracy is required for setting out the curves on long line method for which pre calculation of offsets is necessary. Match casting of segments is required in either method. The cast segments are cured on the bed as well as in stacking yard. Ends of the segments are to be made rough through sand blasting so that gluing of segments can be effective.

The cast segment will be transported on trailers and launched in position through launching girders.

5.4.2 Launching Scheme

Launching girder is specially designed for launching of segments. The suggested launching scheme is designed in such a way that initially the launching girder is erected on pier head at one end of the work. The segments are lifted in sequence and when the lifting is over, they are dry matched while hanging from

the launching girder. After dry matching, the Segments are glued with epoxy and pre-stressed from one end. The girder is lowered on the temporary / permanent bearings after pre-stressing. The launching girder then moves over the launched span to next span and the sequences continue.

5.5 STRUCTURAL SYSTEM OF VIADUCT

5.5.1 Superstructure

The superstructure of a large part of the viaduct comprises of simply supported spans. However at major crossing over or along existing bridge, special steel or continuous unit will be provided.

Normally the Box Girder having a soffit width of about 4.0 m (approx) accommodates the two tracks situated at 4.2m center to center (c/c). The Box Girder superstructure for almost all the simply supported standard spans will be constructed by precast pre-stressed segmental construction with epoxy bonded joints.

The standard spans c/c of piers of simply supported spans constructed by precast segmental construction technique has been proposed as 25.0m. The usual segments shall be 3.0m in length except the Diaphragm segments, which shall be 2.0m each. The other spans (c/c of pier) comprises of 31.0 m, 25.0 m, 22.0 m, 19.0 m & 16.0 m, which shall be made by removing/adding usual segments of 3.0 m each from the center of the span.

The pier segment will be finalized based on simply supported span of 31.0m and the same will be also kept for all simply supported standard span.

For major crossing having spans greater than 31.0m, special continuous units normally of 3 span construction or steel girders have been envisaged.

All these continuous units (in case provided at obligatory location) will be constructed by cast-in-situ balanced cantilever construction technique.

5.5.2 Substructure

The viaduct superstructure will be supported on single cast-in-place RC pier. The shape of the pier follows the flow of forces. For the standard spans, the pier gradually widens at the top to support the bearing under the box webs. At this

preliminary design stage, the size of pier is found to be limited to 1.8m to 2.0 m diameter of circular shape for most of its height so that it occupies the minimum space at ground level where the alignment often follows the central verge of existing roads.

To prevent the direct collision of vehicle to pier, a Jersey Shaped crash barrier of 1.0 m height above existing road level has been provided all around the pier. A gap of 25 mm has also been provided in between the crash barrier and outer face of pier. The shape of upper part of pier has been so dimensioned that a required clearance of 5.5 m is always available on road side beyond vertical plane drawn on outer face of crash barrier. In such a situation, the minimum height of rail above the existing road is 8.4 m.

The longitudinal center to center spacing of elastomeric/pot bearing over a pier would be about 1.8 m. The space between the elastomeric bearings will be utilized for placing the lifting jack required for the replacement of elastomeric bearing. An outward slope of 1:200 will be provided at pier top for the drainage due to spilling of rainwater, if any.

The transverse spacing between bearings would be 3.2 m (to be studied in more details).
The orientation and dimensions of the piers for the continuous units or steel girder (simply supported span) have to be carefully selected to ensure minimum occupation at ground level traffic. Since the vertical and horizontal loads will vary from pier to pier, this will be catered to by selecting the appropriate structural dimensions.

5.5.3 Deck – Simple Spans

Salient features of the precast segmental construction method technique as envisaged for the project under consideration are indicated below:
The superstructure shall be constructed “span by span” sequentially, starting at one end of a continuous stretch and finishing at the other end. Nos. of launching girders may be required so as to work on different stretches simultaneously to enable completion of the project in time.

The number of “breaks” in the stretch can be identified by Nos. of continuous units & stations. The suggested method of erection will be detailed in drawings to be prepared, at the time of detailed design. The launching girder (or, more accurately, the “assembly truss”) is capable of supporting the entire dead load of

one span and transferring it to the temporary brackets attached to the pier. The governing weight of the segments will be of the order of 50t (to be finalized). The launching girder envisaged will be slightly longer than two span lengths. It must be able to negotiate curves in conjunction with temporary brackets.

Transportation of segments from casting yard to the point of erection will be effected by appropriately designed low-bedded trailers (tyre-mounted). The segments can be lifted and erected using erection portal gantry moving on launching girder.

Box girder segments shall be match cast at the casting yard before being transported to location and erected in position. Post-tensioned cables shall be threaded in-situ and tensioned from one end. It is emphasized that for precast segmental construction only one-end pre-stressing shall be used.

The pre-stressing steel and pre-stressing system steel accessories shall be subjected to an acceptance test prior to their actual use on the works. The tests for the system shall be as per FIP Recommendations as stipulated in the special specifications. Only multi-strand jacks shall be used for tensioning of cables. Direct and indirect force measurement device (e.g. Pressure Gauge) shall be attached in consultation with system manufacturer.

The Contractor shall be responsible for the proper handling, lifting, storing, transporting and erection of all segments so that they may be placed in the structure without damage. Segments shall be maintained in an upright position at all times and shall be stored, lifted and/or moved in a manner to prevent torsion and other undue stress. Members shall be lifted, hoisted or stored with lifting devices approved on the shop drawings.

5.5.4 Epoxy Bonded Joints and Shear Keys

A minimum compressive stress of 3 kg/sq cm shall be provided uniformly over the cross-section for the closure stress on the epoxied joint until the epoxy has set. The curing period for application of the compressive stress, method of mixing and application of epoxy and all related aspects including surface preparation shall be as per approved manufacturer's specifications.

The purpose of the epoxy joint, which is about 1mm on each mating surface, shall be to serve as lubricant during segment positioning, to provide

waterproofing of the joints for durability in service conditions and to provide a seal to avoid cross-over of grout during grouting of one cable into other ducts.

The epoxy shall be special purpose and meet requirements of relevant provision of FIP (International Federation of Pre-stressed Concrete)

The temporary compressive stress during the curing period shall be applied by approved external temporary bar pre-stressing (such as Macalloy or Diwidag bar systems or approved equivalent).

5.6 CONSTRUCTION OF STATIONS

It is proposed to construct the elevated stations with elevated concourse over the road at most of the locations to minimize land acquisition. Because of different entity than viaduct, a separate structural configuration is required to be proposed, although this may necessitate the break in the launching operations at each station location

In the cross section of station structure, viaduct column will be located on the median/footpath and other supporting columns/piers shall be provided on other sides of the road within ROW.

In some the cases which suffer space constraint, the cross section is supported by a single viaduct column and cantilever arms to eliminate the columns in the right of way.

5.6.1 Grade of Concrete

It is proposed to carry out construction work with design mix concrete through computerized automatic Batching Plants with following grade of concrete for various members as per design requirement/durability considerations.

i)	Piles	-	M -35
ii)	Pile cap and open foundation	-	M -35
iii)	Piers	-	M -40
iv)	All precast element for viaduct and station	-	M -45
v)	Cantilever piers and portals	-	M -45/M -60
vi)	Other miscellaneous structure	-	M -30

For all the main structures, permeability test on concrete sample is recommended to ensure impermeable concrete.

5.6.2 Reinforcement and pre-stressed Steel

It is proposed to use HYSD 500 or TMT steel as reinforcement bars. For pre-stressing work, low relaxation high tensile steel strands with the configuration 12 K 15 and or 19 K 15 is recommended (confirming to IS:14268).

5.6.3 Road width required during construction

As most of the construction is to be carried out on the middle of the road, central two lanes including median will be required for construction activities. During piling and open foundation work, a width of about 9 m will be required for construction and the same will be barricaded. It is proposed that two lanes are provided for traffic on either side during construction by widening of roads, if necessary. In certain cases, one way traffic may be resorted to.

All these actions will require a minimum period of about 4 to 6 months. During this period, the implementing agency can go ahead with the following preliminary works:

i) Preliminary action for diversion of utility and preparation of estimates thereof.

ii) Reservation of land along the corridor, identification and survey for acquisition.

5.6.4 TUNNELLING

5.6.4.1 TBM

A tunnel boring machine (TBM), also known as a "mole", is a machine used to excavate tunnels with a circular cross section through a variety of soil and rock strata.

Tunnel boring machines are used as an alternative to drilling and blasting methods in rock and conventional "hand mining" in soil. TBMs have the advantages of limiting the disturbance to the surrounding ground and producing a smooth tunnel wall. This significantly reduces the cost of lining the tunnel, and makes them suitable to use in heavily urbanized areas.. This is because tunneling with TBMs is much more efficient and results in shortened completion times.

5.6.4.2 NATM

The New Austrian tunneling method (NATM), also known as sequential excavation method (SEM), is a method of modern tunnel design and construction. The name NATM was intended to distinguish it from the old Austrian tunnelling approach. The fundamental difference between this new method of tunneling, as opposed to earlier methods, comes from the economic advantages made available by taking advantage of the inherent geological strength available in the surrounding rock mass to stabilize the tunnel

5.6.4.3 Cross Passage

Cross-passages are reinforced concrete structures built in between either two tunnels (as in case of twin tunnel systems) or a tunnel and the ground surface. They are provided to serve two primary purposes namely, emergency escape and maintenance work. As a standard practice, during construction of a tunnel system, depending upon length of the tunnel and soil characteristics, a pre-calculated number of cross-passages are planned.

5.6.4.4 Precast Tunnel Segments

Tunnels play a key role in assisting the movement of people, goods, and special services. The functionality of tunnels depends on the structural and durability performance of its **lining system.** Tunnel lining systems act as lines of defense against large overburden loads and complex geotechnical surrounding conditions. The use of **precast concrete tunnel segments** has been escalating due to its efficient and economical installation process compared to that of normal cast in-situ lining practice. Generally, precast concrete tunnel segments are designed for 120 years of service life. Normally, each ring consists of 05 segments and 01 key stone. Each segment weighs about 3 tons.

5.7 GEO-TECHNICAL INVESTIGATIONS

5.7.1 Methodology of Investigation

The investigation has been planned to obtain the subsurface stratification in the proposed project area and collect soil samples for laboratory testing to determine the engineering properties such as shear strength, along with basic engineering classification of the subsurface stratum to arrive at the foundation design parameters.

For Geotechnical investigation work, boring / drilling rig has been installed at the specified borehole locations. Stability of rig was ensured by making the ground level. Boring has been advanced by shell and auger method in soil and sampling carried out at regular interval in the bore hole.

The rig deployed was suitable for and had arrangement for boring, conducting Standard Penetration Test (SPT), collection of Undisturbed Soil Sample (UDS) and Disturbed Soil Samples (DS).

5.7.2 Standard Penetration Tests (SPT)

Standard Penetration Tests (SPT) has been conducted at different depths in these bore holes. SPT split spoon sampler of standard dimensions has been driven into the soil from the borehole bottom using 63.5 kg hammer falling from 75 cm height. The SPT weight was mechanically lifted to the specified height and allowed to fall freely on the anvil with the use of cathead winch with one to one and half turn of the drum. The efficiency of the SPT blows is expected to be 65 to 75% in this system, and can be directly used in most of analyses involving “N” values.

Blow counts for the penetration of every 15 cm were recorded and the N is reported as the blow counts for 30 cm penetration of the sampler leaving the first 15 cm penetration as seating driver.

When the number of blows exceeded 50 to penetrate the first or second 15 cm length of the sampler, the SPT N is regarded as more than 100 as described in IS 2131 – 1981. The test is terminated in such case and a record of penetration of the sampler under 50 blows is made. SPT refusal is recorded when there is no penetration of the sampler at any stage and also when a rebound of the sounding system is recorded. These tests were conducted at close intervals of 1.5m at shallow depths so that a continuous SPT N profile is available.

5.7.3 Disturbed Sampling in Boreholes:

Disturbed soil samples collected in the SPT sampler have been preserved in polythene covers and transported to the laboratory. One more polythene cover was provided to prevent the loss of moisture during the transit period.

5.7.4 Undisturbed Sampling in Boreholes

Undisturbed samples were collected using 100mm dia and 600mm long MS tubes as per IS 2132:1981 provided with sampler head with ball check arrangement. These samples are expected to be “good” for conducing direct shear tests and for determining filed density and void ratio.

The laboratory testing was done following the testing procedures given in the relevant parts of IS-2720. No rock has been encountered near both the corridors of Surat Metro under Phase I

5.7.5 Laboratory Investigation

The following laboratory tests were conducted on the selected samples recovered from the test boreholes:

Tests conducted on Soil Samples

(a) Particle size analysis:

(i) Sieve analysis

(ii) Hydrometer analysis

(b) Consistency tests

(i) Liquid limit

(ii) Plastic limit

(c) Bulk & Dry density

(d) Water content

(e) Specific gravity

(f) Direct shear test

(g) Tri-axial test

All the above laboratory tests were carried out as per relevant Indian Standards. All the samples were identified and classified as per IS: 1498-1970.

5.7.6 General Geology & Related Characteristics

The sites investigated are located in Surat City for both the corridors proposed under Surat Metro Rail network (Phase-I). The general features are described below:

Introduction

Surat is a port city situated on the banks of the Tapi river. Damming of the Tapi caused the original port facilities to close; the nearest port is now in the Magdalla and Hazira area of Surat Metropolitan Region. The city is located at 21°18′N 72°83′E& has a total geographical area of 4109.74 sq. km. The district is situated in the southern part of the state. It is bounded by the Arabian Sea on

the west, Bharuch in the North, the Valsad district in the south, Dangs district in the south-east, and the Tapi district of the in the east. The district has been divided into nine talukas.

Geology

Major geological formations exposed in the district are Quaternary alluvium, Tertiary limestone and sandstones and Deccan Trap basalt. Geographically, the district belongs to the western coastlands of the Deccan peninsula. The main Sahyadri scarp is a little outside the limits of the district towards east, but it gives the district its orientation, landscape features and drainage pattern. Distinct zones, viz., hilly areas, piedmont slopes, alluvial plains and coastal plains.

Drainage

The Tapi is the major river which passes through the central parts of the district and flows towards the west. The river is perennial in nature. It originates in Madhya Pradesh near Betul and has about 62225 sq. Km. of catchment area. The average width of the river upstream of Kathorbridge in about 500m. Downstream of the bridge the average being about 700m.

Soil Type

The basaltic lava flows are covered by black clayey to loamy soil. It is in general ranges in thickness up to one meter. The colour of the soil turns brown due to high iron content at places. In the piedmont slope area the soil is shallow to moderately deep, moderate to severely eroded and non calcareous in nature. The texture is silt clay loam to clay loams. The clay content varies from 30 to 60%. The water holding capacity of the soil is moderate. In the midland and flood plain areas, the soil is deep to very deep, light greyish to yellowish brown in colour. The texture is fine clay loam to sandy loam. The clay content varies from 25 to 60%. In coastal region the soil is deep to very deep dark grey to black colour. The texture is clay loam to silty loam. The area is affected by tide as well as leaching of salts from up land forming saline alkali soils. The content of clay in this is high and permeability is low (mud flats).

Hydrometeorology

The district has the meteorological station in the district town Surat, therefore, the climatological data of Surat IMD station is discussed here. There are many rain gauge stations being monitored by different state government agencies. The Water Resources Investigation Circle (WRI) under the department of Narmada

Water Resources, Govt. of Gujarat, monitors most of the rain gauges stations and also collects and compiles the rainfall data collected by different agencies.

Table:- II IMD Climatological Data

Month	Max Temp	Min Temp	Humidity	Wind Speed	Sunshine	PET	Rainfall
	(°C)	(°C)	(%)	Km/h	(Hours)	(mm/d)	(mm)
January	31.5	14.3	52.0	4.70	9.7	3.9	0.0
February	33.5	16.1	47.5	4.83	10.3	4.7	0.4
March	36.2	20.1	48.0	5.10	9.9	5.7	1.5
April	37.7	23.7	52.0	5.65	10.5	6.6	0.3
May	36.4	26.4	61.5	8.10	10.3	6.9	7.3
June	33.8	26.6	74.5	9.19	7.4	5.5	249.3
July	31.0	25.4	83.5	8.58	4.0	3.9	417.7
August	30.6	25.0	84.0	7.48	4.3	3.8	299.4
September	32.2	24.4	78.0	5.38	6.2	4.2	190.7
October	35.9	23.0	60.5	4.22	9.4	4.8	27.2
November	35.3	19.3	52.0	4.49	9.7	4.3	13.0
December	32.9	16.1	54.0	4.90	9.5	3.9	2.6
Total	-	-	-		-	-	**1209.4**
Average	**33.9**	**21.7**	**62.3**	**6.05**	**8.4**	**4.8**	-

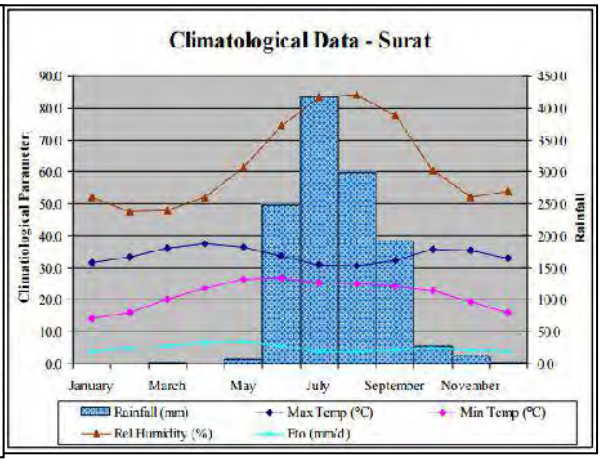


Fig-5.1 Climatological Data

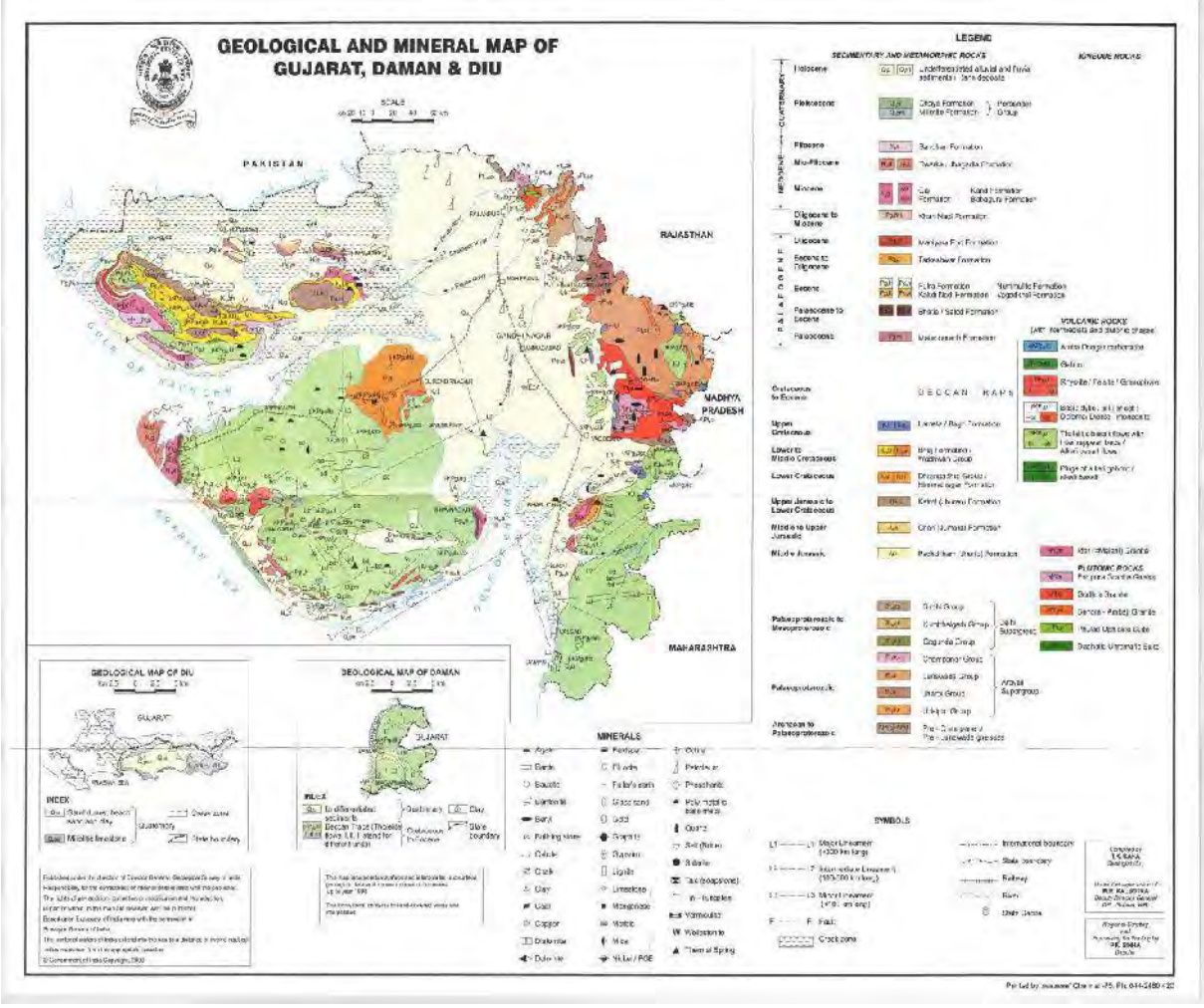


Fig- 5.2: Geological Map of Gujarat & Daman &Diu

5.7.7 Seismicity

The seismic hazard map of India was updated in 2002 by the Bureau of Indian Standards (BIS) as per IS:1893 (Part-1)-2002. The area under study and its surroundings are seismically active falls in Seismic Zone–III. In seismic design Zone factor, Z of 0.16 is recommended. The Seismic map & recent earthquake map is as follows.

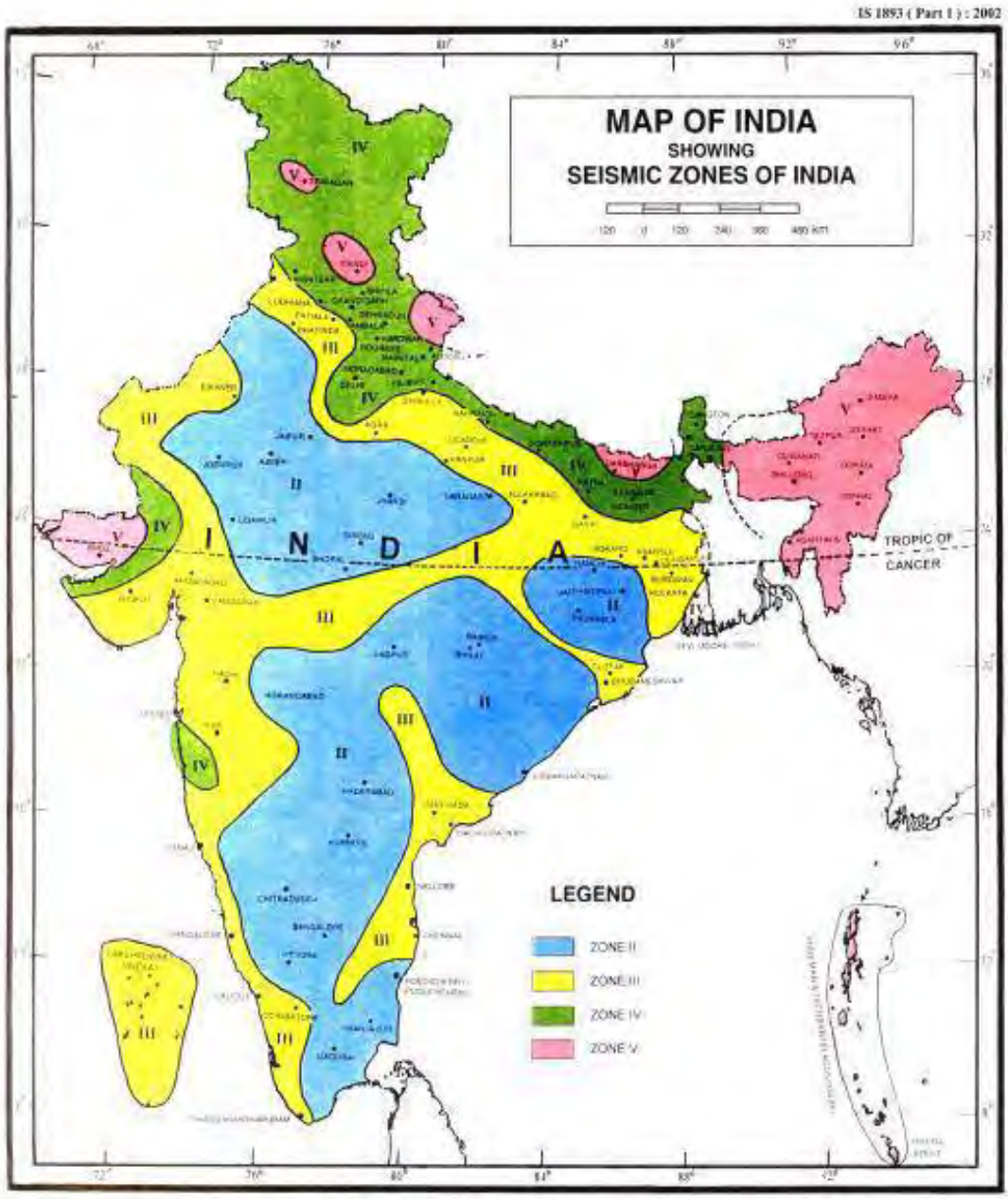


Fig-5.3: Seismic Zone of India (IS: 1893(Part-I)-2002)

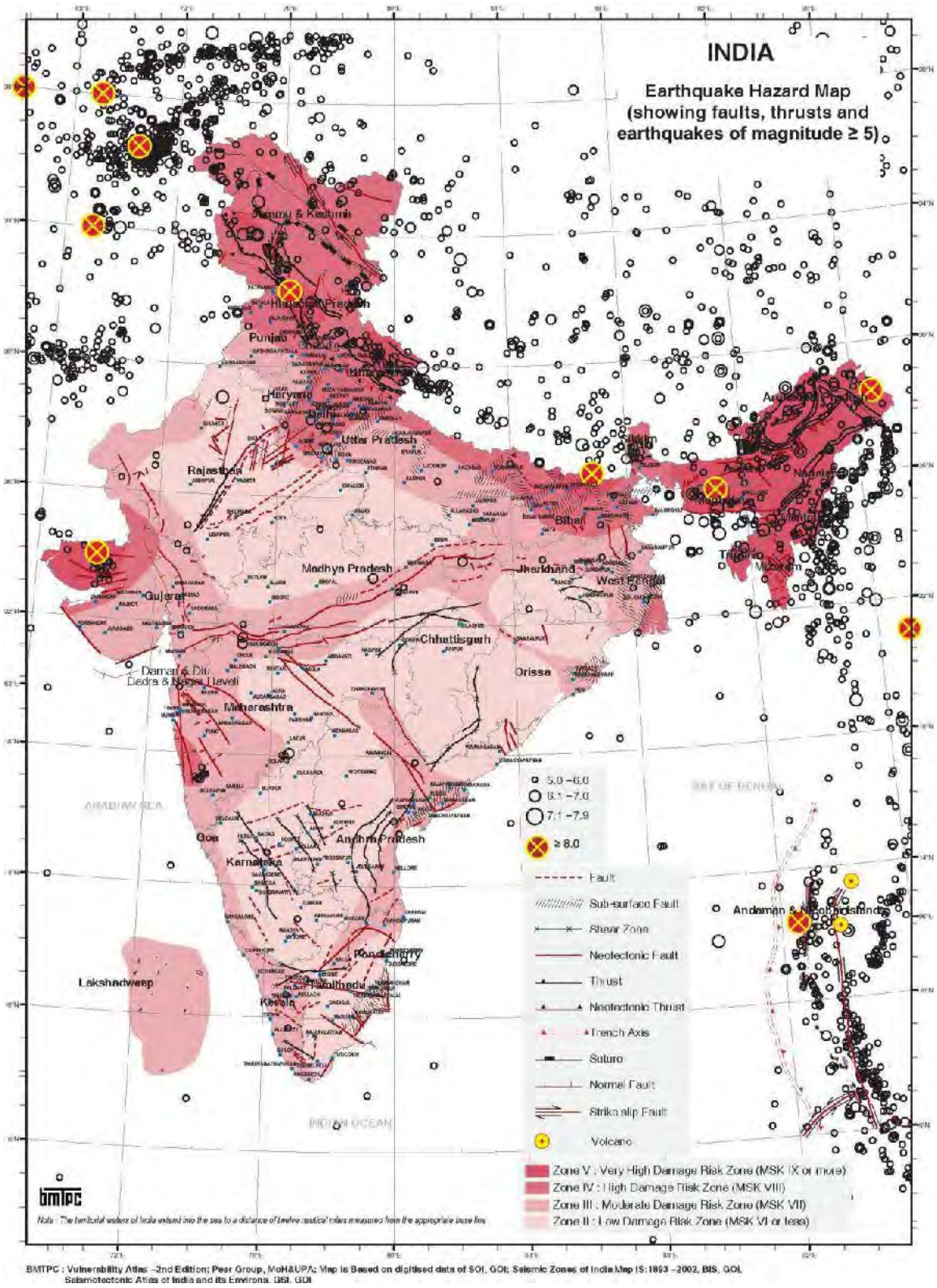


Fig-5.4: Earthquake Hazard Map of India

5.7.8 Field Investigations:

Detailed field investigations have been carried out during 2016-17 along both the corridors under Surat Metro network (Phase-1).

5.7.9 Foundations

Type of Foundation: Considering the nature of soil, type of proposed structures and expected loads on foundations, the recommended type of foundations is Pile Foundation. For the prevailing soil conditions and type of structures, bored cast-in-situ piles shall be adopted.

BRIEF DETAILS OF GEO-TECHNICAL INVESTIGATIONS

5.8 CORRIDOR-I : SARTHANA TO DREAM CITY

Total of 43 boreholes were drilled up to a maximum depth of 30.0-m below existing ground level. A map indicating the locations of boreholes on Sarthana to Dream City Corridor is shown in Figure 5.5.

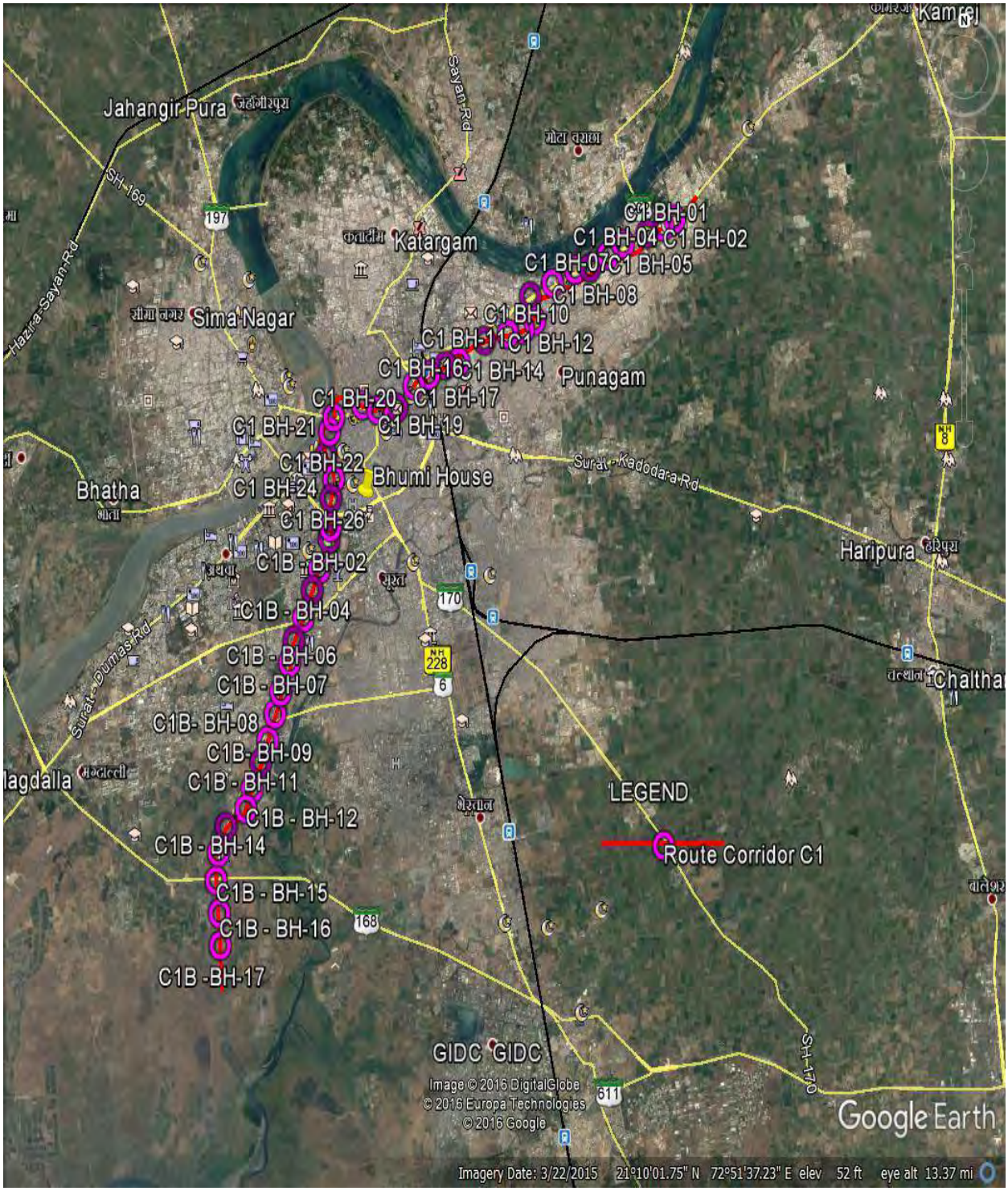


Fig. 5.5: Location of Bore Holes on Sarthana to Dream City, Corridor

5.8.1 Coordinates and other details for Bore Holes on Corridor-1(Sarthana to Dream city) are given in Table 5.1.

Table No. 5.1: Coordinates of Bore Holes on Corridor-I Sarthana-Dream City

S.No.	Corridor No.	Bore hole No.	Location	Ground RL (m)	Investigation Depth (m)	Depth of Water	Coordinates	
							N	E
01	CI	01	SarthanaJakat Naka-overbridge	58.53	30.00	18.00	21°13'49.31"N	72°54'4.69"E
02	CI	02	Nr.deepkamal Complex	59.59	30.00	18.50	21°13'44.74"N	72°53'54.91"E
03	CI	03	Simada Naka	59.27	30.00	17.00	21°13'40.47"N	72°53'40.43"
04	CI	04	Nana Varachha Over Bridge	56.00	30.00	13.00	21°13'34.23"N	72°53'20.41"E
05	CI	05	Foot Over Bridge-Chopati	59.32	30.00	15.00	21°13'27.21"N	72°53'6.84"E
06	CI	06	Nr. CNG Pump	58.58	30.00	16.50	21°13'19.65"N	72°52'51.46"E
07	CI	07	Nr. Kapodra Police Station	58.16	30.00	10.80	21°13'18.58"N	72°52'38.33"E
08	CI	08	Nr. Jalaram Furniture	59.04	30.00	15.30	21°13'10.24"N	72°52'18.76"E
09	CI	09	KapodraChhar Rasta	58.46	30.00	16.00	21°13'2.25"N	72°51'59.43"E
10	CI	10	Nr Rachna Circle	57.13	30.00	15.20	21°12'47.09"N	72°52'3.24"E
11	CI	11	Gayatri Three Road	57.62	30.00	10.50	21 °12'41.84"N	72°51'53.58"E
12	CI	12	Ghanshyam Nagar	57.65	30.00	10.80	21°12'39.12"n	72°51'40.11"E
13	CI	13	Nr. Matawadi Circle-Labheshwar	56.00	30.00	7.60	21°12'35.17"N	72°51'19.91"E
14	CI	14	Nr. VasantBhikhaniwadi	55.50	30.00	9.50	21°12'22.13"N	72°50'57.57"E
15	CI	15	Nr. Bhavani Circle-Afil Tower	55.30.	30.00	13.50	21°12'14.05"N	72°50'46.20"E
16	CI	16	LH Temple Nr. Garnala	54.50	30.00	15.00	21°12'7.75"N	72°50'31.77"E
17	CI	17	Tower Road-Nr. Hotel Ostria	55.60	30.00	N.O	21°12'7.75"N	72°50'18.80"E
18	CI	18	Mahidharpura post office	54.12	30.00	20.00	21°11'54,26"N	72°50'4.46"E
19	CI	19	Bhagat Char Rasta	54.73	30.00	N.O	21°11'52.86"N	72°49'47.69"E

20	CI	20	Lal Gate	55.33	30.00	13.00	21°11'55.82"N	72°49'34.50"E
21	CI	21	Nr. Chowk-ITI College	52.58	30.00	7.80	21°11'48.36"N	72°49'9.83"E
22	CI	22	Near Gandhi Baug	52.25	30.00	4.50	21°11'39.74"N	72°49'6.20"E
23	CI	23	Makaipul to Nanpura road	51.41	30.00	8.30	21°11'25.39"N	72°49.3.23"E
24	CI	24	Nr. SMC Multi level Parking	49.63	30.00	10.00	21°11'25.39"N	72°49'3.23"E
25	CI	25	Sharda Hospital	50.95	30.00	13.50	21°10'59.28"N	72°49'7.73"E
26	CI	26	Nr. Income Tax Office	53.40	30.00	10.00	21°10'41.98"N	72°49'7.32"E
27	CIB	01	Nr. Fire Station-GhodDod Road	53.75	30.00	10.00	21°10'32.70"N	72°49'5.45"E
28	CIB	02	Nr. Canal-Old Raj Empire, SMC, Plot	52.98	30.00	7.00	21°10'16.68"N	72°48'58.01"E
29	CIB	03	Opp. Amar Kunj, Gangor Sweet	52.37	30.00	8.00	21°10'1.70"N	72°48'51.05"E
30	CIB	04	Bhatar Char Rasta	51.57	30.00	11.00	21°9'44.81"N	72°48'42.15"E
31	CIB	05	Nr. Shopping Complex-B/H Bhatar Housing	51.63	30.00	10.00	21°9'36.36"N	72°48'37.82"E
32	CIB	06	Shree Ram Marble	51.08	30.00	6.00	21°9'17.33"N	72°48'32.19"E
33	CIB	07	SohamCircalAlthan canal	51.36	30.00	9.00	21°9'3.12"N	72°48'25.43"E
34	CIB	08	Althan Canal – Blue Arc site	50.46	30.00	9.00	21°8'49.98"N	72°48'20.83"E
35	CIB	09	Nr. Atlantis Shopping-Canal Road, Char Rasta	50.99	30.00	7.10	21°8'34.09"N	72°48'14.58"E
36	CIB	10	Raghuveer-New Khadi Bridge Junction/VIP Gelleria	50.44	30.00	10.00	21°8'20.86"N	72°48'8.69"E
37	CIB	11	Shiv Someshwar Enclave Char Rasta	49.81	30.00	6.50	21°8'5.31"N	72°48'1.52"E
38	CIB	12	B/H. Shiv Resi-Bhimrad-Althan Canal End	48.11	30.00	7.00	21°7'53.38"N	72°47'56.37"E
39	CIB	13	HappAmdavai	47.90	30.00	6.60	21°7'43.32"N	72°47'38.72"E

			Tava Fry-Sarsana					
40	CIB	14	Sarsana Village Road	49.77	30.00	8.00	21°7'27.77"N	72°47'32.60"E
41	CIB	15	NH-06 Junction-KhajodChakdi	46.87	30.00	4.50	21°7'8.52"N	72°47'31.67"E
42	CIB	16	500m towards Dream City	46.39	30.00	5.00	21°6'51.82"N	72°47'31.67"E
43	CIB	17	1 KM towards Dream City	46.57	30.00	5.00	21°6'32.60"N	72°47'35.39"E

5.8.2 Properties of Sub Soil Strata

Borehole wise details of soil strata are available in Detailed Geo-Technical report.

5.8.3 Discussion and Type of Foundation

(a) Bored Cast in situ RCC Pile

Depending on the field and laboratory observations of subsoil strata, test results and the type of structures proposed at site, the most feasible soil-foundation system is recommended as normal bored cast in situ R.C.C. piles foundations of 1.00m & 1.20m diameter at different depths with cut-off level at 2.00m depth below existing Ground level. The safe load carrying capacities of these piles are given in following table.

Table 5.2: Safe Load carrying Capacities of Bored Piles on Corridor-I Sarthana-Dream City

Borehole No.	Capacity Type (Tonne)	Pile Cut off Level (m)	Pile Shaft Length from Cut off level (m)					
			1000mm Dia			1200mm Dia		
			24.00	25.00	26.00	24.00	25.00	26.00
C1-01	Vertical	2.00	294.00	305.00	316.00	523.00	541.00	558.00
	Uplift		190.00	203.00	217.00	261.00	280.00	300.00
	Lateral		12.64			16.00		
C1-02	Vertical	2.00	291.00	303.00	314.00	506.00	524.00	541.00
	Uplift		181.00	195.00	209.00	178.00	192.00	206.00
	Lateral		15.00			21.00		
C1-03	Vertical	2.00	337.00	348.00	359.00	573.00	591.00	607.00
	Uplift		229.00	243.00	256.00	308.00	328.00	347.00
	Lateral		11.00			15.00		

C1-04	Vertical	2.00	364.00	375.00	386.00	606.00	624.00	640.00
	Uplift		255.00	269.00	283.00	245.00	259.00	273.00
	Lateral		12.00			16.00		
C1-05	Vertical	2.00	362.00	374.00	386.00	599.00	617.00	634.00
	Uplift		243.00	257.00	272.00	233.00	247.00	261.00
	Lateral		13.00			18.00		
C1-06	Vertical	2.00	324.00	336.00	347.00	556.00	574.00	591.00
	Uplift		219.00	233.00	246.00	303.00	323.00	342.00
	Lateral		14.00			19.00		
C1-07	Vertical	2.00	357.00	368.00	380.00	593.00	611.00	628.00
	Uplift		244.00	259.00	273.00	320.00	340.00	359.00
	Lateral		12.00			16.00		
C1-08	Vertical	2.00	295.00	306.00	317.00	517.00	535.00	552.00
	Uplift		194.00	207.00	221.00	267.00	287.00	306.00
	Lateral		18.00			24.00		
C1-09	Vertical	2.00	247.00	257.00	268.00	444.00	462.00	479.00
	Uplift		159.00	172.00	184.00	221.00	240.00	258.00
	Lateral		11.00			15.00		
C1-10	Vertical	2.00	266.00	276.00	286.00	469.00	487.00	504.00
	Uplift		205.00	218.00	232.00	275.00	295.00	314.00
	Lateral		10.00			14.00		
C1-11	Vertical	2.00	256.00	267.00	277.00	462.00	480.00	497.00
	Uplift		160.00	173.00	186.00	222.00	241.00	260.00
	Lateral		13.00			17.00		
C1-12	Vertical	2.00	305.00	317.00	328.00	528.00	546.00	563.00
	Uplift		202.00	216.00	230.00	273.00	293.00	312.00
	Lateral		11.00			15.00		
C1-13	Vertical	2.00	277.00	289.00	300.00	494.00	511.00	528.00
	Uplift		161.00	176.00	190.00	221.00	241.00	260.00
	Lateral		13.00			17.00		
C1-14	Vertical	2.00	354.00	365.00	376.00	594.00	612.00	629.00
	Uplift		261.00	274.00	288.00	345.00	365.00	384.00
	Lateral		11.00			14.00		
C1-15	Vertical	2.00	356.00	367.00	379.00	598.00	616.00	633.00
	Uplift		245.00	259.00	273.00	327.00	347.00	366.00

	Lateral		16.00			22.00		
C1-16	Vertical	2.00	341.00	352.00	363.00	579.00	597.00	614.00
	Uplift		237.00	250.00	264.00	317.00	336.00	356.00
	Lateral		9.00			12.00		
C1-17	Vertical	2.00	348.00	359.00	370.00	589.00	607.00	624.00
	Uplift		247.00	260.00	274.00	328.00	347.00	367.00
	Lateral		11.00			15.00		
C1-18	Vertical	2.00	360.00	372.00	383.00	601.00	619.00	636.00
	Uplift		251.00	265.00	279.00	333.00	352.00	372.00
	Lateral		12.00			17.00		
C1-19	Vertical	2.00	349.00	360.00	371.00	588.00	606.00	623.00
	Uplift		246.00	260.00	274.00	330.00	349.00	369.00
	Lateral		12.00			16.00		
C1-20	Vertical	2.00	310.00	321.00	332.00	540.00	558.00	574.00
	Uplift		232.00	246.00	260.00	311.00	331.00	350.00
	Lateral		10.00			13.00		
C1-21	Vertical	2.00	404.00	417.00	429.00	674.00	692.00	709.00
	Uplift		249.00	263.00	278.00	329.00	348.00	368.00
	Lateral		12.00			16.00		
C1-22	Vertical	2.00	352.00	364.00	376.00	599.00	617.00	634.00
	Uplift		219.00	233.00	247.00	289.00	309.00	328.00
	Lateral		10.00			14.00		
C1-23	Vertical	2.00	411.00	423.00	435.00	678.00	696.00	713.00
	Uplift		256.00	270.00	285.00	333.00	353.00	372.00
	Lateral		10.00			13.00		
C1-24	Vertical	2.00	370.00	381.00	393.00	627.00	645.00	662.00
	Uplift		229.00	242.00	256.00	306.00	326.00	345.00
	Lateral		10.00			14.00		
C1-25	Vertical	2.00	411.00	424.00	437.00	678.00	696.00	713.00
	Uplift		256.00	271.00	287.00	328.00	348.00	367.00
	Lateral		15.00			20.00		
C1-26	Vertical	2.00	408.00	421.00	434.00	685.00	703.00	720.00
	Uplift		223.00	238.00	254.00	282.00	302.00	321.00
	Lateral		17.00			23.00		

C1B-01	Vertical	2.00	308.00	319.00	331.00	526.00	544.00	561.00
	Uplift		201.00	215.00	229.00	260.00	280.00	299.00
	Lateral		13.00			17.00		
C1B-02	Vertical	2.000	298.00	309.00	320.00	519.00	537.00	554.00
	Uplift		195.00	209.00	222.00	261.00	281.00	300.00
	Lateral		13.00			18.00		
C1B-03	Vertical	2.00	346.00	358.00	369.00	584.00	602.00	619.00
	Uplift		241.00	255.00	268.00	324.00	343.00	363.00
	Lateral		11.00			15.00		
C1B-04	Vertical	2.00	322.00	333.00	345.00	550.00	568.00	585.00
	Uplift		187.00	201.00	215.00	255.00	274.00	294.00
	Lateral		12.00			17.00		
C1B-05	Vertical	2.00	300.00	311.00	322.00	526.00	544.00	561.00
	Uplift		196.00	210.00	224.00	272.00	292.00	311.00
	Lateral		12.00			16.00		
C1B-06	Vertical	2.00	296.00	307.00	319.00	522.00	540.00	557.00
	Uplift		164.00	178.00	193.00	220.00	240.00	259.00
	Lateral		12.00			16.00		
C1B-07	Vertical	2.00	218.00	209.00	206.00	319.00	307.00	301.00
	Uplift		201.00	213.00	224.00	259.00	274.00	288.00
	Lateral		12.00			16.00		
C1B-08	Vertical	2.00	337.00	349.00	361.00	567.00	585.00	602.00
	Uplift		234.00	248.00	262.00	310.00	330.00	349.00
	Lateral		9.00			12.00		
C1B-09	Vertical	2.00	258.00	269.00	280.00	449.00	466.00	483.00
	Uplift		157.00	170.00	184.00	204.00	224.00	243.00
	Lateral		10.00			14.00		

C1B-10	Vertical	2.00	282.00	292.00	302.00	492.00	510.00	527.00
	Uplift		192.00	205.00	218.00	263.00	281.00	300.00
	Lateral		9.00			12.00		
C1B-11	Vertical	2.00	374.00	386.00	398.00	616.00	634.00	651.00
	Uplift		261.00	276.00	290.00	342.00	362.00	381.00
	Lateral		8.00			11.00		
C1B-12	Vertical	2.00	262.00	273.00	284.00	467.00	485.00	502.00
	Uplift		144.00	158.00	172.00	197.00	217.00	236.00
	Lateral		10.00			13.00		
C1B-13	Vertical	2.00	292.00	303.00	315.00	516.00	534.00	551.00
	Uplift		142.00	157.00	171.00	191.00	211.00	230.00
	Lateral		11.00			15.00		
C1B-14	Vertical	2.00	260.00	272.00	283.00	458.00	476.00	493.00
	Uplift		157.00	171.00	184.00	215.00	234.00	254.00
	Lateral		12.00			17.00		
C1B-15	Vertical	2.00	371.00	384.00	396.00	610.00	628.00	645.00
	Uplift		263.00	278.00	293.00	339.00	359.00	378.00
	Lateral		11.00			15.00		
C1B-16	Vertical	2.00	224.00	234.00	245.00	379.00	396.00	412.00
	Uplift		136.00	149.00	162.00	187.00	205.00	224.00
	Lateral		11.00			14.00		
C1B-17	Vertical	2.00	288.00	300.00	312.00	487.00	505.00	522.00
	Uplift		137.00	152.00	166.00	186.00	205.00	225.00
	Lateral		12.00			16.00		

(b) Open Foundation

Depending on the field and laboratory observations of subsoil strata, test results and the type of structures proposed at site, the types of foundations, depths and net safe bearing capacities recommended for design purposes are given in the following table. The net SBC/SBP in the following table are the lower of the

values obtained from shear failure criterion as per IS: 6403 and settlement failure criterion as per IS: 8009, Part-I.

Table 5.3: Safe Bearing Capacities & Foundation Details (Corridor-I)Sarthana-Dream City

Foundation Details				Safe Bearing Capacity (SBC) t /m^2 (kN/m^2)	Settlement based on SPT-N /Consolidation Values under recommended value of SBC (mm)	Recommended Load (maximum) t /m^2 (kN/m^2)
Borehole No.	Size		Depth from Ground Level (m)			
	Width (m)	Length (m)				
C1-01	2.75	3.00	2.50	23.00 (226.00)	26	**23.00 (226.00)**
	2.50	2.75		23.00 (226.00)	23	**23.00 (226.00)**
	2.25	2.50		23.00 (226.00)	20	**23.00 (226.00)**
C1-02	2.75	3.00	2.50	21.00 (206.00)	29	**21.00 (206.00)**
	2.50	2.75		21.50 (212.00)	27	**21.50 (212.00)**
	2.25	2.50		22.00 (216.00)	25	**22.00 (216.00)**
C1-03	2.75	3.00	2.50	24.00 (235.00)	29	**24.00 (235.00)**
	2.50	2.75		24.00 (235.00)	27	**24.00 (235.00)**
	2.25	2.50		24.00 (235.00)	25	**24.00 (235.00)**
C1-04	2.75	3.00	2.50	24.00 (235.00)	35	**24.00 (235.00)**
	2.50	2.75		24.00 (235.00)	34	**24.00 (235.00)**
	2.25	2.50		24.00 (235.00)	33	**24.00 (235.00)**
C1-05	2.75	3.00	2.50	25.00 (245.00)	43	**25.00 (245.00)**
	2.50	2.75		25.00 (245.00)	39	**25.00 (245.00)**
	2.25	2.50		25.00 (245.00)	35	**25.00 (245.00)**
C1-06	2.75	3.00	2.50	24.00 (235.00)	44	**24.00 (235.00)**
	2.50	2.75		25.00 (245.00)	40	**25.00 (245.00)**
	2.25	2.50		25.00 (245.00)	36	**25.00 (245.00)**
C1-07	2.75	3.00	3.50	26.00 (255.00)	38	**26.00 (255.00)**
	2.50	2.75		26.50 (260.00)	34	**26.50 (260.00)**
	2.25	2.50		27.00 (265.00)	31	**27.00 (265.00)**

C1-08	2.75	3.00	2.50	22.00 (216.00)	34	**22.00 (216.00)**
	2.50	2.75		22.50 (221.00)	32	**22.50 (221.00)**
	2.25	2.50		23.00 (226.00)	28	**23.00 (226.00)**
C1-09	2.75	3.00	5.00	25.00 (245.00)	37	**25.00 (245.00)**
	2.50	2.75		25.50 (250.00)	34	**25.50 (250.00)**
	2.25	2.50		26.00 (255.00)	32	**26.00 (255.00)**
C1-10	2.75	3.00	4.00	23.00 (226.00)	47	**23.00 (226.00)**
	2.50	2.75		23.50 (230.00)	43	**23.50 (230.00)**
	2.25	2.50		24.00 (235.00)	40	**24.00 (235.00)**
C1-11	2.75	3.00	4.00	24.00 (235.00)	28	**24.00 (235.00)**
	2.50	2.75		25.00 (245.00)	25	**25.00 (245.00)**
	2.25	2.50		26.00 (255.00)	23	**26.00 (255.00)**
C1-12	2.75	3.00	3.00	24.00 (235.00)	31	**24.00 (235.00)**
	2.50	2.75		24.00 (235.00)	28	**24.00 (235.00)**
	2.25	2.50		24.00 (235.00)	26	**24.00 (235.00)**
C1-13	2.75	3.00	3.00	25.00 (245.00)	48	**25.00 (245.00)**
	2.50	2.75		26.00 (255.00)	44	**26.00 (255.00)**
	2.25	2.50		26.00 (255.00)	39	**26.00 (255.00)**
C1-14	2.75	3.00	3.00	24.00 (235.00)	34	**24.00 (235.00)**
	2.50	2.75		24.00 (235.00)	31	**24.00 (235.00)**
	2.25	2.50		25.00 (245.00)	29	**25.00 (245.00)**
C1-15	2.75	3.00	3.00	24.00 (235.00)	31	**24.00 (235.00)**
	2.50	2.75		24.00 (235.00)	28	**24.00 (235.00)**
	2.25	2.50		25.00 (245.00)	25	**25.00 (245.00)**
C1-16	2.75	3.00	3.00	22.00 (216.00)	29	**22.00 (216.00)**
	2.50	2.75		22.00 (216.00)	27	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	24	**22.00 (216.00)**

C1-17	2.75	3.00	3.00	24.00 (235.00)	31	**24.00 (235.00)**
	2.50	2.75		24.00 (235.00)	28	**24.00 (235.00)**
	2.25	2.50		25.00 (245.00)	26	**25.00 (245.00)**
C1-18	2.75	3.00	3.00	21.00 (206.00)	49	**21.00 (206.00)**
	2.50	2.75		22.00 (216.00)	44	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	39	**22.00 (216.00)**
C1-19	2.75	3.00	3.00	22.00 (216.00)	49	**22.00 (216.00)**
	2.50	2.75		22.00 (216.00)	45	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	40	**22.00 (216.00)**
C1-20	2.75	3.00	3.00	22.00 (216.00)	27	**22.00 (216.00)**
	2.50	2.75		22.00 (216.00)	25	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	22	**22.00 (216.00)**
C1-21	2.75	3.00	3.00	23.00 (226.00)	46	**23.00 (226.00)**
	2.50	2.75		23.00 (226.00)	42	**23.00 (226.00)**
	2.25	2.50		23.00 (226.00)	37	**23.00 (226.00)**
C1-22	2.75	3.00	3.00	23.00 (226.00)	29	**23.00 (226.00)**
	2.50	2.75		23.50 (230.00)	26	**23.50 (230.00)**
	2.25	2.50		24.00 (235.00)	24	**24.00 (235.00)**
C1-23	2.75	3.00	4.00	16.00 (157.00)	14	**16.00 (157.00)**
	2.50	2.75		16.50 (162.00)	13	**16.50 (162.00)**
	2.25	2.50		17.00 (167.00)	12	**17.00 (167.00)**
C1-24	2.75	3.00	4.50	27.00 (265.00)	45	**27.00 (265.00)**
	2.50	2.75		27.50 (270.00)	42	**27.50 (270.00)**
	2.25	2.50		28.00 (275.00)	38	**28.00 (275.00)**
C1-25	2.75	3.00	3.00	24.00 (235.00)	31	**24.00 (235.00)**
	2.50	2.75		25.00 (245.00)	28	**25.00 (245.00)**
	2.25	2.50		25.00 (245.00)	25	**25.00 (245.00)**

C1-26	2.75	3.00	3.00	24.00 (235.00)	37	**24.00 (235.00)**
	2.50	2.75		25.00 (245.00)	33	**25.00 (245.00)**
	2.25	2.50		25.00 (245.00)	30	**25.00 (245.00)**
C1-B-01	2.75	3.00	2.50	23.00 (226.00)	41	**23.00 (226.00)**
	2.50	2.75		23.00 (226.00)	40	**23.00 (226.00)**
	2.25	2.50		24.00 (235.00)	39	**24.00 (235.00)**
C1-B-02	2.75	3.00	2.50	26.00 (255.00)	24	**26.00 (255.00)**
	2.50	2.75		26.00 (255.00)	22	**26.00 (255.00)**
	2.25	2.50		26.00 (255.00)	19	**26.00 (255.00)**
C1-B-03	2.75	3.00	2.50	23.00 (226.00)	20	**23.00 (226.00)**
	2.50	2.75		23.00 (226.00)	18	**23.00 (226.00)**
	2.25	2.50		23.00 (226.00)	16	**23.00 (226.00)**
C1-B-04	2.75	3.00	3.00	25.00 (245.00)	24	**25.00 (245.00)**
	2.50	2.75		25.00 (245.00)	22	**25.00 (245.00)**
	2.25	2.50		26.00 (255.00)	20	**26.00 (255.00)**
C1-B-05	2.75	3.00	3.00	23.00 (226.00)	27	**23.00 (226.00)**
	2.50	2.75		23.50 (230.00)	25	**23.50 (230.00)**
	2.25	2.50		24.00 (235.00)	22	**24.00 (235.00)**
C1-B-06	2.75	3.00	3.00	23.00 (226.00)	27	**23.00 (226.00)**
	2.50	2.75		24.00 (235.00)	24	**24.00 (235.00)**
	2.25	2.50		24.00 (235.00)	22	**24.00 (235.00)**
C1-B-07	2.75	3.00	3.00	22.00 (216.00)	22	**22.00 (216.00)**
	2.50	2.75		22.50 (221.00)	20	**22.50 (221.00)**
	2.25	2.50		23.00 (226.00)	18	**23.00 (226.00)**
C1-B-08	2.75	3.00	4.00	25.00 (245.00)	28	**25.00 (245.00)**
	2.50	2.75		25.00 (245.00)	26	**25.00 (245.00)**
	2.25	2.50		26.00 (255.00)	24	**26.00 (255.00)**

C1-B-09	2.75	3.00	3.00	22.00 (216.00)	24	**22.00 (216.00)**
	2.50	2.75		23.00 (226.00)	22	**23.00 (226.00)**
	2.25	2.50		23.00 (226.00)	20	**23.00 (226.00)**
C1-B-10	2.75	3.00	5.50	25.00 (245.00)	40	**25.00 (245.00)**
	2.50	2.75		26.00 (255.00)	39	**26.00 (255.00)**
	2.25	2.50		27.00 (265.00)	38	**27.00 (265.00)**
C1-B-11	2.75	3.00	2.50	19.00 (186.00)	39	**19.00 (186.00)**
	2.50	2.75		19.00 (186.00)	38	**19.00 (186.00)**
	2.25	2.50		20.00 (196.00)	37	**20.00 (196.00)**
C1-B-12	2.75	3.00	5.00	23.00 (226.00)	48	**23.00 (226.00)**
	2.50	2.75		24.00 (235.00)	47	**24.00 (235.00)**
	2.25	2.50		24.00 (235.00)	46	**24.00 (235.00)**
C1-B-13	2.75	3.00	3.00	16.00 (157.00)	28	**16.00 (157.00)**
	2.50	2.75		16.00 (157.00)	26	**16.00 (157.00)**
	2.25	2.50		16.00 (157.00)	23	**16.00 (157.00)**
C1-B-14	2.75	3.00	2.50	24.00 (235.00)	29	**24.00 (235.00)**
	2.50	2.75		25.00 (245.00)	26	**25.00 (245.00)**
	2.25	2.50		25.00 (245.00)	23	**25.00 (245.00)**
C1-B-15	2.75	3.00	2.50	23.00 (226.00)	33	**23.00 (226.00)**
	2.50	2.75		23.50 (230.00)	30	**23.50 (230.00)**
	2.25	2.50		24.00 (235.00)	27	**24.00 (235.00)**
C1-B-16	2.75	3.00	3.00	18.00 (176.00)	25	**18.00 (176.00)**
	2.50	2.75		18.00 (176.00)	23	**18.00 (176.00)**
	2.25	2.50		18.00 (176.00)	20	**18.00 (176.00)**
C1-B-17	2.75	3.00	4.00	23.50 (230.00)	26	**23.50 (230.00)**
	2.50	2.75		24.00 (235.00)	24	**24.00 (235.00)**
	2.25	2.50		25.00 (245.00)	22	**25.00 (245.00)**

5.9 CORRIDOR – 2: BHESAN TO SAROLI

Total of 35 boreholes were drilled up to a maximum depth of 30.0-m below existing ground level. A map indicating locations of bore holes on Bhesan to Saroli Corridor is shown in Figure 5-6.

Fig. 5.6: Location of Bore Holes on Bhesan to Saroli Corridor

5.9.1 Co-ordinates and other details for Bore Holes on Corridor-2 (Bhesan to Saroli) are given in Table 5.4.

Table 5.4: Co-ordinates of Bore Holes on Corridor-2 (Bhesan-Saroli)

Sr. No	Corridor No.	Borehole No.	Location	Ground RL (m)	Investigation Depth (m)	Depth of Water Table (m)	Coordinates	
							N	E
01	C2A	01	Bhesan Road -Gaurav Path Junction	52.05	30.00	5.70	21°13'9.00"N	72°46'1.05"E
02	C2A	02	Bhesan Road - GHB site	52.04	30.00	4.80	21°13'13.36"N	72°46'17.51"E
03	C2A	03	Bhesan Road - Canal Junction	52.63	30.00	4.70	21°13'16.62"N	72°46'38.86"E
04	C2A	04	Bhesan Road - Nr. Rander Water Works	51.36	30.00	5.20	21°13'17.26"N	72°46'52.16"E
05	C2A	05	Palanpur road - Curve Road	51.08	30.00	5.60	21°13'0.30"N	72°46'52.62"E
06	C2A	06	Pyramid Marble	51.24	30.00	6.00	21°12'48.86"N	72°46'57.21"E
07	C2A	07	Palanpur Patia	51.28	30.00	6.50	21°12'32.58"N	72°46'57.22"E
08	C2A	08	vidhya Kunj school	51.06	30.00	6.20	21°12'14.58"N	72°46'56.09"E
09	C2A	09	LP Savani School	50.56	30.00	5.50	21°12'3.41"N	72°46'54.43"E
10	C2A	10	Nr. IOCL Pump - Bhagat party plot	51.48	30.00	5.30	21°11'50.51"N	72°47'3.92"E
11	C2A	11	TGB Circle	51.34	30.00	5.50	21°11'37.41"N	72°47'12.54"E
12	C2A	12	Trinity circle to shell pump road	50.93	30.00	6.00	21°11'19.15"N	72°47'23.86"E
13	C2A	13	B/h. Cine police- Star Bazar	49.80	30.00	4.60	21°11'11.31"N	72°47'32.31"E
14	C2A	14	Adajan Village circle	50.66	30.00	5.00	21°11'22.78"N	72°47'45.41"E
15	C2A	15	Badrinath Temple Road	51.06	30.00	4.00	21°11'15.94"N	72°48'4.09"E
16	C2A	16	Athwalines Chopati	52.70	30.00	8.00	21°11'2.41"N	72°48'21.82"E
17	C2A	17	Athwalines Circle	52.33	30.00	7.50	21°11'6.94"N	72°48'38.96"E
18	C2A	18	Opp. RTO - Vanita Vishram	52.42	30.00	10.00	21°10'59.73"N	72°48'49.03"E
19	C2A	19	Civil Hospital Kidny hospital	52.98	30.00	18.00	21°10'48.62"N	72°49'11.00"E
20	C2AB1	20	Center point - Sub Jail	53.65	30.00	19.20	21°10'50.53"N	72°49'29.15"E
21	C2AB1	21	Udhna Darwaja - Gujarat Samachar	54.62	30.00	16.50	21°10'58.39"N	72°49'51.29"E

22	C2-AB3	01	Man Darwaja-Bamba Gate	54.410	30.00	12.00	21°11'9.42"N	72°50'11.85"E
23	C2-AB3	02	KamelaDarwaja-Over Bridge	54.810	30.00	10.00	21°111'16.38"N	72°50'23.87"E
24	C2-AB3	03	Someshwar Textile Market Chokdi-B/H.Millenium	50.840	30.00	8.00	21°11'12.77"N	72°50'41.87"E
25	C-2AB3	04	Nr. Railway Under Bridge	50.400	30.00	10.00	21°11'9.29"N	72°50'51.68"E
26	C2-AB3	05	Anjana Over Bridge – Canal Road	55.200	30.00	8.30	21°11'4.59"N	72°51'3.12"E
27	C2-AB3	06	SamratVidhyalay BRTS Station	54.670	30.00	12.00	21°11'11.17"N	72°51'19.85"E
28	C2-AB3	07	Model Town BRTS Station	54.740	30.00	13.00	21°11'16.14"N	72°51'33.67"E
29	C2-AB3	08	ParvatPatiaChokdi	54.340	30.00	11.00	21°11'27.63"N	72°51'49.64"E
30	C2-AB3	09	Opp.CNG Pump	54.890	30.00	12.00	21°11'25.50"N	72°52'5..42"E
31	C2-AB3	10	Rajput SamajBhavan-Bhakti	54.290	30.00	7.80	21°11'24.85"N	72°52'19.33"E
32	C2-AB3	11	Opp. Landmark Gate	54.230	30.00	9.00	21°11'25.47"N	72°52'32.20"E
33	C2-AB3	12	100m Towards ParvatPatia from Nirali Hospital	54.130	30.00	6.0	21°11'21.20"N	72°52'56.26"E
34	C2-AB3	13	Nr.Saroli Gate	55.080	30.00	13.00	21°11'22.89"N	72°53'14.14"E
35	C2-AB3	14	KumbhariaKhadi Bridge	53.720	30.00	14.00	21°11'21.90"N	72°53'28.07"E

5.9.2 Properties of Sub Soil Strata

Borehole wise details of soil strata are available in Detailed Geo-Technical Report

5.9.3 Discussion and Type of Foundation

5.9.3.1 Bored Cast –in-situ RCC Pile

Depending on the field and laboratory observations of subsoil strata, test results and the type of structures proposed at site, the most feasible soil-foundation system is recommended as normal bored cast in situ R.C.C. piles foundations of 1.00m & 1.20m diameter at different depths with cut-off level at 2.00m depth below existing Ground level. The safe load carrying capacities of these piles are given in following table.

Table 5.5: Safe Load Carrying Capacities of Bored Piles on Corridor-2

Borehole No:	Capacity Type (Tonne)	Pile Cut off Level (m)	Pile Shaft Length from Cut off level (m)					
			1000mm Dia			1200mm Dia		
			24.00	25.00	26.00	24.00	25.00	26.00
C2A-01	Vertical	2.00	337.00	349.00	360.00	568.00	586.00	603.00
	Uplift		228.00	242.00	256.00	303.00	323.00	342.00
	Lateral		11.00			15.00		
C2A-02	Vertical	2.000	247.00	257.00	267.00	445.00	463.00	480.00
	Uplift		201.00	217.00	233.00	279.00	301.00	324.00
	Lateral		11.00			15.00		
C2A-03	Vertical	2.00	259.00	270.00	281.00	463.00	481.00	498.00
	Uplift		149.00	163.00	176.00	202.00	222.00	241.00
	Lateral		14.00			19.00		
C2A-04	Vertical	2.00	314.00	325.00	336.00	537.00	555.00	572.00
	Uplift		213.00	226.00	240.00	281.00	301.00	320.00
	Lateral		11.00			15.00		
C2A-05	Vertical	2.00	325.00	336.00	346.00	560.00	578.00	595.00
	Uplift		238.00	251.00	264.00	318.00	336.00	355.00
	Lateral		12.00					
C2A-06	Vertical	2.00	387.00	399.00	412.00	657.00	675.00	692.00
	Uplift		247.00	262.00	276.00	325.00	345.00	364.00
	Lateral		9.00			12.00		
C2A-07	Vertical	2.00	364.00	376.00	387.00	616.00	634.00	651.00
	Uplift		228.00	243.00	257.00	296.00	316.00	335.00
	Lateral		10.00			13.00		
C2A-08	Vertical	2.00	349.00	361.00	373.00	596.00	613.00	630.00
	Uplift		215.00	230.00	244.00	274.00	294.00	313.00
	Lateral		13.00			18.00		
C2A-09	Vertical	2.00	448.00	461.00	474.00	723.00	741.00	758.00
	Uplift		280.00	296.00	311.00	352.00	371.00	391.00
	Lateral		11.00			12.00		

C2A-10	Vertical	2.00	416.00	429.00	441.00	688.00	706.00	723.00
	Uplift		269.00	284.00	299.00	344.00	364.00	383.00
	Lateral		12.00			16.00		
C2A-11	Vertical	2.00	338.00	350.00	363.00	580.00	598.00	615.00
	Uplift		193.00	208.00	223.00	247.00	266.00	286.00
			9.00			12.00		
C2A-12	Vertical	2.00	370.00	382.00	393.00	616.00	634.00	651.00
	Uplift		260.00	274.00	288.00	344.00	364.00	383.00
	Lateral		9.17			12.00		
C2A-13	Vertical	2.000	411.00	424.00	436.00	677.00	695.00	711.00
	Uplift		268.00	283.00	298.00	347.00	367.00	386.00
	Lateral		19.00			26.00		
C2A-14	Vertical	2.00	341.00	353.00	365.00	576.00	594.00	611.00
	Uplift		213.00	227.00	242.00	281.00	300.00	320.00
	Lateral		13.00			18.00		
C2A-15	Vertical	2.00	414.00	426.00	438.00	681.00	699.00	716.00
	Uplift		247.00	261.00	274.00	330.00	350.00	369.00
	Lateral		10.00			14.00		
C2A-16	Vertical	2.00	217.00	226.00	217.00	310.00	322.00	309.00
	Uplift		191.00	201.00	210.00	239.00	251.00	261.00
	Lateral		15.00			20.00		
C2A-17	Vertical	2.00	183.00	174.00	283.00	259.00	247.00	400.00
	Uplift		155.00	163.00	182.00	194.00	205.00	228.00
	Lateral		11.00			15.00		
C2A-18	Vertical	2.00	285.00	296.00	307.00	487.00	505.00	522.00
	Uplift		170.00	183.00	197.00	220.00	239.00	259.00
	Lateral		11.00			15.00		
C2A-19	Vertical	2.00	360.00	373.00	385.00	618.00	636.00	653.00
	Uplift		224.00	239.00	254.00	299.00	318.00	338.00
	Lateral		13.00			18.00		

C2AB1-20	Vertical	2.00	336.00	347.00	358.00	564.00	582.00	598.00
	Uplift		240.00	254.00	267.00	322.00	341.00	361.00
	Lateral		35.00			47.00		
C2AB1-21	Vertical	2.00	345.00	357.00	368.00	587.00	605.00	622.00
	Uplift		247.00	261.00	275.00	332.00	352.00	371.00
	Lateral		21.00			29.00		
C2AB3-01	Vertical	2.00	274.00	285.00	296.00	485.00	503.00	520.00
	Uplift		146.00	156.00	166.00	200.00	214.00	229.00
	Lateral		13.00			17.00		
C2AB3-02	Vertical	2.000	276.00	286.00	297.00	487.00	505.00	522.00
	Uplift		137.00	147.00	157.00	189.00	203.00	218.00
	Lateral		10.00			13.00		
C2AB3-03	Vertical	2.00	344.00	355.00	367.00	580.00	598.00	615.00
	Uplift		173.00	184.00	194.00	235.00	251.00	266.00
	Lateral		12.00			16.00		
C2AB3-04	Vertical	2.00	350.00	361.00	372.00	593.00	610.00	627.00
	Uplift		198.00	208.00	218.00	265.00	279.00	294.00
	Lateral		11.00			15.00		
C2AB3-05	Vertical	2.00	321.00	333.00	345.00	554.00	572.00	589.00
	Uplift		157.00	168.00	178.00	216.00	231.00	246.00
	Lateral		27.00			36.00		
C2AB3-06	Vertical	2.00	420.00	433.00	446.00	688.00	706.00	723.00
	Uplift		208.00	219.00	231.00	278.00	295.00	311.00
	Lateral		17.00			23.00		
C2AB3-07	Vertical	2.00	284.00	296.00	306.00	485.00	503.00	519.00
	Uplift		167.00	177.00	185.00	212.00	227.00	242.00
	Lateral		10.00			13.00		
C2AB3-	Vertical	2.00	325.00	336.00	348.00	554.00	572.00	589.00

08	Uplift		169.00	180.00	191.00	227.00	242.00	258.00
	Lateral		17.00			23.00		
C2AB3-09	Vertical	2.00	349.00	360.00	372.00	587.00	605.00	622.00
	Uplift		196.00	207.00	217.00	264.00	279.00	294.00
	Lateral		14.00			19.00		
C2AB3-10	Vertical	2.00	376.00	389.00	402.00	622.00	639.00	656.00
	Uplift		169.00	181.00	193.00	232.00	249.00	267.00
	Lateral		15.00			20.00		
C2AB3-11	Vertical	2.00	404.00	417.00	429.00	652.00	670.00	687.00
	Uplift		214.00	225.00	237.00	286.00	302.00	318.00
	Lateral		15.00			21.00		
C2AB3-12	Vertical	2.00	345.00	356.00	367.00	586.00	604.00	621.00
	Uplift		187.00	197.00	207.00	254.00	269.00	284.00
	Lateral		13.00			17.00		
C2AB3-13	Vertical	2.00	369.00	380.00	392.00	613.00	630.00	647.00
	Uplift		200.00	211.00	221.00	268.00	284.00	299.00
	Lateral		11.00			15.00		
C2AB3-14	Vertical	2.00	399.00	411.00	424.00	645.00	663.00	680.00
	Uplift		210.00	222.00	233.00	281.00	298.00	314.00
	Lateral		15.00			20.00		

5.9.3.2 Open Foundation

Depending on the field and laboratory observations of subsoil strata, test results and the type of structures proposed at site, the types of foundations, depths and net safe bearing capacities recommended for design purposes are given in the following table. The net SBC/SBP in the following table are the lower of the valies obtained from shear failure criterian as per IS;6403 and settlement failure criterion as per IS:8009, Part-I.

Table 5.6: Safe Bearing Capacities & Foundation Details (Corridor-2)

Foundation Details				Safe Bearing Capacity (SBC) t /m^2 (kN/m^2)	Settlement based on SPT-N /Consolidation Values under recommended value of SBC (mm)	Recommended Load (maximum) t /m^2 (kN/m^2)
Borehole No.	Size		Depth from Ground Level (m)			
	Width (m)	Length (m)				
C2A-01	2.75	3.00	3.00	21.00 (206.00)	30	**21.00 (206.00)**
	2.50	2.75		21.50 (211.00)	27	**21.50 (211.00)**
	2.25	2.50		22.00 (216.00)	25	**22.00 (216.00)**
C2A-02	2.75	3.00	2.50	20.00 (196.00)	31	**20.00 (196.00)**
	2.50	2.75		20.50 (201.00)	28	**20.50 (201.00)**
	2.25	2.50		21.00 (206.00)	25	**21.00 (206.00)**
C2A-03	2.75	3.00	3.00	18.00 (206.00)	27	**18.00 (206.00)**
	2.50	2.75		18.00 (206.00)	24	**18.00 (206.00)**
	2.25	2.50		19.00 (186.00)	22	**19.00 (186.00)**
C2A-04	2.75	3.00	3.00	19.00 (186.00)	33	**19.00 (186.00)**
	2.50	2.75		20.00 (196.00)	30	**20.00 (196.00)**
	2.25	2.50		20.00 (196.00)	27	**20.00 (196.00)**
C2A-05	2.75	3.00	3.00	21.00 (206.00)	39	**19.00 (186.00)**
	2.50	2.75		21.50 (211.00)	35	**20.00 (196.00)**
	2.25	2.50		22.00 (216.00)	32	**20.00 (196.00)**
C2A-06	2.75	3.00	3.00	21.00 (206.00)	63	**21.00 (206.00)**
	2.50	2.75		22.00 (216.00)	62	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	61	**22.00 (216.00)**
C2A-07	2.75	3.00	3.00	22.00 (216.00)	27	**22.00 (216.00)**
	2.50	2.75		22.00 (216.00)	24	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	22	**22.00 (216.00)**

C2A-08	2.75	3.00	3.00	22.00 (216.00)	41	**22.00 (216.00)**
	2.50	2.75		22.00 (216.00)	37	**22.00 (216.00)**
	2.25	2.50		23.00 (226.00)	33	**23.00 (226.00)**
C2A-09	2.75	3.00	3.00	21.00 (206.00)	32	**21.00 (206.00)**
	2.50	2.75		21.00 (206.00)	29	**21.00 (206.00)**
	2.25	2.50		22.00 (216.00)	26	**22.00 (216.00)**
C2A-10	2.75	3.00	3.00	21.00 (206.00)	28	**21.00 (206.00)**
	2.50	2.75		22.00 (216.00)	25	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	23	**22.00 (216.00)**
C2A-11	2.75	3.00	3.00	22.00 (216.00)	26	**22.00 (216.00)**
	2.50	2.75		23.00 (226.00)	24	**23.00 (226.00)**
	2.25	2.50		23.00 (226.00)	21	**23.00 (226.00)**
C2A-12	2.75	3.00	3.00	21.00 (206.00)	33	**21.00 (206.00)**
	2.50	2.75		21.00 (206.00)	31	**21.00 (206.00)**
	2.25	2.50		22.00 (216.00)	28	**22.00 (216.00)**
C2A-13	2.75	3.00	3.00	21.00 (206.00)	29	**21.00 (206.00)**
	2.50	2.75		22.00 (216.00)	27	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	25	**22.00 (216.00)**
C2A-14	2.75	3.00	3.00	21.00 (206.00)	30	**21.00 (206.00)**
	2.50	2.75		21.00 (206.00)	29	**21.00 (206.00)**
	2.25	2.50		21.00 (206.00)	28	**21.00 (206.00)**
C2A-15	2.75	3.00	3.00	16.00 (157.00)	34	**16.00 (157.00)**
	2.50	2.75		16.00 (157.00)	33	**16.00 (157.00)**
	2.25	2.50		16.00 (157.00)	32	**16.00 (157.00)**
C2A-16	2.75	3.00	3.50	25.00 (245.00)	33	**25.00 (245.00)**
	2.50	2.75		25.00 (245.00)	30	**25.00 (245.00)**
	2.25	2.50		26.00 (255.00)	27	**26.00 (255.00)**

C2A-17	2.75	3.00	3.00	22.00 (216.00)	41	**22.00 (216.00)**
	2.50	2.75		22.00 (216.00)	37	**22.00 (216.00)**
	2.25	2.50		23.00 (226.00)	33	**23.00 (226.00)**
C2A-18	2.75	3.00	2.50	19.00 (186.00)	27	**19.00 (186.00)**
	2.50	2.75		19.00 (186.00)	24	**19.00 (186.00)**
	2.25	2.50		19.00 (186.00)	21	**19.00 (186.00)**
C2A-19	2.75	3.00	2.50	22.00 (216.00)	32	**22.00 (216.00)**
	2.50	2.75		22.00 (216.00)	29	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	26	**22.00 (216.00)**
C2AB1-20	2.75	3.00	2.50	19.00 (186.00)	28	**19.00 (186.00)**
	2.50	2.75		19.00 (186.00)	25	**19.00 (186.00)**
	2.25	2.50		19.00 (186.00)	22	**19.00 (186.00)**
C2AB1-21	2.75	3.00	2.50	23.00 (226.00)	25	**23.00 (226.00)**
	2.50	2.75		24.00 (235.00)	24	**24.00 (235.00)**
	2.25	2.50		24.00 (235.00)	23	**24.00 (235.00)**

Foundation Details				Safe Bearing Capacity (SBC) t /m^2 (kN/m^2)	Settlement based on SPT-N /Consolidation Values under recommended value of SBC (mm)	Recommended Load (maximum) t /m^2 (kN/m^2)
Type	Size		Depth from Ground Level (m)			
	Width (m)	Length (m)				
01	2.75	3.00	3.50	23.00 (226.00)	44	**23.00 (226.00)**
	2.50	2.75		24.00 (235.00)	40	**24.00 (235.00)**
	2.25	2.50		24.00 (235.00)	36	**24.00 (235.00)**
02	2.75	3.00	5.00	27.00 (265.00)	27	**27.00 (265.00)**
	2.50	2.75		28.00 (275.00)	25	**28.00 (275.00)**
	2.25	2.50		29.00 (284.00)	23	**29.00 (284.00)**

03	2.75	3.00	3.50	19.00 (186.00)	26	**19.00 (186.00)**
	2.50	2.75		19.00 (186.00)	25	**19.00 (186.00)**
	2.25	2.50		19.00 (186.00)	24	**19.00 (186.00)**
04	2.75	3.00	3.00	25.00 (245.00)	32	**25.00 (245.00)**
	2.50	2.75		26.00 (255.00)	29	**26.00 (255.00)**
	2.25	2.50		26.00 (255.00)	26	**26.00 (255.00)**
05	2.75	3.00	4.00	24.00 (235.00)	32	**24.00 (235.00)**
	2.50	2.75		24.00 (235.00)	29	**24.00 (235.00)**
	2.25	2.50		24.00 (235.00)	26	**24.00 (235.00)**
06	2.75	3.00	2.50	21.00 (206.00)	31	**21.00 (206.00)**
	2.50	2.75		21.00 (206.00)	30	**21.00 (206.00)**
	2.25	2.50		21.00 (206.00)	29	**21.00 (206.00)**
07	2.75	3.00	2.50	24.00 (235.00)	43	**24.00 (235.00)**
	2.50	2.75		24.00 (235.00)	39	**24.00 (235.00)**
	2.25	2.50		25.00 (235.00)	35	**25.00 (235.00)**
08	2.75	3.00	2.50	22.00 (216.00)	33	**22.00 (216.00)**
	2.50	2.75		23.00 (226.00)	31	**23.00 (226.00)**
	2.25	2.50		23.00 (226.00)	28	**23.00 (226.00)**
09	2.75	3.00	2.50	22.00 (216.00)	31	**22.00 (216.00)**
	2.50	2.75		22.00 (216.00)	28	**22.00 (216.00)**
	2.25	2.50		22.00 (216.00)	25	**22.00 (216.00)**

10	2.75	3.00	2.50	22.00 (216.00)	33	**22.00 (216.00)**
	2.50	2.75		22.00 (216.00)	30	**22.00 (216.00)**
	2.25	2.50		23.00 (226.00)	27	**23.00 (226.00)**
11	2.75	3.00	2.50	25.00 (245.00)	31	**25.00 (245.00)**
	2.50	2.75		25.00 (255.00)	30	**25.00 (245.00)**
	2.25	2.50		26.00 (255.00)	29	**26.00 (255.00)**
12	2.75	3.00	2.50	22.00 (216.00)	30	**22.00 (216.00)**
	2.50	2.75		23.00 (226.00)	27	**23.00 (226.00)**
	2.25	2.50		23.00 (226.00)	24	**23.00 (226.00)**
13	2.75	3.00	2.50	24.00 (235.00)	34	**24.00 (235.00)**
	2.50	2.75		25.00 (245.00)	31	**25.00 (245.00)**
	2.25	2.50		25.00 (245.00)	28	**25.00 (245.00)**
14	2.75	3.00	2.50	23.00 (226.00)	37	**23.00 (226.00)**
	2.50	2.75		23.00 (226.00)	34	**23.00 (226.00)**
	2.25	2.50		23.00 (226.00)	30	**23.00 (226.00)**

5.10 LAND

In order to minimise land acquisitions and to provide good accessibility form either directions, the metro alignments are located mostly along the center of the roads, which lie on the corridor. But, at some locations the geometrics of the roads especially at road turnings may not match with geometric parameters required for metro rail systems. In such cases, either the alignment will be off the road or some properties abutting the road would get affected. Further, some land is required for various purposes as detailed below:

5.10.1 Land Requirement for following Major Components

MRTS Structure (including Route Alignment), Station Building, Platforms, Entry/Exit Structures, Traffic Integration Facilities, Depots, etc.
Receiving/Traction Sub-stations
Radio Towers
Temporary Construction Depots and work sites.
Staff quarters, office complex and operation control centre(OCC)

5.10.2 Land for Underground stretches

No land at surface is required permanently for underground section, except for small areas for entry/exit structures, traffic integration, chilling plant and ventilation shafts at stations. These will be located either on footpath edge or in front marginal open setback of the building along the road.

5.10.3 Land required for elevated stretches

For elevated section, single pier supporting the viaduct will be located on the middle of road so that the existing roads remain in use as usual. Accordingly, necessary permission for using such right-of-way will have to be obtained from the concerned authorities. Elevated station is generally proposed with elevated concourse so that land is required only for locating the entry/exit structures. Traffic integration facilities are provided wherever the same are required and, but no land is proposed for acquisition.

The normal viaduct structure of elevated Metro is about 10 m (edge to edge) wide. Ideally the required right of way is 10m. However, for reasons of safety a clean marginal distance / setback of about 5 m is necessary from either edge of the viaduct (or 10 m on both sides of the centre line) wherein no structures are to be located.. Also, it ensures road access and working space all along the viaduct for working of emergency equipments and fire brigade. In stretches, where the elevated alignment has to be located away from road, a strip of 20-m width is proposed for acquisition.

In view of the constraints on space on ground floor, it is proposed to provide the concourse area exactly below the Station Building at mezzanine level. All the stations in elevated stretch including terminal station are planned with side platforms. Normally, the ideal width required for station building in is 28.0m. The staircase giving access to concourse area from ground will be located at the edge of footpaths or in front marginal open setback of the buildings in the as far

as possible in the open space. Nevertheless it is not possible to find open space at all the locations therefore acquisition of certain private structures is inevitable.

5.10.4 Land for Switch-over Ramps

Switch-over ramps are required for transition from the underground to elevated section or *vice versa*. The ramp covers a stretch at ground for the whole width of structure for two tracks (about 10.5m including the protection works). The length of ramp above ground depends on the existing ground slope and the gradient provided on Metro alignment (normally 3% to 4%). Thus the ramp is to be located in an area where sufficient road width is available or in an open area. On this corridor, three such ramps are provided on the both the corridors.

5.10.5 Land for Traffic integration

As indicated in station planning para certain land is required for traffic integration at the each station. Efforts have been made to identify land required for traffic integration at each station to facilitate park and ride facility, but it is not possible to find open space at all the locations. Hence land for traffic integration has been marked in the drawing wherever is available.

5.10.6 Land for Traction and Receiving Substation and Radio Towers

Two RSS are proposed to be located for Sarthana-Dream City Corridor (i) near Nature Park (4000 Sq.m)and (ii)Kapodra Station (3000 Sq.m)and oneRSS having area as (4000 Sq.m) near Magub Station for Bhesan to Saroli Corridor. Hence, an area of 11000Sq.m has been earmarked for RSS. No additional land proposed for locating radio towers. These will be accommodated in the land already acquired. Land required for RSS will be as under:-

Table 5.7 : Details of Land for RSS

S. No.	Name of Corridor	Area (m^2)	Ownership
1.	Sarthana-Dream City	7000	Government
2.	Bhesan–Saroli	4000	Government
	Total	11000	

5.10.7 Land Requirement for Stations & Running section

As indicated earlier, the ROW of the roads along which the alignment is planned is sufficiently wide and hence no land is required for acquisition as long as the

alignment is straight and in the centre of the road. However, at curved portions, the alignment could not be kept in the centre of the road and land acquisition at such locations is inevitable in spite of introduction of sharper curves.

To the extent possible the Entry and Exit points of stations (underground and elevated) were planned on the foot paths. But, for locating other station facilities such as chiller plants, ventilation shafts, underground water tanks, generator set room etc., land acquisition is proposed.

The details of land permanently required for depot, running sections , stations, ramp, parking, property development etc are indicated in the Tables 5.8 to 5.13

Table 5.8 - Details of Land Required for Depot

S.No.	Location	Area(ha)	Ownership	Purpose
SARTHANA TO DREAM CITY CORRIDOR				
1.	Dream City	24.09	GoG	Depot
BHESAN TO SAROLI				
2.	Bhesan	16.95	SMC	Depot
	Total	41.04		

Table 5.9 – Details of Land Required for Running Section
Corridor-I :Sarthana – Dream City

S.No.	Plot No.	Area (Sqm)	Remarks
1.	RS1	69.25 Sqm.	Government
2.	RS2	124.38 Sqm.	Government
3.	RS	69.00 Sqm.	Government
4.	RS4	140.24 Sqm.	Government
5.	RS5	276.41 Sqm.	Government
6.	RS6	360.82 Sqm.	Government
7.	RS7	279.00 Sqm.	Government
8.	RS8	835.70 Sqm.	Government
9.	RS9	40.70 Sqm.	Government
10.	RS10	281.30 Sqm.	Government
11.	RS11	497.80 Sqm.	Government
12.	RS12	518 Sqm.	Government
	Total	3492.82 Sqm.	

Table 5.10 – Details of Land Required for Running Section

Corridor-II: Bhesan - Saroli

S.No.	Plot No.	Area (Sqm)	Remarks
1.	RS1	45.00 Sqm.	Government
2.	RS2	95.80 Sqm.	Government
3.	RS3	59.30 Sqm.	Government
4.	RS4	77.00 Sqm.	Government
5.	RS5	58.47 Sqm.	Government
6.	RS6	129.00 Sqm.	Government
7.	RS7	159.70 Sqm.	Government
8.	RS8	89.60 Sqm.	Government
9.	RS9	54.00 Sqm.	Government
10.	RS10	106.40 Sqm.	Government
11.	RS11	225.50 Sqm.	Government
12.	RS12	118.67 Sqm.	Government
13.	RS13	58.02 Sqm.	Government
14.	RS14	25.48 Sqm.	Government
	Total	1301.94 Sqm.	

Table 5.11 A - Details of Land Required for Stations

Station No.		Corridor 1 Sarthana to Dream City				
	Station Name	PLOT NO	AREA	LOCATION AS PER TP	OWNERS HIP	REMARKS
1	Sarthana	SN-1	535.9	21 (SARTHANA-SIMADA) FP - 61	Pvt.	Open+shed
		SN-2	574.5	22 (SARTHANA-VALAK) FP - 3	Pvt.	
2	Nature Park	BN-1	505.5	38 (NANAVARACHHA) FP - 23	Pvt.	house+Open
		BN-2	548.1	38 (NANAVARACHHA) FP - 31	Pvt.	maruti bakery+Open

3	Varachha Choupati Garden	JS-1	494.6	20 (NANAVARACHHA-KAPADRA) FP - 90	SMC	Open + School
		JS-2	481.9	20 (NANAVARACHHA-KAPADRA) FP - 24	Pvt.	Open plot
4	Kalakunj Swaminaray an Mandir	MP-1	439.0	16 (KAPADRA) FP - 12	Pvt.	Open plot
		MP-2	466.9	16 (KAPADRA) FP - 13	Pvt.	mamta park society
5	Kapodra	SS-1	180.7	16 (KAPADRA) FP - 44	SMC	open plot
		SS-2	175.1	16 (KAPADRA) FP - 51+52	SMC	open plot
6	Labheshwar Chowk	AS-1	690.1	17 (FULPADA) FP - 6	Pvt.	open+house
		AS-2	562.1	17 (FULPADA) FP - 5	Pvt.	open plot
		AS-3	688.1	17 (FULPADA) FP - 2	Pvt.	open+house
7	Central Warehouse	KN-1	174.8	8(UMARWADA) FP-13	SMC	Open plot
		KN-2	173.3	4(ASHWANIKUMAR-NAVAGAM) FP - M/7	SMC	Open Plot

8	Surat Railway Station	SR-1	185.9	LAMBE HUNUMAN ROAD	SMC	Surat Rly. Stn
		SR-2	180.5	LAMBE HUNUMAN ROAD	SMC	Surat Rly. Open area
9	Maskati Hospital	SMC-1	1902.2	WARD NO - 4,C.S.NO.680 + 681 AREA = 4611 SQ.MT.	GOVT.	police Stn.+police qutr. (to be acquired)
		SMC-2	723.0	WARD NO - 4, C-8 (C.S.NO.693 TO 708,709/P,710/P TO 731/P,732,735/C,75/17) AREA = 5592 SQ.MT.	Pvt.	under acquisition (MOCHINI CHAL)
10	Chowk Bazar	GB-1	166.5	WARD NO - 10,(P-29),C.S.NO.2272/P AREA = 4923.00 SQ.MT.	GOVT.	
		GB-2	162.2	WARD NO - 11,C.S.NO.2038, 2085	S.M.C. (CENTRAL GOVT.)	office+open
		GB-3	1073.7			
11	Kadarsha ni Nal	SC-1	535.9	WARD NO - 2,C.S.NO.1961 AREA - 8811 SQ.MT.	SMC	shopping complex
		SC-2	589.0	TP. NO-2 (NANPURA) F.P.NO-27 14812 SQ.MT.	SMC	water treatment plant
12	Majura Gate	MG-1		TP. NO-2 (NANPURA) F.P.NO-33 AREA 36625	PVT.	
		MG-2			PVT.	
13	Rupali Canal	SN-1	574.5	FINAL T.P.SNO -6 (MAJURA-KHATODARA) FP NO-12 Paikee	SMC	NEAR RUPALI CANAL CHAR RASTA

		SN-2	535.8	ROAD FINAL T.P.SNO -9(MAJURA) NEAR FP NO-167	SMC ROAD	
14	Althan Tenament	SV-1	574.5	PRELIMINARY T.P.SNO -28 (ALTHAN-BHATAR) FP.NO - 108	SUDA	OPEN PLOT
				PRELIMINARY T.P.SNO -28 (ALTHAN-BHATAR) FP.NO-109 Paikee	GRAM PANCHAYAT, BHATAR	
		SV-2	535.8	ROAD DRAFT T.P.SNO -27 (BHATAR-MAJURA) NEAR FP NO-62	SMC ROAD	OPEN PLOT
15	Althan Gam	AN-1	518.6	PRELIMINARY T.P.SNO -28 (ALTHAN-BHATAR) FP.NO - 51 Paikee	SMC	COMMUNITY CENTRE
		AN-2	578.3	PRELIMINARY T.P.SNO - 28 (ALTHAN-BHATAR) FP.NO - 139 Paikee	SMC	OPEN PLOT
16	VIP Road	AP-1	440.4	ROAD , PRELIMINARY T.P.SNO -28 (ALTHAN-BHATAR) NEAR, FP.NO - 10, FP NO-11	SMC ROAD	OPEN PLOT (VIP ROAD)
		AP-2	533.2	FINAL T.P.SNO -37 (ALTHAN-SOUTH) FP.NO 38 Paikee	SMC	OPEN PLOT (P-54)
17	Women ITI	BM-1	535.5	PRELIMINARY T.P.SNO - 42 (BHIMRAD) FP NO -74 Paikee	SMC.	OPEN PLOT
		BM-2	595	FINAL T.P.SNO -43 (BHIMRAD) FP. NO 86 Paikee	SMC.	OPEN PLOT
18	Bhimrad	SW-1	574.5	PRELIMINARY T.P.SNO - 42 (BHIMRAD) FP NO -78	SMC.	OPEN PLOT

		SW-2	535.8	PRELIMINARY T.P.SNO -42 (BHIMRAD) FP. NO 79 FINAL T.P.SNO -43 (BHIMRAD) FP. NO 48	SMC+GOVT.	OPEN PLOT
19	Convention Centre	CC-1	574.5	KHAJOD URBAN DEVELOPMENT AUTHORITY	KHUDA	
		CC-2	535.8	KHAJOD URBAN DEVELOPMENT AUTHORITY	KHUDA	
		CC-3	536.8	KHAJOD URBAN DEVELOPMENT AUTHORITY	KHUDA	
20	DREAM CITY	DC-1	574.5	KHAJOD URBAN DEVELOPMENT AUTHORITY	KHUDA	
		DC-2	535.8	KHAJOD URBAN DEVELOPMENT AUTHORITY	KHUDA	
			21498.8			

Table 5.11 B - Details of Land Required for Stations

Station No.	Station Name	Corridor 2 Bhesan to Saroli				
		PLOT NO	AREA	LOCATION AS PER TP	OWNERSHIP	REMARKS
1	Bhesan	BN-1	499.6	TPS 43 (Jahangirabad) FP 44/P Near Bhesan Treatment Plant	SMC	Open
		BN-2	564.0	TPS 9 (Palanpor Bhesan) FP 188	SMC	Open
2	Botanical Garden	UT1	557.6	T.P.S. NO.- 43 (Jahangirabad) FP 41/p	SMC	Open
		UT-2	536.2	Moje Jahangirabad B.No 26	Pvt.	Open

3	Ugat Varigruh	MS-1	594.1	TPS 42 (Jahangirabad) FP 136/p	Pvt.	Open + Bldg.
		MS-2	536.2	Moje Jahangirabad B.No 240/p and 217/p	Pvt.	Open
4	Palanour Road	RS-1	594.8	T.P.S. NO.- 13 (Adajan) FP 261	SMC	Open
		RS-2	536.2	TPS 13 (Adajan) FP 260/p and 258/p	Pvt.	Apprt. D, G Block
			165.0	On 18 mt TP Road	ROAD	-
5	L P Savani School	LP-1	529.1	T.P.S. NO.- 32 (Adajan) FP 61 & 62	SMC	Front Open
		LP-2	638.3	T.P.S. NO.- 32 (Adajan) FP 61	SMC	Front Open
		-	-	-	-	-
6	Performing Art Centre	GA-1	524.2	T.P.S. NO.- 32 (Adajan) FP 8	Pvt.	Open
		GA-2	477.8	T.P.S. NO.- 32 (Adajan) FP 13	Pvt.	Open + Gate
7	Adajan Gam	MT-1	490.0	T.P.S. NO.- 31 (Adajan) FP 195/p and 13/p	SMC	Open
		MT2	497.4	T.P.S. NO.- 31 (Adajan) FP 84/P &196/P	Pvt.	Open + Under Const. Bldg.
8	Aquarium	SB-1	514.2	T.P.S. NO.- 31(Adajan) FP 40	SMC	Road and Open
		SB2	514.2	T.P.S. NO.- 31(Adajan) FP 189 & 43/P	SMC	Shed + House
9	Badri Narayan Temple	BN-1	531.6	TPS 10 (Adajan) FP 50/p	Pvt.	Bldg. + Road

		BN-2	536.7	Moje Adajan Sr.No 598/p	Pvt.	Shed + Toilet Premvati Uphar gruh Swaminaray an Temple
10	Athwa Chaupati	CH-1	453.4	WARD NO 13 C.S.NO 648/P	SMC	OPEN PLOT
		CH-2	629.0	FINAL TP- 5 ATHWA UMRA FP-428	ROAD	OPEN PLOT
11	Majura Gate	MG-1	464.3	TP NO- 2 (NANPURA), F.P.NO - 33TOTAL AREA 36625 SQMT	Pvt.	dyal ji mandal
12	UDHANA DARWAJA	UD-1	429	WARD NO - 2,C.S.NO - 4 ,5 AREA = 8448	Pvt.	-
		UD-2	537.3	TP-6(MAJURA-KHATODARA),F.P.389	SMC	As discussed in meeting,station to be shifted in P-6(MAJURA-KHATODRA) F.P. NO. - 389 Area - 5686
13	Kamela Darwaja	TP-1	529.5	TP. 7 (ANJANA) F.P.190	Pvt.	open
		TP-2	535.8			open
14	Anjana Farm	AF-1	535.8	TP. 7 (ANJANA) F.P.99	Pvt.	Buld.Front open
		AF-2	487.2	TP. 7 (ANJANA) F.P.98	SMC	Govt.school+ smc toilet
15	Model Town	MT-1	533.0	TP. 33(DUMBHAL) F.P.35	Pvt.	temple+open

		MT-2	440.0	TP. 33(DUMBHAL) F.P.51	Pvt.	open area
16	Magob	MG-1	523.2	TP. 53(MAGOB - DUMBHAL) F.P.64	Pvt.	open plot
		MG-2	475.5	TP. 64(MAGOB - DUMBHAL) F.P.19/A	ROAD	road
17	Bharat Cancer Hospital	SV-1	595.8	Nr.TP. 35,F.P.225/1	ROAD	road row
		SV-2	593.8	TP. 35,F.P.225/2	Pvt.	Buld.Front open
18	Saroli	SL-1	595.9	TP. 35,F.P.187	Pvt.	open
		SL-2	401.2	TP. 35,F.P.188	Pvt.	build.front open
			18596.9			

LAND FOR RAMP

Corridor-I :Sarthana to Dream City

Table 5.12: Details of Land for Ramp

S.No.	Plot No.	Area (Sq.m)	Ownership
1.	RS1	5280	Govt.
2.	RS2	5880	Govt.
Total		11160	

LAND FOR PARKING AND PROPERTY DEVELOPMENT

Table 5.13A Corridor-I : Sarthana to Dream City

Station No.		AREA AVAILABLE FOR						
	Station Name	FOR PARKING			FOR DEVELOPMENT			REMARKS
		NOTATION FOR PARKING	LOCATION	AREA	NOTATION FOR P.D.	LOCATION	AREA	
1	Sarthana							
2	Nature Park							

3	Varachha Choupati Garden	C1P1	TPS NO.-20 (NANAVARACHHA -KAPADRA) F.P. NO.-90	1500				SMC (SCHOOL)
4	Kalakunj Swaminarayan Mandir							
5	Kapodra				C1PD1	TPS NO.-16 (KAPADRA) F.P. NO.-44	2000	SMC (LOCAL COMMERCIAL CENTRE)
		C1P2	TPS NO.-16 (KAPADRA) F.P. NO.- 51 + 52	3000				
6	Labheshwar Chowk							
7	Central Warehouse				C1PD2	TPS NO.- 4 (ASHWANIKUMAR-NAVAGAM) F.P. NO.- M/7	1500	SMC (EXISTING PUMPING STATION)
8	Surat Railway Station							
9	Maskati Hospital	C1P3	WARD NO - 4, C-8 (C.S.NO.693 TO 708,709/P ,710/P TO 731/P,732,735 /C,75/17)	2000	C1PD3	WARD NO - 4, C-8 (C.S.NO.693 TO 708,709/P ,710/P TO 731/P,732,735/C,75/17)	1700	S.M.C. Under acquisition (MOCHINI CHAL)
10	Chowk Bazar							
		C1P4	WARD NO-11 C.S.NO-2085, 2038 P OLD BIRD HOUSE	2000.0				S.M.C. (CENTRAL GOVT.)

11	Kadarsha ni Nal				C1PD4	WARD NO-2 C.S.NO-1958 EXE.VEHICL E DEPOT	2000	S.M.C
		C1P5	TP. NO-2 (NANPURA) F.P.NO-27	500				
12	Majura Gate							
13	Rupali Canal				C1PD5	FINAL T.P.SNO -6 (MAJURA-KHATODAR A) ,FP NO-12 Paikee NEAR RUPALI CANAL	4000	Station development on PPP basis by SMC through metro company
14	Althan Tenament	C1P6	DRAFT T.P.SNO -27 (BHATAR-MAJURA) , FP NO-30 FP NO -63 Paikee (PVT.) NEAR BHATAR CHAR RASTA	2500	C1PD6	PRELIMINARY T.P.SNO -28 (ALTHAN-BHATAR) FP.NO - 108, FP.NO-109 Paikee	5000	Station development on PPP basis by SMC through metro company
15	Althan Gam	C1P7	PRELIMINARY T.P.SNO -28 (ALTHAN-BHATAR), FP.NO - 51 Paikee NEAR SOHAM CIRCLE	2000				SMC
		C1P8	PRELIMINARY T.P.SNO -28 (ALTHAN-BHATAR) FP.NO - 139 Paikee	145.7				

16	VIP Road							
		C1P9	FINAL T.P.SNO -37 (ALTHAN-SOUTH) FP.NO 38 Paikee	3000				SMC
17	Women ITI				C1PD7	PRELIMINARY T.P.SNO -42 (BHIMRAD) , FP NO -74 Paikee	3080	SMC
					C1PD8	FINAL T.P.SNO -43 (BHIMRAD) , FP. NO 86 Paikee	13303	SMC
18	Bhimrad	C1P10	PRELIMINARY T.P.SNO -42 (BHIMRAD) ,FP NO -78	2500				SMC
					C1PD9	PRELIMINARY T.P.SNO -42 (BHIMRAD) FP. NO 79 FINAL T.P.SNO -43 (BHIMRAD) FP. NO 48	5000	SMC+GOVT.
19	Convention Centre							
20	DREAM CITY							
				19145.7			37583.0	

Table 5.13 B Details of Land for Parking & Property Development (Corridor-2)

Station No.	Station Name	AREA AVAILABLE FOR						REMARKS
		FOR PARKING				FOR DEVELOPMENT		
		NOTATION FOR PARKING	LOCATION	AREA	NOTATION FOR P.D.	LOCATION	AREA	
1	Bhesan	C2P1	Near Bhesan STP T.P.S. NO.- 43 (Jahangirabad) F.P 44	2000	C2PD1	Near Bhesan STP T.P.S. NO.- 43 (Jahangiraba d) F.P 44	3000	SMC (Sale for Resi.)
		-	-		C2PD2	TPS 9 (Palanpor Bhesan) FP 188	2500	SMC (sale For Comm.)
2	Botanic al Garden	C2P2	Near Ugat Canal T.P.S. NO.- 43 (Jahangirabad) F.P 41	3000	-	-	-	SMC (Sale for Comm.)
						-	-	
3	Ugat Varigru h	-	-	-	C2PD3	Opp. Ugat WTP Moje - Jahangirabad B.NO 248/P	3900	C-28 WDC reservatio n
		-	-	-				
4	Palanou r Road	C2P3	Opp. Hidayat Police Chowki T.P.S. NO.- 13 (Adajan) F.P 261	1800	-	-	-	SMC (Municipal Ward Off.)
		-	-	-	-	-	-	
		-	-	-	-	-	-	-
5	L P Savani School	C2P4	Pal Transport station T.P.S. NO.- 32 (Adajan) FP 61	2000	-	-	-	SMC (Pal Transport station)
					-	-	-	

		-	-		C2PD4	Opp. Pal Transport station .T.P.S. NO.- 32 (Adajan) FP 75	4836	SMC (Sub Centre)
6	Performing Art Centre	-	-	-	-	-	-	-
		-	-	-	-	-	-	-
7	Adajan Gam	-	-	-	-	-	-	-
		C2P5	Nr. White Orchid T.P.S. NO.- 31 (Adajan) F.P 196	1000	-	-	-	SMC (Sale for Comm. use)
8	Aquarium	C2P6	B/H Star Bazar T.P.S. NO.- 31(Adajan) F.P 189	1000	-	-	-	Station development on PPP basis by SMC through metro company
		-	-	-	-	-	-	-
9	Badri Narayan Temple	-	-	-	-	-	-	-
		-	--	-	-	-	-	-
10	Athwa Chaupati	-			C2PD5	CHAUPATI SWIMMING POOL	1906.6	Station development on PPP basis by SMC through metro company
		-			-	-	-	-
11	Majura Gate	-			-	-	-	
12	UDHANA DARWAJA	-			-	-	-	Pvt.

		C2P7	TP- 6(MAJURA-KHATODRA) F.P. NO. - 389	5148.7	-	-	-	-
13	Kamela Darwaja	-	-	-	-	-	-	-
14	Anjana Farm	C2P8	Canal Road, below flyover bridge	3000	C2PD7	TP. 7 (ANJANA) F.P.98	4000	can be taken up with Tenament Redevelopment Scheme(ppp)
15	Model Town	-	-	-	-	-	-	-
16	Magob	-	-	-	-	-	-	-
17	Bharat Cancer Hospital	-	-	-	-	-	-	-
18	Saroli	-	-	-	-	-	-	-
				18948.7			20142.6	

5.10.8 Land for Staff quarters, office complex and operation control centre (OCC)

A large number of officers and staff will be required to be deployed permanently to take care of project implementation and post construction operational activities. Moreover metro office complex and metro operation control centre will also be required. It is proposed to keep the provision of **2.5 ha** of government

land for this purpose. Exact location of land has not been identified at this stage. It may be decided at the time of project implementation.

5.10.9 Temporary office accommodation

During construction period, huge quantities of construction materials like reinforcing bars, cement, steel sections, shutters, pre-cast segments etc. are to be stored and sufficient land is required for storage of these materials. The areas may be identified based on availability as vacant on date nearer to the corridors. At the time of construction, depending up on theneed the location and size can be reassessed and temporary land acquisitions can be made accordingly.

Since the area of land being acquired permanently at most of the stations is bare minimum, the land required for construction depots purpose has been considered throughout the corridor @ 2000sq m at every 5 km. These sites will be obtained on lease temporarily for the construction period. After completion of construction, these will be handed over back to the land owning agency.

Table 5.14 - Details of Temporary Land office accommodation

S. No.		AREA (m^2)	OWNER-SHIP
1	Sarthana to Dream City	8000	Government
2	Bhesan to Saroli	8000	Government
	Total	16000	

5.10.10 Segment Casting Yard

Large numbers of pre-cast segments are required for construction of elevated/underground structures for which a large open area is required for setting up of casting yards. As far as possible, this area should be close to the site, easily accessible and away from habitation. Considering the various factors, it is proposed to setup five segment casting yards each for Sarthana - Dream City and Bhesan – Saroli Corridors. Accordingly a provision of land has been proposed on temporary basis considering 3-3.50 ha. land for each segment casting yard for a period of a period of four years.

5.10.11 Summary of Land Requirements

Abstract of land requirements for different components of this corridor is given in **Table 5.15.**However, the land requirement is summarized below:

a) Govt. Land permanently required for stations, Depot, Ramp and running section :**52.65ha.**

b) Private Residential and Commercial Required for stations, Ramp and Running section **:1.66 ha.**

c) **Total land required for the metro project** : **54.31 ha.**

d) Govt. Land for Property development (PD)

Corridor-I (Sarthana to Dream City) : 2.98 ha
Corridor-II(Bhesan to Saroli) : 1.72 ha
Development of Stations on PPP mode : 1.07 ha
Total : 5.77 ha

Table 5.15 - Summary of Permanent Land Requirement

(All figures in Sq.m)

S.No.	Description	Sarthana to Dream City		Bhesan to Saroli	
		Govt.	Pvt.	Govt.	Pvt.
1.	Station	15,283.70	6215.10	8245	10351.90
2.	Running Section	3492.82		1301.94	
3.	Ramp	11160.00			
4.	Depot	240900		169500	
5.	Staff Quarter Office Complex and OCC	25000			
6.	RSS	7000		4000	
7.	Mid Shaft	2500			
8	Parking	19145.7	0	18948.70	0
	Total	324482.22	6215.10	201995.64	10351.90 Sq.m

Table 5.16 - Summary of Temporary Land Requirement

S. No.	Name of the Corridor	AREA (m^2)	OWNER-SHIP
1	Sarthana – Dream City	180000	Government
2	Bhesan - Saroli	180000	Government
	Total	3,60,000	

Total land required for temporary acquisition is 36 **ha** which is considered to be government open land.

5.11 UTILITY DIVERSIONS

5.11.1 Introduction

Besides the details of various aspects e.g. transport demand analysis, route alignment, station locations, system design, viaduct structure, geo-technical investigations etc. as brought out in previous paras, there are a number of other engineering issues, which are required to be considered in sufficient details before really deciding on taking up any infrastructure project of such magnitude. Accordingly, following engineering items have been studied and described in this para. Existing underground and at surface utilities and planning for their diversion during construction, if necessary.

5.11.2 Utility and Services

The proposed Metro alignments under Surat Metro Rail network (Phase-1) are passing along major arterial roads of the city road network, which are serving Institutional, Commercial and residential areas. Large number of sub-surface, surface and overhead utility services, viz. Sewers, water mains, storm water drains, telephone cables, electrical transmission lines, electric poles, traffic signals etc. are existing along the proposed alignment. These utility services are essential and have to be maintained in working order during different stages of construction by temporary / permanent diversions or by supporting in position. As such, these may affect construction and project implementation time schedule /costs, for which necessary planning / action needs to be initiated in advance.

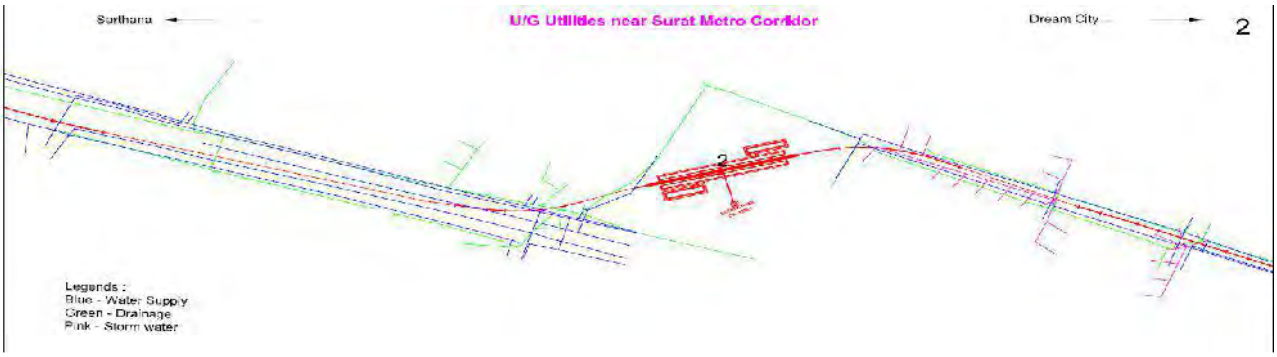
Affected Utilities as indicated hereunder are notional, actual investigation is required before commencement of the field work. A list of available data of Underground Utilities existing near both the Corridors of Surat Metro are shown as hereunder in Fig.5.7 (i) to (xv) and Fig. 5.8: (i) to (xvii):-

A. Corridor-I (Sarthana – Dream City) – Underground Utilities

Fig.5.7 (Underground Utilities available near Sarthana-Dream City Corridor)

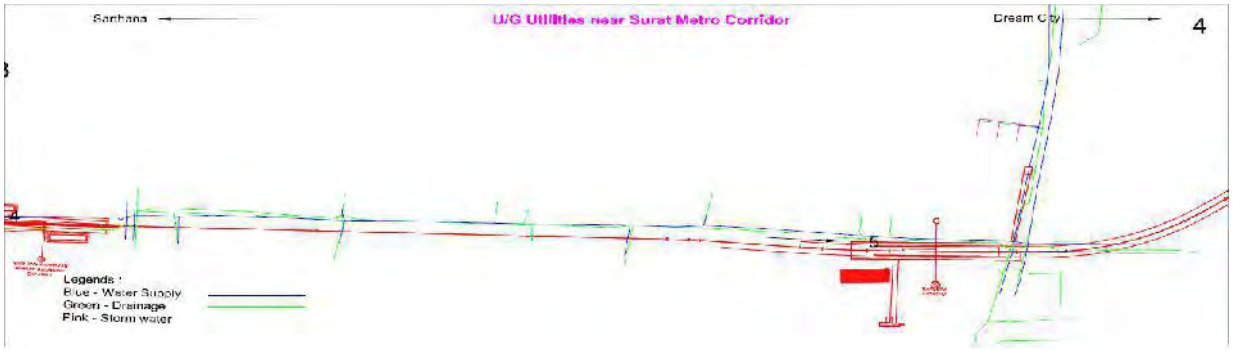
(i)

(ii)

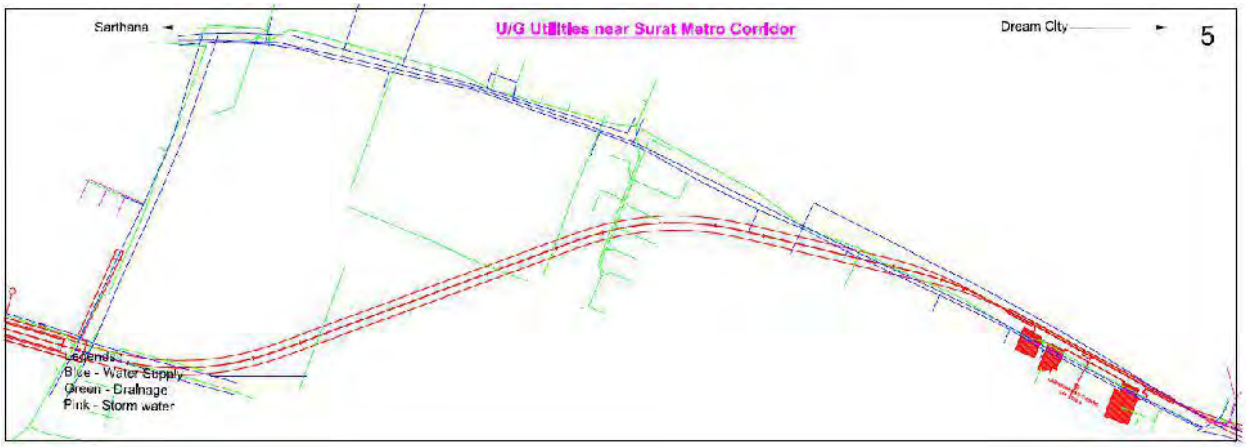


(iii)

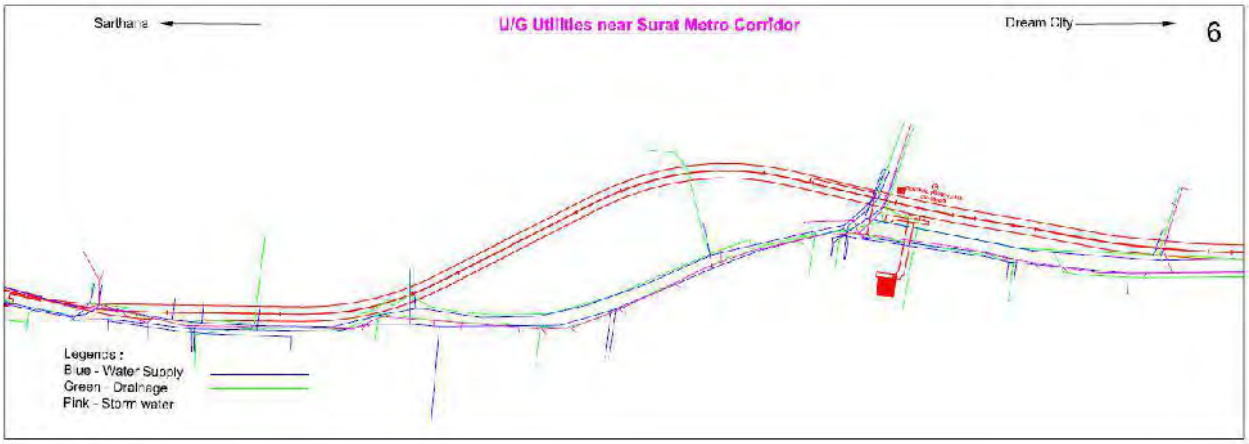
(iv)


Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
4
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

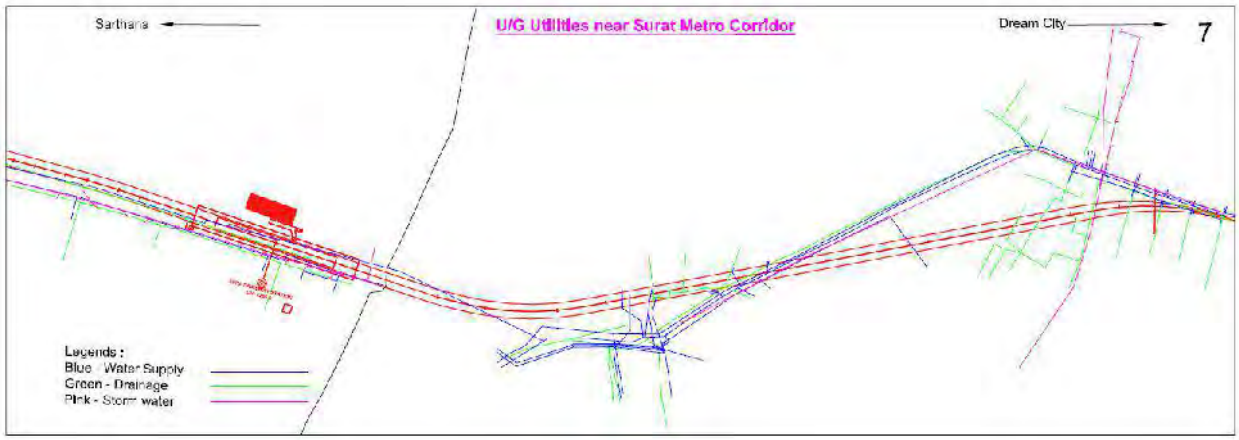
(v)


Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
5
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

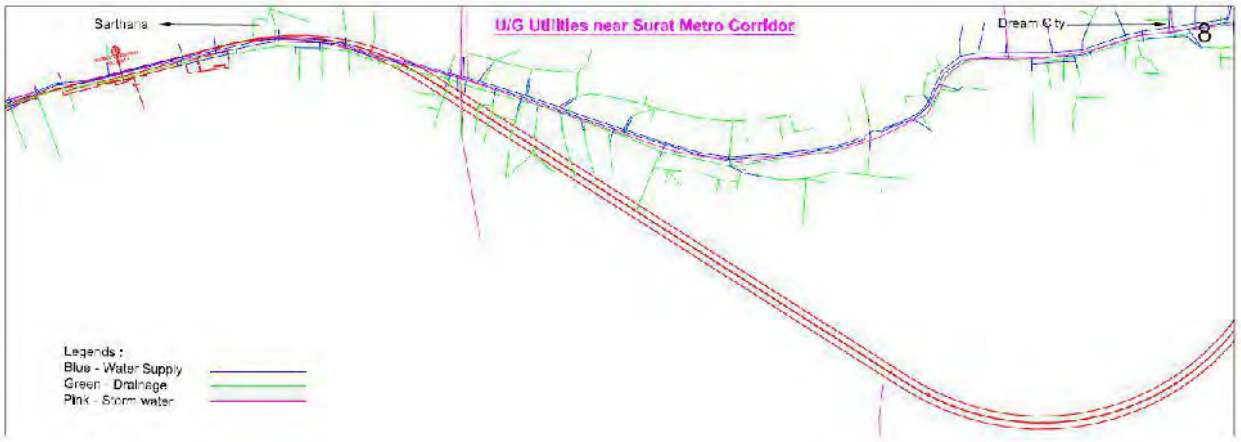
(vi)


Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
6
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

(vii)


Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
7
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

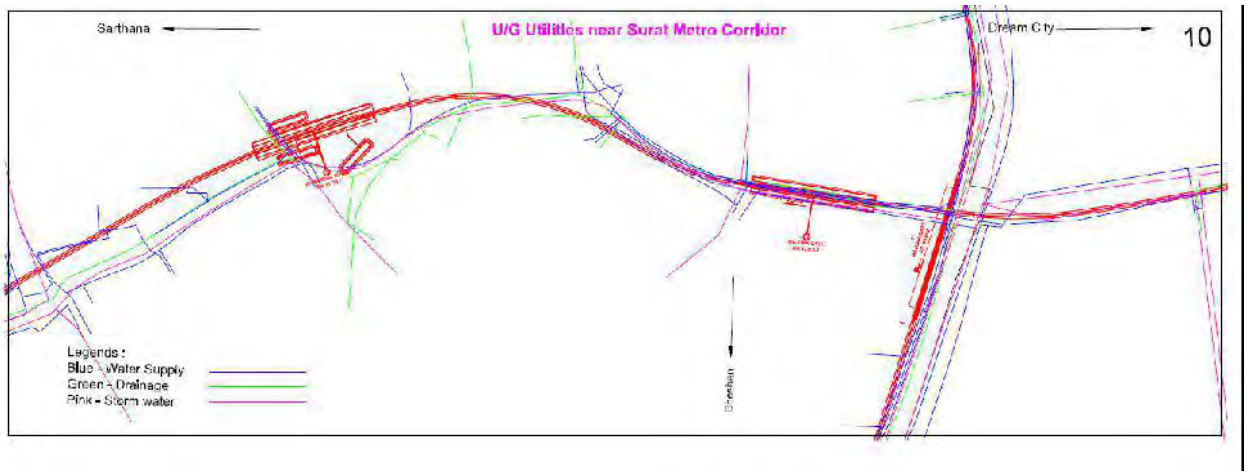
(viii)


Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
8
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

(ix)

Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
9
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

(x)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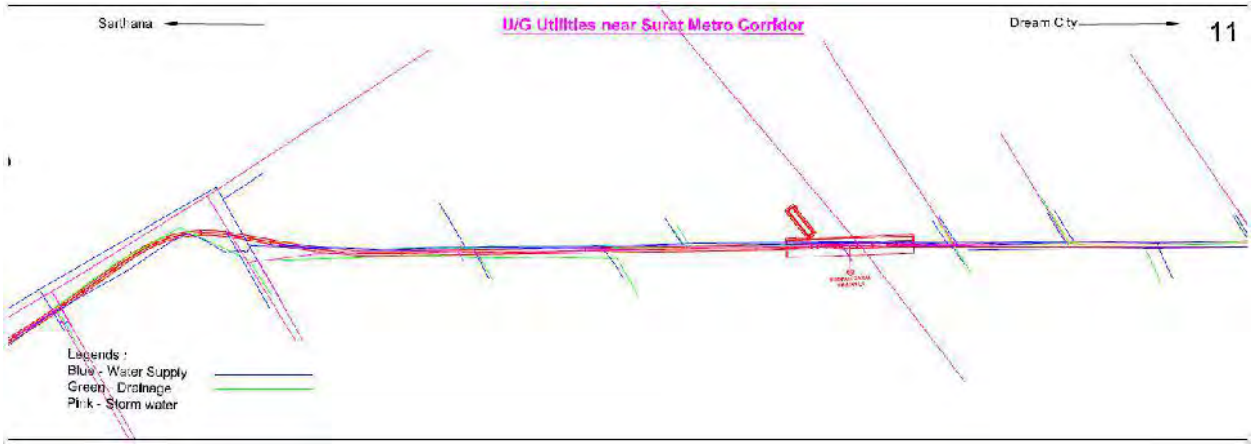
U/G Utilities near Surat Metro Corridor
Dream City
10
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

(xi)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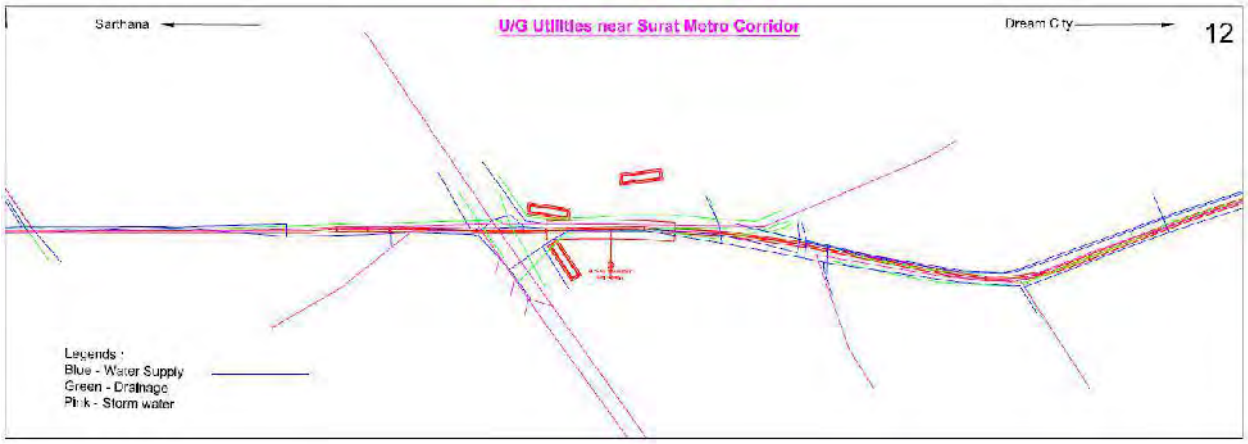
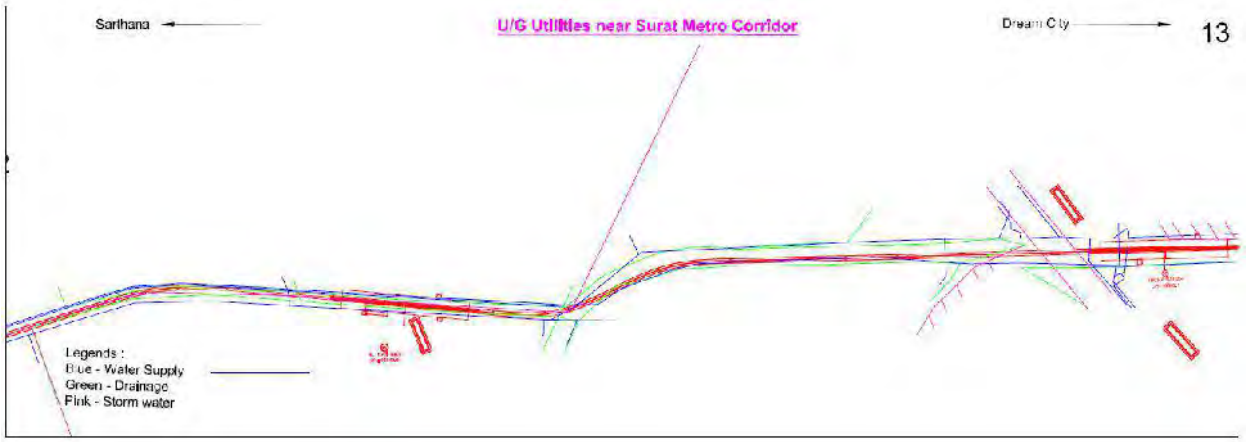
U/G Utilities near Surat Metro Corridor
Dream City
11
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

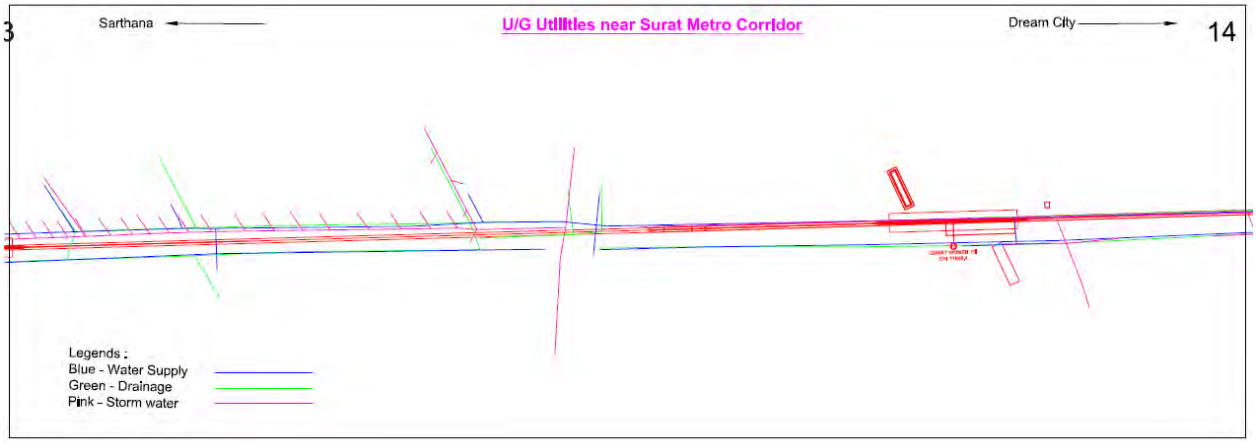
(xii)


Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
12
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

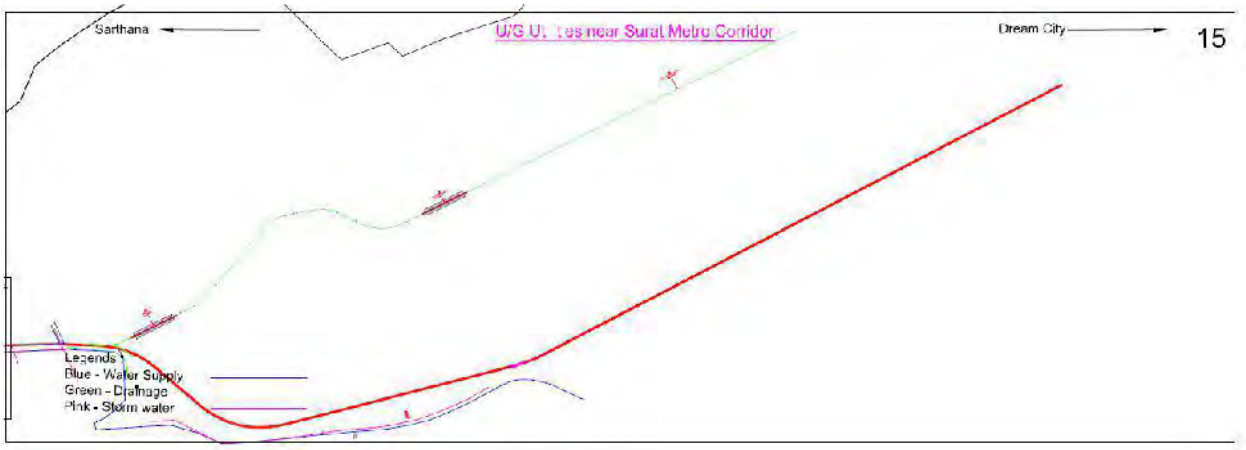
(xiii)


Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
13
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

(xiv)


Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
14
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

(xv)

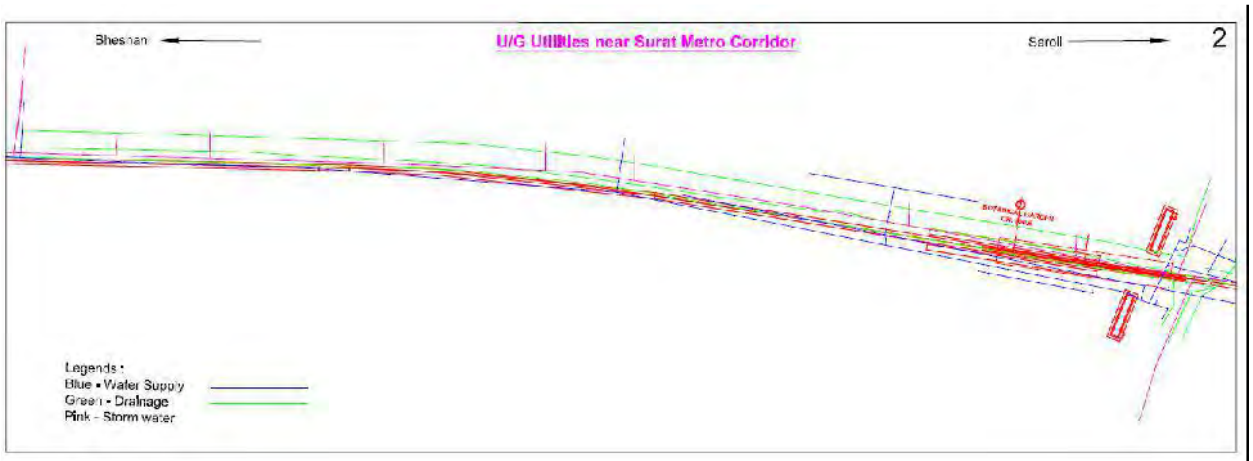

Sarthana
U/G Utilities near Surat Metro Corridor
Dream City
15
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

B. Corridor-2 (Bhesan to Saroli)- Underground Utilities

FIG. 5.8(Underground Utilities available near Bhesan-Saroli Corridor)

(i)

(ii)



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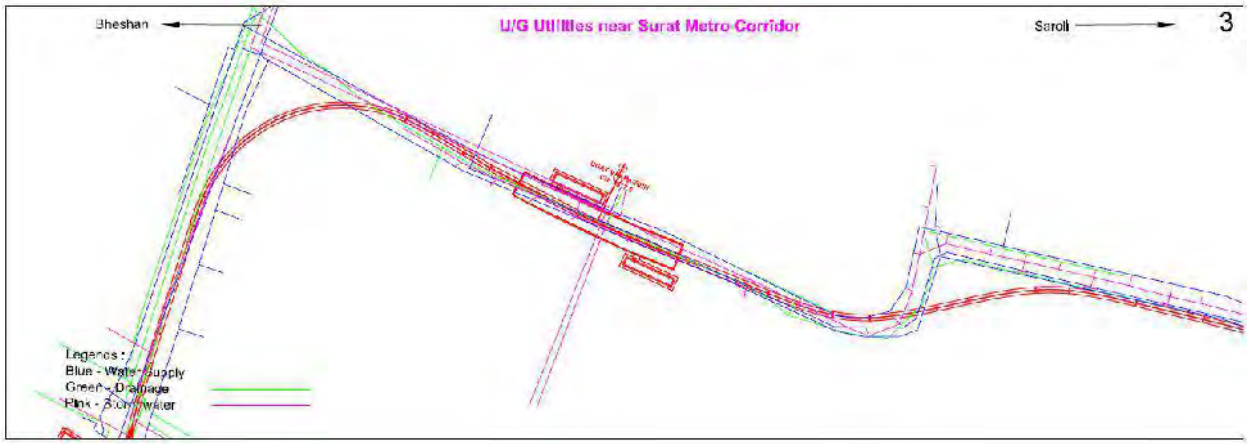

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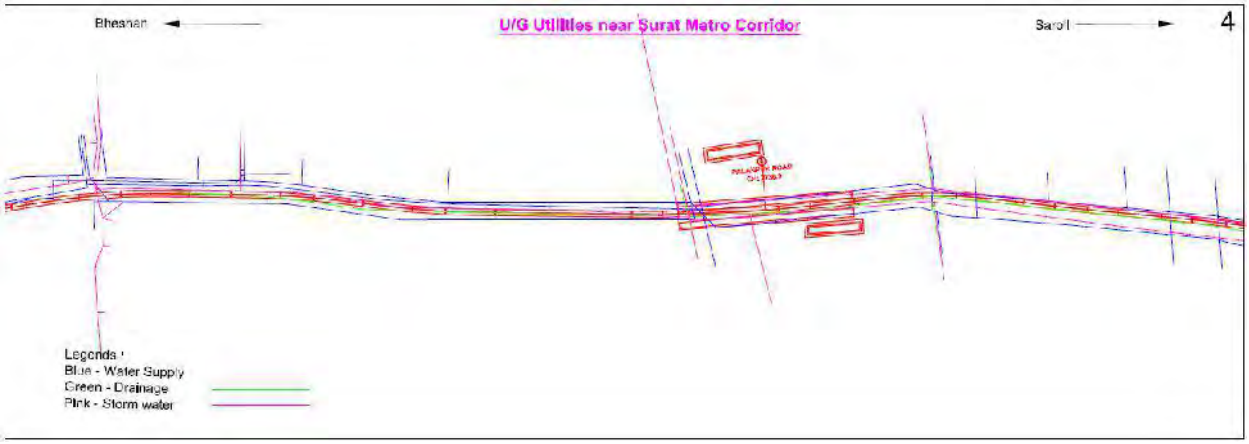
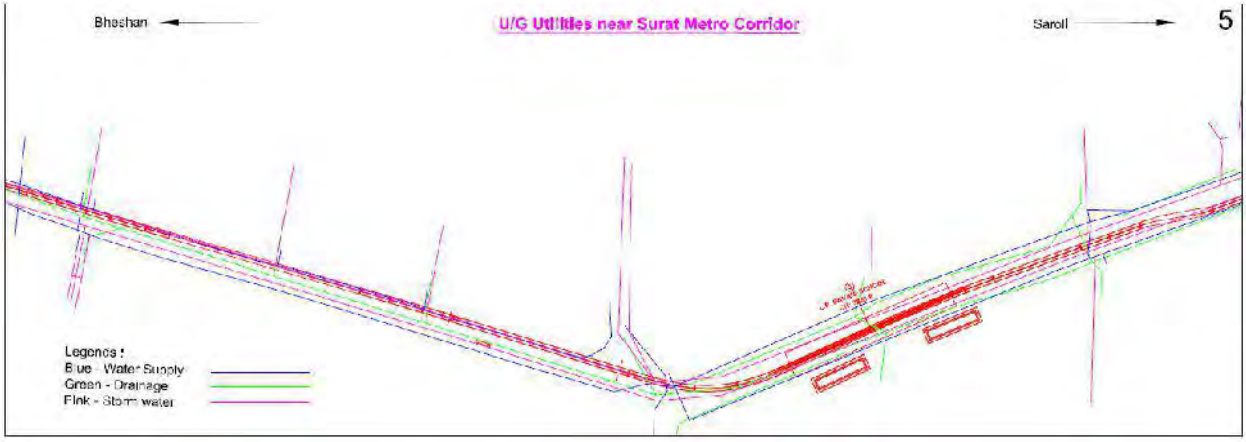
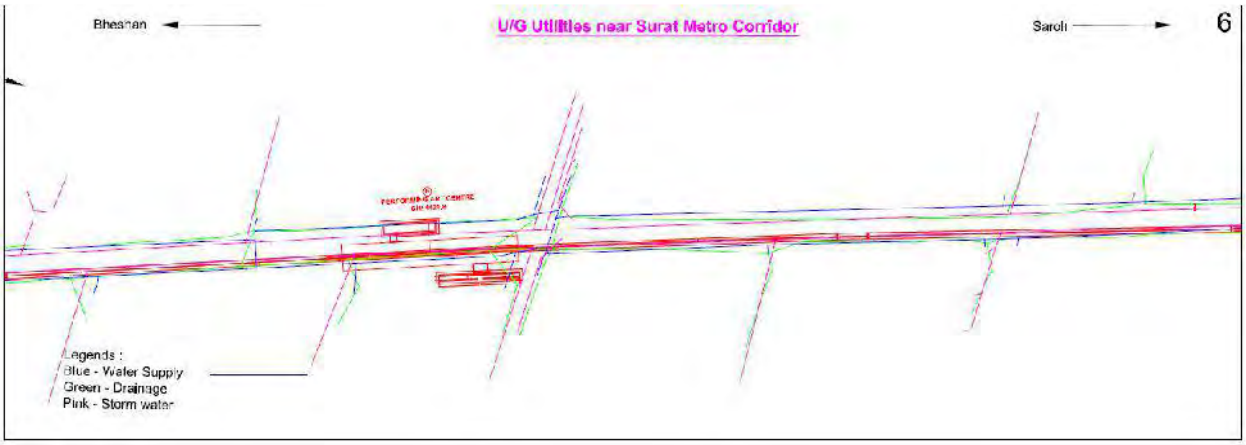

Bhestan
U/G Utilities near Surat Metro Corridor
Saroli
4
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

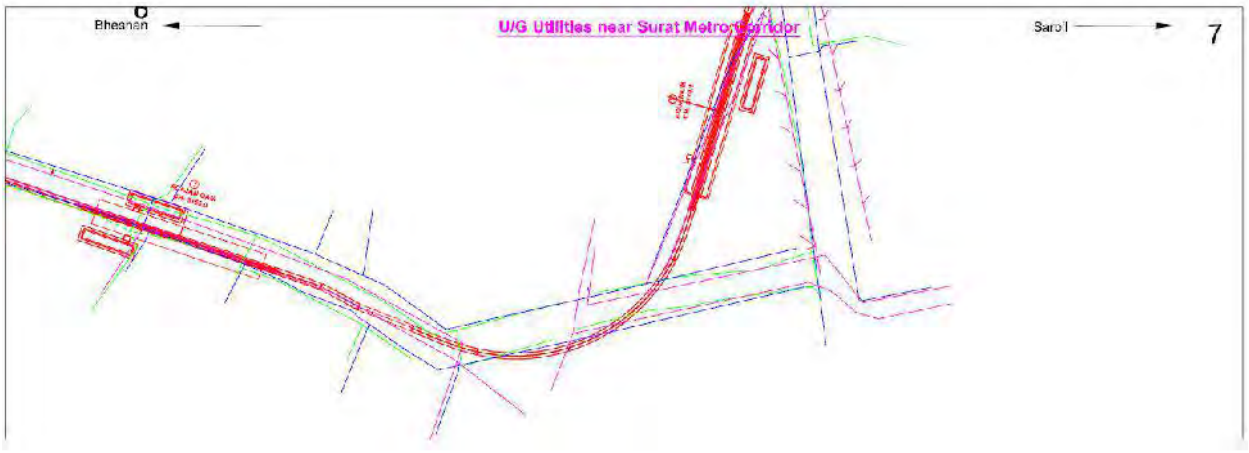
(v)


Bhestan
U/G Utilities near Surat Metro Corridor
Saroli
5
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

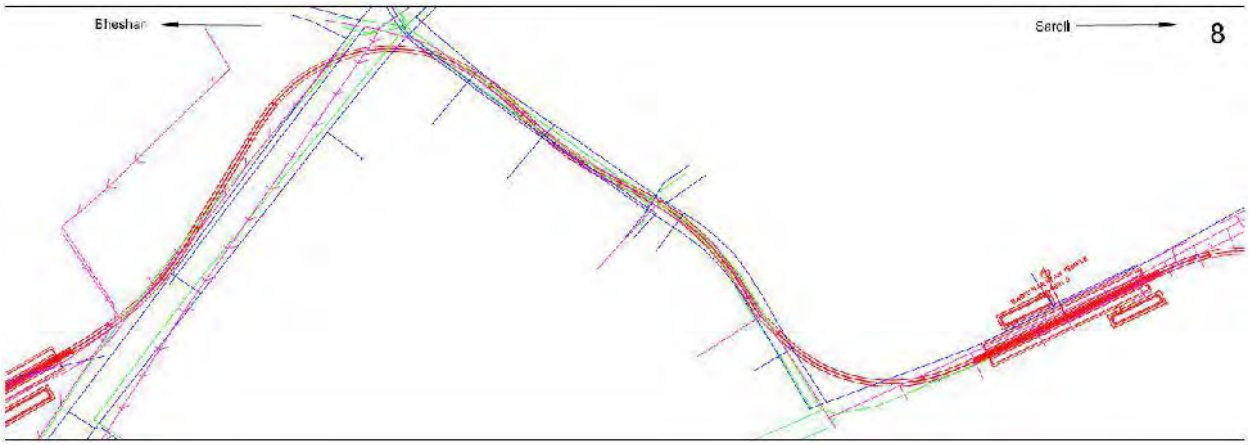
(vi)


Bhestan
U/G Utilities near Surat Metro Corridor
Saroli
6
Legends :
Blue - Water Supply
Green - Drainage
Pink - Storm water

(vii)


Bhestan
U/G Utilities near Surat Metro Corridor
Sarol
7

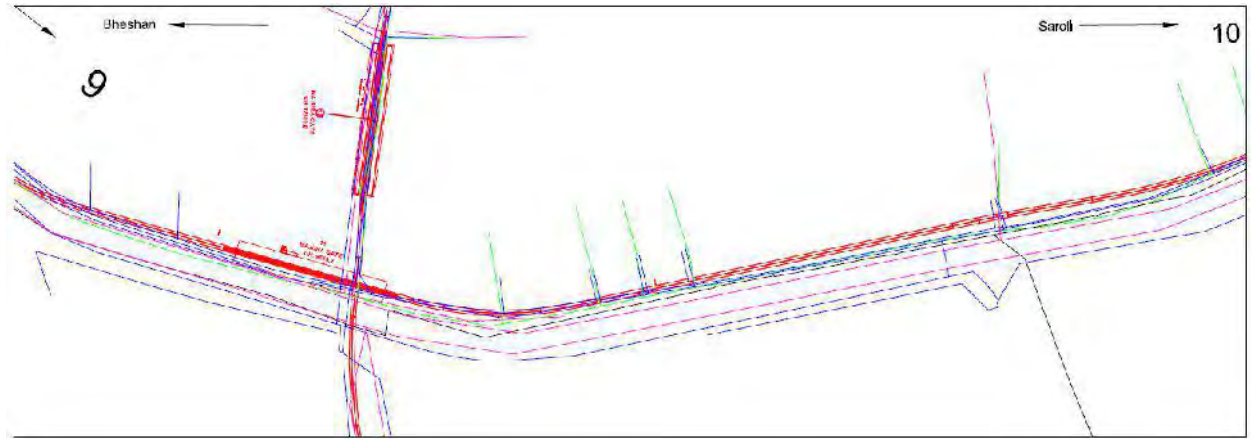
(viii)


Bhestan
Sarol
8

(ix)

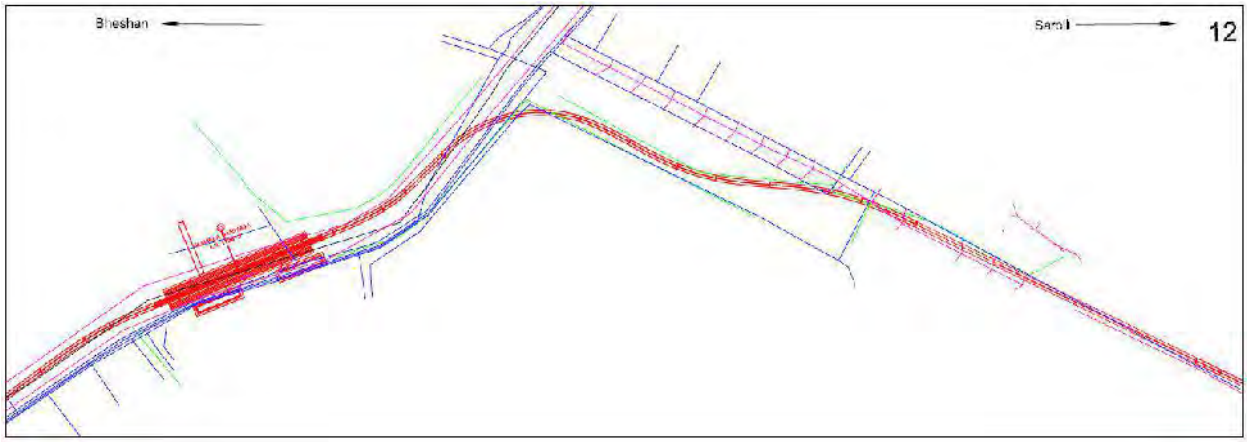
Bhestan
Sarol
9

(x)



(xi)

(xii)



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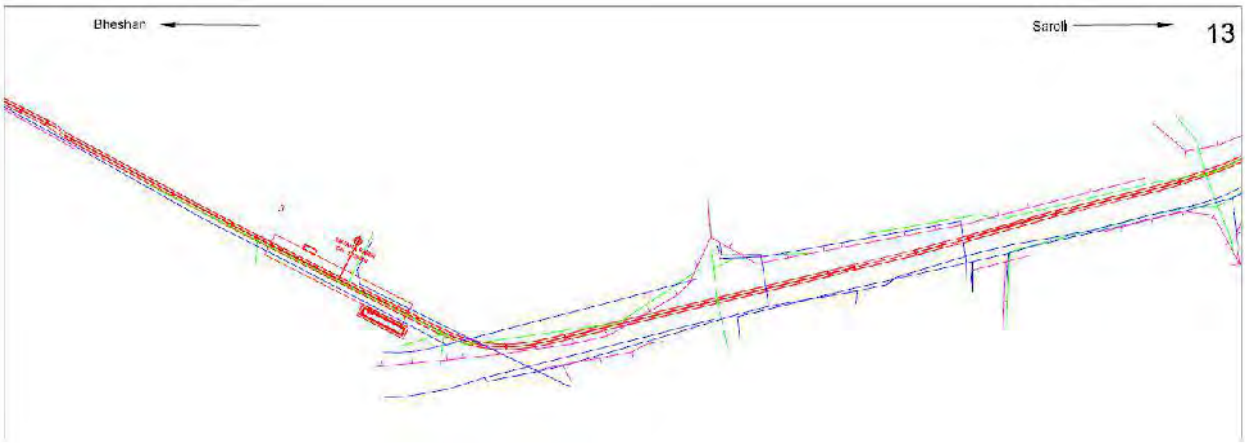
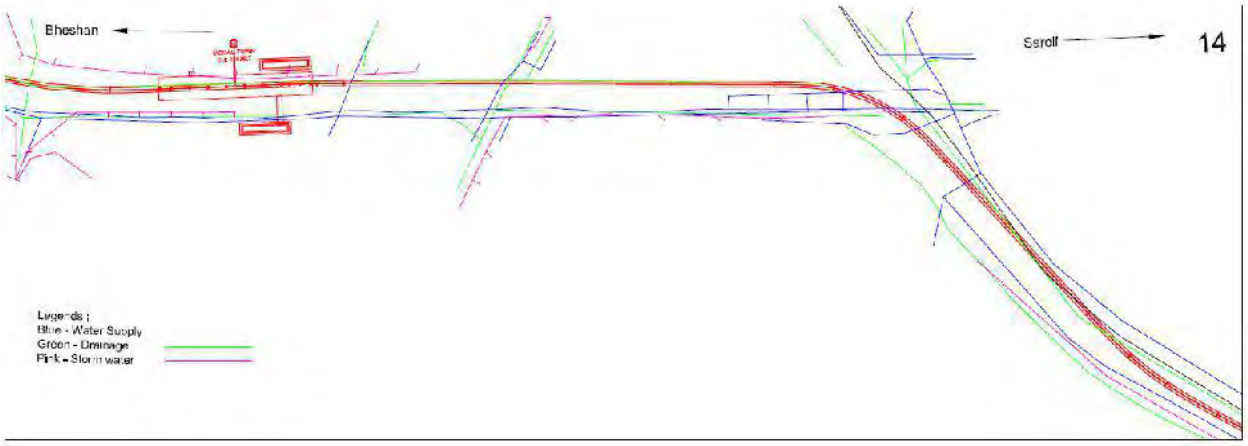
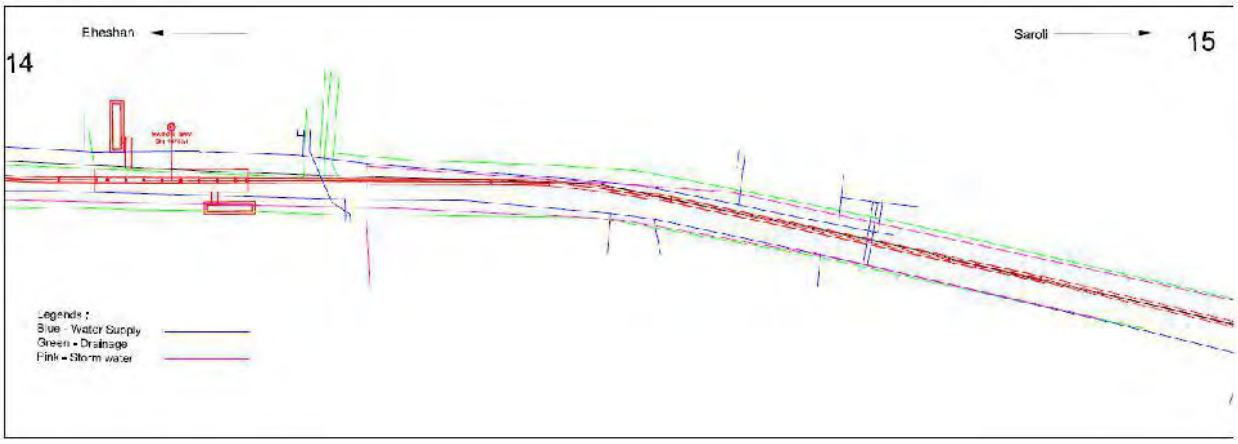

Bheshan
Saroli
13

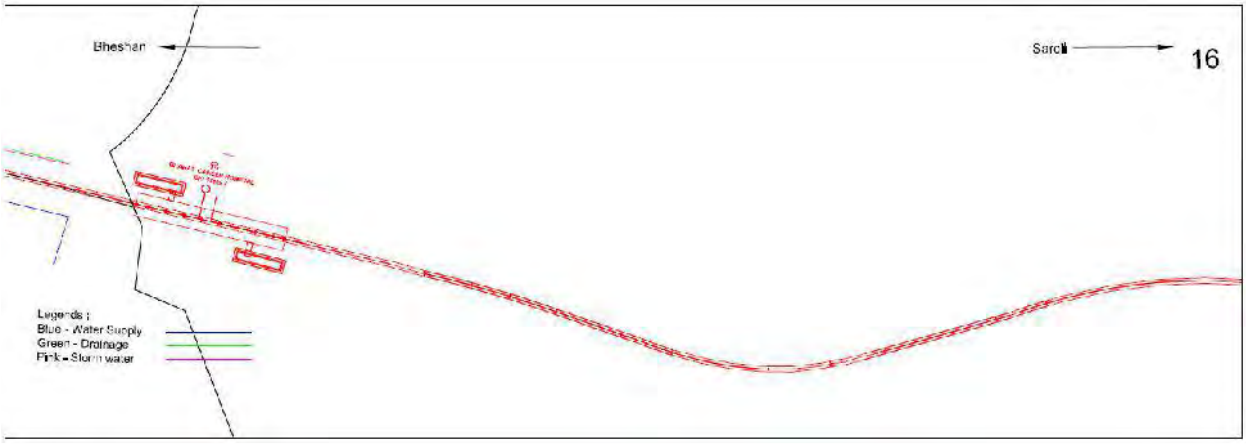
(xiv)


Bheshan
Ssroll
14
Legends ;
Blue - Water Supply
Green - Drainage
Pink - Storm water

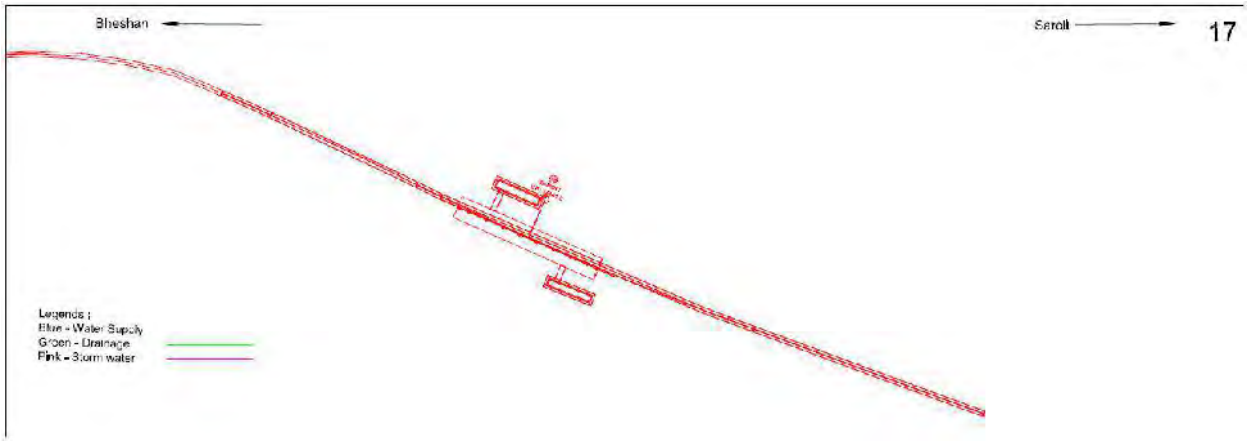
(xv)


Bheshan
Saroli
14
15
Legends ;
Blue - Water Supply
Green - Drainage
Pink - Storm water

(xvi)



(xvii)



C. Organizations / Departments responsible for concerned utility services are provided in subsequent table.

Table 5.17 - Organization responsible for utilities and services

S. No.	Organization/Deptt.	Utility Services
1	Drainage Department of SMC	Sewerage and drainage lines.
2	Hydralic Deptt. Of SMC	Water mains, their service lines, including hydrants, water treatment plants, pumping station etc.
3	Road Deptt. of SMC	Roads & footpaths inside Surat City
4	NHAI	Roads, Surface water drains, Nallah etc.,

S. No.	Organization/Deptt.	Utility Services
		pertaining to National Highway (NH-6)
5	Torrento Power & Dakshin Gujarat Vij Company Ltd.	Power cables and their appurtenances, pole mounted transformers, power cables etc.
6	GETCO	H.T. Lines, their pylons, sub-station, etc.
7	BSNL	Telecommunication cables, junction boxes, telephone posts, O.H. Lines etc.
8	Gujarat Traffic Police	Traffic signal posts, junction boxes and cable connection etc.
9	Reliance Mobile India Ltd.,	Telecommunication cables, junction boxes etc.,
10	Idea, Airtel& Tata Tele services India Ltd.,	Telecommunication cables, junction boxes etc.,
11	Gujarat Gas Co. Ltd.	Gas Pipeline.

Prior to the actual execution of work at site, detailed investigation of all utilities at foundation location will be undertaken well in advance by making trench pit to avoid damage to any utility.

5.11.3 Diversion of Underground Utilities

While planning for diversion of underground utility services viz. sewer lines, water pipelines, cables, etc., during construction of Metro alignment, following guidelines have been adopted:

Utility services have to be kept operational during the entire construction period and after completion of project. All proposals should therefore, ensure their uninterrupted functioning.

Sewer lines and water supply lines are mainly affected in underground cut and cover construction. These services are proposed to be maintained by temporarily replacing them with CI/Steel pipelines and supporting them during construction, these will be encased in reinforced cement concrete after completion of construction and retained as permanent lines

Where permanent diversion of the affected utility is not found feasible, temporary diversion with CI/Steel pipes without manholes is proposed during construction. After completion of construction, these will be replaced with conventional pipes and manholes.

The elevated viaduct does not pose much of a difficulty in negotiating the underground utility services, especially those running across the alignment. The utilities infringing at pier location can be easily diverted away from the pile cap location.

In case a major utility is running along/across the alignment which cannot be diverted or the diversion of which is difficult, time consuming and uneconomical, the spanning arrangement of the viaduct and layout of piles in the foundation may be suitably adjusted to ensure that no foundation needs be constructed at the location, where utility is crossing the proposed alignment. The utility service can also be encased within the foundation piles.

5.11.4 Underground Stretch and Switch Over Ramp

The details of underground sections are given the “Geometric Design and Alignment description chapter. As indicated in the previous paras due to various reasons, the entire length of underground section is proposed to be constructed with tunneling keeping a minimum cover of about 6m above the tunnel, except at stations which will be constructed by cut and cover method. Hence, the utility services existing in above ground or below ground position are not likely to be affected in underground stretch of the alignment except at station location.

5.11.5 Elevated Stretch

Certain length in each corridor is elevated. Elevated length of corridor has been planned in the center of the road except at few locations as detailed in the Alignment description.

5.11.6 Sewer Lines, Storm Water Drains and Water Lines

The sewer/drainage lines generally exist in the service lanes i.e. away from main carriageway. However, in certain stretches, these have come near the central verge or under main carriageway, as a result of subsequent road widening.

The major sewer/drainage lines and water mains running across the alignment and likely to be affected due to location of column foundations are proposed to be taken care of by relocating on column supports of viaduct by change in span or by suitably adjusting the layout of pile foundations. Where, this is not feasible, lines will be suitably diverted. Provision has been made in the project cost estimate towards diversion of utility service lines.

5.11.7 Aboveground Utilities

Above ground utilities namely street light poles, traffic signal posts, telecommunication posts, junction boxes, etc. are also required to be shifted and relocated suitably during construction of elevated viaduct. Since these will be interfering with the proposed alignment. Approximate numbers of affected lamp/ telecom/elect posts & boxes are indicated in **Table** below:

Table 5.18 - Affected Aboveground Services

Corridor	LP	EP	MH	SP	TP	EM	HT	JB	EMH	TF	IGL
Sarthana – Dream City	2864	867	4148	11	28	17	6	1420	18	59	196
Bhesan to Saroli	1067	609	1770	76	8	8	2	332	16	110	68

LP: Light Poles; EP: Electric Pole; MH: Man Hole; SP: Signal Pole; IGL: Gas Pipe Line; HT: High tension Line; JB: Junction Box; TF: Transformer; TP: Telephone Pole; EM: Electric Mast, HM: High Mast Pole

5.12 ISSUES RELATED TO INTERFACE WITH EXTERNAL AGENCIES.

In order to complete the work timely and successfully, for all the corridors under Surat Metro Rail Project (Phase-I) network, interface with external agencies on different issues shall have to be conducted as per details given in **Table 5.19** hereunder:-

Table 5.19 - Interface with External Agencies

S.No.	Name of Agency	Issue
1	SMC	Clearance of ROW for Metro wherever encroached/occupied.
2	Railways	Crossing of Railway lines at some places such as Surat Railway Station and Surat Metro Project (Phase-I)
3	Gujarat Traffic Police	Alignment of various corridors under Surat Metro Project (Phase-I) network shall pass on/along the roads.
4	NHSRC	For high speed corridor

In addition to above, some more external agencies may have to be coordinated during the course of actual construction.

5.13 TRAFFIC DIVERSION

5.13.1 Need

Traffic Diversion Plans are required in order to look for options and remedial measures so as to mitigate any traffic congestion situations arising out due to acquisition of road space during Metro construction of various corridors. Any reduction of road space during Metro construction results in constrained traffic flow. In order to retain satisfactory levels of traffic flow up to the construction time; traffic management and engineering measures need to be taken. They can be road widening exercises, traffic segregation, one-way movements, traffic diversions on influence area roads, acquisition of service lanes, etc.

Various construction technologies are in place to ensure that traffic impedance is done at the minimum. They are:

Some of the corridor length and stations have been proposed by Cut-and-Cover' method for construction of the underground segment. This means that the stretch between two points will have to be blocked during construction. However, temporary decking may be provided by blocking the road carriageway partially to permit traffic movement along the same stretch. Construction of switch-over-ramp also requires some road space.

For elevated section wherever it is passing along the road, the requirement would be mainly along the central verge.

As regards to the alignment cutting across a major traffic corridor, 'Continuous Cantilevered Construction Technology' would be applied to prevent traffic hold-ups or diversions of any kind.

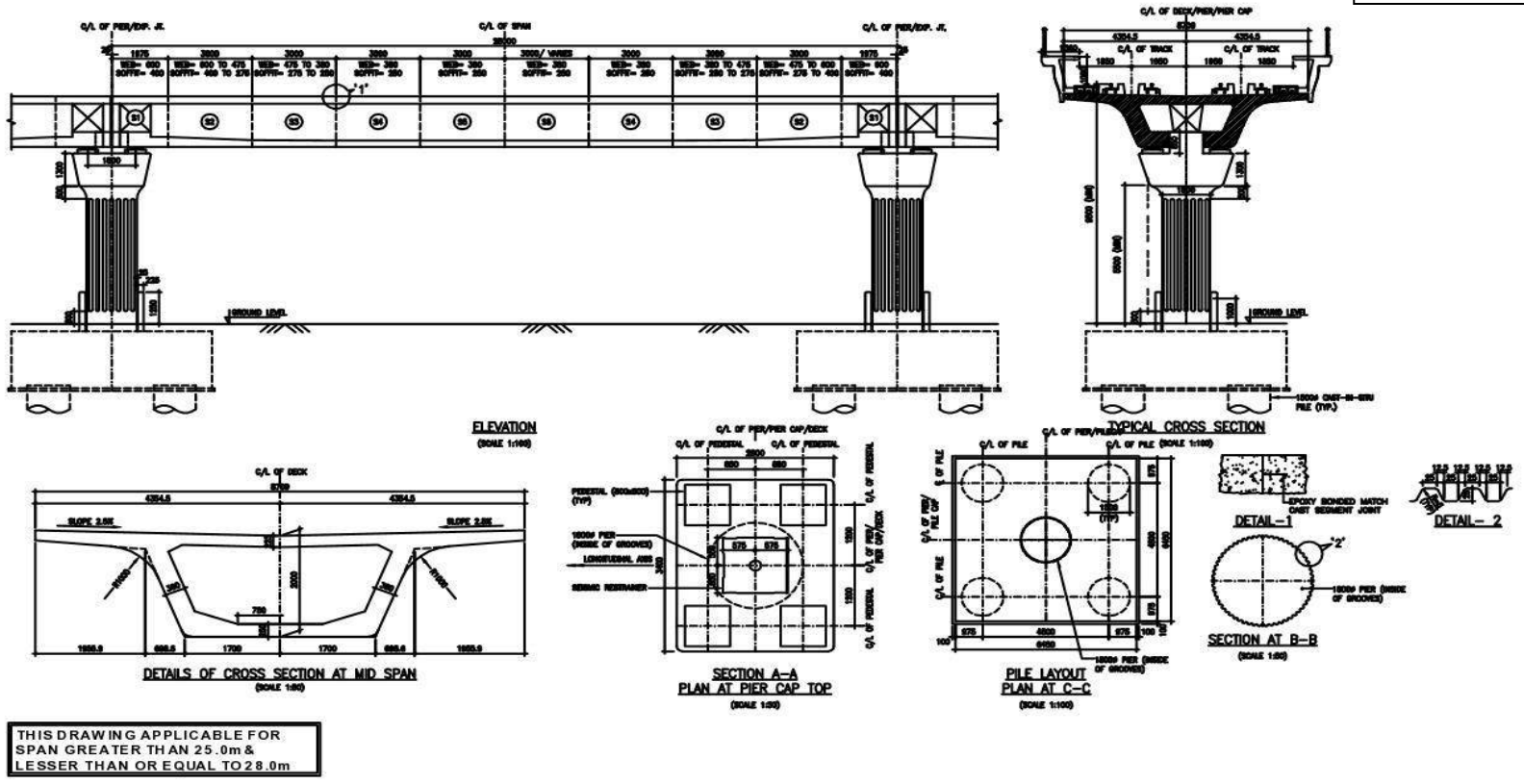
Wherever the stations are isolated, areas available around it should be utilized for road diversion purposes such as lay-byes and service roads.

5.13.2 Traffic Diversion Plans

Only temporary diversion plans will be required during construction of the Metro corridors. At the onset, all encroachments from road ROW will have to be removed. These encroachments vary from 'on-street' parking to informal activities. During the construction of works on underground sections by cut and cover, it is proposed that temporary decking may be provided by blocking the road carriageway partially to permit 'through' as well as right-turning traffic movements. Total blockage of traffic along the section will be done wherever reasonably good alternate road is available.

Keeping in view of future traffic growth and reduction of carriageway due to Metro construction, implementation of traffic management/diversion plans shall become inevitable for ensuring smooth traffic movement.

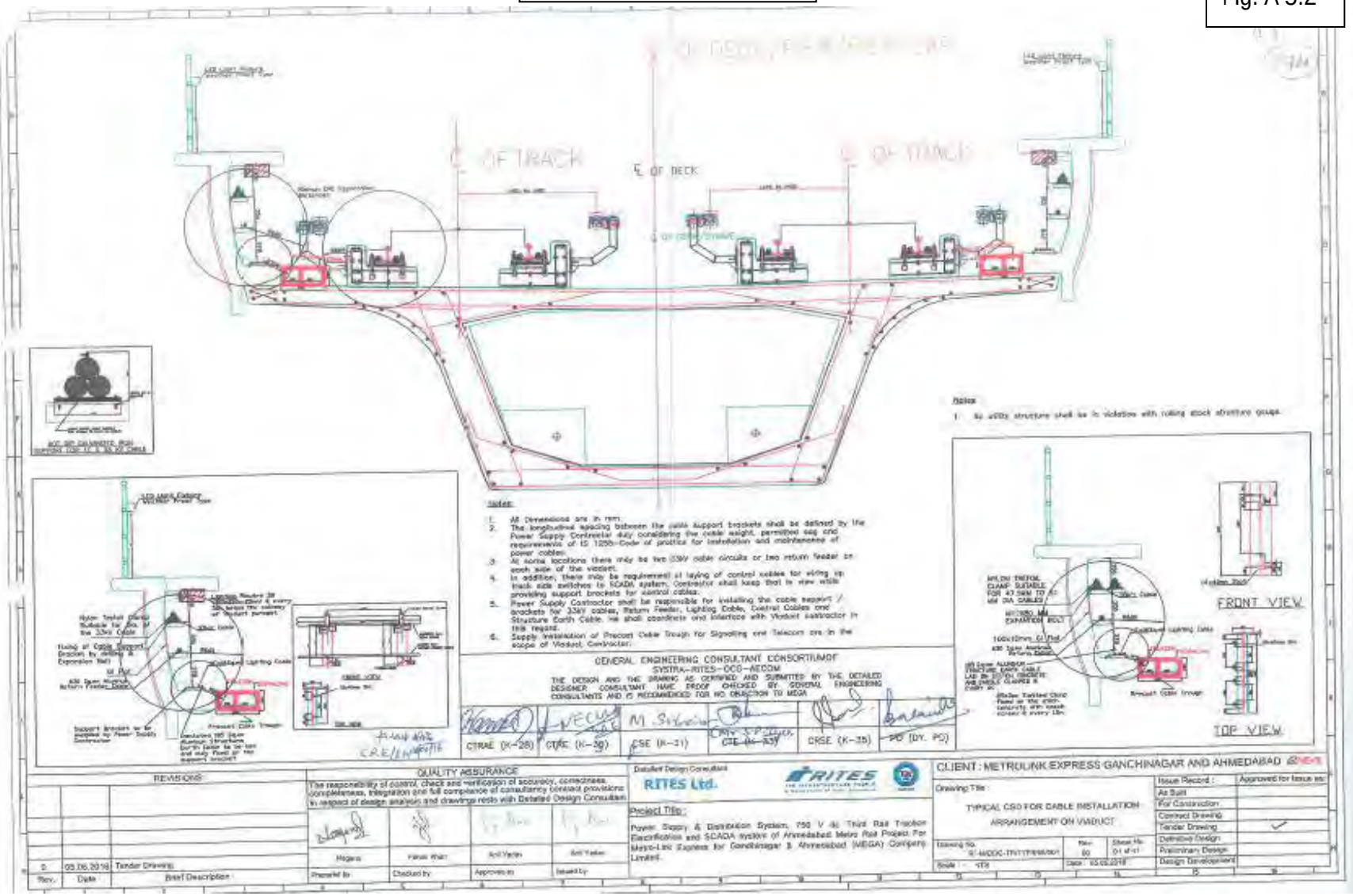
Fig. A 5.1



GENERAL ARRANGEMENT OF STANDARD UNIT (SPAN 28.0m c/c OF PIER)

SECTION OF VIADUCT

Fig. A 5.2



SECTION OF BORED TUNNEL

Fig. A 5.3

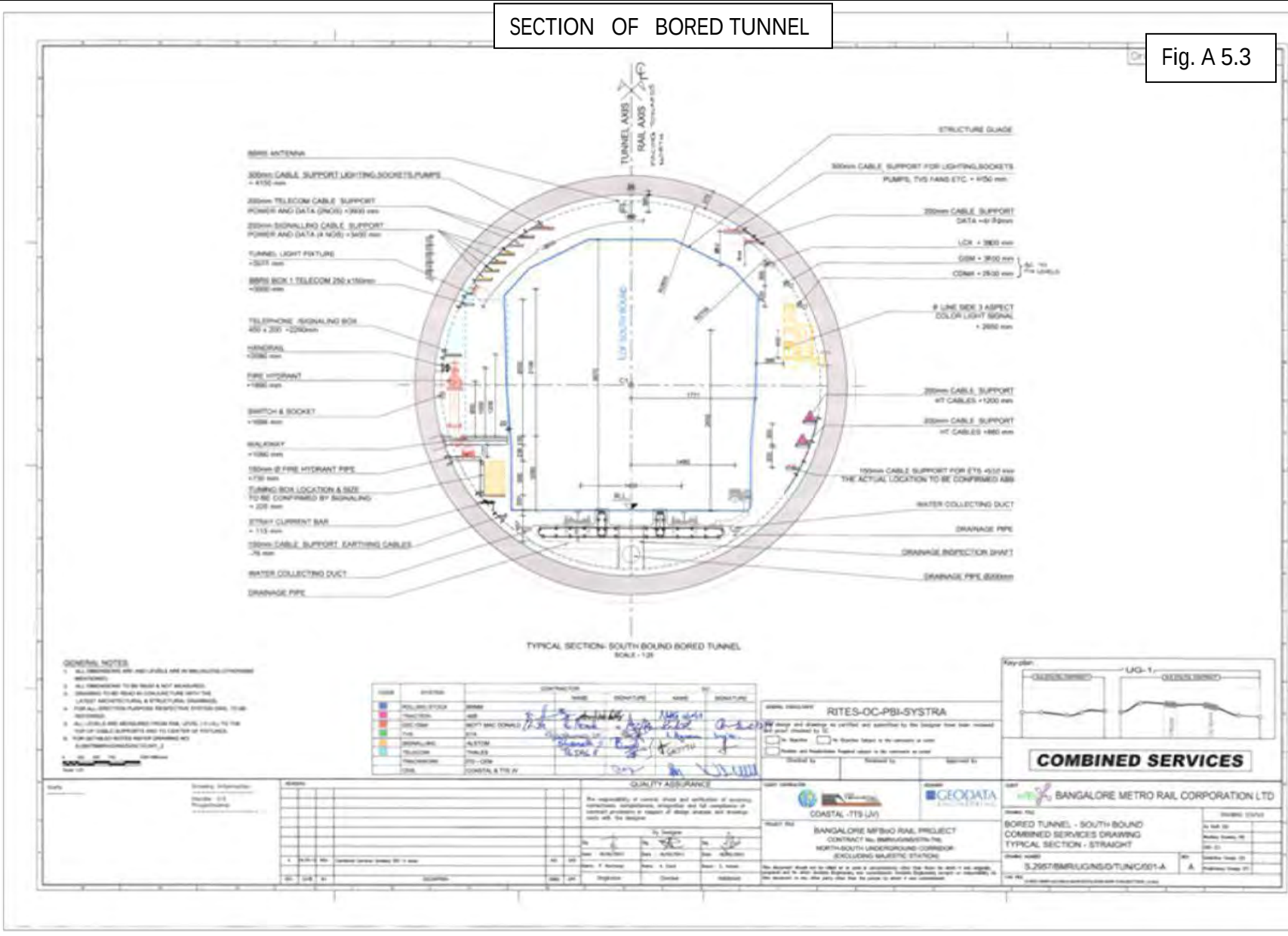


Fig. A 5.4

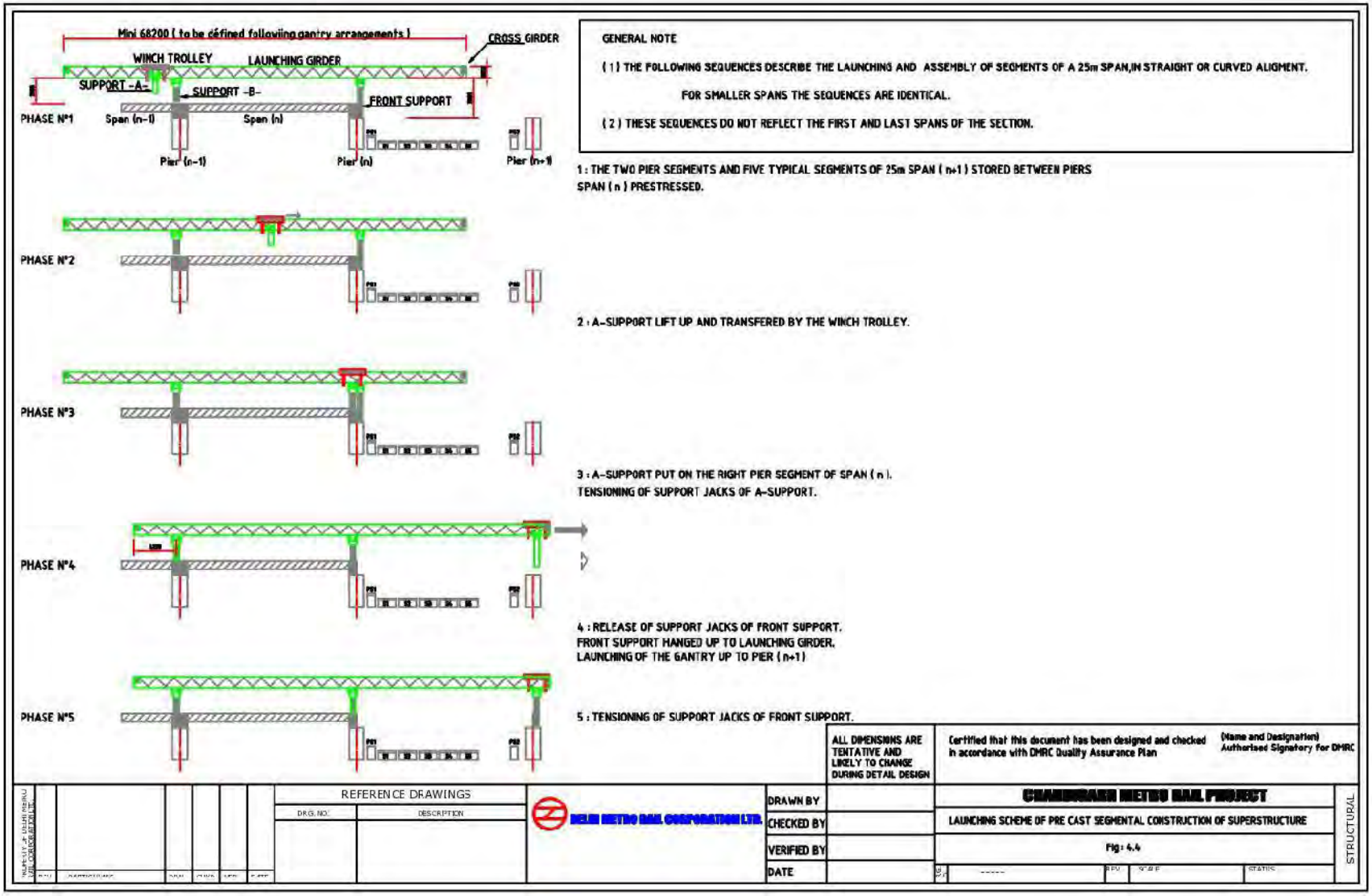
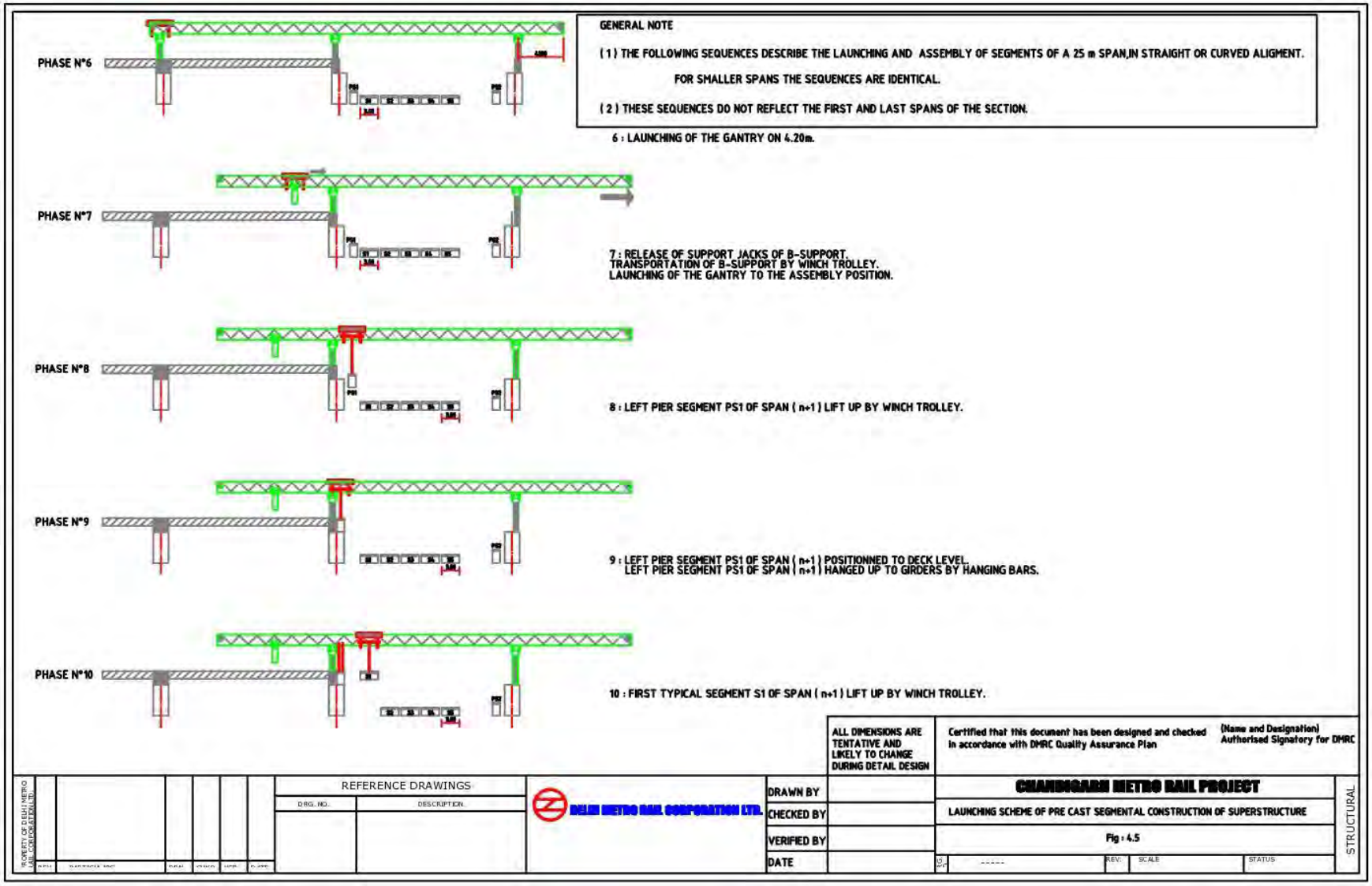

Mini 68200 (to be defined following gantry arrangements)
CROSS GIRDER
WINCH TROLLEY
LAUNCHING GIRDER
SUPPORT -A-
SUPPORT -B-
FRONT SUPPORT
PHASE N°1
Span (n-1)
Span (n)
Pier (n-1)
Pier (n)
Pier (n+1)
PHASE N°2
PHASE N°3
PHASE N°4
PHASE N°5
GENERAL NOTE
(1) THE FOLLOWING SEQUENCES DESCRIBE THE LAUNCHING AND ASSEMBLY OF SEGMENTS OF A 25m SPAN,IN STRAIGHT OR CURVED ALIGMENT.
FOR SMALLER SPANS THE SEQUENCES ARE IDENTICAL.
(2) THESE SEQUENCES DO NOT REFLECT THE FIRST AND LAST SPANS OF THE SECTION.
1 : THE TWO PIER SEGMENTS AND FIVE TYPICAL SEGMENTS OF 25m SPAN (n+1) STORED BETWEEN PIERS SPAN (n) PRESTRESSED.
2 : A-SUPPORT LIFT UP AND TRANSFERED BY THE WINCH TROLLEY.
3 : A-SUPPORT PUT ON THE RIGHT PIER SEGMENT OF SPAN (n).
TENSIONING OF SUPPORT JACKS OF A-SUPPORT.
4 : RELEASE OF SUPPORT JACKS OF FRONT SUPPORT.
FRONT SUPPORT HANGED UP TO LAUNCHING GIRDER.
LAUNCHING OF THE GANTRY UP TO PIER (n+1)
5 : TENSIONING OF SUPPORT JACKS OF FRONT SUPPORT.
ALL DIMENSIONS ARE TENTATIVE AND LIKELY TO CHANGE DURING DETAIL DESIGN
Certified that this document has been designed and checked in accordance with DMRC Quality Assurance Plan
(Name and Designation) Authorised Signatory for DMRC
REFERENCE DRAWINGS
DRG. NO.
DESCRIPTION
DELHI METRO RAIL CORPORATION LTD.
DRAWN BY
CHECKED BY
VERIFIED BY
DATE
LAUNCHING SCHEME OF PRE CAST SEGMENTAL CONSTRUCTION OF SUPERSTRUCTURE
Fig : 4.4
STRUCTURAL

Fig. A 5.5


PHASE N°6
PHASE N°7
PHASE N°8
PHASE N°9
PHASE N°10
GENERAL NOTE
(1) THE FOLLOWING SEQUENCES DESCRIBE THE LAUNCHING AND ASSEMBLY OF SEGMENTS OF A 25 m SPAN,IN STRAIGHT OR CURVED ALIGMENT.
FOR SMALLER SPANS THE SEQUENCES ARE IDENTICAL.
(2) THESE SEQUENCES DO NOT REFLECT THE FIRST AND LAST SPANS OF THE SECTION.
6 : LAUNCHING OF THE GANTRY ON 4.20m.
7 : RELEASE OF SUPPORT JACKS OF B-SUPPORT.
TRANSPORTATION OF B-SUPPORT BY WINCH TROLLEY.
LAUNCHING OF THE GANTRY TO THE ASSEMBLY POSITION.
8 : LEFT PIER SEGMENT PS1 OF SPAN (n+1) LIFT UP BY WINCH TROLLEY.
9 : LEFT PIER SEGMENT PS1 OF SPAN (n+1) POSITIONNED TO DECK LEVEL.
LEFT PIER SEGMENT PS1 OF SPAN (n+1) HANGED UP TO GIRDERS BY HANGING BARS.
10 : FIRST TYPICAL SEGMENT S1 OF SPAN (n+1) LIFT UP BY WINCH TROLLEY.
ALL DIMENSIONS ARE TENTATIVE AND LIKELY TO CHANGE DURING DETAIL DESIGN
Certified that this document has been designed and checked in accordance with DMRC Quality Assurance Plan
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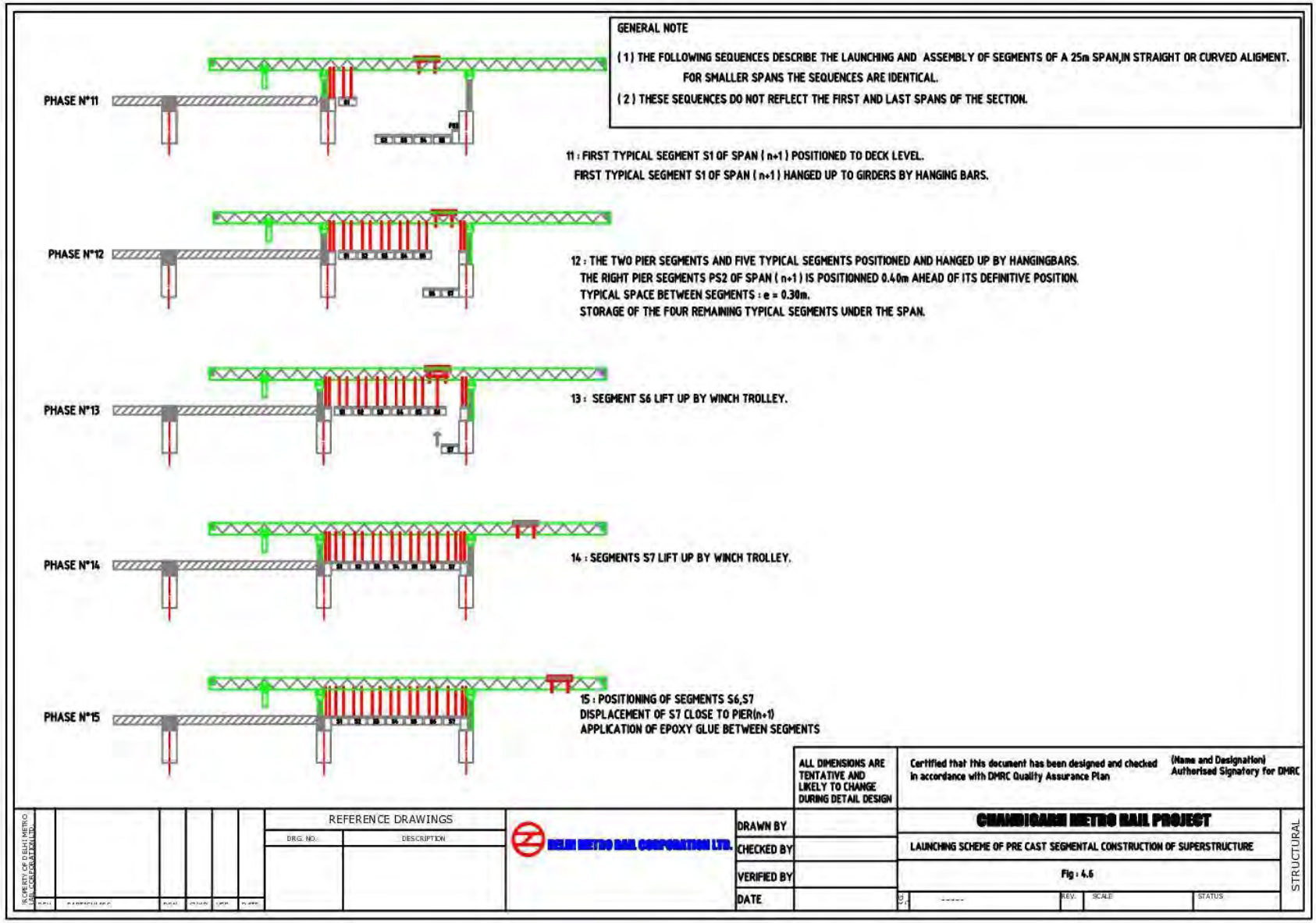
DRG. NO.
DESCRIPTION
DELHI METRO RAIL CORPORATION LTD.
DRAWN BY
CHECKED BY
VERIFIED BY
DATE
CHANDIGARH METRO RAIL PROJECT
LAUNCHING SCHEME OF PRE CAST SEGMENTAL CONSTRUCTION OF SUPERSTRUCTURE
Fig : 4.5
REV:
SCALE
STATUS
STRUCTURAL

Fig. A 5.6


GENERAL NOTE
(1) THE FOLLOWING SEQUENCES DESCRIBE THE LAUNCHING AND ASSEMBLY OF SEGMENTS OF A 25m SPAN,IN STRAIGHT OR CURVED ALIGMENT.
FOR SMALLER SPANS THE SEQUENCES ARE IDENTICAL.
(2) THESE SEQUENCES DO NOT REFLECT THE FIRST AND LAST SPANS OF THE SECTION.
PHASE N°11
11 : FIRST TYPICAL SEGMENT S1 OF SPAN (n+1) POSITIONED TO DECK LEVEL.
FIRST TYPICAL SEGMENT S1 OF SPAN (n+1) HANGED UP TO GIRDERS BY HANGING BARS.
PHASE N°12
12 : THE TWO PIER SEGMENTS AND FIVE TYPICAL SEGMENTS POSITIONED AND HANGED UP BY HANGINGBARS.
THE RIGHT PIER SEGMENTS PS2 OF SPAN (n+1) IS POSITIONNED 0.40m AHEAD OF ITS DEFINITIVE POSITION.
TYPICAL SPACE BETWEEN SEGMENTS : e = 0.30m.
STORAGE OF THE FOUR REMAINING TYPICAL SEGMENTS UNDER THE SPAN.
PHASE N°13
13 : SEGMENT S6 LIFT UP BY WINCH TROLLEY.
PHASE N°14
14 : SEGMENTS S7 LIFT UP BY WINCH TROLLEY.
PHASE N°15
15 : POSITIONING OF SEGMENTS S6,S7
DISPLACEMENT OF S7 CLOSE TO PIER(n+1)
APPLICATION OF EPOXY GLUE BETWEEN SEGMENTS
ALL DIMENSIONS ARE TENTATIVE AND LIKELY TO CHANGE DURING DETAIL DESIGN
Certified that this document has been designed and checked in accordance with DMRC Quality Assurance Plan
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REFERENCE DRAWINGS
DRG. NO.
DESCRIPTION
DELHI METRO RAIL CORPORATION LTD.
DRAWN BY
CHECKED BY
VERIFIED BY
DATE
CHANDIGARH METRO RAIL PROJECT
LAUNCHING SCHEME OF PRE CAST SEGMENTAL CONSTRUCTION OF SUPERSTRUCTURE
Fig : 4.6
REV.
SCALE
STATUS
STRUCTURAL

Fig. A 5.7

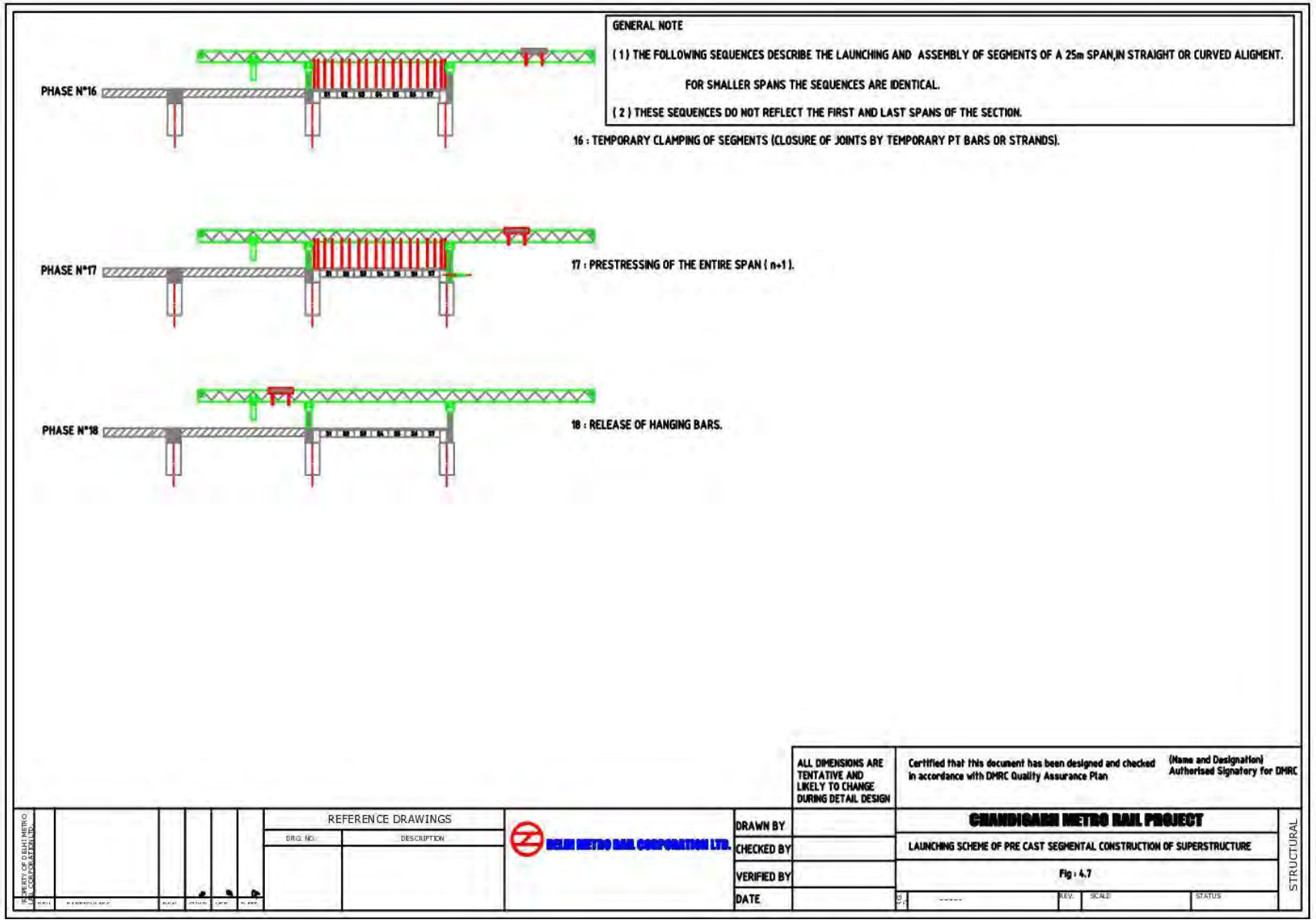
GENERAL NOTE
(1) THE FOLLOWING SEQUENCES DESCRIBE THE LAUNCHING AND ASSEMBLY OF SEGMENTS OF A 25m SPAN,IN STRAIGHT OR CURVED ALIGMENT.
FOR SMALLER SPANS THE SEQUENCES ARE IDENTICAL.
(2) THESE SEQUENCES DO NOT REFLECT THE FIRST AND LAST SPANS OF THE SECTION.
PHASE N°16
16 : TEMPORARY CLAMPING OF SEGMENTS (CLOSURE OF JOINTS BY TEMPORARY PT BARS OR STRANDS).
PHASE N°17
17 : PRESTRESSING OF THE ENTIRE SPAN (n+1).
PHASE N°18
18 : RELEASE OF HANGING BARS.
ALL DIMENSIONS ARE TENTATIVE AND LIKELY TO CHANGE DURING DETAIL DESIGN
Certified that this document has been designed and checked in accordance with DMRC Quality Assurance Plan
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Fig : 4.7
REV.
SCALE
STATUS
STRUCTURAL

Fig. A 5.8

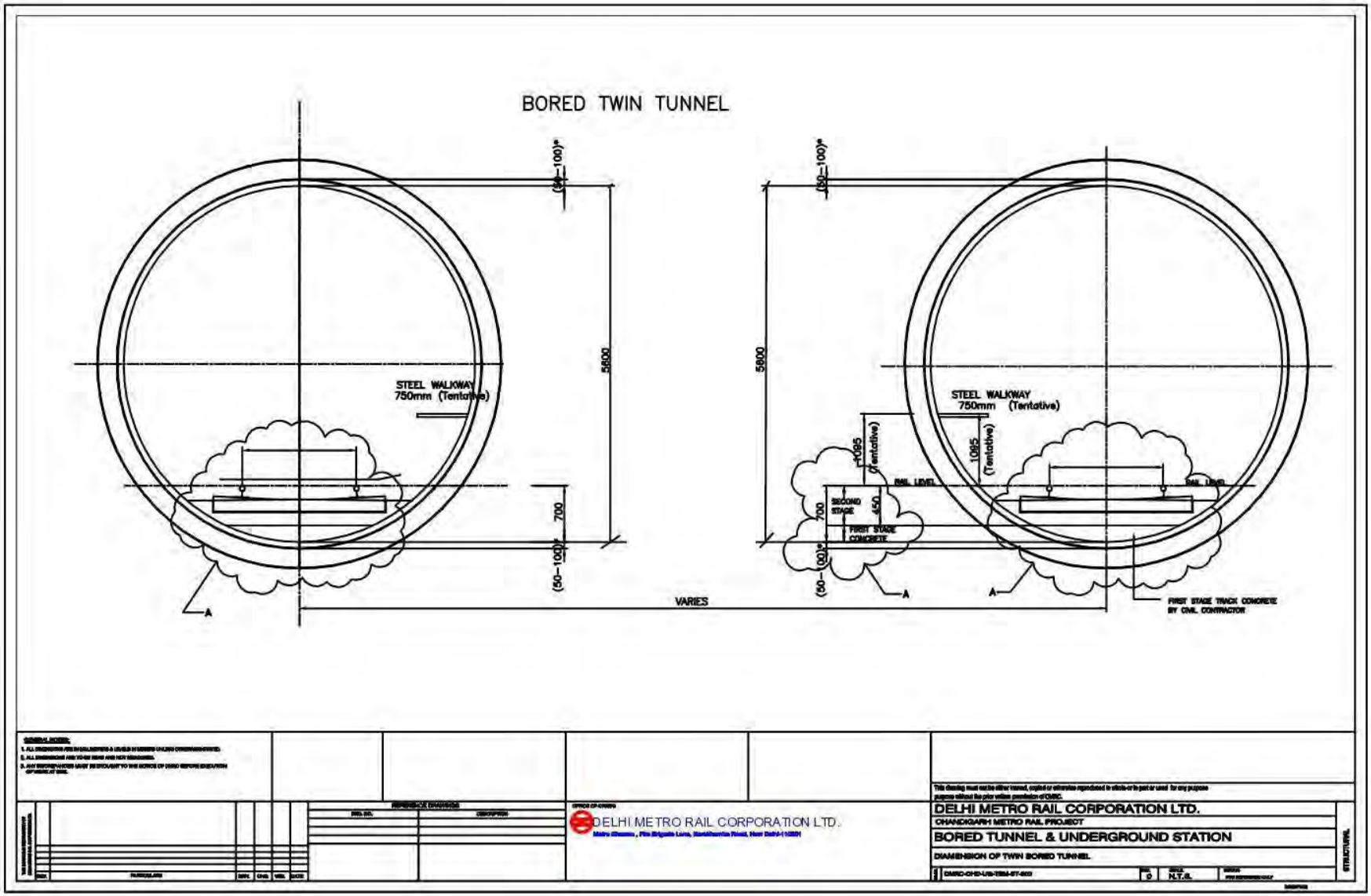

BORED TWIN TUNNEL
STEEL WALKWAY
750mm (Tentative)
5600
700
(50–100)*
VARIES
A
STEEL WALKWAY
750mm (Tentative)
1095
(Tentative)
RAIL LEVEL
SECOND STAGE
450
FIRST STAGE CONCRETE
FIRST STAGE TRACK CONCRETE BY CIVIL CONTRACTOR
DELHI METRO RAIL CORPORATION LTD.
DELHI METRO RAIL CORPORATION LTD.
CHANDIGARH METRO RAIL PROJECT
BORED TUNNEL & UNDERGROUND STATION
DIMENSION OF TWIN BORED TUNNEL
N.T.S.

Fig. A 5.9

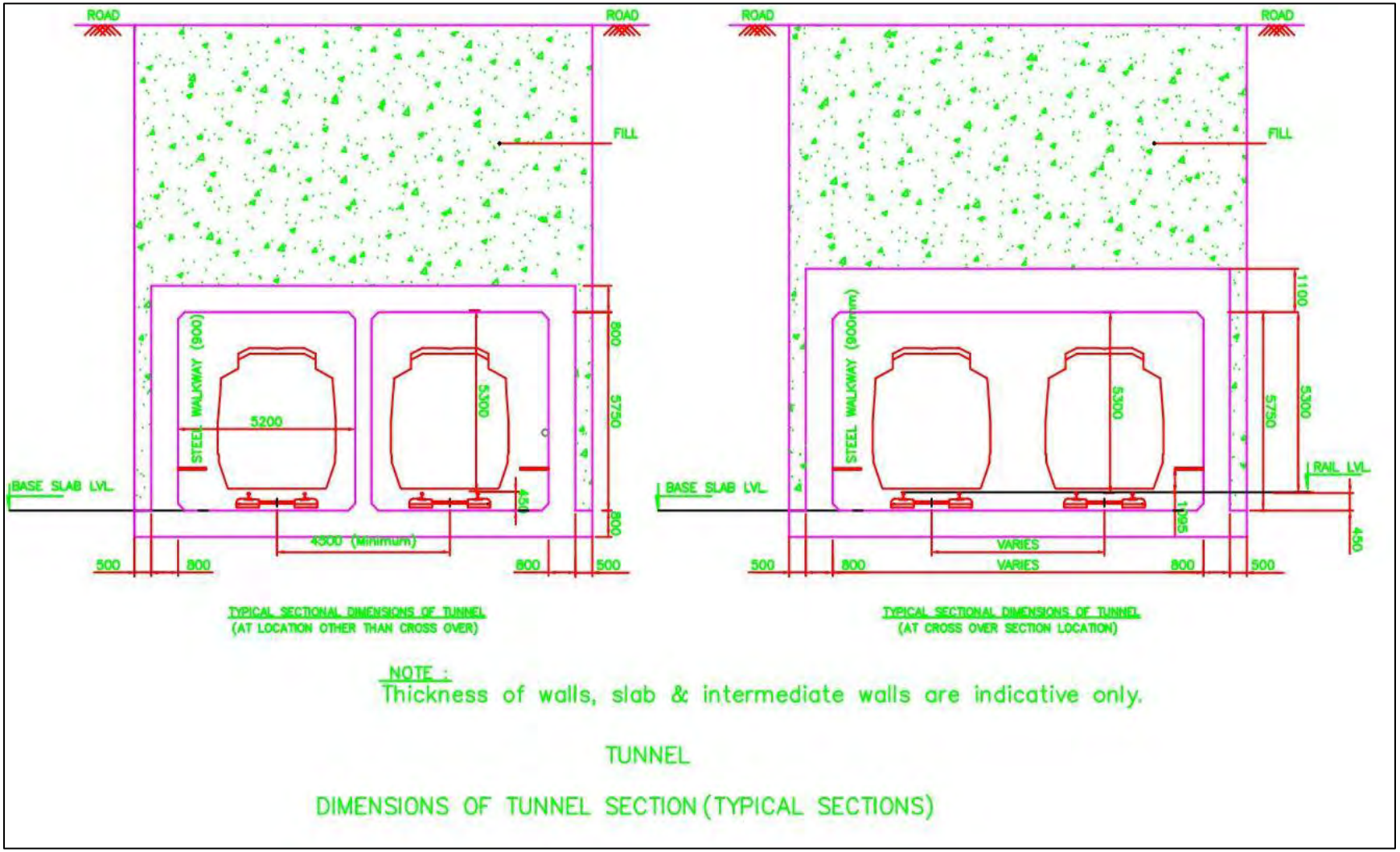

ROAD
ROAD
FILL
STEEL WALKWAY (900)
5200
5300
800
5750
800
BASE SLAB LVL
450
4300 (Minimum)
500
800
800
500
TYPICAL SECTIONAL DIMENSIONS OF TUNNEL
(AT LOCATION OTHER THAN CROSS OVER)
ROAD
ROAD
FILL
STEEL WALKWAY (900mm)
1100
5300
5750
5300
RAIL LVL
BASE SLAB LVL
1295
450
VARIES
VARIES
500
800
800
500
TYPICAL SECTIONAL DIMENSIONS OF TUNNEL
(AT CROSS OVER SECTION LOCATION)
NOTE :
Thickness of walls, slab & intermediate walls are indicative only.
TUNNEL
DIMENSIONS OF TUNNEL SECTION (TYPICAL SECTIONS)

Fig. A 5.10

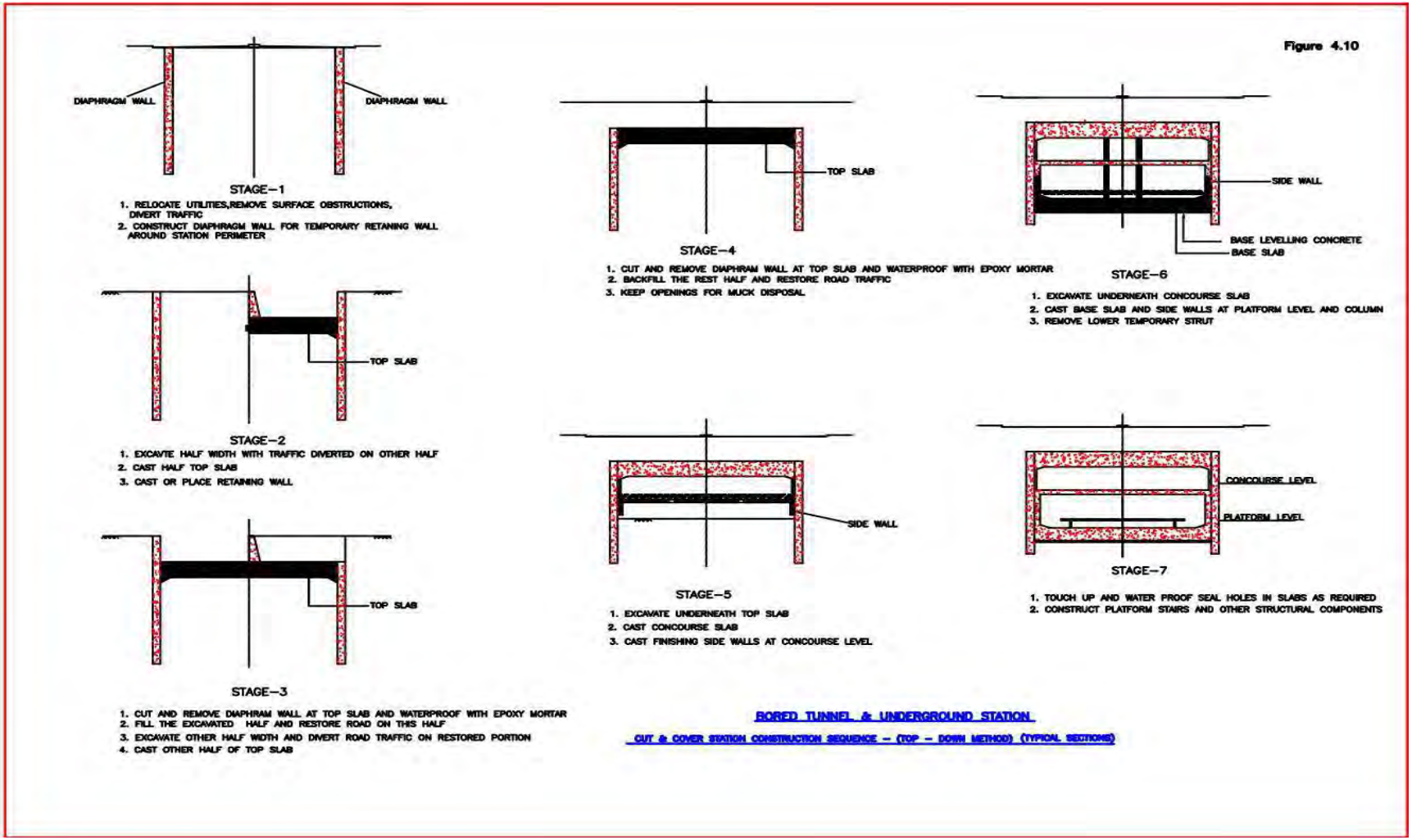


Figure 4.10
DIAPHRAGM WALL
DIAPHRAGM WALL
STAGE–1
1. RELOCATE UTILITIES,REMOVE SURFACE OBSTRUCTIONS, DIVERT TRAFFIC
2. CONSTRUCT DIAPHRAGM WALL FOR TEMPORARY RETAINING WALL AROUND STATION PERIMETER
TOP SLAB
STAGE–2
1. EXCAVTE HALF WIDTH WITH TRAFFIC DIVERTED ON OTHER HALF
2. CAST HALF TOP SLAB
3. CAST OR PLACE RETAINING WALL
TOP SLAB
STAGE–3
1. CUT AND REMOVE DIAPHRAM WALL AT TOP SLAB AND WATERPROOF WITH EPOXY MORTAR
2. FILL THE EXCAVATED HALF AND RESTORE ROAD ON THIS HALF
3. EXCAVATE OTHER HALF WIDTH AND DIVERT ROAD TRAFFIC ON RESTORED PORTION
4. CAST OTHER HALF OF TOP SLAB
TOP SLAB
STAGE–4
1. CUT AND REMOVE DIAPHRAM WALL AT TOP SLAB AND WATERPROOF WITH EPOXY MORTAR
2. BACKFILL THE REST HALF AND RESTORE ROAD TRAFFIC
3. KEEP OPENINGS FOR MUCK DISPOSAL
SIDE WALL
STAGE–5
1. EXCAVATE UNDERNEATH TOP SLAB
2. CAST CONCOURSE SLAB
3. CAST FINISHING SIDE WALLS AT CONCOURSE LEVEL
SIDE WALL
BASE LEVELLING CONCRETE
BASE SLAB
STAGE–6
1. EXCAVATE UNDERNEATH CONCOURSE SLAB
2. CAST BASE SLAB AND SIDE WALLS AT PLATFORM LEVEL AND COLUMN
3. REMOVE LOWER TEMPORARY STRUT
CONCOURSE LEVEL
PLATFORM LEVEL
STAGE–7
1. TOUCH UP AND WATER PROOF SEAL HOLES IN SLABS AS REQUIRED
2. CONSTRUCT PLATFORM STAIRS AND OTHER STRUCTURAL COMPONENTS
BORED TUNNEL & UNDERGROUND STATION
CUT & COVER STATION CONSTRUCTION SEQUENCE – (TOP – DOWN METHOD) (TYPICAL SECTIONS)

CHAPTER 6

STATION PLANNING

6.1 GENERAL

Stations on the Line

The proposed Metro Rail at Surat has two corridors covering an approximated distance of about 40.35 Kilometer. The Corridor-I is partially Underground and partially Elevated having 20 stations and Corridor-II is entirely Elevated having 18 stations thus a total of 38 stations are proposed in entire Phase-I. Almost all the stations are on the road stations Kadarsha-Ni-Nal station is a part of development plan and is designed on the model of TOD (Transit Oriented Development) station having Property Development around the station.

Corridor-I run from Sarthana, Nature Park, Varchha Chopati Garden, Shri Swaminarayan Mandir Kalakunj, Kapodara, Labheshwar Chowk, Central Warehouse, Surat Railway Station, Maskati Hospital, Gandhi Baug, Kadarsha Ni Nal, Majura Gate, Rupali Canal, Althan Tenament, Althan Gam, VIP Road, Women ITI, Bhimrad Convention Centre and Dream City.

The corridor-I covers a total distance of 21.61 Km from dead end to dead end of which about 6.47km is underground excluding Ramps on either side and balance corridor is elevated. It has 20 stations out of which 6 stations are underground and 14 stations are elevated.

Corridor II runs from Bheshan, Botanical Garden, Ugat Vaarigruh, Palanpur Road, L.P. Swani School, Performing Art Centre, Adajan Gam, Aquarium, Badri Narayan Temple, Athwa Chopati, Majura Gate, Udhana Darwaja, Kamela Darwaja, Anjana Farm, Model Town, Magob, Bharat Cancer hospital and Saroli.

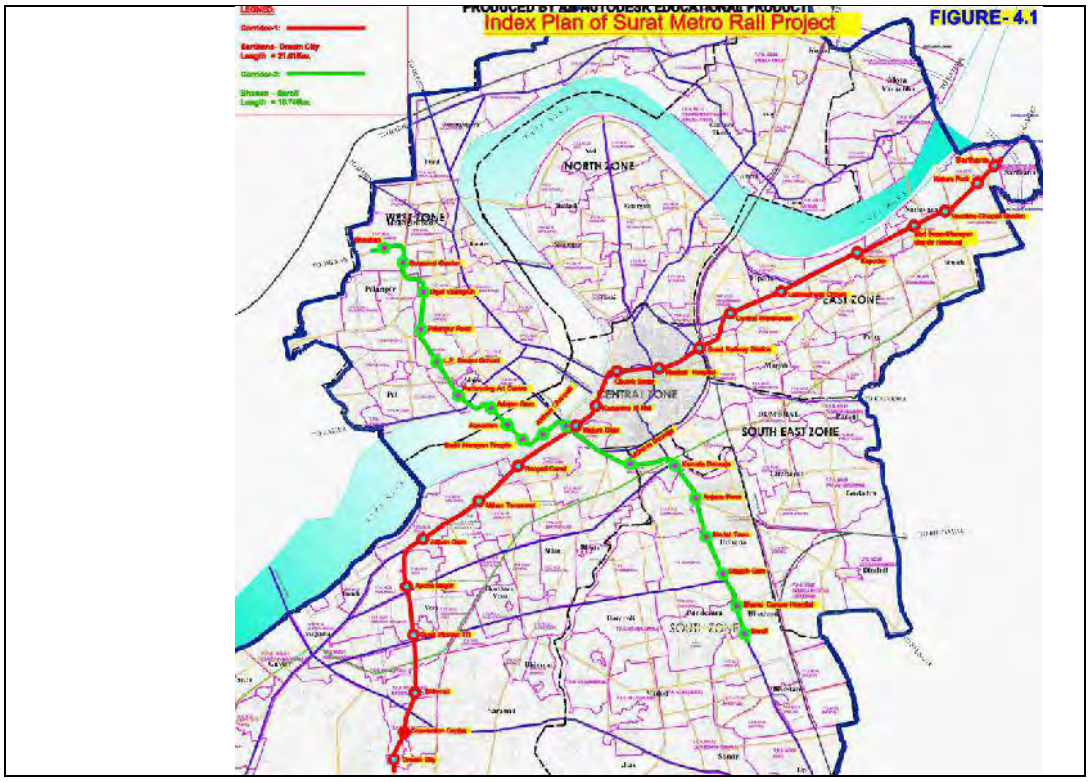
It covers a total distance of about 18.74Kilometer from Dead End to End of Track and the entire corridor is elevated stations having 18 stations.

The Stations have been located so as to serve passenger requirements and to enable convenient integration with other modes of transport. Efforts have been

made to propose station locations at a uniform inter-station distance as feasible. Average inter-station distance is ~1 Kilometer, though it varies from 0.5867 Km to 1.8028km due to land-use and topographic reasons.

The Minimum Inter-station distance in corridor-I between Central Warehouse Station and Surat Railway Station is 586.70 meters and in corridor-II 590.2 meters is between L.P. Swani School and Performing Art Centre. The Maximum distance between stations on corridor-I is between Maskati Hospital and Gandhi Baug and 1802.80 meters and in corridor-II between Bharat Cancer Hospital and Saroli1442.40 meters.

Key plan Surat Metro Corridor-I & II



KEY PLAN

6.2 RAIL LEVELS AND ALIGNMENT

The rail levels in the stations are decided to provide sufficient clearance (Headroom) for the traffic on the road. In the Elevated section, Rail Level is generally about 13 meter above Ground in order to maintain a clearance of 5.50 meter between the Road and the Station Structure. The alignment is planned generally in middle of the road in order to keep the land acquisition to

minimum, and a two-level station design has been proposed. Entry/exit structures to the proposed stations and traffic integration areas have been planned in the open spaces available on the road sides.

6.3 PLATFORMS

The stations have been planned with side platforms to avoid the viaduct structure from flaring in and out at stations, which obstructs the road traffic below. Care has been taken to locate stations on straight alignment. However, in some stations, site constraints have become the deciding criteria and a curve of 1000m radius has been introduced in platform.

6.4 SEQUENCE OF STATIONS

The sequence of stations along with their respective chainages, site and platform characteristics are presented in the Table 6.4.1.and 6.4.2.

Table 6.4.1
STATION LOCATION CHARACTERISTICS CORRIDOR-I

CORRIDOR-I : SARTHANA TO DREAM CITY : SURAT METRO					
S.No	Station Name	Chainage(m)	Inter Distance Between Two Stations.	TYPE OF STATION UNDERGROUND / ELEVATED	Remark's
0	DEAD END	(-) 450			
1	Sarthana	0.0	450.00	ELEVATED	Side Platform
2	Nature Park	1238.1	1238.10	ELEVATED	Side Platform
3	VarchaChopati Garden	2108.5	870.40	ELEVATED	Side Platform
4	Shri Swaminarayan Mandir Kalakunj	3196.6	1088.10	ELEVATED	Side Platform
5	Kapodara	4237.0	1040.40	U/G	Island Platform
6	Labheshwar Chowk	5506.60	1269.60	U/G	Side Platform Platform above each other
7	Central Warehouse	6668.9	1162.30	U/G	Island Platform
8	Surat Railway Station	7255.6	586.70	U/G	Island Platform
9	Maskati Hospital	8487.00	1231.40	U/G	Side Platform Platform above each other
10	Chowk Bazar	10289.8	1802.80	U/G	Island Platform
11	Kadarshna Ni Nal	11721.7	1431.90	ELEVATED	Side Platform

12	Majura Gate	12313..8	592.10	ELEVATED	Side Platform
13	Rupali Canal	13611.5	1297.70	ELEVATED	Side Platform
14	Althan Tenament	14696.2	1084.70	ELEVATED	Side Platform
15	Althan Gam	15843.5	1147.30	ELEVATED	Side Platform
16	VIP Road	16699.1	855.60	ELEVATED	Side Platform
17	Surat Women ITI	17808.4	1109.30	ELEVATED	Side Platform
18	Bhimarad	18659.5	851.10	ELEVATED	Side Platform
19	Convention Centre	19829.4	1169.90	ELEVATED	Side Platform
20	Dream City	20710.3	880.90	ELEVATED	Side Platform
	DEAD END	21160.3	450.00		

Table 6.4.2

STATION LOCATION CHARACTERISTICS CORRIDOR-II

CORRIDOR-II : BHESHAN TO SAROLI: SURAT METRO					
S.No	Station Name	Chainage(m)	Inter Distance Between Two Stations.	TYPE OF STATION UNDERGROUND / ELEVATED	Remark's
0	DEAD END	-949.6			
1	Bheshan	0	949.60	ELEVATED	Side Platform
2	Botanical Garden	950.6	950.60	ELEVATED	Side Platform
3	Ugat Vaarigruh	1617.5	666.90	ELEVATED	Side Platform
4	Palanpur Road	2738.9	1121.40	ELEVATED	Side Platform
5	L.P.Savani school	3831.6	1092.70	ELEVATED	Side Platform
6	Performing Art Centre	4421.8	590.20	ELEVATED	Side Platform
7	Adajan Gam	5155	733.20	ELEVATED	Side Platform
8	Aquarium	5770.1	615.10	ELEVATED	Side Platform
9	Badri Narayan Temple	6991.5	1221.40	ELEVATED	Side Platform
10	Athwa Chaupati	7863.5	972.00	ELEVATED	Side Platform
11	Majura Gate	9019.2	1155.70	ELEVATED	Side Platform
12	Udhana Darwaja	10646.2	16270.00	ELEVATED	Side Platform
13	Kamela Darwaja	11247.2	601.00	ELEVATED	Side Platform
14	Anjana Farm	12593.9	1346.70	ELEVATED	Side Platform
15	Model Town	13636.3	1042.40	ELEVATED	Side Platform

16	Magob	14763.10	1126.80	ELEVATED	Side Platform
17	Bharat Cancer Hospital	15899.4	1155.40	ELEVATED	Side Platform
18	Saroli	17341.8	1442.40	ELEVATED	Side Platform
	DEAD END	17791.851	450.05		

6.5 PLANNING AND DESIGN CRITERIA FOR STATIONS

1. The stations can be divided into public and non-public areas (those areas where access is restricted). The public areas can be further subdivided into paid and unpaid areas.

2. The platform level has adequate assembly space for passengers for both normal operating conditions and a recognized abnormal scenario.

3. The platform level at elevated stations is determined by a critical clearance of 5.5m under the concourse above the road intersection, allowing 3.3m for the concourse height, about 0.8m for concourse floor and 1.8 m for structure of tracks above the concourse. Further, the platforms are 1.09-m above the tracks. This would make the rail level in an elevated situation at least 13 meters above ground.

4. In the underground stations, platform level is determined by a critical clearance of 2.0m - 2.50-m above the station box, which would be 13.7 high. Allowing about 80 cm for the box structure, ~70 cm for rails /supporting structure and 1.09 m for rail to platform ht, would make the platforms in an underground situation at least 13.5-m below ground.

5. In the case of Up and Down Line Platforms Decked above each other the Rail level of First Platform is about -11.5 m and that of the Second Platform is about -20.5m below the ground.

6. The concourse contains automatic fare collection system in a manner that divides the concourse in two distinct areas. The “Unpaid Area” is where passengers gain access to the system, obtain travel information and purchase tickets. On passing through the ticket gates, the passenger enters the “Paid Area”, which includes access to the platforms.

7. The arrangement of the concourse is assessed on a station-by-station basis and is determined by site constraints and passenger access requirements. However, it is planned in such a way that maximum surveillance can be achieved by the ticket hall supervisor over ticket machines, automatic fare collection (AFC) gates, stairs and escalators. Ticket machines and AFC gates are positioned to minimize cross flows of passengers and provide adequate circulation space.

8. Sufficient space for queuing and passenger flow has been allowed at the ticketing gates.

9. Station entrances are located with particular reference to passenger catchment points and physical site constraints allowing for 30 meter right-of-way in order to provide a minimum of 3 lane road under the station building on either side of the median.

10. Office accommodation, operational areas and plant room space is required in the non-public areas at each station. The functions of such areas are given below in Table 5.5.1

11. The DG set, Bore Well, Pump House and Underground Water Tanks would be located generally in one area on ground within the Entry / Exit structures.

12. The system is being designed to maximize its attraction to potential passengers and the following criteria have been observed:

    Minimum distance of travel to and from the platform and between platforms for transfer between lines.

    Adequate capacity for passenger movements.

    Convenience, including good signage relating to circulation and orientation.

    Safety and security, including a high level of protection against accidents.

13. Following requirements have been taken into account:

    Minimum capital cost is incurred consistent with maximizing passenger attraction.

    Minimum operating costs are incurred consistent with maintaining efficiency and the safety of passengers.

Flexibility of operation including the ability to adapt to different traffic conditions changes in fare collection methods and provision for the continuity of operation during any extended maintenance, repair period, etc.

Provision of good visibility of platforms, fare collection zones and other areas, thus aiding the supervision of operations and monitoring of efficiency and safety.

Provision of display of passenger information and advertising.

14. The numbers and sizes of staircases/escalators are determined by checking the capacity against AM and PM peak flow rates for both normal and emergency conditions such as delayed train service, fire etc.

15. In order to transfer passengers efficiently from street to platforms and vice versa, station planning has been based on established principles of pedestrian flow and arranged to minimize unnecessary walking distances and cross-flows between incoming and outgoing passengers.

16. Passenger handling facilities comprise of stairs/escalators, lifts and ticket gates required to process the peak traffic from street to platform and vice-versa (these facilities must also enable evacuation of the station under emergency conditions, within a set safe time limit).

6.5.1 Typical Elevated Station:

The station is generally located on the road median. Total length of the station is ~140-m. All the stations are two-level stations. The concourse is planned along the whole length of the platform with staircases leading from either side of the road. The maximum width of the station at concourse is ~21-m. Passenger facilities like ticketing, information, etc as well as operational areas are provided at the concourse level.

Typically, the concourse is divided into public and non-public zones. The non-public zone or the restricted zone contains station operational areas such as Station Control Room, Station Master's Office, Waiting Room, Meeting Room, UPS& Battery Room, Auxiliary Service Station, Signal and Telecom Rooms, Train Crew Room & Supervisor's Office, Security Room, Station Store Room, Staff Toilets and Public Toilets, etc. The public zone is further divided into paid

and unpaid areas. Water Tanks, Pump Room and DG room is provided on the ground under the entry/ exit structure.

Since the station is in the middle of the road, minimum vertical clearance of 5.5-m has been provided under the concourse. Platforms are at a level of about 14.0 meters from the road. To reduce physical and visual impact of the elevated station, stations have been designed as cantilevered structures with single column located at the central verge of the road.

With respect to its spatial quality, an elevated Metro structure makes a great impact on the viewer as compared to an At-grade station. The positive dimension of this impact has been accentuated to enhance the acceptability of an elevated station and the above ground section of tracks. Structures that afford maximum transparency and are light looking have been envisaged. A slim and ultra-modern concrete form is proposed, as they would look both compatible and modern high-rise environment as well as the lesser-built, low-rise developments along some parts of the metro corridors.

Platform roofs, that can invariably make a structure look heavy, have been proposed to be of steel frame with Galvalume Sheets and aluminum (ACP) cladding to achieve a light look. Platforms would be protected from the heat and rains by providing an overhang of the roof and sidewalls are avoided, thereby enhancing the transparent character of the station building.

It is proposed to install solar panels on the station roof to reduce energy demand of the station and to recharge the entire Rain water of the station and the viaduct for recharging the Underground aquifers.

6.5.2 Typical Underground Station: Type- I (UG)

The typical underground station is a two-level station with platforms at the lower level and concourse on the upper level. Concourses are provided at the two ends to provide two alternate routes for evacuation. The upper level has, in addition to the concourse, all the passenger amenities, ECS plant rooms, electrical and S&T equipment rooms, station operation areas such as Station Control Room, Station Master’s Office, Waiting Room, Meeting Room, UPS& Battery Room, Signaling& Train Crew Room, Train Crew Supervisor's Office, Security & Station Store Room, Staff Toilets, etc. Lower level has platforms, tracks, seepage sump, pump room and similar ancillary spaces beyond the platforms on either side.

Ventilation shafts, equipment hatch, entrances and chiller plants for ECS plant are aboveground structures associated with the underground station and are being provided on the open spaces by the road side. Generally four entrances have been provided to the station, two at each end (one each from either side of the road). Other aboveground structures are suitably located near the station.

Structure of the underground station is essentially a concrete box about 20~21m wide, 14.6-m high and 190 or as per design long with an intermediate slab. Sides of the box are made of 1.2 m thick RCC. The roof of station box is about 1.2 m thick.

6.5.3 Typical Underground Station: Type-II (UG)

Generally an underground station is a two-level station with platforms at the lower level and concourse on the upper level however, it is proposed to construct an underground station with both the platforms decked on each other and provide all the passenger facilities and technical rooms on a separate building adjoining the station building thereby reducing the width of the station box.

in this type of configuration both the tunnels (Up-line and Dn-Line)enter and exit the station building one over the other thus the platforms are at two different levels instead of at same level. The tunnels travel outside the station at the same level till they are sufficiently apart to maintain the safe distance from each other to ensure its structural safety of each other before they start parallel to each other at the same level. The nozzles have been proposed at the ends of both the platforms and the tunnel ventilation has been ensured with booster fans provided at the ends of platforms suspended from the ceiling of the platform. The seepage sump, sewage sump and pump rooms have been proposed in the undercroft along with the cable trays etc.

The Concourses are provided in two separate buildings outside the station box at the two ends to provide two alternate routes for evacuation. The lower levels of the concourse has, in addition to the concourse, all the passenger amenities, ECS plant rooms, electrical and S&T equipment rooms, station operation areas such as Station Control Room, Station Master’s Office, Waiting Room, Meeting Room, UPS& Battery Room, Signaling & Train Crew Room, Train Crew Supervisor's Office, Security & Station Store Room, Staff Toilets, etc.

The Ventilation shafts, equipment hatch, entrances and chiller plants for ECS plant DG Sets, pump room, Water Tanks for fire safety, HVAC and other station functions and similar ancillary spaces are proposed in these buildings

at ground level and above ground levels. Generally two entrances have been provided to the station, one at each end from the road. Other aboveground structures are suitably located near the station.

Structure of this type of underground station is essentially a concrete box about 10~11m wide, 20.5 ~ 22.0 m high and 190 or as per design long with an intermediate slab separating two Platforms. Sides of the box are made of about 1 ~1.2 m thick RCC D/Wall. The roof of station box is about 1.2 m thick. With about 2m overburden for essential services like drainage, water supply, electrical cables, telecommunication etc to cross the road above station structure.

6.5.4 Typical Interchange Station

An interchange station is a station two or more lines meet or cross each other and provide flexibility to the passenger to change from one line to the other in order to save time and money. The interchange station provides great utility and flexibility for the system as a whole, and it will decrease the time required for travel within the city. The interchange passenger movements require larger areas for circulation than normal stations

Majura gate station has an interchange type configuration, since it is located at the intersection of Crridor-1 and corridor-2 a large number of passengers will change from one line to the other from this station.

The station is located at an important crossing of Majura Gate flyover and Ring Road flyover surrounded by an Engineering college, Surat Civil hospital, Income tax colony, BSNL colony and Kailash Nagar and a very important commercial area of the city having a number of Malls, shopping complexes, Hotels, offices and high rise housing complexes. The corridor-I station is located parallel to the ROB from Majura Gate leading to Udhana Darwaza and the Corridor-II station is perpendicular to this station located near Dayalji Ashram as both the stations are Elevated stations.

The area is very congested and a number of high rise buildings on one side of the road and a ROB is running parallel to the alignment of corridor-1 thereby making it out of bound for locating services across the road. On corridor-II there is a very old charitable Ashram property and on other side a number of High rise commercial buildings are existing making it extremely difficult to place any entry/exit or ancillary structure.

The Lower concourse of the corridor-II station is divided in Paid and Un-paid concourse thus, making it convenient for the passengers from both the lines to transfer form one line to the other without coming out of the system. Thus the

passengers coming out of the corridor-I can go to corridor-II and vice versa through common concourse at lower level.

The passengers entering this station may go directly to either line platform in Dream City direction, Sarthana direction on corridor-1 or can board the train on corridor-2 for Bheshan direction or Saroli station direction. Similarly Passenger boarding at Majura Gate station on corridor-2 line will also have the option to change the train for Dream city direction or Sarthana direction after alighting at this station and interchanging from the lower concourse of the station without going out of the station.

Table 6.3 STATION ACCOMMODATION

For Elevated and Underground Stations	
1. Station Control Room	2. Cleaner's Room
3. Station Master's Office	4. Security Room
5. Information & Enquiries	6. First Aid Room
7. Ticket Office	8. Miscellaneous Operations Room
9. Ticket Hall Supervisor & Access Fare Collection (AFC gates)	10. Platform Supervisor's Booth
11. Cash and Ticket Room	12. Auxiliary Substation / DG Room
13. Staff Area	14. Fire Tank and Pump Room
15. Staff Toilets	16. Commercial Outlets and Kiosks
17. Station Store Room	18. UPS and Battery Room
19. Refuse Store	20. Signaling / Communication Room
21. Tunnel Ventilation Room	22. Tunnel Ventilation Shafts
23. ECS Plant Room	24. ECS Supply and Exhaust Shafts
25. Chiller Room	26. Space for Cooling Towers
27. Water softening Plant Room	28. Sump, Seepage & Pump Room

6.6 PASSENGER AMENITIES

Passenger amenities such as ticketing counters/automatic ticket vending machines, RCTVM, ticketing gates, etc. are provided in the concourse. Uniform numbers of these facilities have been provided for system wide uniformity, although the requirement of the facilities actually varies from station to station. The same applies to provision of platform widths and staircase/escalators. Maximum capacity required at any station in the year 2026, 2036 and 2046 have been calculated and the most critical year for emergency operation has been adopted for all stations.

For this purpose, peak minute traffic is assumed to be 2% of the peak hour traffic. For checking the adequacy of platform area, stair widths and requirement additional of emergency evacuation stairs, a maximum accumulation of passengers in the station has been considered to be comprising waiting passengers at the platform (including two missed headways) and section load expected to be evacuated at the station in case of an emergency.

6.6.1 Concourse

Concourse forms the interface between street and platforms. In elevated stations, this is contained along the full length of the station. This is where all the passenger amenities are provided. The concourse contains automatic fare collection system in a manner that divides the concourse into distinct paid and unpaid areas. The 'unpaid area' is where passengers gain access to the system, obtain travel information and purchase tickets. On passing through the ticket gates, the passenger enters the 'paid area', which includes access to the platforms.

The concourse is planned in such a way that maximum surveillance can be achieved by the ticket hall supervisor over ticket machines, automatic fare collection (AFC) gates, stairs and escalators. Ticket machines &AFC gates are positioned to minimize cross flows of passengers and provide adequate circulation space. Sufficient space for queuing and passenger flow has been allowed in front of the AFCs.

6.6.2 Ticketing Gates (AFC Gates)

Ticketing gates' requirement has been calculated taking the gate capacity as 28 persons per minute per gate. Passenger forecast for the horizon year 2046 has been used to compute the maximum design

capacity. At least two ticketing gates (one for Normal passenger and one for Wheel chair bound passenger) shall be provided at any station even if the design requirement is satisfied with only one gate. Uniform space has been provided in all stations where gates can be installed as and when required.

6.6.3 Ticket Counters and Ticket Issuing Machines (TIMs), RCTVMs

It is proposed to deploy manual ticket issuing in the beginning of the operation of the line. At a later stage, automatic TIMS would be used for which space provision has been made in the concourse. Capacity of manual ticket vending counters is taken to be 10 passengers per minute and it is assumed that only 40% of the commuters would purchase tickets at the stations while performing the journey. The rest are expected to buy prepaid tickets or prepaid card, etc. Accordingly, the requirement of ticket counters has been calculated and the same provided for in the plans.

6.6.4 Platforms

A uniform platform width of 3.60 m wide side platforms have been proposed in all Elevated stations. These platform widths also have been checked for holding capacity of the platform for worst-case scenario.

6.6.5 Stairs, Escalators and Lifts

Provision has been made for escalators in the paid area i.e. from concourse to platforms. On each platform, one escalator has been proposed. In addition, two staircases with a combined width of 4.8 m are provided on each platform connecting to the concourse.

These stairs and escalator together provide an escape capacity adequate to evacuate maximum accumulatedpassengers in emergency from platforms to concourse in 5.5 minutes. Lifts have been provided one each on either platform, to provide access for elderly and disabled. Since the rise to road from the concourse is about 8m, it is proposed to provide escalators and lifts in addition to stairs for vertical movement of passengers from Road to Concourse level also.

6.6.6 Fire Fighting Measures

Fire fighting provisions in Elevated metro stations are in accordance with the National Building Code of India 1983 (part IV, Fire protection) amendment no. 3 under Fire protection Annexure II.

National Building Code (clause 6.4.8). Fire protection and fire fighting system for metro stations stipulates: -

Wet riser system

Main and diesel pump of 1800 Liters/min capacity to support 3 to 4 hydrant at a time [station building is split into two halves. It is presumed that fire will not break in the two parts simultaneously. There are 3 hydrants in one part. Therefore, pump capacity as above are proposed Jockey pump 180 l/min shall also have DG back up.

Internal Hydrant

The internal hydrant is provided with 2 no. RRL hose pipes of 38 mm Ø with 63 mm standard instantaneous coupling along with associated branch pipe and cabinet and a first aid hose reel of 25 mm Ø length 45m fitted with 6.5 mm nozzle. One hydrant each at Ground Level, Passage Level and Platform Level in each half of the station building and so located that every part of station is within 30 m radius.

Sprinklers are provided in the property development area only. Additional sprinkler pump is not provided as these are not required being the integral part of the station. The two pumps already provided will take care of sprinkler flow requirements.

Detectors are provided in the operational areas only, and above false ceiling if the gap is > 750 mm.

One manual call box at each level in each half of the station building is provided.

The HT panels, LT panels, main LT distribution board and essential power panels shall be provided with linear heat sensing tubes with CO2 cylinder.

A two way fire brigade inlet at ground level on each rising main for hydrants is provided.

Draw off connection is provided on the fine water tank for fire brigade.

Water tank of 50,000 liters capacity if planned since commercial development is restricted to 250 sqm.

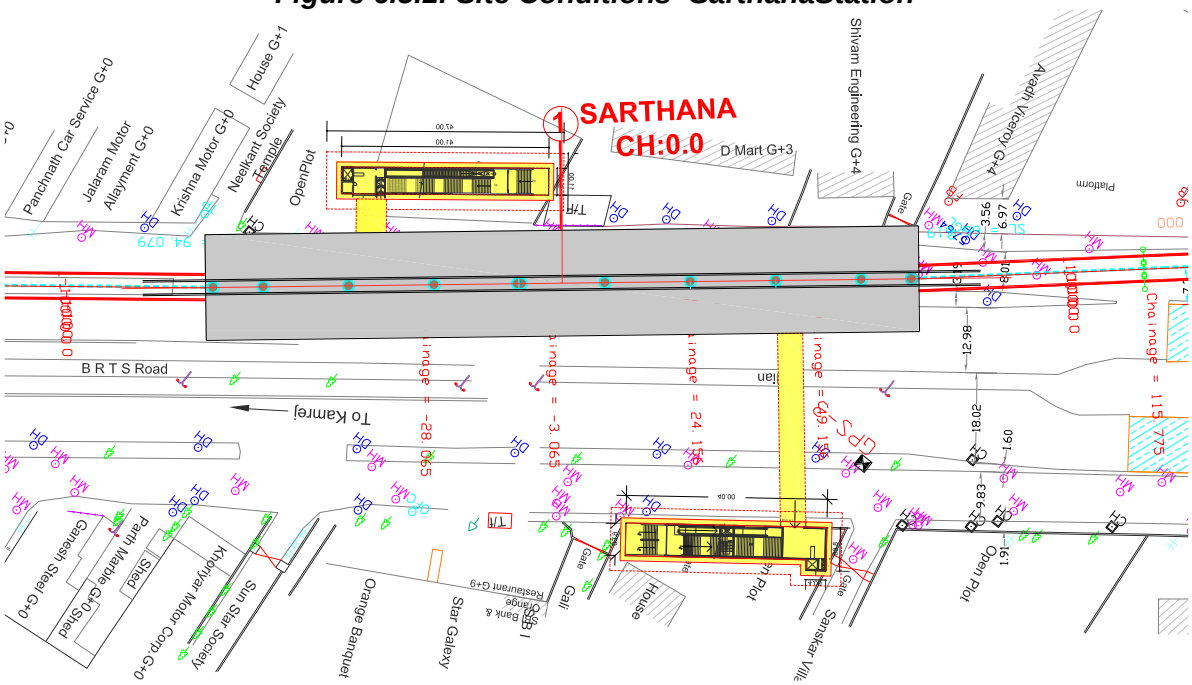
Portable fire extinguishers (CO2) a set of two is provided in each of the equipment room.

Corridor-I: Sarthana to Dream City

1.SarthanaStation

Chainage	:	0.00
Inter station Distance	:	450 m (from Dead End)
Rail Level	:	13.86m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at BRT road near Sarthana Nature Park BRT station
Catchment Area	:	The station is situated at main BRT road near Dr.Shyama Prasad Mukerjee Zoological Garden and surrounded by a number of Apartments and Commercial buildings which will be the main source of passengers to the station

Figure 6.5.1: Site Conditions- SarthanaStation



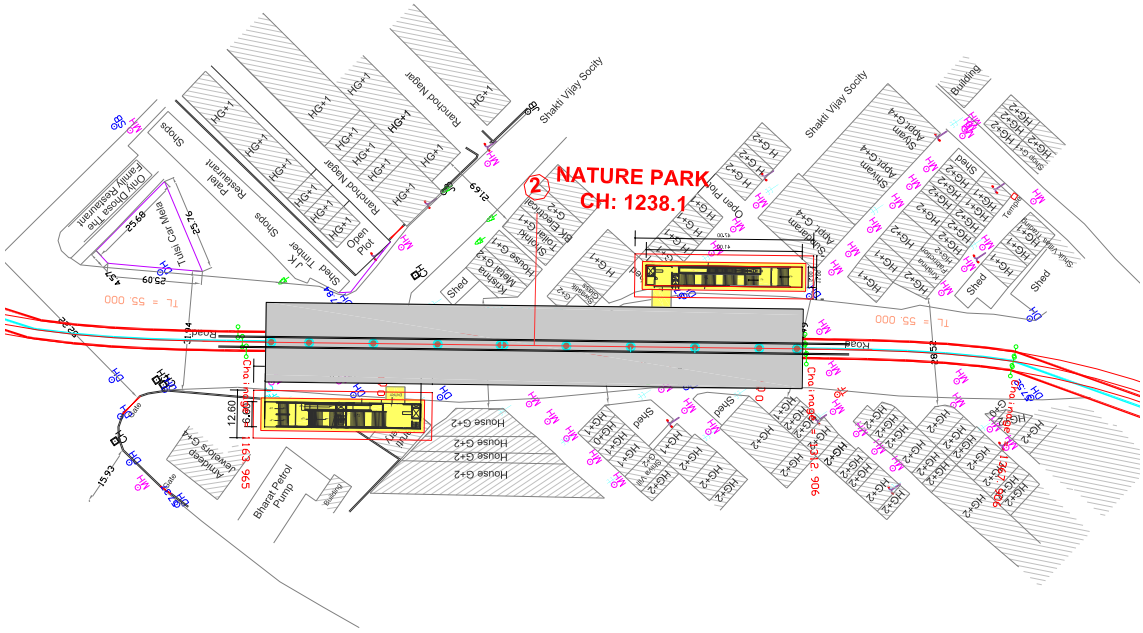
Corridor-I : Sarthana to Dream City

2.Nature Park Station

Chainage	:	1238.10
Inter station Distance	:	1238.10 m
Rail Level	:	13.77m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at main BRT Road.
Catchment Area	:	

The station is situated at near BRT road in Bajrang Nagar which is a densely populated residential area and the Commercial buildings will form the main source of passengers to the station

Figure 6.5.2: Site Conditions- Nature Park Station

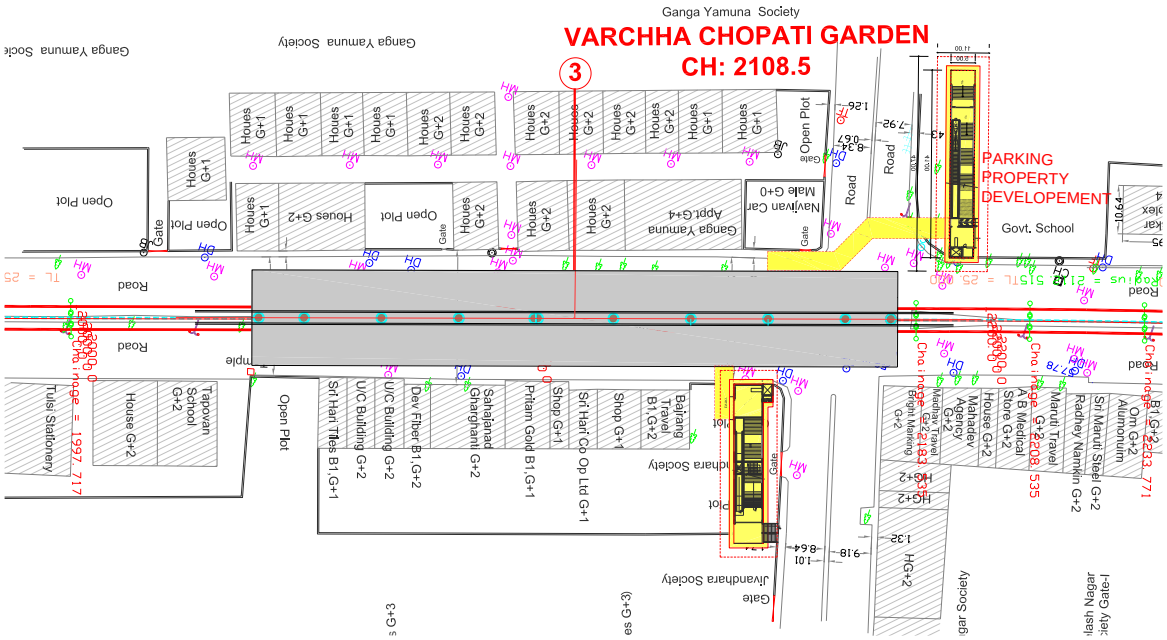


Corridor-I: Sarthana to Dream City

3.VarchaChopati Garden Station

Chainage	**:**	2108.50
Inter station Distance	**:**	870.40 m
Rail Level	:	13.64 m above road level
Station type	**:**	Elevated
Entry / Exit	**:**	Proposed on both sides of the Road
Location	**:**	The station is located near Tapovan Circle.
Catchment Area	**:**	The station is situated at near BRT road in Jeewandhara Society, Chukiwadi and Nana Varchha residential area and the Commercial areas will form the main source of passengers.

Figure 6.5.3: Site Conditions- VarchaChopati Garden Station

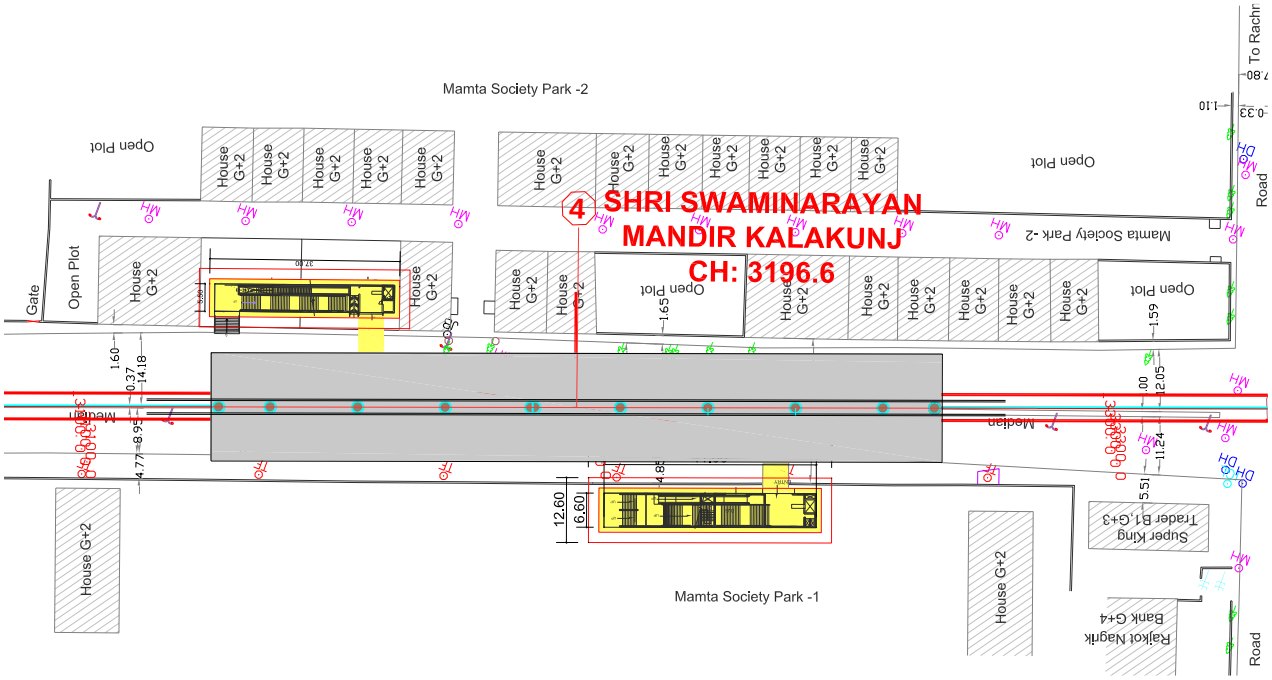


Corridor - I :Sarthana to Dream City

4.ShriSwaminarayanMandir-KalakunjStation

Chainage	:	3196.60
Inter station Distance	:	1088.10m
Rail Level	:	13.63m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at Mamta Park society near P S Swani Heart Institute
Catchment Area	:	Mamta Park Society is a very posh Residential area of the city the Residents of Mamta Park, Chikuwadi and visitors to PS Swani Heart Institute will constitute the main passengers to the station.

Figure 6.5.4: Site Conditions- ShriSwaminarayanMandir-KalakunjStation

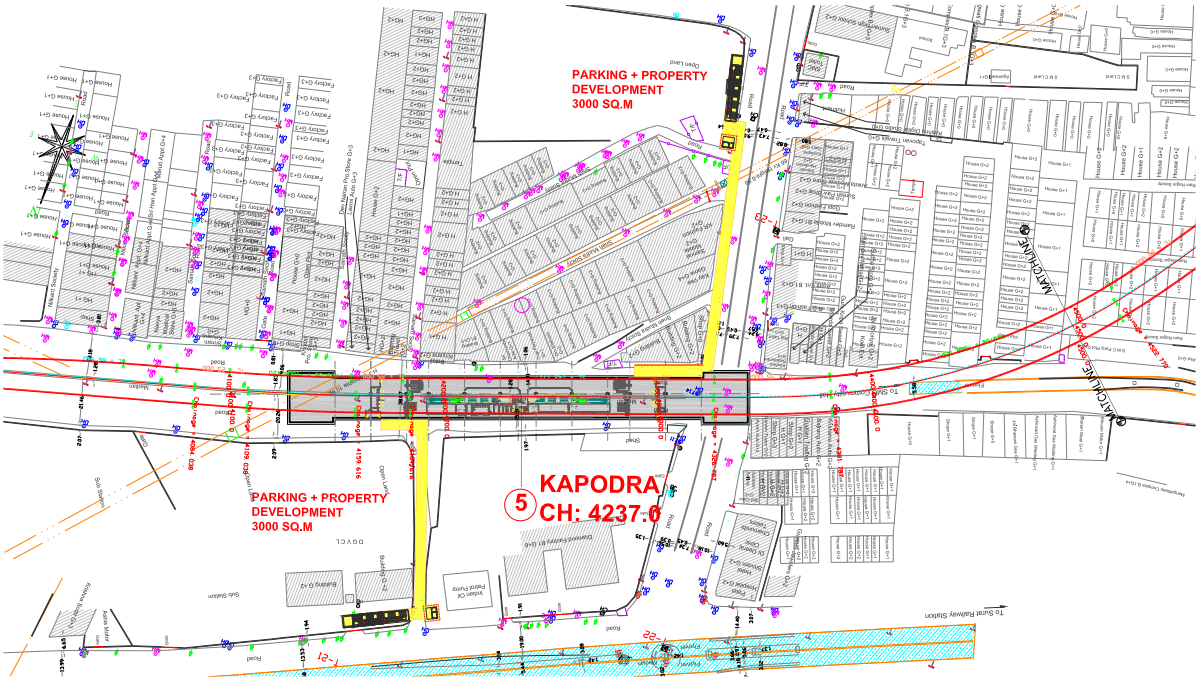


Corridor-I :Sarthana to Dream City

5.Kapodara Station

Chainage	:	4237.00
Inter station Distance	:	1040.40m
Rail Level	:	-14.059m below Ground level
Station type	:	Under Ground
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at Somnath Society near Diamond Factory.
Catchment Area	:	The station is situated in an area having an number of Diamond factories, Hotels, guest houses and residential areas which will form the main source of passengers to the station.

Figure 6.5.5: Site Conditions - KapodaraStation

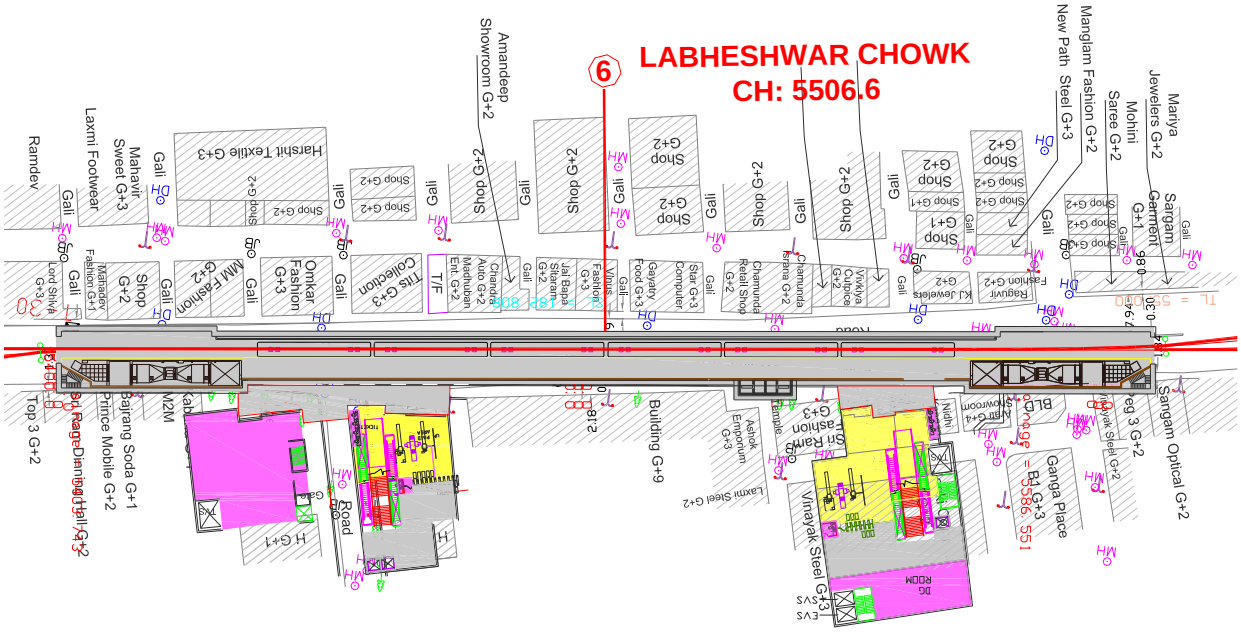


Corridor-I :Sarthana to Dream City

6.LabheshwarChowkStation

Chainage	**:**	5506.60
Inter station Distance	**:**	1269.60 m
Rail Level	**:**	**-**11.47 m and -20.45m below Ground Level
Station type	**:**	Underground
Entry / Exit	**:**	Proposed on one side of the Road
Location	**:**	The station is located at the main road in Anuradha society Market.
Catchment Area	**:**	The station is situated in a very old commercial area of the city having densely populated residential and commercial areas which will form the main source of passengers.

Figure 6.5.6: Site Conditions- LabheshwarChowkStatation

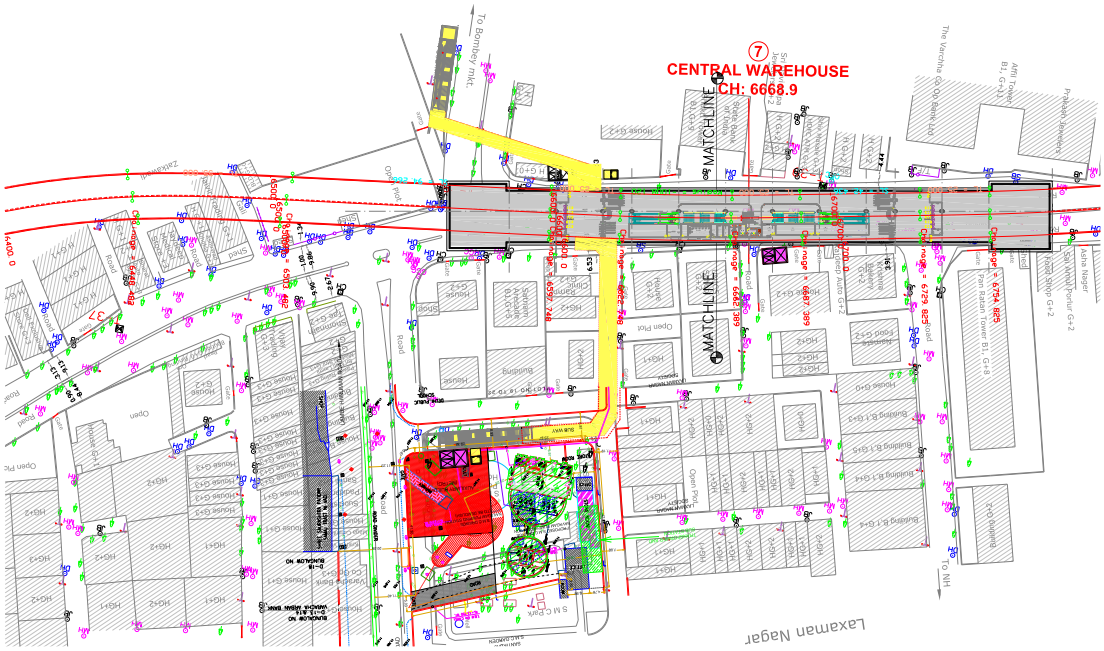


Corridor-I :Sarthana to Dream City

7.Central Warehouse Station

Chainage	**:**	6668.9
Inter station Distance	**:**	1162.30m
Rail Level	**:**	-13.79m below Ground level.
Station type	**:**	Underground
Entry / Exit	**:**	Proposed on both sides of the Road
Location	**:**	The station is located near central warehouse.
Catchment Area	**:**	The Kalidas nagar is a High end residential area Of Surat city having plotted colonies with G+2 and G+3 houses. The residents of Sham Nagar, Kalidas Nagar, Laxman Nagar and other commercial establishments will form main source of passengers to the station.

Figure 6.5.7: Site Conditions- Central Warehouse Station

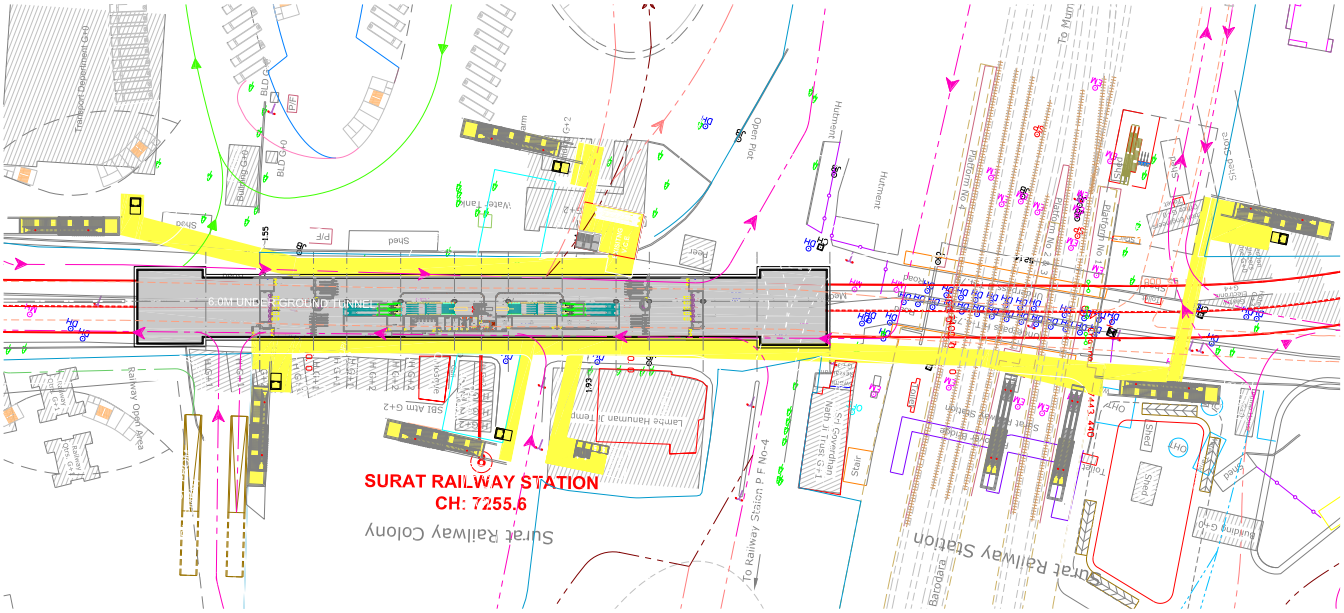


Corridor-I : Sarthana to ream City

8.Surat Railway Station

Chainage	:	7255.60
Inter station Distance	:	586.7 m
Rail Level	:	-20.0m below Road level
Station type	:	Underground
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at the Surat Railway station
Catchment Area	:	The area around the Surat Railway station is being developed as a very density commercial and residential complex by MMTH. The proposed Metro station will be integrated with Transport and commercial scheme which will form the main source of passengers to this station.

Figure 6.5.8: Site Conditions- Surat RailwayStation

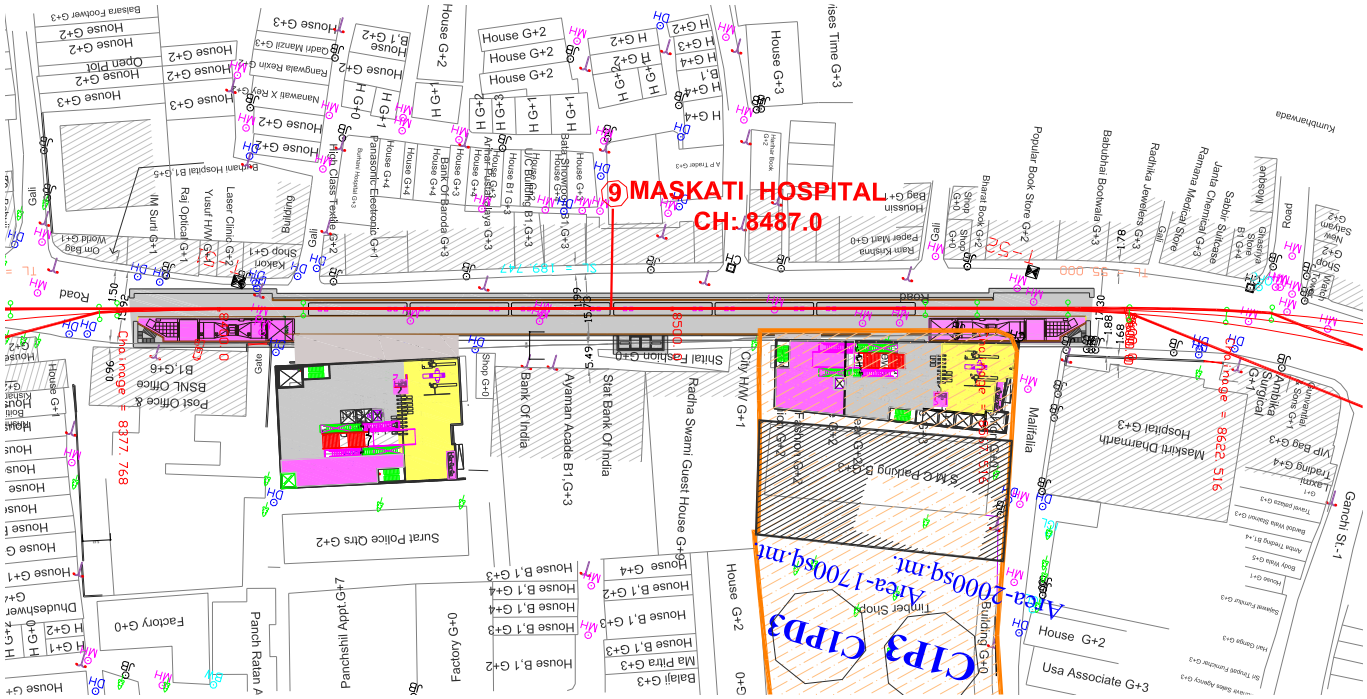


Corridor-I :Sarthana to Dream City

9. Maskati Hospital Station

Chainage	:	8447.00
Inter station Distance	:	1231.40 m
Rail Level	:	-11.07 m and -20.379 below Road level
Station type	:	Underground
Entry / Exit	:	Proposed on one side of the Road.
Location	:	The station is located near SMC Hospital.
Catchment Area	:	Chauta Pul, Sahpore, Sodagrawad, Mughal Sarai, SMC hospital and Maskati Hospital are the areas in the near vicinity of the station which will form the passenger base to the station.

Figure 6.5.9: Site Conditions- Maskati Hospital Station

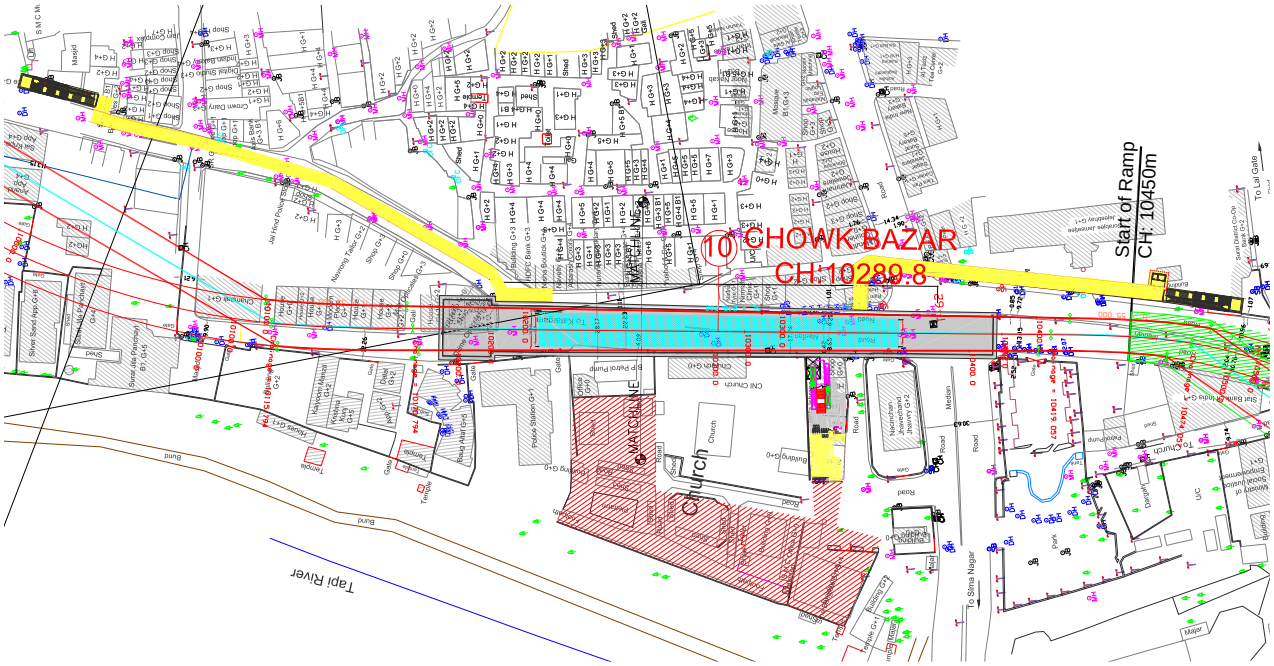


Corridor-I :Sarthana to Dream City

10. Chowk Bazar Station

Chainage	:	10289.80
Inter station Distance	:	1802.80 m
Rail Level	:	-11.606 m below Road level
Station type	:	Underground
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located Near Old Surat Fort, Nehru Bridgeand historical Church.
Catchment Area	:	Situated near Old Fort area having a number of important historical buildings the station is located just outside the old city of Surat which has highest density of population as the main source of passengers.

Figure 6.5.10: Site Conditions- Chowk BazarStation

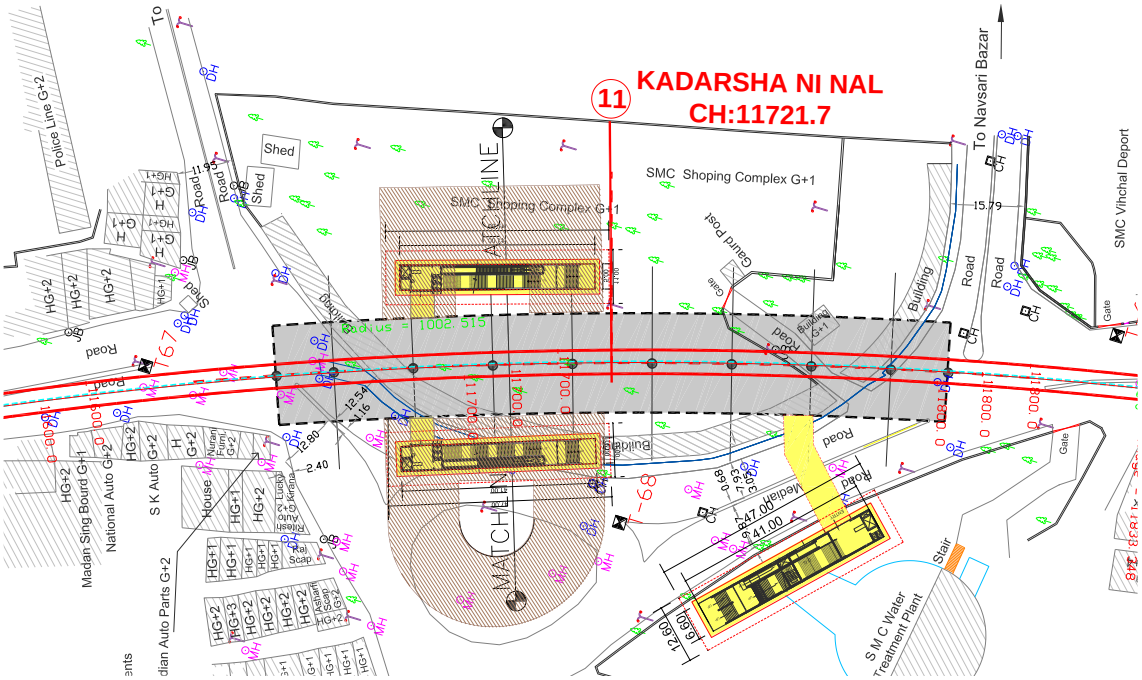


Corridor-I :Sarthana to Dream City

11. Kadarshna Ni Nal Station

Chainage	:	11721.70
Inter station Distance	:	1431.90m
Rail Level	:	14.782m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located near SMC Shopping complex.
Catchment Area	:	It is proposed to develop the open land around The station also as commercial cum office complex. The commercial and residential areas as Garden colony, Rehmatpura, Timliwad and Dharmora will form the main source of passengers.

Figure 6.5.11: Site Conditions- Kadarshna Ni Nal Station

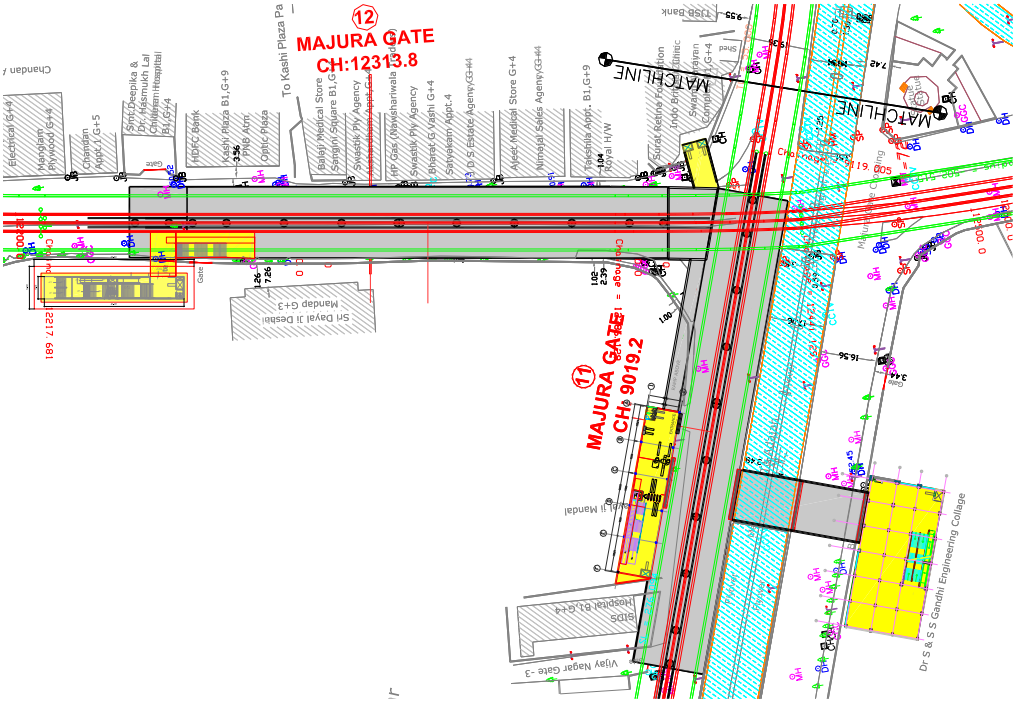


Corridor-I : Sarthana to Dream City

12. Majura Gate Station

Chainage	:	12313.80
Inter station Distance	:	592.10m
Rail Level	:	21.500 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at Majura Gate Flyover.
Catchment Area	:	The station is an interchange station connecting two lines being proposed for Surat Metro Rail. It is located in a very posh locality and commercial area of Surat city. BSNL, Income Tax colony Kailashnagar and Civil Hospital shall be the main source of passengers.

Figure 6.5.12: Site Conditions- MajuraGateStation

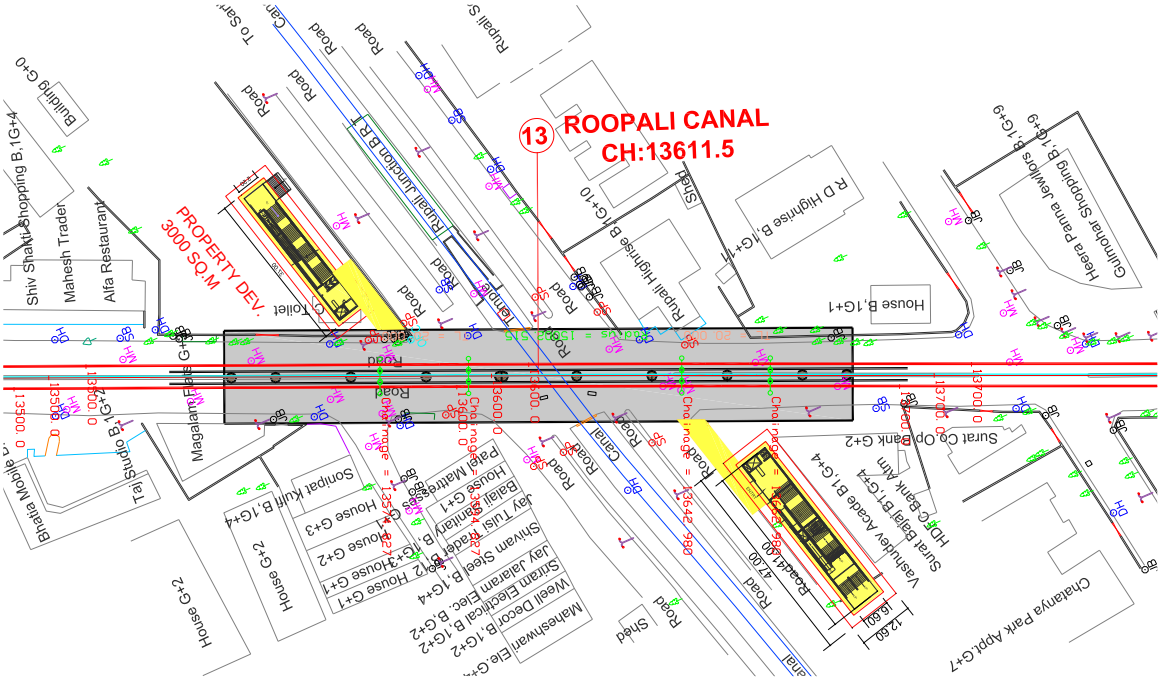


Corridor-I : Sarthana to Dream City

13. Roopali Canal Station

Chainage	:	13611.50
Inter station Distance	:	1297.70m
Rail Level	:	13.675 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at Rupali Junction.
Catchment Area	:	The main source of passengers to the station shall be the residents of ONGC colony, Amba Nagar, Khatodra Wadi, Subhash Nagar and Fame Raj Empire Multiplex.

Figure 6.5.13: Site Conditions- ***Rupali CanalStation***

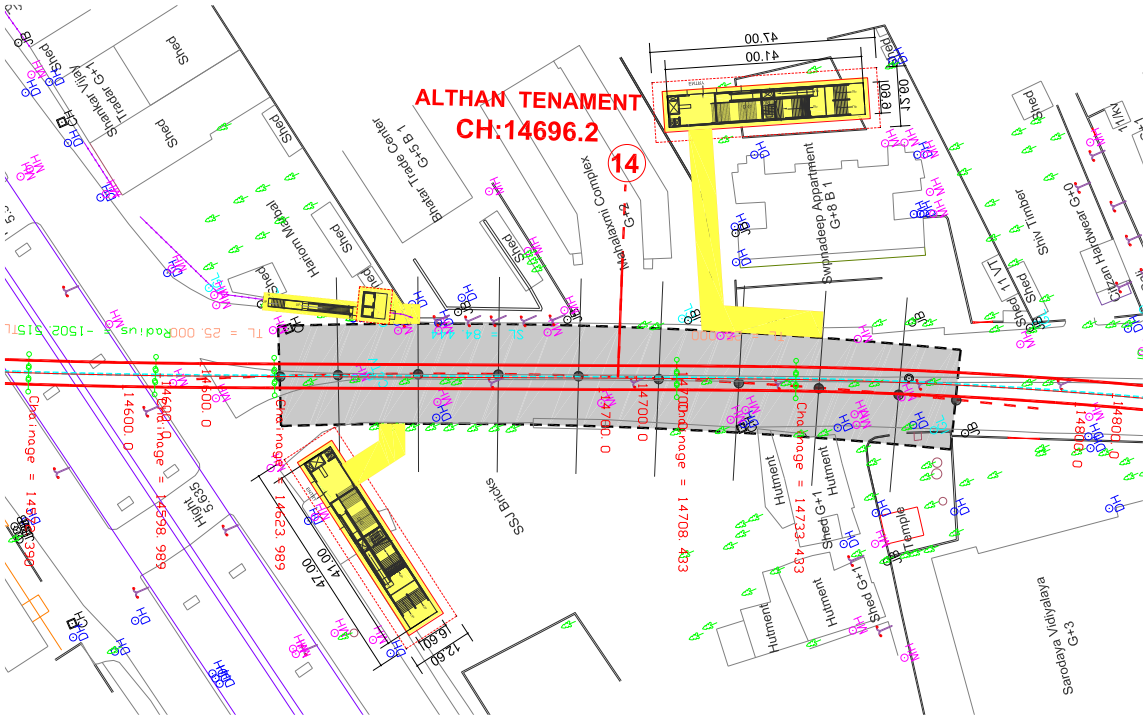


Corridor-I :Sarthana to Dream City

14. AlthanTenament Station

Chainage	:	14696.2
Inter station Distance	:	1084.70m
Rail Level	:	13.608 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located Near Mahalaxmi complex.
Catchment Area	:	The residents of Vishal co operative Housing complex, Ravidarshan society, Bhatar char Rasta and patrons of Mahalaxmi complex will be the main source of passengers to the station.

Figure 6.5.14: Site Conditions- AlthanTenamentStation

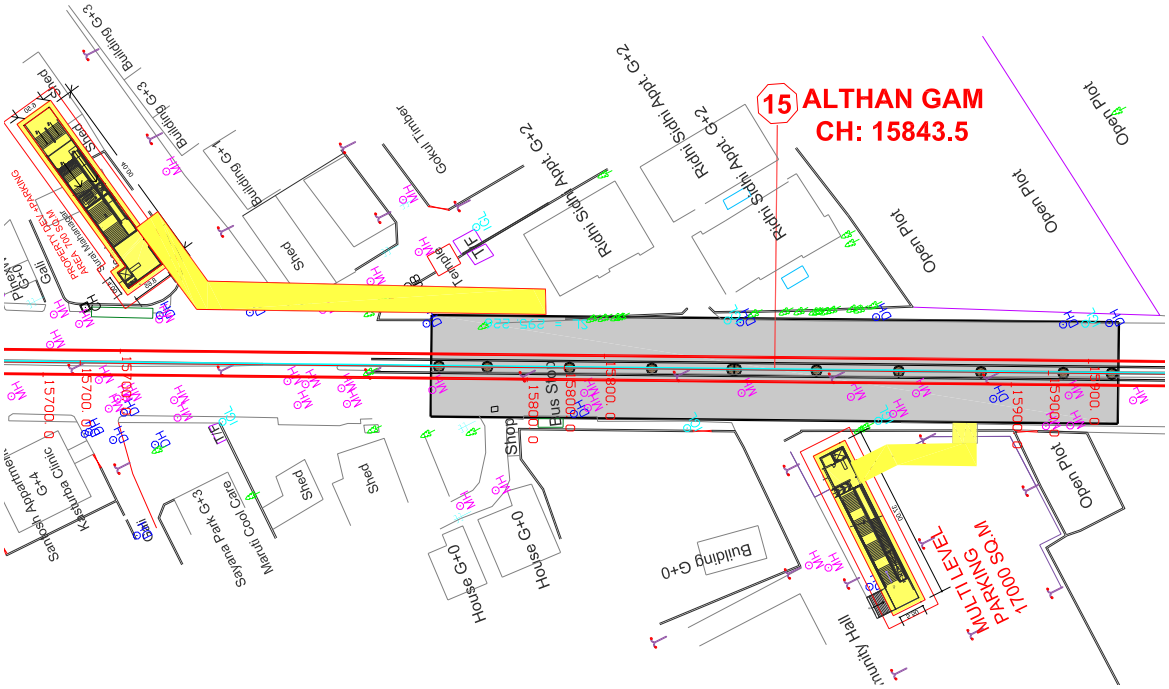


Corridor-I :Sarthana to Dream City

15. Althan Gam Station

Chainage	:	15843.5
Inter station Distance	:	1147.30m
Rail Level	:	13.733 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located near Ridhi Sidhi apartments.
Catchment Area	:	The area is being developed a very well planned new city wherein a large number of Residential apartments complexes have come up with ultra modern amenities that will form the main passenger base for the station.

Figure 6.5.15: Site Conditions- Althan Gam Station

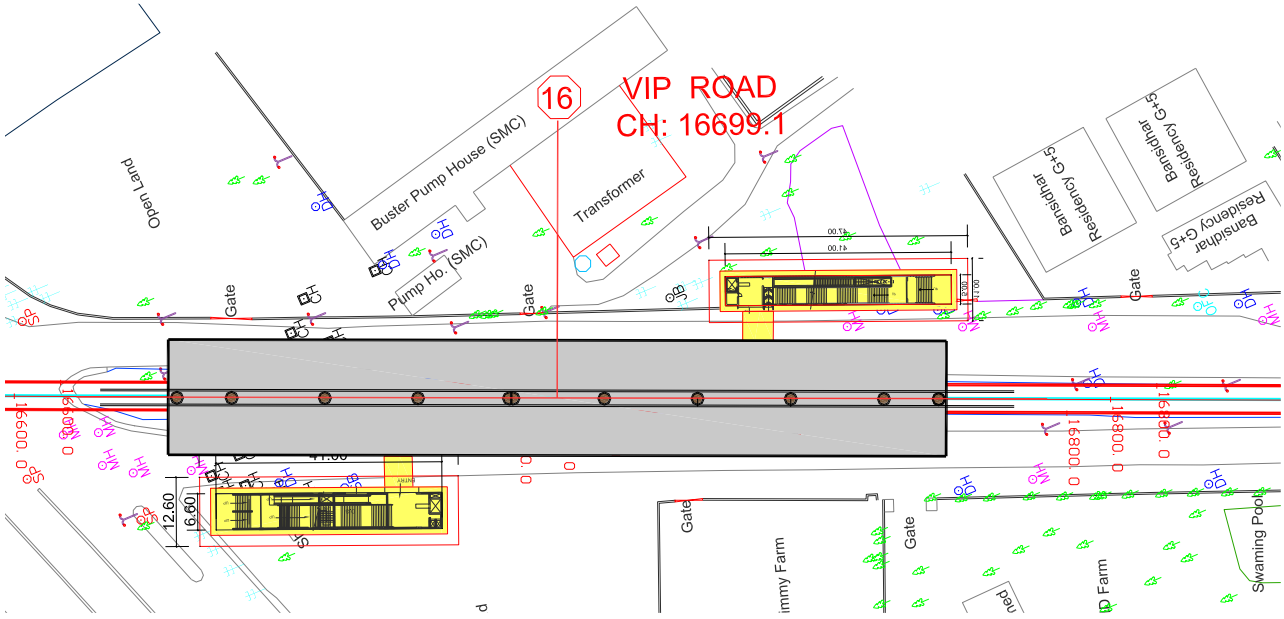


Corridor-I :Sarthana to Dream City

16. VIP Road Station

Chainage	:	16699.10
Inter station Distance	:	855.60m
Rail Level	:	13.556 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located near SMC Booster Pump House Apcha Nagar.
Catchment Area	:	A large number of Residential apartment Complexes like Anad Park, Ravi Nagar Althan andApcha Nagar will form the main passenger base for the station

Figure 6.5.16: Site Conditions- VIP Road Station

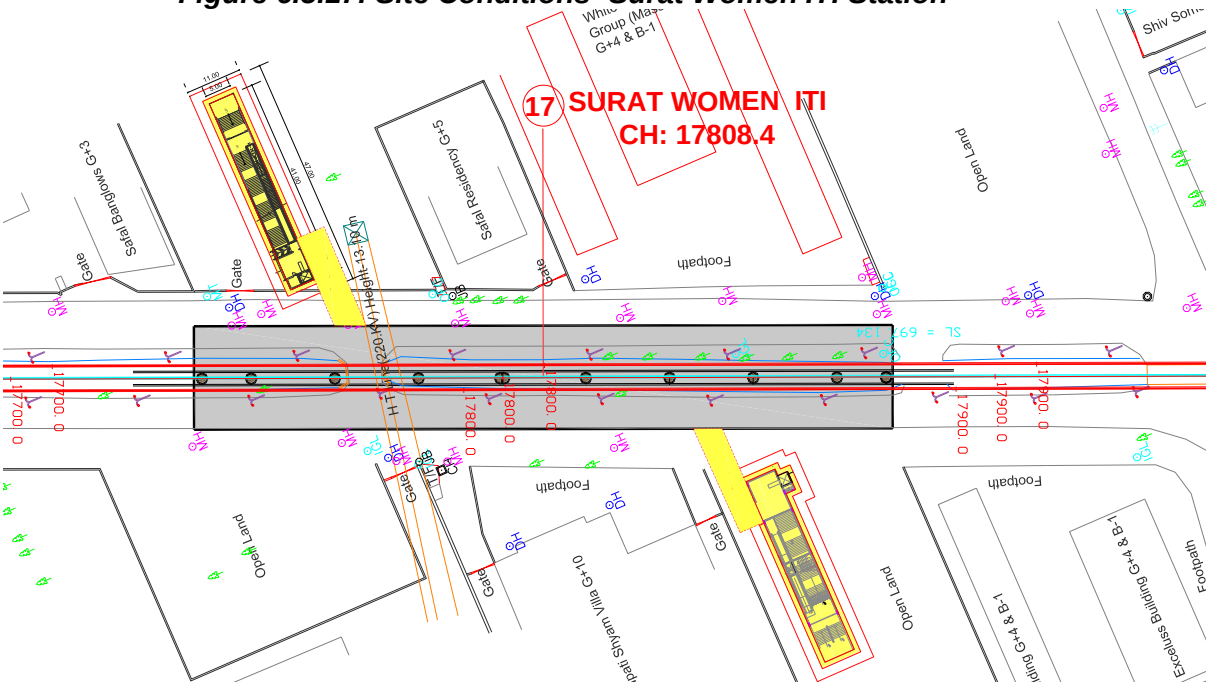


Corridor-I :Sarthana to Dream City

17. Surat Women ITI Station

Chainage	:	17808.40
Inter station Distance	:	1109.30m
Rail Level	:	13.656m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located Near Bahgwan Mahavir Engineering college.
Catchment Area	:	The main source of passengers to the station will be the students of the Bahgwan Mahavir Engineering college, Surat Women ITI and residents of the area commuting to the City Centre Surat.

Figure 6.5.17: Site Conditions- Surat Women ITI Station

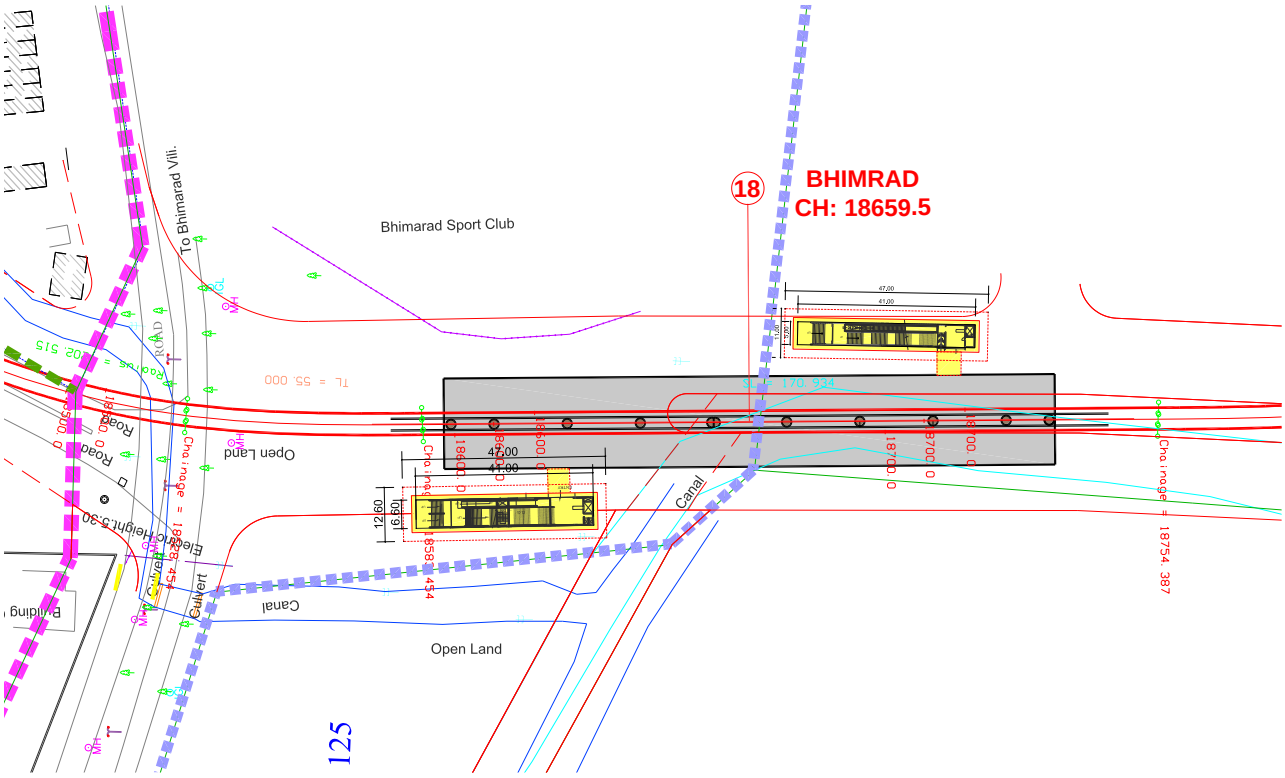


Corridor-I :Sarthana to Dream City

18.Bhimarad Station

Chainage	:	18659.50
Inter station Distance	:	851.10m
Rail Level	:	13.500 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located near Women ITI Surat.
Catchment Area	:	The area is under development but in future it is planned to be development as an important residential and commercial hub with all modern facilities which will form the passenger base to the station.

Figure 6.5.18: Site Conditions- BhimaradStation

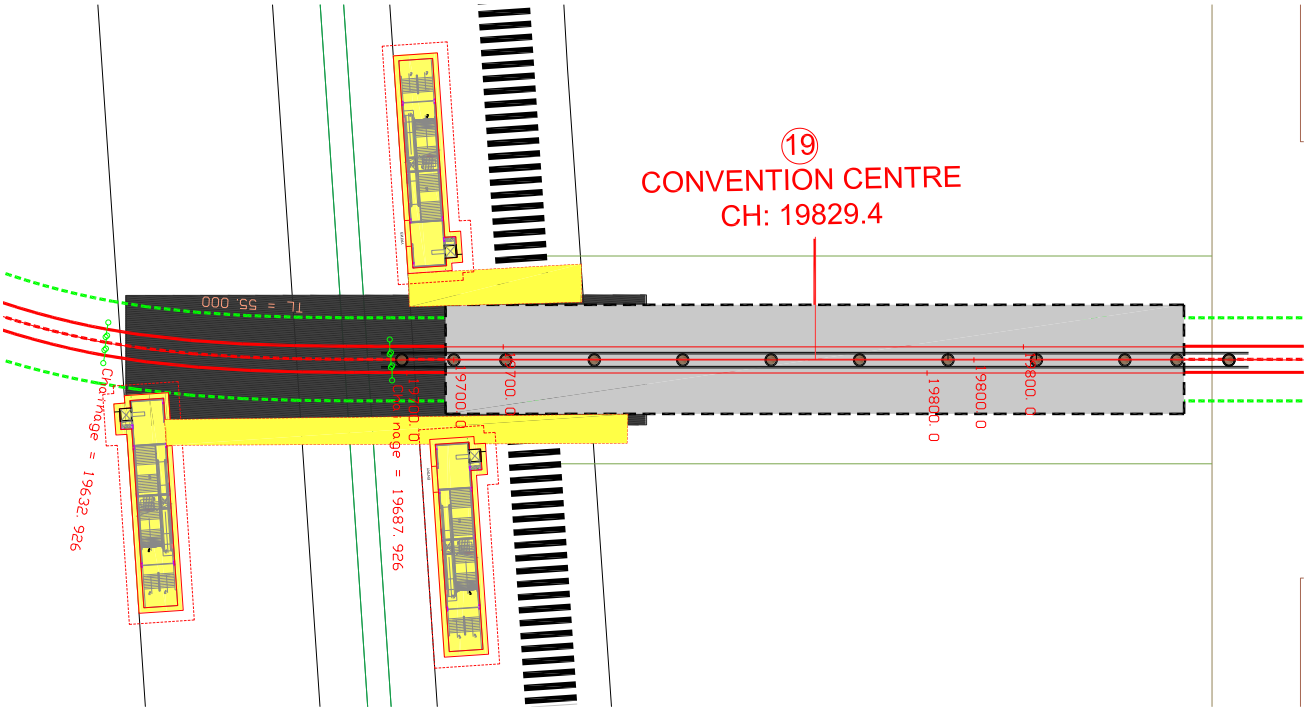


Corridor-I :Sarthana to Dream City

19.Convention Centre Station

Chainage	**:**	19829.40
Inter station Distance	**:**	1169.90m
Rail Level	**:**	13.50 m above road level
Station type	**:**	Elevated
Entry / Exit	**:**	Proposed on both sides of the Road
Location	**:**	The station is located at Convention Centre near Ring road.
Catchment Area	**:**	The main source of passengers to the station shall be the people coming to convention centre from various part of the city.

Figure 6.5.19: Site Conditions- Convention Centre Station

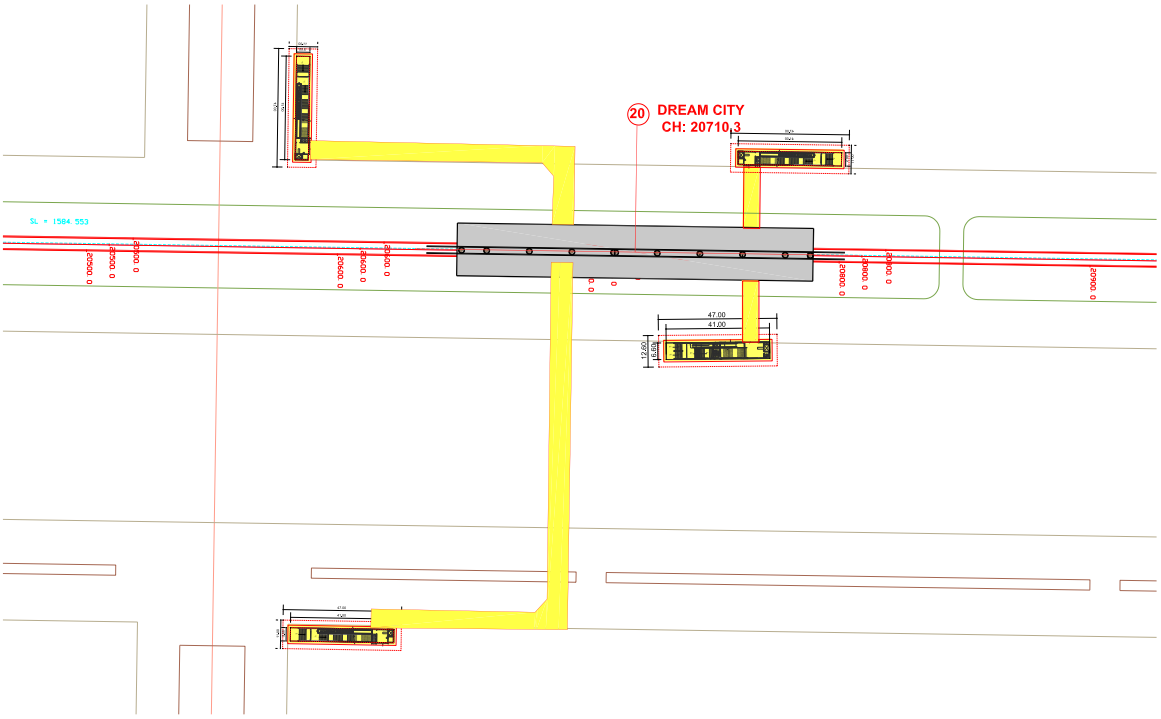


Corridor-I :Sarthana to Dream City

20. Dream City Station

Chainage	:	20710.30
Inter station Distance	:	880.90m
Rail Level	:	13.5m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at proposed Dream City.
Catchment Area	:	Dream city is an important Sub-city being developed as a model Ultra Modern town having high end residential and commercial areas and other amenities residents of which will form the passenger base to station.

Figure 6.5.20: Site Conditions- Dream City Station

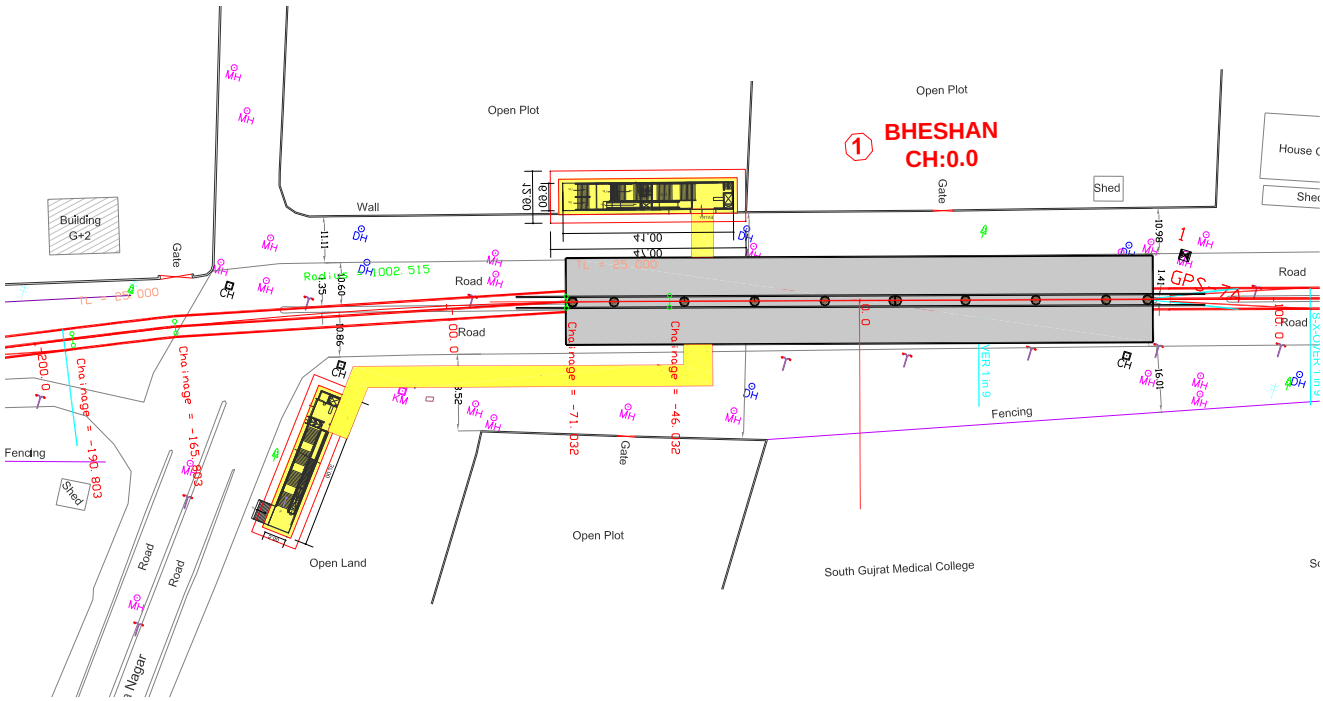


Corridor-II :Bhesan to Saroli

1. Bheshan Station

Chainage	:	0.00
Inter station Distance	:	949.63 m
Rail Level	:	13.538 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at South Gujrat Medical college.
Catchment Area	:	People visiting South Gujrat Medical college and hospital and the resident of Bhesan will be main passengers to this station.

Figure 6.5.1: Site Conditions- BheshanStation

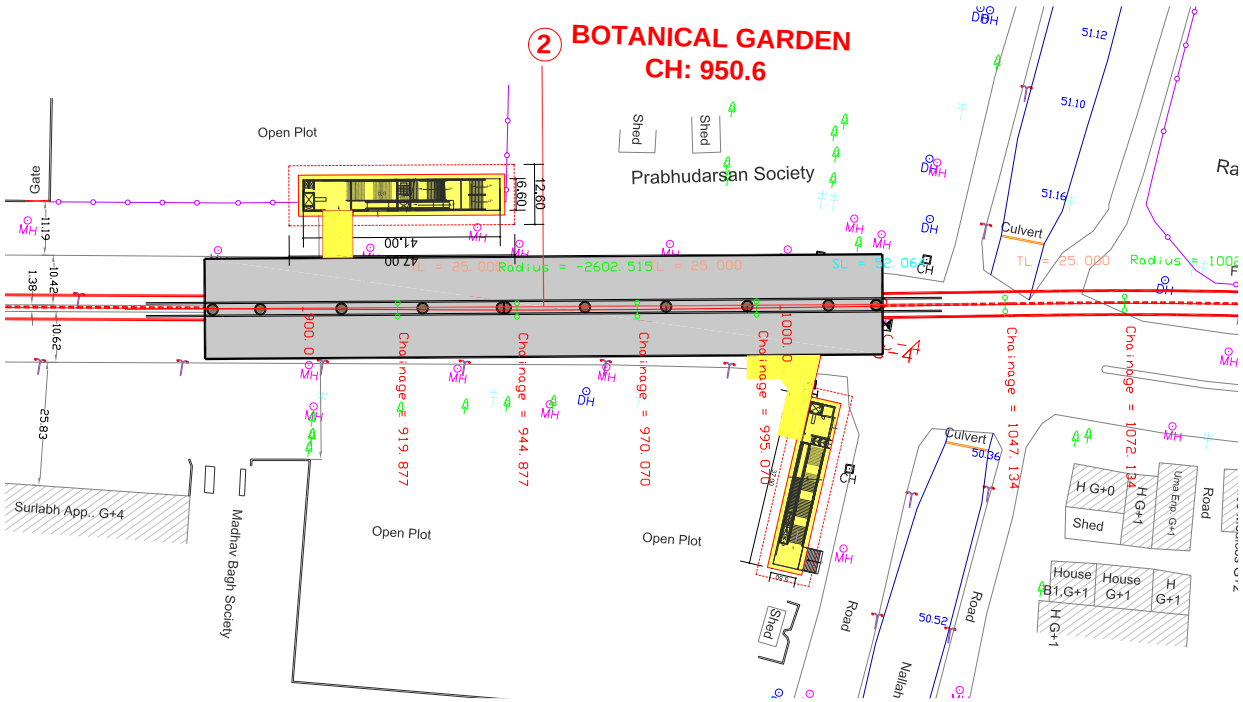


Corridor-II :Bhesan to Saroli

2. Botanical Garden Station

Chainage	:	950.60
Inter station Distance	:	950.60m
Rail Level	:	13.785 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at Prabhudarshan Society.
Catchment Area	:	Residents of Prabhudarshan Society, Ugat town, Radhe Shyam Park and Kalyan society will be main passengers of this station.

Figure 6.5.2: Site Conditions- Botanical GardenStation

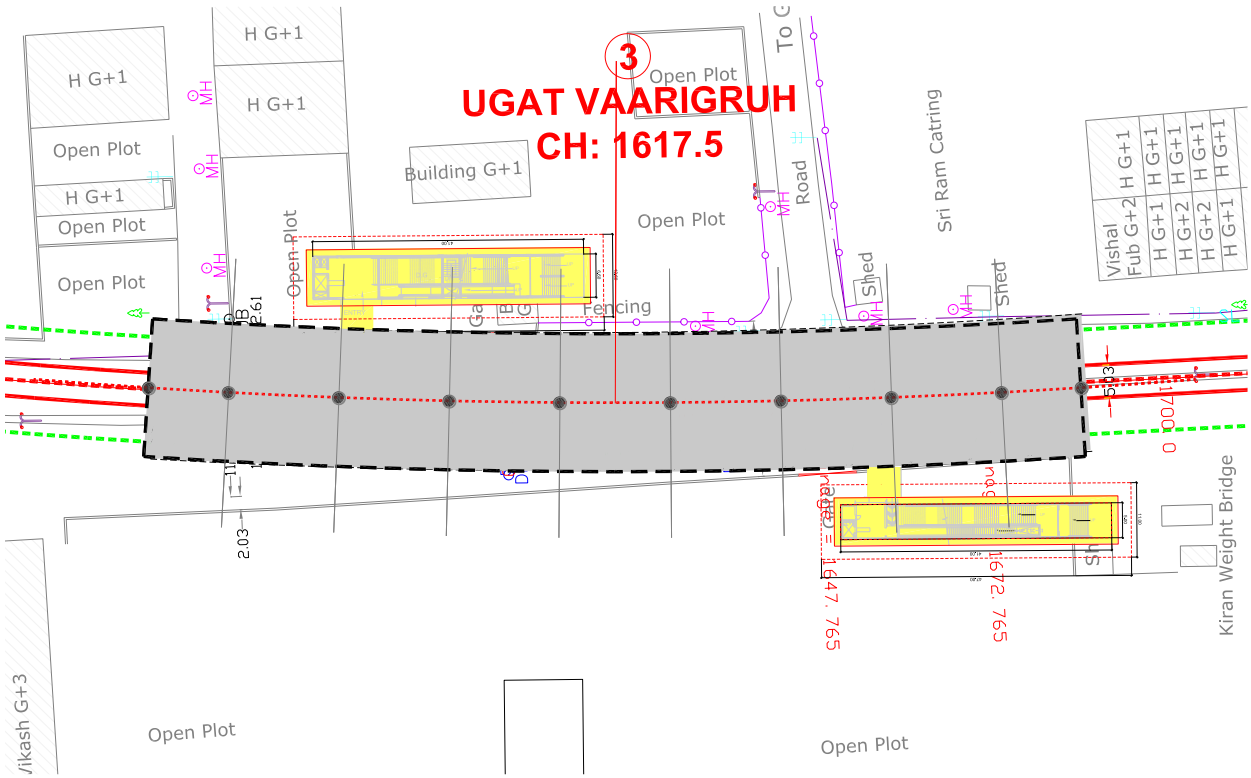


Corridor-II : Bhesan to Saroli

3. UgatVaarigruh Station

Chainage	:	1617.50
Inter station Distance	:	666.90 m
Rail Level	:	13.718 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located SMC water treatment plant
Catchment Area	:	The main source of passengers to this station shall be the residents of Ganganagar Society, Dattatrey Society and Prashant Nagar.

Figure 6.5.3: Site Conditions- UgatVaarigruh Station

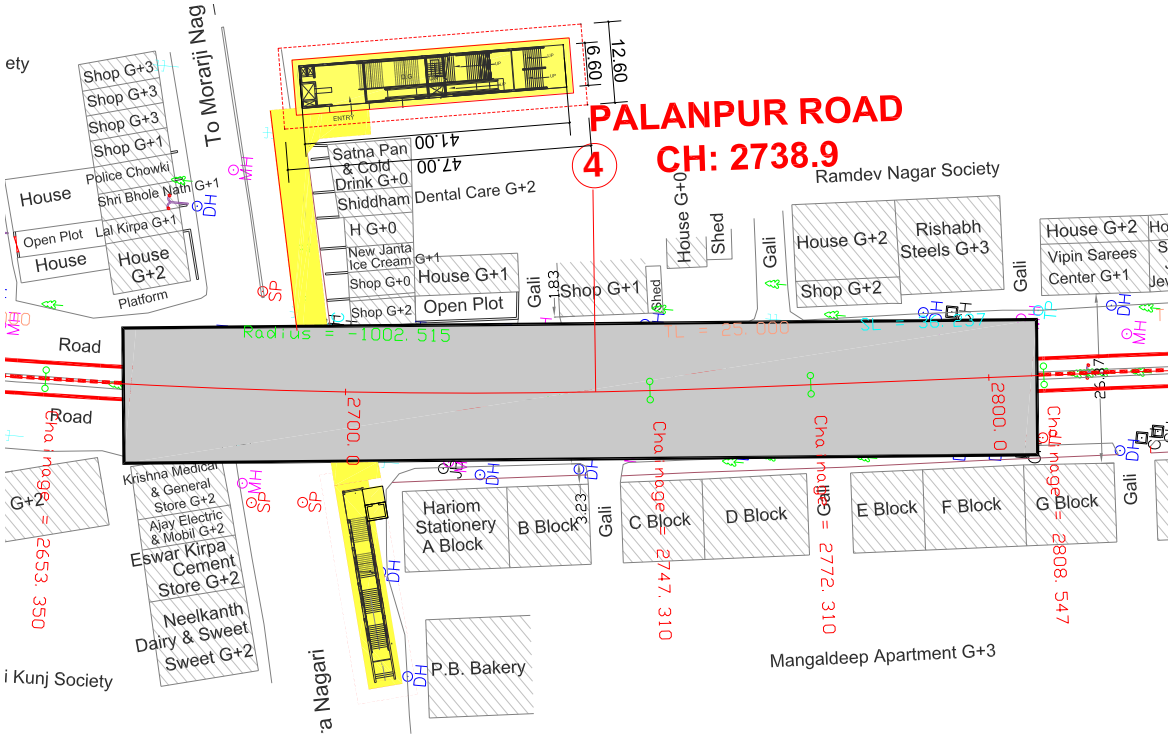


Corridor-II :Bhesan to Saroli

4. Palanpur Road Station

Chainage	:	2738.90
Inter station Distance	:	1121.40m
Rail Level	:	13.518m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at Ramdev Nagar.
Catchment Area	:	The station is located in predominantly Middle income group housing area such as Thakurdwar Sciety, Shanti Kunj Society, Ramdev Nagar, Mangal Deep Apartments etc. which will provide the ridership to the station.

Figure 6.5.4: Site Conditions- Palanpur Road Station

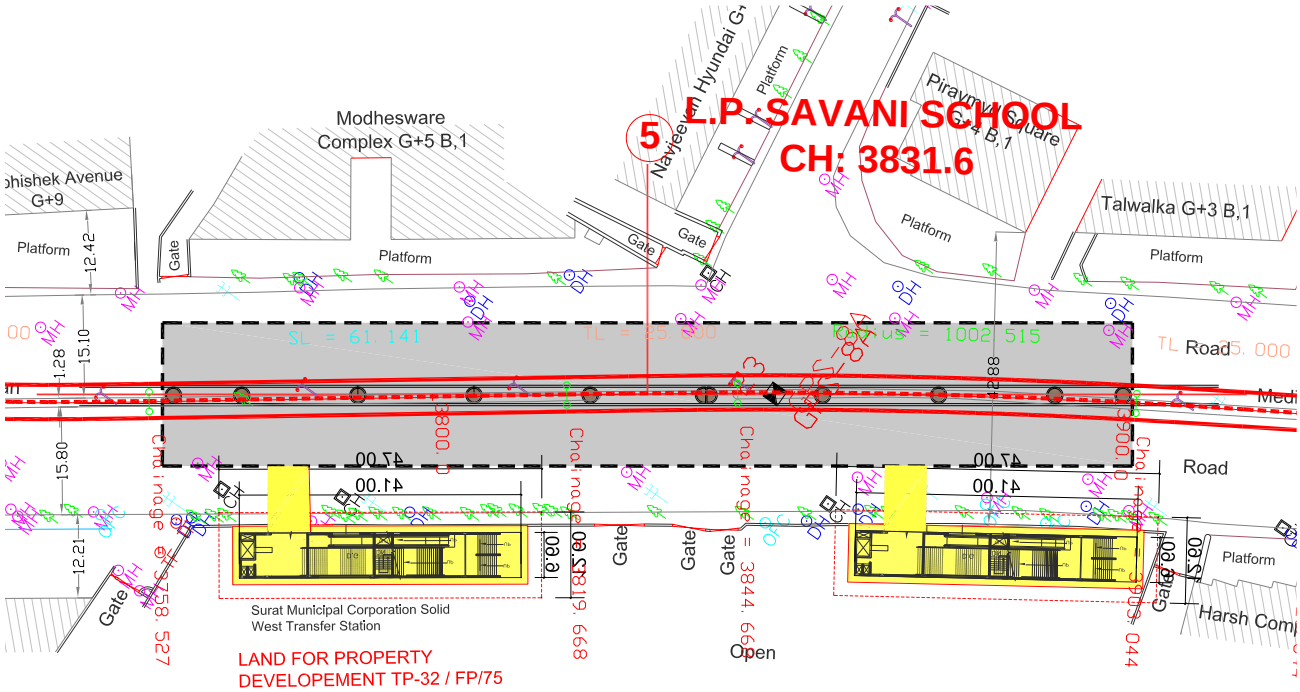


Corridor-II :Bhesan to Saroli

5.L.P. Savani School Station

Chainage	:	3831.60
Inter station Distance	:	1092.70 m
Rail Level	:	13.581m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at L P Swani School.
Catchment Area	:	The main source of passengers to the station shall be the students of various Schools and Institutes of L P Swani Trust and residents of western City, Sitaram Nagar, Govindham Society

Figure 6.5.5: Site Conditions- L.P. SavaniSchoolStation

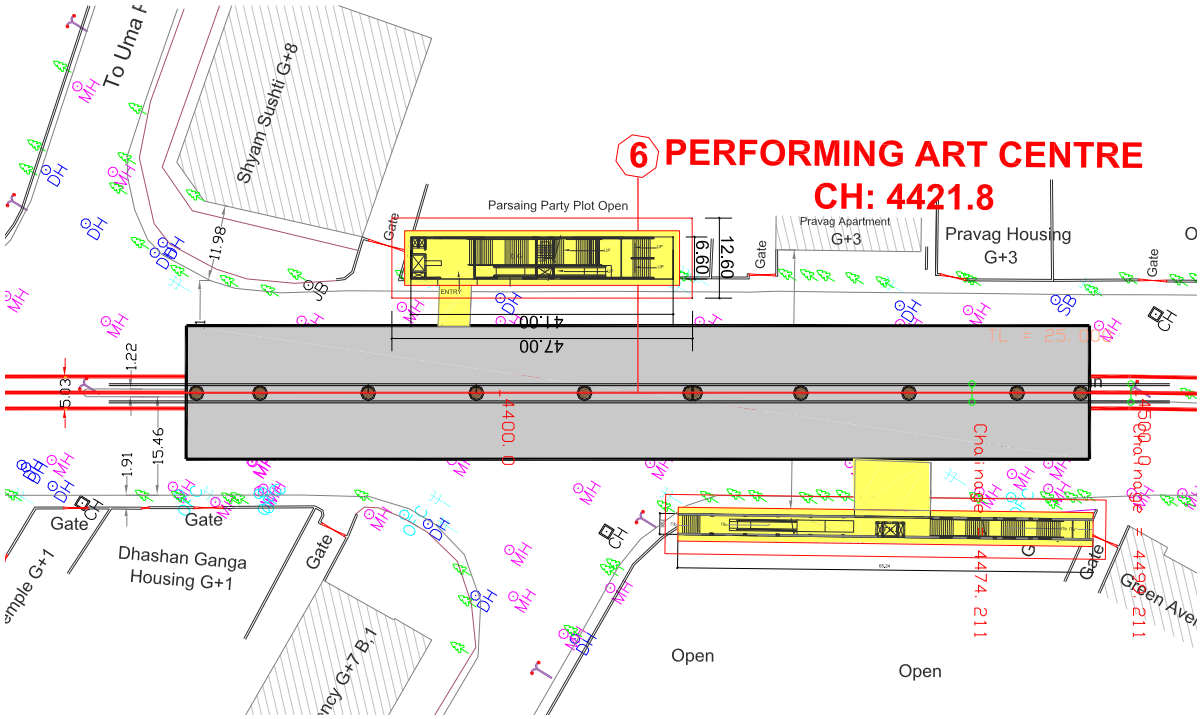


Corridor-II :Bhesan to Saroli

6. Performing Art Centre Station

Chainage	:	4421.80
Inter station Distance	:	590.20 m
Rail Level	:	13.896 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located Near Green Avenue Appt.
Catchment Area	:	The main source of passengers to the station shall be Residents of the apartments in the surrounding and the institutes/Commercial area.

Figure 6.5.6: Site Conditions- Performing Art Centre Station

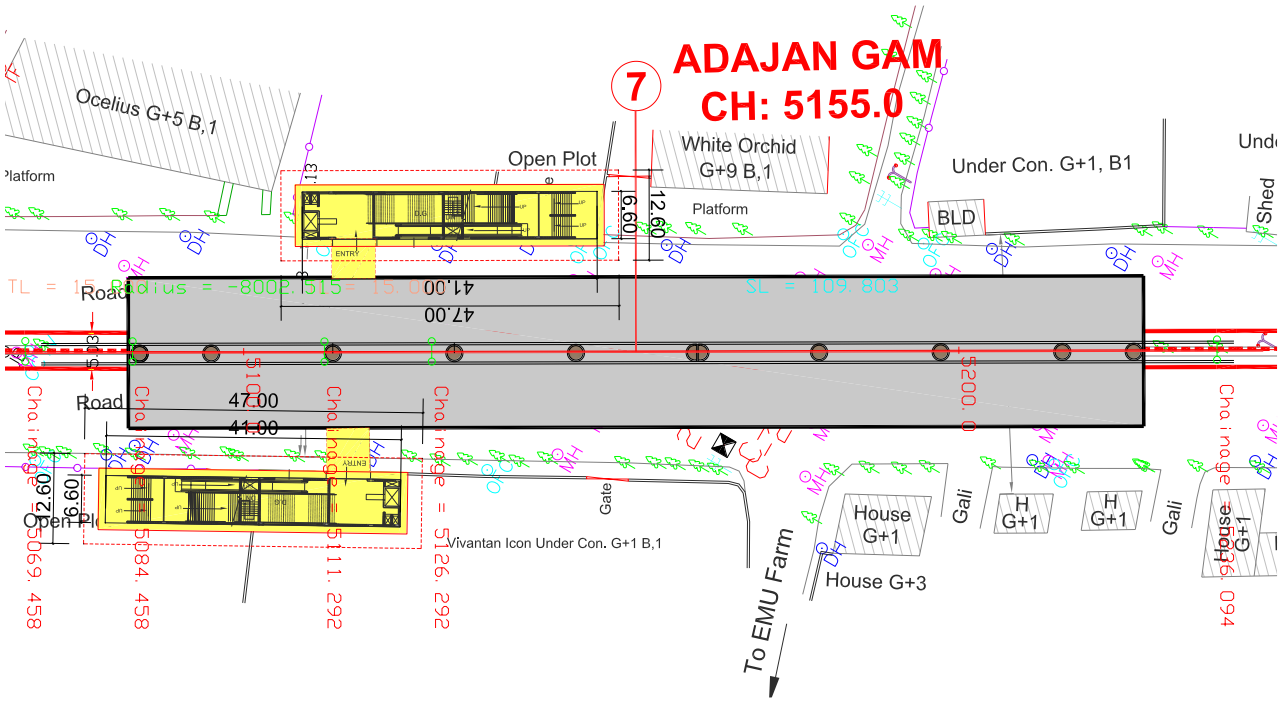


Corridor-II :Bhesan to Saroli

7. Adajan Gam Station

Chainage	:	5155.00
Inter station Distance	:	733.20 m
Rail Level	:	14.053 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at MataniTekri.
Catchment Area	:	The station is located inMataniTekri area and residents of Saurabh Society, Grurukrupa Society, Srinath society and Adjan Gam will use this station for commuting.

Figure 6.5.7: Site Conditions- Adajan Gam Station

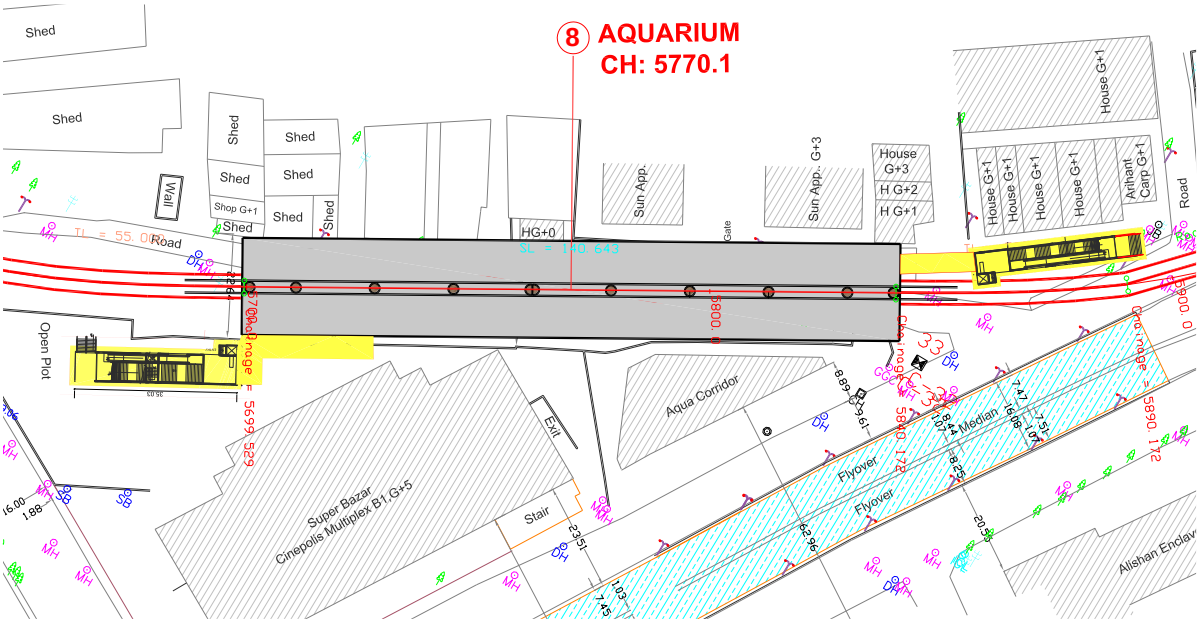


Corridor-II :Bhesan to Saroli

8. Aquarium Station

Chainage	:	5770.10
Inter station Distance	:	615.10 m
Rail Level	:	13.591m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located atCinepolis Multiplex.
Catchment Area	:	The station is located apposite Cinnepolis Multiplex in MataniTekri area which in tern will be the main catchment area for station to attract passengers.

Figure 6.5.8: Site Conditions- Aquarium Station

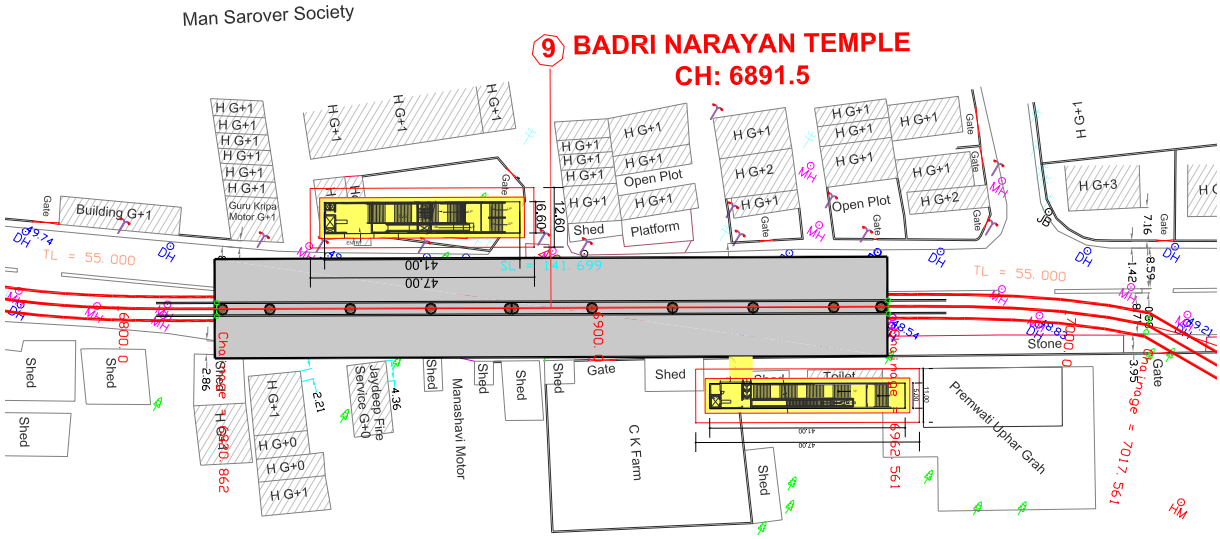


Corridor-II :Bhesan to Saroli

9. Badri Narayan Temple Station

Chainage	:	6891.50
Inter station Distance	:	1121.40m
Rail Level	:	13.686 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located near Mansarover Society.
Catchment Area	:	The residents of Mansarover Society, Lalji Nagar Society, Jalaram Society and other residential areas will be the main commuters at this station.

Figure 6.5.9: Site Conditions- Badri Narayan TempleStation

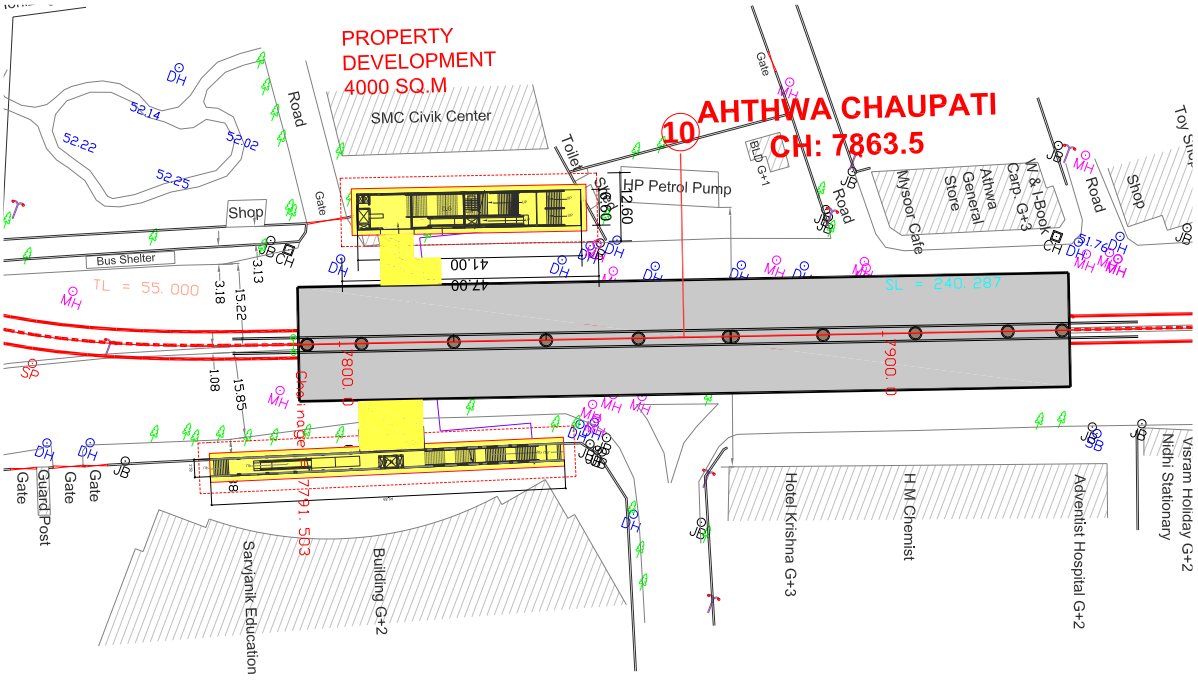


Corridor-II :Bhesan to Saroli

10. Athwa Chaupati Station

Chainage	:	7863.50
Inter station Distance	:	972.00 m
Rail Level	:	13.771 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at ChaupatiAthva Lines.
Catchment Area	:	The station is located in Civic Centre having Hotels, Hospitals, Shopping Malls, SMC Civic Centre and Education institutes. All these will form the passenger base to this station.

Figure 6.5.10: Site Conditions- AthwaChaupatiStation

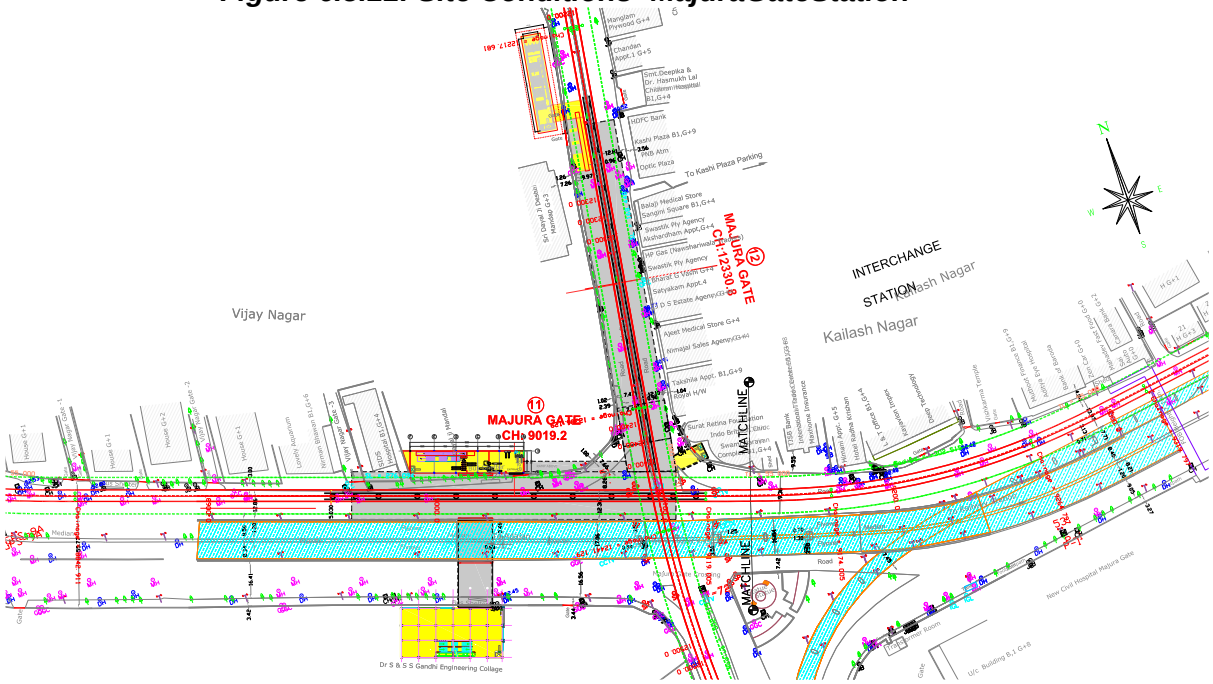


Corridor-II :Bhesan to Saroli

11. Majura Gate Station

Chainage	**:**	9019.20
Inter station Distance	**:**	1155.70m
Rail Level	**:**	09.910m above road level
Station type	**:**	Elevated
Entry / Exit	**:**	Proposed on both sides of the Road
Location	**:**	The station is located across Majura Gate flyover.
Catchment Area	**:**	The station is an interchange station connecting two lines being proposed for Surat Metro Rail. It is located in a very posh locality and commercial area of Surat city. BSNL, Income Tax colony Kailashnagar and Civil Hospital shall be the main source of passengers.

Figure 6.5.11: Site Conditions- MajuraGateStation

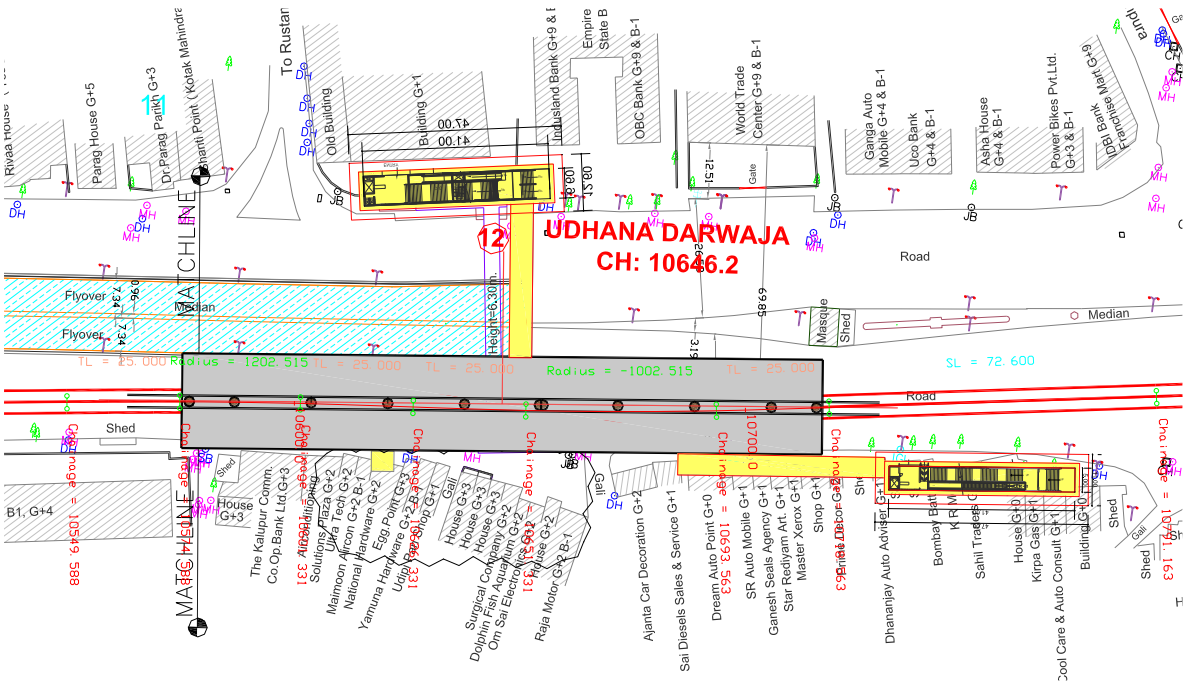


Corridor-II :Bhesan to Saroli

12.Udhana Darwaja Station

Chainage	:	10646.20
Inter station Distance	:	1627.00 m
Rail Level	:	13.896 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located Near UdhanaDarwaja Bus station of BRT Corridor.
Catchment Area	:	The station falls in the commercial area and very close to BRT corridor. The area also has Upper Middle class residential area which will provide the essential ridership to the station.

Figure 6.5.12: Site Conditions- UdhanaDarwajaStation



Corridor-II :Bhesan to Saroli

13.Kamela Darwaja Station

Chainage	:	11247.20
Inter station Distance	:	601.00 m
Rail Level	:	13.46 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on one side of the Road as there is flyover on the other side.
Location	:	The station is located opposite Torrent Power Ltd
Catchment Area	:	The station falls in the commercial area having 3-4 star hotels, Guest Houses, Office complexes and mercantile units. The area also has Upper Middle class residential area which will provide the essential ridership to the station.

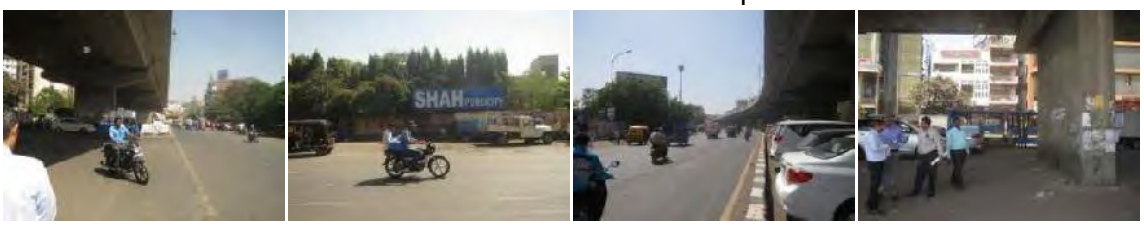
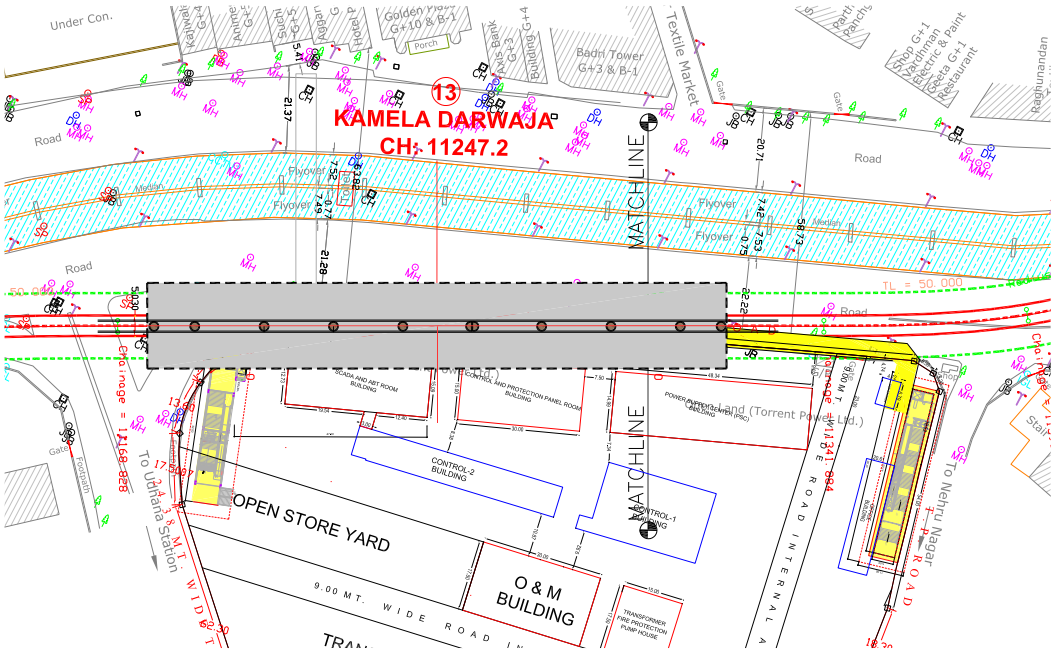


Figure 6.5.13: Site Conditions- KamelaDarwaja Station

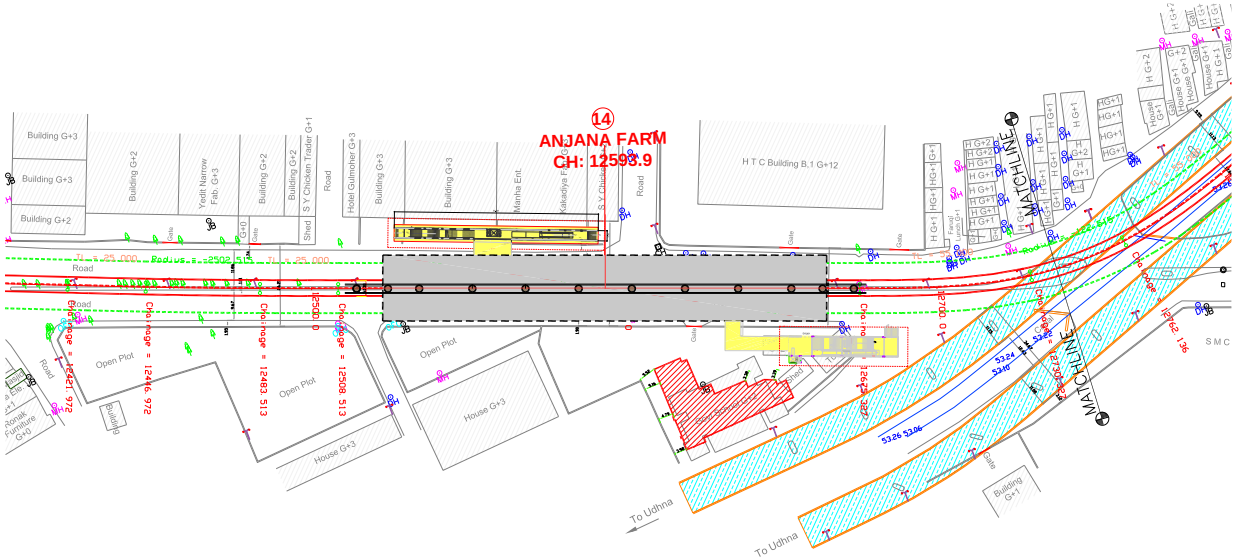


Corridor-II :Bhesan to Saroli

14. Anjana Farm Station

Chainage	**:**	12593.90
Inter station Distance	**:**	1346.70 m
Rail Level	**:**	13.865 m above road level
Station type	**:**	Elevated
Entry / Exit	**:**	Proposed on both sides of the Road
Location	**:**	The station is located near Government School
Catchment Area	**:**	The area is a low density residential area with mix land use having residential areas, factories and industrial units, commercial and trading centers that will form the main passenger .

Figure 6.5.14: Site Conditions- AnjanaFarmStation

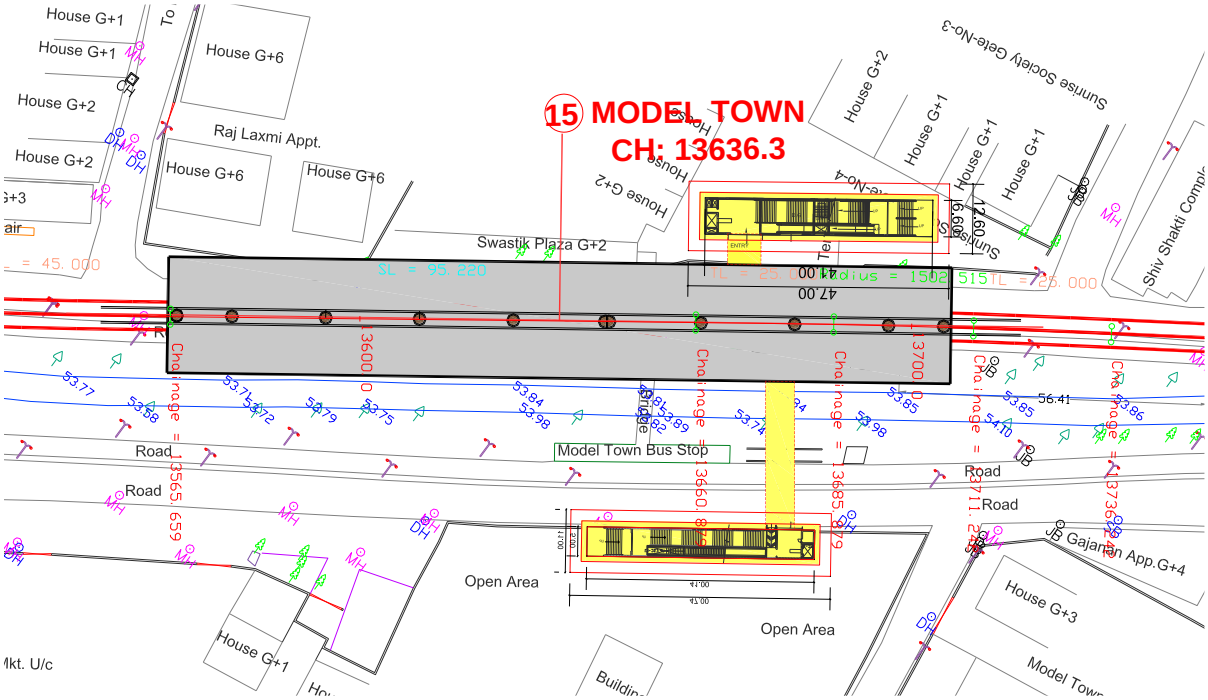


Corridor-II : Bhesan to Saroli

15. Model Town Station

Chainage	:	13636.40
Inter station Distance	:	1042.40 m
Rail Level	:	13.565 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at Model Town.
Catchment Area	:	The station is proposed in the mix land use area having Residential, commercial area with educational institutes and schools. The residents of Samrat Township, Salasar Nagar, Srinathji Society and Model Town will be the main passengers.

Figure 6.5.15: Site Conditions- Model TownStation

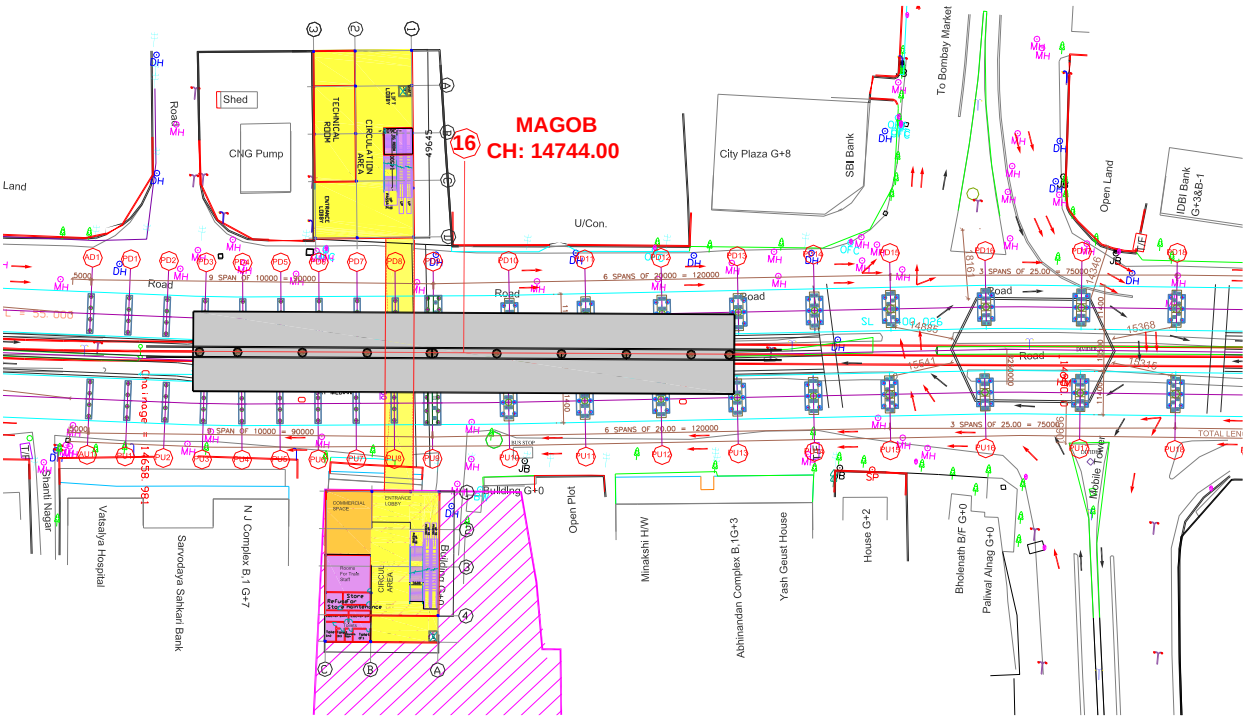


Corridor-II :Bhesan to Saroli

16. Magob Station

Chainage	:	14763.10
Inter station Distance	:	1126.80m
Rail Level	:	20.808 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at BRT road Megum Gam.
Catchment Area	:	The station is located on the BRT corridor in the area basically on the out skirts of the city. Presently having basically trading units and Banks, Travel agents and transporters shall form the basic passenger base to the station.

Figure 6.5.16: Site Conditions- Magob Station

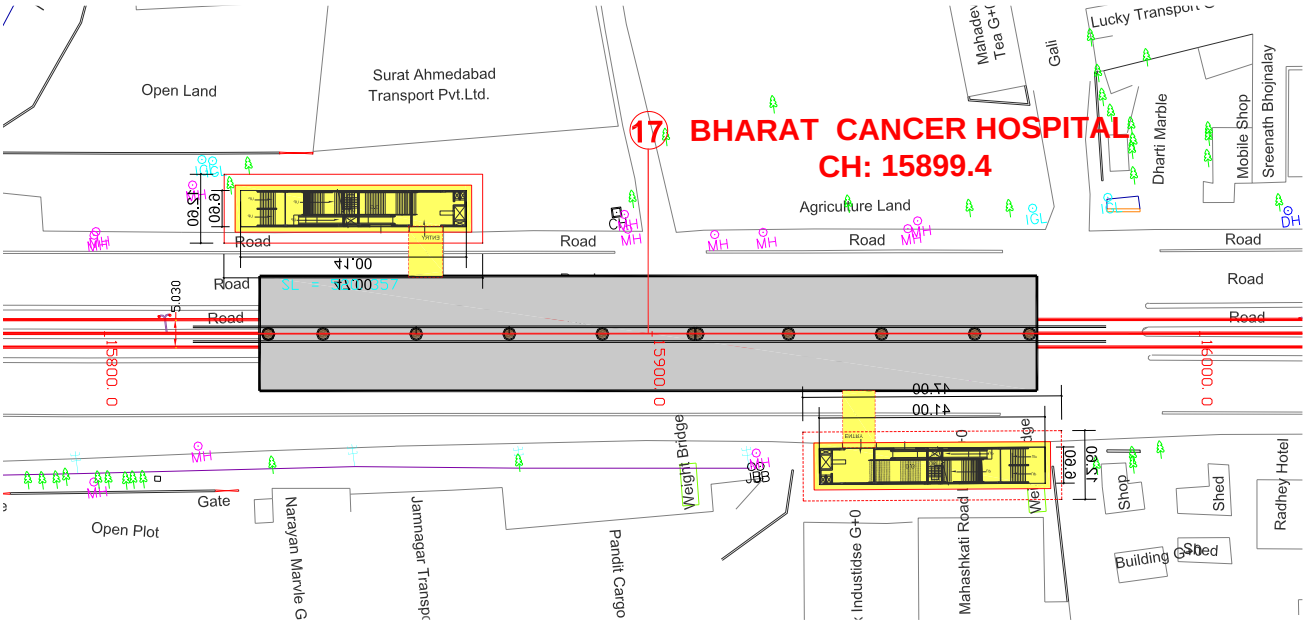


Corridor-II :Bhesan to Saroli

17. Bharat Cancer Hospital Station

Chainage	:	15899.40
Inter station Distance	:	1136.30m
Rail Level	:	13.386m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located near Shanti Vihar Society.
Catchment Area	:	The station is located on the BRT corridor in the area on the out skirts of the city. Presently having basically trading units and Banks, Travel agents and transporters shall form the basic passenger base to the station.

Figure 6.5.17: Site Conditions- Bharat Cancer Hospital



Corridor-II :Bhesan to Saroli

18. Saroli Station

Chainage	:	17341.80
Inter station Distance	:	1442.40 m
Rail Level	:	13.386 m above road level
Station type	:	Elevated
Entry / Exit	:	Proposed on both sides of the Road
Location	:	The station is located at Saroli.
Catchment Area	:	The station is located on the BRT corridor in the area on the out skirts of the city. Presently having textile marketing and trading units, Travel agents and transporters which will form the passenger base to the station.

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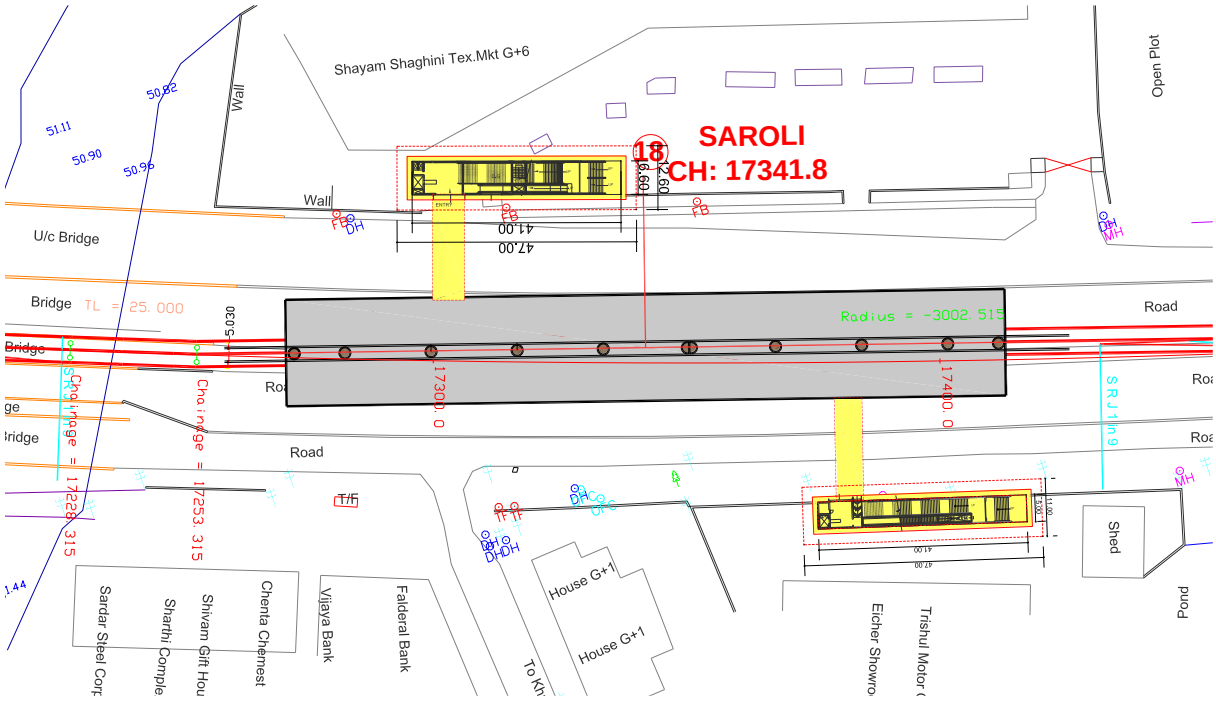
Figure 6.5.18: Site Conditions- SaroliStation



DETAILED PROJECT REPORT - SURAT METRO

Corridor 1 - DREAM CITY TO SARTHANA - YEAR 2021

Assumptions

1 No. of cars per train = 3 cars as per train operation plan

2 Passenger capacity of 3 cars train = 764 @ 6 pers/sqm

3 Platform length = 140 m

4 Boarding & Alighting figures for Sarsoli Station (station no. 18 of Corridor 2) are taken as 35% of the provided value as per explanation by Ms Shalini Associate Professor, Center of Excellence in Urban Transport, CEPT University) in email to DMRC dated 15/01/2017

S NO	STATION NAME	Peak Hour Boarding (D > S)	Peak Hour Alighting (D > S)	Peak Hour Boarding (S > D)	Peak Hour Alighting (S > D)	Peak Minute Boarding (P)	Peak Alighting per Service Interval	Train Service Interval (SI)	P * SI	Platform width (Normal Operation)	Platform width (Delayed service)	Proposed platform width (not less than 3.0 m)	Egress capacity required as per calculations	Proposed Egress capacity (Staircase width)	AFC gates required (as per evacuation requirement)	Requirement for Entry AFC gates for Normal Operation	Requirement for Exit AFC gates for Normal Operation	Total AFC gates	TOM/TVM
1	Dream City	95	0	2	58	2.9	7.5	6.50	13	0.65	1.15	3.0	2.63	3	5	2	2	5	2
2	Convention Center	27	0	2	27	0.5	3.5	6.50	4	0.61	1.11	3.0	2.54	3	5	2	2	5	2
3	Bamroli	20	0	5	53	0.5	3.2	6.50	4	0.61	1.11	3.0	2.54	3	5	2	2	5	2
4	Womens ITI	20	0	2	25	0.5	3.3	6.50	4	0.61	1.11	3.0	2.54	3	5	2	2	5	2
5	V P Road	1465	5	5	1567	29.7	203.7	6.50	197	1.29	1.92	3.0	4.40	5	7	3	9	12	3
6	Althan Gam	508	1	1	543	10.2	70.6	6.50	27	0.84	1.58	3.0	3.16	4	5	2	4	6	2
7	Althan Terminal	480	1	1	490	9.8	68.1	6.50	14	0.61	1.11	3.0	2.54	3	5	2	4	6	2
8	Rajpath Canal	302	2	1	376	6.0	48.5	6.50	10	0.78	1.26	3.0	2.90	3	5	2	3	5	2
9	Mahila Gate-1	1472	104	29	1421	25.8	202.7	6.50	188	1.27	1.90	3.0	4.35	5	7	3	9	12	3
10	Radhesha Nilla	382	7	20	287	7.7	57.3	6.50	30	0.78	1.51	3.0	2.99	3	5	2	3	5	2
11	Gandhi Baug	1981	221	275	2025	38.6	268.0	6.50	248	1.54	2.41	3.0	5.54	6	8	3	11	14	4
12	Makkai Hospital	843	233	59	642	16.9	57.5	6.50	15	0.76	1.48	3.0	2.95	3	5	2	4	8	2
13	Surat Railway Station	686	1085	1439	663	23.8	141.1	6.50	188	1.27	1.90	3.0	4.35	5	7	3	7	10	3
14	Central Warehouse	187	669	885	57	13.7	84.4	6.50	30	0.92	1.64	3.0	3.79	4	6	2	5	7	2
15	Lalbhadwar Chowk	131	986	1367	83	27.3	125.6	6.50	178	1.24	1.85	3.0	4.25	5	7	2	6	8	3
16	Kapodra	108	2057	1863	39	33.7	267.4	6.50	219	1.38	2.03	3.0	4.66	5	7	3	11	14	4
17	Shri Swaminarayan Mandir (Kaliawadi)	26	930	954	12	19.1	126.3	6.50	125	1.05	1.63	3.0	3.73	4	6	2	6	8	3
18	Vaccine Depot Garden	0	417	447	0	8.8	57.0	6.50	28	0.84	1.44	3.0	3.30	4	5	2	4	6	2
19	Nature Park	0	959	681	0	13.6	122.1	6.50	89	0.92	1.47	3.0	3.38	4	6	2	6	8	2
20	Sarthana	0	667	581	0	11.2	86.7	6.50	73	0.86	1.40	3.0	3.22	4	5	2	5	7	2

Corridor 2 - BHESAN TO SARSOLI

S NO	STATION NAME	Peak Hour Boarding (D > S)	Peak Hour Alighting (D > S)	Peak Hour Boarding (S > D)	Peak Hour Alighting (S > D)	Peak Minute Boarding (P)	Peak Alighting per Service Interval	Train Service Interval (SI)	P * SI	Platform width (Normal Operation)	Platform width (Delayed service)	Proposed platform width (not less than 3.0 m)	Egress capacity required as per calculations	Proposed Egress capacity (Staircase width)	AFC gates required (as per evacuation requirement)	Requirement for Entry AFC gates for Normal Operation	Requirement for Exit AFC gates for Normal Operation	Total AFC gates	TOM/TVM
1	Bhesan	42	0	0	37	0.1	17.4	10.00	12	0.64	1.14	3.0	2.62	3	5	2	2	5	2
2	Botanical Garden	187	0	2	190	3.7	38.0	10.00	38	0.74	1.45	3.0	2.88	3	5	2	3	5	2
3	Utran Village	392	0	2	436	8.8	87.1	10.00	59	0.85	1.52	3.0	3.48	4	6	2	5	7	2
4	Palanpur Bhesan	325	0	5	423	9.5	94.6	10.00	56	0.84	1.51	3.0	3.45	4	6	2	5	7	2
5	LP Savani School	540	0	2	681	10.8	97.4	10.00	106	0.99	1.55	3.0	3.56	4	6	2	5	7	2
6	Performing Art Centre	272	6	1	235	5.4	47.0	10.00	55	0.80	1.53	3.0	3.04	4	5	2	3	5	2
7	Anjanjar Gam	292	7	5	272	5.8	54.2	10.00	59	0.82	1.54	3.0	3.08	4	5	2	3	5	2
8	Amusement	429	29	28	584	8.5	58.6	10.00	45	0.90	1.66	3.0	3.34	4	6	2	4	6	2
9	Shadri Narayan Temple	278	20	13	335	5.6	67.0	10.00	56	0.80	1.53	3.0	3.05	4	5	2	4	6	2
10	Atri Wa Chopadi	331	28	31	170	6.7	34.0	10.00	67	0.86	1.58	3.0	3.16	4	5	2	3	5	2
11	Mahila Gate - 2	1063	409	460	2520	21.0	204.0	10.00	210	1.35	1.99	3.0	4.57	5	7	2	9	11	3
12	Udhna Darwaja	975	567	459	696	11.5	133.2	10.00	115	1.01	1.58	3.0	3.63	4	6	2	6	8	2
13	Kamala Darwaja	570	415	548	448	11.4	89.6	10.00	114	1.01	1.58	3.0	3.62	4	6	2	5	7	2
14	Majura Farm	208	279	330	202	6.0	55.8	10.00	60	0.82	1.55	3.0	3.59	4	5	2	3	5	2
15	Model Town	280	389	259	412	5.0	82.4	10.00	56	0.80	1.53	3.0	3.05	4	5	2	4	6	2
16	Majura	917	1364	1118	986	26.4	276.8	10.00	264	1.54	2.22	3.0	5.10	6	8	2	11	13	3
17	Bharat Cancer Hospital	0	67	61	1	2.2	13.4	10.00	13	0.65	1.15	3.0	2.63	3	5	2	2	5	2
18	Sarsoli	0	1243	1106	0	22.1	220.6	10.00	220	1.49	2.14	3.0	4.68	5	7	2	10	12	3

DETAILED PROJECT REPORT - SURAT METRO

Corridor 1 - DREAM CITY TO SARTHANA - YEAR 2026

Assumptions

1 No. of cars per train – 3 cars as per train operating plan

2 Passenger capacity of 3 cars train – 764 @ 6 pass/sq m

3 Platform length – 140 m

4 Coordinate & Alignment data for Corridor 2 are taken as 10% of the provided values as per explanation by M/S Sital Mahajan Associate Professor, Center of Excellence in Urban Transport, CEPT University, in email to DMRC dated 15/06/2017

S NO	STATION NAME	Peak Hour Peak Hour Boarding (D > S)	Peak Hour Peak Hour Alighting (D > S)	Peak Hour Peak Hour Boarding (S > D)	Peak Hour Peak Hour Alighting (S > D)	Peak Minute Boarding (P)	Peak Minute Alighting (P)	Peak Alighting per Service Interval	Train Service Interval (Sec)	P * Si	Platform width (Normal Operation)	Platform width (Delayed service)	Proposed platform width (not less than 3.0 m)	Egress capacity required as per calculations	Proposed Egress capacity (Staircase width)	AFC gates required (as per evacuation requirement)	Requirement for Entry AFC gates for Normal Operation	Requirement for Exit AFC gates for Normal Operation	Total AFC gates	TOM/TVM
1	Dream City	1476	0	25	0	3435	2825	100.5	3.50	100	2.97	1.56	3.0	3.52	4	6	3	5	8	3
2	Convention Center	347	0	0	0	345	659	24.4	3.50	25	2.64	1.20	3.0	4.75	4	5	2	2	5	2
3	Bhimrad	148	0	2	0	218	3.0	15.3	3.50	11	2.61	1.14	3.0	2.61	3	5	2	2	5	2
4	Vidhan Sabha	101	0	2	0	108	2.8	10.4	3.50	10	2.64	1.13	3.0	2.60	3	5	2	2	5	2
5	VIP Road	2731	98	85	0	2868	743.2	207.8	3.50	196	1.78	1.91	3.0	4.37	5	7	3	4	12	5
6	Athwa Gate	1230	11	8	0	1387	258	97.1	3.50	91	2.93	1.48	3.0	3.40	4	6	2	5	7	3
7	Athwa Tenament	839	17	23	0	965	158.8	60.2	3.50	59	2.81	1.39	3.0	3.08	4	5	2	4	6	3
8	Rupali Canal	723	29	18	0	918	145	64.3	3.50	51	2.73	1.33	3.0	3.00	4	5	2	4	6	2
9	Majura Gate - 1	3402	436	533	0	3710	640	260.0	3.50	248	3.04	2.12	3.0	4.85	5	7	4	11	15	6
10	Kadarsha Ni Vav	959	36	70	0	731	132	51.2	3.50	68	2.84	1.38	3.0	3.17	4	5	2	3	5	3
11	Gandhi Baug	1819	696	821	0	1423	956	305.6	3.50	338	3.81	2.56	3.0	5.82	6	8	5	13	18	7
12	Mahakali Hospital	934	426	437	0	1057	187	74.0	3.50	66	2.84	1.35	3.0	3.15	4	5	2	4	6	3
13	Surat Railway Station	1864	1980	431	0	1886	462	133.0	3.50	164	3.24	1.70	3.0	4.10	5	7	3	6	9	4
14	Central Warehouse	963	1162	1074	0	883	213	81.3	3.50	76	2.87	1.46	3.0	3.25	4	6	2	4	6	3
15	Ladies Wal Chowk	322	1701	2939	0	172	418	123.3	3.50	171	3.21	1.82	3.0	4.18	5	7	3	6	9	4
16	Kapodra	271	4385	3818	0	111	70.4	33.4	3.50	266	3.50	2.24	3.0	5.14	6	10	4	14	18	6
17	Shri Swaminarayan Mandir Kalkhum	76	3671	3244	0	45	643	215.0	3.50	225	3.46	2.06	3.0	4.71	5	7	4	9	13	5
18	Varcha Chopati Garden	0	1367	1955	0	25	35.4	95.7	3.50	102	3.08	1.55	3.0	3.51	4	6	3	5	8	3
19	Nature Park	0	2692	2372	0	0	47.4	188.4	3.50	167	3.20	1.83	3.0	4.16	5	7	3	8	11	4
20	Sarthana	0	2373	2134	0	0	44.2	166.0	3.50	154	3.14	1.73	3.0	3.98	4	6	3	7	10	4

Corridor 2 - BHESAN TO SARTHANA

S NO	STATION NAME	Peak Hour Peak Hour Boarding (D > S)	Peak Hour Peak Hour Alighting (D > S)	Peak Hour Peak Hour Boarding (S > D)	Peak Hour Peak Hour Alighting (S > D)	Peak Minute Boarding (P)	Peak Minute Alighting (P)	Peak Alighting per Service Interval	Train Service Interval (Sec)	P * Si	Platform width (Normal Operation)	Platform width (Delayed service)	Proposed platform width (not less than 3.0 m)	Egress capacity required as per calculations	Proposed Egress capacity (Staircase width)	AFC gates required (as per evacuation requirement)	Requirement for Entry AFC gates for Normal Operation	Requirement for Exit AFC gates for Normal Operation	Total AFC gates	TOM/TVM
1	Bhesan	334	0	5	0	468	67	73.3	3.00	54	2.73	1.32	3.0	3.03	4	5	2	4	6	2
2	Bodakdi Kal Garden	686	0	15	0	602	137	96.5	3.00	110	3.03	1.56	3.0	3.58	4	6	2	5	7	2
3	Nilesh Vasangadh	1114	0	15	0	1160	227	160.7	3.00	180	3.20	1.84	3.0	4.24	5	7	3	8	11	4
4	Palanpur Road	649	0	12	0	845	138.0	135.2	3.00	101	2.97	1.56	3.0	3.52	4	6	2	6	8	2
5	S.P. Savani School	1148	3	2	0	1078	243.6	172.5	3.00	181	3.25	1.88	3.0	4.31	5	7	2	8	10	3
6	Performing Art Centre	616	5	2	0	585	123.3	94.1	3.00	99	2.95	1.52	3.0	3.48	4	6	2	5	7	2
7	Adajan Gam	723	18	18	0	728	155	113.9	3.00	124	3.04	1.62	3.0	3.72	4	6	2	6	8	2
8	Aquarium	800	33	58	0	653	150.0	104.5	3.00	128	3.06	1.69	3.0	3.76	4	6	2	5	7	2
9	Badri Na Vyas Temple	770	55	76	0	728	154	126.2	3.00	124	3.04	1.62	3.0	3.72	4	6	2	6	8	2
10	Ketwas Chorahi	546	40	88	0	375	109	87.0	3.00	88	2.91	1.47	3.0	3.37	4	6	2	5	7	2
11	Majura Gate - 2	1558	1394	1675	0	1530	335	244.8	3.00	262	3.55	2.26	3.0	5.14	6	8	3	10	13	4
12	Ulna Darwaja	927	1245	841	0	1000	185	139.2	3.00	149	3.13	1.73	3.0	3.97	4	6	2	9	11	3
13	Kamela Darwaja	747	765	976	0	579	155	122.1	3.00	157	3.15	1.76	3.0	4.05	5	6	2	6	8	3
14	Anjana Farm	784	980	641	0	765	116	117.0	3.00	165	3.18	1.84	3.0	3.42	4	6	2	6	8	2
15	Kodali Town	241	881	532	0	375	106	141.0	3.00	86	2.91	1.46	3.0	3.35	4	6	2	7	9	2
16	Magob	1462	2395	2387	0	1403	417	383.2	3.00	382	3.85	2.75	3.0	6.26	7	9	3	15	18	4
17	Bharat Cancer Hospital	0	229	226	0	4	8.5	36.6	3.00	37	2.73	1.25	3.0	2.87	3	5	2	3	5	2
18	Sarthana	0	1745	1685	0	0	33.7	278.2	3.00	270	3.55	2.25	3.0	5.16	6	8	3	11	14	4

DETAILED PROJECT REPORT - SURAT METRO

Corridor 1 - DREAM CITY TO SARTHANA - YEAR 2036

Assumptions

1 No. of cars per train = 3 cars as per train operation plan

2 Passenger carrying capacity of 3 cars train = 764 @ 6 persons per sq.m

3 Platform length = 140 m

4 Boarding & Alighting figures for Sensitive Station (station no. 18 of Corridor 2) are taken as 35% of the provided value as per explanation by Ms Sheila Inl (Associate Professor, Centre of Excellence in Urban Transport, CEPT University) in email to DMRC dated 25/04/2017

S.NO	STATION NAME	Peak Hour Peak Hour Boarding (5 > D)	Peak Hour Peak Hour Alighting (D > 5)	Peak Hour Peak Hour Boarding (5 > D)	Peak Hour Peak Hour Alighting (5 > D)	Peak Hour Peak Minute Boarding (F)	Peak Alighting per Service Interval	Train Service Interval (S)	F * S1	Platform width (Normal Operation)	Platform width (Delayed service)	Proposed platform width (not less than 3.0 m)	Egress capacity required as per calculations	Proposed Egress Capacity (Staircase width)	AFC gates required (as per evacuation requirement)	Requirement for Entry AFC gates for Normal Operation	Requirement for Exit AFC gates for Normal Operation	Total AFC gates	TOM/TVM
1	Dream City	4711	0	1647	0	78.5	2745	2.25	231	1.41	2.08	3.0	4.47	5	7	5	5	10	4
2	Convention Center	1416	0	495	0	23.6	825	2.25	66	0.50	0.37	3.0	3.35	4	5	3	4	7	3
3	Bhimrad	658	8	230	3	11.0	383	2.25	32	0.71	0.23	3.0	2.52	3	5	2	3	5	2
4	Varachha Road	667	8	233	3	11.1	389	2.25	31	0.71	0.22	3.0	2.51	3	5	2	3	5	2
5	VIP Road	4239	477	1483	167	70.7	2472	2.25	197	1.29	1.92	3.0	4.41	5	7	5	10	15	7
6	Althan Gam	2400	43	840	15	40.0	1400	2.25	108	0.99	1.55	3.0	3.56	4	6	3	6	9	4
7	Althan Tenement	1797	9	629	3	30.0	1048	2.25	81	0.99	1.44	3.0	3.50	4	5	3	4	7	4
8	Rusali Canal	1315	249	460	87	21.9	767	2.25	62	0.82	1.35	3.0	3.10	4	5	2	3	5	3
9	Mahura Gate - 1	4516	1650	1581	578	75.3	2633	2.25	206	1.33	2.07	3.0	4.51	5	7	5	8	13	7
10	Kacareshwar Nal	1416	252	496	88	23.6	825	2.25	64	0.53	0.37	3.0	3.13	4	5	3	4	6	3
11	Ganchi Saus	8091	1610	2832	564	134.9	4717	2.25	365	1.30	3.55	3.0	6.37	7	9	7	13	25	11
12	Maldhari Hospital	1260	649	441	227	21.0	735	2.25	58	0.99	1.22	3.0	3.02	4	5	2	3	5	3
13	Surat Railway Station	2027	5525	710	1934	33.8	1182	2.25	169	1.20	1.82	3.0	4.16	5	7	4	10	14	6
14	Central Warehouse	804	1894	281	663	13.4	469	2.25	77	0.88	1.12	3.0	3.26	4	5	3	5	8	4
15	Labheshwar Chowk	429	1848	150	647	7.2	250	2.25	118	1.02	1.60	3.0	3.86	4	5	3	5	8	5
16	Kadodra	547	7416	191	2595	9.1	319	2.25	285	1.22	2.31	3.0	5.30	6	8	5	13	19	9
17	Shri Swaminarayan Mandir Kalsak.nj	252	3324	88	1163	4.2	147	2.25	166	1.19	2.80	3.0	4.13	5	7	4	7	11	6
18	Vanche Utkarsh Candidate	0	1680	0	588	0.0	0	2.25	95	0.53	1.48	3.0	3.40	4	5	3	4	7	4
19	Nature Park	0	5654	0	1979	0.0	0	2.25	208	1.34	3.38	3.0	4.55	5	7	5	11	15	7
20	Sarthana	0	5252	0	1838	0.0	0	2.25	202	1.39	3.04	3.0	4.58	5	7	5	10	15	7

Corridor 2 - BHESAN TO SAROLI

S.NO	STATION NAME	Peak Hour Peak Hour Boarding (5 > D)	Peak Hour Peak Hour Alighting (D > 5)	Peak Hour Peak Hour Boarding (5 > D)	Peak Hour Peak Hour Alighting (5 > D)	Peak Hour Peak Minute Boarding (F)	Peak Alighting per Service Interval	Train Service Interval (S)	F * S1	Platform width (Normal Operation)	Platform width (Delayed service)	Proposed platform width (not less than 3.0 m)	Egress capacity required as per calculations	Proposed Egress Capacity (Staircase width)	AFC gates required (as per evacuation requirement)	Requirement for Entry AFC gates for Normal Operation	Requirement for Exit AFC gates for Normal Operation	Total AFC gates	TOM/TVM
1	Bhesan	2732	0	956	0	45.5	1638	3.75	169	1.20	2.32	3.0	4.16	5	7	5	8	12	4
2	Botanical Garden	3471	0	1215	0	57.9	2083	3.75	261	1.53	2.21	3.0	5.37	6	8	6	8	12	6
3	Ujgar Vastigrut	2276	0	797	0	37.9	1366	3.75	171	1.21	2.32	3.0	4.18	5	7	5	8	12	4
4	Pal Amroli Road	1642	0	575	0	27.4	985	3.75	122	1.04	1.72	3.0	3.72	4	5	4	5	9	3
5	L.P.Savani School	2382	8	834	3	39.7	1429	3.75	178	1.21	1.86	3.0	4.26	5	7	5	7	12	4
6	Performing Art Centre	2374	7	831	2	39.6	1424	3.75	179	1.24	1.86	3.0	4.26	5	7	5	8	13	4
7	Adajan Gam	1439	28	504	10	24.0	863	3.75	115	1.20	1.38	3.0	3.61	4	5	3	4	7	3
8	Appu Ham	1850	225	648	79	30.8	1110	3.75	140	1.10	1.69	3.0	3.58	4	6	3	5	8	4
9	Bapa Narayan Temple	1213	255	425	89	20.2	728	3.75	91	0.53	1.48	3.0	3.40	4	5	3	4	9	3
10	Mitwa Chopati	1687	273	591	96	28.1	1012	3.75	127	1.25	1.84	3.0	3.75	4	6	3	4	7	4
11	Mahura Gate - 2	2317	2520	811	882	38.6	1390	3.75	207	1.34	1.38	3.0	4.54	5	7	5	11	16	5
12	Udhna Darwaja	2317	3330	811	1166	38.6	1390	3.75	228	1.41	2.07	3.0	4.74	5	7	4	12	16	5
13	Kamela Darwaja	1612	1792	564	627	26.9	968	3.75	175	1.22	2.83	3.0	4.20	5	7	5	5	8	4
14	Angana Farm	959	1206	336	422	16.0	576	3.75	108	0.99	1.55	3.0	3.56	4	5	3	5	8	3
15	Model Town	281	1779	98	623	4.7	169	3.75	50	0.78	1.21	3.0	2.59	3	5	2	6	8	2
16	Majoth	5206	5127	1822	1794	86.8	3124	3.75	410	2.17	2.38	3.0	6.83	7	10	6	15	21	9
17	Nherat Cancer Hospital	0	4914	0	1720	0.0	0	3.75	247	0.96	1.40	3.0	3.71	4	5	4	6	10	4
18	Saroli	0	5375	0	1881	0.0	0	3.75	426	2.13	2.93	3.0	6.21	7	10	6	15	21	8

DETAILED PROJECT REPORT - SURAT METRO

Corridor 1 - DREAM CITY TO SARTHANA - YEAR 2046

Assumptions

1 No. of Seats per Unit = 8 Seats as per Urban Transport plan

2 Crowd per capacity of Train remains = 764 @ 6 persons per sq.m

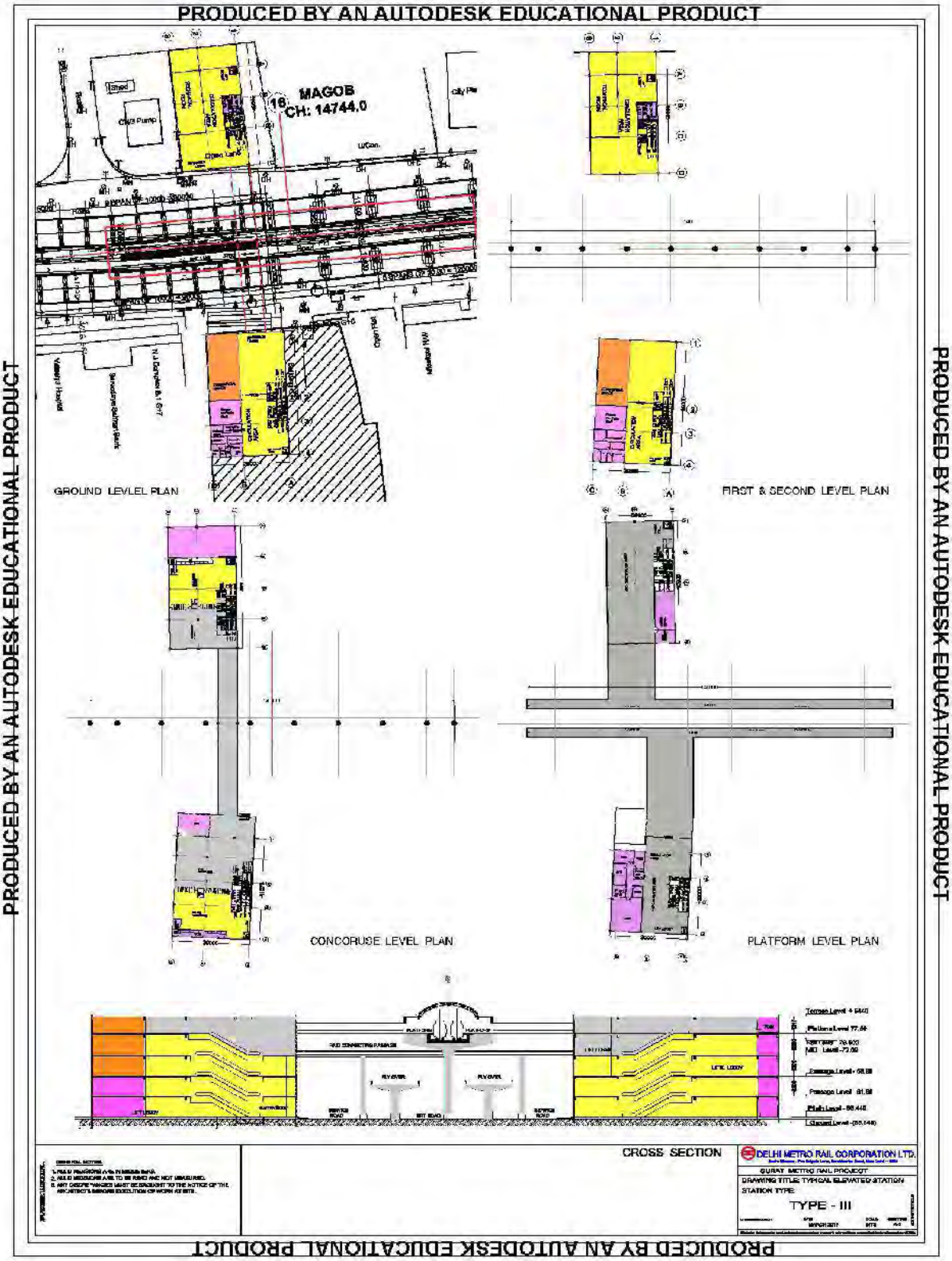
3 Platform length = 140 m

4 Boarding & Alighting figures for Saroli Station, station no 15 for Corridor 2 are taken as 35% of the provided values as per explanation by Mr. Sharad Nikam, Associate Professor, Center of Excellence in Urban Transport, CEPT University, in email to DMRC dated 25/04/2017

S.NO	STATION NAME	Peak Hour Peak Direction Boarding (D > S)	Peak Hour Peak Direction Alighting (D > S)	Peak Hour Peak Direction Boarding (S > D)	Peak Hour Peak Direction Alighting (S > D)	Peak Minute Boarding (P)	Peak Minute Alighting (P)	Peak Alighting per Service Interval	Train Service Interval (Sec)	F * SI	Platform width (Normal Operation)	Platform width (Delayed services)	Proposed platform width (not less than 9.0 m)	Egress capacity required as per calculations	Proposed Egress capacity (Staircase width)	AFC gates required (as per evacuation requirement)	Requirement for Entry AFC gates for Normal Operation	Requirement for Exit AFC gates for Normal Operation	Total AFC gates	TOM/TVM
1	Dream City	4515	0	0	0	451.5	0.0	2258	120	266	15.96	22.34	9.0	5.32	6	8	7	10	17	11
2	Convention Center	2251	0	0	0	225.1	0.0	300	120	79	0.88	1.23	9.0	3.28	4	6	3	4	7	4
3	Dhirubhai	1027	12	0	0	102.7	1.2	137	120	37	0.41	0.57	9.0	2.87	3	5	2	3	5	3
4	Kadodara	1011	12	11	0	101.1	1.2	135	120	36	0.40	0.56	9.0	2.86	3	5	2	3	5	3
5	VIP Road	1517	509	518	171	255.6	90.3	266.5	120	153	1.17	1.77	9.0	4.07	5	7	5	11	16	7
6	Althan Gam	2556	79	28	9	127.8	4.0	133.3	120	90	0.99	1.48	9.0	3.78	4	6	1	3	8	5
7	Althan Terminus	2169	42	123	42	108.4	14.7	113.1	120	76	0.84	1.26	9.0	3.25	4	6	3	3	6	4
8	Rajpath Circle	2963	196	175	59	148.1	19.6	154.5	120	105	0.98	1.47	9.0	3.53	4	6	4	3	7	5
9	Vaijnath Gate - 3	4650	2723	2090	1224	232.5	136.1	242.4	120	164	1.14	1.72	9.0	4.10	5	7	5	5	10	7
10	Kadarshi Nihal	1856	196	376	196	92.8	19.6	96.8	120	58	0.81	1.21	9.0	3.07	4	5	3	3	6	4
11	Ganesh Baug	2150	3348	4193	1173	107.5	167.4	112.0	120	253	1.55	2.32	9.0	4.98	5	8	7	13	20	10
12	Vaishali Hospital	2246	567	641	181	112.3	28.4	117.1	120	79	0.83	1.25	9.0	3.28	4	6	3	3	6	4
13	Surat Railway Station	2259	6035	5220	3438	112.9	301.8	117.7	120	183	1.25	1.88	9.0	4.30	5	7	5	9	14	8
14	Central Warehouse	1045	1915	2872	1014	52.2	95.7	54.4	120	101	0.95	1.42	9.0	3.50	4	6	4	4	8	5
15	Lakhanshivar Chowk	458	2392	2770	958	22.9	119.6	23.9	120	91	0.94	1.41	9.0	3.46	4	6	1	4	7	5
16	Kapodra	1301	6312	7068	196	65.1	315.6	67.8	120	248	1.49	2.23	9.0	4.94	5	8	7	9	16	10
17	Shri Swaminarayan Vaudi Kakadiya	492	2753	5978	171	24.6	137.6	25.6	120	210	1.33	1.98	9.0	4.57	5	7	6	5	11	9
18	Varacha Chopati Garden	0	1023	0	0	0.0	51.1	0.0	120	85	0.91	1.36	9.0	3.14	4	6	1	4	7	4
19	Nature Park	0	9281	4652	0	0.0	464.0	0.0	120	163	1.13	1.70	9.0	4.10	5	7	5	13	18	7
20	Sarthana	0	7083	6817	0	0.0	354.1	0.0	120	239	1.65	2.47	9.0	4.85	5	7	6	10	16	10

Corridor 2 - BHESAN TO SAROLI

S.NO	STATION NAME	Peak Hour Peak Direction Boarding (D > S)	Peak Hour Peak Direction Alighting (D > S)	Peak Hour Peak Direction Boarding (S > D)	Peak Hour Peak Direction Alighting (S > D)	Peak Minute Boarding (P)	Peak Minute Alighting (P)	Peak Alighting per Service Interval	Train Service Interval (Sec)	F * SI	Platform width (Normal Operation)	Platform width (Delayed services)	Proposed platform width (not less than 9.0 m)	Egress capacity required as per calculations	Proposed Egress capacity (Staircase width)	AFC gates required (as per evacuation requirement)	Requirement for Entry AFC gates for Normal Operation	Requirement for Exit AFC gates for Normal Operation	Total AFC gates	TOM/TVM
1	Bhesan	3351	0	0	0	335.1	0.0	2682	120	202	1.32	1.98	9.0	4.49	5	7	4	11	15	8
2	Bodakdal Garden	4468	0	0	0	446.8	0.0	2005	120	269	1.55	2.32	9.0	5.15	6	8	5	9	14	7
3	Jilla Vasahat Ji	3669	0	0	0	366.9	0.0	3207	120	221	1.38	2.07	9.0	4.67	5	7	4	13	17	8
4	Bhesan Junction	1752	0	0	0	175.2	0.0	1248	120	104	0.92	1.38	9.0	3.72	4	6	1	6	9	4
5	Sai Ashram School	2552	7	8	2	255.2	0.7	1460	120	153	1.13	1.70	9.0	4.04	5	6	3	7	10	5
6	Parbhu Ni Aai Centre	3359	9	9	3	335.9	0.9	1282	120	202	1.32	1.98	9.0	4.49	5	7	4	6	10	6
7	Kadam Gam	1927	47	27	9	192.7	4.7	1927	120	116	1.01	1.52	9.0	3.64	4	6	3	4	7	4
8	Kapesthan	2158	261	458	152	215.8	26.1	1027	120	132	1.07	1.61	9.0	3.80	4	6	3	5	8	4
9	Bapji Narayan Temple	1060	176	332	110	106.0	17.6	1655	120	85	0.83	1.25	9.0	3.14	4	5	2	5	7	3
10	Bhesan Chowki	2121	241	440	146	212.1	24.1	1324	120	128	1.06	1.64	9.0	3.76	4	6	3	4	7	4
11	Vaijnath Gate - 2	3439	3439	3440	3440	343.9	343.9	2580	120	206	1.34	1.99	9.0	4.53	5	7	4	10	14	8
12	Jolina Darwaja	2868	5298	4076	3133	286.8	529.8	1433	120	245	1.58	2.37	9.0	4.92	5	8	4	13	17	8
13	Kamela Darwaja	1753	1535	2550	1393	175.3	153.5	927	120	150	1.15	1.73	9.0	4.01	5	6	3	5	8	5
14	Bajas Farm	644	1498	152	673	64.4	149.8	1288	120	92	0.93	1.40	9.0	3.41	4	6	3	5	8	3
15	Model Town	1002	2051	878	1672	100.2	205.1	1337	120	81	0.82	1.23	9.0	3.10	4	5	2	6	8	3
16	Vaijnath	5364	5727	5585	3757	536.4	572.7	4021	120	348	1.97	2.96	9.0	6.78	7	9	6	18	24	9
17	Bharat Cancer Hospital	0	1394	1462	0	0.0	69.8	0.0	120	88	0.91	1.37	9.0	3.37	4	6	3	4	7	3
18	Saroli	0	7371	8225	0	0.0	368.5	0.0	120	304	2.36	3.54	9.0	7.36	8	11	7	17	24	11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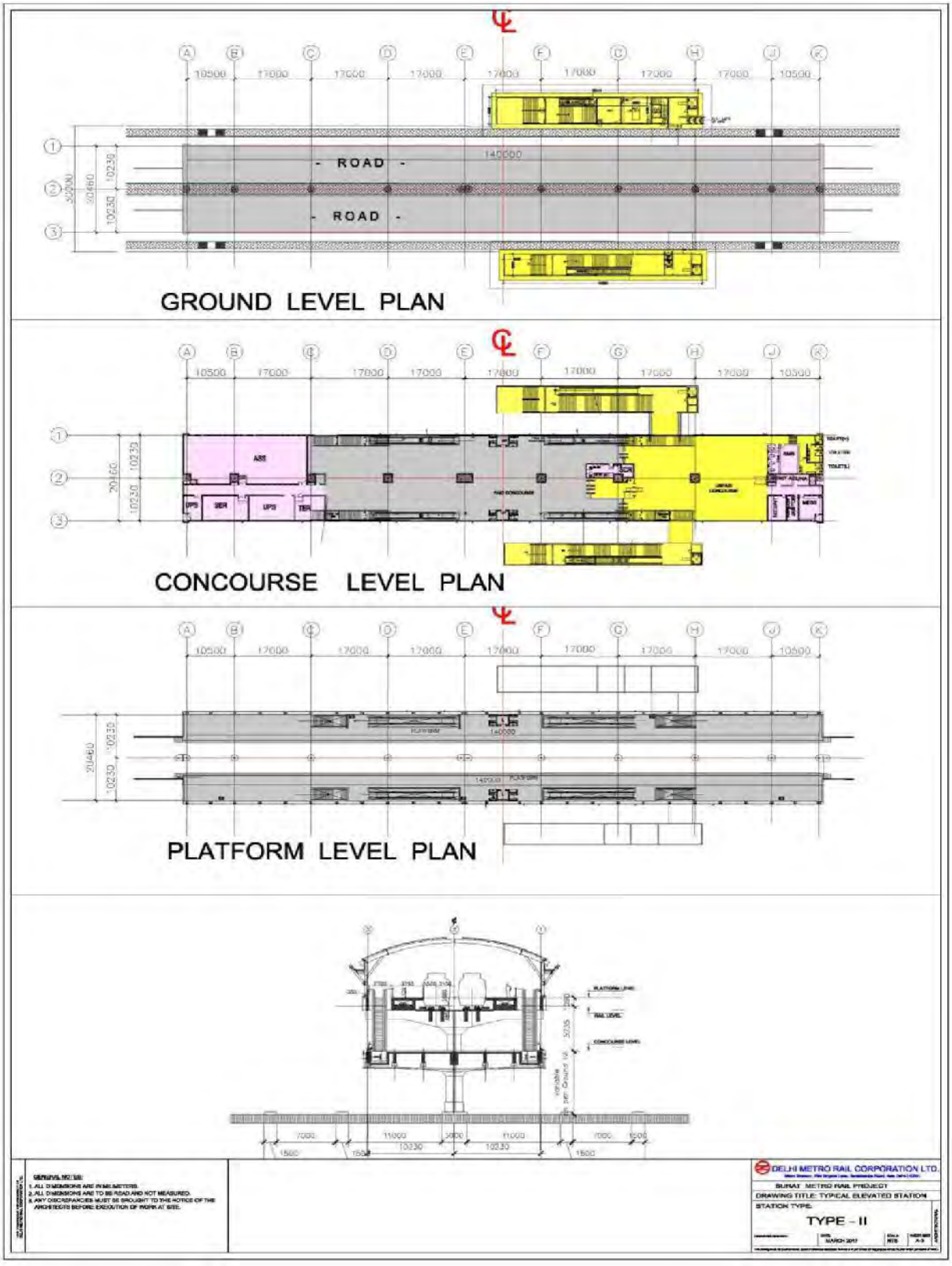
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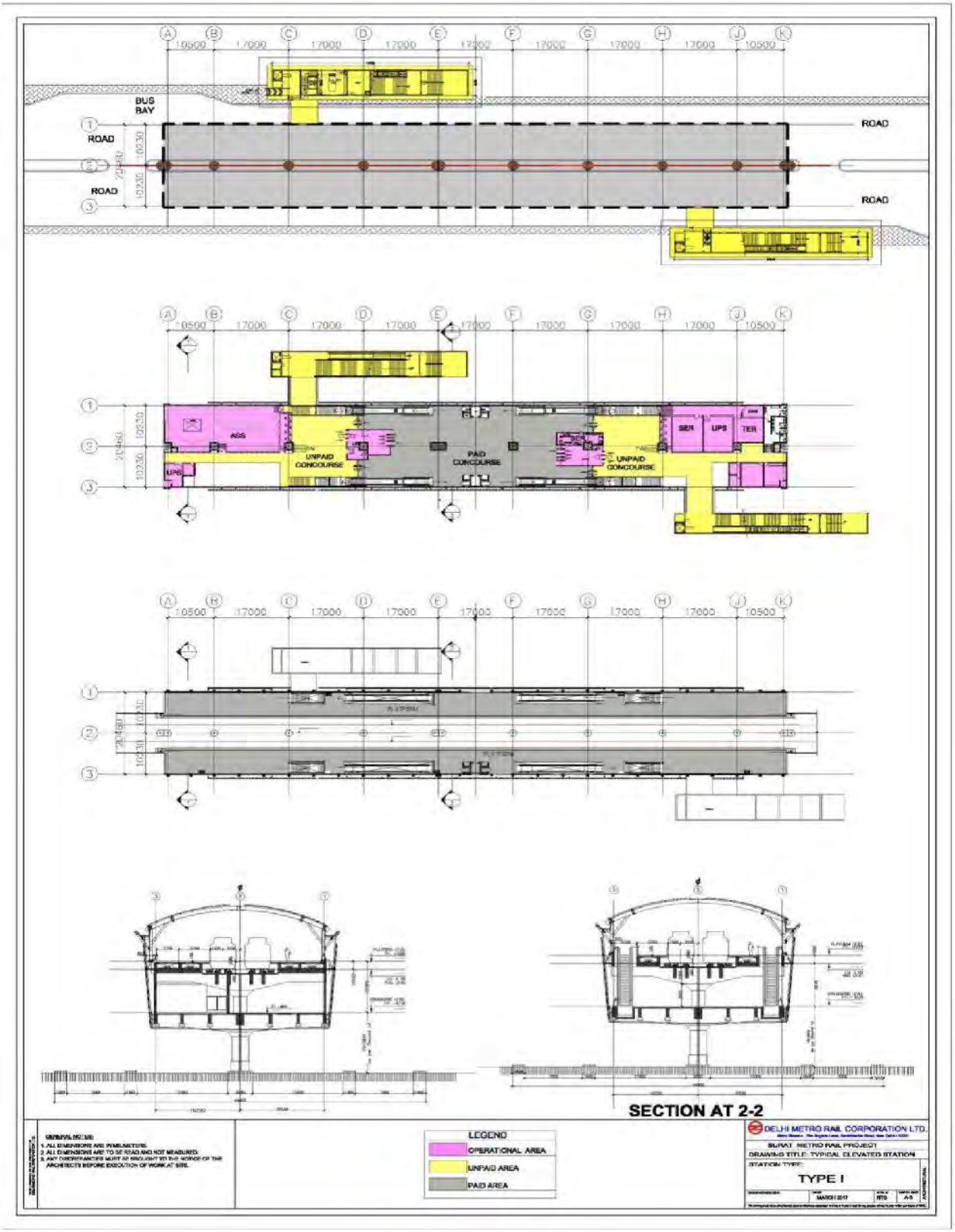
STATION TYPE
TYPE – II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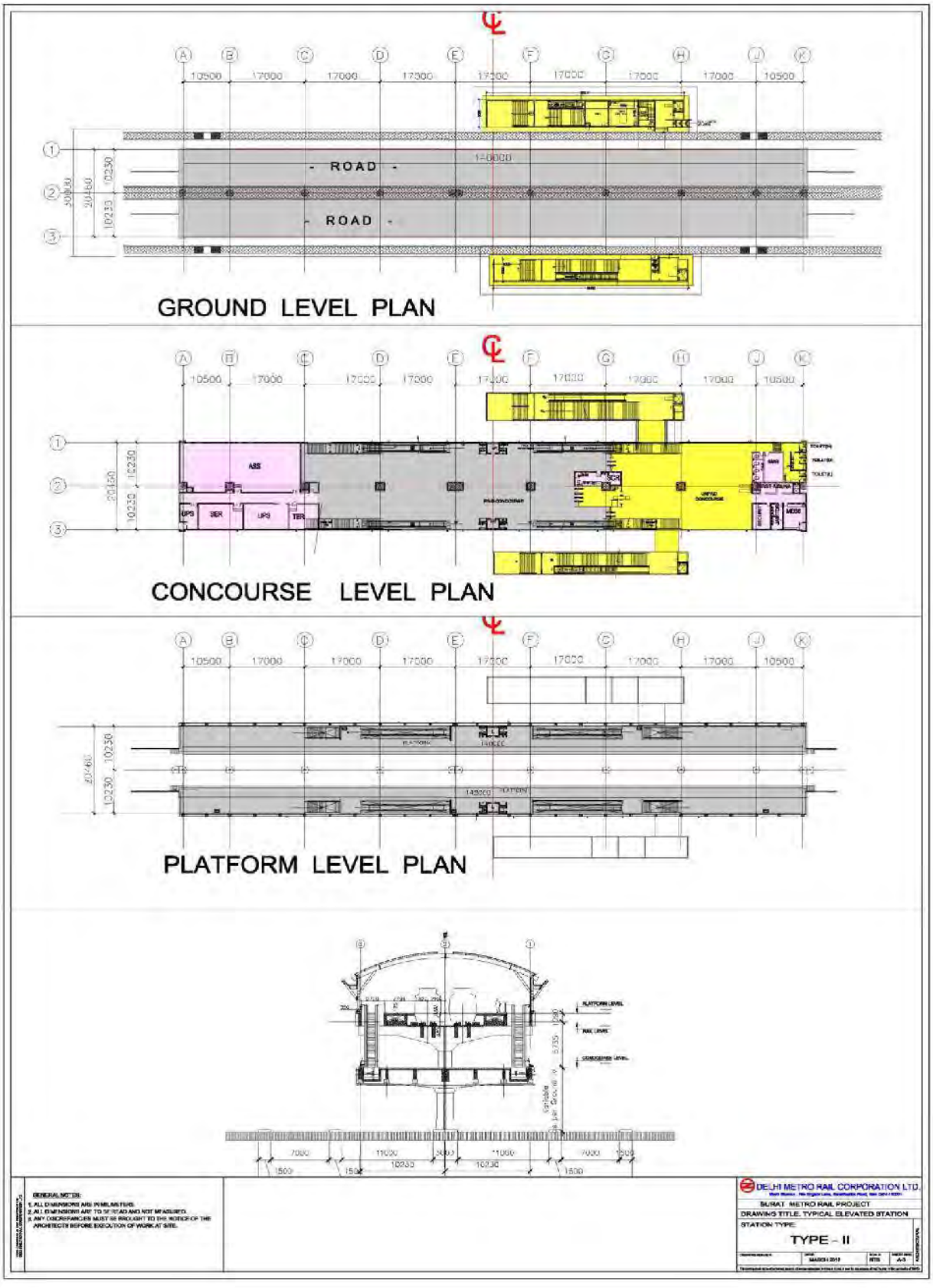
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CONCOURSE LEVEL PLAN
PLATFORM LEVEL PLAN
DELHI METRO RAIL CORPORATION LTD.
SURAT METRO RAIL PROJECT
DRAWING TITLE: TYPICAL ELEVATED STATION
STATION TYPE:
TYPE – II

CHAPTER 7

TRAIN OPERATION PLAN & ROLLING STOCK

7.1 Operation Philosophy

The underlying operation philosophy is to make the MRT System more attractive and economical, the main features being:

Selecting the most optimum frequency of Train services to meet sectional capacity requirement during peak hours on most of the sections.

Economical & optimum train service frequency not only during peak period, but also during off-peak period.

A short train consists of 3 coaches.

Multi-tasking of train operation and maintenance staff.

7.2 Stations

List of stations for the two Corridors of Surat Metro are given below:-

TABLE :7.1 : STATIONS

CORRIDOR-I: DREAM CITY – SARTHANA				
S. No	**Name of Station**	**Chainage (in m)**	**Inter – Station Distance (in m)**	**Remarks**
0	Dead End	(-) 450		
1	Sarthana	0.0	450	Elevated
2	Nature park	1238.1	1238.1	Elevated
3	VarachhaChopati Garden	2108.5	870.4	Elevated
4	Shri Swaminarayan Mandir Kalakunj	3196.6	1088.1	Elevated
5	Kapodra	4237.0	1040.4	U/G
6	Labheshwar Chowk	5506.6	1269.6	U/G
7	Central Warehouse	6668.9	1162.3	U/G
8	Surat Railway Station	7255.6	586.7	U/G
9	Maskati Hospital	8487.0	1231.4	U/G
10	Chowk Bazar	10289.8	1802.8	U/G
11	Kadarsha Ni Nal	11721.7	1431.9	Elevated
12	Majura Gate-1	12313.8	592.1	Elevated
13	Rupali Canal	13611.5	1297.7	Elevated

CORRIDOR-I: DREAM CITY – SARTHANA				
S. No	Name of Station	Chainage (in m)	Inter – Station Distance (in m)	Remarks
14	Althan Tenament	14696.2	1084.7	Elevated
15	Althan Gam	15843.5	1147.3	Elevated
16	VIP Road	16699.1	855.6	Elevated
17	Surat Women ITI	17808.4	1109.3	Elevated
18	Bhimrad	18659.5	851.1	Elevated
19	Convention Centre	19829.4	1169.9	Elevated
20	Dream City	20710.3	880.9	Elevated
	Dead End	21160.3	450.0	

CORRIDOR-II BHESAN – SAROLI				
S. No	Name of Station	Chainage (in m)	Inter – Station Distance (in m)	Remarks
0	Dead End	(-) 949.630		
1	Bhesan	0.0	949.63	Elevated
2	Botanical Garden	950.6	950.6	Elevated
3	UgatVaarigruh	1617.5	666.9	Elevated
4	Palanpur Road	2738.9	1121.4	Elevated
5	L.P. Savani School	3831.6	1092.7	Elevated
6	Peforming Art Centre	4421.8	590.2	Elevated
7	Adajan Gam	5155.0	733.2	Elevated
8	Aquerium	5770.1	615.1	Elevated
9	Badri Narayan Temple	6891.5	1121.4	Elevated
10	AthwaChaupati	7863.5	972.0	Elevated
11	Majura Gate-2	9019.2	1155.7	Elevated
12	Udhana Darwaja	10646.2	1627.0	Elevated
13	Kamela Darwaja	11247.2	601.0	Elevated
14	Anjana Farm	12593.9	1346.7	Elevated
15	Model Town	13636.3	1042.4	Elevated
16	Magob	14763.1	1126.8	Elevated
17	Baharat Cancer Hospital	15899.4	1136.3	Elevated
18	Saroli	17341.8	1442.4	Elevated
	Dead End	17791.9	450.1	

7.3 Train Operation Plan: Salient Features

Running of services for 19 hours of the day (5 AM to Midnight) with a station dwell time of 30 seconds,

Make up time of 5-10% with 8-12% coasting.

Scheduled speed for these corridors has been considered as:

Corridor-I: Dream City – Sarthana

- 'Dream City to Sarthana' section: 33kmph

Corridor-II Bhesan – Saroli

- 'Bhesan to Saroli' section: 33kmph

7.4 Traffic Demand

Peak hour peak direction traffic demands (PHPDT) for the Surat Metro 'Corridor-I Dream City-Sarthana' & 'Corridor-II Bhesan-Saroli' for the year 2021, 2026, 2036and 2046 for the purpose of planning are indicated in Attachment I/A1, B1, C1, D1 and Attachment I/A2, B2, C2, D2 respectively.

7.5 Train formation

To meet the above projected traffic demand, the possibility of running trains with composition of 3 cars with different headway has been examined.

Composition

DMC : Driving Motor Car
TC : Trailer Car

Capacity (@ 6 passengers per square meter of standee area)

Driving Motor Car (DMC)	- 247 (43 seated + 204 standing)
Trailer Car (TC)	- 270 (50 seated + 220 standing)
3 Car Train	- 764 (136 seated + 628 standing)

7.6 Train Operation Plan

Based on the projected PHPDT demand, Train operation plan with train carrying capacity @ 6 persons per square meter of standee area for the Surat Metro 'Corridor-I: Dream City-Sarthana' & 'Corridor-II: Bhesan-Sroli' for the year 2021, 2026, 2036 and 2046 are given below:

1. Corridor-I: Dream City –Sarthana

Train Operation Plan for Corridor-I: Dream City-Sarthana has been planned in such a way that there is Single loop of train operation end to end.

i) Year 2021:

Train operation in 'Dream City to Sarthana' Loop is at 6.5 min headway with 3-Car train. This results in following train operation:

'Dream City to Sarthana' Section (Refer Attachment I/A1)

6.5 min Effective Headway with 3-car train.

Available Peak Hour Peak Direction Capacity of 7052@ 6 persons per square meter of standee area.

Available Peak Hour Peak Direction Capacity of 8972 @ 8 persons per square meter of standee area under dense loading conditions.

The maximum PHPDT demand of 6992 is in the Section between Maskati Hospital to Surat Railway Station and demand in the remaining sections is in the range of 6609 to 119 only. The planned capacity of 7052 (8972 under dense loading) is more than the PHPDT demand.

Traffic demand and train capacity for this corridor is the year 2021istabulated and represented on a chart enclosed as Attachment I/A1.

ii) Year 2026:

Train operation in 'Dream City to Sarthana' Loop is at 3.5 min headway with 3-Car train. This results in following train operation:

'Dream City to Sarthana' Section (Refer Attachment I/B1)

3.5 min Effective Headway with 3-car train.

Available Peak Hour Peak Direction Capacity of 13097@ 6 persons per square meter of standee area

Available Peak Hour Peak Direction Capacity of 16663 @ 8 persons per square meter of standee area under dense loading conditions.

The maximum PHPDT demand of 13536 is in the Section between Surat Railway Station to Central Warehouse and demand in the remaining sections is in the range of 13495 to 615 only. The planned capacity of 13097 (16663 under dense loading) is slightly less than the PHPDT demand in only two (zero, with dense loading capacity) sections out of nineteen sections.

Traffic demand and train capacity for this corridor is the year 2026is tabulated and represented on a chart enclosed as Attachment I/B1.

iii) Year 2036:

Train operation in 'Dream City to Sarthana' Loop is at 2.25 min headway with 3-Car train. This results in following train operation:

'Dream City to Sarthana' Section (Refer Attachment I/C1)

2.25 min Effective Headway with 3-car train.

Available Peak Hour Peak Direction Capacity of 20373@ 6 persons per square meter of standee area.

Available Peak Hour Peak Direction Capacity of 25920 @ 8 persons per square meter of standee area under dense loading conditions.
The maximum PHPDT demand of 20856 is in the Section between Chowk Bazar to Maskati Hospital and demand in the remaining sections is in the range of 20677 to 2150 only. The planned capacity of 20373 (25920 under dense loading) is slightly less than the PHPDT demand in only two (zero, with dense loading capacity) sections out of nineteen sections.

Traffic demand and train capacity for this corridor is the year 2036 is tabulated and represented on a chart enclosed as Attachment I/C1.

iv) Year 2046:

Train operation in 'Dream City to Sarthana' Loop is at 1.75 min headway with 3-Car train. This results in following train operation:

'Dream City to Sarthana' Section (Refer Attachment I/D1)
1.75 min Effective Headway with 3-car train.
Available Peak Hour Peak Direction Capacity of 26194@ 6 persons per square meter of standee area.
Available Peak Hour Peak Direction Capacity of 33326 @ 8 persons per square meter of standee area under dense loading conditions.
The maximum PHPDT demand of 26587 is in the Section between Surat Railway Station to Central Warehouse and demand in the remaining sections is in the range of 25236 to 3095 only. The planned capacity of 26194 (33326 under dense loading) is slightly less than the PHPDT demand in only one (zero, with dense loading capacity) section out of nineteen sections.

Traffic demand and train capacity for this corridor is the year 2046is tabulated and represented on a chart enclosed as Attachment I/D1.

2. Corridor-II: Bhesan –Saroli

Train Operation Plan for Corridor-II: Bhesan-Saroli has been planned in such a way that there is Single loop of train operation end to end.

i) Year 2021:

Train operation in 'Bhesan to Saroli' Loop is at 10 min headway with 3-Car train. This results in following train operation:

'Bhesan to Saroli'Section (Refer Attachment I/A2)
10 min Effective Headway with 3-car train.
Available Peak Hour Peak Direction Capacity of 4584@ 6 persons per square meter of standee area.

Available Peak Hour Peak Direction Capacity of 5832 @ 8 persons per square meter of standee area under dense loading conditions.
The maximum PHPDT demand of 3918 is in the Section between Udhana Darwaja to Kamela Darwaja and demand in the remaining sections is in the range of 3848 to 245 only. The planned capacity of 4584 (5832 under dense loading) is more than the PHPDT demand.

Traffic demand and train capacity for this corridor is the year 2021istabulated and represented on a chart enclosed as Attachment I/A2.

ii) Year 2026:

Train operation in 'Bhesan to Saroli' Loop is at 8 min headway with 3-Car train. This results in following train operation:

'Bhesan to Saroli' Section (Refer Attachment I/B2)
8 min Effective Headway with 3-car train.
Available Peak Hour Peak Direction Capacity of 5730@ 6 persons per square meter of standee area.
Available Peak Hour Peak Direction Capacity of 7290 @ 8 persons per square meter of standee area under dense loading conditions.
The maximum PHPDT demand of 5698 is in the Section between Udhana Darwaja to Kamela Darwaja and demand in the remaining sections is in the range of 5666 to 629 only. The planned capacity of 5730 (7290 under dense loading) is more than the PHPDT demand.

Traffic demand and train capacity for this corridor is the year 2026istabulated and represented on a chart enclosed as Attachment I/B2.

iii) Year 2036:

Train operation in 'Bhesan to Saroli' Loop is at 3.75 min headway with 3-Car train. This results in following train operation:

'Bhesan to Saroli' Section (Refer Attachment I/C2)
3.75 min Effective Headway with 3-car train.
Available Peak Hour Peak Direction Capacity of 12224@ 6 persons per square meter of standee area.
Available Peak Hour Peak Direction Capacity of 15552 @ 8 persons per square meter of standee area under dense loading conditions.
The maximum PHPDT demand of 12573 is in the Section between Badri Narayan Temple to Athwa Chaupati and demand in the remaining sections is in the range of 12098 to 2649 only. The planned capacity of 12224 (15552 under dense loading) is slightly less than the PHPDT demand in only one (zero, with dense loading capacity) section out of seventeen sections.

Traffic demand and train capacity for this corridor is the year 2036is tabulated and represented on a chart enclosed as Attachment I/C2.

iv) Year 2046:

Train operation in 'Bhesan to Saroli' Loop is at 3 min headway with 3-Car train. This results in following train operation:

'Bhesan to Saroli' Section (Refer Attachment I/D2)

3 min Effective Headway with 3-car train.

Available Peak Hour Peak Direction Capacity of 15280@ 6 persons per square meter of standee area.

Available Peak Hour Peak Direction Capacity of 19440 @ 8 persons per square meter of standee area under dense loading conditions.

The maximum PHPDT demand of 15448 is in the Section between Badri Narayan Temple to Athwa Chaupati and demand in the remaining sections is in the range of 15175 to 3487 only. The planned capacity of 15280 (19440 under dense loading) is slightly less than the PHPDT demand in only one (zero, with dense loading capacity) section out of seventeen sections.

Traffic demand and train capacity for this corridor is the year 2046is tabulated and represented on a chart enclosed as Attachment I/D2.

The above Train Operation Plan is based on calculations on the basis of available traffic data. In case of any mismatch in the capacity provided and the actual traffic, the capacity can be moderated suitably by adjusting the Headway.

The PHPDT capacity provided on the two corridors in different years of operation is tabulated below:

TABLE :7.2

Capacity Provided for Corridor-I: Dream City – Sarthana

Sections	Year	Headway (min)	No. of Rakes	Rake Consist	No. of Coaches	Max. PHPDT Demand	PHPDT Capacity Available
Dream City to Sarthana	2021	6.5	15	3-car	45	6992	7052 (8972*)
Dream City to Sarthana	2026	3.5	26	3-car	78	13536	13097 (16663*)

Dream City to Sarthana	2036	2.25	41	3-car	123	20856	20373 (25920*)
Dream City to Sarthana	2046	1.75	52	3-car	156	26587	26194 (33326*)

TABLE :7.3

Capacity Provided for Corridor-II:Bhesan-Saroli

Sections	Year	Headway (min)	No. of Rakes	Rake Consist	No. of Coaches	Max. PHPDT Demand	PHPDT Capacity Available
Bhesan to Saroli	2021	10	9	3-car	27	3918	4584 (5832*)
Bhesan to Saroli	2026	8	11	3-car	33	5698	5730 (7290*)
Bhesan to Saroli	2036	3.75	21	3-car	63	12573	12224 (15552*)
Bhesan to Saroli	2046	3	26	3-car	78	15448	15280 (19440*)

* @ 8 persons per square meter of standee area

7.7 Train frequency

TABLE:7.4

Train Frequency Corridor-I: Dream City-Sarthana

Section	2021		2026		2036		2046	
	Peak Hour Head-way	Lean Hour Head-way	Peak Hour Head-way	Lean Hour Head-way	Peak Hour Head-way	Lean Hour Head-way	Peak Hour Head-way	Lean Hour Head-way
Dream City to Sarthana	6.5 min	12 to 32 min	3.5 min	10 to 24 min	2.25 min	10 to 24 min	1.75 min	10 to 24 min

TABLE: 7.5

Train Frequency Corridor-II:Bhesan-Saroli

Section	2021		2026		2036		2046	
	Peak Hour Head-way	Lean Hour Head-way	Peak Hour Head-way	Lean Hour Head-way	Peak Hour Head-way	Lean Hour Head-way	Peak Hour Head-way	Lean Hour Head-way
Bhesan to Saroli	10 min	12 to 32 min	8 min	10 to 24 min	3.75 min	10 to 24 min	3 min	10 to 24 min

No services are proposed between 00:00 hrs to 5.00 hrs, which are reserved for maintenance of infrastructure and rolling stock.

7.8 Hourly Train Operation plan

The hourly distribution of daily transport capacity is presented in **Table 1.1A, 1.2A, 1.3A, 1.4A**for 'Dream City to Sarthana' Section (Corridor-I) and **Table 1.1B, 1.2B, 1.3B, 1.4B**for 'Bhesan to Saroli' Section (Corridor-II) respectively for years 2021, 2026, 2036 and 2046 enclosed as **Attachment II.**

Number of train trips per direction per day for 'Dream City to Sarthana' Section (Corridor-I) is worked out as 95 in the year 2021, 155 in the year 2026, 209 in the year 2036 and 257 in the year 2046 respectively. Number of train trips per directions per day for 'Bhesan to Saroli' Section (Corridor-II) is worked out as 74 in the year 2021, 95 in the year 2026, 146 in the year 2036 and 170 in the year 2046 respectively. The directional splits for Corridor-I: Dream City-Sarthana and Corridor-II: Bhesan-Saroli is presented in **Table 2.1 and 2.2** enclosed as **Attachment III**.

7.9 Vehicle Kilometer

Based on above planning, after considering maintenance period and assuming 340 days in service in a year, Vehicle Kilometers for Surat Metro Rail Network is given in **Table 3.1** for Corridor-I: Dream City-Sarthana and **Table 3.2** for Corridor-II: Bhesan-Saroli enclosed as **Attachment IV**.

7.10 Year wise rake Requirement

Based on Train formation and headway as decided above to meet Peak Hour Peak Direction Traffic Demand, Rake requirement has been calculated and enclosed as Attachment V.

Requirements of coaches is calculated based on following assumptions-

Assumptions –

(i) Train Composition planned as under

3 car Train Composition (with 66.7% Powering)	: DMC +TC +DMC
Train Carrying Capacity of 3 Car Train (@6 passengers per square meter of standee area)	: 764 passengers @6 standee/sqm

(ii) Coach requirement has been calculated based on headway during peak hours.
(iii) Traffic reserve is taken as one train to cater to failure of train on line and to make up for operational time list.
(iv) Repair and maintenance reserve has been estimated as 8 % of total requirement (Bare).
(v) The calculated number of rakes in fraction is rounded off to next higher number.
(vi) Schedule speed is taken as: 33kmph
(vii) Total Turn round time is taken as 6 min at terminal stations.

7.11 Cost Estimate

The estimated cost per coach at Jan 2017 Price level exclusive of taxes and duties may be assumed as INR 10.2 Crores per Coach. Total 45+27 = 72 coaches are required in year 2021 for the two Corridors in Surat Metro Rail Network. Hence budget provision of Rs. 734.4 Crores is to be kept in the Estimate for Rolling stock.

ROLLING STOCK

7.12 INTRODUCTION

The required transport demand forecast is the governing factor for the choice of the Rolling Stock. The forecasted Peak Hour Peak Direction Traffic calls for a Mass Rapid Transit System (MRTS).

7.13 OPTIMIZATION OF COACH SIZE

The following optimum size of the coach has been chosen for this corridor as mentioned in Table 7.5.

Table 7.5 - Size of the coach

	Length*	Width	Height
Driving Motor Car (DMC)	21.64 m	2.9 m	3.9 m
Trailer Car (TC)	21.34 m	2.9 m	3.9 m

**Maximum length of coach over couplers/buffers = 22 to 22.6 m*

7.14 PASSENGER CARRYING CAPACITY

In order to maximize the passenger carrying capacity, longitudinal seating arrangement shall be adopted. The whole train shall be vestibuled to distribute the passenger evenly in all the coaches. Criteria for the calculation of standing passengers are 3 persons per square meter of standing floor area in normal state and 6 persons in crush state of peak hour.

Therefore, for the Medium Rail Vehicles (MRV) with 2.9 m maximum width and longitudinal seat arrangement, conceptually the crush capacity of 43 seated, 204 standing thus a total of 247 passengers for a Driving motor car, and 50 seated, 220 standing thus a total of 270 for a Trailer car is envisaged.

Following train composition is recommended:
3-car Train: DMC+TC+DMC

Table 7.6 shows the carrying capacity of Medium Rail Vehicles.

Table 7.6 - Carrying Capacity of Medium Rail Vehicles

Particulars	Driving Motor car		Trailer car		3 Car Train	
	Normal	Crush	Normal	Crush	Normal	Crush
Seated	43	43	50	50	136	136
Standing	102	204	110	220	314	628
Total	145	247	160	270	450	764

NORMAL-3 Person/sqm of standee area

CRUSH -6 Person/sqm of standee area

7.15 WEIGHT

The weights of driving motor car and trailer car have been estimated as in Table 7.7, referring to the experiences in Delhi Metro. The average passenger weight has been taken as 65 kg.

Table 7.7 - Weight of Light Rail Vehicles (TONNES)

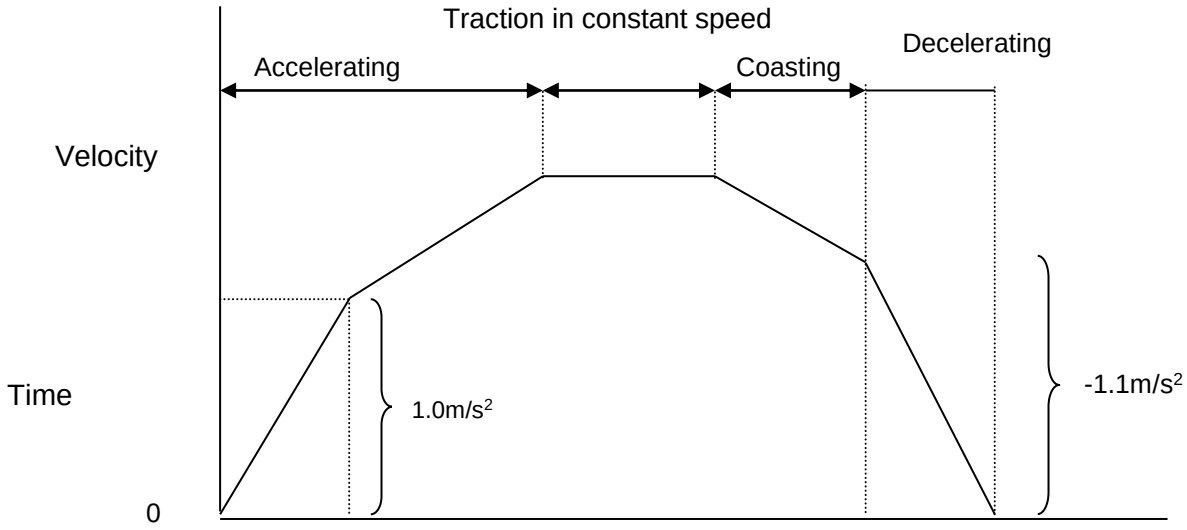
	DMC	TC	3 Car Train
TARE (maximum)	40	40	120
Passenger			
(Normal)	9.425	10.4	29.25
(Crush @6p/sqm)	16.055	17.55	49.66
(Crush @8p/sqm)	20.475	22.295	63.245
Gross			
(Normal)	49.425	50.4	149.25
(Crush @6p/sqm)	56.055	57.55	169.66
(Crush @8p/sqm)	60.475	62.295	183.23
Axle Load @6 person/sqm(AW2 Load)	14.014	14.388	
Axle Load @8 person/sqm(AW3 Load)	15.119	15.574	

The axle load @ 6persons/sqm of standing area works out in the range of 14.014T to 14.388T. Heavy rush of passenger, having 8 standees per sq. meter can be experienced occasionally. It will be advisable to design the coach with sufficient strength so that even with this overload, the design will not result in over stresses in the coach. Coach and bogie should, therefore, be designed for **16 T axle load.**

7.16 PERFORMANCE PARAMETERS

The recommended performance parameters are:

Maximum Design Speed: 90 kmph
Maximum Operating Speed: 80 kmph
Max. Acceleration: 1.0 m/s^2 (with AW3 load)
1.2 m/s^2 (with AW2 load)
Max. Deceleration: 1.1 m/s^2 (with AW3 load)
1.1 m/s^2 (with AW2 load)
>1.3 m/s^2 (Emergency brake)



7.17 COACH DESIGN AND BASIC PARAMETERS

The important criteria for selection of rolling stock are as under:

(i) Proven equipment with high reliability
(ii) Passenger safety feature
(iii) Energy efficiency
(iv) Light weight equipment and coach body
(v) Optimized scheduled speed
(vi) Aesthetically pleasing Interior and Exterior
(vii) Low Life cycle cost
(viii) Flexibility to meet increase in traffic demand
(ix) Anti-telescopic

The controlling criteria are reliability, low energy consumption, lightweight and high efficiency leading to lower annualized cost of service. The coach should have high rate of acceleration and deceleration.

7.18 SELECTION OF TECHNOLOGY

Low life cycle cost

Low life cycle cost is achieved by the way of reduced scheduled and unscheduled maintenance and high reliability of the sub-systems. It is possible to achieve these objectives by adopting suitable proven technologies. Selection of following technologies has been recommended to ensure low life cycle cost-.

7.18.1 Car body

In the past carbon high tensile steel was invariably used for car bodies. In-fact almost all the coaches built by Indian Railways are of this type. These steel bodied coaches need frequent painting and corrosion repairs, which may have to be carried out up to 4-5 times during the service life of these coaches. It is now a standard practice to adopt stainless steel or aluminum for carbody.

The car bodies with aluminum require long and complex extruded sections which are still not manufactured in India. Therefore, aluminum car body has not been considered for use. Stainless steel sections are available in India and therefore stainless steel car bodies have been specified. No corrosion repair is necessary on stainless steel cars during their service life.

Stainless steel car body leads to energy saving due to its lightweight. It also results in cost saving due to easy maintenance and reduction of repair cost from excellent anti corrosive properties as well as on improvement of riding comfort and safety in case of a crash or fire.

7.18.2 Bogies

Bolster less lightweight fabricated bogies with helical coil spring/rubber springs are now universally adopted in metro cars. These bogies require less maintenance and overhaul interval is also of the order of 4,20,000km. Use of air spring at secondary stage is considered with a view to keep the floor level of the cars constant irrespective of passenger loading unlike those with coil spring. Perturbation from the track are also dampened inside the car body on account of the secondary air spring along with suitable Vertical Hydraulic Damper. The primary suspension system improves the curve running performance by reducing lateral forces through application of helical coil spring/ conical rubber spring. Helical springs is preferred over conical rubber spring based upon DMRC experience. A smooth curving performance with better ride index is being ensured by provision of above type of bogies.

7.18.3 Braking System

The brake system shall consist of –

(i) An electro-pneumatic (EP) service friction brake
(ii) A fail safe, pneumatic friction emergency brake
(iii) A spring applied air-release parking brake
(iv) An electric regenerative service brake
(v) Provision of smooth and continuous blending of EP and regenerative braking

The regenerative braking will be the main brake power of the train and will regain the maximum possible energy and pump it back to the system and thus fully utilize the advantage of 3 phase technology. The regenerative braking should have air supplement control to bear the load of trailer car. In addition, speed sensors mounted on each axle, control the braking force of the axles with anti-skid valves, prompting re-adhesion in case of a skid. The brake actuator shall operate either a tread brake or a wheel disc brake, preferably a tread brake.

7.18.4 Propulsion System Technology

In the field of Electric Rolling Stock, DC series traction motors have been widely used due to its ideal characteristics and good controllability for traction applications. But these required intensive maintenance because of commutator and electro-mechanical contactors, resistors etc.

The brush less 3 phase induction motors has now replaced the D.C. Series motors in traction applications. The induction motor, for the same power output, is smaller and lighter in weight and ideally suited for rail based Mass Rapid Transit applications. The motor tractive effort and speed is regulated by 'Variable Voltage and Variable frequency' control and can be programmed to suit the track profile and operating requirements. Another advantage of 3 phase a.c. drive and VVVF control is that regenerative braking can be introduced by lowering the frequency and the voltage to reverse the power flow and to allow braking to very low speed.

For this corridor, three phase a.c. traction drive that are self-ventilated, highly reliable, robust construction and back up by slip/slid control have been recommended for adoption.

The DC voltage from the 3rd Rail is stepped up through a 'STEP up Chopper' to DC link voltage, which feeds Inverter operated with Pulse Width Modulation (PWM) control technology and using insulated Gate Bipolar Transistors (IGBT). Thus three-phase variable voltage variable frequency output drives the traction motors for propulsion.

Recently advanced IGBT has been developed for inverter units. The advanced IGBT Contains an Insulated Gate Bipolar Transistor (IGBT) and gate drive circuit and protection. The advanced IGBT incorporates its own over current protection, short circuit protection, over temperature protection and low power supply detection. The IGBT has internal protection from over current, short circuit, over temperature and low control voltage. The inverter unit uses optical fiber cable to connect the control unit to the gate interface. This optical fiber cable transmits the gate signals to drive the advanced IGBT via the gate interface. This optical fiber cable provides electrical isolation

between the advanced IGBT and the control unit and is impervious to electrical interference. These are recommended for adoption in Trains of MRTS.

7.18.5 Interior and Gangways

Passenger capacity of a car is maximized in a Metro System by providing longitudinal seats for seating and utilizing the remaining space for standing passenger. Therefore, all the equipments are mounted on the under frame for maximum space utilization. The gangways are designed to give a wider comfortable standing space during peak hours along with easy and faster passenger movement especially in case of emergency.

Interior View

7.18.6 Passenger Doors

For swift evacuation of the passenger in short dwell period, four doors of adequate width, on each side of the coach have been considered. These doors shall be of such dimensions and location that all the passenger inside the train are able to evacuate within least possible time without conflicting movement. As the alignment passes through elevated section above ground, automatic door closing mechanism is envisaged from consideration of passenger safety. Passenger doors are controlled electrically by a switch in Driver cab. Electrically controlled door operating mechanism has been preferred over pneumatically operated door to avoid cases of air leakage and sluggish operation of doors.

The door shall be of Bi-parting external sliding type as in the existing coaches of DMRC.

Passenger Doors

7.18.7 Air–conditioning

With heavy passenger loading of 6 persons/sqm for standee area and doors being closed from consideration of safety and with windows being sealed type to avoid transmission of noise, air conditioning of coaches has been considered essential. Each coach shall be provided with two air conditioning units capable of cooling, heating and dehumidifying and thus automatically controlling interior temperature throughout the passenger area at 25°C with 60% RH all the times under varying ambient conditions up to full load. For emergency situations such as power failure or both AC failures etc, ventilation provision supplied from battery will be made. Provision shall be made to shut off the fresh air intake and re-circulate the internal air of the coach, during an emergency condition, such as fire outside the train causing excessive heat and smoke to be drawn in to the coach.

7.18.8 Cab Layout and Emergency Detrainment Door.

The modern stylish driver panel shall be FRP moulded which give maximum comfort and easy accessibility of different monitoring equipments to the driver along with clear visibility. The driver seat has been provided at the left side of the cabin.

Driving cab

In Standard Gauge (2.9 m wide stock) cars, emergency evacuation for easy detrainment of the passenger will be done through side saloon doors from which passengers usually board and alight.

7.18.9 Communication

The driving cab of the cars are provided with continuous communication with base Operational Control Center and station control for easy monitoring of the individual train in all sections at all the time.

Public Address and Passenger Information Display System is provided in the car so that passengers are continuously advised of the next stoppage station, final destination station, interchange station, emergency situations if any, and other messages. The rolling stock is provided with Talk Back Units inside the cars, which permit conversation between passengers and the drivers in case of any emergency.

7.18.10 Noise and Vibration

The trains will pass through heavily populated urban area. The noise and vibration for a metro railway become an important criterion from public acceptance view point. The sources of noise are (i) rail-wheel interaction (ii) noise generated from equipment like Blower, Compressor, air conditioner, door, Inverter etc. (iii) traction motor in running train. For elimination and reduction of noise following feature are incorporated: -

Provision of anti-drumming floor and noise absorption material.
Low speed compressor, blower and air conditioner.
Mounting of under frame equipments on anti-vibration pad

Smooth and gradual control of door.

Provision of GRP baffle on the via-duct for elimination of noise transmission.

Provision of sound absorbing material in the supply duct and return grill of air conditioner.

Sealing design to reduce the aspiration of noise through the gap in the sliding doors and piping holes.

Provision of wheel flange and top of rail lubrication to reduce squealing noise.

Provision of noise attenuators (Hypno dampers) on wheels to reduce noise due to rail wheel interaction

The lower vibration level has been achieved by provision of bolster less type bogies having secondary air spring.

7.18.11 Passenger Safety Features

(i) ATP

The rolling stock is provided with Continuous Automatic Train Protection to ensure absolute safety in the train operation. It is an accepted fact that 60-70% of the accidents take place on account of human error. Adoption of this system reduces the possibility of human error.

(ii) Fire

The rolling stock is provided with fire retarding materials having low fire load, low heat release rate, low smoke and toxicity inside the cars. The electric cables used are also normally low smoke zero halogen type which ensures passenger safety in case of fire.

(iii) Emergency door

In Standard Gauge (2.9 m wide stock) cars, in case of emergency evacuation will be done through side saloon doors to ensure well directed evacuation of passengers in case of any emergency including fire in the train. Some suitable bridging mechanism like foldable ladder shall be provided to de-board the passengers onto viaduct.

(iv) Crash worthiness features

The rolling stock is provided with inter car couplers having crashworthiness feature which reduces the severity of injury to the passengers in case of accidents.

(v) Gangways

Broad gangways are provided in between the cars to ensure free passenger movement between cars in case of any emergency.

Gangways

Attachment - II/A1

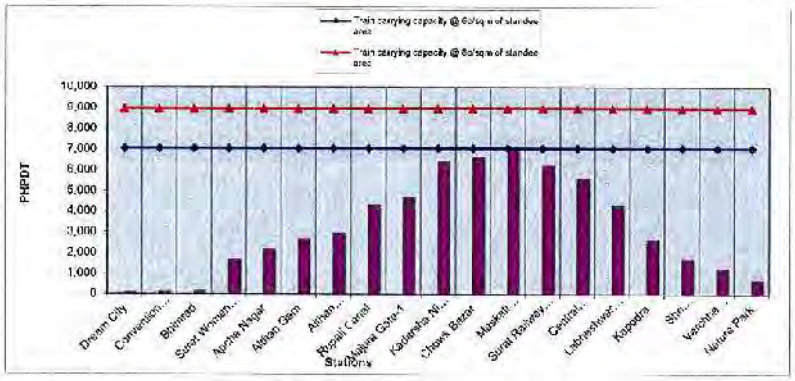
PHPDT Demand and Capacity Chart

Corridor- I (Dream City- Sarthana)

Year: 2021
No. of cars per train 3
Passenger Capacity @ 6 persons/sqm of a 3-Car Train: 764
Passenger Capacity @ 8 persons/sqm of a 3-Car Train: 972

Headway (min) 6.50

S.N	FROM	TO	Traffic Demand in PHPDT	Train carrying capacity @ 6p/sqm of standee area	Train carrying capacity @ 8p/sqm of standee area
1	Dream City	Convention Centre	119	7,052	8,972
2	Convention Centre	Bhimrad	145	7,052	8,972
3	Bhimrad	Surat Women ITI	169	7,052	8,972
4	Surat Women ITI	Apcha Nagar	1,649	7,052	8,972
5	Apcha Nagar	Althan Gam	2,158	7,052	8,972
6	Althan Gam	Althan Tenament	2,645	7,052	8,972
7	Althan Tenament	Rupali Canal	2,945	7,052	8,972
8	Rupali Canal	Majura Gate-1	4,283	7,052	8,972
9	Majura Gate-1	SMC Shopping Complex	4,659	7,052	8,972
10	Kadarsha Ni Nal	Chowk Bazar	5,420	7,052	8,972
11	Chowk Bazar	Maskati Hospital	5,609	7,052	8,972
12	Maskati Hospital	Surat Railway station	6,992	7,052	8,972
13	Surat Railway station	Central Warehouse	6,217	7,052	8,972
14	Central Warehouse	Labheshwar Chowk	5,594	7,052	8,972
15	Labheshwar Chowk	Kapodra	4,280	7,052	8,972
16	Kapodra	Shri Swaminarayan Mandir Kalakunj	2,626	7,052	8,972
17	Shri Swaminarayan Mandir Kalakunj	Varchha Chopati Garden	1,684	7,052	8,972
18	Varchha Chopati Garden	Nature Park	1,242	7,052	8,972
19	Nature Park	Sarthana	667	7,052	8,972



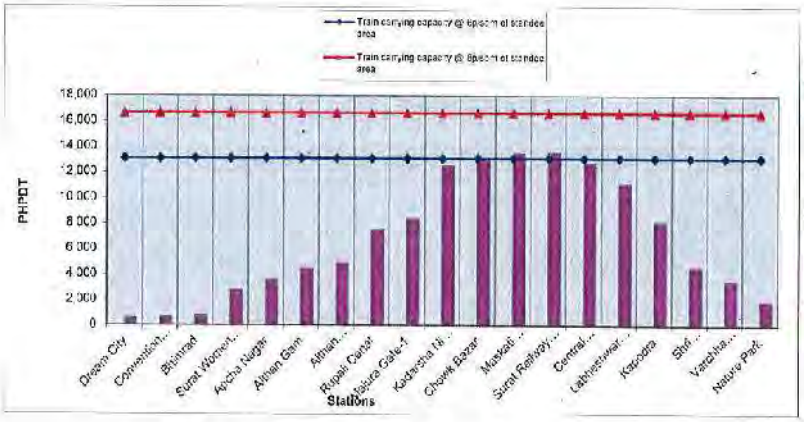
Attachment - I/B1

PHPDT Demand and Capacity Chart

Corridor- I (Dream City- Sarthana)

Year:	2026
No. of cars per train	3
Passenger Capacity @ 6 persons/sqm of a 3-Car Train:	764
Passenger Capacity @ 8 persons/sqm of a 3-Car Train:	972
Headway (min)	3.50

S.N	FROM	TO	Traffic Demand in PHPDT	Train carrying capacity @ 6p/sqm of standee area	Train carrying capacity @ 8p/sqm of standee area
1	Dream City	Convention Centre	615	13,097	16,663
2	Convention Centre	Bhimrad	656	13,097	16,663
3	Bhimrad	Surat Women ITI	745	13,097	16,663
4	Surat Women ITI	Apcha Nagar	2,807	13,097	16,663
5	Apcha Nagar	Althan Gam	3,553	13,097	16,663
6	Althan Gam	Althan Tenament	4,447	13,097	16,663
7	Althan Tenament	Rupali Canal	4,918	13,097	16,663
8	Rupali Canal	Majura Gate-1	7,515	13,097	16,663
9	Majura Gate-1	SMC Shopping Complex	8,380	13,097	16,663
10	Kadarsha Ni Nal	Chowk Bazar	12,525	13,097	16,663
11	Chowk Bazar	Maskati Hospital	13,035	13,097	16,663
12	Maskati Hospital	Surat Railway station	13,495	13,097	16,663
13	Surat Railway station	Central Warehouse	13,536	13,097	16,663
14	Central Warehouse	Labheshwar Chowk	12,770	13,097	16,663
15	Labheshwar Chowk	Kapodra	11,089	13,097	16,663
16	Kapodra	Shri Swaminarayan Mandir Kalakunj	8,060	13,097	16,663
17	Shri Swaminarayan Mandir Kalakunj	Varchha Chopati Garden	4,550	13,097	16,663
18	Varchha Chopati Garden	Nature Park	3,455	13,097	16,663
19	Nature Park	Sarthana	1,873	13,097	16,663



Attachment - I/C1

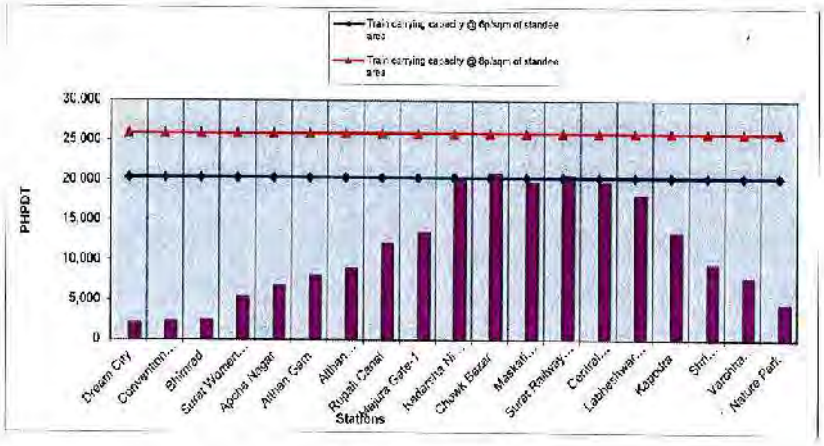
PHPDT Demand and Capacity Chart

Corridor- I (Dream City- Sarthana)

Year:	2036
No. of cars per train	3
Passenger Capacity @ 6 persons/sqm of a 3-Car Train:	764
Passenger Capacity @ 8 persons/sqm of a 3-Car Train:	972
Headway (min)	2.25

Train carrying capacity @ 8p/sqm of sta

S.N	FROM	TO	Traffic Demand In PHPDT	Train carrying capacity @ 6p/sqm of standee area	Train carrying capacity @ 8p/sqm of standee area
1	Dream City	Convention Centre	2,150	20,373	25,920
2	Convention Centre	Bhimrad	2,277	20,373	25,920
3	Bhimrad	Surat Women ITI	2,538	20,373	25,920
4	Surat Women ITI	Apcha Nagar	5,456	20,373	25,920
5	Apcha Nagar	Althan Gam	6,844	20,373	25,920
6	Althan Gam	Althan Tenament	8,101	20,373	25,920
7	Althan Tenament	Rupali Canal	8,963	20,373	25,920
8	Rupali Canal	Majura Gate-1	12,165	20,373	25,920
9	Majura Gate-1	SMC Shopping Complex	13,524	20,373	25,920
10	Kadarsha Ni Nal	Chowk Bazar	20,062	20,373	25,920
11	Chowk Bazar	Maskati Hospital	20,856	20,373	25,920
12	Maskati Hospital	Surat Railway station	19,781	20,373	25,920
13	Surat Railway station	Central Warehouse	20,677	20,373	25,920
14	Central Warehouse	Labheshwar Chowk	19,743	20,373	25,920
15	Labheshwar Chowk	Kapodra	18,153	20,373	25,920
16	Kapodra	Shri Swaminarayan Mandir Kalakunj	13,440	20,373	25,920
17	Shri Swaminarayan Mandir Kalakunj	Varchha Chopati Garden	9,548	20,373	25,920
18	Varchha Chopati Garden	Nature Park	7,879	20,373	25,920
19	Nature Park	Sarthana	4,507	20,373	25,920



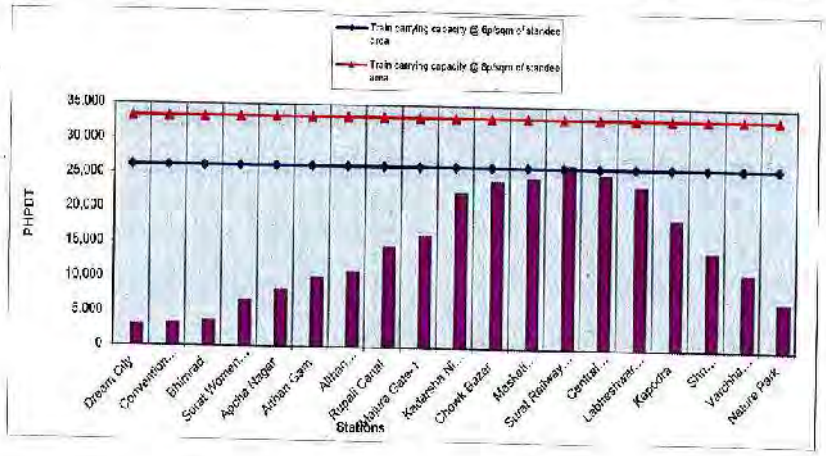
Attachment - I/D1

PHPDT Demand and Capacity Chart

Corridor- I (Dream City- Sarthana)

Year:	2046
No. of cars per train	3
Passenger Capacity @ 6 persons/sqm of a 3-Car Train:	764
Passenger Capacity @ 8 persons/sqm of a 3-Car Train:	972
Headway (min)	1.75

S.N	FROM	TO	Traffic Demand in PHPDT	Train carrying capacity @ 6p/sqm of standee area	Train carrying capacity @ 8p/sqm of standee area
1	Dream City	Convention Centre	3,095	26,194	33,326
2	Convention Centre	Bhimrad	3,272	26,194	33,326
3	Bhimrad	Surat Women ITI	3,646	26,194	33,326
4	Surat Women ITI	Apcha Nagar	6,598	26,194	33,326
5	Apcha Nagar	Althan Gam	8,251	26,194	33,326
6	Althan Gam	Althan Tenament	10,056	26,194	33,326
7	Althan Tenament	Rupali Canal	10,903	26,194	33,326
8	Rupali Canal	Majura Gate-1	14,599	26,194	33,326
9	Majura Gate-1	SMC Shopping Complex	16,238	26,194	33,326
10	Kadarsha Ni Nal	Chowk Bazar	22,530	26,194	33,326
11	Chowk Bazar	Maskati Hospital	24,247	26,194	33,326
12	Maskati Hospital	Surat Railway station	24,763	26,194	33,326
13	Surat Railway station	Central Warehouse	26,587	26,194	33,326
14	Central Warehouse	Labheshwar Chowk	25,236	26,194	33,326
15	Labheshwar Chowk	Kapodra	23,681	26,194	33,326
16	Kapodra	Shri Swaminarayan Mandir Kalakuni	18,837	26,194	33,326
17	Shri Swaminarayan Mandir Kalakuni	Varchha Chopati Garden	14,145	26,194	33,326
18	Varchha Chopati Garden	Nature Park	11,145	26,194	33,326
19	Nature Park	Sarthana	6,976	26,194	33,326



Attachment- II

TABLE 1.1 A
Hourly Train Operation Plan
Dream City to Sarthana (Corridor-I)
Year- 2021

6.5 min Headway

Time of Day	Headway in Minutes	No. of Trains per day	
		UP	DN
5 to 6	32	2	2
6 to 7	24	3	3
7 to 8	12	5	5
8 to 9	**6.5**	**10**	**9**
9 to 10	**6.5**	**10**	**9**
10 to 11	**6.5**	**10**	**9**
11 to12	12	5	5
12 to 13	24	3	3
13 to 14	32	2	2
14 to 15	32	2	2
15 to 16	24	2	2
16 to 17	12	5	5
17 to 18	**6.5**	**9**	**10**
18 to 19	**6.5**	**9**	**10**
19 to 20	**6.5**	**9**	**10**
20 to 21	12	5	5
21 to 22	24	2	2
22 to 23	32	1	1
23 to 24	40	1	1
Total No. of train trips per direction per day		95	95

TABLE 1.2 A
Hourly Train Operation Plan
Dream City to Sarthana (Corridor-I)

Year- 2026

3.5 min Headway

Time of Day	Headway in Minutes	No. of Trains per day	
		UP	DN
5 to 6	24	3	3
6 to 7	20	3	3
7 to 8	10	6	6
8 to 9	**3.5**	**18**	**17**
9 to 10	**3.5**	**18**	**17**
10 to 11	**3.5**	**18**	**17**
11 to12	10	6	6
12 to 13	20	3	3
13 to 14	24	3	3
14 to 15	24	3	3
15 to 16	20	3	3
16 to 17	10	6	6
17 to 18	**3.5**	**17**	**18**
18 to 19	**3.5**	**17**	**18**
19 to 20	**3.5**	**17**	**18**
20 to 21	10	6	6
21 to 22	20	3	3
22 to 23	24	3	3
23 to 24	30	2	2
Total No. of train trips per direction per day		155	155

TABLE 1.3 A
Hourly Train Operation Plan
Dream City to Sarthana (Corridor-I)

Year- 2036

2.25 min Headway

Time of Day	Headway in Minutes	No. of Trains per day	
		UP	DN
5 to 6	24	3	3
6 to 7	20	3	3
7 to 8	10	6	6
8 to 9	**2.25**	**27**	**26**
9 to 10	**2.25**	**27**	**26**
10 to 11	**2.25**	**27**	**26**
11 to12	10	6	6
12 to 13	20	3	3
13 to 14	24	3	3
14 to 15	24	3	3
15 to 16	20	3	3
16 to 17	10	6	6
17 to 18	**2.25**	**26**	**27**
18 to 19	**2.25**	**26**	**27**
19 to 20	**2.25**	**26**	**27**
20 to 21	10	6	6
21 to 22	20	3	3
22 to 23	24	3	3
23 to 24	30	2	2
Total No. of train trips per direction per day		209	209

TABLE 1.4 A
Hourly Train Operation Plan
Dream City to Sarthana (Corridor-I)

Year- 2046

1.75 min Headway

Time of Day	Headway in Minutes	No. of Trains per day	
		UP	DN
5 to 6	24	3	3
6 to 7	20	3	3
7 to 8	10	6	6
8 to 9	**1.75**	**35**	**34**
9 to 10	**1.75**	**35**	**34**
10 to 11	**1.75**	**35**	**34**
11 to12	10	6	6
12 to 13	20	3	3
13 to 14	24	3	3
14 to 15	24	3	3
15 to 16	20	3	3
16 to 17	10	6	6
17 to 18	**1.75**	**34**	**35**
18 to 19	**1.75**	**34**	**35**
19 to 20	**1.75**	**34**	**35**
20 to 21	10	6	6
21 to 22	20	3	3
22 to 23	24	3	3
23 to 24	30	2	2
Total No. of train trips per direction per day		257	257

Attachment III

TABLE 2.1
Dream City to Sarthana (Corridor-I)
Directional split of PHPDT for the year 2021

S.No	From Station	To Station	Peak hour Load	Directional Split to Dream City	Directional Split to Sarthana
1	Dream City	Convention Centre	119	50%	50%
2	Convention Centre	Bhimrad	145	50%	50%
3	Bhimrad	Surat Women ITI	169	50%	50%
4	Surat Women ITI	Apcha Nagar	1,649	50%	50%
5	Apcha Nagar	Althan Gam	2,158	50%	50%
6	Althan Gam	Althan Tenament	2,645	50%	50%
7	Althan Tenament	Rupali Canal	2,945	50%	50%
8	Rupali Canal	Majura Gate-1	4,283	50%	50%
9	Majura Gate-1	Kadarsha Ni Nal	4,659	50%	50%
10	Kadarsha Ni Nal	Chowk Bazar	6,420	50%	50%
11	Chowk Bazar	Maskati Hospital	6,609	50%	50%
12	Maskati Hospital	Surat Railway station	**6,992**	50%	50%
13	Surat Railway station	Central Warehouse	6,217	50%	50%
14	Central Warehouse	Labheshwar Chowk	5,594	50%	50%
15	Labheshwar Chowk	Kapodra	4,280	50%	50%
16	Kapodra	Shri Swaminarayan Mandir Kalakunj	2,626	50%	50%
17	Shri Swaminarayan Mandir Kalakunj	Varchha Chopati Garden	1,684	50%	50%
18	Varchha Chopati Garden	Nature Park	1,242	50%	50%
19	Nature Park	Sarthana	667	50%	50%

TABLE 2.1
Dream City to Sarthana (Corridor-I)
Directional split of PHPDT for the year 2026

S.No	From Station	To Station	Peak hour Load	Directional Split to Dream City	Directional Split to Sarthana
1	Dream City	Convention Centre	615	50%	50%
2	Convention Centre	Bhimrad	656	50%	50%
3	Bhimrad	Surat Women ITI	745	50%	50%
4	Surat Women ITI	Apcha Nagar	2,807	50%	50%
5	Apcha Nagar	Althan Gam	3,553	50%	50%
6	Althan Gam	Althan Tenament	4,447	50%	50%
7	Althan Tenament	Rupali Canal	4,918	50%	50%
8	Rupali Canal	Majura Gate-1	7,515	50%	50%
9	Majura Gate-1	Kadarsha Ni Nal	8,380	50%	50%
10	Kadarsha Ni Nal	Chowk Bazar	12,525	50%	50%
11	Chowk Bazar	Maskati Hospital	13,035	50%	50%
12	Maskati Hospital	Surat Railway station	13,495	50%	50%
13	Surat Railway station	Central Warehouse	**13,536**	50%	50%
14	Central Warehouse	Labheshwar Chowk	12,770	50%	50%
15	Labheshwar Chowk	Kapodra	11,089	50%	50%
16	Kapodra	Shri Swaminarayan Mandir Kalakunj	8,060	50%	50%
17	Shri Swaminarayan Mandir Kalakunj	Varchha Chopati Garden	4,550	50%	50%
18	Varchha Chopati Garden	Nature Park	3,455	50%	50%
19	Nature Park	Sarthana	1,873	50%	50%

TABLE 2.1
Dream City to Sarthana (Corridor-I)
Directional split of PHPDT for the year 2036

S.No	From Station	To Station	Peak hour Load	Directional Split to Dream City	Directional Split to Sarthana
1	Dream City	Convention Centre	2,150	50%	50%
2	Convention Centre	Bhimrad	2,277	50%	50%
3	Bhimrad	Surat Women ITI	2,538	50%	50%
4	Surat Women ITI	Apcha Nagar	5,456	50%	50%
5	Apcha Nagar	Althan Gam	6,844	50%	50%
6	Althan Gam	Althan Tenament	8,101	50%	50%
7	Althan Tenament	Rupali Canal	8,963	50%	50%
8	Rupali Canal	Majura Gate-1	12,165	50%	50%
9	Majura Gate-1	Kadarsha Ni Nal	13,524	50%	50%
10	Kadarsha Ni Nal	Chowk Bazar	20,062	50%	50%
11	Chowk Bazar	Maskati Hospital	20,856	50%	50%
12	Maskati Hospital	Surat Railway station	19,781	50%	50%
13	Surat Railway station	Central Warehouse	20,677	50%	50%
14	Central Warehouse	Labheshwar Chowk	19,743	50%	50%
15	Labheshwar Chowk	Kapodra	18,153	50%	50%
16	Kapodra	Shri Swaminarayan Mandir Kalakunj	13,440	50%	50%
17	Shri Swaminarayan Mandir Kalakunj	Varchha Chopati Garden	9,548	50%	50%
18	Varchha Chopati Garden	Nature Park	7,879	50%	50%
19	Nature Park	Sarthana	4,507	50%	50%

TABLE 2.1
Dream City to Sarthana (Corridor-I)
Directional split of PHPDT for the year 2046

S.No	From Station	To Station	Peak hour Load	Directional Split to Dream City	Directional Split to Sarthana
1	Dream City	Convention Centre	3,095	50%	50%
2	Convention Centre	Bhimrad	3,272	50%	50%
3	Bhimrad	Surat Women ITI	3,646	50%	50%
4	Surat Women ITI	Apcha Nagar	6,598	50%	50%
5	Apcha Nagar	Althan Gam	8,251	50%	50%
6	Althan Gam	Althan Tenament	10,056	50%	50%
7	Althan Tenament	Rupali Canal	10,903	50%	50%
8	Rupali Canal	Majura Gate-1	14,599	50%	50%
9	Majura Gate-1	Kadarsha Ni Nal	16,238	50%	50%
10	Kadarsha Ni Nal	Chowk Bazar	22,530	50%	50%
11	Chowk Bazar	Maskati Hospital	24,247	50%	50%
12	Maskati Hospital	Surat Railway station	24,763	50%	50%
13	Surat Railway station	Central Warehouse	26,587	50%	50%
14	Central Warehouse	Labheshwar Chowk	25,236	50%	50%
15	Labheshwar Chowk	Kapodra	23,681	50%	50%
16	Kapodra	Shri Swaminarayan Mandir Kalakunj	18,837	50%	50%
17	Shri Swaminarayan Mandir Kalakunj	Varchha Chopati Garden	14,145	50%	50%
18	Varchha Chopati Garden	Nature Park	11,145	50%	50%
19	Nature Park	Sarthana	6,976	50%	50%

Attachment IV

TABLE 3.1
Dream City to Sarthana, Corridor-I
Vehicle Kilometer

Year	2021	2026	2036	2046
Section Length	20.71	20.71	20.71	20.71
No of cars per train	3	3	3	3
No of working Days in a year	340	340	340	340
Number of Trains per day each Way	95	155	209	257
Daily Train -KM	3935	6421	8657	10645
Annual Train - KM (10^5)	13.38	21.83	29.43	36.19
Annual Vehicle - KM (10^5)	40.14	65.49	88.30	108.58

Attachment V

RAKE REQUIREMENT

Surat Metro Corridor-I(Dream City-Sarthana) 3 car option

Year 2021

S. No.	Section		Length (km)	Gauge	Schedule speed (kmph)	Year	Headway (min)	Run time (min)	Turn round time (min)	Any other time to be considered* (min)	Total round time+any other time	Total round trip time (min)	Rake Requirement					Total cars
	From	To											Bare	Bare	Traffic Reserve	R&M	Total (a)	
1	Dream City	Sarthana	20.710	SG	33	2021	5.50	37.65	3	0	6	81.30	12.51	13	1	1	15	45

Year 2026

S. No.	Section		Length (km)	Gauge	Schedule speed (kmph)	Year	Headway (min)	Run time (min)	Turn round time (min)	Any other time to be considered* (min)	Total round time+any other time	Total round trip time (min)	Rake Requirement					Total cars
	From	To											Bare	Bare	Traffic Reserve	R&M	Total (a)	
1	Dream City	Sarthana	20.710	SG	33	2026	3.50	37.65	3	0	6	81.30	23.23	23	1	2	26	78

Year 2036

S. No.	Section		Length (km)	Gauge	Schedule speed (kmph)	Year	Headway (min)	Run time (min)	Turn round time (min)	Any other time to be considered* (min)	Total round time+any other time	Total round trip time (min)	Rake Requirement					Total cars
	From	To											Bare	Bare	Traffic Reserve	R&M	Total (a)	
1	Dream City	Sarthana	20.710	SG	33	2036	2.25	37.65	3	0	6	81.30	36.13	36	1	4	41	123

Year 2046

S. No.	Section		Length (km)	Gauge	Schedule speed (kmph)	Year	Headway (min)	Run time (min)	Turn round time (min)	Any other time to be considered* (min)	Total round time+any other time	Total round trip time (min)	Rake Requirement					Total cars
	From	To											Bare	Bare	Traffic Reserve	R&M	Total (a)	
1	Dream City	Sarthana	20.710	SG	33	2046	1.75	37.65	3	0	6	81.30	46.46	46	1	5	52	156

* Passenger capacity @ 6p/sqm for a train of 3 car configuration 754.00

R&M has been considered as 10% of (Bare+ Traffic reserve)

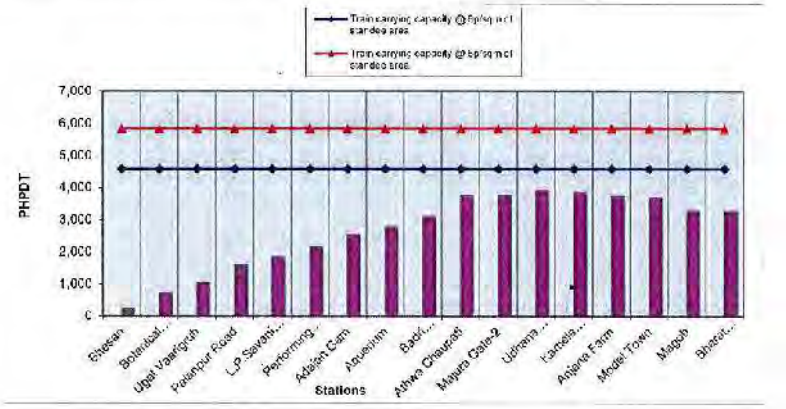
Attachment - IA2

PHPDT Demand and Capacity Chart

Corridor-II (Bheshan-Saroli)

Year:	2021
No. of cars per train	3
Passenger Capacity @ 6 persons/sqm of a 3-Car Train:	764
Passenger Capacity @ 8 persons/sqm of a 3-Car Train:	972
Headway (min)	10.00

S.N	FROM	TO	Traffic Demand in PHPDT	Train carrying capacity @ 6p/sqm of standee area	Train carrying capacity @ 8p/sqm of standee area
1	Bhesan	Botanical Garden	245	4,584	5,832
2	Botanical Garden	Ugat Vaarigruh	737	4,584	5,832
3	Ugat Vaarigruh	Palanpur Road	1,003	4,584	5,832
4	Palanpur Road	L.P.Savani School	1,600	4,584	5,832
5	L.P.Savani School	Performing Art Centre	1,866	4,584	5,832
6	Performing Art Centre	Adajan Gam	2,151	4,584	5,832
7	Adajan Gam	Aquerium	2,545	4,584	5,832
8	Aquerium	Badri Narayan Temple	2,794	4,584	5,832
9	Badri Narayan Temple	Athwa Chaupati	3,211	4,584	5,832
10	Athwa Chaupati	Majura Gate-2	3,750	4,584	5,832
11	Majura Gate-2	Udhana Darwaja	3,763	4,584	5,832
12	Udhana Darwaja	Kamela Darwaja	3,918	4,584	5,832
13	Kamela Darwaja	Anjana Farm	3,848	4,584	5,832
14	Anjana Farm	Model Town	3,739	4,584	5,832
15	Model Town	Magob	3,693	4,584	5,832
16	Magob	Bharat Cancer Hospital	3,265	4,584	5,832
17	Bharat Cancer Hospital	Saroli	3,265	4,584	5,832



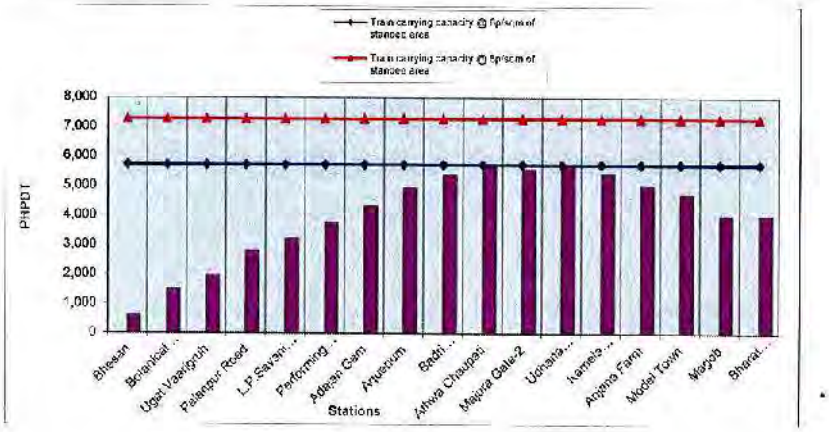
Attachment - I/B2

PHPDT Demand and Capacity Chart

Corridor-II (Bheshan-Saroli)

Year:	2026
No. of cars per train	3
Passenger Capacity @ 6 persons/sqm of a 3-Car Train:	764
Passenger Capacity @ 8 persons/sqm of a 3-Car Train:	972
Headway (min)	8.00

S.N	FROM	TO	Traffic Demand in PHPDT	Train carrying capacity @ 6p/sqm of standee area	Train carrying capacity @ 8p/sqm of standee area
1	Bhesan	Botanical Garden	629	5,730	7,290
2	Botanical Garden	Ugat Vaarigruh	1,490	5,730	7,290
3	Ugat Vaarigruh	Palanpur Road	1,970	5,730	7,290
4	Palanpur Road	L.P.Savani School	2,795	5,730	7,290
5	L.P.Savani School	Performing Art Centre	3,226	5,730	7,290
6	Performing Art Centre	Adajan Gam	3,737	5,730	7,290
7	Adajan Gam	Aquerium	4,346	5,730	7,290
8	Aquerium	Badri Narayan Temple	4,945	5,730	7,290
9	Badri Narayan Temple	Athwa Chaupati	5,379	5,730	7,290
10	Athwa Chaupati	Majura Gate-2	5,666	5,730	7,290
11	Majura Gate-2	Udhana Darwaja	5,581	5,730	7,290
12	Udhana Darwaja	Kamela Darwaja	5,698	5,730	7,290
13	Kamela Darwaja	Anjana Farm	5,415	5,730	7,290
14	Anjana Farm	Model Town	5,033	5,730	7,290
15	Model Town	Magob	4,734	5,730	7,290
16	Magob	Bharat Cancer Hospital	4,013	5,730	7,290
17	Bharat Cancer Hospital	Saroli	4,013	5,730	7,290



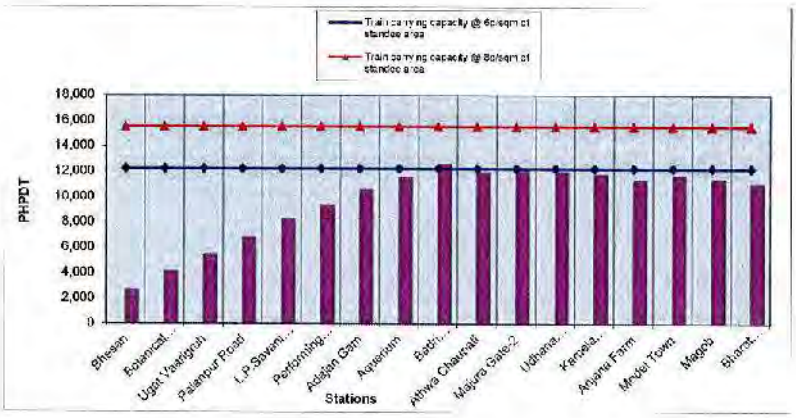
Attachment - I/C2

PHPDT Demand and Capacity Chart

Corridor-II (Bheshan-Saroli)

Year:	2036
No. of cars per train	3
Passenger Capacity @ 6 persons/sqm of a 3-Car Train:	764
Passenger Capacity @ 8 persons/sqm of a 3-Car Train:	972
Headway (min)	3.75

S.N	FROM	TO	Traffic Demand in PHPDT	Train carrying capacity @ 6p/sqm of standee area	Train carrying capacity @ 8p/sqm of standee area
1	Bhesan	Botanical Garden	2,649	12,224	15,552
2	Botanical Garden	Ugat Vaarigruh	4,150	12,224	15,552
3	Ugat Vaarigruh	Palanpur Road	5,432	12,224	15,552
4	Palanpur Road	L.P.Savani School	6,821	12,224	15,552
5	L.P.Savani School	Performing Art Centre	8,281	12,224	15,552
6	Performing Art Centre	Adajan Gam	9,309	12,224	15,552
7	Adajan Gam	Aquerium	10,544	12,224	15,552
8	Aquerium	Badri Narayan Temple	11,532	12,224	15,552
9	Badri Narayan Temple	Athwa Chaupati	12,573	12,224	15,552
10	Athwa Chaupati	Majura Gate-2	11,963	12,224	15,552
11	Majura Gate-2	Udhana Darwaja	12,098	12,224	15,552
12	Udhana Darwaja	Kamela Darwaja	11,998	12,224	15,552
13	Kamela Darwaja	Anjana Farm	11,778	12,224	15,552
14	Anjana Farm	Model Town	11,295	12,224	15,552
15	Model Town	Magob	11,690	12,224	15,552
16	Magob	Bharat Cancer Hospital	11,430	12,224	15,552
17	Bharat Cancer Hospital	Saroli	11,013	12,224	15,552



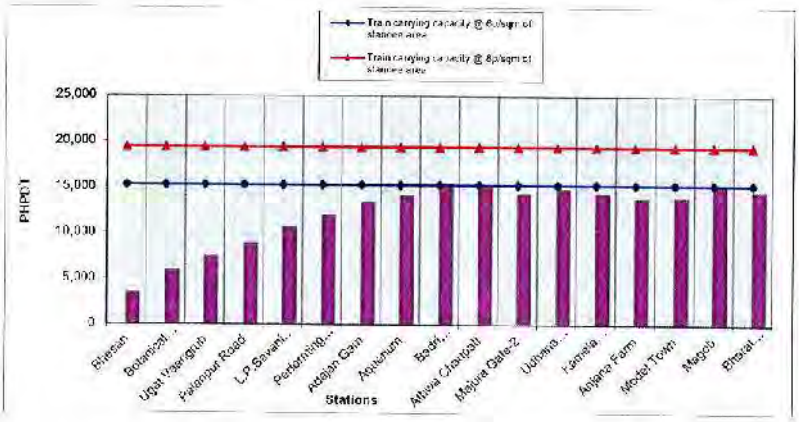
Attachment - I/D2

PHPDT Demand and Capacity Chart

Corridor-II (Bheshan-Saroli)

Year:	2046
No. of cars per train	3
Passenger Capacity @ 6 persons/sqm of a 3-Car Train:	764
Passenger Capacity @ 8 persons/sqm of a 3-Car Train:	972
Headway (min)	3.00

S.N	FROM	TO	Traffic Demand in PHPDT	Train carrying capacity @ 6p/sqm of standee area	Train carrying capacity @ 8p/sqm of standee area
1	Bhesan	Botanical Garden	3,487	15,280	19,440
2	Botanical Garden	Ugat Vaarigruh	5,883	15,280	19,440
3	Ugat Vaarigruh	Palanpur Road	7,418	15,280	19,440
4	Palanpur Road	L.P.Savani School	8,825	15,280	19,440
5	L.P Savani School	Performing Art Centre	10,714	15,280	19,440
6	Performing Art Centre	Adajan Gam	11,970	15,280	19,440
7	Adajan Gam	Aquerium	13,396	15,280	19,440
8	Aquerium	Badri Narayan Temple	14,112	15,280	19,440
9	Badri Narayan Temple	Athwa Chaupati	15,448	15,280	19,440
10	Athwa Chaupati	Majura Gate-2	15,096	15,280	19,440
11	Majura Gate-2	Udhana Darwaja	14,356	15,280	19,440
12	Udhana Darwaja	Kamela Darwaja	14,842	15,280	19,440
13	Kamela Darwaja	Anjana Farm	14,381	15,280	19,440
14	Anjana Farm	Model Town	13,790	15,280	19,440
15	Model Town	Magob	13,857	15,280	19,440
16	Magob	Bharat Cancer Hospital	15,175	15,280	19,440
17	Bharat Cancer Hospital	Saroli	14,584	15,280	19,440



Attachment- II

TABLE 1.1 B
Hourly Train Operation Plan
Bhesan to Saroli (Corridor-II)
Year- 2021

10 min Headway

Time of Day	Headway in Minutes	No. of Trains per day	
		UP	DN
5 to 6	32	2	2
6 to 7	24	3	3
7 to 8	12	5	5
8 to 9	**10**	**6**	**6**
9 to 10	**10**	**6**	**6**
10 to 11	**10**	**6**	**6**
11 to12	12	5	5
12 to 13	24	3	3
13 to 14	32	2	2
14 to 15	32	2	2
15 to 16	24	2	2
16 to 17	12	5	5
17 to 18	**10**	**6**	**6**
18 to 19	**10**	**6**	**6**
19 to 20	**10**	**6**	**6**
20 to 21	12	5	5
21 to 22	24	2	2
22 to 23	32	1	1
23 to 24	40	1	1
Total No. of train trips per direction per day		**74**	**74**

TABLE 1.2 B
Hourly Train Operation Plan
Bhesan to Saroli (Corridor-II)
Year- 2026

8 min Headway

Time of Day	Headway in Minutes	No. of Trains per day	
		UP	DN
5 to 6	24	3	3
6 to 7	20	3	3
7 to 8	10	6	6
8 to 9	**8**	**8**	**7**
9 to 10	**8**	**8**	**7**
10 to 11	**8**	**8**	**7**
11 to12	10	6	6
12 to 13	20	3	3
13 to 14	24	3	3
14 to 15	24	3	3
15 to 16	20	3	3
16 to 17	10	6	6
17 to 18	**8**	**7**	**8**
18 to 19	**8**	**7**	**8**
19 to 20	**8**	**7**	**8**
20 to 21	10	6	6
21 to 22	20	3	3
22 to 23	24	3	3
23 to 24	30	2	2
Total No. of train - trips per direction per day		**95**	**95**

Attachment III

TABLE 2.2
Bhesan to Saroli (Corridor-II)
Directional split of PHPDT for the year 2021

S.No	From Station	To Station	Peak hour Load	Directional Split to Bhesan	Directional Split to Saroli
1	Bhesan	Botanical Garden	245	50%	50%
2	Botanical Garden	Ugat Vaarigruh	737	50%	50%
3	Ugat Vaarigruh	Palanpur Road	1,063	50%	50%
4	Palanpur Road	L.P.Savani School	1,600	50%	50%
5	L.P.Savani School	Performing Art Centre	1,866	50%	50%
6	Performing Art Centre	Adajan Gam	2,151	50%	50%
7	Adajan Gam	Aquerium	2,545	50%	50%
8	Aquerium	Badri Narayan Temple	2,794	50%	50%
9	Badri Narayan Temple	Athwa Chaupati	3,111	50%	50%
10	Athwa Chaupati	Majura Gate-2	3,750	50%	50%
11	Majura Gate-2	Udhana Darwaja	3,763	50%	50%
12	Udhana Darwaja	Kamela Darwaja	**3,918**	50%	50%
13	Kamela Darwaja	Anjana Farm	3,848	50%	50%
14	Anjana Farm	Model Town	3,739	50%	50%
15	Model Town	Magob	3,693	50%	50%
16	Magob	Bharat Cancer Hospital	3,265	50%	50%
17	Bharat Cancer Hospital	Saroli	3,265	50%	50%

TABLE 2.2
Bhesan to Saroli (Corridor-II)
Directional split of PHPDT for the year 2026

S.No	From Station	To Station	Peak hour Load	Directional Split to Bhesan	Directional Split to Saroli
1	Bhesan	Botanical Garden	629	50%	50%
2	Botanical Garden	Ugat Vaarigruh	1,490	50%	50%
3	Ugat Vaarigruh	Palanpur Road	1,970	50%	50%
4	Palanpur Road	L.P.Savani School	2,795	50%	50%
5	L.P.Savani School	Performing Art Centre	3,226	50%	50%
6	Performing Art Centre	Adajan Gam	3,737	50%	50%
7	Adajan Gam	Aquerium	4,346	50%	50%
8	Aquerium	Badri Narayan Temple	4,945	50%	50%
9	Badri Narayan Temple	Athwa Chaupati	5,379	50%	50%
10	Athwa Chaupati	Majura Gate-2	5,666	50%	50%
11	Majura Gate-2	Udhana Darwaja	5,581	50%	50%
12	Udhana Darwaja	Kamela Darwaja	**5,698**	50%	50%
13	Kamela Darwaja	Anjana Farm	5,415	50%	50%
14	Anjana Farm	Model Town	5,033	50%	50%
15	Model Town	Magob	4,734	50%	50%
16	Magob	Bharat Cancer Hospital	4,013	50%	50%
17	Bharat Cancer Hospital	Saroli	4,013	50%	50%

TABLE 2.2
Bhesan to Saroli (Corridor-II)
Directional split of PHPDT for the year 2036

S.No	From Station	To Station	Peak hour Load	Directional Split to Bhesan	Directional Split to Saroli
1	Bhesan	Botanical Garden	2,649	50%	50%
2	Botanical Garden	Ugat Vaarigruh	4,150	50%	50%
3	Ugat Vaarigruh	Palanpur Road	5,432	50%	50%
4	Palanpur Road	L.P.Savani School	6,821	50%	50%
5	L.P.Savani School	Performing Art Centre	8,281	50%	50%
6	Performing Art Centre	Adajan Gam	9,309	50%	50%
7	Adajan Gam	Aquerium	10,544	50%	50%
8	Aquerium	Badri Narayan Temple	11,532	50%	50%
9	Badri Narayan Temple	Athwa Chaupati	**12,573**	50%	50%
10	Athwa Chaupati	Majura Gate-2	11,963	50%	50%
11	Majura Gate-2	Udhana Darwaja	12,098	50%	50%
12	Udhana Darwaja	Kamela Darwaja	11,998	50%	50%
13	Kamela Darwaja	Anjana Farm	11,778	50%	50%
14	Anjana Farm	Model Town	11,295	50%	50%
15	Model Town	Magob	11,690	50%	50%
16	Magob	Bharat Cancer Hospital	11,430	50%	50%
17	Bharat Cancer Hospital	Saroli	11,013	50%	50%

TABLE 2.2
Bhesan to Saroli (Corridor-II)
Directional split of PHPDT for the year 2046

S.No	From Station	To Station	Peak hour Load	Directional Split to Bhesan	Directional Split to Saroli
1	Bhesan	Botanical Garden	3,487	50%	50%
2	Botanical Garden	Ugat Vaarigruh	5,883	50%	50%
3	Ugat Vaarigruh	Palanpur Road	7,418	50%	50%
4	Palanpur Road	L.P.Savani School	8,825	50%	50%
5	L.P.Savani School	Performing Art Centre	10,714	50%	50%
6	Performing Art Centre	Adajan Gam	11,970	50%	50%
7	Adajan Gam	Aquerium	13,396	50%	50%
8	Aquerium	Badri Narayan Temple	14,112	50%	50%
9	Badri Narayan Temple	Athwa Chaupati	**15,448**	50%	50%
10	Athwa Chaupati	Majura Gate-2	15,096	50%	50%
11	Majura Gate-2	Udhana Darwaja	14,356	50%	50%
12	Udhana Darwaja	Kamela Darwaja	14,842	50%	50%
13	Kamela Darwaja	Anjana Farm	14,381	50%	50%
14	Anjana Farm	Model Town	13,790	50%	50%
15	Model Town	Magob	13,857	50%	50%
16	Magob	Bharat Cancer Hospital	15,175	50%	50%
17	Bharat Cancer Hospital	Saroli	14,584	50%	50%

Attachment IV

TABLE 3.2
Bhesan to Saroli, Corridor-II
Vehicle Kilometer

Year	2021	2026	2036	2046
Section Length	17.34	17.34	17.34	17.34
No of cars per train	3	3	3	3
No of working Days in a year	340	340	340	340
Number of Trains per day each Way	74	95	146	170
Daily Train -KM	2567	3295	5064	5896
Annual Train - KM (10^5)	8.73	11.20	17.22	20.05
Annual Vehicle - KM (10^6)	26.18	33.61	51.65	60.14

Attachment V

RAKE REQUIREMENT

Surat Metro Corridor-II(Bhesan-Saroli) 3 car option

Year 2021

S. No.	Section		Length (km)	Gauge	Schedule speed (kmph)	Year	Headway (min)	Run time (min)	Turn round time (min)	Any other time to be considered* (min)	Total round time+any other time	Total round trip time (min)	Rake Requirement					Total cars
	From	To											Bare	Bare	Traffic Reserve	R&M	Total (a)	
1	Bhesan	Saroli	17.340	SG	33	2021	10.00	31.53	3	0	6	69.06	6.91	7	1	1	9	27

Year 2026

S. No.	Section		Length (km)	Gauge	Schedule speed (kmph)	Year	Headway (min)	Run time (min)	Turn round time (min)	Any other time to be considered* (min)	Total round time+any other time	Total round trip time (min)	Rake Requirement					Total cars
	From	To											Bare	Bare	Traffic Reserve	R&M	Total (a)	
1	Bhesan	Saroli	17.340	SG	33	2026	8.00	31.53	3	0	6	69.06	8.63	9	1	1	11	33

Year 2036

S. No.	Section		Length (km)	Gauge	Schedule speed (kmph)	Year	Headway (min)	Run time (min)	Turn round time (min)	Any other time to be considered* (min)	Total round time+any other time	Total round trip time (min)	Rake Requirement					Total cars
	From	To											Bare	Bare	Traffic Reserve	R&M	Total (a)	
1	Bhesan	Saroli	17.340	SG	33	2036	3.75	31.53	3	0	6	69.06	18.42	18	1	2	21	63

Year 2046

S. No.	Section		Length (km)	Gauge	Schedule speed (kmph)	Year	Headway (min)	Run time (min)	Turn round time (min)	Any other time to be considered* (min)	Total round time+any other time	Total round trip time (min)	Rake Requirement					Total cars
	From	To											Bare	Bare	Traffic Reserve	R&M	Total (a)	
1	Bhesan	Saroli	17.340	SG	33	2046	3.00	31.53	3	0	6	69.06	23.02	23	1	2	26	78

Passenger capacity @ 6p/sqm for a train of 3 car configuration 764.00

R&M has been considered as 10% of (Bare+ Traffic reserve)

Attachment-VI

Salient Features of 2.9 m wide SG Rolling Stock for MRTS for Corridor – I & II

S.No.	Parameter	Standard Gauge 2.9m wide stock
1	**Gauge (Nominal)**	1435mm Standard Gauge
2	**Traction system**	
2.1	Voltage	750 V DC
2.2	Method of current collection	Third Rail Bottom Current Collection System
3	**Train composition**	
3.2	For 3 car train	DMC+TC+DMC
4	**Coach Body**	Stainless Steel/Aluminum
5	**Coach Dimensions**	
5.1	Height	3.9m
5.2	Width	2.9m
5.3	Length over body (approx.)	
	- Driving Motor Car (DMC)	21.81m
	- Trailer Car (TC)	21.34m
	Maximum length of coach over couplers/buffers:	*22 to 23m (depending upon Kinematic Envelop and SOD)*
5.4	Locked down Panto height	4048mm
5.5	Floor height	1100mm
6	**Designed - Passenger Loading**	
6.1	Design of Propulsion equipment	8 Passenger/ m^2
6.2	Design of Mechanical systems	10 Passenger/ m^2
7	**Carrying capacity- @ 6 standees/sqm**	
7.1	Coach carrying capacity	
	DMC	247 (seating - 43 ; standing - 204)
	TC	270 (seating - 50 ; standing - 220)
7.2	Train Carrying capacity	
	3 car train	764 (seating - 136 ; standing - 628)

8	**Weight (Tonnes)**	
8.1	Tare weight (maximum)	* The maximum tare weight for 3-car unit shall be 120 T.
	DMC	40
	TC	40
8.2	Passenger Weight in tons	@ 0.065 T per passenger
	DMC	16.05(@ 6 persons per sqm of standee area)
	TC	17.55(@ 6 persons per sqm of standee area)
8.3	Gross weight in tons	
	DMC	56.05
	TC	57.55
9	**Axle load(T)(@ 8 persons per sqm of standee area)**	**16**
		System should be designed for 16T axle load
10	**Maximum Train Length (approx.) (3 car configuration)**	67.8m
11	**Speed**	
10.1	Maximum Design Speed	90kmph
10.2	Maximum Operating Speed	80kmph
12	**Wheel Profile**	UIC 510-2/RDSO Profile
13	**Traction Motors Ventilation**	Self
14	**Acceleration on level tangent track**	1.0m/sec^2 @ AW3(AW3 means seating +8 passengers per sqm of standing area) 1.2 m/sec^2 @ AW2(AW2 means seating +6 passengers per sqm of standing area)
15	**Deacceleration on level tangent track**	1.1 m/sec^2 @ AW3 1.1 m/sec^2 @ AW2 (>1.3 m/sec^2 during emergency)
16	**Type of Bogie**	Fabricated
17	**Secondary Suspension springs**	Air

18	**Brakes**	- An electro-pneumatic (EP) service friction brake - An electric regenerative service brake - Provision of smooth and continuous blending of EP and regenerative braking - A fail safe, pneumatic friction emergency brake - A spring applied air-release parking brake - Tread brakes
19	**Coupler**	
	Between cars of same Unit	Semi-permanent couplers
	Driving Cab end of cars (DMC)	Automatic coupler with mechanical & pneumatic coupling but without electrical coupling head. Jumper cable may be provided in cab for communication between train operators during train to train coupling.
20	**Detrainment Door**	Side evacuation through saloon doors
21	**Type of Doors**	Bi-parting external sliding type
22	**Lighting**	LED
23	**Cooling**	
23.1	CI & SIV	Self/Forced
23.3	TM	Self-ventilated
24	**Control System**	Train based Monitor & Control System (TCMS)
25	**Traction Motors**	3 phase VVVF controlled
26	**Temperature Rise Limits**	
26.1	Traction Motor	Temperature Index minus 70 deg C
26.2	CI & SIV	10 deg C temperature margin for Junction temperature
27	**HVAC**	- Cooling& Heating - Automatic controlling of interior temperature throughout the passenger area at 25°C all the times under varying ambient conditions up to full load.
28	**PA/PIS**	Required
29	**Passenger Surviellance (CCTV)**	Required
30	**Battery**	Ni-Cd
31	**Headlight type**	LED
32	**Train Operation**	UTO(GOA4)

8	**Weight (Tonnes)**	
8.1	Tare weight (maximum)	* The maximum tare weight for 3-car unit shall be 120 T
	DMC	40
	TC	40
8.2	Passenger Weight in tons	@ 0.065 T per passenger
	DMC	16.05 (@ 6 persons per sqm of standee area)
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15	**Deacceleration on level tangent track**	1.1 m/sec^2 @ AW3 1.1 m/sec^2 @ AW2 (>1.3 m/sec^2 during emergency)
16	**Type of Bogie**	Fabricated
17	**Secondary Suspension springs**	Air

18	**Brakes**	- An electro-pneumatic (EP) service friction brake - An electric regenerative service brake - Provision of smooth and continuous blending of EP and regenerative braking - A fail safe, pneumatic friction emergency brake - A spring applied air-release parking brake - Tread brakes
19	**Coupler**	
	Between cars of same Unit	Semi-permanent couplers
	Driving Cab end of cars (DMC)	Automatic coupler with mechanical & pneumatic coupling but without electrical coupling head
20	**Detrainment Door**	Front
21	**Type of Doors**	Sliding
22	**Lighting**	LED
23	**Cooling**	
23.1	Transformer	Forced
23.2	CI & SIV	Self/Forced
23.3	TM	Self-ventilated
24	**Control System**	Train based Monitor & Control System (TCMS)
25	**Traction Motors**	3 phase VVVF controlled
26	**Temperature Rise Limits**	
26.1	Traction Motor	Temperature Index minus 70 deg C
26.2	CI & SIV	10 deg C temperature margin for Junction temperature
26.3	Transformer	IEC specified limit minus 20 deg C
27	**HVAC**	- Cooling & Heating - Automatic controlling of interior temperature throughout the passenger area at 25°C all the times under varying ambient conditions up to full load.
28	**PA/PIS**	Required
29	**Passenger Surviellance (CCTV)**	Required
30	**Battery**	Ni-Cd
31	**Headlight type**	LED
32	**Train Operation**	UTO(GoA4)

CHAPTER – 8

TRAIN MAINTENANCE DEPOTS

DEPOT FOR CORRIDOR-I

8.1 Corridor: Surat Metro Network Corridor-I comprises of following corridor:

Corridor	Route length(Km)
Dream City to Sarthana	20.71

8.2 Depot-cum- Workshop

8.2.1 It is proposed to establish one depot- cum- workshop with following functions:

(i) Major overhauls of all the trains.

(ii) All minor schedules and repairs.

(iii) Lifting for replacement of heavy equipment and testing thereafter.

(iv) Repair of heavy equipments

8.2.2 The Depot planning is based on following assumptions:

(i) Enough space should be available for establishment of a Depot- cum - workshop.

(ii) All inspection lines, workshop lines, stabling lines are designed to accommodate two train sets of 3-Car each and space earmarked for future provision.

(iii) All Stabling lines are designed to accommodate two trains of3- Car each.

(iv) All stabling lines are planned in the proposed depot-cum-workshop assuming adequate space availability. In case of space constraints, if any, stabling facilities may need to be created at terminal stations or elsewhere to cater to the required stability facilities.

(v) In case of space constraint for depot two storeyed Stabling lines can also be planned.

In broad terms, based on the planned Rolling Stock requirements, this chapter covers conceptual design on following aspects and will work as a guide for detailed design later:

Layout of Stabling-shed, Inspection-shed, minor repairs and heavy repair over hauling workshop and cleaning of Rolling Stock.
Operational and functional safety requirements.
Ancillary buildings for other maintenance facilities.
Electrical &Mechanical Services, power supply and distribution system.
Water Supplies, Drainage & Sewerage.

8.3 MAINTENANCE PHILOSOPHY

Monitoring of the performance of all key Rolling Stock equipment by suitable advanced condition monitoring techniques available. The concept is to evolve the need based maintenance regime, which can be suitably configured in the form of schedules like daily check, “A” checks, “B” type checks, “IOH” and “POH”.

Labour intensive procedures are kept to the minimum. Automation with state of the art machinery to ensure quality with reliability.

Increase in the periodic maintenance intervals with predictive maintenance based on condition monitoring.

Multi skilling of the Maintenance staff to ensure quality and productivity in their performance.

Periodic review of maintenance practices to update replacement cycle of critical components based on experience.

Energy conservation is given due attention

8.4 ROLLINGSTOCK MAINTENANCENEEDS

8.4.1 Maintenance Schedule

The following maintenance schedule has been envisaged for conceptual design of depots assuming approx.264 kms running per train per day, taking in consideration the passenger load of 2021,2026, 2036 & 2046 respectively.

Table-8.1

Type of Schedule	Interval	Work Content	Locations
Daily	Daily	Check on the train condition and function at every daily service completion. Interval cleaning/mopping of floor and walls with Vacuum cleaner.	Stabling Lines
"A" Service Check	5,000 Km(approx.15 days)	Detailed inspection and testing of sub- systems, under frame, replacement/topping up of oils & lubricants.	Inspection Bays
"B" Service Check	15,000Km (approx.45 days)	Detailed Inspection of 'A 'type tasks plus items at multiples of 15,000Km ('B' type tasks)	Inspection Bays
Intermediate Overhaul (IOH)	420,000cKm, (3and half Years approx.) whichever is earlier	Check and testing of all sub-assemblies (Electrical +Mechanical). Overhaul of pneumatic valves, Compressor. Condition based maintenance of sub- systems to bring them to original condition. Replacement of parts and rectification, trial run.	Workshop
Periodical Overhaul (POH)	840,000 Km, (7 Years approx.) whichever isearlier	Dismantling of all sub-assemblies, bogies suspension system, traction motor, gear, control equipment, air-conditioning units etc. Overhauling to bring them to original condition. Checking repair and replacement as necessary. Inspection and trial.	Workshop
Heavy Repairs	-	Changing of heavy item such as bogies, traction motor, wheel sets/axles, gear cases& axle boxes etc.	Workshop

The above Schedule may need slight revision based on the actual earned kilometers per train and the specific maintenance requirements of Rolling Stock finally procured.

8.4.2 Washing Needs of Rolling Stock: Cleanliness of the trains is essential.

Following schedules are recommended for Indian environment:-

Table 8.2

S.N.	Kind Inspection	Maint. Cycle	Time	Maintenance Place
1.	Outside cleaning(wet washing on automatic washing plant)	3 Days	10 mins.	Single Pass through Automatic washing plant of Depot
2.	Outside heavy Cleaning (wet washing on automatic washing plant and Front Face, Vestibule/ Buffer area. Floor, walls inside/outside of cars and roof. Manually)	30 days	2 –3 hrs.	Automatic washing plant & intensive cleaning shed

8.5 Year-wise planning of maintenance facility set up at depot cum workshop based on planned Rolling Stock requirement in TOP is tabulated below:

Traffic data are available up to year 2046 only; hence space to be earmarked for future expansion beyond 2046 year stabling inspection and workshop line.

(i) Planned rakes as per TOP:

Table-8.3

Year	No. of Rakes	No. of coaches
2021	15	45
2026	26	78
2036	41	123
2046	52	156

ii) Requirement of Stabling Lines (SBL), Inspection Lines (IBL) and Workshop Lines (WSL) in the Depot-cum -Workshop.

Stabling and Inspection lines

Table-8.4

Year	No. of Rakes	SBLs	IBLs
2021	15	6 lines x two train of 3-car	One bay of 3 lines each with two train of 3-cars.
2026	26	12 lines x two train of 3-car	No additional requirement.IBL For the year 2021 is also catering to 26 rakes up to year 2026
2036	41	16 lines x two train of 3-car	Additional one bay of 3 lines each with two train of 3- cars is required for year 2036.
2046	52	21 lines x two train of 3-car	No additional requirement. Available 2 bays of 3 lines each shall also cater to requirements upto year 2046.

Workshop lines

Table -8.5

Year	No. of Rakes	WSLs
2021	15	Two bays of 2 lines each with two trains of 3-cars is to be required for the years 2021 & 2026 and additional one bay of 2 lines each with two trains of 3-cars is to be kept for future provision from 2036, which will cater to the requirements upto year 2046.
2026	26	
2036	41	
2046	52	

8.6 Requirement of maintenance/Inspection lines for depot-cum-workshop:

Table -8.6

Schedule	Maintenance Requirement (no. of Cars)	Lines Needed
i) Year2021- Maximum no. of rake holding is (15TS x3=45 Cars)		
'A' Checks (5000km) approx. 15days	(15X3) Cars =45 Cars	1 Line x two train of 3-Cars (with Sunken Floor)
'B' Checks (15000km) approx. 45days.	(15X3) Cars=45 Cars	1 Line x two train of3-Cars(with Sunken Floor)
Unscheduled line & Adjustment lines	For minor repairs, testing and after IOH /POH adjustments	1 Line x two train of 3-Cars(with Sunken Floor)
Requirement		1 bay of 3 lines
ii) Year2026-Maximumno.of rake holding is (26TS x3= 78 Cars)		
'A' Checks (5000km) 15days	(26X 3)Cars=78 Cars	No additional requirement. IBL for the year 2021 is also catering to 26 rakes up to year 2026.
'B' Checks (15000km) 45 days	(26X 3)Cars=78 Cars	
Unscheduled line& Adjustment lines	For minor repairs, testing and after IOH /POH adjustments	
Requirement		No additional requirement. IBL for the year 2021 is also catering to 26 rakes up to year 2026.
iii)Year 2036- Maximum no. of rake holding is (41TS x3= 123 Cars)		
'A' Checks (5000 km) approx.15 days	(41X3)Cars=123 Cars	1 Line x two train of 3-Cars (with Sunken Floor)
'B' Checks (15000km) approx. 45 days.	(41X3)Cars=123 Cars	1 Line x two train of 3-Cars (with Sunken Floor)
Unscheduled line & adjustment lines	For minor repairs, testing and after IOH /POH	1 Line x two train of 3-Cars (with Sunken Floor)

Requirement		Additional 1bay of 3 lines each with two trains of cars is required for year 2036.
iv) Year 2046- Maximum no. of rake holding is (52TS x3= 156 Cars)		
'A' Checks (5000km) approx. 15days	(52X3) Cars = 156 Cars	No additional requirement
'B' Checks (15000km) approx. 45days.	(52X3) Cars=156 Cars	
Unscheduled line & adjustment lines	For minor repairs, testing and after IOH/POH adjustments	
Requirement		No additional requirement. Available2baysof3 lines each shall also cater to requirements up-to year 2046

8.7 Inspection requirements at depot:

Facilities for carrying out inspection activities shall be provided in the inspection bay for following Systems /Equipments of a train:

- Electronics; PA /PIS
- Mechanical components, couplers etc.
- Batteries
- Air conditioner
- Brake modules
- Bogie
- Traction Motor
- Vehicle doors, windows and internal fittings
- Power system including converter, circuit breaker etc.

These activities shall be grouped into "A" checks and "B" checks. The minor scheduled inspections ("A" checks) shall be carried out during the day off peak and night .Since "B" checks take longer time, these can not be completed in the off- peak times .Certain inspection lines will be nominated for "A" checks. For "B" checks, separate line will be nominated where the rakes may be kept for long time

8.8 Design of Depot-cum-Workshop Facilities

8.8.1Stabling lines at depot:

As per advised dimensions of the Rolling Stock, the length of 3-Car train would be Approx. 67.8 mts for the design of the stabling lines in the depot and terminal stations or elsewhere (as maybe required), following approximates lengths have been taken in consideration:

(i) Length of one 3- Car rake=67.8

(ii) Gap between two trains 3-car rakes = 10m

(iii) Free length at outer ends of two trains of 3-Car rakes (for cross pathway, Signal and Friction buffers) = 10 m each side

(iv) Total length of Stabling lines = (iii)+(i)+(ii)+(i)+(iii)=

10 +67.8+10+67.8+10=165.6 m approx 166m.

Looking to the car width of 2900 mm on SG 5 m "Track Centre" is proposed for all the stabling lines. Thus, space between stabling shall be sufficient to include1m wide path way to be constructed between tracks to provide access for internal train cleaning and under carriage inspection with provision of following facilities:

a) Each Stabling line to have water connection facility so that local cleaning, if required, is facilitated.

b) Platforms at suitable points at each end of stabling lines to enable train operators to board or de- board conveniently.

8.8.2 Inspection Bay at depot-cum-workshop:

The length of Inspection shed is computed as below:

(i) Length of one 3- Car rake=67.8

(ii)Stairs on both pit ends=2.5m

(iii) Gap between two train of 3-cars= 10

(iv) Embedded track at entry side upto IBL pit= 7m

(v) Embedded track at exit side up to IBL pit= 11.5m

(vi) Total length of Inspection lines = (iv)+(ii)+(i)+(iii)+(i)+(ii)+(v)=

7+2.5+67.8+10+67.8+2.5+11.5=169.1m approx 169m.

The width of the Inspection bay is computed as below:

(i) Centre–to- centre spacing between the lines=6.5m

(ii) Centre line of outer lines to column of Shed=4.5m

(iii) Width ofa3 line Inspection Bay=(ii)+(i)+(i)+(ii)=4.5+6.5+ 6.5+4.5= 22m

a) There shall be one inspection bay of 169m X 22m size each with provision of accommodating three inspection lines each having sunken floor and overhead roof inspection platforms at each of the line. The floor will be sunken by 1100mm. The track spacing between the adjacent IBLs shall be 6.5m.

b) Roof Inspection platforms of 1.2m width and walkways for roof inspection supported on the columns shall be provided. There would be lighting below the rail level to facilitate the under-frame inspection. Ramps of1:8 slopes,3 meter wide should be provided with sunken floor system for movement of material for the cars. Further,10m cross pathways are left at each end for movement of material by fork lifter/Leister/Hand trolley. 415V 3phase 50Hz, 230V 1phase50 Hz AC supply and Pneumatic supply shall also be made available on each inspection shed columns. Air-circulators shall be provided on each column. The inspection bay shall be provided with EOT crane of 1.5T to facilitate lifting of equipment.

Roof and walls shall be of such design that optimum natural air ventilation occurs all the time and sufficient natural light is also available. Each Inspection bay will also have arrangement close by for cleaning of HVAC filter under high pressure water jet.

8.8.3 Workshop Shed at Depot:

Requirement of workshop lines is planned as under:

Table -8.7

<table>
<tr><th>Year</th><th>IOH &POH</th><th>Major Overhauling</th><th>Unscheduled repairs</th><th>Total</th><th>Remarks</th></tr>
<tr><td>2021</td><td>1</td><td>2 line3-Car train and free space of 3-car length for storage of other equipment.</td><td>1line x 2 train of 3 Car train length.</td><td>Two bays of 2 lines each with two trains of 3-cars is to be required for</td><td>The size of workshop shall be166X21m for one working bay</td></tr>
<tr><td>2026</td><td></td><td></td><td></td><td rowspan="3">The years
2021& 2026 and additional one bay of 2 lines each with two trains of 3-cars is to be kept for future provision for year 2036 & 2046.</td><td rowspan="3">Comprising of two lines capable of accommodating one 3-Car rake with Bogie turn table facility, one lineof3- Car rake length with free space of 3-Car rake length for storage of wheel/bogie/ equipments etc.</td></tr>
<tr><td>2036</td><td></td><td rowspan="2">1line3-Car train and free space of
3-car length for storage of other equipment.</td><td rowspan="2">1line x 2 train of 3 Car train length.</td></tr>
<tr><td>2046</td><td></td></tr>
</table>

(a) Each bay shall comprise of two lines (as detailed in 'Remarks' above). Size of the workshop bay is proposed to be166m x21m. The unscheduled lifting and heavy repair line shall be fitted with jack system capable to lift the 3-Car unit simultaneously for quick change of bogie, thereby saving down time of Rolling Stock. The arrangement of jack system shall be such that lifting of any coach in train formation for replacement of bogie/ equipments is also individually possible. Space on one line shall be available for stocking of Bogies and wheels. These lines are to be provided with pits at regular intervals for inspection of under carriage and lines are to be interconnected by turn tables. Each workshop bay shall be equipped with two 15T/2Tand 5Toverhead cranes, each spanning the entire length of the workshop bay.

(b) There shall be space provided for repairs of HVAC, Door, and Traction motor etc. repairs. Distinct spaces shall be earmarked for dismantling /repairs/ assembling and testing of each of these equipments. Related machinery for Overhauling/Repairs & testing activities of every equipment are also to be housed in the space earmarked.

(c) There shall be washing and cleaning equipments on the workshop floor. Bogie test stand shall be provided in the workshop. Other heavy machinery shall also be suitably installed on the workshop floor. Air-circulators, lights, Power supply points and compressed air supply line shall be provided on each workshop column.

(d) Workshop lines shall be inter-linked through turn tables, each suitable for movement of a train in AWo (unloaded) condition and shall also be capable to rotate with a fully loaded bogie on it. Repair of heavy equipments such as air conditioners shall be so located so that it does not affect the movement inside workshop.

(e) There shall be walk ways on columns for roof inspections, along the workshop lines. These walkways shall not infringe with cars being lifted/ lowered by means of mobile jacks. Suitable space between the nearest exterior of a car and farthest edge of the walkway has to be ensured to avoid conflict in lifting and lowering of cars.

(f) The small component, bogie painting and battery maintenance cells will be located in the workshop with arrangement that fumes are extracted by suitable exhaust systems.

(g) Workshop will have service building with array of rooms along its length. Total size is proposed to be 166x8m. These can be made by column and beam structure and architecture made of brick works. These shall cater for overhauling sections, offices, costly store item, locker rooms, toilets etc. Two opposite sides width wise shall be open

to facilitate natural air circulation and cross ventilation besides the egress & Ingress for coaches. The side walls shall also have sufficient width of louvers for providing adequate ventilation.

(h) There shall be space for bogie/ axle repair shop with necessary infrastructure for disassembly, overhead, assembly and testing of mechanical components of bogies/axle. The repair shop shall be easily approachable from with the workshop for transportation of components.

Following equipment repair/overhaul facilities are planned in the workshop and wheel repairs shop at the workshops:

1. Body furnishing
2. Bogie
3. Wheels
4. Traction Motors
5. Axle Box and Axle Bearing
6. Carbon Pad assembly.
7. Transformer, converter/inverter, circuit breaker
8. Battery
9. Air Compressor
10. Air-conditioner
11. Brake Equipment
12. Door actuators
13. Control and measuring equipments
14. Pneumatic equipment
15 Damper sandprings
16. Couplers/Gangways
17. Coach Paining (Applicable only for Aluminum coaches, If any)

8.9 Car Delivery Area

There shall be rail connectivity between the Depot-cum-Workshop and main line and all trains due for scheduled/unscheduled works shall reach the depot-cum- Workshop by rail.

However, in case of newly procured coaches, which are transported by road, these shall reach the Depot-cum Work shop by the road on trailers. To unload the coaches and bring them to the track, provision of space, along the side of shunting neck, has to be made for unloading of cars and other heavy

materials. This area shall have an insulated track embedded in the floor facilitating the movement of road trawler, which brings in the cars. The length of the track embedded area shall be about 50 mlong. The unloading bay should be of 50mX30m and the bearing capacity of the floor should be 15-20MT/m^2. There should be enough space available for movement of heavy cranes for lifting of coaches. The unloading area should be easily accessible for heavy duty hydraulic trailers and minimum turning radius for the trailer movement should be 20-23 m. In case of space limitation a point lifting jack system can be installed.

8.10 Operational Features

The rake induction and withdrawal to main line will be primarily from the stabling shed. Further, provisions are there for direct rake induction and withdrawal to main line from Inspection Shed/ workshop area. Movement from depot to the main line is so planned that the main line train operation is not affected. Simultaneous receipt and dispatch of trains from depot to main line is feasible in the present site scenario. Both of these activities will be done effectively without effecting the train operation on the main line. The stabling lines would be interlocked with the main line thereby induction of train from the stabling would be safe and without loss of time. The proposition for a transfer track on the incoming line as well as on the outgoing line to facilitate the movement of rake in the depot by Operation Control Centre (OCC) even though the further path in side the depot is not clear shall be explored in the detailed design stage depending on the actual availability of land.

An emergency line is also provided from which an emergency rescue vehicle may be dispatched to mainline in the event of emergency if necessary.

8.11 Infrastructure Facilities

I. Inspection and Workshop facilities:

As indicated in 8.8.2 & 8.8.3above.

II. Stabling Lines in Depot:

a) The requirement of lines shall be in accordance with the details indicated in para 8.8.1 above. A part of the stabling siding in the depot shall be covered with a roof in order to facilitate testing of air conditioning of trains and their pre-cooling under controlled condition of temperature.

b) Separate toilets adjustment to stabling lines shall be provided with small room for keeping cleaning aids and for utilization by the working staff.

III. Automatic Coach Washing Plant (AWP)

Provision to be made for Rolling Stock exterior surfaces to be washed using a fully automated Train Washing System, with a throughput capacity of approximately ten trains per hour. The AWP shall be situated at such a convenient point on the incoming route so that incoming trains can be washed before entry to the depot and undesirable movement/shunting over ingress and egress routes with in the depot is avoided. Additional space for plant room for AWP system shall be earmarked alongside the washing apron as indicated at S. No.6 of Table 8.8.

IV. Train Operators Booking Office

Suitable office facility adjacent to the stabling lines at each depot should be provided so that train operators reporting 'On' duty or going 'Off 'duty can obtain updates regarding 'Special Notices' ,'Safety Circulars' and other technical updates/ information in vogue. These offices should have an attached a cycle/scooter/car stand facility for convenience of the train operating staff.

V. Test Track

A test track of 1000 mts. In length covered & fenced should be provided beside workshop in the depot. It shall be equipped with signaling equipments (ATP/ATO). It shall be used for the commissioning of the new trains, their trials and testing of the trains after the IOH and POH. Entry into the test track shall be planned for a 3-Car train. In compliance to safety norms, the boundary of the track shall be completely fenced to prevent unauthorized trespassing across or along the track.

VI. Heavy Cleaning Shed

Monthly heavy cleaning of interior walls, floors, seats, windows glasses etc, outside heavy cleaning, Front/rear Face, Vestibule/ Buffer area, outside walls and roof shall be done manually in the interior cleaning plant designed for cleaning of one at a time. A line adjacent to inspection shed should be so provided that placement of rakes is possible from workshop or inspection lines &vice–versa conveniently and with ease.

VII. Power Supply

Auxiliary substations are planned for catering to the power supply requirement of the whole depot and workshop. Details of connected load feeder shall be worked out. Taking diversity factor of 0.5 the

maximum demands shall be computed. Two Auxiliary substations are proposed, as the demand by machines in Workshop area would be very large. The

Standby power supply is proposed through DG set with AMF panel. The capacity of DG set will be adequate to supply all essential loads without over loading.

VIII. Compressed Air Supply

Silent type compressor units shall be suitably installed inside the depots at convenient location for the supply of compressed air to workshop and Inspection sheds. Thus, the pneumatic pipeline shall run within the workshop, inspection and intensive cleaning bays as to have compressed air supply line at all convenient points.

IX. Water Supply, Sewerage and Drainage Works

In house facilities shall be developed for the water supply of each depot. Sewerage, storm water drainage shall be given due care while designing the depots for efficient system functioning. Past records of Municipal Corporation shall be used to design the drainage system. Rain water harvesting would be given due emphasis to charge the under-ground reserves.

X. Ancillary Workshop

This workshop will have a line at floor level with provision of pits. Arrangement for repairs of Shutters, Rail Road Vehicles and other ancillary vehicles will be provided. These vehicles will also be housed here itself. Heavy lifting works can be carried out in main workshop.

Ancillary workshop will be used for storing Third Rail assembly parts and their maintenance/ repair for restoration of 750 VDC feed system.

XI. Watch Towers

There shall be provision of adequate number of watch towers for the vigilance of depot boundary.

XII. Administrative Building

An administrative building close to the main entrance is planned. It can be suitably sized and architecturally designed at the detailed design stage. A time and security office is also provided close to main entrance. It shall be equipped with suitable Access control system for all the staff working in the complex.

XIII. Parking Facilities

a) Ample parking space shall be provided for the two wheelers and four wheelers at the following points.

i) Close to the depot entry.
ii) Close to the stabling lines.
iii) Close to the Workshop /IBL.

b) Space for parking of road cum rail vehicle equipped with re-railing and rescue item:

Enough space for parking of road vehicle/trailers/ trucks etc. Enough space will also have to be earmarked adjacent to workshops. Similarly, provision of space for parking of road cum rail vehicle equipped with re- railing and rescue item will have to be made close to the main exit gate of the Depot.

XIV. Shed and Buildings

The shed and buildings normally provided in the depot with their sizes and brief functions are indicated at Para 8.12.1. At the detailed design stage depending upon the land availability, the decision to locate these buildings can be taken. These can then be architecturally and functionally grouped.

XV. Plant and Machinery

a) A separate building is planned for housing pit wheel lathe (PWL) of50 mX16m approachable from workshop, inspection bay and stabling lines through rail and road for placement of cars for re-profiling of wheels within the depot along with space for depot of scrap.

b) Requirement of buildings and major plants and machinery, is given at Paras 8.12.1& 8.12.2.

8.11.1 Following Safety features should be incorporated in the design of the Maintenance Depot-cum -Workshop:

a) 1.5 EOT cranes in the inspection bay should be interlocked with750VDCin such a way that, the cranes become operational only when the traction supply is isolated and grounded.

b) Red flasher lights should be installed along the inspection lines at conspicuous location to indicate the 750V DC is ‘Live’.

c) Multi level wheel and TM stacking arrangement should be an in built feature at the end of Workshop Lines.

d) Pillars in the inspection bay & workshop should have provision for power sockets.

e) Placement of rakes from inspection /workshop lines on to washing lines for interior cleaning on their own power should be possible. Linking of OHE and its isolation at the cleaning area should be provided. Necessary requirements of safety should be kept in view.

f) The roof inspection platform should have open-able doors to facilitate staff to go up the roof for cleaning of roof. Suitable safety interlock should be provided to ensure maintenance staff are enabled to climb on the roof inspection platform only after the OHE is isolated.

g) Control Centre, PPIO & store depot must be close to Workshop.

h) Width of the doors of the sections where in repairs of equipments are done should be at least 2 meters wide to allow free passage of equipment through them.

i) Provision of water hydrants should be done in workshops & stabling yards also.

j) Compressed air points along with water taps should be available in interior of buildings for cleaning.

k) Ventilation arrangement inside the inspection shed and workshop should be ensured. Arrangement for natural cross ventilation from one side to another of inspection & workshop bays to be incorporated along with optimum availability of natural light at floor level.

8.12 List of Buildings & List of Plants Equipments at Depot-cum-workshop:

8.12.1 List of Buildings at Depot-cum-workshop:

Table -8.8

S.No	Nameof Building	Size	Remarks
1.	Inspection Shed	169mx 22m(each inspection bay)	Servicing of Cars for 15 days & 45days inspection. Space for future provision of One bay(for three lines) to be kept for year 2036 and upto year2046
	Workshop Shed	166m x 21m(each inspection bay)	Major repair & overhaul of rolling stocks, diesel shunters , electric tractors, Tower wagons. All heavy lifting jobs. Space for future provision of one bay(for two lines) to be kept for year2036and upto year 2046

	Associated Sections	166mx8m	Rooms for carrying out the inspection & workshop activity.
	Stabling line shed	166mx 30m	Provisional for total area as per requirement of stabling of 52 rakes during year2046 is to be made (with initial provision for 15 rakes only).
2.	Stores Depot &Offices including Goods Plat form with Ramp	45mx45m	i. Stocking of spares for regular& emergency requirement including consumable items ii. This store caters for the requirement of depot for rolling stock& other disciplines. .iii. To be provided with computerized inventory control.
3.	Elect. Substation & DG set room	20mx15m	To cater for normal and emergency Power supply for depot, workshop, service and all other ancillary buildings, essential power supply for essential loads and security light.
4.	Traction repair depot and E &M repair shop	80m x 30m (partly double storey)	Stabling and routine maintenance of Shunting engine etc. &Traction maintenance depot. For maintenance of lifts/escalators and other General service
5.	Cycle/Scooter/Car Parking	100mx6m 60mx6m	i) Close to depot entry ii). Close to the stabling lines.
6.	Auto coach washing plant	60mx10m	For automatic washing of coaches. Provision of Washing apron for collection of dripping water and its Proper drainage to been ensured.
	Auto coach washing platform	20m X 10m	
7.	Washing apron for Interior Cleaning	160mx6.5m	Heavy wet washing of rakes from inside, under frame, roof at 30 days interval.
8.	Blow down plant	30m X 5m (additional to intensive cleaning)	Heavy cleaning of under frame and roof through compressedairat30 days interval.
9.	P-way office, store & Workshop including Welding plant	80mx20m	i. Fortrackmaintenanceofsection and depot. ii. To weldrails for construction periodonly.
10.	ETU Building	30m X 15m	For parking of CMV and OHE machinery.
11.	Security office & Time Office Garages (4 Nos.)	15mx8m	For security personnel. For time punching For parking vehicle jeep, truck etc.
12.	Check Post (2Nos.)	5mx3m	For security check of incoming/outgoing staff material and coaches.
13.	Depot control centre & Crew booking centre	25mx20m (double storey)	To control movement of trains in and out of the depot and for crew booking.
14.	O.H raw water Tank	1,00,000Ltrs.Capacity	For Storage of water.
15.	Pump house Bore well	7.3mx5.4m (200mm bore)	Submersible type pump planned with 200 mm diameter bore well.

16.	Dangerous goods Store	15mx10m	For Storage of paints, inflammables &Lubricants
17.	a)Traction 750 V DC	a)120mx80m	Traction Power Supply
	b)Feeding Post	b)15mx30m	
18.	Waste Collection Bin	10mx10m	Garbage dumping
19.	Repair shops for S &T	40mx20m	For the AFC gates, Signaling and Telecom equipment.
20.	Workshop Manager Office	30mx20m	Office of Depot in charge
21.	ATP &ATO Room	10mx8m	To keep equipments of ATP/ATO
22.	Waste Water Treatment Plant	12mx6m	For treating the discharge waters of the depot and remove the oil, acids etc.before discharging into the river. with U/G tank.
23.	Canteen	200 sqm.	To cater staff of depot and workshop. Should be in a separate building with modern kitchen ware and facilities.
24.	Toilets -Gents -Ladies	 10mx7m 10mx7m	These toilets shall be approachable both from workshop as well as from Inspection bay and ladies toilets shall be completely insulated from gent's toilet.

8.12.2 List of Plants & Equipments at Depot-cum-Workshop*:

Table - 8.9

S. No.	Description	Approx. Unit Price (INR Lakh)	Corridor-I Depot	
			IBL-3 Lines WSL-4Lines	
			Qty.(set)	Approx.Cost Provision (INR Lakh)
1	Under floor Pit Wheel lathe	621.00	**1**	**621.00**
2	Syn.PitJacks-for**3 cars** unit	521.00	**1**	**521.00**
3	Automatic Train Washing Plant	268.00	**1**	**268.00**
4	Battery Shunting Loco	349.89	**1**	**349.89**
5	Electric Tractors(RRM)	199.00	**1**	**199.00**
6	Electric Tractors(RRM)for UFWL	184.00	**1**	**184.00**
7	Syn. Mobile Jacks for **3 cars** unit	136.40	**1**	**136.40**
8	Bogie Turn Table	27.86	**4**	**111.44**
9	Re-railing& Rescue equipment(set)	81.99	**1**	**81.99**
10	Rail cum Road Vehicle(RRV)	42.26	**1**	**42.26**

11	Blow Down Plant	196.00	**1**	**196.00**
12	Mobile jib Cranes(1T Manual)	1.53	**1**	**1.53**
13	Under track Mobile Lifting Table(1T for IBL)	2.75	**3**	**8.25**
14	CI/SIV Mobile Lifting Table(3TforWS)	3.62	**1**	**3.62**
15	Arial Work Lift Platform	19.61	**1**	**19.61**
16	High Pressure Wash Pumps	5.37	**2**	**10.74**
17	AC Filter cleaning machine	23.50	**1**	**23.50**
18	Mobilecompressor-10bar	4.20	**1**	**4.20**
19	HPcompressor-17bar	3.36	**1**	**3.36**
20	EMU Battery Charger	5.56	**2**	**11.12**
21	Box Container for re-railing equipment	5.12	**1**	**5.12**
22	Wooden blocks	0.77	**1**	**0.77**
23	Auxiliary truck	1.82	**1**	**1.82**
24	Road Truck	10.56	**1**	**10.56**
25	Battery operated Platform Truck for WS andDCOS	6.33	**2**	**12.65**
26	Welding & Cutting Equipment	2.24	**1**	**2.24**
27	Work Benches	0.54	**10**	**5.39**
28	Vertical Carousal storage system	35.00	**1**	**35.00**
29	Weighing scales	2.65	**1**	**2.65**
30	Storage Bins	7.59	**1**	**7.59**
31	Pallet Trucks	0.39	**5**	**1.93**
32	Fork LiftTruck-3T(Elect)	10.35	**1**	**10.35**
33	Stackers(1T for DCOS)	9.50	**1**	**9.50**
34	Mobile Safety Steps & Ladders	5.12	**LS**	**5.12**
35	Set of Pallets	15.00	**LS**	**15.00**
36	Storage racks for DCOS stores	62.50	**LS**	**62.50**
37	Storage racks for workshop, tool room	31.50	**LS**	**31.50**
38	Electric and Pneumatic Tools	33.50	**LS**	**33.50**
39	Measuring & calibration Instruments	62.51	**LS**	**62.51**

40	Special Jigs and Fixtures	60.29	**LS**	**60.29**
41	Industrial Furniture	90.00	**LS**	**90.00**
42	Miscellaneous/other machinery	93.71	**LS**	**93.71**
43	Display boards inside depot	7.50	**1**	**7.50**
44	Industrial vacuum Cleaners (heavy duty+dry/wet)	4.83	**1**	**4.83**
45	Small Part Cleaner	1.00	**1**	**1.00**
46	Polyester Web Sling+B51+B51:B70:BB51:B70	1.00	**LS**	**1.00**
47	750VDC Live Indicators	2.60	**3**	**7.80**
48	Wheel Gauges/Templates	3.80	**LS**	**3.80**
49	Ultrasonic Flaw Detector	2.18	**1**	**2.18**
50	Memory Recorder /Chart recorder etc.	10.00	**1**	**10.00**
51	Induction heater	8.40	**1**	**8.40**
52	Hyd .Axle Bearing puller	9.60	**1**	**9.60**
53	Training equipment/diagnostic software/ Camera, Projector, computer equipment/laptop etc.	16.79	**LS**	**16.79**
54	Industrial Videoscope	10.13	**2**	**20.26**
55	Lifting jacks for Aircon (two post lift)	4.37	**1**	**4.37**
56	Auto wheel profile meters and attachments for dia. And back to back	19.18	**1**	**19.18**
57	Coupler back log Gauge	8.15	**1**	**8.15**
58	25TonHydraulicCFramePress	7.44	**1**	**7.44**
59	Hydraulic work bench for Gear Box	16.00	**1**	**16.00**
60	Hydraulic work bench for couplers	7.60	**1**	**7.60**
61	Special tools for coupler	17.50	**LS**	**17.50**
62	Other tools /equipment as per RS contractor	50.00	**LS**	**50.00**
63	Bogie Test Stand	388.00	**1**	**388.00**
64	Wheel Press (300T)	467.50	**1**	**467.50**
65	Vertical turret lathe	131.00	**1**	**131.00**
66	Damper testing machine	52.98	**1**	**52.98**
67	Spring testing machine	166.10	**1**	**166.10**
68	Rail fed Bogie wash plant	188.10	**0**	**0.00**
69	Heating oven for TM	5.88	**1**	**5.88**
70	High Voltage test set	2.00	**1**	**2.00**
71	SS cage for HV Test Set	3.20	**1**	**3.20**

72	Impulse Tester for TMs	11.05	**1**	**11.05**
Overall Total				**4807.72**

8.13 Cost Estimate: The total estimated cost at Jan'17 price level may be assumed as Rs.48 Crores. This would be required for Rolling Stock M& P equipment for one depot at Dream City for Corridor-I.

8.14 Un-Attended train operation(UTO):

(i) Proper segregation for UTO and non-UTO zone shall be earmarked while finalizing of depot layout.

(ii) Train Operator (TO) plat form of 10mX12m (LXB) shall be provided in each overlap zone of UTO and non-UTO.

(iii) Bulb fencing of depot tracks shall bed one upto1.5m height from T.O.R. (Top of the Rail) level.

(iv) Gates should be provided in some interval with locking facility.

(v) All crossing roads which cross the track should have gates with locking facilities.

vi) All stabling lines should have SPK (Staff Protection Key) to avoid unauthorized entry.

DEPOT FOR CORRIDOR-II

8.15 Corridor: Surat Metro Network Corridor-II comprises of following corridor:

Corridor	Route length(Km)
Bhesan to Saroli	17.34

8.16 Depot-cum- Workshop

8.16.1 It is proposed to establish one depot- cum- workshop with following functions:

(i) Major overhauls of all the trains.

(ii) All minor schedules and repairs.

(ii) Lifting for replacement of heavy equipment and testing thereafter.

(iv) Repair of heavy equipments.

8.16.2 The Depot planning is based on following assumptions:

(i) Enough space should be available for establishment of a Depot-Cum- workshop.

(ii) All inspection lines, workshop lines, stabling lines are designed to accommodate two train sets of 3-Car each and space earmarked for future provision.

(iii) All Stabling lines are designed to accommodate two trains of 3- Car each.

(iv) All stabling lines are planned in the proposed depot-cum-workshop assuming adequate space availability. In case of space constraints, if any, stabling Facilities may need to be created at terminal stations or else where to cater to the required stability facilities.

(v) In case of space constraint for depot two storeyed Stabling lines can also be planned.

In broad terms, based on the planned Rolling Stock requirements, this chapter covers conceptual design on following aspects and will work as a guide for detailed design later:

- Layout of Stabling-shed, Inspection-shed, minor repairs and heavy repair Overhauling workshop and cleaning of Rolling Stock.
- Operational and functional safety requirements
- Ancillary buildings for other maintenance facilities.
- Electrical &Mechanical Services, power supply and distribution system.
- Water Supplies, Drainage & Sewerage.

8.16.3 MAINTENANCE PHILOSOPHY

Monitoring of the performance of all key Rolling Stock equipment by suitable advanced condition monitoring techniques available. The concept is to evolve the need based maintenance regime, which can be suitably configured in the form of schedules like daily check, “A” checks, “B” type checks, “IOH” and “POH”.

Labour intensive procedures are kept to the minimum. Automation with state of the art machinery to ensure quality with reliability.

Increase in the periodic maintenance intervals with predictive maintenance based on condition monitoring.

Multi skilling of the Maintenance staff to ensure quality and productivity in their performance.

Periodic review of maintenance practices to update replacement cycle of critical components based on experience.

Energy conservation is given due attention

8.16.4 ROLLINGSTOCK MAINTENANCE NEEDS

8.16.4.1 Maintenance Schedule

The following maintenance schedule has been envisaged for conceptual design of depots assuming approx. 318 kms running per train per day, taking in consideration the passenger load of 2021,2026, 2036 &2046 respectively.

Table 8.10

Type of	Interval	Work Content	Locations
Daily	Daily	Check on the train condition and function at every daily service completion. Interval cleaning/mopping of floor and walls with vacuum cleaner.	Stabling Lines
"A" Service Check	5,000 Km (approx.15 days)	Detailed inspection and testing of sub-systems, under frame, replacement/Topping up of oils & lubricants.	Inspection Bays
"B" Service Check	15,000 Km (approx. 45 days)	Detailed Inspection of 'A' type tasks plus items at multiplesof15,000Km ('B' type tasks)	Inspection Bays
Intermediate Overhaul (IOH)	420,000Km ,(3and half Years approx.)which ever is earlier	Check and testing of all sub-assemblies (Electrical +Mechanical). Overhaul of pneumatic valves, Compressor. Condition based maintenance of sub-systems to bring them to original condition. Replacement of parts and rectification, trial run.	Workshop
Periodical Overhaul (POH)	840,000 Km, (7Years approx.) whichever is earlier	Dismantling of all sub-assemblies, bogies Suspension system, traction motor, gear, Control equipment, air-conditioning units etc. Overhauling to bring them to original condition. Checking repair and replacement as necessary. Inspection and trial.	Workshop
Heavy Repairs	-	Changing of heavy item such as bogies, traction motor, wheel sets/axles, gear cases& axle boxes etc.	Workshop

The above Schedule may need slight revision based on the actual earned kilometers per train and the specific maintenance requirements of Rolling Stock finally procured.

8.17 Washing Needs of Rolling Stock

Cleanliness of the trains is essential. Following schedules are recommended for Indian environment

Table -8.11

S.N.	Kind of Inspection	Maint. Cycle	Time	Maintenance Place
1.	Outside cleaning (wet washing on automatic washing plant)	3Days	10 mins.	Single Pass through Automatic washing plant of Depot
2.	Outside heavy Cleaning (wet washing on automatic washing plant and Front Face, Vestibule/Buffer area. Floor, walls inside/outside of cars and roof. Manually)	30days	2 –3 hrs.	Automatic washing plant & intensive cleaning shed

8.18 Year-wise planning of maintenance facility setup at depot cum workshop based on planned Rolling Stock requirement in TOP is tabulated below:

Traffic data are available upto year 2046 only, hence space to be earmarked For future expansion beyond 2046 year for stabling, inspection and workshop line.

(i) Planned rakes as per TOP:

Table-8.12

Year	No. of Rakes	No. of coaches
2021	9	27
2026	11	33
2036	21	63
2046	26	78

ii) Requirement of Stabling Lines (SBL), Inspection Lines (IBL) and Workshop Lines (WSL) in the Depot-cum -Workshop.

Stabling and Inspection lines

Table -8.13

Year	No. of Rakes	SBLs	IBLs
2021	9	4 lines x two train of 3-car	One bay of 3 lines each with two train of 3-cars
2026	11	4 lines x two train of 3-car i.e. (No additional requirement)	No additional requirement. One bay of three lines with two trains of 3- cars is also catering to requirements of 26 rakes upto year 2046.
2036	21	10 lines x two train of 3-car	
2046	26	12 lines x two train of 3-car	

Workshop lines

Table -8.14

Year	No. of Rakes	WSLs
2021	9	One bay of 2 lines each with two trains of 3-cars is to be required for the Years 2021 & 2026 and additional one bay of 2 lines each with two trains of 3- cars is to be kept for future provision .From 2036,which will cater to the requirements upto year 2046.
2026	11	
2036	21	
2046	26	

8.19 Requirement of maintenance/Inspection lines for depot-cum-workshop:

Table -8.15

Schedule	Maintenance Requirement	Lines Needed
i) Year 2021- Maximum no. of rake holding is(9TS x3= 27 Cars)		
'A' Checks (5000km) approx.15days	(9X3)Cars=27 Cars	1 Line x two train of3-Cars (with Sunken Floor)
'B' Checks(15000 km) approx. 45days.	(9X3) Cars=27 Cars	1 Line x two train of 3-Cars (with Sunken Floor)
Unscheduled line& Adjustment lines	For minor repairs, testing and after IOH/POH	1 Line x two train of3-Cars (with Sunken Floor)
Requirement		1 bay of 3 lines
ii) Year 2026- Maximum no. of rake holding is (11TS x3= 33 Cars)		
'A' Checks(5000km) 15days	(11X 3)Cars=33 Cars	No additional requirement
'B' Checks (15000 km) 45days	(11X 3)Cars=33 Cars	
Unscheduled line& Adjustment lines	For minor repairs, testing and after IOH/POH adjustments	
Requirement		No additional requirement. Available One bayof3 lines with two trains of 3-cars is Also catering to requirements of 26 rakes upto year 2046.
iii) Year 2036- Maximum no. of rake holding is(21TS x3= 63 Cars)		
'A' Checks(5000km) approx.15days	(21X3)Cars=63 Cars	No additional requirement
'B' Checks(15000 km) approx. 45days.	(21X3)Cars=63 Cars	
Unscheduled line& Adjustment lines	For minor repairs, testing and after IOH/POH adjustments	

Requirement	No additional requirement. Available One bay of 3 lines with two trainsof3-cars is also catering to requirements of26rakesupto year2046.

iv)Year 2046-Maximum no. of rake holding is(26TS x3= 78 Cars)		
'A' Checks(5000km) approx.15days	(26X3)Cars=78 Cars	No additional requirement
'B' Checks(15000 km) approx. 45days.	(26X3)Cars=78 Cars	
Unscheduled line& Adjustment lines	For minor repairs, testing and after IOH/POH adjustments	
Requirement		No additional requirement. Available One bay of 3lineswith two trains of 3-cars is also catering to requirements of 26 rakes upto year2046.

8.20 Inspection requirements at depot:

Facilities for carrying out inspection activities shall be provided in the inspection bay for following Systems/ Equipments of a train:

- Electronics; PA/PIS
- Mechanical components, couplers etc
- Batteries
- Air conditioner
- Brake modules Bogie
- Traction Motor
- Vehicle doors, windows and internal fittings
- Power system including converter, circuit break etc.

These activities shall be grouped into "A" checks and "B" checks. The minor scheduled inspections ("A" checks) shall be carried out during the day off peak and night. Since "B" checks take longer time, these cannot be completed in the off peak times. Certain inspection lines will be nominated for "A" checks. For "B" checks, separate line will be nominated where the rakes may be kept for long time.

8.21 Design of Depot-cum-Workshop Facilities

8.21.1 Stabling lines at depot:

As per advised dimensions of the Rolling Stock, the length of 3-Car train would be Approx.67.8 mts. For the design of the stabling lines in the depot and terminal stations or elsewhere (as may be required), following approximates lengths have been taken in consideration:

(i) Length of one 3- Car rake=67.8m

(ii) Gap between two trains 3-car rakes=10m

(iii) Free length at outer ends of two trains of 3-Car rakes (for cross path way, Signal and Friction buffers)= 10m each side

(iv)Total length of Stabling lines=(iii)+(i)+(ii)+(i)
+(iii)=10+67.8+10+67.8+10=165.6m approx 166m.

Looking to the car width of 2900 mm on SG, 5m “Track Centre” is proposed for all the stabling lines. Thus, space between stabling shall be sufficient to include1 m wide pathway to be constructed between tracks to provide access for internal train cleaning and under carriage inspection with provision of following facilities:

a) Each Stabling line to have water connection facility so that local cleaning, if required, is facilitated.

b) Platforms at suitable points at each end of stabling lines to enable train operators to board or de- board conveniently.

8.21.2 Inspection Bay at depot-cum-workshop:

The length of Inspection shed is computed as below:

(i) Length of one 3- Car rake=67.8m

(ii)Stairs on both pit ends=2.5m

(iii)Gap between two train of 3-cars= 10m

(iv)Embedded track at entry side upto IBL pit= 7m

(v)Embedded track at exit side up to IBL pit= 11.5m

(vi) Total length of Inspection lines = (iv)+(ii)+(i)+(iii)+(i)+(ii)+(v)

=7+2.5+67.8+10+67.8+2.5+11.5=169.
1m approx 169m.

The width of the Inspection bay in computed as below:

(i) Centre–to- centre spacing between the lines=6.5m

(ii) Centre line of outer lines to column of Shed=4.5m

(iii) Width of a 3 line Inspection Bay=(ii)+(i)+(i)+(ii)=4.5+6.5+ 6.5+4.5= 22m

a) There shall be one inspection bay of 169mX22m size each with provision of accommodating three inspection lines each having sunken floor and over head roof inspection platforms at each of the line. The floor will be sunken by 1100mm.The track spacing between the adjacent IBLs shall be 6.5m.

b) Roof Inspection platforms of 1.2 m width and walk ways for roof inspection supported on the columns shall be provided. There would be lighting below the rail level to facilitate the under-frame inspection. Ramps of1:8 slopes,3 meter wide should be provided with sunken floor system for movement of material for the cars. Further,10m cross pathways are left at each end for movement of material by fork lifter/Leister/Hand trolley. 415V3phase50Hz,230V 1phase 50Hz AC supply and Pneumatic supply shall also be made available on each inspection shed columns. Air-circulators shall be provided on each column. The inspection bay shall be provided with EOT crane of 1.5T to facilitate lifting of equipment.

Roof and walls shall be of such design that optimum natural air ventilation occurs all the time and sufficient natural light is also available. Each Inspection bay will also have arrangement close by for cleaning of HVAC filter under high pressure water jet.

8.21.3 Workshop Shed at Depot:

Requirement of workshop lines is planned as under:

Table 8.16

Year	IOH &POH	Major Overhauling	Unscheduled repairs /lifting	Total	Remarks
2021		1line3-Car train and free space of3-car length for storage of other equipment.	1line x2 train of3 Car train length.	Onebayof2 Lines each with two trains of3-cars is to be required for the years 2021& 2026 and additional onebayof2 lines	The size of workshop shall be166 X21m for one working bay Comprising of two lines capable of accommodating
2026					

2036	1	2line3-Car train and free space of3-car length for storage of other equipment.	1linex2 train of3 Car train length.	each with two trains of3-cars is to be kept for future provision fromyear2036 which will cater to the requirements up to year 2046.	one3-Car rake with Bogie turn Table facility, one lineof3-Car rake length with free space of 3-Carrake length for Storage of wheel/bogie/ equipment's etc.
2046					

(a) Each bay shall comprise of two lines (as detailed in 'Remarks' above).Size of the workshop bay is proposed to be 166mx21m. The unscheduled lifting and heavy repair line shall be fitted with jack system capable to lift the 3-Car unit simultaneously for quick change of bogie, there by saving down time of Rolling Stock. The arrangement of jack system shall be such that lifting of any coach in train formation for replacement of bogie/equipments is also individually possible. Space on one line shall be available for stocking of Bogies and wheels. These lines are to be provided with pits at regular intervals for inspection of under carriage and lines are to be inter connected by turn tables. Each workshop bay shall be equipped with two 15T/2T and 5T overhead cranes, each spanning the entire length of the workshop bay.

(b)There shall be space provided for repairs of HVAC, Door, and Traction motor etc. repairs. Distinct spaces shall be earmarked for dismantling/repairs/ assembling and testing of each of these equipment's. Related machinery for Overhauling/Repairs & testing activities of every equipment are also to be housed in the space earmarked.

(c)There shall be washing and cleaning equipments on the workshop floor. Bogie test stand shall be provided in the workshop. Other heavy machinery shall also be suitably installed on the workshop floor. Air-circulators, lights, Powers supply points and compressed air supply line shall be provided on each workshop column.

(d) Workshop lines shall be inter-linked through turn tables, each suitable for movement of a train in AWo (unloaded) condition and shall also be capable to rotate with a fully loaded bogie on it. Repair of heavy equipments such as air conditioners shall be so located so that it does not affect the movement inside workshop.

(e)There shall be walk ways on columns for roof inspections, along the workshop lines. These walkways shall not infringe with cars being lifted/ lowered by means of mobile jacks. Suitable space between the nearest exterior of a car and farthest edge of the walkway has to be ensured to avoid conflict in lifting and lowering of cars.

(f) The small component ,bogie painting and battery maintenance cells will be located in the workshop with arrangement that fumes are extracted by suitable exhaust systems.

(g)Workshop will have service building with array of rooms along its length. Total size is proposed to be 166x8m. These can be made by column and beam structure and architecture made of brick works. These shall cater for overhauling sections, offices, costly store item, locker rooms, toilets etc. Two opposite sides widthwise shall be open to facilitate natural air circulation and cross ventilation besides the egress & ingress for coaches. The side walls shall also have sufficient width of louvers for providing adequate ventilation.

(h)There shall be space for bogie /axle repair shop with necessary infrastructure for disassembly, overhead, assembly and testing of mechanical components of bogies/ axle. The repair shop shall be easily approachable from with the workshop for transportation of components.

Following equipment repair/overhaul facilities are planned in the workshop and wheel repairs shop at the workshops:

1. Body furnishing

2. Bogie

3. Wheels

4. Traction Motors

5. Axle Box and Axle Bearing

6. Carbon Pad assembly.

7. Transformer, converter/inverter, circuit breaker

8. Battery

9. Air Compressor

10.Air-conditioner

11.Brake Equipment

12.Dooractuators

13.Control and measuring equipments

14.Pneumatic equipment

15.Dampers and Springs

16.Couplers/Gangways

17.Coach Painting (Applicable only for Aluminum coaches, if any)

8.22 Car Delivery Area

There shall be rail connectivity between the Depot-cum-Workshop and mainline and all trains due for scheduled/unscheduled works shall reach the depot-cum- Workshop by rail.

However, in case of newly procured coaches, which are transported by road, these shall reach the Depot-cum Workshop by the road on trailers. To unload the coaches and bring them to the track, provision of space, along the side of shunting neck, has to be made for unloading of cars and other heavy materials.

This area shall have an insulated track embedded in the floor facilitating the movement of road trawler, which brings in the cars. The length of the track embedded area shall be about 50 m long. The unloading bay should be 50m X 30m and the bearing capacity of the floor should be 15-20MT/sqm . There should be enough space available for movement for heavy cranes for lifting of coaches. The unloading area should be easily accessible for heavy duty hydraulic trailers and minimum turning radius for the trailer movement should be 20-23m. In case of space limitation a point lifting jack system can be installed.

8.23 Operational Features

The rake induction and withdrawal to main line will be primarily from the stabling shed. Further, provisions are there for direct rake induction and withdrawal to main line from Inspection Shed/workshop area. Movement from depot to the main line is so planned that the main line train operation is not affected. Simultaneous receipt and dispatch of trains from depot to main line is feasible in the present site scenario. Both of these activities will be done effectively without effecting the train operation on the mainline. The stabling lines would be interlocked with the main line thereby induction of train from the stabling would be safe and without loss of time. The proposition for a transfer track on the incoming line as well as on the outgoing line to facilitate

the movement of rake in the depot by Operation Control Centre(OCC) even though the further path inside the depot is not clear shall be explored in the detailed design stage depending on the actual availability of land.

An emergency line is also provided from which an emergency rescue vehicle may be dispatched to main line in the event of emergency If necessary

8.24 Infrastructure Facilities

I. Inspection and Workshop facilities:

As indicated in 8.21.2 & 8.21.3 above.

II. Stabling Lines in Depot:

a) The requirement of lines shall be in accordance with the details indicated in para 8.21.1 above. A part of the stabling siding in the depot shall be covered with a roof in order to facilitate testing of air conditioning of trains and their pre-cooling under controlled condition of temperature.

b) Separate toilets adjustment to stabling lines shall be provided with small room for keeping cleaning aids and for utilization by the working staff.

III. Automatic Coach Washing Plant (AWP)

Provision to be made for Rolling Stock exterior surfaces to be washed using a fully automated Train Washing System, with a throughput capacity of approximately ten trains per hour. The AWP shall be situated at such a convenient point on the incoming route so that incoming trains can be washed before entry to the depot and undesirable movement/ shunting over ingress and egress routes with in the depot is avoided. Additional space for plant room for AWP system shall be earmarked along side the washing apron as indicated at S. No.6 of Table 8.8

IV. Train Operators Booking Office

Suitable office facility adjacent to the stabling lines at each depot should be provided so that train operators reporting ‘On’ duty or going ‘Off ’duty can obtain updates regarding ‘Special Notices’, ‘Safety Circulars’ and other technical updates/information in vogue. These offices should have an attached a cycle/scooter/car stand facility for convenience of the train operating staff.

V. Test Track

A test track of 1000 mts in length covered & fenced should be provided beside workshop in the depot. It shall be equipped with signaling equipments (ATP/ATO). It shall be used for the commissioning of the new trains, their trials and testing of the trains after the IOH and POH .Entry into the test track shall be planned for a 3-Car train. In compliance to safety norms, the boundary of the track shall be completely fenced to prevent unauthorized trespassing across or along the track.

VI. Heavy Cleaning Shed

Monthly heavy cleaning of interior walls, floors, seats, windows glasses etc, out side heavy cleaning, Front/rear Face, Vestibule/Buffer area, outside walls and roof shall be done manually in the interior cleaning plant designed for cleaning of one at a time. A line adjacent to inspection shed should be so provided that placement of rakes is possible from workshop or inspection lines & vice–versa conveniently and with ease.

VII. Power Supply

Auxiliary substations are planned for catering to the power supply requirement of the whole depot and workshop. Details of connected load feeder shall be worked out. Taking diversity factor of 0.5 the maximum demands shall be computed. Two Auxiliary substations are proposed, as the demand by machines In Workshop area would be very large. The standby power supply is proposed through DG set with AMF panel. The capacity of DG set will be adequate to supply all essential loads without overloading.

VIII. Compressed Air Supply

Silent type compressor units shall be suitably installed inside the depots at convenient location for the supply of compressed air to workshop and Inspection sheds. Thus, the pneumatic pipeline shall run with in the workshop inspection and intensive cleaning bays as to have compressed air supply line at all convenient points.

IX. Water Supply, Sewerage and Drainage Works

In house facilities shall be developed for the water supply of each depot. Sewerage, storm water drainage shall be given due care while designing the depots for efficient system functioning. Past records of Municipal Corporation shall be used to design the drainage system.

Rain water harvesting would be given due emphases to charge the underground reserves.

X. Ancillary Workshop

This workshop will have a line at floor level with provision of pits. Arrangement for repairs of Shunters, Rail Road Vehicles and other ancillary vehicles will be provided. These vehicles will also be housed here it self. Heavy lifting works can be carried out in main workshop.

Ancillary workshop will be used for storing Third Rail assembly parts and their maintenance/ repair for restoration of750 VDC feed system.

XI. Watch Towers

There shall be provision of adequate number of watch towers for the vigilance of depot boundary.

XII. Administrative Building

An administrative building close to the main entrance is planned. It can be suitably sized and architecturally designed at the detailed design stage. A time and security office is also provided close to main entrance. It shall be equipped with suitable Access control system for all the staff working in the complex.

XIII. Parking Facilities

a) Ample parking space shall be provided for the two wheelers and four wheelers at the following points.

i) Close to the depot entry.
ii) Close to the stabling lines.
iii) Close to the Workshop/IBL.

b) Space for parking of road cum rail vehicle equipped with re-railing and rescue item:

Enough space for parking of road vehicle/trailers/trucks etc. Enough space will also have to be earmarked adjacent to workshops. Similarly, provision of space for parking of road cum rail vehicle equipped with re- railing and rescue item will have to be made close to the main exit gate of the Depot.

XIV. Shed and Buildings

The shed and buildings normally provided in the depot with their sizes and brief functions are indicated at Para 8.26.1 .At the detailed design stage depending upon the land availability, the decision to locate

these buildings can be taken. These can then be architecturally and functionally grouped.

XV. Plant and Machinery

a) A separate building is planned for housing pit wheel lathe (PWL), approachable from workshop, inspection bay and stabling lines through rail and road for placement of cars for re-profiling of wheels within the depot along with space for depot of scrap.

b) Requirement of buildings and major plants and machinery, is given at paras 8.26.1& 8.26.2.

8.25 Following Safety features should be incorporated in the design of the Maintenance Depot-cum-Workshop:

a) 1.5EOTcranes in the inspection bay should be interlocked with750VDC in such a way that, the cranes become operational only when the traction supply is isolated and grounded.

b) Red flasher lights should be installed along the inspection lines at conspicuous location to indicatethe750V DC is 'Live'.

c) Multi level wheel and TM stacking arrangement should be an inbuilt feature at the end of Work shop Lines.

d) Pillars in the inspection bay &workshop should have provision for power Sockets.

e) Placement of rakes from inspection/workshop lines on to washing lines for interior cleaning on their own power should be possible. Linking of OHE and its isolation at the cleaning area should be provided. Necessary requirements of safety should be kept in view.

f) The roof inspection platform should have open-able doors to facilitate staff to go up the roof for cleaning of roof. Suitable safety interlock should be provided to ensure maintenance staff are enabled to climb on the roof inspection platform only after the OHE is isolated.

g) Control Centre, PPIO & store depot must be close to Workshop.

h) Width of the doors of the sections where in repairs of equipments are done should be at least 2meters wide to allow free passage of equipment through them.

i) Provision of water hydrants should be done in workshops & stabling yards also.

j) Compressed air points along with water taps should be available in interior of buildings for cleaning.

k) Ventilation arrangement inside the inspection shed and workshop should be ensured. Arrangement for natural cross ventilation from one side to another of inspection & workshop bays to be incorporated along with optimum availability of natural light at floor level.

8.26 List of Buildings & List of Plants & Equipments at Depot-cum-workshop:

8.26.1 List of Buildings at Depot-cum-workshop

Table 8.17

Sl.No	Name of Building	Size	Remarks
1.	Inspection Shed	169m x 22m (each inspection	Servicing of Carsfor15days&45days inspection.
	Workshop Shed	166m x 21m(each inspection bay)	Major repair &overhaul of rolling stocks, diesel shunters, electric tractors, tower wagons. All heavy lifting jobs. Space for future provision of one bay(for two lines)to be kept for year 2036and uptoyear2046
	Associated Sections	166mx8m	Rooms for carrying out the inspection& workshop activity.
	Stabling line shed	166mx 20m	Provisional for total area as per Requirement of stabling of 26 rakes duringyear2046is to be made(with initial provision for 9 rakes only).
2.	Stores Depot & Offices including Goods Plat form with Ramp	45mx45m	i. Stocking of spares for regular &emergency requirement including consumable items. ii. This store caters for the Requirement of depot for rolling stock& other disciplines. iii. To be provided with computerized inventory control. iv. Loading/Unloading of material received by road.
3.	Elect. Substation & DG set room	20mx15m	To cater for normal and emergency power supply for depot, workshop, service and all other ancillary buildings, essential power supply for essential loads and security light.
4.	Traction repair depot and E &M repair shop	80m x 30m (partly double story)	Stabling and routine maintenance of shunting engine etc .&Traction maintenance depot. For maintenance of lifts/escalators and other General service works
5.	Cycle /Scooter/Car Parking	100mx6m 60mx6m	i. Close to the depot entry. ii. Close to the stabling lines.
6.	Auto coach washing plant	60mx10m	For automatic washing of coaches. Provision of Washing apron for Collection of dripping water and its proper drainage to be ensured.
	Auto coach washing platform	20m X 10m	

7.	Washing apron for Interior Cleaning	160mx6.5m	Heavy wet washing of rakes from inside, under frame, roof at 30 days interval.
8.	Blow down plant	30m X 5m (additional to intensive cleaning)	Heavy cleaning of under frame and roof through compressed air at 30 days interval.
9.	P-way office, store &Workshop including Welding plant	80mx20m	i. Fort rack maintenance of section and depot. ii. To weld rails for construction period only. iii. To stable track Tamping machine.
10.	ETU Building	30m X 15m	For parking of CMV and OHE machinery.
11.	Security office &Time Office Garages (4 Nos.)	15mx8m	For security personnel. For time punching For parking vehicle jeep ,truck etc.
12.	Check Post (2Nos.)	5mx3m	For security check of incoming/outgoing staff material and coaches.
13.	Depot control centre &Crew booking centre	25mx20m (double storey)	To control movement of trains in and out of the depot and for crew booking.
14.	O. H raw water Tank	1,00,000Ltrs. Capacity	For Storage of water.
15.	Pump house Bore well	7.3mx5.4m(200mm bore)	Submersible type pump planned with 200 mm dia meter bore well.
16.	Dangerous goods Store	15mx10m	For Storage of paints, inflammables & Lubricants
17.	a)Traction750 V DC b)Feeding Post	a)120mx80m b)15mx30m	Traction Power Supply
18.	Waste Collection Bin	10mx10m	Garbage dumping
19.	Repair shops for S & T	40mx20m	For the AFC gates, Signaling and telecom equipment.
20.	Workshop Manager Office	30mx20m	Office of Depot in charge
21.	ATP & ATO Room	10mx8m	To keep equipments of ATP/ATO
22.	Waste Water Treatment Plant	12mx6m	For treating the discharge waters of the Depot and remove the oil, acids etc. before discharging into the river, with U/G tank.
23.	Canteen	200 sqm.	To cater staff of depot and workshop. Should be in a separate building with modern kitchen ware and facilities. Obligatory as per statutory requirements.
24.	Toilets -Gents -Ladies	 10mx7m 10mx7m	These toilets shall be approachable Both from workshop as well as from inspection bay and ladies toilets shall be completely insulated from gent's toilet.

8.26.2 List of Plants & Equipments at Depot-cum-Workshop*:

Table -8.18

S. No.	Description	Approx. Unit Price (INR Lakh)	Corridor-2 Depot	
			IBL-3 Lines	
			WSL-2 Lines	
			Qty.(set)	Approx. Cost Provision (INR Lakh)
1	Under floor Pit Wheel lathe	621.00	**1**	**621.00**
2	Syn. Pit Jacks-for **3cars** unit	521.00	**1**	**521.00**
3	Automatic Train Washing Plant	268.00	**1**	**268.00**
4	Battery Shunting Loco	349.89	**1**	**349.89**
5	Electric Tractors (RRM)	199.00	**1**	**199.00**
6	Electric Tractors (RRM) for UFWL	184.00	**1**	**184.00**
7	Syn. Mobile Jacks for **3 cars** unit	136.40	**1**	**136.40**
8	Bogie Turn Table	27.86	**2**	**55.72**
9	Re-railing & Rescue equipment (set)	81.99	**1**	**81.99**
10	Rail cum Road Vehicle (RRV)	42.26	**1**	**42.26**
11	Blow Down Plant	196.00	**1**	**196.00**
12	Mobile jib Cranes (1T Manual)	1.53	**1**	**1.53**
13	Under track Mobile Lifting Table (1T for IBL)	2.75	**3**	**8.25**
14	CI/SIV Mobile Lifting Table(3TforWS)	3.62	**1**	**3.62**
15	Arial Work Lift Platform	19.61	**1**	**19.61**
16	High Pressure Wash Pumps	5.37	**2**	**10.74**
17	AC Filter cleaning machine	23.50	**1**	**23.50**
18	Mobile compressor-10bar	4.20	**1**	**4.20**
19	HPcompressor-17bar	3.36	**1**	**3.36**
20	EMU Battery Charger	5.56	**2**	**11.12**
21	Box Container for re-railing equipment	5.12	**1**	**5.12**
22	Wooden blocks	0.77	**1**	**0.77**

23	Auxiliary truck	1.82	**1**	**1.82**
24	Road Truck	10.56	**1**	**10.56**
25	Battery operated Platform Truck for WS and DCOS	6.33	**2**	**12.65**
26	Welding & Cutting Equipment	2.24	**1**	**2.24**
27	Work Benches	0.54	**10**	**5.39**
28	Vertical Carousal storage system	35.00	**1**	**35.00**
29	Weighing scales	2.65	**1**	**2.65**
30	Storage Bins	7.59	**1**	**7.59**
31	Pallet Trucks	0.39	**5**	**1.93**
32	Fork LiftTruck-3T(Elect)	10.35	**1**	**10.35**
33	Stackers(1T for DCOS)	9.50	**1**	**9.50**
34	Mobile Safety Steps & Ladders	5.12	**LS**	**5.12**
35	Set of Pallets	15.00	**LS**	**15.00**
36	Storage racks for DCOS stores	62.50	**LS**	**62.50**
37	Storage racks for workshop, tool room	31.50	**LS**	**31.50**
38	Electric and Pneumatic Tools	33.50	**LS**	**33.50**
39	Measuring & calibration Instruments	62.51	**LS**	**62.51**
40	Special Jigs and Fixtures	60.29	**LS**	**60.29**
41	Industrial Furniture	90.00	**LS**	**90.00**
42	Miscellaneous/other machinery	93.71	**LS**	**93.71**
43	Display boards inside depot	7.50	**1**	**7.50**
44	Industrial vacuum Cleaners(heavy duty+ dry/wet)	4.83	**1**	**4.83**
45	Small Part Cleaner	1.00	**1**	**1.00**
46	Polyester Web Sling+B51+B51:B70:BB51:B70	1.00	**LS**	**1.00**
47	750VDC Live Indicators	2.60	**3**	**7.80**
48	Wheel Gauges /Templates	3.80	**LS**	**3.80**
49	Ultra sonic Flaw Detector	2.18	**1**	**2.18**

50	Memory Recorder/Chart recorder etc.	10.00	**1**	**10.00**
51	Induction heater	8.40	**1**	**8.40**
52	Hyd .Axle Bearing puller	9.60	**1**	**9.60**
53	Training equipment/diagnostic software/ Camera, Projector, computer equipment/lap top etc.	16.79	**LS**	**16.79**
54	Industrial Video scope	10.13	**2**	**20.26**
55	Lifting jacks for Aircon (two post lift)	4.37	**1**	**4.37**
56	Auto wheel profile meters and attachments for dia. and back to back	19.18	**1**	**19.18**
57	Coupler back log Gauge	8.15	**1**	**8.15**
58	25Ton Hydraulic C frame Press	7.44	**1**	**7.44**
59	Hydraulic work bench for Gear Box	16.00	**1**	**16.00**
60	Hydraulic work bench for couplers	7.60	**1**	**7.60**
61	Special tools for coupler	17.50	**LS**	**17.50**
62	Other tools/equipment as per RS contractor	50.00	**LS**	**50.00**
63	Bogie Test Stand	388.00	**0**	**0.00**
64	Wheel Press (300T)	467.50	**0**	**0.00**
65	Vertical turret lathe	131.00	**0**	**0.00**
66	Damper testing machine	52.98	**0**	**0.00**
67	Spring testing machine	166.10	**0**	**0.00**
68	Rail fed Bogie wash plant	188.10	**0**	**0.00**
69	Heating oven for TM	5.88	**1**	**5.88**
70	High Voltage test set	2.00	**1**	**2.00**
71	SS cage for HV Test Set	3.20	**1**	**3.20**
72	Impulse Tester for TMs	11.05	**1**	**11.05**
	OVERALLTOTAL			**3546.42**

8.27 Cost Estimate: The total estimated cost at Jan'17 price level may be assumed as Rs.35.5Crores.This would be required for Rolling Stock M&P equipment for one depot at Bhesan for Corridor-II.

8.28 Un-Attended train operation (UTO):

(i) Proper segregation for UTO and non-UTO zone shall be earmarked while finalizing of depot layout.

(ii) Train Operator (TO) platform of 10mX12m (LXB) shall be provided in each overlap zone of UTO and non-UTO.

(iii) Bulb fencing of depot tracks shall be done upto1.5m height from T.O.R. (Top of the Rail) level.

(iv) Gates should be provided in some interval with locking facility.

(v) All crossing roads which crosses the track should have gates with locking facilities.

(vi) All stabling lines should have SPK (Staff Protection Key) to avoid unauthorized entry.

CHAPTER- 9

POWER SUPPLY ARRANGEMENTS

9.1 Power Requirements

Electricity is required for operation of Metro system for running of trains, station services (e.g. lighting, lifts, escalators, signalling & telecom, fire fighting etc) and workshops, depots & other maintenance infrastructure within premises of metro system. The power requirements of a metro system are determined by peak-hour demands of power for traction and auxiliary applications. Broad estimation of auxiliary and traction power demand is made based on the following requirements:-

(i) Specific energy consumption of rolling stock 60 kWh/1000 GTKM in case of 750V dc Traction at Pantograph level is considered as per MOUD guideline vide letter No. 14011/9/2014-UT.II-Part I, dated 21.04.2017.

(ii) Elevated/at –grade station load – initially 250 kW, which will increase to 300 kW in the year 2046.

(iii) Underground station load – initially 2000 kW, which will increase to 2200 kW in the year 2046.

(iv) Depot & Metro Bhawan auxiliary load - initially 2000 kW (each), which will increase to 2200 KW in the year 2046.

Keeping in view of the train operation plan and demand of auxiliary and traction power, power requirements projected for the year 2021, 2026, 2036 and 2046 are summarized in table 9.1 below:-

Table 9.1 Power Demand Estimation (MVA)

Corridor	Load	Year			
		2021	2026	2036	2046
Sarthana to Dream City Corridor-1 20 Stations (21.61 km)	Traction	6.16	10.83	16.16	20.44
	Auxiliary	24.09	24.76	25.51	26.93
	Sub-total	**30.25**	**35.59**	**41.67**	**47.37**
Bhesan to Saroli Corridor-2 18 Stations (18.74 km)	Traction	3.96	5.00	9.15	11.16
	Auxiliary	8.03	8.31	8.71	9.39
	Sub-total	**11.99**	**13.31**	**17.86**	**20.55**

The detailed calculations of power demand estimation are attached at annexure 9.1.

9.2 Need for High Reliability of Power Supply

The proposed Surat metro system is being designed to handle about 26,587 passengers per direction during peak hours when trains are expected to run at 1.75 minutes intervals. Incidences of any power interruption, apart from affecting train running, will cause congestion at stations. Interruption of power at night is likely to cause alarm and increased risk to traveling public. Lack of illumination at stations, non-visibility of appropriate signages, disruption of operation of lifts and escalators is likely to cause confusion, anxiety and ire in commuters, whose tolerance level are low on account of stress. Effect on signal and communication may affect train operation and passenger safety as well. Therefore, uninterrupted power supply is mandatory for efficient metro operations.

To ensure reliability of power supply, it is essential that both the sources of Supply and connected transmission & distribution networks are reliable and have adequate redundancies built in. Therefore, it is desirable to obtain power supply at high grid voltage of 220 kV, 132 kV or 66 kV from stable grid sub-stations and further transmission & distribution is done by the Metro Authority itself.

9.3 Sources of Power Supply

The high voltage power supply network of Surat city was studied in brief. The city has 220 kV and 66 kV network to cater to various types of demand in vicinity of the proposed corridors. A meeting was held on dated 05.04.2017 & 06.04.2017 with M/s Torrent Power, GETCO & Surat Municipal Corporation official, and various sub-stations sites had been inspected to finalize the Input Power Supply sources & Supply Voltage.

Keeping in view the reliability requirements, Four Receiving Sub-stations (RSS) are proposed to be set up for corridor – 1 (2 RSS) and corridor – 2 (2 RSS). This is an economical solution without compromising reliability. Based on the discussions in meeting with M/s Torrent Power, M/s GETCO & Surat Municipal Corporation, it is proposed to avail power supply for traction as well as auxiliary services from the following grid sub-stations at 66 kV voltage through cable feeders: -

Table 9.2 Sources of Power Supply

S. No.	Corridor	Grid sub-station of Power Supply Authority (Input voltage)	Location of RSS of Metro Authority	Approx. length of cables
1.	Sarthana to Dream City Corridor-1	Two no. 66 kV bays from 220/66 kV Uttaran Grid Sub Station of M/s GETCO **or** Two no. 66 kV bays from 220/66 kV Puna Grid substation of Torrent	Near Nature Park Station **or** Near Kapodra station	5 km or 4km
2.		Two no. 66 kV bays from Khajod Grid Sub-station (Proposed) (M/s GETCO) **or**	At Dream City Depot **or**	2 km **or**
		Two no. 66 kV bays from Bhatar Grid Sub-station (M/s Torrent Power)	Near Althan Tenement	2 km
3.	Bhesan to Saroli Corridor-2	Two no. 66 kV bays from 220/66 kV Ved Grid Sub Station (M/s Torrent Power) **or** Two no. 66 kV bays from 220/66 kV Ichchapore Grid substation of GETCO	At Bhesan Depot	7 km or 8 km
4.		Two no. 66 kV bays from 220/66 kV Puna Grid substation of Torrent	Near Magob Station	2 km

M/s GETCO letter no. TCN/Tech-2/1921,dated:-10.04.2017 and M/s Torrent Power letter no. Technical/397,dated:-11.04.2017 have assured that reliable power supply from their 66 kV Sub-station will be provided (annexure 9.2). In view of this, during the details design stage, the locations of RSS and GSS may be reviewed/ fine tuned and finalized based on the updated status of power supply/ Sub-stations of M/s Torrent Power & M/s GETCO. The summary of expected power demand at various sources is given in table 9.3.

Table 9.3 – Power Demand projections for various sources (in MVA)

Corridor	Input Source	Peak demand - Normal		Peak demand** - Emergency	
		Year (2021)	Year (2046)	Year (2021)	Year (2046)
Sarthana to Dream City Corridor-1	**RSS near Nature Park/ Kapodara Station**				
	Traction	2.53	9.39	6.16	20.44
	Auxiliary	9.57	10.75	24.09	26.93
	Sub-total (A)	**12.10**	**20.14**	**30.25**	**47.37**
	RSS at Dream City Depot/ Near Althan Tenement				
	Traction	3.63	11.05	6.16	20.44
	Auxiliary	14.51	16.18	24.09	26.93
	Sub-total (B)	**18.14**	**27.23**	**30.25**	**47.37**

Bhesan to Saroli Corridor-2	**RSS at Bhesan Depot**				
	Traction	2.53	6.41	3.96	11.16
	Auxiliary	5.25	6.05	8.03	9.39
	Sub-total (C)	**7.78**	**12.46**	**11.99**	**20.55**
	RSS Near Magob Station				
	Traction	1.43	4.75	3.96	11.16
	Auxiliary	2.78	3.34	8.03	9.39
	Sub-total (C)	**4.21**	**8.09**	**11.99**	**20.55**

** Incase of failure of other source of power

The 66 kV power supply will be stepped down to 33 kV level at the RSS's of metro authority. The 33 kV power will be distributed along the alignment through separate 33 kV Ring main cable network for feeding traction as well as auxiliary loads. These cables will be laid in dedicated ducts/cable brackets along the viaduct and tunnel.

Interconnection of 33 kV power supply between the two corridors has been planned at the Interchange station of Majuragate which can be used for transfer of power from one corridor to other in the emergency situations. In case of tripping of One RSS of either corridor on fault or input supply failure, train services can be maintained from stand-by source of the same line or by feed extension from RSS of other line. But if one more RSS fails, only curtailed services can be catered to. However, in case of total grid failure, all trains may come to a halt but station lighting, fire and hydraulics & other essential services can be catered to by stand-by DG sets. However, no train services can be run with power supply received from DG Sets. Therefore, while the proposed scheme is expected to ensure adequate reliability in normal and emergency situation, it would cater to grid failure as well, except for the train running.

Typical High Voltage Receiving Sub-station

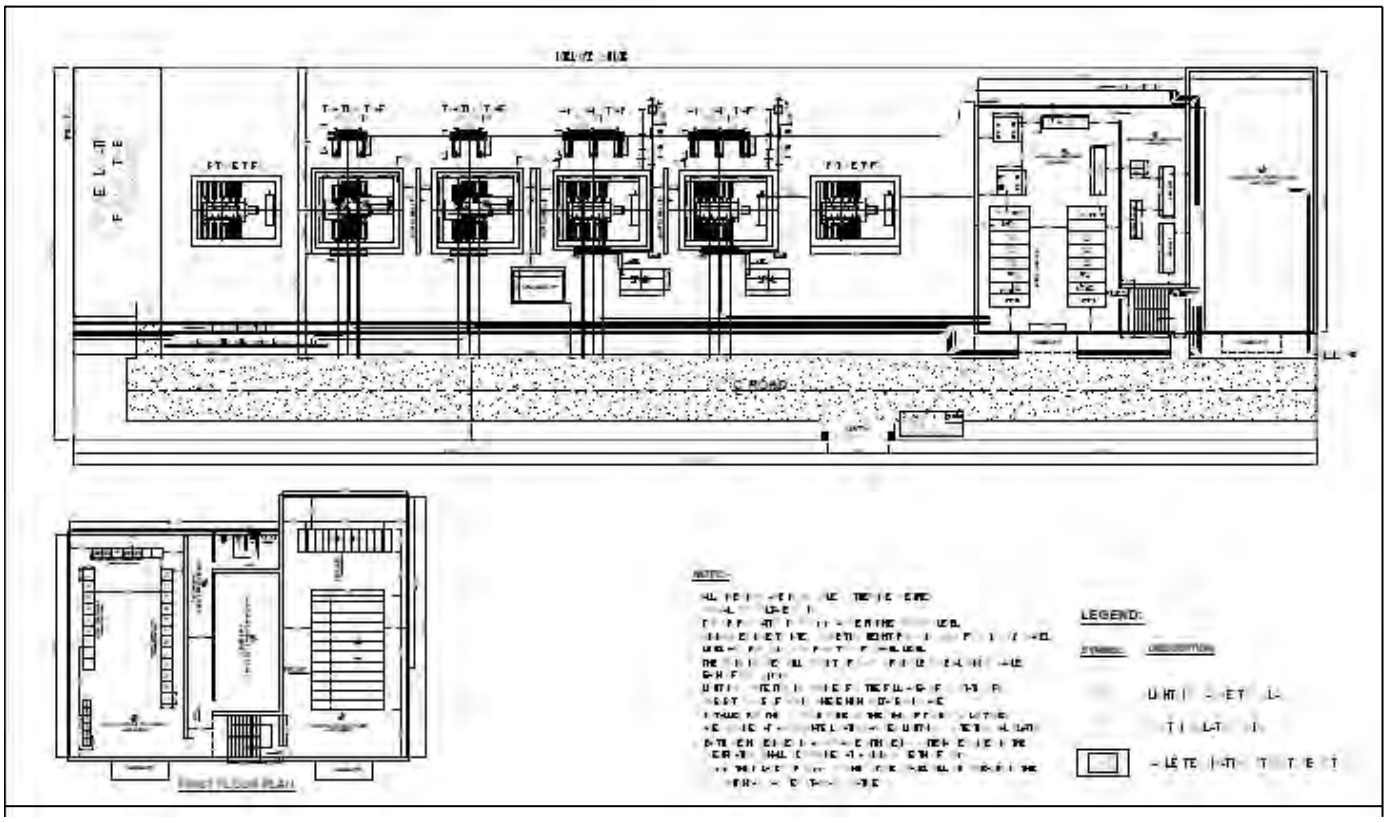
Typical High Voltage Gas Insulated Sub-Station (GIS)

The 66 kV cables will be laid through public pathways from M/s Torrent Power and M/s GETCO Sub-stations to RSS of Metro Authority. RSS near Nature Park/ Kapodara Station & at Dream City Depot or near Althan Tenement shall be provided with 2 nos. (one as standby) 66/33 kV, 40/50 MVA (ONAN/ONAF) three phase Transformers for feeding Traction as well as auxiliary loads for

corridor-1.For corridor-2, RSS at Bhesan Depot & near Magob Station shall be provided with 2 nos. (one as standby) 66/33 kV, 20/30 MVA (ONAN/ONAF) three phase Transformers. The capacity of transformers may be reviewed considering the load requirement / Power distribution of both the corridors at the time of detailed design stage.

Gas Insulated Switchgear (GIS) type Switchgear will be planned for Surat Metro due to less space and reduced maintenance. Land plot area 40m x 30m (1200 sq. m) is required.

The typical GIS RSS layout is given in figure 9.1



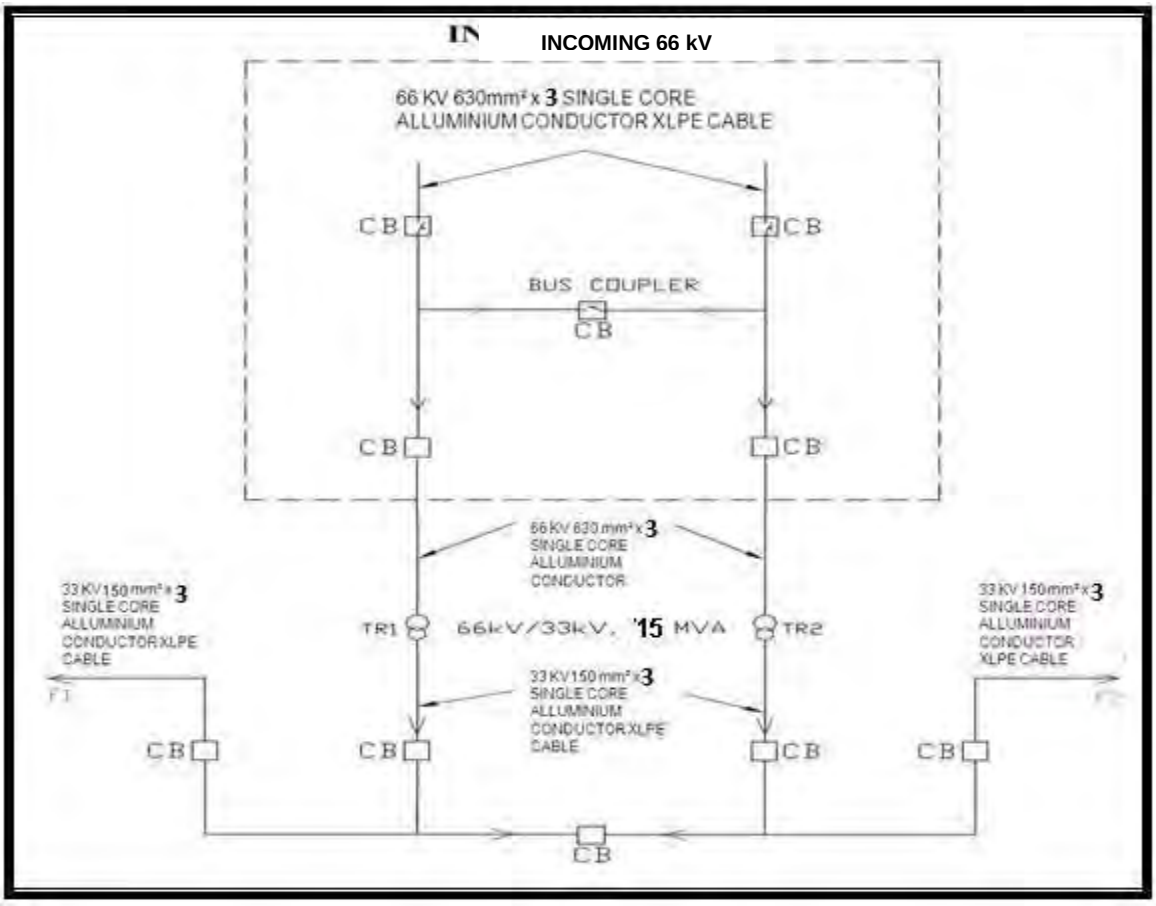


FIGURE 9.1 – TYPICAL 66/33kV RSS LAYOUT

9.4 Various options of Traction system

There are three options available for power supply system for MRTS:-

25 kV & 2x25 kV AC Overhead Catenary system.

750 V dc third rail system.

1500 V dc Overhead Catenary system

A sub- committee set up by “Ministry of Urban Development” on Traction system for metro railway has studies various aspects of merits and demerits of various traction system. The following are the highlights of Report:-

Merits and Demerits of various traction systems:

a) 25 kV ac with OCS (Flexible/rigid):- Merits

Reduced cost – Unlike dc traction, this system does not require substations at frequent intervals due to high voltage, reduced current levels and lower voltage drops as a result, there is substantial reduction in cost. Cost of 25 kV ac traction systems is about 30% less as compared to 750V DC 3rd rail traction system.

Energy regeneration & line losses- Energy regeneration is more than 30% in 25 kV AC traction system as compared to 18% in 750V DC 3rd rail traction system. In 25 kV AC traction system line losses are 12% less as compared to 750V DC 3rd rail traction system

Cost of rolling stock- The cost of rolling stock & maintenance cost of traction system are comparable.

Capacity – In future, the system can cater to traffic needs even in excess of 75000 PHPDT, which, however, is restricted on account of other constraints.

Easy of capacity enhancement – Capacity enhancement can be easily achieved by simply enhancing the transformer and its associated equipment at the receiving substation.

Higher efficiency of operation – The efficiency of regeneration is substantially more than dc systems and line losses are very less of the order of 5%.100% recovery of regenerated energy is possible in the case of 25 KV ac traction compared to a figure of 75% in the case of 1500 V dc systems and 60% in the case of 750 V dc systems.

Less Fire hazards-AC system poses lesser fire hazards as current levels are much lower than DC system.

Stray current - There are no problem of stray currents and hence nearby metallic structures are not affected by corrosion. However there are problems of EMC / EMI which can be controlled by using return conductor & screened cables in signaling applications & fiber optic cable in telecommunication system without using booster transformer as per recent developments. This also helps in avoiding use of booster transformer which causes 2% line loss and excessive voltage drops besides involving maintenance & reliability issues.

Traction equipments in 25kV AC system are standardized & mostly indigenously available. In D.C. traction system it is mostly imported.

Though in underground section higher size tunnel diameter is required.

b) 600-850 V dc third rail traction system:-Demerits

High operating currents and High voltage drops necessitating reduction in spacing of sub-station- This leads to larger voltage drops along the Third Rail distribution system, which necessitates

closer spacing of sub- stations at an interval of almost every 2 Km, leading to higher costs of construction.

Low levels of regeneration- The regeneration is 18%, because 60% of re-generated energy in a 750 V dc system is possible to be retrieved.

Safety hazards with use of medium voltage at ground level- Due to existence of the "live" third rail at ground level, this system can be hazardous to safety of commuters and maintenance personnel if they fail to adopt safety precautions.

Line losses- Line losses are more due to higher current. Transmission line losses on 750 V dc traction system are around 21% as against 5% of 25 kV ac traction system.

Phenomenon of stray current- In a third rail system, where the running rails are used as a return path, a part of the return current leaks into track structure. This current is called stray current. It is necessary to manage the stray current to ensure minimal corrosion effect and consequent damages to metallic components in the track structure as well as metallic reinforcement and metal pipes of building of metro and public areas adjacent to the Metro alignment.

c) 1500 V dc system with Overhead Catenary System:-Demerits

Higher maintenance requirement and costs as compared to 750V dc third rail system.

Theoretical traffic capacity with 1500V traction system is less as compared to 25KV ac system.

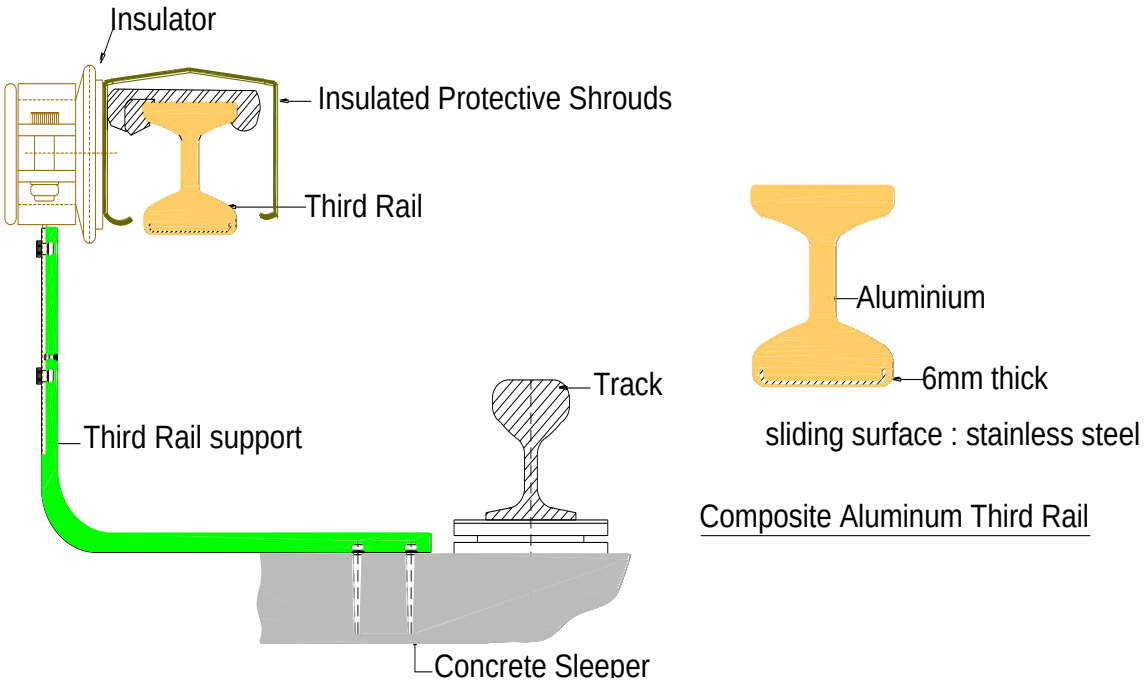
Line losses are more due to higher current as compared to 25KV ac. It may be in the range of 10 to 12% as against 5% of 25KV ac system.

In view of above techno- economic considerations, DMRC recommends 25 kV AC Traction power supply. This is 95% available indigenously. However due to the low PHPDT of the network, lesser network lengths and aesthetics use of 750V DC system as used in Ahmedabad Metro by MEGA, another similar Metro system in close vicinity is also suitable and justified. The DC suppliers are located outside India with 95% import content. Final decision may be taken by the client in this regard.

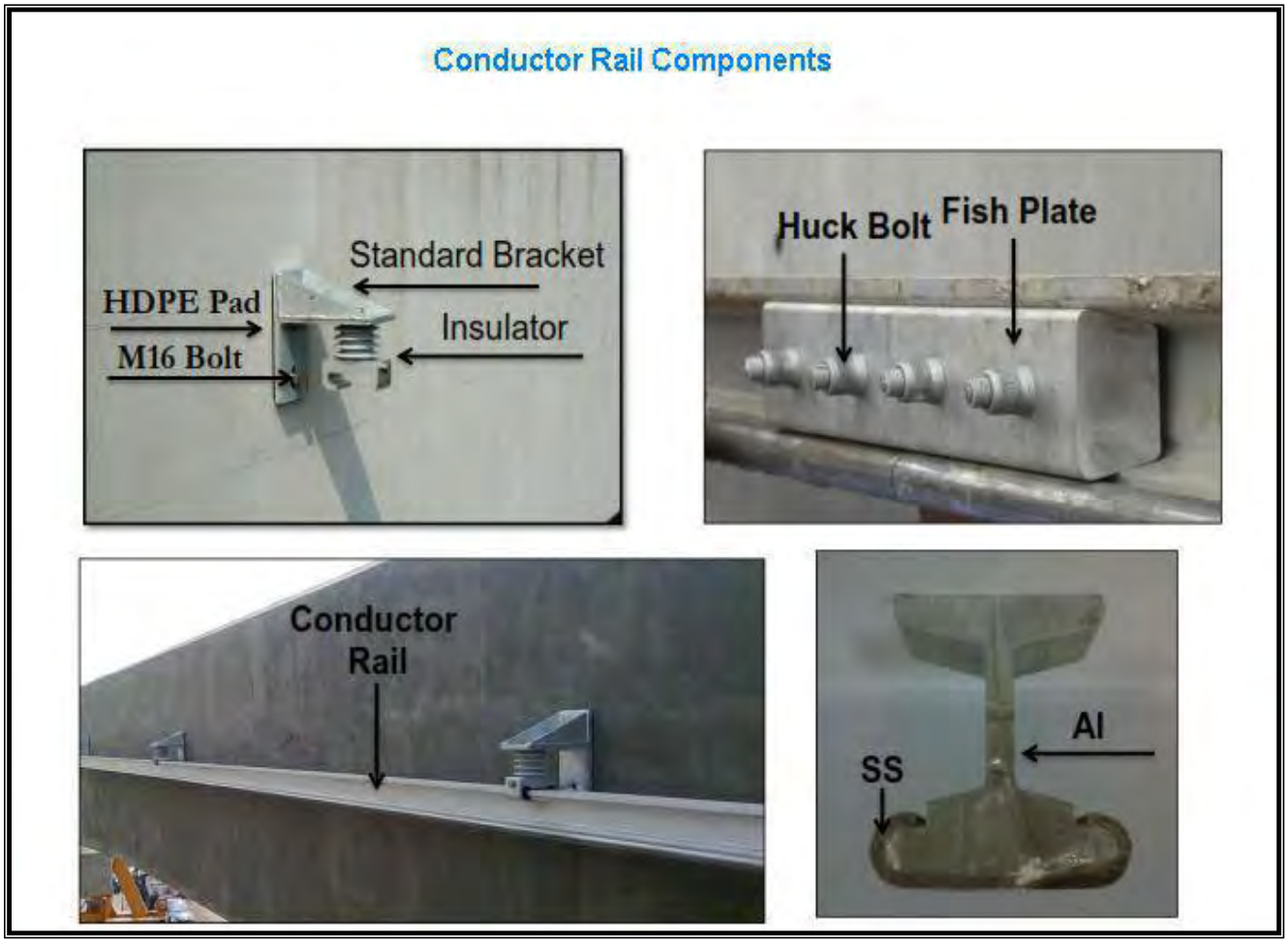
9.5 750V dc Third Rail Current Collection System

For the 750V dc Third Rail Current Collection System, Bottom current collection with the use of composite Aluminum steel third rail on main lines is envisaged from reliability and safety considerations (figure below). Low carbon steel third

rail available indigenously is proposed for the depot because of reduced current requirements.



750V dc Third Rail Current Collection System

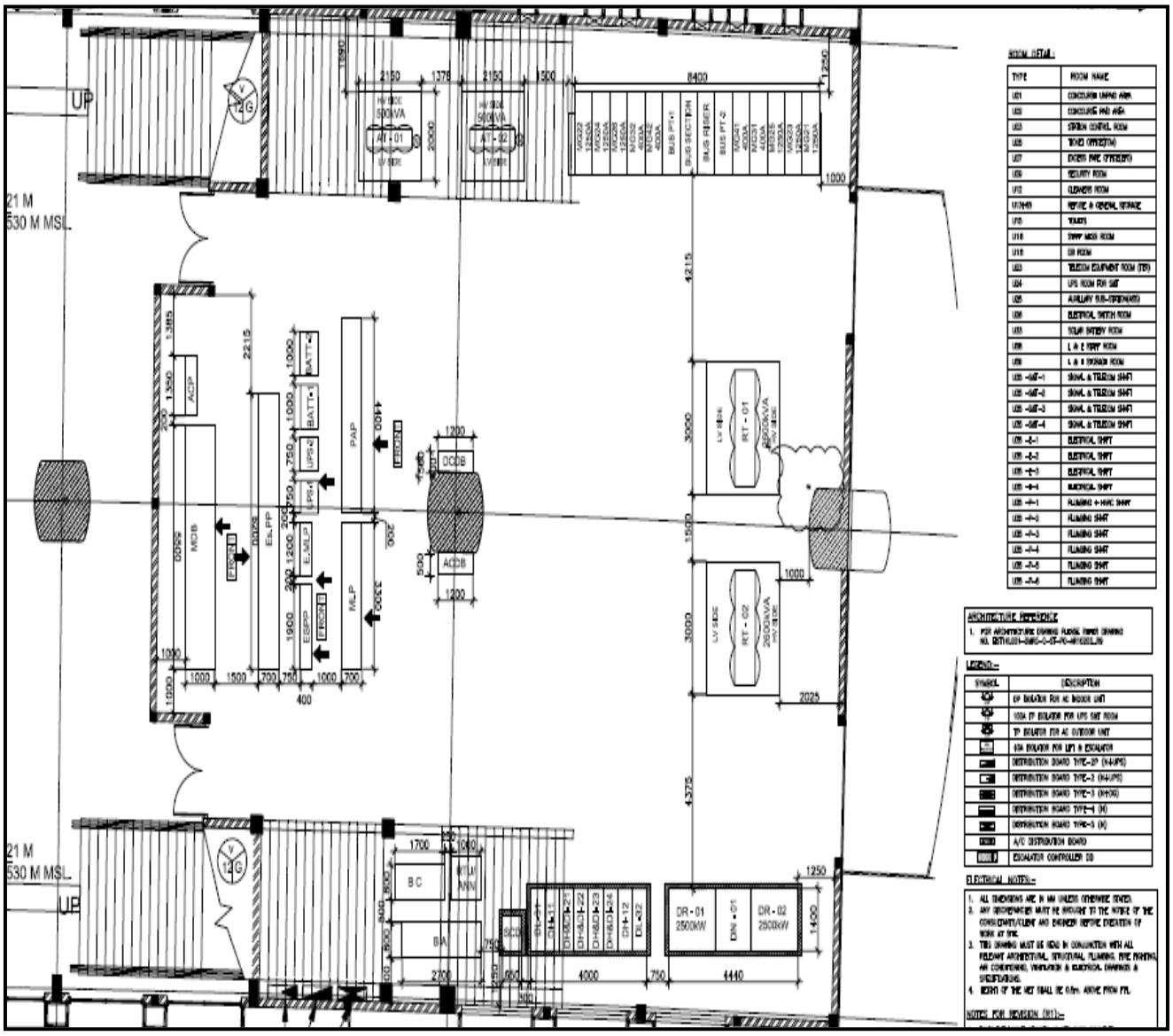


The cross-section of third rail will be about 5000 mm^2. The longitudinal resistance of composite and steel third rail is about 7 and 20 m-ohm/km respectively. The life of composite and steel third rail is expected to be 25-30 years.

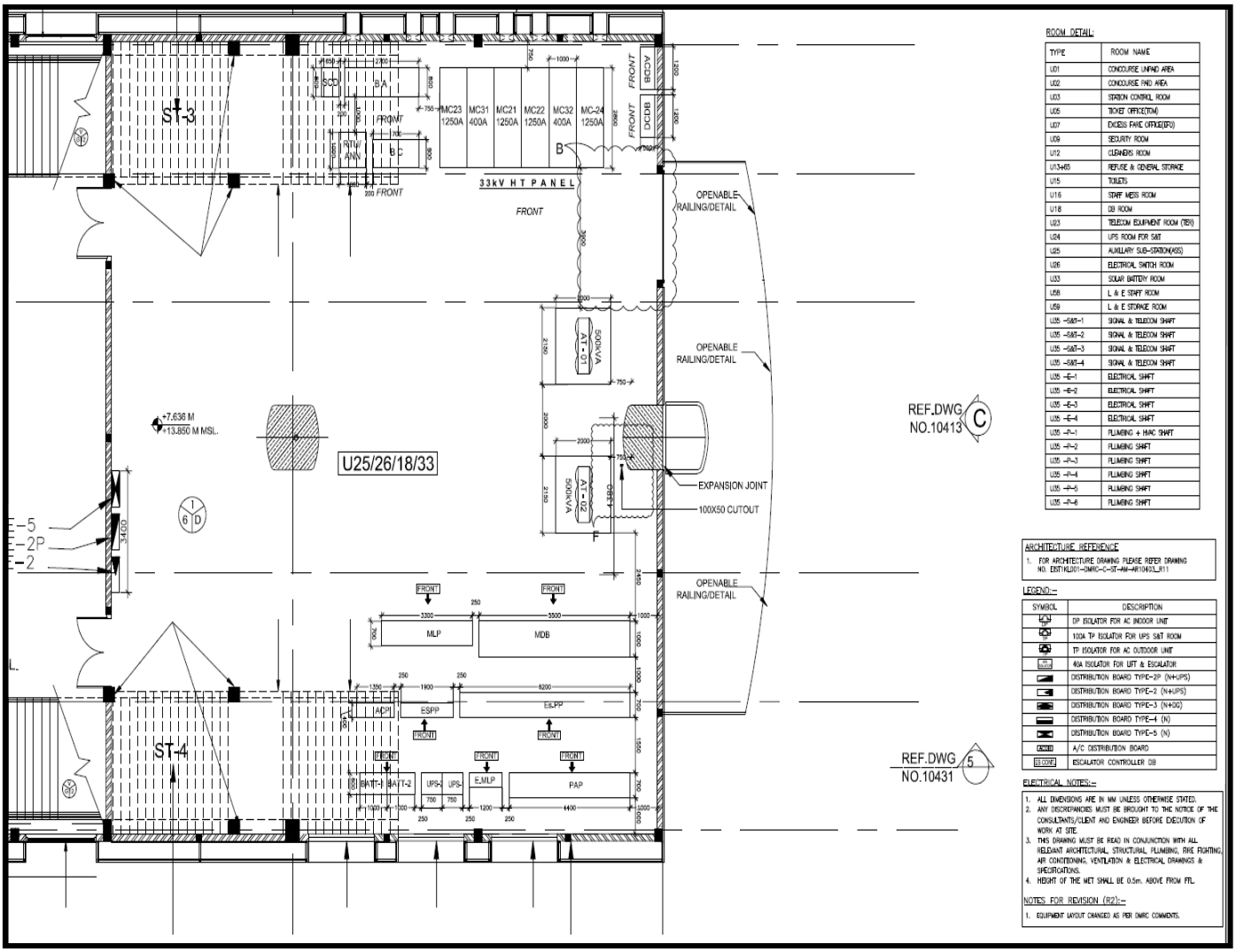
9.6 Traction Sub-stations (33kV/750V dc)

Traction sub-stations (33kV/750V dc) are required to be set up for feeding 750V dc power supply to the third rail. In order to cater to traction load as per train operation plan, it is envisaged to provide traction sub-stations (TSS) at alternate stations. The location of TSS though proposed to be at alternate stations, shall be decided on DC simulation studies. Thus, space provision for TSS shall be kept in all stations during planning stage. The TSS along with Auxiliary Sub-Stations (ASS) will be located at station building itself at mezzanine or platform level inside a room. The typical layouts for TSS & ASS are given below. The requirement comes to 10 TSS for Corridor – 1 and 9 TSS for Corridor – 2. An

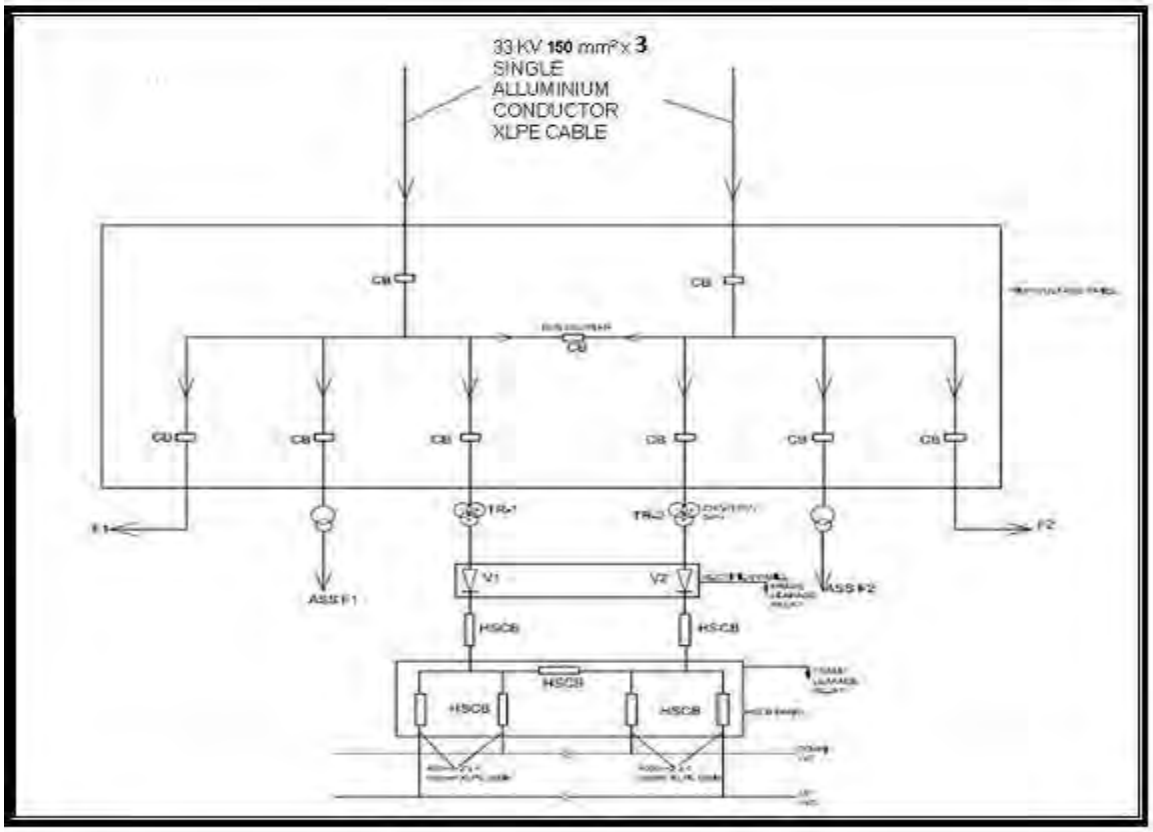
additional TSS will be located in maintenance depot. Thus the total requirement of TSS works out to 11 and 10 for the Corridor – 1 and Corridor – 2 respectively.



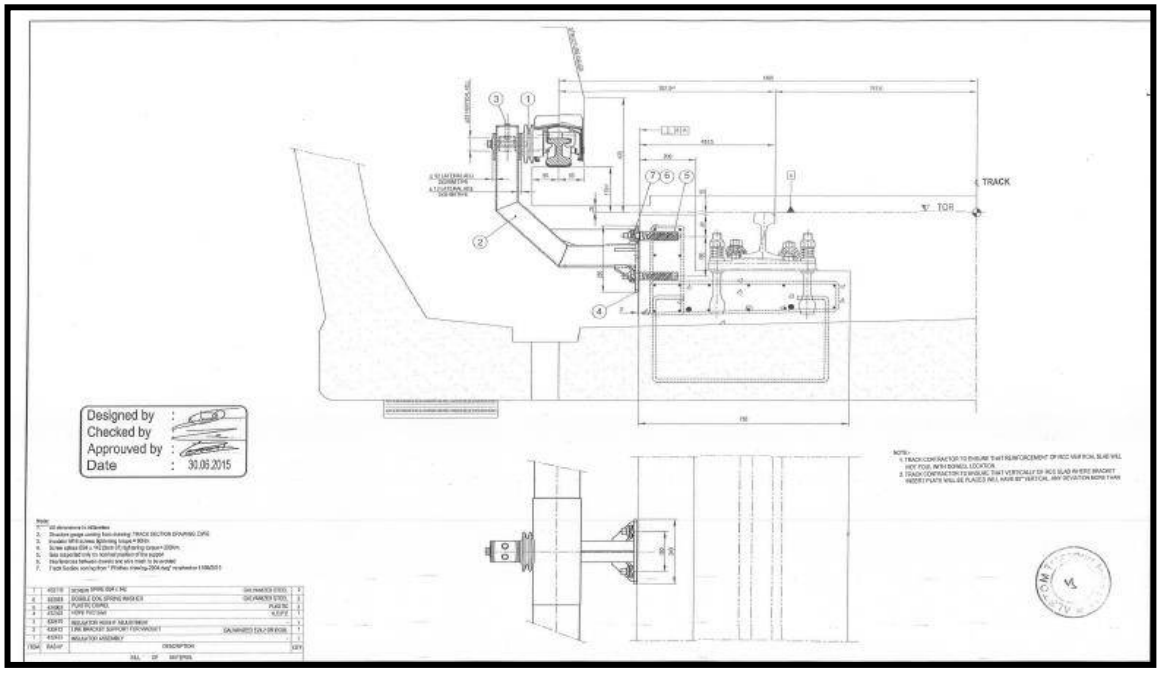
TYPICAL DRAWING OF ASS & TSS



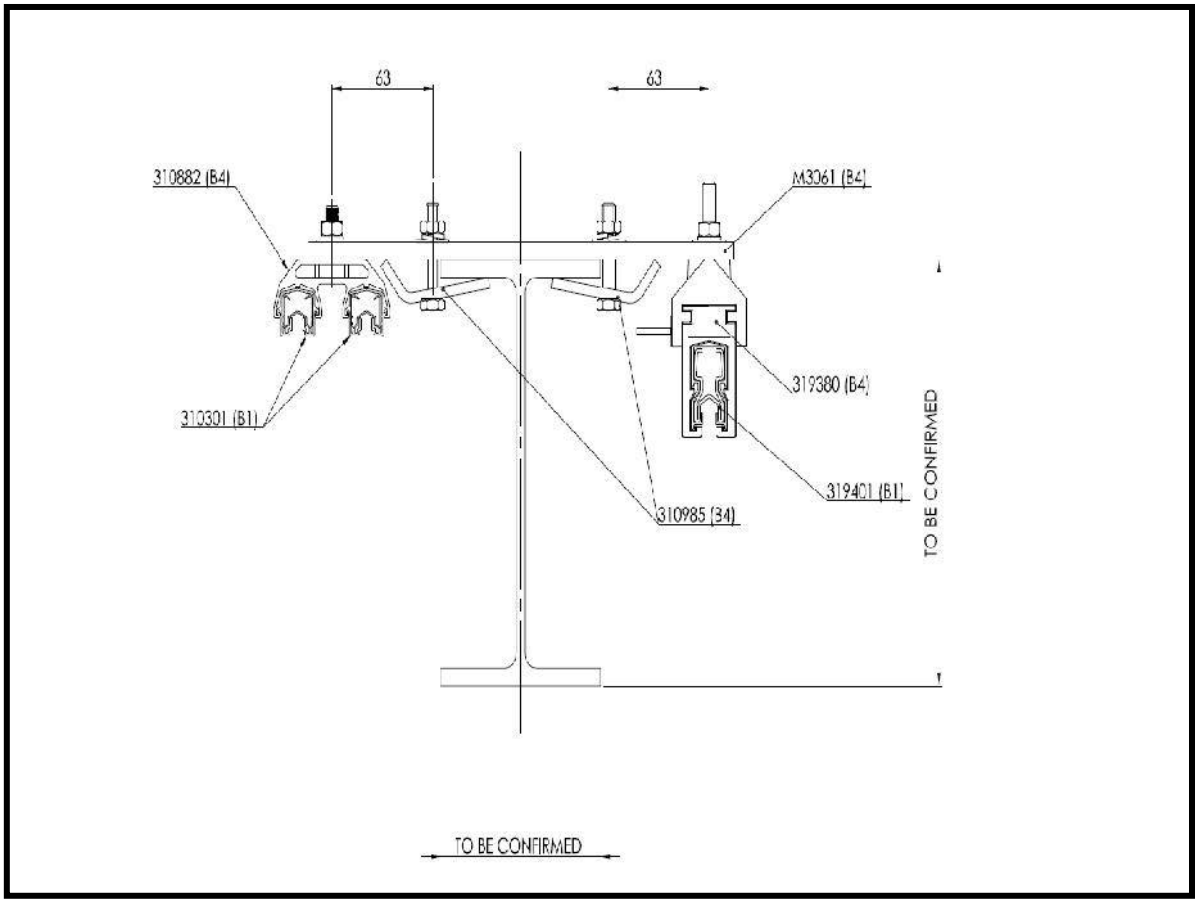
TYPICAL DRAWING OF ASS



Schematic Diagram of DC Traction



LINE BRACKET SUPPORT ASSEMBLY FOR VIADUCT



SUSPENSION ASSEMBLY IN DEPOT

9.6.1 Stray Current Corrosion Protection Measures

Concept of dc Stray Current Corrosion

In dc traction systems, bulk of return current finds its path back to the traction sub-station via the return circuit i.e. running rails. The running rails are normally insulated to minimize leakage of currents to the track bed. However, due to leaky conditions, some current leakage takes place, which is known as 'stray current'. The current follows the path of least resistance. Return current deviates

from its intended path if the resistance of the unintended path is lower than that of intended path. The stray current may flow through the unintended path of metallic reinforcements of the structure back to the sub-station. It

is also possible that part of the stray current may also flow into soil, where it may be picked up by metallic utilities and discharged back to soil and then to near the sub-station.

The dc stray currents cause metal detraction in watery electrolytes as per the following chemical reactions:-

Stray current enters in the metal

$2H_2O + 2e^-$ $H_2 + 2OH^-$ (development of Hydrogen gas)

Stray current exits from metal

Fe $Fe^{2+} + 2e^-$ (Fe^{2+} ions migrate away from the metal)

That is how, dc stray currents cause corrosion of metallic structure where it leaves the metal. This is shown in figure 9.4. Pitting and general form of corrosion are most often encountered on DC Electrified Railways.

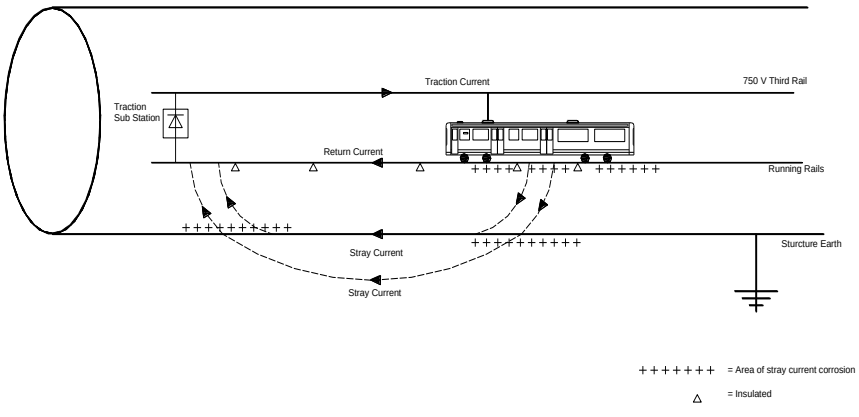


FIGURE 8.4 -- DC TRACTION SYSTEM : STRAY CURRENT CORROSION CONCEPT

Effect of Corrosion

Detraction rate of metals can be calculated by Faraday's First Law:

m = c.i.t

Where m = mass (kg)
c = Coefficient of detraction (kg/Amp.year)
i = Current (Amp)
t = time (year)
c = 2.90 for Aluminium
= 33.80 for Lead
= 9.13 for Iron
= 10.4 for Copper

That means dc stray current of 1 – ampere flowing continuously can eat away approx. 9 kg of steel in a year. If 5000 amperes of current flows for one year to power the trains on a transit system, and that 2 percent of this current (100 amperes) leaks as stray current, the amount of steel metal loss is 0.9 ton per year. Therefore, the safety implications are considerable for structural reinforcements. In addition, corrosion may also affect neighboring infrastructure components such as buried pipelines and cables.

Earthing & Bonding

The earthing installation at the RSS, stations, ASS+TSS at the stations and depots shall be provided to cover all the buildings, structures, passenger ways and connected structures during operation or non-operation hours to achieve;

1. Safety of operating personnel and other persons from electrical shock.
2. Minimum of electrical interference between the electrical power supply and other electrical and electronic systems.
3. Minimum of disturbance to existing statutory services and parts of the Metro network system due to any electrolytic corrosion effects arising from dc traction currents flowing to and from the general body of the earth.

Main Line Earthing for DC traction system

1. Floating system (i.e. traction system with floating negative) is to be adopted for all sections except depot area. Under this system electrical isolation from the earth shall be provided with an insulation gasket where ever required.
2. The running rails shall be adequately insulated as per EN 50122-2.
3. Stray current cables: Stray collection cable shall be provided along the viaduct and in the tunnel. The stray collection cable shall enter each traction substation building and terminate on a stray current bus bar adjacent to the

DC negative bus bar. The stray current bus bar shall be insulated to earth and shall not be connected to either the rails or to the traction negative bus bar. It shall be connected or bonded at intervals to track plinth sacrificial bars or stray current collection mat, made of continuous longitudinal steel cage, installed within the track support concrete. The stray current collection cable preferably of aluminum. Stray current collection can never be used as a general earth.

4. In addition, provisions shall be made for continuous monitoring of the stray current as per EN-50122 at multiple locations through SCADA system.
5. Trunk earth conductor/System/structural earth cable: A trunk earth conductor of appropriate cross section preferably ACSR shall be provided throughout the viaduct & in the tunnels on up and down lines and to the places where it is required for the earthing without endangering the safety of person.
6. The DC feeder protection, switchgear and return current bonding shall be specified, designed, and maintained so that all short circuits from the conductor rails to the running rails and items bonded to the running rails are interrupted in a sufficiently short time that touch voltages at all points in the system, as functions of time, do not exceed the maximum levels specified in EN 50122-1.
7. The metallic or steel structures like cable trays, pipe lines, railings, trusses, foot over bridges, road over bridges, advertising structures, equipment with metallic frames that can come in contact with the commuters and are in the influence zone must ensure electric isolation every 10-12 meters using insulation gasket for providing electrical isolation and separately earthed. The chances of accidental potential difference under fault condition due to separate earthing points should be minimized.

Earthing System for AC supplies

1. The Earthing system for AC power supplies shall be maintained distinct and isolated from the DC system for the Main line. The AC earthing shall be provided at RSS, ASS at stations, depots and OCC.

Depot

2. An earthed system shall be used for Depot area.
3. A separate TSS shall be provided for depot so as to facilitate isolation of depot traction supply from mainlines in order to prevent the leakage of return currents to depot area.

4. For safety reasons, the system in depot is negative earthed one therefore, the mainline tracks and OCS supply shall be isolated from depot tracks through double insulted rail joints and sectioning in order to minimize the stray currents, even during the movement of the trains. Remote operated sectionalizing switches will be provided for operational exigencies.
5. Special measures for safety of person in wheel lathe area and washing lines area be provided and at the same time minimize the stray currents as leaky conditions exists in such areas. The non-electrified tracks shall be provided with the IRJs

Measures for Protection against Stray Current Corrosion

Earthing & bonding and protection against stray current corrosion are inter-related and conflicting issues. Therefore, suitable measures are required to suppress the stray currents as well as the presence of high touch potentials. Safety of personnel is given preference even at a cost of slightly increased stray currents. Following measures are required to restrict the stay current:-

(i) Decreasing the resistance of rail-return circuit
(ii) Increasing the resistance of rail to ground insulation

Whenever buried pipes and cables are in the vicinity of dc systems, efforts shall be made to ensure that metal parts are kept away as far as practicable to restrict stray current. A minimum distance of 1 meter has been found to be adequate for this purpose.

Generally, three types of earthing arrangements (viz. Earthed System, Floating System & Hybrid Earthing System) are prevalent on metros World over for protection against stray current corrosion. Traditionally, earthed system was used by old metros. Hybrid earthing system is being tried on experimental basis on few new metros. Floating system has been extensively used by recent metros. As per global trends, floating system (i.e. traction system with floating negative) is preferred. It reduces the dc stray current considerably. The arrangement shall comply with the following latest CENELEC standards:-

EN 50122-1:- Railway Applications (fixed installations) protective provisions relating to electrical safety & earthing

EN 50122-2:- Railway Applications (fixed installations) protective provisions against the effects of stray currents caused by dc traction system

The conceptual scheme of the proposed floating system is described as follows:-

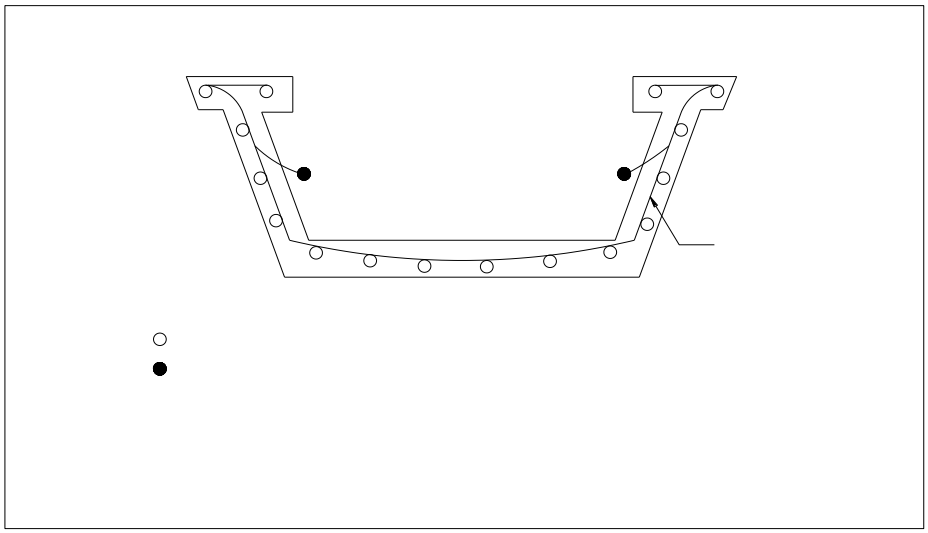
i) The running rails shall be adequately insulated as per EN50122-2. The recommended conductance per unit length for single track sections are as under:-

Elevated section: - 0.5 Siemens/Km

Tunnel section: - 0.1 Siemens/Km.

ii) Stray Current Collector Cables {commonly known as structural earth (SE) cable} (2x200 mm^2 copper) shall be provided along the viaduct and all the metallic parts of equipment, cable sheath, viaduct reinforcement, signal post etc. shall be connected to SE cable.

iii) The continuity of the reinforcement bars of the viaduct as well as track slabs has to be ensured along with a tapping point for connection with SE cable in order to drain back the stray current. The typical arrangement of connecting the reinforcements of viaduct is shown in figure 9.5.



iv) A provision shall be made to earth the running rail (i.e. negative bus) in case of rail potential being higher than limits prescribed (120V) in relevant standard (EN 50122-1) in order to ensure safety of personnel. This will be achieved by providing track earthing panel (TEP) at stations close to platform and at traction sub-stations.

v) In addition, provisions shall be made for connection of SE cable to negative return path through diode only for the purpose of periodical monitoring of stray currents. Under normal operations, switch provided for this connection will be in normally open (NO) position and switch will be closed for monitoring of stray current once or twice in a year as required.

vi) The third rail installation shall be kept adequately away from the drain for the seepage water in the tunnel section.

The proposed scheme is shown in figure 9.6

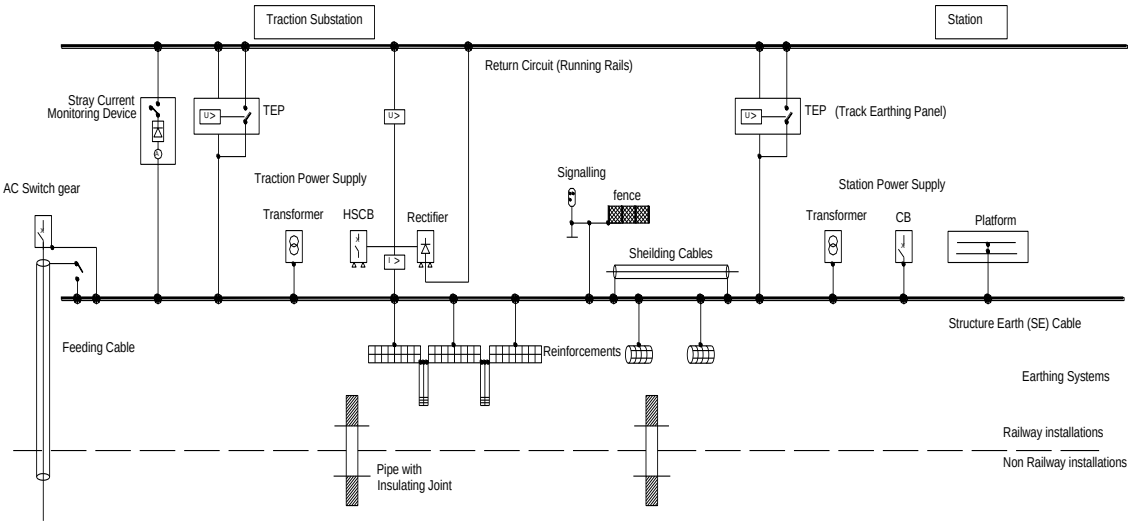


FIGURE 8.6 -- BASIC DIGRAM FOR EARTHING , BONDING AND STRAY CURRENT PROTECTION MEASURES

Special Arrangements in Depot

A separate traction sub-station (TSS) shall be provided for the depot so as to facilitate isolation of depot traction supply from mainlines in order to prevent the leakage of return currents to depot area. Tracks of Depot area shall also be isolated from mainline through insulated rail joints (IRJ). Remote operated sectionalizing switches shall be provided to feed power from depot to mainline and vice-versa in case of failure of TSS.

The prescribed limit of highest touch potential in depot is 60V as per EN50122-1 and therefore Track Earthing Panels (TEP) shall be provided at suitable locations to earth the rail in case the rail potential exceeds this limit. In areas, where leaky conditions exist (e.g. washing lines, pit wheel lathe etc.), insulated rail joints (IRJ) shall be provided with power diodes to bridge the IRJ to facilitate passage of return current.

A detailed scheme shall be developed during the design stage.

9.6.2 Electromagnetic Interference (EMI) and Electromagnetic Compatibility (EMC)

AC traction currents produce alternating magnetic fields that cause voltages to be induced in any conductor running along the track. However, dc traction currents do not cause electromagnetic induction effect resulting in induced voltages and magnetic fields.

The rectifier-transformer used in dc traction system produces harmonic voltages, which may cause interference to telecommunications and train control/protection systems. The rectifier-transformer shall be designed with the recommended limits of harmonic voltages, particularly the third and fifth harmonics. 12-pulse rectifier-transformer has been proposed, which reduces the harmonics level considerably. Detailed specification of equipment e.g. power cables, rectifiers, transformer, E&M equipment etc shall be framed to reduce conducted or radiated emissions as per appropriate international standards. The Metro system as a whole (trains, signaling & telecomm, traction power supply, E&M system etc) shall comply with the EMC requirements of international standards viz. EN50121, EN50123, IEC61000 series etc. A detailed EMC plan will be required to be developed during project implementation stage.

9.6.3 Auxiliary Supply Arrangements for Stations

Auxiliary sub-stations (ASS) are envisaged to be provided at each station. The ASS will be located at mezzanine or platform level inside a room. Wherever TSS is required, ASS & TSS will be housed together inside a room. The auxiliary load requirements have been assessed at 300 kW for elevated/at-grade stations and 2200 kW for underground station. Accordingly, two dry type cast resin transformers (33/0.415kV) of 315 kVA capacity are proposed to be installed at the stations (one transformer as standby) for elevated station. Metro Bhawan & OCC to be located at Dream city Depot, one ASS with 2x2.5 MVA transformer will be required. The underground station will be provided with two ASS's having 1x2.5 MVA transformer.

2 x 2.5 MW transformer-rectifier for corridor-1 and 2 x 2.0 MW for corridor-2 transformer-rectifier set for feeding traction load shall be provided in each TSS with space provisions for an additional set to be accommodated in future as and when the cars per train composition is increased.

Self-cooled, cast resin dry type rectifier-transformer is proposed, which is suitable for indoor application. From the traction sub-stations, 750V dc cables will be laid up to third rail and return current cables will be connected to running rails.

9.6.4 Auxiliary Supply Arrangements for DEPOT

Following are the major plant and machinery has been used in Depot.

RRV for carrying re railing equipments

Road vehicles (pick up van/ truck)

Flat wagon for carrying material.

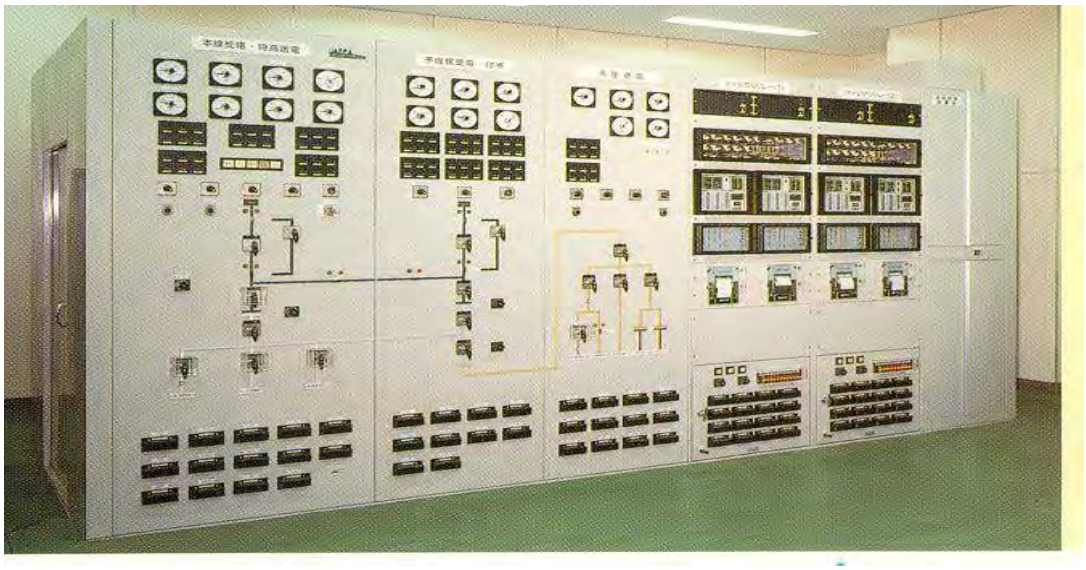
Diesel/Electric battery powered locomotive with traction battery charger.

Under floor Pit wheel lathe, chip crusher and conveyor for lathe on pit, Electric tractor for movement over under floor wheel lathe.

Travelling O/H crane workshop 15T/3T,1.5T capacity(IBL),ETU shed 5T crane

Mobile Jib crane

For feeding above load, a separate ASS is required at each depot and metro bhawan. The Depot ASSs will also be provided with 2x2500 kVA auxiliary transformers.



Typical Indoor Auxiliary Sub-station

9.6.5 Rating of Major Equipment

Based on emergency demand expected at each RSS as shown in Table 9.3, and expected power demand during emergency, RSS near Nature Park Station/ Kapodara and at Dream City depot/ near Althan Tenement shall be provided with 2 nos. of (One to be in service and one as standby) 66/33 kV, 40/50 MVA three

phase transformers for feeding traction as well as auxiliary loads. RSS at Bhesan depot & near Magob Station will be provided with 2 nos. of (one to be in service and one as standby) 66/33kV, 20/30 MVA (ONAN/ONAF) three phase auxiliary transformers for feeding traction and auxiliary loads. The incoming cable 3-phase single core XLPE insulated with 630 mm^2 Aluminum conductors for corridor-1 and 500 mm^2 Aluminum conductors for corridor-2 shall be to meet the normal & emergency loading requirements and fault level of the 66 kV supply.

Traction transformer-rectifier set (33 kV/750 V dc) shall be of 2.5 MW (for corridor-1) and 2.0 MW (for corridor-2) rated capacity with overload requirement of 150% for 2 hours with four intermittent equally spaced overloads of 300% for 1 minute, and with one 450% full load peak of 15 seconds duration at the end of 2 hour period. The traction transformer - rectifier set shall produce 750 V dc nominal output voltage with 12-pulse rectification so as to minimize the ripple content in the output dc voltage. The IEC 850 international standard envisages the minimum and maximum voltages of 500V and 900V respectively for 750 V dc traction system and therefore, the dc equipment shall be capable of giving desired performance in this voltage range.

33kV cable network shall be adequately rated to transfer requisite power during normal as well as emergency situations and to meet the fault current requirement of the system. FRLS Cable for Elevated section and FRLSOH cable for U/G section are proposed to be used. Accordingly, proposed 33kV cables sizes are as under:-

3, Single core x 240 mm^2 Copper conductor (double run) XLPE insulated for 33kV ring main cable network for corridor-1.

3, Single core x 400 mm^2 Copper conductor (single run) XLPE insulated for 33kV ring main cable network for corridor-2.

Adequate no. of cables are required for transfer of power from TSS to third rail. Single-phase XLPE insulated cables with 300mm^2 copper conductor are proposed for 750V dc as well as return current circuit. Based on current requirements, 3 cables are required for each of the Four circuits to feed power to third rail.

The above capacities of transformers, cables etc. have been worked out based on the conceptual design. Therefore, these may be required to be revised for better accuracy during design stage of project implementation.

9.7 MV/LV system

Following major E&M Equipments/system shall be required for elevated stations:-

MV/LV panels

DG set
UPS & Battery system
Lifts (Typical four per station)
Escalators (Typical four per station)
Fire suppression, detection system, Fire Pump and water pump.
Lights & fans
Air conditioning system
BMS system
Lightning protection system
Earthing system

Panels shall be front operated front access cubical type indoor duty floor mounted totally enclosed dust and vermin proof with neoprene gaskets fabricated from CRCA sheet with powder coated finish suitable for 415 V 3 Phase 4 wire 50Hz system.

Power for PD area will be given through separate feeder in case of large area is given through station supplies using separate meters for small loads.

9.8 Standby Diesel Generator (DG) Sets

In the unlikely event of simultaneous tripping of all the input power sources or grid failure, the power supply to stations as well as to trains will be interrupted. It is, therefore, proposed to provide a standby DG set of 160 kVA capacity at the elevated stations and 2x900 kVA at underground station to cater to the following essential services:

(i) Essential lighting
(ii) Signaling & telecommunications
(iii) Fire fighting system
(iv) Lift operation
(v) Fare collection system
(vi) Ventilation system
(vii) Smoke management system

Silent type DG sets with low noise levels are proposed, which do not require a separate room for installation.

9.9 Solar Photo Voltaic (PV) Power system

In solar PV power system are installed at various sites in RESCO (Renewable Energy Service Company) model. In DMRC Stations and Depots 14 MWp solar PV power system has been installed in RESCO model.

Solar PV Power panel

"RESCO Model" means where the developers intend to provide solar power system on rooftop/sites owned by DMRC on mutually agreed terms and conditions from DMRC and enters into the PPA (Power purchase agreement) with DMRC for supply of Solar power for 25 years from the date of Commissioning of project.

In elevated stations about 50 kWp to 150 kWp capacity of Solar PV power system can be provided depending upon type of roof availability, shadow free roof area, orientation of stations. In DMRC receiving sub-station 20 kWp to 50 kWp capacity Solar PV systems are generally provided. In DMRC Depot area, approx.1000 kWp to 1500 kWp of solar capacity has been provided. Solar PV system in station parking area can also be planned as per availability of area.

9.10 Sewage Treatment System using Integrated Constructed wetlands(ICW)

For RSS installation, the sewage shall be connected to the city sewage system if available. If not available, sewage treatment system using integrated constructed wetland (ICW) will be installed. Following are the objectives for providing Sewage Treatment System using Integrated Constructed Wetlands (ICW):-

1) To establish an effective option for treatment of wastewater that is generated from campus.

2) Establish an onsite treatment solution which is effective and cost effective option without producing any by products.
3) To establish a sustainable and environmental friendly solution with minimal maintenance.
4) The treated water can be reused for various non-portable applications landscaping, flushing and cleaning.

9.11 Constructed Wetlands

The objective of Constructed Wetlands is to utilize the decomposable organic matter present in sewage, which can be disposed of into the environment without causing health hazards or nuisance. The degree of treatment to be adopted would meet the regulatory agencies (surface water discharge standards).

Constructed wetlands (CW) are complex and modular system provides an efficient and sustainable purification treatment method that is applicable to practically all pollutant sources and in all climate and environmental conditions. CW relies on Constructed Wetlands, and is based on the activity of plants together with microorganism communities in the root zone. Together they degrade, accumulate, extract, and volatilize contaminants of all kinds in water, soil and the air, resulting in clean and purified outflow.

In DMRC Faridabad RSS 1 KLD capacity Sewage Treatment System provided through integrated constructed wetland method.

9.12 Supervisory Control and Data Acquisition (SCADA) System

The entire system of power supply (receiving, traction & auxiliary supply) shall be monitored and controlled from a centralized Operation Control Centre (OCC) through SCADA system. Modern SCADA system with intelligent remote terminal units (RTUs) shall be provided. Optical fiber provided for telecommunications will be used as communication carrier for SCADA system.

Digital Protection Control System (DPCS) is proposed for providing data acquisition, data processing, overall protection control, interlocking, inter-tripping and monitoring of the entire power supply system consisting of 33kV ac switchgear, transformers, 750V dc switchgear and associated electrical equipment. DPCS will utilize microprocessor-based fast-acting numerical relays & Programmable Logic Controllers (PLCs) with suitable interface with SCADA system.

9.13 Energy Saving Measures

Energy charges of any metro system constitute a substantial portion of its operation & maintenance (O & M) costs. Therefore, it is imperative to incorporate energy saving measures in the system design itself. The auxiliary power consumption of metros is generally more than the traction energy consumed by train movement during initial years of operation. Subsequently, traction power consumption increases with increase in train frequency/composition in order to cater more traffic. The proposed system of Surat Metro includes the following energy saving features:

(i) Modern rolling stock with 3-phase VVVF drive and lightweight stainless steel coaches has been proposed, which has the benefit of low specific energy consumption and almost unity power factor.

(ii) Rolling stock has regeneration features and it is expected that 18% of total traction energy will be regenerated and fed back to 750 V dc third rail to be consumed by nearby trains.

(iii) Effective utilization of natural light is proposed. In addition, the lighting system of the stations will be provided with different circuits (33%, 66% & 100%) and the relevant circuits can be switched on based on the requirements (day or night, operation or maintenance hours etc).

(iv) Machine-room less type lifts with gearless drive has been proposed with 3-phase VVVF drive. These lifts are highly energy efficient.

(v) The proposed heavy-duty public services escalators will be provided with 3-phase VVVF drive, which is energy efficient & improves the power factor. Further, the escalators will be provided with infrared sensors to automatically reduce the speed (to idling speed) when not being used by passengers.

(vi) The latest state of art and energy efficient electrical equipment (e.g. transformers, motors, light fittings etc) has been incorporated in the system design.

(vii) Efficient energy management is possible with proposed modern SCADA system by way of maximum demand (MD) and power factor control.

(viii) LED lights to be used in the station area and Depot area.

9.14 Electric Power Tariff

The cost of electricity is a significant part of Operation & Maintenance (O&M) charges of the Metro System, which constitutes about 30-38% of total annual

working cost. Therefore, it is the key element for the financial viability of the Project. The annual energy consumption is assessed to be about 73.81 million units for Corridor-1 and 30.39 Million units for Corridor-2 in initial years (2021), which will be about 109.64 Million Units and 48.50 Million Units in the year 2046 respectively. In addition to ensuring optimum energy consumption, it is also necessary that the electric power tariff be kept at a minimum in order to contain the O& M costs. Therefore, the power tariff for Surat Metro should be at effective rate of purchase price (at 66 kV voltage level) plus nominal administrative Charges i.e. on a no profit no loss basis. The power tariff off Gujarat Electricity Regulatory Commission for the railways for financial Year 2016-17 demand charges Rs.180/KVA/month and energy charges Rs 5.00/Kwh. This is expected to be in the range of Rs 5.25 to Rs 5.50 per unit. It is proposed that Government of Gujarat takes necessary steps to fix power tariff for Surat Metro at “No Profit No Loss” basis. Similar approach has been adopted for Delhi Metro. For keeping the Electricity cost at the minimum & optimum level, following suggestion are indicated:

Surat Muncipal Corporation should not levy any municipal taxes on Electricity for Surat Metro.

DISCOMS to be pursued to give Electricity on cost to serve basis. A separate category of Electricity tariff for Metro system may be finalized by the Regulatory Commission.

DISCOMS shall not object or levy charge/ Gross-subsidy wheeling charges, any other charges etc. if Surat Metro decides to go for open access to reduce the cost of Energy and thereby the operating expenses of Metro.

For obtaining power connection from DICOMS GSS, Metro normally pays only for the bay commission charges. Torrent/GETCO have indicated to charge all expenditure relating to modification/ addition/ alteration will be recovered from the Surat Metro to this project. This may be dealt by Surat Municipal Corporation separately for arriving at a reasonable connection charges.

SURAT METRO **Annexure-9.1**

CORRIDOR-1 SARTHANA TO DREAM CITY

POWER (Traction & Auxiliary)			750V DC Traction System			
S.No.	**Particulars**	**Unit**	**2021**	**2026**	**2036**	**2046**
A	**Traction Power Requirement**	1	2	3	4	5
1	No. of cars	(2DMC+1 TC)	3	3	3	3
2	Passenger Weight	T	63.18	63.18	63.18	63.18
3	Train Tare Weight	T	128.00	128.00	128.00	128.00
4	Total Train Weight	T	191.18	191.18	191.18	191.18
5	Section Length	km	21.61	21.61	21.61	21.61
6	Headway	mts	6.50	3.50	2.25	1.75
7	SEC at Pantograph/ current Collector (As per MOUD guideline vide letter No. 14011/9/2014-UT.II-Part I, dated 21.04.2017)	KWhr/ 1000 GTKM	60	60	60	60
8	No. of Trains/hr in both directions	Nos.	18	34	53	69
9	Peak Traction Power Requirement	MW	4.58	8.50	13.22	17.00
11	Depot Power Requirements	MW	1.00	1.30	1.40	1.50
12	No. of Depot	No	1	1	1	1
13	Total Traction Power Requirement	MW	5.58	9.80	14.62	18.50
	Total Traction Power Requirement (MVA) assuming 5% energy losses and 0.95 pf	**MVA**	**6.16**	**10.83**	**16.16**	**20.44**
B	**Aux. Power Requirement**					
1	Elevated/at-grade Station Power Consumption	MW	0.25	0.26	0.28	0.30
2	Underground station Power Consumption	MW	2.00	2.05	2.10	2.20
3	No. of Elevated/at-grade Stations	Nos.	14	14	14	14
4	No. of Underground stations	Nos.	6	6	6	6
5	Total Station Aux Power Requirement	MW	15.50	15.94	16.45	17.40
6	Depot Aux Power Requirement	MW	2.00	2.05	2.10	2.20
7	Metro Bhawan & OCC Building	MW	2.00	2.05	2.10	2.20
8	No. of Depot	No.	1	1	1	1
9	No. Metro Bhawan & OCC Building	No.	1	1	1	1
10	Total Aux Power Requirement	MW	19.50	20.04	20.65	21.80
	Total Aux. Power Requirement (MVA) assuming 5% energy losses and 0.85 pf for aux loads	**MVA**	**24.09**	**24.76**	**25.51**	**26.93**
C (A+B)	**Total Traction & Aux. Power Requirement (MVA)**	**MVA**	**30.25**	**35.59**	**41.67**	**47.37**

Note: The requirement of PD load is not considered in estimation of power calculation.

SURAT METRO

CORRIDOR-2 BHESHAN TO SAROLI

POWER (Traction & Auxiliary)			750V DC Traction System			
S.No.	**Particulars**	**Unit**	**2021**	**2026**	**2036**	**2046**
A	**Traction Power Requirement**	1	2	3	4	5
1	No. of cars	(2DMC+1 TC)	3	3	3	3
2	Passenger Weight	T	63.18	63.18	63.18	63.18
3	Train Tare Weight	T	128.00	128.00	128.00	128.00
4	Total Train Weight	T	191.18	191.18	191.18	191.18
5	Section Length	km	18.74	18.74	18.74	18.74
6	Headway	mts	10.00	8.00	3.75	3.00
7	SEC at Pantograph/ current Collector (As per MOUD guideline vide letter No. 14011/9/2014-UT.II-Part I, dated 21.04.2017)	KWhr/ 1000 GTKM	60	60	60	60
8	No. of Trains/hr in both directions	Nos.	12	15	32	40
9	Peak Traction Power Requirement	MW	2.58	3.22	6.88	8.60
11	Depot Power Requirements	MW	1.00	1.30	1.40	1.50
12	No. of Depot	No	1	1	1	1
13	Total Traction Power Requirement	MW	3.58	4.52	8.28	10.10
	Total Traction Power Requirement (MVA) assuming 5% energy losses and 0.95 pf	**MVA**	**3.96**	**5.00**	**9.15**	**11.16**
B	**Aux. Power Requirement**					
1	Elevated/at-grade Station Power Consumption	MW	0.25	0.26	0.28	0.30
2	Underground station Power Consumption	MW	2.00	2.05	2.10	2.20
3	No. of Elevated/at-grade Stations	Nos.	18	18	18	18
4	No. of Underground stations	Nos.	0	0	0	0
5	Total Station Aux Power Requirement	MW	4.50	4.68	4.95	5.40
6	Depot Aux Power Requirement	MW	2.00	2.05	2.10	2.20
7	No. of Depot	No.	1	1	1	1
8	Total Aux Power Requirement	MW	6.50	6.73	7.05	7.60
	Total Aux. Power Requirement (MVA) assuming 5% energy losses and 0.85 pf for aux loads	**MVA**	**8.03**	**8.31**	**8.71**	**9.39**
C (A+B)	**Total Traction & Aux. Power Requirement (MVA)**	**MVA**	**11.99**	**13.31**	**17.86**	**20.55**

Note: The requirement of PD load is not considered in estimation of power calculation.

SURAT METRO

CORRIDOR-1 SARTHANA TO DREAM CITY

ENERGY CONSUMPTION		750V DC Traction System				
S.No.	**Particulars**	**Unit**	**2021**	**2026**	**2036**	**2046**
A	**Traction Energy**	1	2	3	4	5
1	Section Length	KM	21.61	21.61	21.61	21.61
2	No. of Trains per direction in a day*	Nos.	94	153	207	255
3	Weight of Train & Passenger	T	191.18	191.18	191.18	191.18
4	SEC at Pantograph/ current Collector (As per MOUD guideline vide letter No. 14011/9/2014-UT.II-Part I, dated 21.04.2017)	KWH/ 1000 GTKM	60	60	60	60
	Yearly Traction Energy consumption with 365 days working	**million units**	**17.01**	**27.69**	**37.46**	**46.14**
B	**Auxiliary Energy**					
1	Elevated/at-grade Station	MW	0.25	0.26	0.28	0.30
2	Underground Station	MW	2.00	2.05	2.10	2.20
3	No. of Elevated/at-grade Stations	Nos.	14	14	14	14
4	No. of Underground Stations	Nos.	6	6	6	6
5	Total Station Aux. Power Requirement	MW	15.50	15.94	16.45	17.40
6	Depot Aux power requirement	MW	2.00	2.05	2.10	2.20
7	Metro Bhawan & OCC Building	MW	2.00	2.05	2.10	2.20
8	No. of Depot	No.	1	1	1	1
9	No. Metro Bhawan & OCC Building	No.	1	1	1	1
10	Total Aux. Power Requirement	MW	19.50	20.04	20.65	21.80
11	Total Aux. Power Requirement (MVA) assuming 5% energy losses and 0.85 pf for Aux. loads	**MVA**	**24.09**	**24.76**	**25.51**	**26.93**
12	Diversity Factor of Aux. loads		0.40	0.40	0.40	0.40
	Yearly Aux. Energy Consumption 19 hrs/day and 365 days working (million units)	**million units**	**56.80**	**58.37**	**60.15**	**63.50**
C (A+B)	**Net Annual Energy Consumption (Traction & Aux.)**	**million units**	**73.81**	**86.06**	**97.61**	**109.64**

Note: The requirement of PD load is not considered in energy calculation.

SURAT METRO

CORRIDOR-2 BHESHAN TO SAROLI

ENERGY CONSUMPTION		750V DC Traction System				
S.No.	**Particulars**	**Unit**	**2021**	**2026**	**2036**	**2046**
A	**Traction Energy**	1	2	3	4	5
1	Section Length	KM	18.74	18.74	18.74	18.74
2	No. of Trains per direction in a day*	Nos.	73	93	144	168
3	Weight of Train & Passenger	T	191.18	191.18	191.18	191.18
4	SEC at Pantograph/ current Collector (As per MOUD guideline vide letter No. 14011/9/2014-UT.II-Part I, dated 21.04.2017)	KWH/ 1000 GTKM	60	60	60	60
	Yearly Traction Energy consumption with 365 days working	**million units**	**11.46**	**14.59**	**22.60**	**26.36**
B	**Auxiliary Energy**					
1	Elevated/at-grade Station	MW	0.25	0.26	0.28	0.30
2	Underground Station	MW	2.00	2.05	2.10	2.20
3	No. of Elevated/at-grade Stations	Nos.	18	18	18	18
4	No. of Underground Stations	Nos.	0	0	0	0
5	Total Station Aux. Power Requirement	MW	4.50	4.68	4.95	5.40
6	Depot Aux power requirement	MW	2.00	2.05	2.10	2.20
7	No. of Depot	No	1	1	1	1
8	Total Aux. Power Requirement	MW	6.50	6.73	7.05	7.60
9	Total Aux. Power Requirement (MVA) assuming 5% energy losses and 0.85 pf for Aux. loads	**MVA**	**8.03**	**8.31**	**8.71**	**9.39**
10	Diversity Factor of Aux. loads		0.40	0.40	0.40	0.40
	Yearly Aux. Energy Consumption 19 hrs/day and 365 days working (million units)	**million units**	**18.93**	**19.60**	**20.53**	**22.14**
C (A+B)	**Net Annual Energy Consumption (Traction & Aux.)**	**million units**	**30.39**	**34.20**	**43.13**	**48.50**

Note: The requirement of PD load is not considered in energy calculation.

Annexure-9.2

torrent POWER

Ref. : Technical/ 397 11.04.2017

To,
Executive Engineer,
Light & Energy Efficiency Cell
Surat Municipal Corporation
Muglisara
Surat 395003

Dear Sir,

Sub : Providing Power Supply to Surat Metro Rail Project

We refer to the discussion we had during the meeting held on 6th April'2017 and the subsequent mail received from the executive engineer on 7th April'2017 regarding the feasibility of the power supply from the existing sources for the Surat metro rail. In continuation to the same we confirm / clarify as under,

1. It is conveyed that the power requirement for both the corridors 1 & 2 shall be 43 MVA and 26 MVA respectively, totalling to 69 MVA in a span of Five years.

2. Surat Metro rail will apply for four connection points from our various sources (220KV/66KV Substations) located at VED, PUNA and Bhatar.

3. The connection points (point of supply) shall be at our 220kV substation. Surat metro shall lay 2 parallel cables from the 66KV Bays provided at our 220 kV substation to its respective RSS located at Althan Gam, Kapodra, Botanical Garden and Magob.

4. The Tentative RSS locations as mentioned in your mail shall have the connection points as under;

TORRENT POWER LIMITED
CIN : L31200GJ2004PLC044068
New Registered Office
"Tapovan" 600 Tapovan, Ambawadi,
Ahmedabad-380015, India.
Torrent House, Station Road, Surat - 395 003. Gujarat. India. Phone : +91 261 2400240, 98240 09930 Fax : +91 2422171
Reg. Office : Torrent House, Off Ashram Road, Ahmedabad - 380 009. India. Phone : +91 79 26585090 www.torrentpower.com

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Feeding Source GSS	RSS (Metro rail substation)	No. of Bays
Bhatar 220/66KV	Althan Gam / Tenement	2
Puna 220KV/66KV	Kapodara	2
Puna 220KV/66KV	Magob	2
Ved 220KV/66KV	Botanical Garden	2

5. To meet the above requirements, TPL will have to undergo a major augmentation/ modification in terms of creating space for the additional bays, capacity enhancement to cater to the required demand, strengthening the upstream network to ensure reliability of the supply etc. In case necessity arises for the requirement of land to do augmentation/ alteration/ addition in the system Surat Municipal Corporation shall allocate the land as per the requirement.

6. Prima facie, TPL will be able to provide the supply, however it is subject to the required modification/ augmentation in the system and the final technical feasibility.

7. The cost associated in the modification/ augmentation/ addition/ alteration will be recovered from the Surat Metro.

8. The operational philosophy (i.e. to run the bays for the respective corridor in parallel) will be based on the outcome of the technical feasibility study of the system which will be taken up by TPL after receiving confirmation from your end. In case of failure of any RSS, the entire load will be on the other RSS, meaning TPL will have to design both the bays to cater to the full contract demand for the respective corridor. SOP to operate the system shall be devised accordingly.

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Ahmedabad-380015, India.
Torrent House, Station Road, Surat - 395 003. Gujarat. India. Phone : +91 261 2400240, 98240 09930 Fax : +91 2422171
Reg. Office : Torrent House, Off Ashram Road, Ahmedabad - 380 009, India. Phone : +91 79 26585090 www.torrentpower.com

torrent POWER

9. This is only the preliminary evaluation of the system. TPL will have to conduct detailed study of its system and will come up with the requirements based on its outcome.

10. The Contract demand for each connection shall be defined individually and billed accordingly. However, we may discuss this aspect at an appropriate stage and adapt a suitable commercial modal available across the country, to match with our system requirements and in line with the regulatory requirements.

We give our confirmation to provide power supply to the corridors at 66 kV level from our Ved, Puna & Bhatar (220 / 66 kV) Receiving Stations for the Surat metro rail project subject to the above clarifications & requirements.

We hope the above will suffice the requirement.

Thanking you,

Yours faithfully,
For Torrent Power Limited

B. D. Mistry
General Manager (Distribution)

Copy to :

Shri S. C. Jindal, IRSEE
Executive Director
Delhi Metro Rail Corporation Ltd.
Office of Director Business Development
25, Ashoka Road
New Delhi-110001

New Registered Office
"Tapovan" 600 Tapovan, Ambawadi,
Ahmedabad-380015, India.

TORRENT POWER LIMITED
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Reg. Office : Torrent House, Off Ashram Road, Ahmedabad - 380 009. India. Phone : +91 79 26585090 www.torrentpower.com

wcs to: (1) Chief Engr.(Proj) GETCO, Corporate Office, Vadodara
(2) CE(TR) GETCO, Corporate Office, Vadodara
(3) Addl. Chief Engineer(TR) Zonal Office Bharuch

Copy to: (1) Ex.Engr.(Const/Civil) GETCO, Navsari
(2)Ex.Engr(TR) Ichhapore

CHAPTER-10

VENTILATION AND AIR-CONDITIONING SYSTEM

10.1 Introduction:

This chapter covers the Ventilation and Air-conditioning (VAC) system requirements for the underground sections of the proposed Surat Metro. It includes the following:

- Station Air-conditioning System
- Ventilation System for station plant rooms
- Station Smoke Management System
- Tunnel Ventilation System

10.2 Alignment:

The alignment of corridor-I from Sarthana to Dream City is of 21.61kms consist of 20 stations, of which six are underground and 14 are elevated. The underground section starts from Kapodara and passes through Labheshwar Chowk, Central Warehouse, Surat Railway Station, Maskati Hospital and Chowk Bazar. The inter-station distances between underground stationsvary from 586.7meter to 1802.8meter.

10.3 Need for Ventilation and Air Conditioning

The underground stations of the Metro Corridor are built in a confined space. A large number of passengers occupy concourse and the platforms, especially at the peak hours. The platform and concourse areas have a limited access from outside and do not have natural ventilation, it is therefore, essential to provide forced ventilation and air-conditioning in the stations and inside the tunnel for the purpose of:

- Supplying fresh air for the physiological needs of passengers and the authority's staff;
- Removing body heat, obnoxious odours and harmful gases like carbon dioxide exhaled during breathing;
- Preventing concentration of moisture generated by body sweat and seepage of water in the sub-way;

- Removing large quantity of heat dissipated by the train equipment like traction motors, braking units, compressors mounted below the under-frame, lights and fans inside the coaches, A/c units etc.;
- Removing vapour and fumes from the battery and heat emitted by light fittings, water coolers, Escalators, Fare Gates etc. working in the stations;
- Removing heat from air conditioning plant and sub-station and other equipment, if provided inside the underground station.

This large quantity of heat generated in M.R.T. underground stations cannot be extracted by simple ventilation. It is, therefore, essential to provide mechanical cooling in order to remove the heat to the maximum possible extent. As the passengers stay in the stations only for short periods, a fair degree of comfort conditions, just short of discomfort are considered adequate.

10.4 External Environment Conditions and Weather data

The analysis of Surat weather suggests that the summer season for Surat is generally between November to April. The maximum dry bulb temperature is seen to have reached up to 38deg C during end of April. The monsoon months can be assumed from May to October when the maximum temperatures are around 31deg C.

Air-Quality (Environmental control) in public places like MRT stations is required to be maintained for city like Surat. Therefore, it requires consideration of appropriate measures for air-pollution control in Metro stations, while designing the VAC system. The design weather data from the ASHRAE handbooks have been used to arrive at the design criteria. For VAC system, it is suggested that 1% criteria would be acceptable on techno economic reasons.

10.5 Sub Soil Temperature

The temperature conditions of sub-soil play a vital role in the system design of the underground stations. and is vital for facilitating adequate heat exchange between the tunnel structures and soil. It is proposed that water table surrounding the underground alignment shall be reviewed. The sub soil temperature of Surat is estimated to be 25 °c (approx.).It is to be obtained or to be measured at site.

10.6 Internal Design conditions in Underground Stations

With hot and humid ambient conditions of Surat during the summer and monsoon months, it is essential to maintain appropriate conditions in the underground stations in order to provide a ‘comfort’ and pollution-free

environment. The plant capacity and design of VAC system needs to be optimized for the “Designed inside Conditions”.

The patrons will stay for much shorter durations in the underground stations. The comfort of a person depends on rapidity of dissipation of his body heat, which in turn depends on temperature, humidity and motion of air in contact with the body. Body heat gets dissipated by the process of evaporation, convection and conduction. Evaporation prevails at high temperature. Greater proportion of heat is dissipated by evaporation from the skin, which gets promoted by low humidity of air. The movement of air determines the rate of dissipation of body heat in the form of sensible and latent heat.

There are different comfort indices recognized for this purpose. The ‘Effective Temperature’ criterion used in selecting the comfort condition is a function of temperature and the air velocity experienced by a person. More recently a new index named RWI(Relative Warmth Index) has been adopted for metro designs worldwide. This index depends upon the transient condition of the metabolic rate and is evaluated based on the changes to the surrounding ambient of a person in a short period of about 6 to 8 minutes. It is assumed that during this period human body adjusts its metabolic activities, therefore in a subway system where the train headway is expected to be six minutes or less, RWI is the preferred criterion.

10.7 Design parameters for VAC system

Based on the above, the following VAC system design parameters are assumed in the present report.

(1) Outside ambient conditions:

This is based upon Indian Weather Data 2017 issued by ISHRAE recommended design conditions of Surat for 1% criteria, as under

	1% Criteria
Summer :	36.4 DB, 22.9 WB
Monsoon:	31.4 DB, 27.7WB

For Surat Metro Underground Corridor it is suggested to use 1% criteria, which is defined as the conditions, when the DB or WB temperatures are likely to exceed for only 1% of the total time.

(2) Inside design conditions:

Platform/Concourse areas - 27 deg. C at 55 % RH

(3) Tunnel design conditions

	Normal conditions	–	Max. DB 40 deg. C
	Congested conditions	--	Max. DB 45 deg. C
(4)	Minimum fresh air (In station public areas).	-	10 % or 18 cmh / person

10.8 Design Concepts for VAC system

There are various VAC design concepts technically feasible in a subway system that can provide and maintain acceptable subway environment conditions under different requirement and constraints. These are: Open type/ Closed type and Platform Screen Doors (PSDs).

The station premises (public areas) are equipped with separate air-conditioning system during the summer and monsoon months to provide acceptable environment for patrons.

There shall be provision of Track way Exhaust System (TES) by which platform air can be re-circulated. The train cars reject substantial heat inside subway. The TES is installed in the train ways of each station to directly capture heat rejected by the vehicle propulsion, braking, auxiliary and air conditioning systems as the train dwells in the station. When the trains dwell at the stations TES would capture a large portion of heat released by the train air conditioners mounted on the roof tops and under gear heat because of braking, before it is mixed with the platform environment. The TES includes both an Under Platform Exhaust (UPE) duct and an Over-Track way Exhaust (OTE) duct. The TES uses ducts formed in the under platform void and over the track way. Exhaust intakes are to be located to coincide with the train-borne heat sources.

Fig. 10.1: Track way Exhaust Fan

10.9 Station Air Conditioning:

The platform and concourse areas will be air-conditioned using supply 'Air Handling Units' located in Environmental Control System (ECS) plant rooms throughout the station. Each platform will be served by at least two air handling units (AHU's) with the distribution systems combined along each platform to ensure coverage of all areas in the event of single equipment failure. Based on the initial estimation about 4 units of 15 m^3/s each would be needed for the full system capacity.

Fig. 10.2: Air Handling Unit

These air conditioning systems mix return air with a desired quantity of outside air. The outside air requirement is based on occupancy, with a minimum of 5 liters per second per person or 10% of circulated air volume, whichever is greater. The provision of free cooling by a simple two-position economizer control system will be included, with the use of enthalpy sensors to determine the benefits of using return air or outside air. This will signal the control system to operate dampers between minimum and full fresh air, so as to minimize the enthalpy reduction needed to be achieved by the cooling coil. This mixture of outside and return air is then filtered by means of suitable filters and then cooled by a cooling coil before being distributed as supply air via high level insulated ductwork to diffusers, discharging the air into the serviced space in a controlled way to minimize draughts. Return air to the platform areas is extracted via the Track way Exhaust System and either returned to the AHUs or exhausted as required.

UVC Emitters can also be installed in the AHUs for the reduction of molds and fungus growth on the coil and keeps the surface clean, eliminating need for coil cleaning programme and improve the overall coil efficiency. Following are the advantages of UVC emitter:

(a) UVC emitter kills or in activates the surface and air borne microorganism that contribute to poor indoor air quality or spread of infectious diseases.

(b) UVC emitter doesn't allow bio film to form on cooling coil surface and lowers HVAC costs by improving heat transfer capacity.

(c) Increase in air flow results in better air conditioning in the public area hence reduced requirement of additional cooling through AC plant.

Water-cooled chiller units with screw compressors, which are energy efficient are recommended to be provided at each station. These units can be installed in a chiller plant room at ground level. Based on the initial concept design, the estimated capacity for a typical station would be around 600TR, hence three units of 200TR or 2 units of 300TRmay be required for full system capacity (i.e. design PHPDT traffic requirement). During the detail design stage, the chiller capacity for individual station will be frozen through SES analysis depending on the heat loads.

Fig. 10.3: Chiller

10.10 Ventilation and Air Conditioning of Ancillary Spaces

Ancillary spaces such as Staff Room, Equipment Room, will be mechanically ventilated or air conditioned in accordance with the desired air change rate, temperatures and humidity.

All ancillary areas that require 24-hour air conditioning will be provided with Fan Coil Units (FCUs),main Chilled Water plant for running during the revenue hours and with Air Cooled Chillers for running during the non-revenue hours. Return air will be circulated through washable air filters.

Fig. 10.4: Air Cooled Chiller

Where fresh air is required it will be supplied to the indoor unit via a fresh air supply system, complete with filters, common to a group of ancillary areas.

10.11 Station Smoke Management System

The Track way Exhaust and Concourse Smoke Extract Fans or the combined fans for both will be provided for smoke extract purposes from the public areas and will operate in various modes depending on the location of the fire. The control of this system in fire mode will be fail-safe. These exhaust fans will be provided with “essential” power supplies, with automatic changeover on loss of supply.

Down stand beams will be provided underneath the ceiling around floor openings for stairs and escalators, so that a smoke reservoir is formed on the ceiling. The smoke will be contained in this reservoir at ceiling level and exhausted to atmosphere. By controlling smoke in this manner, it is possible to maintain a relatively smoke clear layer above human head height and to protect the escape route, giving sufficient time for evacuation. The stations will be designed to accommodate the full smoke exhaust volumes and thus prevent the reservoir from completely filling with smoke. To provide an additional barrier against smoke migration, the overall smoke management system would be designed to provide a draught of fresh air through entrances and escape routes, to assist in protecting those routes from smoke.

10.12 Space Requirement for VAC System

The station air conditioning equipment plant rooms are normally located at each end of the concourse for the two level stations. The approximate area for air handling equipment room would be 400$m^2$to 500m^2at each end of the station. There shall be supply shafts and exhaust shafts of about 8m^2 to 10m^2 each at each end of the stations. The space requirements for Labheshwar Chowk and Maskati Hospital U/G stations will require special considerations as the platform for UP and DN Line are at different level i.e. One-Over other.

10.13 Tunnel Ventilation Systems (TVS)

The TVS is provided in a Subway system essentially to carry out the following functions:

(a) Provide a tenable environment along the path of egress from a fire incident in enclosed stations and enclosed train ways.
(b) Produce airflow rates sufficient to prevent back layering of smoke in the path of egress within enclosed train ways.
(c) Be capable of reaching full operational mode within 180 seconds.
(d) Accommodate the maximum number of trains that could be between ventilation shafts during an emergency.

Fig. 10.5: Tunnel Ventilation Fan

Tunnel ventilation fans will be installed in each of the fan rooms near vent shafts. There shall be two fans in a fan room at each end of the station. The fan capacity depends on the inter-station distances and may vary from 60 m^3/s to 100 m^3/s depending upon the length and cross section of the tunnel. The exact capacity will be obtained through the simulation during detailed

design stage. If necessary, nozzle type structures made up of concrete or steel may also be constructed to achieve desired airflow and air velocity in the tunnel sections. Alternatively booster fans (jet fans) may be installed to direct the flow in the desired direction. These fans may also be used for emergency ventilation at crossover locations.

The Track way Exhaust System (part of Tunnel Ventilation System) will have two fans of each 13.5 m^3/sec. for each platform. The connections to tunnels and shafts will be through damper units that may be either electrically or pneumatic actuated.

There are various operating modes (scenarios) for the Tunnel Ventilation system. These are described as under:

10.14 Normal Conditions

Normal condition is when the trains are operating to timetable throughout the system, at prescribed headways and dwell times, within given tolerances. The primary source of ventilation during normal conditions is generated by the movement of trains operating within the system and, in some cases, the track way exhaust system.

During summer and the monsoon seasons, the system will be functioning essentially with the station air conditioning operating. The vent shafts to the surface will enable the tunnel heat to be removed due to train movements. The platform air captured by the track way exhaust system shall be cooled and recirculated in the station. For less severe (i.e. cool) environmental conditions (or in the event of an AC system failure), station air conditioning will not be used and ventilation shafts will be open to atmosphere (open system) with the track way exhaust system operating.

10.15 Congested Conditions

Congested conditions occur when delays cause disruption to the movement of trains. It is possible that the delays may result in the idling of a train in a tunnel section. Without forced ventilation, excessive tunnel temperatures may result reduced performance of coach air conditioners that may lead to passenger discomfort.

During congested operations, the tunnel ventilation system is operated to maintain a specific temperature in the vicinity of the car air conditioner condenser coils (i.e. allowing for thermal stratification). The open system congested ventilation shall be via a ‘push-pull’ effect where tunnel vent fans behind the train are operated in supply and tunnel vent fans ahead of the trains are operated in exhaust mode. Nozzles or booster (jet) fans will be used to direct air into the desired tunnel, if required.

10.16 Emergency Conditions

Emergency conditions are when smoke is generated in the tunnel or station track way. In emergency conditions, the tunnel ventilation system would be set to operate to control the movement of smoke and provide a smoke-free path for evacuation of the passengers and for the fire fighting purposes. The ventilation system is operated in a 'push-pull' supply and exhaust mode with jet fans or nozzles driving tunnel flows such that the smoke is forced to move in one direction, enabling evacuation to take place in the opposite direction depending upon the location of Fire on the train.

10.17 Design Concepts for TVS system

There are various TVS design concepts technically feasible in a subway system that can provide and maintain acceptable subway environment conditions under different requirement and constraints. These are: Open type; Closed type; Use of jet fans; use of mid-shafts; etc. The experience available from the design of TVS system for Delhi Metro also provides key guidelines.

Under the normal train running the train heat generated inside the tunnel sections would be removed by the train piston action. It is envisaged that for the design outside conditions, it may not be necessary to provide forced ventilation using Tunnel Ventilations Fans for normal operating conditions. Two tunnel ventilation shafts would be provided at the end of the stations. These end-shafts at the stations also serve as Blast Relief Shafts i.e. the piston pressure is relieved to the atmosphere before the train reaches the station. All these shafts are connected to the tunnels through dampers. The dampers are kept open when the exchange of air with the atmosphere is permitted (Open Mode). For the Closed Mode system the shaft dampers are in closed mode and the displaced air is dumped in the adjacent tunnel.

Generally each tunnel ventilation shaft is connected to a fan room in which there are two reversible tunnel ventilation fans (TVF) are installed with isolation dampers. These dampers are closed when the fan is not in operation. There is a bypass duct around the fan room, which acts as a pressure relief shaft when open during normal conditions, and enables the flow of air to bypass the TV fans, allowing air exchange between tunnel with flows generated by train movements. Dampers are also used to close the connections to tunnels and nozzles under different operating conditions. The details for the shaft sizes, airflow exchange with the atmosphere, fan capacities can be estimated in a more accurate manner with the help of Computer Simulations during the detailed design stage.

Fig. 10.6: Tunnel Ventilation Dampers

10.18 Pressure Transients

The movement of trains within the underground system induces unsteady air motion in the tunnels and stations. Together with changes in cross section, this motion of air results in changes in air pressure within trains and for wayside locations. These changes in pressure or 'pressure transients' can be a source of passenger discomfort and can also be harmful to the wayside equipment and structures. Two types of transient phenomenon are generally to be examined:

(a) Portal Entry and Exit Pressure Transients – As a train enters a portal, passengers will experience a rise in pressure from when the nose enters until the tail enters. After the tail enters the pressure drops. Similarly, as the nose exits a portal, pressure changes are experienced in the train. There is two location of the portal between Shri Swami Narayan Mandir Kalakunj - Kapodara and Chowk Bazar-Kadarsha Ni nal for corridor-I.

b) Wayside Pressure Transients – As trains travel through the system they will pass structures, equipment and patrons on platforms. Equipment would include cross passage doors, lights, dampers, walkways etc. Pressures are positive for the approaching train and negative for retreating trains. Most rapid changes occur with the passage of the train nose and tail. The repetitive nature of these pressures may need to be considered when considering fatigue in the design of equipment.

The detailed analysis to assess the effect of pressure transients will be done during the design stage. For the portal entry/exits the effect of higher train speed may pose discomfort to the passengers. Although, based on experience, a design train speed of 90kmph would not be of major concern.

The estimation of Way-side transients during design stage would be necessary to select design mechanical strength of the trackside fixtures, e.g. false ceilings, light fittings etc at the platform levels.

10.19 Space Requirement for Tunnel Ventilation System

The tunnel ventilation equipment plant rooms are normally located at each end of the concourse for the two level stations. The approximate area for tunnel ventilation fan room would be 500 sq. m. respectively at each end of the station. The tunnel vent shafts of approximately 20 sq. m. area will be constructed at each end of the stations. There shall be supply shaft and exhaust shafts of similar dimensions at the stations. The space requirements for Labheshwar Chowk and Maskati Hospital U/G stations will require special considerations as the platform for UP and DN Line are at different level i.e. One-Over other.

The inter – station distance between Maskati Hospital and Chowk Bazar is1.8 km and considering the ultimate headway of 1.75 min in the corridor, two trains may come in a ventilation zone. The system will either be required to be designed to handle two trains in a ventilation zone or mid shaft to be provided between these two stations. This may be decided at detail design stage of the project.

10.20 Control and monitoring Facilities

For the underground stations the control and monitoring of station services and systems such as station air-conditioning, ventilation to plant rooms, lighting, pumping systems, lifts & Escalators, etc shall be performed at Station Control Room (SCR). However, the operation and control of Tunnel Ventilation as well as Smoke Management system will normally be done through OCC. All these systems shall be equipped with automatic, manual, local and remote operation modes. The alarms and signals from the equipment at stations shall be transmitted to the OCC via communication network (such as FOTS).

There shall be an Auxiliary Power Controller at OCC who will be monitoring these services and systems. The command signals will be initiated at OCC and relayed up to the relevant equipment for operation. The feedback signal is received through SCADA whether the command is implemented or not. The control from OCC is generally performed using ‘Mode Tables’ for each system. This table defines the sequence of the desired equipment that needs to be operated based on the event. The abnormal conditions such as train congestion, emergency, fire in subway would be detected by various components and the emergency response thereto will be activated based on the mode tables. In the event that remote control is not possible due to any reason, the local control via SCR would be performed. In case the control at

work station in SCR is also not available, the manual overriding provisions shall be provided through Ventilation Control Panel (VCP) place in the SCR.

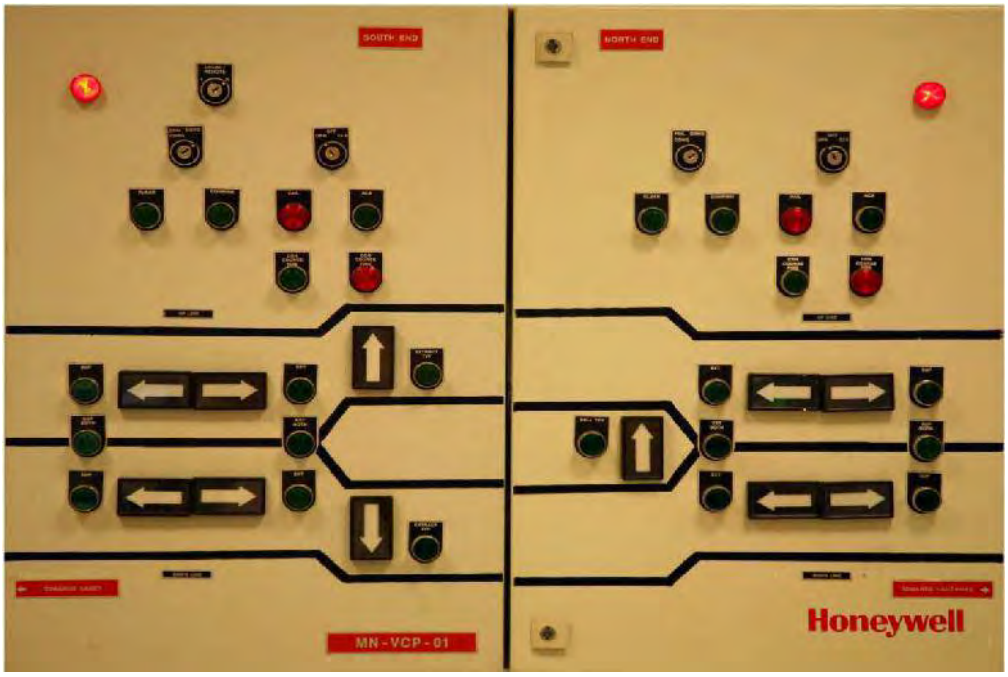

Fig. 10.7: Ventilation Control Panel

The OCC will also be used for logging the alarm status, fault occurrences, and other maintenance related data for the above systems.

10.21 Codes and Standards

The concept VAC design is guided by the following codes and standards:

(a) SEDH – Subway Environment Design Handbook
(b) ASHRAE – Handbook, current series.
(c) CIBSE – relevant document.
(d) NFPA – 130.
(e) ECBC- Energy Conservation Building Code

CHAPTER 11

SIGNALLING AND TRAIN CONTROL SYSTEM

11.1. INTRODUCTION:

The Signaling and Train Control System shall provide the highest security level for means of an efficient Train Control, ensuring safety in train movements. It assists in optimization of rail infrastructure investment and running of efficient train services on the network.

This chapter provides the main design features of the signaling and train control for the operation of Corridor-1 i.e. from Sarthana to Dream City and Corridor-2 i.e. from Bhesan to Saroli of Surat Metro Rail Corridors taking into account the proven and advance system being used worldwide.

The Proposed Corridor-1 i.e. from Sarthana to Dream City and Corridor-2 i.e. from Bhesan to Saroli of Surat Metro Rail Corridors are planned to be operated at maximum safe speed of 90 Km/hr.

Corridor-1 i.e. from Sarthana to Dream City and Corridor-2 i.e. from Bhesan to Saroli of Surat Metro Rail corridors, trains are to be maintained headway at every about 100 seconds. However, the signaling System shall be designed at minimum 90 second headway in one direction.

11.2. SIGNALLING

The Signaling shall provide the highest security level to ensure that the operational activities are developed following strict safety requirements. At the same time, it shall meet the requirements for efficient train operations and high quality of service.

The proposed signalling system design for this metro line will cater the following:

Continuous Automatic Train Control System (CATC)

Unattended Train Operation (UTO) System

Automatic Train Operation (ATO)System

Communication based Automatic Train Control (ATC) System

Automatic Train Protection (ATP) System

On board Equipment

Cab Signalling

Fall-Back Block Working System
Interlocking device (Computer based Interlocking)
Track side Radio equipment
Track Vacancy Detection System
Electric Point Machine
Track side Signals
Centralized Traffic Control System
Power Supply of signalling
Cable for signalling
Half Height Integrated Platform Gate (PG)
Display of CCTV images from Train to OCC
Onboard Radio Antennas, Large Video Screen, MMIs etc

11.3. OVERVIEW OF SIGNALLING SYSTEM

It is expected to carry large number of passengers by maintaining shorter spacing between trains requiring a very high level of safety enforcement and reliability. At the same time heavy investment in infrastructure and Rolling stock necessitates optimization of its capacity to provide the best services to the people.

The requirements of the Surat Metro Corridor planned to be achieved by adopting following basic principles of signaling System: -

The Train Control and Monitoring shall be ensured from Centralized Traffic control System located at Operation Control Centre (OCC). OCC equipment shall be connected to station equipment room through optical fiber network.

The CBTC (Communication based Train Control) based system shall be provided in main line & depot (except workshop area) for train operation & primary mode of detection. Secondary detection shall be through Axle Counter.

Computer Based Interlocking System shall be designed on failsafe philosophy. In case of failure of any equipment, the equipment shall fail on safe side or more restrictive state. In such case the signalling System shall authorized movement of train in normal and degraded operations.

Track side equipment shall be connected through Electronic Interlocking (to Station Equipment Room) by secure links to ensure safe movement of train.

Provide high level of safety with trains running at shorter headways ensuring continuous safe train separation.

Eliminate accidents due to driver passing Signal at Danger by continuous speed monitoring and automatic application of brake in case of disregard of signal / warning by the driver.

Provide safety and enforce speed limit on the sections having permanent and temporary speed restrictions.

Improve capacity with safer and smoother operations. Driver will have continuous display of Target Speed in his cab enabling him to optimize the speed potential of the track section. It provides signal / speed status in the cab even in bad weather.

Increased productivity of rolling stock by increasing line capacity and train speeds, and enabling train to arrive at its destination sooner. Hence more trips will be possible with the same number of rolling stock.

Improve maintenance of Signaling and Telecommunication equipment by monitoring System status of trackside and train borne equipment and enabling preventive maintenance.

Signaling & Train Control System on the line shall be designed to meet the required headway during peak hours.

For monitoring inside train saloon, signaling system shall provide radio transmission media to transfer live streams to OCC controller on large video screen & MMI.

To avoid any accident at platform, Integrated Passenger Gate shall be provided, which will be a barrier between the track and platform accessible to passengers. Signaling and Rolling Stock interfaces shall be provided for Passenger Gate System.

11.4. SYSTEM DESCRIPTION AND SPECIFICATIONS

The requirements of the metro are planned to be achieved by adopting 'CATC' (Continuous Automatic Train Control System) based on "CBTC" (Communication based Train Control System) which includes UTO (Unattended Train Operation), ATO (Automatic Train Operation), ATP (Automatic Train Protection) and ATS (Automatic Train Supervision) sub-systems using radio communication between Track side and Train.

Wireless communication system shall be used for communication between the wayside and train borne CBTC system. Radio for CBTC shall work in License free ISM band.

The Signalling and Train Control system shall be as below and Sub-system/ components will conform to international standards like CENELEC, IEC, IEEE, IS, ITU-T etc:

11.4.1. Continuous Automatic Train Control (CATC)

Continuous Automatic Train Control based on CBTC will consist of - UTO, ATO, ATP, and ATS sub-systems. The Train- borne Automatic Train Control System will consist of Unattended Train Operation (UTO), Automatic Train Operation (ATO) and Automatic Train Protection (ATP).

This vital system maintains the safety of the train operations on the principle of moving block including separation of trains enforcement of speed restrictions and safe operation through interlocking.

11.4.2. Automatic Train Protection (ATP)

Automatic Train Protection is the primary function of the train control systems. This sub-system will be inherently capable of achieving the following objectives in a fail-safe manner. Line side signals will be provided at diverging routes (i.e. at points & crossings) as well as other required locations, which shall serve as backup signalling in case of failure of ATP system. ATP mode shall be the normal mode of operation in event of failure of ATO-UTO Mode. In this mode, the train control and signaling system shall

Provide Cab Signalling.

Determine continuously and protect the train in excess of MSS & LOMA.

Track Related Speed Profile generation based on line data and train data continuously along the track.

Continuous monitoring of braking curve with respect to a defined target point.

Monitoring of maximum permitted speed on the line and speed restrictions in force.

Detection of over-speed with audio-visual warning and application of brakes, if necessary.

Maintaining safety distance between trains.

Monitoring of stopping point.

Monitoring of Direction of Travel and Rollback.

Enable opening of train doors as per stopping platform when train is docked.

The cab borne equipment will be of modular sub-assemblies for each function for easy maintenance and replacement. The ATP assemblies will be fitted in the vehicle integrated with other equipment of the rolling stock.

11.4.3. Automatic Train Operation (ATO)

This system shall operate the trains automatically from station to station while remaining within the safety envelope of ATP & open the train doors. Driver will close the train doors and press a button when ready to depart. In conjunction with ATP/ ATS, ATO can control dwell time at stations and train running in accordance with headway / timetable. ATO Mode shall be normal mode of operation in the event of failure of UTO Mode. In ATO mode the train control and signalling system shall carry out the following function:

Accelerate and decelerate the train by applying traction power, coasting, and applying and removing brakes.

Automatically control speed, acceleration, preventing unnecessary braking and stopping.

Automatic operation of train between stations and stop the train at stations.

Provide all indications necessary to operate the train.

Determine continuously the Maximum Safe Speed (MSS) and Limit of Movement Authority (LOMA) with ATP function.

Train doors open indication on the correct side when the train is docked if permitted by the ATP door release.

Prevent the train from starting if train doors are not detected closed.

11.4.4. Unattended Train Operation (UTO)

In this mode, the train shall operate full driverless (without any crew member onboard), it shall operate under the supervision and control of ATP function. This shall be operated unmanned or with attendant under fully automated conform to Grade of Automation 3 (GOA3) / Grade of Automation 4 (GOA4) as defined in IEC 62290-12006.

The train shall be automatically driven by UTO sub-system under monitoring & full control of OCC operator. On receipt of OCC operator's start-up command, ATS shall send wake up command to onboard ATC. Initialisation of UTO operation after system start-up or recovery after a system failure shall be without any manual intervention in the train or OCC operator's command. However, OCC operator can be able to stop/hold any operation in the train in case of emergency, if required.

In this mode doors will be opened as well as closed automatically including Platform Screen Door. This mode shall be available everywhere on the line and the depot except for the workshop lines.

UTO mode shall be the normal mode of operation. Transit between UTO &

ATO/ATP/RM/ROS modes must be possible continuously and anywhere on the running line and in the yards.

11.4.5. Automatic Train Supervision (ATS)

A train supervision system will be installed to facilitate the monitoring of train operation and also remote control of the station. The train supervision will log each train movement and display it on the workstations with each Traffic Controller at the OCC and on one workstation placed in the Station Control room (SCR) with each Station Controller.

The centralized system will be installed in the Operation Control Centre. The OCC will have a direct line projection display panel showing a panoramic view showing the status of tracks, points, signals and the vehicles operating in the relevant section / whole system. ATS will provide following main functionalities:

Train movement control (Automatic Route setting, train dispatch, inter station stop, platform/system hold & release dwell time, skip-stop, auto crew/RS management etc.).

Automatic Train Regulation.

Continuous Tracking of train position.

Display Panel & Workstation interface.

Link to Passenger Information Display System for online information.

Computation of train schedules & Timetable.

Event & fault logging.

System distinguishes between a train ready signal in ATP and a train ready signal in ATP / ATO mode.

11.4.6. Automatic Train Reversal / Turn Back (ATB)

To minimize the turn back time at terminal and intermediate station, Automatic Turn Back mode (Cycle mode/Sequence mode) is introduced to automatically operate more than one route one after the other in a sequential manner.

ATB function is a part of ATO/UTO mode function. At the terminal station or intermediate station the train shall be operated automatically by the onboard ATC to the turn back track and back to the terminal station without driver.

11.4.7. Restricted Mode (RM)/ Run of Site Mode (ROS)

This mode shall be available only when On Board ATC in operation. If the On-Board ATC does not receive ATP information, train shall be manually driven by driver using ATP (CBTC on board) with constant speed supervision. If train

speed is exceeded to 25 Km/h, On Board ATC shall apply the emergency break. The On Board ATC give cab signal as soon as the train get ATP information or train reach a track position where normal running can be resumed. RM mode shall be operated in depot.

11.4.8. Cut Off Mode

This mode shall used in case of On Board ATC failure. In this mode, the train speed is controlled entirely by the Train Operator in accordance with line side signals and verbal instruction from controller. The rolling stock provides equipment that limits speed less than 25 Kmph. If safety cut out switch is handled, On Board ATC power supply is shut down.

11.4.9. Computer Based Interlocking System(CBI)

The entire line including turn back track, transfer track, sidings will be equipped with CBI system for operation of points & crossings and setting of routes. The setting of the route and clearing of the signals will be done by workstation, which can be either locally (at station) operated or operated remotely from the OCC.

This CBI system is used for controlling vehicle movements into or out of stations automatically from a workstation. Interlocking stations having points and crossings will be provided with workstations for local control. Track occupancy, point position, etc. will be clearly indicated on the workstation. It will be possible to operate the workstation locally, if the central control hands over the operation to the local station. The interlocking system design will be based on fail-safe principle.

The equipment will withstand tough environmental conditions encountered in a Mass Transit System. Suitable IS, IRS, BS standards or equivalent international standards will be followed in case wiring, installation, earthing, cabling, power supply and for material used in axle counters, relays, point operating machines, power supply etc.

11.4.10. Track Vacancy Detection

Primary mode of track vacancy detection system on main line shall be through Radio and secondary detection can be through Axle Counter.

The Axle counters have been used in vital train detection schemes on a large scale in Europe and outside of Europe. Also, an Axle counter is a cost-effective alternative to track circuits when applied correctly and are available from several manufacturers. In view of advantages of Axle counter, the track detection system by Axle Counter is recommended for secondary detection system.

Axle counter is used to detect the track occupancy and to count the number of axles and which train detection is discontinuously performed. It is not affected by weather conditions, and achieves reliable train detection. It interfaces with interlocking system in order to respond to functional errors on the basis of self-diagnosis as well as to transmit the information.

The axle counter consists of the following equipment: -

Detection Point (or counting head)

Evaluator

11.4.11. Wayside Signals

Multi Aspect Color Light (LED) type Line side signals shall be installed on the Main Line at stations with point & crossing for point protection catering for bidirectional working & depot entry / exit.

11.4.12. Cab Signaling

Cab signaling is a railway safety system that communicates track status information to the cab, crew compartment or driver's compartment of a train. The train driver can see the information continuously. The DMI (Driver Machine Interface Display) is the device that displays driving information in the driver cab. Information is transmitted by the wayside radio equipment to on board Radio equipment & vice versa. The data is computed by the on-board equipment and displayed on a screen on DMI for monitoring/controlling the running of train.

Vital information concerning the safe working of the train is displayed directly in the driving cab on the DMI. The DMI displays:

Brake details: distance to first brake application.

Speed information: current train speed, permitted speed, target speed on circular speed gauge with speed pointer preferably with disguise color.

Auxiliary driving information: state of brakes (service brake, emergency brake), state of the connection between the on-board and the track side equipment.

The DMI is also the interface between the driver and the on-board equipment to get driver information, train characteristics or request for shunting operation.

11.4.13. Point Machines

Non-Trailable Electrical Point Machine capable of operating with 3-phase, 50 Hz. 380V AC will be used on main line and the depot point machine will be

trailable type electrical point machine capable of operating with either 3 phase, 50 Hz. 380V AC or 110V DC.

11.4.14. Train Depot: Signaling

All depot lines except the workshop area shall be interlocked. A workstation shall be provided in the Depot Control Centre for electrical operation of the points, signals and routes of the depot yard. Track vacancy detection using Radio & Axle Counter will be used in the depot as well.

The Depot shall be equipped with all mode of train operation including UTO mode except depot workshop line. The trains shall be controlled from DCC and OCC as well.

A test track with similar Signaling and Train control system as adopted in Main Line shall be provided at Depot.

11.5. SIGNALLING MODE OF OPERATION

There are six signaling modes of operation which shall be available but only one single signaling mode shall be active at any one instant of time. These five Modes are mentioned as under:-

a) Restricted Manual (RM) Mode for Depot.

b) Automatic Train Protection (ATP) Mode

c) Automatic Train Operation (ATO) Mode

d) Unattended Train Operation (UTO) Mode

e) Run on Sight Mode (ROS) Mode

f) Automatic Train Reversal / Turn Back (ATB) Mode

11.6. HALF HEIGHT INTEGRATED PLATFORM GATE (PG)

The Integrated Platform Gate system shall provide a barrier between the track and the platform accessible to passengers. The system shall improve the safety of passengers by isolating the platforms from the track unless there is a train stopped at its correct position. PG system shall be around 1.5 Metre heights and it shall consist of sets of bi-parting doors installed along the full length of platform.

The PG system shall comprise Automatic Sliding Gates (PGs), Platform End Doors (PEDs), Emergency Escape Doors (EEDs) and Fixed Screens (FSS) to form a barrier along the edge of the platform adjacent to the track. Platform Gates shall correspond to the location of each of the train doors when the train has berthed at its correct position. Each platform end shall be closed by a Platform End Door. The remaining portion of PG facade shall be

provided with manually openable Emergency Escape Doors and Fixed Screens.

The PG system shall be integrated with structure and architecture of the station and operationally with Signaling System as well as Rolling stock System. The interface between Signaling System and PG shall be designed to fail safe signaling standards and according to relevant International standard. All vital control and detection circuits of PG system shall be double cut.

Opening and closing of PG and Train doors shall be synchronous. Train movement should not be permitted until it is confirmed that both Train doors and Platform Gates are properly closed. The PG shall be quiet in operation and all the elements of the PG installations (fixed and moving) shall be sufficiently rigid to avoid generation of noise by panel excitation.

11.7. Display of CCTV images from Train to OCC

For monitoring of train saloons, there shall be provision at OCC for displaying live video streams for the onboard CCTV cameras on each train as selected by the OCC operator on large video screen & MMIs. Onboard camera shall be provided by rolling stock, Signaling shall provide the radio infrastructure for transmission of CCTV live stream from train to OCC through use of one of the available ISM band (preferably 5.8 GHz band) frequency. Data transmission network of CCTV & CBTC shall be separate and redundancy in radio units.

Provision shall be made for displaying a minimum of four live streams from a train at OCC. Signaling shall also provide sufficient size large video screen display and MMI at OCC as per requirement. Display shall be sufficient good quality of operator to view required simultaneous live streams. However, actual bandwidth requirement, number of live streams per train, size of large video screen, etc. shall be finalized as per requirement during design stage.

11.8. CENTRALIZED TRAFFIC CONTROL (CTC)

The Metro operation shall be managed from the Central Traffic Control that located in Operation control Centre (OCC) that is in charge of managing real time traffic, safety of movement, rolling stock, on-board staffing, and work maintenance. The primary objective of the OCC system is to operate the train in UTO/ATO mode (in CBTC) and construct the routes of the trains from the origin up to the destination automatically / manually under normal / abnormal conditions, the OCC system will provide effectively alternatives to minimize the delay of the train.

It supplies all the information required to the centralized traffic control operator in order to check the normal operations of the trains. The OCC

system interfaces to the external systems (interlocking, Radio equipment, SCADA, PIDS, PAS, etc.) to monitor and control the traffic and to ensure the safe operations of trains.

The OCC system shall meet the following requirements:

The systems and communication lines shall be in redundant configuration and will ensure reliability and safety through continuous operations of the system.

The OCC is interfaced with signaling devices set along the track and allow the operator to access different functionalities for traffic management with a man-machine interface (MMI).

MMI allows the command acquisition, alarms display, and the viewing of control images.

Each equipment units used for servers and industrial MMIs will be suitable for the metro rail environment with high MTBF. The servers for the OCC shall be self diagnostic and fault noticing functions.

Operation during emergency situation can be planed through the back-up CTC (BCC) when OCC is not functional.

The suitable software for each operator workstation and server is configured to achieve the convenience of the operation.

Operational Room at OCC shall monitor the train operations and control the operations of train so that the trains can operate safely & efficiently. The functions of the operating room will be supported by the LDP (Large Display Panel), and Workstations for the operators. The LDP in the operating room shows the entire track line of the Metro Rail in real-time so as to monitor it any time.

11.9. FALL- BACK BLOCK WORKING SYSTEM

A Fall-Back block working system shall be used by using secondary detection (axle counters) & Track Side Signals in case of failure of CBTC System or wayside communication link become unavailable.

The Fall-Back Block working system shall follow fixed block working, it can temporarily be worked to maintain safety and smooth operation with the help of Line Side Signals provided at each station / interlocking. When the Fall-Back Block working system is operated, it is necessary to check no other trains exist in the protection area to keep safety operation at first on priority.

11.10. ROOMS for SIGNALLING at OCC & STATIONS

The OCC is composed of several rooms that have specific functions. In a basic

configuration, four rooms are directly concerned by the Signalling System: the Operation Room, the Central Signaling Equipment room, the maintenance room and power supply room**.**

A) Operation Room:

The Operational Room is the place from where the operators can monitor and control the traffic on the Line, using dedicated workstations and Direct Line large projection Display.

B) Central Signalling Equipment Room

This room includes all equipment managing the Signalling System included in the OCC control area.

As per site requirement, additional devices (other than signaling i.e Telecom and AFC System) can be considered in the Central signalling room. Also, this room shall be available at Central location as well as interlocking stations (SER, Station Equipment Room).

C) Maintenance room

All signalling devices information and technical alarms are displayed on workstations and manual or automatic commands are possible from these workstations. This room shall be available at Central location as well as interlocking stations.

D) Power supply room

The room contains Uninterrupted Power Supply (UPS) necessary for the signaling technical room, the maintenance room and the operation room. The power supply arrangement is designed in order to provide uninterrupted power in case of general power breakdown. It includes all the equipment that provides power supply for OCC rooms (Signaling, Telecom, AFC equipments). UPS room should be available adjacent to Signalling Equipment rooms at Stations and OCC. UPS room shall be available at Central location at all stations.

The minimum surface areas required for each room at stations are:

The signaling technical room: 40m2

The maintenance room : 30m2

The power supply room : 50m2

At the OCC, BCC and the Depot, the areas required shall be as per final configuration of the equipment and network configuration keeping space for further expansion.

11.11. BACK UP OF THE OCC (BCC)

In order to decrease the risks of disruption due to a local disaster such as fire, flood, building collapse, etc., a Main CTC (OCC) and a fall back CTC (BCC) shall be provided & both shall be located in different areas.

The OCC may be located at terminal station inside the premises of the station or in Depot. The BCC could be located around other terminal station / locations / Depot. The BCC shall be similar to OCC, and also, BCC shall provide full redundancies of all systems and communications.

The OCC is normally on-line and used by the Operators to control the Metro Line traffic. Operation & Maintenance Control.

The BCC is normally off-line. It will be used to control the Line only in case the OCC is accidentally unavailable. Besides this BCC being normally off-line, will be also available for other purposes such as training, testing, replay without disturbing the live traffic.

11.12. POWER SUPPLY

Uninterrupted Power Supply provision is must for the Signalling System to have high availability. The concept of the power supply system is of immense importance, because the availability of the Signalling System entirely depends on its power supply.

All devices along the line are computerized devices and therefore need to be fed with low-voltage power supply. The low-voltage power supply shall be designed in a way to ensure the quality and reliability of the supply to all components of the Signalling System. The Uninterrupted Power Supply System shall have sufficient backup time and in built redundancies to ensure very high of Availability and reliability.

The solutions that are usually implemented on metro rail include:

Redundancy of mains feeder (delivery from multiple cables/sources),

Back-up of the AC supply by means of uninterrupted power supplies and associated batteries,

Batteries capacity based on system consumption and autonomy with Back-up time requirements.

Architecture and dimension of the system allowing failures and/or maintenance without service disruption (possibility to switch off one or several converters or other modules without impact).

The Uninterrupted Power Supply system shall be designed for use of Signalling equipments, Telecom equipments, AFC equipment & PSD equipments as per requirement.

11.13. STANDARDS

The following standards will be adopted with regard to the Signalling system.

Table 11.1

Description	Standards
Train protection system	Train Protection system shall be based on CBTC (Communication based Train Control) System. The system architecture shall provide in redundancy. The system will conform to IEEE 1474 standards.
Interlocking	Computer based Interlocking adopted for station having switches and crossing. All related equipment as far as possible will be centralised in the equipment room at the station. The depot shall be interlocked except for lines mainly used for workshop lines etc.
Block Working	Moving Block working concept may be followed in CBTC System and Fixed Block working in failure of CBTC system.
Default Mode of Operation	The system shall be capable of Unattended Train Operation (UTO), however, the mode of operation may be decided / finalized by metro at detail design stage.
Maximum Safe Speed of Operation	90 Km/h
Grade of Automation	GOA4 (UTO) / GOA3 (DTO)
Operation of Points	Non-Trailable Electrical Point Machine capable of operating with 3-phase, 50 Hz. 380V AC will be used on main line and the depot point machine will be trailable/ non -trailable type electrical point machine capable of operating with either 3 phase, 50 Hz. 380V AC or 110V DC.

Description	Standards
Track Vacancy Detection System	Primary mode for track vacancy detection system on main line and in depot (except workshop line) may be through radio (CBTC System) and secondary detection it may be through Axle Counter.
Signals at Stations with point & crossings	Line Side signals to protect the points (switches). LED type signals for reliability and less maintenance cost.
Uninterrupted power Supply at stations as well as for OCC	Uninterrupted Power Supply System is Common for Signalling, Telecommunications, AFC and PSD systems.
Train Describer System	Automatic Train Supervision system. Movement of all trains to be logged on to a central computer and displayed on workstations in the Operational Control Centre and at the SCR. Remote control of stations from the OCC. The system architecture shall provide in redundancy.
Fall Back CTC	Backup OCC (BCC)
Platform Gate	Integrated Platform Gate System
On board CCTV Stream	Display of CCTV images from Train to OCC on Large Video Screen and MMIs
Cables	Outdoor cables will be steel armoured as far as possible.
Fail Safe Principles	SIL-4 safety levels as per CENELEC standard for Signal and Train Control System.
Immunity to External Interface.	All data transmission on telecom cables/OFC/Radio. All Signalling and telecom cables will be separated from power cables as per standard. CENELEC standards to be implemented for EMC.
Train Working under emergency	Running on site with line side signal with speed automatically restricted between 15-25 Kmph.

Description	Standards
Environmental Conditions	Air-conditioners for all equipment rooms.
Maintenance philosophy	Philosophy of continuous monitoring of system status and preventive & corrective maintenance of Signalling equipment shall be followed. Card / module / sub-system level replacement shall be done in the field and repairs under taken in the central laboratory/ manufacturer's premises.

11.14. SPACE REQUIREMENT FOR SIGNALLING INSTALLATIONS

Adequate space for proper installations of all Signaling equipment and Platform screen doors at each of the stations has to be provided keeping in view the case of maintenance and use of instrumentation set up for regular testing and line up of the equipment/system.

The areas required at Interlocking stations for Signalling Equipment Room shall be generally 40 sqm. For UPS Room (common for Signalling, Telecommunication, AFC and PSD systems) at all stations, the area required shall be approximately minimum 50 sqm.

At Non-interlocking stations, Signaling & PSD Equipments shall be installed in the Telecommunication Equipment Room (TER) available at the station.

At the OCC and the Depot, the areas required shall be as per the final configuration of the equipment and network configuration keeping space for further expansion.

11.15. MAINTENANCE PHILOSOPHY FOR SIGNALLING SYSTEMS

The philosophy of continuous monitoring of system status and preventive & corrective maintenance of Signalling and Telecommunication equipment shall be followed. Card / module / sub-system level replacement shall be done in the field. Maintenance personnel shall be suitably placed at intervals and they shall be trained in multidisciplinary skills. Each team shall be equipped with a fully equipped transport vehicle for effectively carrying out the maintenance from station to station.

The defective card/ module / sub-system taken out from the section shall be sent for diagnostic and repair to a centralized S&T repair lab suitably located in the section/depot. This lab will be equipped with appropriate diagnostic and test equipment to rectify the faults and undertake minor repairs. Cards / modules / equipment requiring major repairs as specified in suppliers documents shall be sent to manufacturer's workshop.

CHAPTER 12

TELECOMMUNICATION & AUTOMATIC FARE COLLECTION SYSTEMS

12.0 TELECOMMUNICATION SYSTEM

12.1. INTRODUCTION:

The Telecommunication system acts as the communication backbone for Signaling systems and other systems such as SCADA, AFC etc and provides Telecommunication services to meet operational and administrative requirements of the metro network.

12.2. OVERVIEW

The Telecommunication facilities proposed are helpful in meeting the requirements for operation of trains:

1. Supplementing the Signalling system for efficient train operation.
2. Exchange of managerial information
3. Crisis management during emergencies
4. Passenger information system

The proposed Telecom system will cater to the following requirements:

Radio System

Backbone network using Optical Fiber Cable (OFC)

Ethernet & WAN Network.

Station to Station dedicated communication

Telephone System with Telephone Exchanges, Telephones and their Recording

Centralized Recording System (CDRS)

Centralized Clock System

Closed Circuit Television (CCTV) System

Passenger Information & Display System within the station & trains and from Central Control to each station, Integrated Passenger

Announcement System

Train Traffic Control, Maintenance Control, Emergency Control, Assistance to Train Traffic Control.

Data Channels for Signalling, SCADA, Automatic Fare Collection

Power Supply of Telecommunications, and

Cables for Telecommunications etc.

12.3. TELECOMMUNICATION SYSTEM AND TRANSMISSION MEDIA

12.3.1. Fibre Optic System (FOTS) - Main Telecommunication Bearer

The main bearer of the bulk of the Telecommunication network is proposed with optical fiber cable system. An OFC system shall provide a transmission network of Voice, Data, Ethernet, Video, and Signals among all Stations, Depot and OCC with sufficient transmission bandwidth to cater for the operational need of Metro line. The size of the OFC will fully meet with the applications need of the Metro line and commercial exploitation of the Telecommunication Network of Metro line. A minimum 96 / or 144 Fibers optical fiber cable with redundancy (cable on both side of track) is proposed to be laid. The optical fiber cable shall provide common transmission backbone network for Telecom and other systems which are formed by the two outdoor single mode optical fiber cables, one laying along the up-track and other one along the down-track. Additional 144 fiber optical fiber cables may be laid along track as per present commercial requirement for revenue.

12.3.2. Gigabit Ethernet Network (WAN)

A totally IP Based High Capacity, highly reliable and fault tolerant, MPLS Ethernet Network shall be provided. IP network shall have important data therefore Ethernet requires high reliability. Considering the rapidly increased demand during the operation for top-level backbone network 10Gbps Equipment is proposed. The communications network shall be configured as LAN and WAN – LAN shall be responsible for train operations and maintenance tasks within each passenger station and WAN shall be responsible for mutual communications between the stations and between depot and the central computer system. To maximize the reliability and survivability, each equipment and transmission line are configured as a dual system.

Redundant Layer-3, Layer-2 switches and Routers at each station, depot & OCC shall be provided to meet requirement of other Telecom systems (like CCTV, AFC system, maintenance management system and Wi-Fi network at station, PA, Clock, PIDS , Telephone System, SCADA etc) and to support comparatively unimportant facilities for the operation.

Layer-3 Core switch at OCC shall be provided to cover all requirements for Centralized Management and Control facility of all equipment used in line. Data lines of sufficient quantity and bandwidth shall be provided to other systems between Central Terminal Unit and Remote Terminal Unit.

12.4. TELEPHONE EXCHANGE

The System shall be IP Based with some of the extensions being Analog. For an optimized cost effective solution small exchanges of 30 port each shall be planned at each station and a 60 Port Exchange at the Terminal Stations and Depots shall be provided. The station exchanges will be connected to the Centre OCC main exchange. The Exchanges will serve the subscribers at all the stations and Central Control. The exchanges will be interconnected at the channel level on optical backbone. The exchanges shall be software partitioned for EPABX and Direct Line Communication from which the phones shall be extended to the stations. For the critical control communication, the Availability & Reliability should be high. Alternatively, only for non-operational (other than Direct Line Communication) a separate IP Based Phone System can be implemented.

12.5. MOBILE RADIO COMMUNICATION

Mobile Radio communication system having minimum 8 logical channels is proposed for on-line emergency communication between Motorman (Front end and Rear end) of moving train and the Central Control. The system shall be based on Digital Trunk Radio Technology to TETRA International standard. All the stations, depots and the OCC will be provided with fixed radio sets. Mobile communication facility for maintenance parties and Security Personnel will be provided with handheld sets. These persons will be able to communicate with each other as well as with central control.

The frequency band for operation of the system will be in 400/800 MHz band, depending on frequency availability. The system shall provide instant mobile radio communication between the motorman of the moving cars from any place and the Central Control. The motorman can also contact any station in the network through the central control, besides intimating the approaching trains about any emergency like accident, fire, line blocked etc., thus improving safety performance.

To provide adequate coverage, based on the RF site survey to be carried out during detailed Design stage, base stations for the system will be located at sites conveniently selected after detailed survey. Tentatively minimum 5 sites with rooftop towers with Base Stations and minimum 3 base station for coverage in U/G feeding LCX cable with repeaters shall be required along the proposed Corridor-1 i.e Sarthana to Dream City and Tentatively minimum 5

sites with rooftop towers with Base Stations shall be required along the proposed Corridor-2 i.e. from Bheshan to Saroli of Surat Metro Rail corridors.

12.6. PASSENGER ANNOUNCEMENT SYSTEM (PAS)

The PAS shall be provided to broadcast voice messages to passengers /staff in all stations/ Depot from the locally as well as from OCC. It includes a network of amplifier and speakers linked to the station. The system capable of announcements from Station level will have over-riding priority in case of emergency announcements. The System shall be linked to Signalling System for automatic train actuated announcements.

The PAS and Passenger Information Display System (PIDS) shall be coordinated automatically to provide real time passenger audio broadcast and visual information at each station. Live audio broadcast relating to emergency, fire and evacuation messages from OCC and Station Control Room shall be recorded in the Centralised digital recording system at OCC. FOTS WAN network shall be used for transportation of data from Station/Depot to OCC vice versa.

12.7. PASSENGER INFORMATION DISPLAY SYSTEM (PIDS)

These shall be located at convenient locations at all stations to provide bilingual visual indication of the status of the running trains and will typically indicate information such as destination, arrival/departure time, and also special messages in emergencies. The boards shall be provided at all platforms and concourses of all stations. The System shall be integrated with the PA system and available from same MMI. For the Platform Area, high intensity LED Boards will be used in Evaluated Section. For all the concourses and Platform Area of underground Stations, HDLED Panels shall be used, which can also provide Audio/Visual Advertisements apart from Trains running status.

12.8. CENTRALIZED CLOCK SYSTEM

This will ensure an accurate display of time through a synchronization system of slave clocks driven from the GPS Based Master Clock at the Operation Control Center. The Master Clock signal shall also be required for synchronization of FOTS, Exchanges, Radio, Signaling, etc. The System will ensure identical display of time at all locations. Clocks are to be provided at platforms, concourse, Station Master's Room, Depots and other service establishments.

12.9. CLOSED CIRCUIT TELEVISION (CCTV) SYSTEM

The CCTV system shall provide video surveillance and recording function for the operations to monitor each station. The monitoring shall be possible both locally at each station and remotely from the OCC on the Video Wall.

The CCTV System shall be end to end IP based Full HD IP cameras using backbone of FOTS WAN network and shall consist of a mix of Fixed Cameras and Pan/Tilt/Zoom (PTZ) Cameras. Cameras shall be extended /located at areas where monitoring for security, safety and crowd control purpose is necessary. All Videos shall be extended at Video Wall located at security control room at OCC.

Intelligent Video Analytic (Track protections, abandoned object detection, Perimeter protection, Movement detection, Platform track protection from falling object, Camera Tempering, Overcrowding / Consation detection, Excessive Queuing, Rule based detection, Face detection & tracking features etc) shall be provided in cameras of specific locations like Platforms, Vulnerable locations, etc. Alarm shall be generated and relevant data and video shall be transfer to OCC/Stations/Security Rooms through optical fiber network.

12.10. ACCESS CONTROL SYSTEM

An Access Control System shall be provided for entering into important areas like SCR, SER, TER, OCC, DCC, TOM Rooms, etc. The System shall provide the Access only to the Authorized Personnel in operational rooms and shall not allow the same Card for Travel in metro. The System Shall be controlled and monitored centrally from the OCC.

12.11. NETWORK MONITORING AND MANAGEMENT

For efficient and cost effective maintenance of the entire communication network, it is proposed to provide an Integrated Network Control System, which will help in diagnosing faults immediately from a central location and attending the same with least possible delay, thus increasing the operational efficiency and reduction in manpower requirement for maintenance. The proposed NMS system will be covering Radio communication, Optical Fiber Transmission, Telephone Exchange and summary alarms of PA/PIDS, CCTV and Clock System. The Integrated NMS will collect and monitor status and alarms from the individual NMS of the respective sub-systems and display on a common Work Station.

12.12. TECHNOLOGY

The Technologies proposed to be adopted for Telecommunication systems are shown in Table below:

Table 12.1 TECHNOLOGIES FOR TELECOMMUNICATION SYSTEMS

System	Standards
Transmission Media	Optical Fibre system as the main bearer for bulk of the Telecommunication network
Telephone Exchange	PABX of minimum 30 ports is to be provided at all Stations, an Exchange of 60 Ports to be provided at Terminal Station
Train Radio System	Digital Train radio (TETRA) communication between motorman of moving cars, stations, maintenance personnel and central control.
Train Destination Indicator System	LED based boards with adequate visibility on Elevated and LED Panels in concourse to be provided at convenient location at all stations to provide bilingual visual indication of the status of the running trains, and also special messages in emergencies.
Centralized clock System	Accurate display of time through a synchronization system of slave clocks driven from a GPS master clock at the OCC and sub – master clock in station. This shall also be used for synchronization other systems.
Passenger Announcement System	Passenger Announcement System covering all platform and concourse areas with local as well as Central Announcement.
Redundancy (Major System)	Redundancy on Radio's in the Base Stations, Path Redundancy for Optical Fibre Cable by provisioning in ring configuration.
Environmental Conditions	All equipment rooms to be air-conditioned.
Maintenance Philosophy	System to have, as far as possible, automatic switching facility to alternate routes/circuits in the event of failure. Philosophy of preventive checks of maintenance to be

System	Standards
	followed. System networked with NMS for diagnosing faults and co-ordination. Card/module level replacement shall be done in the field and repairs undertaken in the central laboratory/manufacture's premises.

12.13. Space Requirement for Telecom Installations

Adequate space for proper installations of all Telecommunication equipment at each stations has to be provided keeping in view the case of maintenance and use of instrumentation set up for regular testing and line up of the equipment/system. The areas required at each of the stations for Telecommunication equipments shall be approximately 40 sqm. The Telecommunication Room shall be used for Signaling, Telecommunication, AFC & PSD systems equipments at non- interlocking stations. In interlocking station, Telecommunication Room shall be used for Telecommunication, AFC & PSD systems equipments. Uninterrupted Power Supply (UPS) System shall be common for Signaling, Telecommunication, AFC & PSD systems equipments at input stage and installed in UPS room at every station, depot and OCC which is approximately 50 sqm at station. These areas shall also cater to local storage and space for maintenance personnel to work.

At the OCC, the areas required shall be as per the final configuration of the equipment and network configuration keeping space for further expansion.

12.14. Maintenance Philosophy for Telecom Systems

The philosophy of continuous monitoring of system status and preventive & corrective maintenance of Signaling and Telecommunication equipments shall be followed. Card / module / sub-system level replacement shall be done in the field. Maintenance personnel shall be suitably placed at intervals and they shall be trained in multidisciplinary skills. Each team shall be equipped with a fully equipped transport vehicle for effectively carrying out the maintenance from station to station.

The defective card/ module / sub-system taken out from the section shall be sent for diagnostic and repair to the existing centralized S&T repair lab suitably located on the section. This lab will be equipped with appropriate diagnostic and test equipments to rectify the faults and undertake minor repairs. Cards / modules / equipment requiring major repairs as specified in suppliers documents shall be sent to manufacturer's workshop.

AUTOMATIC FARE COLLECTION

12.15. Introduction

Metro System handles large number of passengers. Ticket issue and fare collection play a vital role in the efficient and proper operation of the system. To achieve this objective, ticketing system shall be simple, easy to use / operate and maintain, easy on accounting facilities, capable of issuing single / multiple journey tickets, amendable for quick fare changes and require overall less manpower. In view of the above computer based automatic fare collection system is proposed**.** Seamless ticketing is now being thought of for Surat Metro Rail.

Automatic Fare Collection system is recommended to be adopted as this will enable the commuters to travel hassle free by different modes of transport viz. Metro, suburban trains, buses, water transport (whenever introduced) and even taxies without purchasing multiple tickets for each mode separately.

Automatic fare collection systems have the following advantages:

1. Less number of staff required.
2. Less possibility of leakages of revenue due to 100% ticket check by control gates.
3. Recycling of ticket fraudulently by staff avoided.
4. Efficient and easy to operate.
5. System is amenable for quick fare changes.
6. Management information reports generation is easy.
7. System has multi operator capabilities. Same Smart Card can be used for other applications also.
8. AFC systems are the world wide accepted systems for Metro environment.

The proposed AFC system shall be of Contactless Smart Token / Card type. For multiple journeys, the stored value smart card shall be utilized and for the single journey, the smart media shall be as utilized as contactless smart token. The equipments for the same shall be provided at each station counter / booking offices and at convenient locations and will be connected to a local area network with a computer in the Station Master's room. Equipment and installation cost of Contactless Smart Card / Token based AFC system is similar to magnetic ticket based AFC system, but Contactless system proves cheaper due to reduced maintenance, less wear and tear and less prone to dusty environment.

It is proposed, the smart NCMC (National Common Mobility card) standard model for implementation of AFC system in Surat Metro. The AFC system as per the guidelines issued by Govt of India shall enable seamless travel by

different metros and other transport systems across the city besides retail shopping and purchases.

The AFC system shall support the EMV (Europay, MasterCard, and Visa) and RuPay based open loop ticketing following the NCMC standard model for interoperability with other operators by use of non-proprietary standard so that the interface is scalable to other networks (transit operator/ retail outlets/parking/Toll etc) in Surat. The AFC equipments shall support EMV, RuPay, QR, NFC (Near field communication) based ticketing, integration of clearing house, smart card host system of Financial Institutions and integration of mobile application with AFC system.

12.16. Gate

Retractable Flap Type/Paddle Type Control Gates are proposed which offer high throughput, require less maintenance and are latest in modern systems internationally. All these gates will have a functionality of Auto Top on smart cards in case balance goes below the threshold value (as per choice / business rule).

The gate should also capable to NFC enabled Mobile Tickets or any latest type of Ticket media at the time of procurement/installation. The AFC system shall provide access control solutions, offering both access control devised and hardware which can be tailored to accept any ticket media readily available in market (Barcode, QR code, NFC etc).

12.16.1. Gate Function

a) Gate arrays shall be the normal-means of controlling entry to and exit from the paid areas. Control shall be by means of actuating a physical barrier on recognition of a valid ticket or card by the gate. The barrier may be a bi-parting leaves, centre flaps, end flaps or other configuration however the use of tripod or turnstile type gates is not acceptable. The gate shall be capable of operating either in normally open or normally closed mode.

b) Where required, barriers shall be provided to separate paid and unpaid areas of the concourse. The barriers shall meet local public safety requirements and be aesthetically merged with station engineering.

12.16.2. Features

a) **Power Failure** - In the event of a total power failure to the gates, the gates shall open to allow unrestricted user access. All latch gates shall automatically unlatch where electric locks are installed.

b) **Concourse Emergency Mode** - All AFC gates shall open whenever the Concourse Operating Mode is in emergency. An Emergency

Push Button independent of the SC shall be provided in each Excess Fare Office.

c) **Ergonomics** - The engineering of the gate arrays should be such that the passenger uses reader placed on the right hand side while passing through the gate. The display and Contact less Smart Card (CSC) reader associated with each gate shall be grouped such that they bias the passenger towards the aisle through which the passenger should pass.

12.16.3. Types of Gates

(a). **Passenger Entry Gate**: - The Passenger Entry Gate shall control the entry of passengers into the paid area by validating the fare media.

(b). **Passenger Exit Gate**: - The Passenger Exit Gate shall control the exit from the paid area by validating the fare media.

(c). **Passenger Reversible Gate**: - The Passenger Reversible Gate shall combine the features of the Entry and Exit gates. It shall be capable of being switched by the Station Computer from entry mode to exit mode and vice-versa depending on the operational requirements of passenger flow. Reversible Gates shall also function automatically, based on the side from where the Passenger approaches first.

(d). **Staff / Emergency Gate**: - Normally situated adjacent to the Excess fare Office and kept open during emergency situations.

12.16.4. Spacing

Spacing for passenger gates shall be based generally on the following dimensional criteria:

a) Gate Centre spacing: - Standard gates 880mm

b) Aisle width: - Standard gates 465 - 580mm

12.16.5. Gate Enclosure

a) The gate enclosure shall be fabricated of stainless steel. The gate shall be finished to conform to the architectural requirements of the station.

b) The degree of protection provided by the enclosure against dust, splashing, intrusion of foreign objects shall meet or exceed the standard IP54 (IP43 for token acceptor slot, if any), as defined by British Standards.

12.16.6. Tail Gating Prevention:

Minimum distance for detection shall be less than 20 cm and methodology shall be in accordance with that being used in AFC operations.

12.17. Ticket Vending Machine (TVM)

The TVM should provide the convenience for the passengers to procure ticket on their own, without the need to queue at the ticket sale counter.

At all stations, Passenger Operated Ticket Vending Machines (Automatic Ticket Vending Machines) are proposed. The TVM's will provide convenience to passengers to avoid standing in queues at ticket booths and provide them international standard service. This will be used for

1. Dispensing Smart Tokens for single journey
2. Add Value in Smart card by paying money using Bank Notes or through Credit Card /Debit card /pre-Paid card.
3. Return the remaining money through Bank Notes and Coins (Min 2 types)

12.17.1. Function

a) Enable passengers to purchase tickets for journey.

b) The touch-point including the screen interface should be customizable in terms of the text, graphics and video. It should be able to support the promotion of any preferred products.

c) The machines shall accept payment in the form of bank notes, coins and credit / debit cards and shall interact with the passengers via a touch screen display and receipt printer.

d) A reject button shall be provided to enable a passenger to abort a transaction before a token issue cycle has commenced.

e) The bank note reader shall accept notes inserted in any orientation (any way up or round) and change shall be provided via a combination of note and coin re-circulating mechanism, which minimises the number of times the station staffs need to replenish the machines with change.

12.17.2. Physical

The TVM's hardware and peripherals should come equipped with durable housing. It shall be made from stainless steel and shall be freestanding or recessed into the walls of the TVM rooms as required by the station architecture. Separate tamper-proof coin boxes and note vaults shall be provided.

Minimum 2 TVM machines shall be provided at every entry to station to dispense journey ticket.

12.18. Types of Ticket

(a) The system shall provide, or be capable of processing, the following types of ticket:

Single Journey Ticket (SJT)
Daily Pass
Staff/Employee Pass (EP)
Stored Value (SV) (at least 16 configurable types)
Period Pass (PP)

(b) Each ticket type shall be capable of being associated with at least four fare tables (One full fare and two concession fares).

12.18.1. Ticket Media

(a) CSC (for Stored Value, Employees Pass etc.)

Contactless media shall be to ISO/IEC 14443 & ISO 18092 standard (minimum EAL4 Security Criteria for CSC) and also to support common mobility card specifications of Ministry of Urban Development (GOI).

The system must support minimum standard & specification for CSC & devices and recommendations to be used for the implementation of National Common Mobility program in the multimodal and multi operator environment within the practical limits of transport.

(b) Other Media (for Single Journey Tickets)

Media for Single Journey Tickets shall be determined at design stage, which can be a token. Choice of SJT media shall take financial and usage constraints into account.

12.18.2. Ticket Reader/Add Value Machines

These machines will be used to know the Card/Token balance and can also be used as Add value device in case payment for Card top up is made through alternate Internet based channel like net banking, Credit/Debit card (Payment gateway) etc.

12.18.3. Recharge Card Terminal Machine (RCTM)

RCTM will be used to recharge the Card using Credit Card /Debit card /Pre Paid card as well as bank Note

12.19. Security

12.19.1. Revenue Protection

The AFC machines shall resist tampering by either passengers or unauthorized staff.

12.19.2. Revenue Security

(a) The AFC machines and system shall provide a complete audit trail of all transactions, transfers of cash and other payments.

(b) Cash handling equipment and systems shall be an integral part of the audit trail.

(c) Data & Revenue Security shall be ensured by a Key Management System (KMS) which needs to issue a Hardware SAM for each AFC equipment in use in the system. The SAM shall be used to authenticate the equipment and the transaction integrity.

12.19.3. Data Security

(a) In the event the SC fails, each item of equipment shall be able to operate autonomously without loss of data.

(b) Security of communications between the AFC equipment, SC and CC system shall ensure no loss of data in transmission.

12.20. Station Computer (SC)

a) Station Computer (SC) enables the overall control and monitoring of each item of AFC equipment within the station and transfer of data to the Central Computer (CC).

b) The SC shall include the power and data communication links to each item of AFC equipment and CC system interface.

c) It shall enable printing of reports at stations. The reports shall include accounting and statistical information. It shall include any other reports required for AFC operation.

d) The SC shall be able to download data to the AFC machines individually or as groups.

e) The SC shall receive maintenance data from AFC equipment and transmit the same to CC for monitoring and use of the same as an effective maintenance tool.

f) The SC shall be able to monitor certain critical functions of the AFC

system and collect data for warnings and alarms.

g) If there is loss of communication between the SC and AFC equipment (Gates, TOM etc.) then the equipment shall operate in stand-alone mode utilizing the most recent data from the SC. AFC equipment (Gates, TOM etc.) shall store data up to seven days for transmission when SC communication is restored.

h) In the event of loss of communication with the CC the SC shall utilize the most recent operational data received from the CC and shall be capable of storing at least thirty days of transaction data.

12.20.1. Equipment Control

The normal method of control of the equipment shall be by the SC. The SC shall enable all AFC equipment control (put in service, taken out of service and initiated etc.) without the requirement for communication with the CC.

12.21. Central Computer System

Central Computer System shall be redundant configuration and placed at OCC. It is connected to Station Computer and equipments via redundant secured link provided in Telecom Chapter.

a) The Central Computer System (CC) shall collect and analyze information received from the station computers. It shall produce network-wide revenue and traffic data and monitor the performance of all AFC equipment.

b) A Central Computer (CC) System shall generate the necessary management reports from the CST, CSC and transaction information received from the Station Computer Systems..

c) The CC shall hold and download CST and CSC parameters, Configuration Data (CD), AFC device software and fare table information to each SC from where they shall be distributed to the station AFC equipment.

d) The CC shall automatically collate all CST, CSC and usage data (UD) from the SC to provide accurate audit and traffic statistics for the line.

e) The CC shall be located in a dedicated computer room in the Administration Building or Operations Control Centre.

f) The CC shall maintain a blacklist of invalid tickets. Blacklisted tickets shall be rejected by the AFC Gates.

g) The CC shall support a Fare Table with adequate number of stations.

12.22. CENTRAL CLEARING HOUSE SYSTEM (CCHS)

The CCHS system shall be installed at OCC for sharing revenue between different operators. The Central Clearing House System (CCHS) shall handle all transactions for multiple applications and seamlessly integrate AFC System with different operators. The CCHS shall have all the functionalities required, thereof, for clearing and settlement between different registered Operators.

The CC should be able to interface with Central clearing house system (CCHS). The CCHS system already installed & working for other metro rail may be used for Surat Metro Rail requirement, which may be decided at the detailed design stage.

12.23. AFC Equipment Requirement

The AFC equipment required at various locations of Surat Metro Corridor 1 (Sarthana to Dream City) and Corridor 2 (Bheshan to Saroli) are tabulated at Annexure 1 & Annexure 2 for projection years 2021, 2026, 2036, 2046.

However, the exact number and type shall depend on the final station layout and traffic being catered to.

12.24. Standards:

The standard proposed for AFC systems are as under:

Table 12.2

Standards	Description
Fare media	a) Contactless Smart Token – For single journey. Token are captured at the exit gate. b) Contactless Smart Card – For multiple journeys. Contactless readers shall be as per ISO 14443 standards. The system should also capable to NFC enabled Mobile Tickets(ISO 18092 or equivalent) or any latest type of Ticket media(Barcode, QR code etc)
Gates	Computer controlled retractable flap type automatic gates at entry and exit. There will be following types of gates : - Entry - Exit

Standards	Description
	- Reversible The System shall support the EMV and RuPay based open loop ticketing following the NCMC standard model for interoperability.
Station computer, central computer and AFC Network	All the Fare Collection Equipment shall be connected in a local area network with a station server controlling the activities of all the machines. The station servers will be linked to the AFC central computer situated in the operational control center through the optic fiber communication channels. The centralized control of the system shall provide real time data of earnings, passenger flow analysis, blacklisting of specified cards etc.
Ticket office machine (TOM/ EFO)	Manned Ticked Office Machines shall be installed in the station for selling cards / token to the passengers. Also TVM's shall be provided for Automatic Ticket Vending.
Ticket Readers	Ticket Reader shall be installed near EFO for passengers to check information stored in the token / cards.
Portable ticket decoder(PTD)	PTD will be used to check the card/token during travel
Recharge card terminal machine	RCTM will be used to recharge the card using bank note/debit card/credit card/pre paid card
UPS	Common UPS of S&T system will be utilized.
Maintenance philosophy	Being fully Contactless system, manpower requirement for maintenance is much less compared to system with magnetic tickets. However, adequate facilities to be provided similar to that of S & T systems.

12.25. Integration of AFC with other Lines and Modes of Transport:

In Surat, different mode of transport are being constructed and operated by different operators. In view of passenger convenience and operational efficiency, it is proposed that AFC for different metro lines should be integrated and smart card based fare products should be inter-operable. AFC system shall take into account revenue sharing mechanism among different operators based on journeys performed at each system. The single ride tickets (tokens) may not be inter-operable and may be limited to each operators system.

The proposed AFC system shall provide interfaces to other operators such as Suburban Rail, Bus, Waterway, Parking, Toll etc so that these systems may also be integrated with common smart card based fare products. This will facilitate the passengers as they need not carry different cards for different applications.

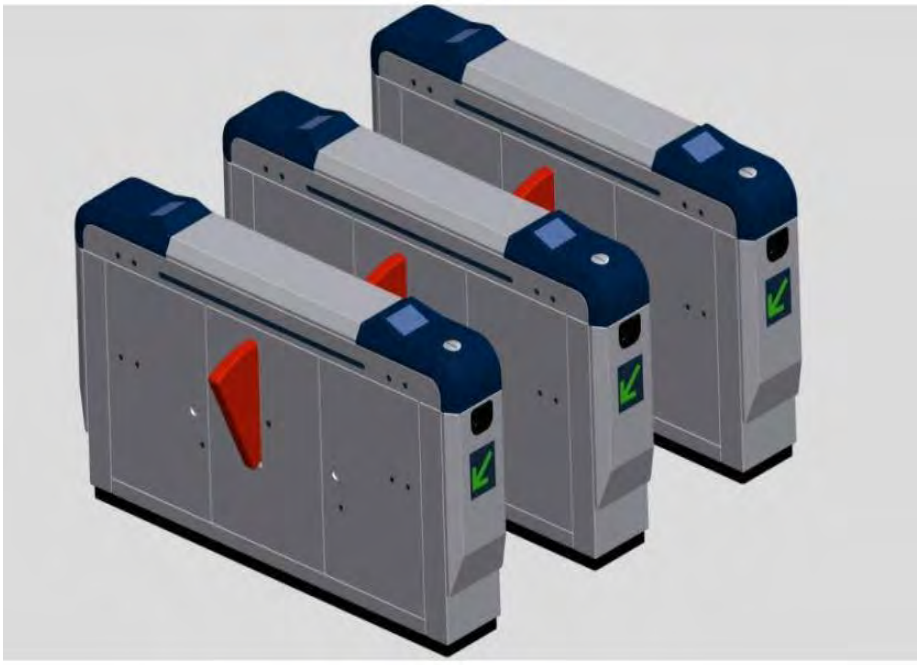

Fig 12.1 Entry/Exit Gates

Fig 12.2: Ticket Office Machine

Fig 12.3: Ticket vending machine

Fig. 12.4 Ticket Reader/Add Value Machine

Annexure 1

AFC Equipments for Surat Metro Corridor-1 from Sarthana to Dream City (Projection for 2021)

Sr. No.	Station Name	Hourly Boarding	Hourly Alighting	Peak min. Boarding	Peak min. Alighting	Gate		TOM	EFO	TR	TVM	RCTM
						Entry	Exit					
1	Dream City	93	58	2	1	2	2	2	2	4	2	2
2	Convention Center	27	27	1	1	2	2	2	2	4	2	2
3	Bhimrad	26	63	1	1	2	2	2	2	4	2	2
4	Woman ITI	24	25	1	1	2	2	2	2	4	2	2
5	VIP Road	1490	1572	25	26	2	2	2	2	4	2	2
6	Althan Gam	510	544	9	9	2	2	2	2	4	2	2
7	AlthanTenament	489	451	8	8	2	2	2	2	4	2	2
8	Rupali Canal	303	378	5	6	2	2	2	2	4	2	2
9	Majura Gate	1571	1725	26	29	2	2	2	2	4	2	2
10	Kadarsha Ni Nal	403	294	7	5	2	2	2	2	4	2	2
11	Gandhi Baug	2256	2244	38	37	2	2	2	2	4	2	2
12	Maskati Hospital	402	655	7	11	2	2	2	2	4	2	2
13	Surat Railway Station	2125	1748	35	29	2	2	2	2	4	2	2
14	Central Warehouse	867	711	14	12	2	2	2	2	4	2	2
15	Labheshwar Chowk	1498	1019	25	17	2	2	2	2	4	2	2
16	Kapodra	1791	3740	30	62	2	2	2	2	4	2	2
17	Shri Swaminarayan Mandir Kalakunj	980	922	16	15	2	2	2	2	4	2	2
18	VarchaChopati Garden	442	407	7	7	2	2	2	2	4	2	2
19	Nature Park	681	939	11	16	2	2	2	2	4	2	2
20	Sarthana	561	667	9	11	2	2	2	2	4	2	2
	TOTAL					40	40	40	40	80	40	40

AFC Equipments for Surat Metro Corridor-1 from Sarthana to Dream City (Projection for 2026)

Sr. No.	Station Name	Hourly Boarding	Hourly Alighting	Peak min. Boarding	Peak min. Alighting	Gate		TOM	EFO	TR	TVM	RCTM
						Entry	Exit					
1	Dream City	538	615	9	10	2	2	2	2	4	2	2
2	Convention Center	41	41	1	1	2	2	2	2	4	2	2
3	Bhimrad	39	90	1	2	2	2	2	2	4	2	2
4	Woman ITI	37	40	1	1	2	2	2	2	4	2	2
5	VIP Road	2264	2424	38	40	2	2	2	2	4	2	2
6	Althan Gam	755	847	13	14	2	2	2	2	4	2	2
7	AlthanTenament	922	803	15	13	2	2	2	2	4	2	2
8	Rupali Canal	503	738	8	12	2	2	2	2	4	2	2
9	Majura Gate	3024	3238	50	54	2	2	3	2	4	2	2
10	Kadarsha Ni Nal	935	716	16	12	2	2	2	2	4	2	2
11	Gandhi Baug	4920	4414	82	74	3	3	4	2	4	2	2
12	Maskati Hospital	849	1324	14	22	2	2	2	2	4	2	2
13	Surat Railway Station	3728	3463	62	58	2	2	3	2	4	2	2
14	Central Warehouse	1294	2872	22	48	2	2	2	2	4	2	2
15	Labheshwar Chowk	2108	1420	35	24	2	2	2	2	4	2	2
16	Kapodra	3335	3984	56	66	2	3	3	2	4	2	2
17	Shri Swaminarayan Mandir Kalakunj	3609	3419	60	57	2	2	3	2	4	2	2
18	VarchaChopati Garden	1095	975	18	16	2	2	2	2	4	2	2
19	Nature Park	1736	1937	29	32	2	2	2	2	4	2	2
20	Sarthana	1719	1873	29	31	2	2	2	2	4	2	2
	TOTAL					41	42	46	40	80	40	40

AFC Equipments for Surat Metro Corridor-1 from Sarthana to Dream City (Projection for 2036)

Sr. No.	Station Name	Hourly Boarding	Hourly Alighting	Peak min. Boarding	Peak min. Alighting	Gate		TOM	EFO	TR	TVM	RCTM
						Entry	Exit					
1	Dream City	1746	2150	29	36	2	2	2	2	4	2	2
2	Convention Center	126	127	2	2	2	2	2	2	4	2	2
3	Bhimrad	116	266	2	4	2	2	2	2	4	2	2
4	Woman ITI	108	112	2	2	2	2	2	2	4	2	2
5	VIP Road	3925	3858	65	64	3	3	3	2	4	2	2
6	Althan Gam	1425	1582	24	26	2	2	2	2	4	2	2
7	AlthanTenament	1312	1153	22	19	2	2	2	2	4	2	2
8	Rupali Canal	1114	1038	19	17	2	2	2	2	4	2	2
9	Majura Gate	4351	3906	73	65	3	3	4	2	4	2	2
10	Kadarsha Ni Nal	1501	1212	25	20	2	2	2	2	4	2	2
11	Gandhi Baug	8387	7373	140	123	6	5	7	2	4	2	2
12	Maskati Hospital	1172	1432	20	24	2	2	2	2	4	2	2
13	Surat Railway Station	4311	6501	72	108	3	4	4	2	4	2	2
14	Central Warehouse	1710	1393	29	23	2	2	2	2	4	2	2
15	Labheshwar Chowk	2211	1433	37	24	2	2	2	2	4	2	2
16	Kapodra	5310	5750	89	96	4	4	4	2	4	2	2
17	Shri Swaminarayan Mandir Kalakunj	4072	3357	68	56	3	2	3	2	4	2	2
18	VarchaChopati Garden	1669	1356	28	23	2	2	2	2	4	2	2
19	Nature Park	3547	3941	59	66	2	3	3	2	4	2	2
20	Sarthana	4331	4507	72	75	3	3	4	2	4	2	2
	TOTAL					51	51	56	40	80	40	40

AFC Equipments for Surat Metro Corridor-1 from Sarthana to Dream City (Projection for 2046)

Sr. No.	Station Name	Hourly Boarding	Hourly Alighting	Peak min. Boarding	Peak min. Alighting	Gate		TOM	EFO	TR	TVM	RCTM
						Entry	Exit					
1	Dream City	2384	3095	40	52	2	2	2	2	4	2	2
2	Convention Center	174	177	3	3	2	2	2	2	4	2	2
3	Bhimrad	174	382	3	6	2	2	2	2	4	2	2
4	Woman ITI	139	155	3	3	2	2	2	2	4	2	2
5	VIP Road	4685	5292	78	88	3	4	4	2	4	2	2
6	Althan Gam	1708	1955	28	33	2	2	2	2	4	2	2
7	AlthanTenament	1388	645	23	11	2	2	2	2	4	2	2
8	Rupali Canal	1672	1189	28	20	2	2	2	2	4	2	2
9	Majura Gate	5286	5805	88	97	4	4	4	2	4	2	2
10	Kadarsha Ni Nal	1808	1405	30	23	2	2	2	2	4	2	2
11	Gandhi Baug	9093	9805	152	163	6	7	8	2	4	2	2
12	Maskati Hospital	2161	1289	36	21	2	2	2	2	4	2	2
13	Surat Railway Station	5523	8336	92	139	4	6	5	2	4	2	2
14	Central Warehouse	2540	1644	42	27	2	2	2	2	4	2	2
15	Labheshwar Chowk	2320	1974	39	33	2	2	2	2	4	2	2
16	Kapodra	6377	4978	106	83	4	3	5	2	4	2	2
17	Shri Swaminarayan Mandir Kalakunj	6220	3141	104	52	4	2	5	2	4	2	2
18	VarchaChopati Garden	2058	1449	34	24	2	2	2	2	4	2	2
19	Nature Park	4168	7225	69	120	3	5	3	2	4	2	2
20	Sarthana	6976	6920	116	115	5	5	6	2	4	2	2
	TOTAL					57	60	64	40	80	40	40

Annexure 2

AFC Equipments for Surat Metro Corridor-2 from Bheshan to Saroli (Projection for 2021)

Sr. No.	Station Name	Hourly Boarding	Hourly Alighting	Peak min. Boarding	Peak min. Alighting	Gate		TOM	EFO	TR	TVM	RCTM
						Entry	Exit					
1	Bheshan	57	87	1	2	2	2	2	2	4	2	2
2	Botanical Garden	187	190	3	3	2	2	2	2	4	2	2
3	UgatVaarigruh	492	436	8	7	2	2	2	2	4	2	2
4	Palanpur Road	326	423	5	7	2	2	2	2	4	2	2
5	L.P. Savani School	542	490	9	8	2	2	2	2	4	2	2
6	Performing Art Centre	273	241	5	4	2	2	2	2	4	2	2
7	Adajan Gam	301	278	5	5	2	2	2	2	4	2	2
8	Aquerium	452	373	8	6	2	2	2	2	4	2	2
9	Badri Narayan Temple	291	364	5	6	2	2	2	2	4	2	2
10	AthwaChopati	365	188	6	3	2	2	2	2	4	2	2
11	Majura Gate	1528	1429	25	24	2	2	2	2	4	2	2
12	Udhna Darwaja	1010	1218	17	20	2	2	2	2	4	2	2
13	Kamela Darwaja	1088	863	18	14	2	2	2	2	4	2	2
14	Anjana Farm	509	481	8	8	2	2	2	2	4	2	2
15	Model Town	489	801	8	13	2	2	2	2	4	2	2
16	Magob	2295	2230	38	37	2	2	2	2	4	2	2
17	Bharat Cancer Hospital	61	68	1	2	2	2	2	2	4	2	2
18	Saroli	3160	3265	53	54	2	2	3	2	4	2	2
	TOTAL					36	36	37	36	72	36	36

AFC Equipments for Surat Metro Corridor-2 from Bheshan to Saroli (Projection for 2026)

Sr. No.	Station Name	Hourly Boarding	Hourly Alighting	Peak min. Boarding	Peak min. Alighting	Gate		TOM	EFO	TR	TVM	RCTM
						Entry	Exit					
1	Bheshan	185	260	3	4	2	2	2	2	4	2	2
2	Botanical Garden	444	418	7	7	2	2	2	2	4	2	2
3	UgatVaarigruh	861	870	14	15	2	2	2	2	4	2	2
4	Palanpur Road	481	640	8	11	2	2	2	2	4	2	2
5	L.P. Savani School	830	746	14	12	2	2	2	2	4	2	2
6	Performing Art Centre	439	416	7	7	2	2	2	2	4	2	2
7	Adajan Gam	525	483	9	8	2	2	2	2	4	2	2
8	Aquerium	714	568	12	9	2	2	2	2	4	2	2
9	Badri Narayan Temple	720	617	12	10	2	2	2	2	4	2	2
10	AthwaChopati	527	308	9	5	2	2	2	2	4	2	2
11	Majura Gate	2492	2257	42	38	2	2	2	2	4	2	2
12	Udhna Darwaja	1471	1758	25	29	2	2	2	2	4	2	2
13	Kamela Darwaja	1531	1197	26	20	2	2	2	2	4	2	2
14	Anjana Farm	759	743	13	12	2	2	2	2	4	2	2
15	Model Town	650	995	11	17	2	2	2	2	4	2	2
16	Magob	3295	3444	55	57	2	2	3	2	4	2	2
17	Bharat Cancer Hospital	138	150	2	3	2	2	2	2	4	2	2
18	Saroli	3876	4013	65	67	3	3	3	2	4	2	2
	TOTAL					37	37	38	36	72	36	36

AFC Equipments for Surat Metro Corridor-2 from Bheshan to Saroli (Projection for 2036)

Sr. No.	Station Name	Hourly Boarding	Hourly Alighting	Peak min. Boarding	Peak min. Alighting	Gate		TOM	EFO	TR	TVM	RCTM
						Entry	Exit					
1	Bheshan	907	1298	15	22	2	2	2	2	4	2	2
2	Botanical Garden	1742	1406	29	23	2	2	2	2	4	2	2
3	UgatVaarigruh	1501	2728	25	45	2	2	2	2	4	2	2
4	Palanpur Road	1109	1292	18	22	2	2	2	2	4	2	2
5	L.P. Savani School	1573	1297	26	22	2	2	2	2	4	2	2
6	Performing Art Centre	1467	1130	24	19	2	2	2	2	4	2	2
7	Adajan Gam	1070	795	18	13	2	2	2	2	4	2	2
8	Aquerium	1571	1106	26	18	2	2	2	2	4	2	2
9	Badri Narayan Temple	1518	1156	25	19	2	2	2	2	4	2	2
10	AthwaChopati	1374	836	23	14	2	2	2	2	4	2	2
11	Majura Gate	2993	3915	50	65	2	3	3	2	4	2	2
12	Udhna Darwaja	3604	4031	60	67	2	3	3	2	4	2	2
13	Kamela Darwaja	2822	2109	47	35	2	2	2	2	4	2	2
14	Anjana Farm	1356	1226	23	20	2	2	2	2	4	2	2
15	Model Town	655	2050	11	34	2	2	2	2	4	2	2
16	Magob	8282	8589	138	143	6	6	7	2	4	2	2
17	Bharat Cancer Hospital	428	415	7	7	2	2	2	2	4	2	2
18	Saroli	11013	9605	184	160	7	6	9	2	4	2	2
	TOTAL					45	46	50	36	72	36	36

AFC Equipments for Surat Metro Corridor-2 from Bheshan to Saroli (Projection for 2046)

Sr. No.	Station Name	Hourly Boarding	Hourly Alighting	Peak min. Boarding	Peak min. Alighting	Gate		TOM	EFO	TR	TVM	RCTM
						Entry	Exit					
1	Bheshan	1286	1976	21	33	2	2	2	2	4	2	2
2	Botanical Garden	2201	1887	37	31	2	2	2	2	4	2	2
3	UgatVaarigruh	2396	3556	40	59	2	2	2	2	4	2	2
4	Palanpur Road	1122	1407	19	23	2	2	2	2	4	2	2
5	L.P. Savani School	1592	1501	27	25	2	2	2	2	4	2	2
6	Performing Art Centre	2139	1407	36	23	2	2	2	2	4	2	2
7	Adajan Gam	1311	959	22	16	2	2	2	2	4	2	2
8	Aquerium	1871	1278	31	21	2	2	2	2	4	2	2
9	Badri Narayan Temple	1481	1296	25	22	2	2	2	2	4	2	2
10	AthwaChopati	1768	1041	29	17	2	2	2	2	4	2	2
11	Majura Gate	4609	4727	77	79	3	3	4	2	4	2	2
12	Udhna Darwaja	4336	5066	72	84	3	3	4	2	4	2	2
13	Kamela Darwaja	2939	2018	49	34	2	2	2	2	4	2	2
14	Anjana Farm	1458	1454	24	24	2	2	2	2	4	2	2
15	Model Town	1254	2255	21	38	2	2	2	2	4	2	2
16	Magob	8510	10292	142	172	6	7	7	2	4	2	2
17	Bharat Cancer Hospital	531	666	9	11	2	2	2	2	4	2	2
18	Saroli	14584	12661	243	211	10	8	12	2	4	2	2
	TOTAL					50	49	55	36	72	36	36

Assumptions :

A. Each Station has only 2 access

B. Minimum AFC equipments at a station with "2 access-1 for entry , 1 for exit": 2 entry gates, 2 exit gates, 2 EFO, 2 TOM, 4 AVM/TR, 2 TVM

C. Thoughput of gate: 25 passengers per minute, TOM : One per access, D. 50% passenger are assumed on Smart card and 50% on single journey token

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CHAPTER– 13

DISABLED FRIENDLY FEATURES

13.1 INTRODUCTION

The objective of making this chapter is to create a user-friendly mass transport system in India which can ensure accessibility to persons with disabilities, people travelling with small children or are carrying luggage, as well as people with temporary mobility problems (e.g. a leg in plaster) and the elderly persons.

The design standards for universal access to Public Transport Infrastructure including related facilities and services, information, etc. would benefit people using public transport.

The access standards given here are extracted from Indian Roads Congress Code, IRC 103: 2012, Guidelines for Pedestrian Facilities; Model Building Bye-Laws, 2011 and National Building Code, 2005. Central Public Works Department's (CPWD) "Space Standards for Barrier Free Built Environment for Disabled and Elderly Persons", 1998 and 2013 edition (under revision by MoUD), and international best practices / standards

Further, it has also been attempted to provide guidelines/ standards for alighting and boarding area, approach to station, car parking area, drop-off and pick-up areas, taxi/auto rickshaw stand, bus stand/stop, footpath (sidewalk), kerb ramp, road intersection, median/pedestrian refuge, traffic signals, subway and foot over bridge etc. to achieve a seamless development around metro stations.

13.2 CONTENT

1. Rail Transport

2. Metro Rail Station

Way finding
Signage

Automated Kiosks
Public Dealing Counters
Audio-visual Displays
Public Telephones
Rest Areas/Seating
Tactile Paving - Guiding & Warning
Doors
Steps & Stairs
Handrails
Ramps
Lifts/Elevators
Platform/Stair Lift
General and Accessible toilets
Drinking Water Units
Visual Contrasts
Emergency Egress/Evacuation

3. Street Design

Footpath (Sidewalk)
Kerb Ramp
Road Intersection
Median/Pedestrian Refuge
Traffic Signals
Subway and Foot Over Bridge

4. Alighting and Boarding Area

Approach
Car Park
Drop-off and Pick-up Areas
Taxi/Auto Rickshaw Stand
Bus Stand/Stop

13.3 RAIL TRANSPORT

1. General

- Whether over-ground or underground, rail travels is a highly effective mode of transport.

- Every train should contain fully accessible carriages.
- Staff should be trained in methods of assistance and be at hand on request.
- Stations for all rail travel should be fully accessible with extra wide turnstiles where possible alongside wheelchair accessible doorways
- Staff should be on hand to assist persons with disabilities and elderly to enter or exit through convenient gates.
- All new railway stations should be designed to be fully accessible.
- For persons with hearing impairments, an electronic sign board (digital display) should be displayed on each platform at conspicuous location for all announcements made by the railways.
- For persons with visual impairments audio system announcing the station names and door location should be available.

2. Accessible Railway Cars

The railway carsshould have the following features:

- Railway car doors should be at least 900 mm wide;
- The gap between the car doors and the platform should preferably be less than 12 mm;
- Identification signage should be provided on the doors of wheelchair accessible coach
- If the car door and the platform cannot be at the same level, then at least one car doors should have apparatus such as a hydraulic lift or pull-out ramp installed in the doorway for wheelchair users.

3. Wheel Chair Space

- Space for a wheel chair should be available at the side of the door:-
- The space should be indicated inside and outside the car by using the international symbol of access; and
- Wheel stoppers and ring-strap or other appropriate safety grip should be provided for wheelchair users.

4. Seats

- An appropriate number of designated seats for passengers with disabilities and elderly people should be provided near the doors.

5. Aisles

- Aisles should be at least 900 mm wide.

13.4 INFORMATION SIGNS AND ANNOUNCEMENTS

A map of train routes should be installed. This should be in Braille/raised numbers as well. In each car, there should be an announcement and provision of a visual display of the names of stations route. This display should be in raised numbers with sharp contrast from the background.

13.5 METRO RAILWAY STATIONS

1. LEVEL APPROACH

Approach route should not have level differences. If the station is not on the same level as the walkway or pathway, it should a ramp.
Walkway surfaces should be non-slip.
Approach walkway should have tactile pavements for persons with visual impairments.

2. STATION ENTRANCES AND EXITS

These should have a minimum width of 1800mm and is level or ramped.

3. RESERVATION AND INFORMATION COUNTERS

Should have clear floor space of at least 900 mm x 1200 mm in front of the counters;
There should be at least one low counter at a height of 750 mm to 800 mm from the floor with clear knee space of 750 mm high by 900 mm wide by 480 mm deep.
At least one of the counters should have an induction loop unit to aid people with hearing impairments; and
The counters should have pictographic maps indicating all the services offered at the counter and at least one of the counter staff should be sign language literate.

4. TOILET FACILITIES

There should be at least one unisex accessible toilet
Ticket Gates

At least one of the ticket gates should:

Be minimum 900 mm wide to allow a wheelchair user through; and
Have a continuous line of guiding paver for people with visual impairments.

5. PLATFORMS

The Platforms should:

Have a row of warning paver installed 600mm before the track edge (photo 6);
Have non-slip and level flooring;
Have seating areas for people with ambulatory disabilities;
Be well illuminated lux level 35 to 40;
There should be no gap or difference in level between the train entry door and the platform.
All platforms should inter-connect by means of an accessible routes or lifts; and provide accessible level entrance to the train coach.

6. WAY FINDING

Way finding references should be available at decision points.
Colour can be used to identify routes and provide assistance in locating doors, walls and hazards. Proper colour contrast between different elements greatly improves visibility for all users and is critical for persons with low vision. For example, colour contrasting of door frames can assist in locating doors, and likewise floors should be contrasted with walls. In addition, furniture should contrast with walls and floors so as not to create an obstacle.
Structural elements such as columns should be colour contrasted or brightly marked so as to be visible to those who may have a visual disability.
Generally, patterns on flooring should be avoided or else should be minimal and small to avoid visual confusion.
In addition to identifying hazards or warnings, tactile floor surfaces can also be used to inform that there is a change in area (e.g. leaving a corridor and entering a boarding area).
Tactile systems should be consistent throughout the building. For example, terminals should not have carpeting in some boarding areas and

tile in others as this may create confusion for those who rely on tactile surfaces to guide them to their destination.

Good lighting assists those with a visual disability to see better and allows people who have a hearing impairment to lip read easier. However, care should be taken to properly direct lighting and to use matte finishes on floors, walls and signage, so as not to create glare which may create difficulties for all travelers.

Blinds can be used to adjust lighting levels in areas where the natural lighting changes significantly throughout the day.

7. SIGNAGE

Signs must be clear, concise, and consistent. All travelers need clear information about the purpose and layout of terminals to maintain a sense of direction and independent use of all facilities. Using internationally and nationally established symbols and pictograms with clear lettering and Braille ensures universal accessibility cutting across regional/cultural and language barriers. A cohesive information and signage system can provide visual (e.g. signs, notice boards), audible (e.g. public address and security systems, induction loops, telephones, and infrared devices), and/ or tactile information (e.g. signs with embossed lettering or Braille).

8. SIGN DESIGN SPECIFICATIONS

The sign should be in a prominent position.

The face of the sign should be well-illuminated by natural or artificial light.

Letters should be simple such as Arial, Helvetica medium, and san serif or similar and numbers should be Arabic.

The colour of the text should be in a colour that contrasts with the sign board.

The sign board should also contrast with the wall on which it is mounted.

The surface of the sign should not be reflective.

Some signs such as those adjacent to or on a toilet door may be embossed so that they can be read by touch.

Illuminated signs should not use red text on a dark background.

Signs should be supplemented by Braille where possible.

Fig. 13.1 - Way finding signage

Fig. 13.2 - International Symbol Accessibility

9. AUTOMATED KIOSKS

Automated kiosks should be accessible for wheelchair users.
Should be clearly marked with international symbol of accessibility.
Should have Braille buttons and audio announcement system for persons with vision impairments.
Operations should be easy to understand and operate for persons with learning disabilities, intellectual disabilities, and elderly persons.

10. PUBLIC DEALING COUNTERS

Ticketing, Information, Check-in, Help desk, Restaurants, Shops, etc. should have public dealing counters.
Information or help desks should be close to the terminal entrance, and highly visible upon entering the terminal. In addition, they should be clearly identified and accessible to both those who use wheelchairs and those who stand.
It should provide information in accessible formats, viz. Braille leaflets for persons with vision impairments.
Ideally, these desks should have a map of the facility that desk attendants can view with passengers, when providing directions.
Staff manning the counters should know sign language.
Information desk acoustics should be carefully planned and controlled as a high level of background noise is confusing and disorienting to persons with hearing impairment.

Lighting should be positioned to illuminate the receptionist/person manning the counter and the desk top without creating glare.
Lighting should not create shadows over the receptionist staff, obscuring facial detail and making lip reading difficult.
There should be a hearing enhancement system such as a loop induction unit, the availability of which is clearly indicated with a symbol.
One of the counters should not be more than 800mm from the floor, with a minimum clear knee space of 650mm high and 280mm- 300mm deep .

11. AUDIO-VISUAL DISPLAYS

Terminal maps should be placed so that they are readily visible to persons who are standing and persons who use wheelchairs. They should also be accessible to persons with a visual disability (i.e. tactile maps). Other alternatives include electronic navigation systems or audio maps.
Enable captioning at all times on all televisions and other audiovisual displays that are capable of displaying captions and that are located in any portion of the terminal.
The captioning must be in high contrast for all information concerning travel safety, ticketing, check-in, delays or cancellations, schedule changes, boarding information, connections, checking baggage, individuals being paged by bus railway or airlines, vehicle changes that affect the travel of persons with disabilities, and emergencies (*e.g.*, fire, bomb threat).

12. REST AREAS/SEATING

Seating area / benches should be provided along the circulation path at regular intervals so that passengers do not need to walk more than 50 to 60 metres before being able to sit and rest.
Where seating is provided, designated seating for passengers with disabilities is to be provided at boarding gates and departure areas within viewing distance of communication boards and/or personnel and identified by the symbol of access.
Public transit operators should provide seating in passenger service areas where there may be long waiting lines or times, including at ticket sales counters, check-in counters, secured screening and during inter-country travel in customs areas and baggage retrieval areas.
Designated seating should be provided for at boarding gates and departure areas within viewing distance of communication boards, and within hearing range of audio announcements as well. Such seating areas

should be identified by the symbol of accessibility and shelter should be provided where this seating is outdoors.

In outdoor settings, seating should be provided along with the planned hawker spaces.

At waiting lounges for persons with disabilities chairs should have armrests and backrest.

13. TACTILE PAVING- GUIDING & WARNING[1]

(a) Tactile Guiding Paver (Line-Type)

It is recommended to install a row of tactile guidance paver along the entire length of the proposed accessible route for visual impaired persons. Care must be taken to ensure that there are no obstacles, such as wall, pillar, uneven surfaces, Soffit (underside /open area under the stairs, along the route traversed by the guidance paver. Also, there should be clear headroom of at least 2.1 meters height above the tactile guidance paver, free of protruding objects such as overhanging advertisement panel and signage, along the entire length of the walk.

(b) Tactile Warning Paver (Dot-Type)

Indicate an approaching potential hazard or a change in direction of the walkway, and serve as a warning of the approaching danger to persons with visual impairments, preparing them to tread cautiously and expect obstacles along the travel path, traffic intersections, doorways, stairs, etc. They are used to screen off obstacles, drop-offs or other hazards, to discourage movement in an incorrect direction, and to warn of a corner or junction. Two rows of tactile warning paver should be installed across the entire width of the designated accessible passenger pathway at appropriate places such as before intersections, terminal entrances, obstacles such as signage, and each time the walkway changes direction.

14. PLACES TO INSTALL WARNING PAVER

In front of an area where traffic is present.

In front of an entrance/exit to and from a staircase or multi-level crossing facility.

Entrances/exits at public transport terminals or boarding areas.

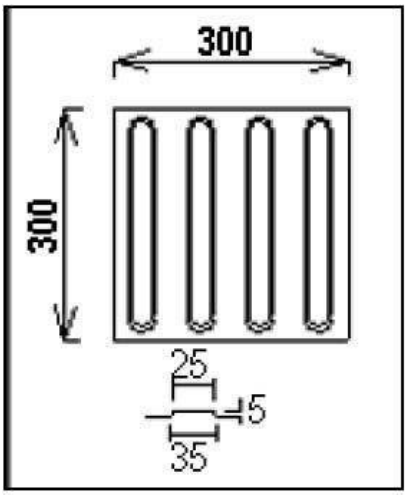


Fig. 13.3 - Guiding paver

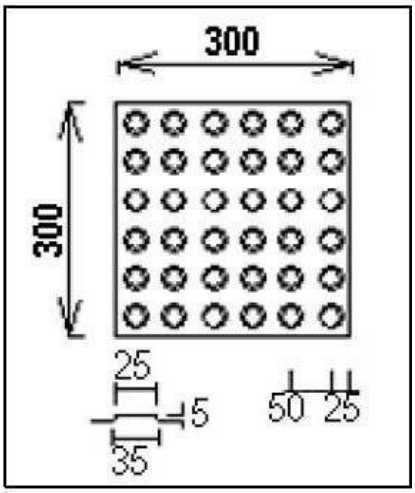


Fig. 13.4 - Warning paver

15. DOORS

Whatever the type of entrance door, it must be wide enough to accommodate passenger traffic comfortably.

The recommended minimum clear opening width of an internal door is 900mm minimum.
Where doors comprise two leaves (i.e. double doors), each leaf should be 900mm min. wide, so that persons carrying large items and people using wheelchairs do not have to open both leaves.
Manual doors should incorporate kick plates 300-400mm high to withstand impact of wheelchair footrest (this is especially important where doors are glazed).

- Also be fitted with vision panels at least between 900mm and 1500mm from floor level.
- Be color contrasted with the surrounding wall and should not be heavier than 22N to open.
- Lever handles and push type mechanisms are recommended . When a sliding door is fully open, handles should be usable from both sides.

Where revolving doors or turnstiles are used, an alternative wheelchair-accessible entrance must also be provided.
A distance of 400mm should be provided beyond the leading edge of door to enable a wheelchair user to maneuver and to reach the handle.
To ensure maximum clarity for persons with visual impairments, the entrance should be easily distinguishable from its surroundings by the effective use of landscaping, signage, colour (preferably yellow/orange), tonal contrast and tactile surfacing.
Door hardware should be positioned between 900-1000mm above floor (figure 28).
Operable devices such as handles, pulls, latches and locks should:

- Be operable by one hand
- Not require fine finger control, tight grasping, pinching or twisting to operate

Glazed doors and fixed glazed areas should be made visible by use of a clear, colour and tone contrasted warning or decorative feature that is effective from both inside and outside and under any lighting conditions, e.g.

a logo, of minimum dimensions 150mm by 150mm (though not necessarily square), set at eye level.

16. STEPS & STAIRS

Steps should be uniform with the tread not less than 300mm and the risers 150mm.
The risers should not be open.
The steps should have an unobstructed width of 1200mm minimum.
All steps should be fitted with a permanent colour and tone contrasting at the step edge, extending the full width of the step, reaching a minimum depth of 50mm on both tread and riser.
Have continuous handrails on both sides including the wall (if any) at two levels
Warning paver to be placed 300mm at the beginning and at the end of all stairs.
Nosing to be avoided.
The staircase should be adequately and uniformly illuminated during day and night (when in use). The level of illumination should preferably fall between 100-150 lux.
The rise of a flight between landings must be no more than 1200mm.
There should be no more than 12 risers in one flight run.
The stair covering and nosing should be slip-resistant, non-reflective, firmly-fixed and easy to maintain.
Soffit (underside /open area under the stairs) of the stairs should be enclosed or protected.

17. HANDRAILS

Handrails should be circular in section with a diameter of 38-45mm and formed from materials which provide good grip such as timber, nylon or powder coating, matt finish metal finishes.
The handrail should contrast in colour (preferably yellow/orange) with surrounding surfaces.
At least 50mm clear of the surface to which they are attached and should be supported on brackets which do not obstruct continuous hand contact with the handrail.
The handrail should be positioned at two levels- 760mm and 900mm above the pitch-line of a flight of stairs.
Handrail at foot of the flight of stairs should extend 300mm beyond the stairs in the line of travel and returning to the wall or floor or rounded off, with a

positive end that does not project into the route of travel.

18. RAMPS

Ramps gradient should ideally be 1 in 20 and no greater than 1 in 12.
Width of the ramp should not be less than 1200mm and preferred width is 1800mm.
The steeper the gradient, the shorter the length of ramp between landings.
On long ramps, a horizontal resting space should be provided every 6 meters.
Surface materials should be slip-resistant, non-reflective, firmly-fixed and easily maintained
The edge of the ramp should have an edge protection with a minimum height of 100mm.
Landings every 750mm of vertical rise.
A tapping or lower rail should be positioned so that its bottom edge is no higher than 200mm above ground level.
Handrails on the ramps should be on both sides at two levels: upper at 900mm and lower at 760mm; both end to be rounded and grouted; extend 300 mm beyond top and bottom of ramp .
A row of tactile warning paver should be placed 300mm beginning and end of each run.
Landings should be provided at regular intervals as indicated in the table (Table 13.1).

Table 13.1 - Specifications for Ramps

Level difference	Minimum gradient of Ramp	RampWidth	Handrail on both sides	Comments
≥ 150 mm ≤ 300 mm	1:12	1200 mm	√	
≥ 300 mm ≤ 750 mm	1:12	1500 mm	√	Landings every 5 meters of ramp run.
≥ 750 mm ≤ 3000mm	1:15	1800 mm	√	Landings every 9 meters of ramp run.
≥ 3000 mm	1:20	1800 mm	√	Landings every 9 meters of ramp run.

19. LIFTS/ELEVATORS

A carefully designed lift makes a huge contribution to the accessibility of a multi-storied terminal building for persons with disabilities.

Lift locations should be clearly signposted from the main pedestrian route and recognizable through design and location.

The colour and tone of the lift doors should contrast with the surrounding wall finish to assist in their location. Lift doors with metallic finishes such as steel grey and silver should be avoided as they are difficult to identify by persons with low vision.

The lift lobby shall be of an inside measurement of 1800mm X 2000mm or more. A clear landing area in front of the lift doors of minimum dimensions 1500mm x 1500mm should be provided.

By making the landing area distinguishable by floor surface and contrast, it will aid location and recognition of core areas. This could comprise a change in floor finish from thin carpet to vinyl/PVC, or cement/mosaic floor to carpet.

Changes in floor finish must be flushed. There should be no level difference between lift door and the floor surface at each level; the gap if unavoidable should not be more than 12mm.

The floor level/location should be indicated on the wall adjacent to or just above the call buttons, and opposite the lift doors where possible.

20. LIFT DIMENSIONS

Provisions of at least one lift shall be made for people using wheelchairs with the following car dimensions:

- Clear internal depth -1500 mm minimum
- Clear internal width - 1500 mm minimum
- Entrance door width - 900 mm minimum

21. LIFT CONTROLS

The lift call button should be wall-mounted adjacent to the lift and should contrast with wall finish, either by using a contrasting panel, or a contrasting border around the button panel.

The call buttons should be located within the range 800-1000mm above floor finish.

Buttons should not be touch sensitive, but should require a light positive

pressure and should ideally be large enough to be operable by the palm of the hand if required.

The control buttons inside the lift should be positioned on the side wall rather than front wall to allow access from the back and front of the lift car, by mobility aid users like wheelchair users.

The control buttons should contrast with their surroundings and illuminate when pressed and should incorporate highly visible tactile embossed (NOT engraved) characters and in Braille.

Time of closing of an automatic door should be more than 5 seconds and the closing speed should not exceed 25 meters per second. There should be a provision of censor enabled closing.

In larger lifts, controls should be positioned on both side walls, at least 400mm from front wall and between 800-1000mm above floor level.

22. CAR DESIGN

Internal walls should have a non-reflective, matt finish in a colour and tone contrasting with the floor, which should also have a matt, non-slip finish.

Use of reflective materials such as metal (stainless steel for example) can be problematic in creating sufficient contrast with control buttons, emergency telephone cabinet, etc. for persons with low vision and the use of such materials should be avoided wherever possible.

A mirror (750mm above floor level) on the rear wall can be useful to persons using wheelchairs and other mobility aids should they need to reverse safely out of the lift car or view the floor numbers.

Internal lighting should provide a level of illumination of minimum 100 lux (approximately 50-75 lux at floor level), uniformly distributed, avoiding the use of spotlights or down lighters.

A grab bar should be provided along both sides and the back wall, 900mm above floor level.

Handrails should be of tubular or oval cross section, in order to be easily gripped and capable of providing support.

Handrails should be positioned so that there is a clear space behind the handrail to allow it to be grasped i.e. knuckle space should be 50mm.

13.6 INFORMATION SYSTEMS

Lifts should have both visual and audible floor level indicators

Audible systems are also usually capable of incorporating additional messages, such as door closing, or, in the case of an emergency, reassurance (with manual over-ride allowing communication with lift

occupants).
Announcement system should be of 50 decibel.
The display could be digital or segmented LED, or an appropriate alternative. A yellow or light green on black display is preferred to a red on black display as it is easier to read.

13.7 General and Accessible Toilets

1. Signages

All signage of general toilets should be in bold and contrasting colors.
For persons with low vision and vision impairments: male pictogram in triangle and female pictogram in circle, marked on plates along with Braille & raised alphabets, to be mounted on wall next to door near the latch side, at a height between 1400mm-1600mm.
Warning strip/ thin rubber door mat to be provided 300mm before and after the toilet entrance.
Tactile paver to be provided for urinals, WC and washbasins for persons with vision impairments.

2. Accessible toilets

Should have the international symbol of accessibility displayed outside for wheelchair access.
The toilet door should be an outward opening door or two way opening or a sliding type and should provide a clear opening width of at least 900mm.
It should have a horizontal pull-bar, at least 600mm long, on the inside of the door, located so that it is 130mm from the hinged side of the door and at a height of 1000mm.

3. WC Compartment Dimensions

The dimensions of a unisex toilet are critical in ensuring access. The compartment should be at least 2200mm and 2000mm. This will allow use by both manual and motorized wheelchair users.
Layout of the fixtures in the toilet should be such that a clearing maneuvering space of 1500mm x 1500mm in front of the WC and washbasin.

4. Water Closet (WC) Fittings

Top of the WC seat should be 450-480mm above finished floor level,

preferably be of wall hung or corbel type as it provides additional space at the toe level.

An unobstructed space 900mm wide should be provided to one side of the WC for transfer, together with a clear space 1200mm deep in front of the WC.

WC should be centred 500mm away from the side wall, with the front edge of the pan 750mm away from the back wall. Have a back support. The WC with a back support should not incorporate a lid, since this can hinder transfer.

L-shape grab bar at the adjacent wall and on the transfer side (open side) swing up grab bar shall be provided.

The cistern should have a lever flush mechanism, located on the transfer side and not on the wall side and not more than 1000mm from the floor.

5. GRAB BARS

Grab bars should be manufactured from a material which contrasts with the wall finish (or use dark tiles behind light colored rails), be warm to touch and provide good grip.

It is essential that all grab rails are adequately fixed, since considerable pressure will be placed on the rail during maneuvering. Grab bars should sustain weight of 200kgs minimum.

A hinged type moveable grab bar should be installed adjacent to the WC on the transfer side. This rail can incorporate a toilet tissue holder. A distance of 320mm from the centre line of the WC between heights of 200-250mm from the top of the WC seat. It should extend 100-150mm beyond the front of the WC.

A fixed wall-mounted L- shape grab bar (600mm long horizontal and 700mm long vertical) on the wall side should be provided. It should be placed at a height of 200-250mm above the WC seat level.

6. WASHBASINS

Hand washbasins should be fitted on cantilevered brackets fixed to the wall.

The basin should be fixed no higher than 750mm above the finished floor level.

Be of dimensions 520mm and 410mm, mounted such that the top edge is between 800- 900mm from the floor; have a knee space of at least 760mm wide by 200mm deep by 650-680mm high.

The position of the basin should not restrict access to the WC i.e. it should be located 900mm away from the WC.

A lever operated mixer tap fitted on the side of the basin closest to the WC is

useful as it allows hot and cold water to be used from a seated position on the WC.
The hand drying facilities should be located close to the hand washbasin between 1000-1200mm.
Lever type handles for taps are recommended.
Mirror's bottom edge to be 1000mm from the floor and may be inclined at an angle.

7. FIXTURES AND FITTINGS

Contrast between fittings and fixtures and wall or floor finishes will assist in their location. For example, using contrasting fittings, or dark tiles behind white hand washbasins and urinals, contrasting soap dispensers and toilet roll holders. Contrast between critical surfaces, e.g. floors, walls and ceilings helps to define the dimensions of the room.
Towel rails, rings and handrails should be securely fixed to the walls and positioned at 800-1000mm from the floor.
The mirror should be tilted at an angle of 300 for better visibility by wheelchair users.
It should have lower edge at 1000mm above floor finish and top edge around 1800mm above floor finish.
Hooks should be available at both lower-1200mm and standard heights-1400mm, projecting not more than 40mm from the wall.
Where possible, be equipped with a shelf of dimensions 400mm x 200mm fixed at a height of between 900mm and 1000mm from the floor.
Light fittings should illuminate the user's face without being visible in the mirror. For this reason, most units which have an integral light are unsatisfactory.
Large, easy to operate switches are recommended, contrasting with background to assist location, at a maximum height of 1000mm above floor finish.
All toilet facilities should incorporate visual fire alarms.
Alarms must be located so that assistance can be summoned both when on the toilet pan i.e. at 900mm height and lying on the floor i.e. at 300mm, from floor surface. Alarms should be located close to the side wall nearest the toilet pan, 750mm away from rear wall and at 900mm and 200mm above floor finish

8. SIGNAGE OF ACCESSIBLE TOILETS

All unisex accessible toilets to have access symbol in contrast colours. A distinct audio sound (beeper/clapper) may be installed above the entrance door for identification of the toilets.

Fig. 13.5 - Signage for accessible washroom

9. ACCESSIBLE URINAL

At least one of the urinals should have grab bars to support ambulant persons with disabilities (for example, people using mobility aids like crutches).

A stall-type urinal is recommended.

Urinals shall be stall-type or wall-hung, with an elongated rim at a maximum of 430mm above the finish floor. This is usable by children, short stature persons and wheelchair users.

Urinal shields (that do not extend beyond the front edge of the urinal rim) should be provided with 735mm clearance between them.

Grab bars to be installed on each side, and in the front, of the urinal.

The front bar is to provide chest support; the sidebars are for the user to hold on to while standing.

13.8 DRINKING WATER UNITS

Drinking water fountains or water coolers shall have up front spouts and control .

Drinking water fountains or water coolers shall be hand-operated or hand and foot-operated.

Conventional floor mounted water coolers may be convenient to individuals in wheelchairs if a small fountain is mounted on the side of the cooler 800mm above the floor.

Fully recessed drinking water fountains are not recommended.

Leg and knee space to be provided with basin to avoid spilling of water . This

allows both front and parallel access to taps for persons using mobility aids like wheel chair, crutches etc.

13.9 VISUAL CONTRASTS

Visual contrasts means adequate contrast created by difference of at least 30 LRV (Light Reflectance Value) of the two surfaces/ objects and it helps everyone especially persons with vision impairments.
Visual contrast should be provided between:

- Critical Surfaces (walls, ceiling and floor),
- Signage and background sign frame/ wall,
- Step edges and risers/ treads on steps,
- Handrails and background walls,
- Doors and surrounding walls,
- Switches/ sockets and background wall,
- Toilet fixtures and critical surfaces in toilet.

Barriers and hazards should be highlighted by incorporating colours and luminance contrast.

13.10 EMERGENCY EGRESS/EVACUATION

Placement (accessibility) and visibility of such devices is very important. The following is to be considered for the installation of such alarm devices; fire alarm boxes, emergency call buttons and lit panels should be installed between heights of 800mm and 1000mm from the furnished floor surface. These should be adequately contrasted from the background wall and should be labelled with raised letters and should also be in Braille.

A pre-recorded message, alerting an emergency to the control room or reception should be installed in the telephone and this should be accessible by a ‘hotkey’ on the phone keypad. This ‘hotkey’ should be distinct from the rest of the keypad.

13.11 ALERTING SYSTEMS

In emergency situations, it is critical that people are quickly alerted to the situation at hand, for persons with disability the following needs to be considered.

Consider having audible alarms with ‘voice instructions’ that can help guide them to the nearest emergency exit. As an alternative to the pre-recorded

messages, these alarms may be connected to the central control room for on-the-spot broadcasts.

Non-auditory alarms (visual or sensory) to alert persons with hearing impairments should be installed at visible locations in all areas that the passengers may use (including toilet areas, etc).

Non-auditory alarms include:

Flashing beacons
Vibrating pillows and vibrating beds.
Pagers or mobile phones that give out a vibrating alarm along with a flashing light (these may be issued to persons with vision or hearing impairments at the time of check-in or boarding the vehicle.)

13.12 WRITTEN EVACUATION PROCEDURE

A written evacuation procedure that details the egress plan for people with disability should be installed behind the entrance door in the accessible rest rooms. The evacuation procedure should be detailed in large print letters that contrast strongly against the background. Where possible, it should also incorporate raised letters and Braille. The evacuation route should be displayed on a high contrast tactile map for benefit of persons with vision impairments.

13.13 EMERGENCY EVACUATION ROUTE

Designate routes that are at least 1200mm wide, to ensure that a person using a wheelchair and a non disabled person are able to pass each other along the route. The route should be free of any steps or sudden changes in level and should be kept free from obstacles such as furniture, coolers, AC units and flower pots.

Use Exit signage along the route. Orientation and direction signs should be installed frequently along the evacuation route and these should preferably be internally illuminated. The exit door signage should also be internally illuminated.

A ‘way guidance lighting system’ consisting of low mounted LED strips to outline the exit route (with frequent illuminated direction indicators along the route) should be installed along the entire length of the evacuation route.

Way guidance systems allow persons with vision impairments to walk significantly faster than traditional overhead emergency lighting. Moreover, emergency exit lights in green color and directional signals mounted near the floor have been found to be useful for all people in cases where a lot of smoke is present.

13.14 WAY GUIDANCE SYSTEM

Luminance on the floor should be 1lux minimum provided on along the centre line of the route and on stairs.
Install clear illuminated sign above exit and also directional signage along the route.
The directional exit signs with arrows indicating the way to the escape route should be provided at a height of 500mm from the floor level on the wall and should be internally illuminated by electric light connected to corridor circuits.

13.15 FIRE RESISTANT DOORS

Fire resistant doors and doors used along the emergency evacuation route are generally heavy and the force required to open these is much higher than 25 Newtons, making it difficult for people with disability to negotiate these doors independently. There are, however, magnetic and other types of door holders available that can be connected to fire alarms so that they will hold the doors open normally but will release the doors when the fire alarm is activated.

13.16 STREET DESIGN

(a) Footpath (Sidewalk)

Footpaths should be regarded as a transportation system which is connected and continuous, just like roadways and railways. They should not be sporadically placed where ever convenient, but instead should be provided consistently between all major attractions, trip generators, and other locations where people walk.

Footpath should:

Be along the entire length of the road;
Have height of a standard public step riser i.e. 150 mm maximum;
Be at least 1800 mm wide;

Have non-slip surface;
Have tactile guiding paver for persons with visual impairments;
Preferably have well defined edges of paths and routes by use of different colours and textures;
Have no obstacles or projections along the pathway. If this is unavoidable, there should be clear headroom of at least 2200 mm from the floor level;
The minimum 1.8m (width) x 2.2m (Height) Walking Zone should be clear of all obstructions – both horizontally and vertically.

Footpath should have:

Have kerb ramps where ever a person is expected to walk into or off the pathway; and
Have tactile warning paver installed next to all entry and exit points from the footpath.

(b) Kerb Ramp

Kerb should be dropped, to be flush with walk way, at a gradient no greater than 1:10 on both sides of necessary and convenient crossing points. Width should not be less than 1200mm. If width (X) is less than 1200mm, then slope of the flared side shall not exceed 1:12.
Floor tactile paving- Guiding & Warning paver shall be provided to guide persons with vision impairment so that a person with vision impairment does not accidentally walk onto the road.
Finishes shall have non-slip surface with a texture traversable by a wheel chair.

(c) Road Intersections

Pedestrian crossings should be equipped with traffic control signal.
Traffic islands to reduce the length of the crossing are recommended for the safety of all road users.
Warning pavers should be provided to indicate the position of pedestrian crossings for the benefit of people with visual impairments.
Table tops (raised road level to the sidewalk height) are helpful in reducing the speed of traffic approaching the intersection

(d) Median/Pedestrian Refuge

Raised islands in crossings should:

Cut through and level with the street; or
Have kerb ramps on both the sides and have a level area of not less than 1500 mm long in the middle; and
A coloured tactile marking strip at least 600 mm wide should mark the beginning and end of a median/ pedestrian refuge to guide pedestrian with visual impairments to its location.

13.17 TRAFFIC SIGNALS

Pedestrian traffic lights should be provided with clearly audible signals for the benefit of pedestrians with visual impairments;
Acoustic devices should be installed on a pole at the point of origin of crossing and not at the point of destination;
The installation of two adjacent acoustic devices such as beepers is not recommended in order to avoid disorientation;
The time interval allowed for crossing should be programmed according to the slowest crossing persons; and
Acoustical signals encourage safer crossing behaviour among children as well.

13.18 SUBWAY AND FOOT OVER BRIDGE

Subways and foot over bridges should be accessible for people with disabilities. This may be achieved by:

Provision of signage at strategic location;
Provision of slope ramps or lifts at both the ends to enable wheelchair accessibility ;
Ensuring that the walkway is at least 1500 mm wide;
Provision of tactile guiding and warning paver along the length of the walkway;
Keeping the walkway; free from any obstructions and projections; and
Providing for seats for people with ambulatory disabilities at regular intervals along the walkway and at landings.

13.19 ALIGHTING AND BOARDING AREAS

- All areas and services provided in the Mass Rapid Transit System (Metro/subway), bus terminuses, etc. that are open to the public should be accessible.

13.20 APPROACH

Passenger walkways, including crossings to the bus stops, taxi stands, terminal / station building, etc. should be accessible to persons with disabilities.

Uneven surfaces should be repaired and anything that encroaches on corridors or paths of travel should be removed to avoid creating new barriers. Any obstructions or areas requiring maintenance should be white cane detectable2.

Access path from plot entry and surface parking to terminal entrance shall have even surface without any steps.

Slope, if any, shall not have gradient greater than 5%. The walkway should not have a gradient exceeding 1:20. It also refers to cross slope.

Texture change in walk ways adjacent to seating by means of tactile warning paver should be provided for persons with vision impairment.

Avoid gratings in walks.

13.21 CAR PARK

(A) SIGNAGE

International symbol of accessibility (wheelchair sign) should be displayed at approaches and entrances to car parks to indicate the provision of accessible parking lot for persons with disabilities within the vicinity.

Directional signs shall be displayed at points where there is a change of direction to direct persons with disabilities to the accessible parking lot.

Where the location of the accessible parking lot is not obvious or is distant from the approach viewpoints, the directional signs shall be placed along the route leading to the accessible parking lot.

Accessible parking lot should be identifiable by the International Symbol of Accessibility. The signs should not be obscured by a vehicle parked in the designated lot.

Vertical signs shall be provided, to make it easily visible, the sign should be at a minimum height of 2100 mm .

(B) SYMBOL

International Symbol of Accessibility should be clearly marked on the accessible parking lot for drivers/riders with disabilities only.

A square with dimensions of at least 1000 mm but not exceeding 1500 mm in length;

Be located at the centre of the lot; and

The colour of the symbol should be white on a blue background.

(C) CAR PARK ENTRANCE

The car park entrance should have a height clearance of at least 2400 mm.

LOCATION

Accessible parking lots that serve a building should be located nearest to an accessible entrance and / or lift lobby within 30 meters. In case the access is through lift, the parking shall be located within 30 meters.

The accessible route of 1200 mm width is required for wheelchair users to pass behind vehicle that may be backing out.

(D) ACCESSIBLE CAR PARKING LOT

The accessible car parking lot should:

Have minimum dimensions 5000 mm × 3600 mm;

Have a firm, level surface without aeration slabs;

Wherever possible, be sheltered;

Where there are two accessible parking bays adjoining each other, then the 1200 mm side transfer bay may be shared by the two parking bays. The transfer zones, both on the side and the rear should have yellow and while cross-hatch road markings;

Two accessible parking lots shall be provided for every 25 no of car spaces.

(E) DROP OFF AND PICK UP AREAS

Designated drop-off and pick-up spaces, to be clearly marked with international symbol of accessibility.

Kerbs wherever provided, should have kerb ramps.

CHAPTER – 14

ENVIRONMENT AND SOCIAL IMPACT ASSESSMENT

14.1 LEGAL, POLICY AND INSTITUTIONAL FRAME WORK

The available national and state level legal Acts and Legislation referred during the study are:

The Water (Prevention and Control of Pollution) Act, 1974 (Amendment 1988).
The Water (Prevention and Control of Pollution) Cess Act 1977, (Amendment), 2003.
The Water (Prevention and Control of Pollution) Cess Rules, 1978, 1991.
The Air (Prevention and Control of Pollution) Act 1981, amended 1987.
The Air (Prevention and Control of Pollution) (Union Territories) Rules, 1982, 1983
Noise Pollution (Regulation and Control) Rules, 2000 amendment 2002, 2006.
Municipal Solid Waste Rules, 2000
The Environment (Protection) Act, 1986, amended 1991.
The Environment (Protection) Rules,1986.
The Indian Forest Act, 1927.
Forest (Conservation) Act, 1980, amended 1988.
Forest (Conservation) Rules, 2003.
Gujarat (Urban Area) Protection of Trees Act 1975
The Wild Life (Protection) Act 1972, Amendment, 2002

14.1.1 Environmental Categorization

The proposed project does not passes through any Wildlife Sanctuary, National Park, or any other environmentally sensitive or protected areas. The proposed project comprises two alignments viz., between Beshan to Saroli and Sarthana to Dream City. The alignments are proposed mostly on the centre of the road. Although, the proposed project will bring in many benefits

to the area, there is potential for environmental impacts on the ambient environment.

Requirement of Environmental Clearance

As per provisions of the EIA Notification, 14 September 2006 as amended up to 1 December 2009, any person who desires to undertake any new project in any part of India or the expansion or modernization of any existing industry or project listed in Schedule-I of the said notification shall submit an application to the Ministry of Environment and Forests, Government of India in accordance with the guidelines issued by the Central Government in the Ministry of Environment and Forests from time to time. Metro Rail project is not included in the Schedule-I of the EIA Notification, 2006. Thus, the project does not require an environmental clearance certificate from the Ministry of Environment and Forests, Government of India.

Requirement of Forest Clearance

As per Indian "Forests Conservation Act (1980), every project requiring diversion of forest land for non-forestry purposes require forest clearance from MoEF. The forestry clearance is granted through two-stage process: Stage 1 refers, in principle agreement, to the project proposal in which usually the conditions relating to transfer, mutation and declaration as RF/ PF under the Indian Forest Act, 1972, of equivalent non-forest land for compensatory afforestation and funds for raising compensatory afforestation thereof are stipulated. Stage II involves formal approval under the Act after receipt of compliance report from the State Government in respect of the stipulated conditions. Since alignment is not passing through any forest land and no diversion of forest land is involved in the proposed project, no forest clearance is required for this project.

Required Clearances/Permissions

For the proposed project, required clearances/ permissions related to environment have been summarized below.

Table 14.1: Permissions/Clearances Required for the Project

S. No.	Permissions/ clearances	Acts/Rule/Notifications/ Guidelines	Concerned Agency	Responsibility
A. Pre-construction Stage				
1	Permission for felling of trees	Forest Conservation Act (1980) Procedural Guidelines developed by the Department of Environment, GoM; Tree removal will be guided as per state government rules.	SMC / District Collector	SMC
B. Implementation Stage				
2	Consent to operate hot mix plant, crushers, batching plant	Air (Prevention and Control of Pollution) Act 1981	Gujarat State Pollution Control Board	Contractor
3	Permission for withdrawal of groundwater	Environment (Protection) Act, 1986	Central Ground Water Authority	Contractor
4	Permission for sand mining from river bed	Environment (Protection) Act, 1986	Mining Department/ MoEF	Contractor
5	Authorization for Disposal of Hazardous Waste	Hazardous Waste (Management and Handling) Rules 1989	Gujarat State Pollution Control Board	Contractor
6	Disposal of bituminous and other wastes	Hazardous Waste (Management and Handling) Rules 1989	Local civic body to use local solid waste disposal	Contractor
7	Consent for disposal of sewage from labour camps.	Water (Prevention and Control of Pollution) Act 1974	Gujarat State Pollution Control Board	Contractor
8	Pollution Under Control Certificate	Central Motor and Vehicle Act 1988	Department of Transport, Govt. of Gujarat authorised testing centres	Contractor
9	Roof Top Rain Water Harvesting (RWH)	Central Groundwater Authority (CGWA) Guidelines	Central Ground Water Authority/ SCM	Contractor

10	Permission for groundwater extraction for drinking purpose	Environment (Protection) Act, 1986	CGWA	Contractor
11	Employing Labour/ workers	The Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996	District Labour Commissioner	Contractor

14.1.2 Objective and Scope of the Study

The objective of the Environment and Social Impact Assessment study is to facilitate the Surat Municipal Corporation(SMC) evaluate the environmental impacts of its proposed activity. SMC proposes to apply for loan to seek financial support from multilateral funding agencies. Thus, the objective of the study is to conduct Environmental Impact Assessment as per requirement of multilateral funding agencies. The scope of EIA includes the impacts resulting from pre-construction, during construction and operation phases of the proposed metro alignment in Surat. In addition, it is proposed to establish environmental baseline and safeguard measures for protection of environment for sustainable development during project cycles.

14.1.3 Approach and Methodology

The SMC has considered different alternative corridors. The underlying principles for evaluation for each corridor, without affecting the overall usefulness of the corridor, are minimum private land acquisition, least disturbance to properties, minimal disturbance to ecology/biodiversity. In the analysis of alternatives, a comparison of scenario with and without the project has also been made. The final alternative was fixed based on Technical Feasibility, Socio-economic acceptability, and Environmental sustainability for Metro Corridors. The environmental study is carried out for the alignment proposed by SMC. The approach is to follow the sequence of steps adopted in an EIA study. The basic concept is to ascertain the existing baseline conditions and assess the impacts as a result of construction and operation of the project. The changes likely to occur in different components of the environment viz. physical, biological / ecological, environmental and socio-economic etc. have been studied, analyzed and quantified, wherever possible. The identification of parameters for data generation and impact assessment are important. The analysis of assessment depends upon the reliable data generated/ available on environmental attributed. This study has

documented the baseline data for various parameters of physical, ecological and environmental pollution (air, water and noise). The impacts are assessed for various phases of project cycle namely:

Impacts due to project location,
Impacts due to project design,
Impacts due to project construction, and
Impacts due to project operation.

The impacts are categorized as negative and positive. The cost of management and monitoring programs were estimated and budgeted for.

14.2 PROJECT AREA

The metro project in Surat city of Gujarat between Sarthana and Dream City. The second alignment is proposed between Beshan and Saroli. The proposed alignment would serve the city by providing better connectivity. The project area also includes the maintenance depot and construction depots in addition to viaduct and station areas. This metro corridor is proposed in Surat to cater the requirement of the city for a length of about 40.352 Km. Corridor-1 between Sarthana and Dream City will be partially elevated and partially underground. The alignment-2 between Beshan and Saroli will be elevated entirely. The Metro corridor will have standard Guage alignment.

14.3 ENVIRONMENTAL SCOPING

Baseline environmental status in and around the proposed project depicts the existing environmental conditions of the location. Baseline data was collected for various/environmental attributes so as to compute the impacts that are likely to arise due to proposed project.

The scope of the present study includes detailed characterization of following environmental components, which are most likely to be influenced by the proposed project:

Land Environment
Water Quality (Surface + Ground water)
Meteorological conditions
Ambient Air Quality
Noise Levels
Biodiversity
Socio Economic studies.

The information presented in this chapter has been acquired from various sources. Data on land environment has been collected and compiled from various reports and field surveys. The data on water, air, noise quality, and biodiversity were collected through field studies, sampling in January and March 2017. Climatological data was collected from India meteorological Department. Efforts have been made to compile the available data from literature, books, maps and reports. The methodology adopted for data collection is highlighted wherever necessary. Environmental Attributes and Frequency of Baseline Survey is presented in **Table 14.2**.

Table 14.2 Environmental Attributes and Frequency of Monitoring

S. No		Attribute	Parameter	No. of Samples	Source
	LAND ENVIRONMENT				
1		Geology	Geological Status	---	Literature review
2		Seismology	Seismic Hazard	---	Literature review
	WATER ENVIRONMENT				
3		Ground Water	Physical, Chemical and Biological parameters	7	Sampling locations
4.		Surface Water	Physical, Chemical and Biological parameters	3	Sampling locations
	AIR, NOISE AND METEOROLOGY				
4		Ambient Air Quality	PM_{10}, SO_2, NO_x	36	Sampling/Monitoring locations
5		Noise	Noise levels in dB (A) Leq, Lmax, Lmin, L_{10}, L_{50}, L_{90}	36	Sampling/Monitoring locations
6.		Soil	Physico-chemical parameters	7	Sampling Locations
7		Vibration	PPV mm/s, V_{rms} mm/s, VdB_{rms} (wrt $2.54x10^{-5}$ mm/s)	18	Sampling/ Monitoring Locations
	SOCIO-ECONOMIC				
8		Socio-economic aspects	Socio-economic profile	Once	Field Studies, Literature review.
	Ecology				
9		Trees	Number	Once	Filed Studies

Sampling locations of Water Quality, Noise Levels, and Ambient Air Qaulity are depicted in **Fig. 14.1**.

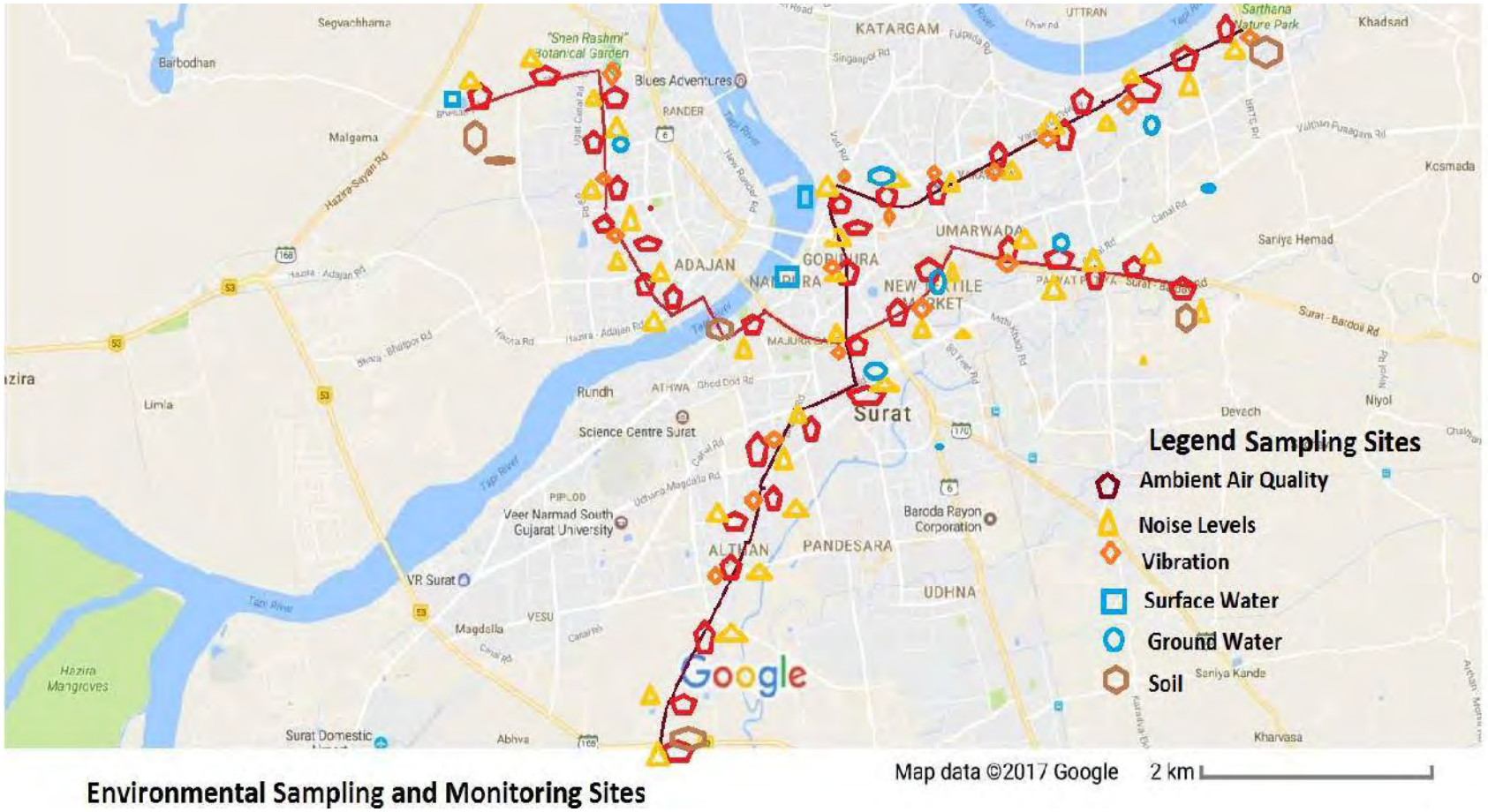


Fig. 14.1 Air Quality, Noise level and Water Sampling/Monitoring Sites

Table 14.3 Sampling / Monitoring Locations:

S. No	Station Name	Air	Noise	Vibration	Ground Water	Surface Water	Soil
	No. of Samples	39	39	18	7	3	7
	Corridor-1	**Sarthana to Dream City**					
1	Sarthana	Neelkanta Society	Neelkanta Society	Neelkanta Society			Sarthana
2	Nature park	Krishna Hotel	Krishna Hotel				
3	Varchha chopati garden	Ganga Yamuna Aptt	Ganga Yamuna Aptt	Ganga Yamuna Aptt	Ganga Yamuna Aptt		
4	Shri swaminarayan mandir kalakunj	Mamta Park-2	Mamta Park-2				
5	Kapodra	Balaji Kiriyana	Balaji Kiriyana	Balaji Kiriyana			
6	Labheshwar Chowk	Star Computers	Star Computers	Star Computers			
7	Central Warehouse	SBI	SBI	SBI			
8	Surat Railway Station	Lamba Hanuman Ji Temple	Lamba Hanuman Ji Temple	Lamba Hanuman Ji Temple	Nr Temple	Tapi Nr. Zila Panchayat	
9	Maskati Hospital	Sheetal Fashions	Sheetal Fashions	Sheetal Fashions			
10	Chowk Bazar	Church	Church	Church		Tapi Nr. Machiwar	
11	Kadarsha Ni Nal	Guard Post	Guard Post				
12	Majura Gate	DayalJi Mandap	DayalJi Mandap	DayalJi Mandap			
13	Roopali Canal	Rajdarshan Aptt	Rajdarshan Aptt		Rajdarshan Aptt		
14	Althan Tenament	ICICI Bank	ICICI Bank				
	Sarodaya Vidiyalaya,	Citizen Hardware	Citizen Hardware	Citizen Hardware			
15	Althan Gam	Ridhi-Sidhi Aptt.	Ridhi-Sidhi Aptt.	Ridhi-Sidhi Aptt.			
16	Apcha Nagar	Nr Transformer	Nr Transformer				
17	Surat Women ITI	Safal residency	Safal residency	Safal residency			
18	Bhimrad	Nr. Police Post	Nr. Police Post				
19	Convention Centre	Nr Market	Nr Market				
20	Dream City	Nr. OH Tank	Nr. OH Tank				Dream City
	Corridor-2	**Beshan to Saroli**					
1	Bheshan	Nr STP	Nr STP			Bheshan Pond	Bheshan
2	Botanical Garden	Nr Temple	Nr Temple				
3	Ugat Vaarigruh	Jyoti Marble	Jyoti Marble	Jyoti Marble			
4	Palanpur Road	D Block Mangalam Aptt	D Block Mangalam Aptt		Manglam Aptt.		
5	L.P. Savani school	Drishti Complex	Drishti Complex	Drishti Complex			
6	Performing Art Centre	Pravag Aptt.	Pravag Aptt.	Pravag Aptt.			
7	Adajan Gam	White Orchid	White Orchid				

8	Aquarium	Sun Aptt	Sun Aptt	Sun Aptt	Sun Aptt		
9	Badri Narayan Temple	Temple	Temple				
10	Ahthwa Chaupati	Hotel Krishna	Hotel Krishna				Chaupati
11	Majura Gate	Lovely Aquarium	Lovely Aquarium				
12	Udhana Darwaja	SB Centre	SB Centre				
13	Kamela Darwaja	Supr tex Tower	Supr tex Tower	Supr tex Tower			
14	Anjana Farm	Kakadiya Fabs	Kakadiya Fabs				
15	Modal Town	Swastik Plaza	Swastik Plaza	Swastik Plaza			
16	Magob Gam	Dharmendra Complex	Dharmendra Complex				
17	Bharat Cancer Hospital	Pandit Cargo	Pandit Cargo				
18	Saroli	Eicher Showroom	Eicher Showroom				Saroli

14.3.1 Land Environment

The Project area is situated in Surat city Gujarat. The elevation of the project area is ranging between 7m near to 22 m above the mean sea level (a-MSL). Parameters involved in land environment are, physiography, geology and soils, and seismicity. These are discussed in the following paragraphs.

14.3.1.1 Geography, Geology and Soil

Surat is a port city situated on the banks of the Tapi river. Damming of the Tapi caused the original port facilities to close; the nearest port is now in the Magadalla and Hazira area of Surat Metropolitan Region. The city is located at 21°10′N 72°50′E.[23] It has an average elevation of 13 meters. The Surat district is surrounded by the Bharuch, Narmada, Navsari, to the west is the Gulf of Cambay and the surrounding districts. The climate is tropical and monsoon rainfall is abundant (about 2,500 mm a year). Surat has grown in area since the early 1700s. The oldest part of the city developed in the area between the train station and the area known as Athwa lines. Since the 1970s most of the new development, including the most desirable location for the city's burgeoning middle and upper class, is the area between Athwa lines and the coast at Dumas.

Geographically, the district belongs to the western coastlands of the Deccan peninsula. The main Sahyadri scarp is a little outside the limits of the district towards east, but it gives the district its orientation, landscape features and drainage pattern. Distinct zones, viz., hilly areas, piedmont slopes, alluvial plains and coastal plains.

Hilly areas: The north-eastern parts of the district fall in this category. Here the general elevations are more than 100m amsl. The topography is rugged with low to moderate high hills and steep hill slopes. These parts are poorly populated and are infested by dense jungle of teak and bamboo.

Piedmont slopes: East-central parts of the district fall in this category. Here the elevations range between 60 and 100m.amsl. These parts show a gentle slope towards west. Topography is mainly plain with moderate to deep cutting river valleys and occasional hillocks.

Alluviual plains: Alluvial plains towards situated in the central parts of the district are characterized by flood plains of the Tapi, Kim and Purna rivers. Tapi has a meandering channel entrenched fairly deep and has cut deep terraces. The topography is generally plain with gentle slope towards west. The general elevations are below 60 m.amsl, the lowest elevation being 45m.amsl near Madhi.

Coastal Plains: The alluvial plains towards west merge into a dry barren sandy coastal plain fringed by marshy shore line. All the rivers form estuarine mouths. There are sand bars and spits near the shore.

Major geological formations exposed in the district are Quaternary alluvium, Tertiary limestone and sandstones and Deccan Trap basalt.

Geological Succession

Era	Period	Epoch /Series	Description
Cenozoic	Quaternary	Holocene	Soil, Younger alluvium Blown and Fluvio-marine deposits.
	Tertiary	Pleistocene	Older alluvium, conglomerate
		Miocene	Ferruginous sandstone (Gaj)
		Eocene	Numulitic limestone
		Palaeocene	Sub-Numulitic limestone, gypseous shale,Supra- Trappean sediments lateritised and having bauxite pockets.
Mesozoic	Lower Tertiary to Upper Cretaceous	Early Eocene To Upper	Deccan Trap basalt with dykes

14.3.1.2 Seismicity

The country has been classified into different zones indicating the intensity of damage or frequency of earthquake occurrences. Surat sits on a seismically active zone owing to the presence of 23 fault lines in the vicinity. Surat falls in zone III according to IS 1893: 2002 which means an

earthquake upto magnitude 6.5 on Richer scale may be expected (Figure 14.2).

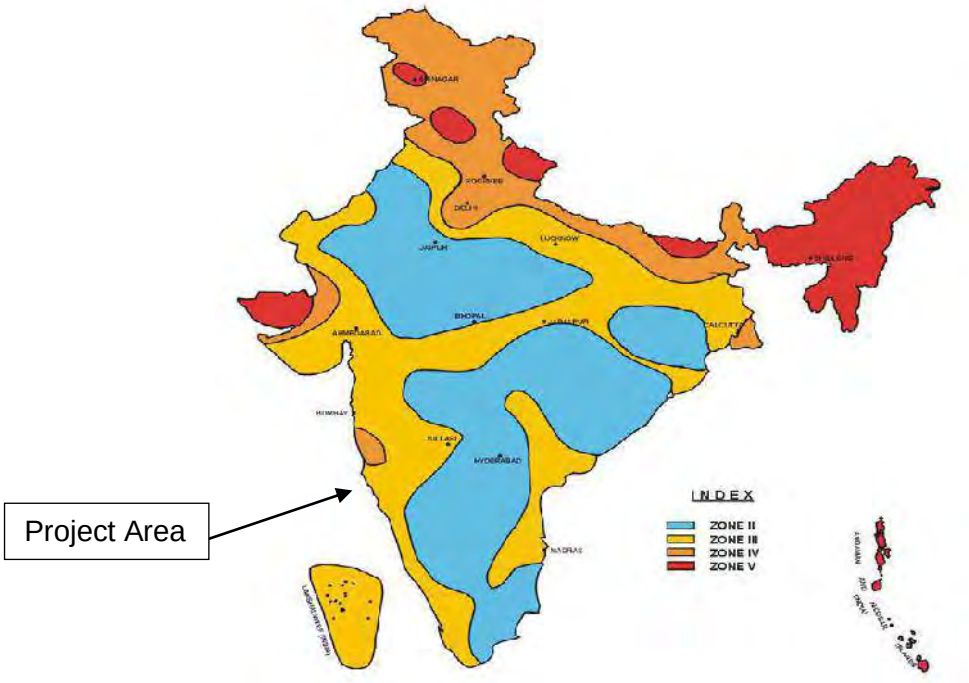


Figure – 14.2 Seismic Zones in India

14.4 WATER ENVIRONMENT

Water environment consists of water resources and its quality. Its study is important from the point of view of assessing the sufficiency of water resources for the needs of the project in its various stages of the project cycle and also to assess the impact of the project on water environment. In the proposed project, ground water is proposed to be used during operations to meet out domestic water requirements of the project in case water is not made available by Surat Municipal Corporation. Hence its quality has been tested to evaluate its suitability for the intended purpose. Anticipated impacts of the proposed project on water environment have also been addressed.

14.4.1 Water Resources

The Tapi is the major river which passes through the central parts of the district and flows towards the west. The river is perennial in nature. It originates in Madhya Pradesh near Betul and has about 62225 sq. Km. of

catchment area. The average width of the river upstream of Kathor bridge in about 500m. Downstream of the bridge the average being about 700m. Pickup weir was constructed on the Tapi River in 1954 at Kakarapar about 56km west of Surat. Ukai dam, constructed in 1965, is situated about 25km upstream of Kakarapar weir. Other prominent rivers draining the district are Kim, and Purna.

The basaltic lava flows are covered by black clayey to loamy soil. It is in general ranges in thickness up to one meter. The colour of the soil turns brown due to high iron content at places. In the piedmont slope area the soil is shallow to moderately deep, moderate to severely eroded and non calcareous in nature. The texture is silt clay loam to clay loams. The clay content varies from 30 to 60%. The water holding capacity of the soil is moderate. In the midland and flood plain areas, the soil is deep to very deep, light greyish to yellowish brown in colour. The texture is fine clay loam to sandy loam. The clay content varies from 25 to 60%. In coastal region the soil is deep to very deep dark grey to black colour. The texture is clay loam to silty

14.4.2 Hydrogeology and Ground Water

Aquifer System

The hydro geological frame work of the area is essentially governed by geological setting, distribution of rainfall fall and facilities of circulation and movement of water through inter connected primary and secondary porosity of the geological units forming the aquifers. The major aquifers in the district are formed by alluvium and Deccan Trap basalt with Tertiary formations occupying a small patch. The alluvium occurs in the western part of the district and along the streams whereas in eastern parts weathered and fractured basalt form aquifers. Hydro geological units:

1. Fissure Formations:
2. Porous Formation:

Fissure Formations

Deccan Traps from the aquifers in north-eastern, eastern and south eastern parts of the district comprising Mangrol, Mandvi, Vyara, Valod Mahuva, Songadh, Uchchhal and Nizar talukas. The ground water occurs in unconfined to semi-confined conditions. The occurrence and movement of ground water is governed by the thickness and extent of weathered zone and presence of fractures and joints. At places, dykes act as ground water barriers and restrict

the flow of ground water. On the other, at places, the dykes themselves from good aquifers owing to deeper levels of weathering in them.

Depths of dug wells in Deccan traps ranges from less than 2m to 25m with maximum number of wells, about 80%, being 5 to 15m deep. The depths to water levels in the dug wells range from 1 to 20m. However, in 90% of the wells, the water levels are less than 10m. bgl. Yields of the dug wells are low to moderate in central parts whereas they are low to very low on the hill slopes. The dug wells generally sustain pumping for 4 to 5 hours at the yields of 50 to 150 m^3/day. Recuperation of water levels is generally slow.

Alluvium

The western and north western parts of the district comprising Choryasi, Olpad, Kamrej, Palsana and parts of Bardoli talukas is covered by Alluvium. This aquifer can broadly be demarcated into two zones namely newer alluvium and older alluvium. The newer alluvium is present along the river courses and comprises fine to coarse grained sand trap wash with cay intercalations. The sand is unconsolidated but shows some degree of cohesion at places. Water levels are in general deeper in newer alluvium. Older alluvium is present in inter river plains and comprises sand, clay, kankar, grave and silt. The ground water occurs mostly under unconfined conditions but at places semi-confined conditions are also observed, probably due to presence of clay lenses.

The depths of the wells in alluvium generally range from 3.0 to 30.0m with some of the wells having bores down to 50m below bottom of the wells. The maximum number of wells, about 80% are 5 to 20m deep. Depths to water levels in the wells range from 0.5 to 15m bgl about 90% wells shows water levels less than 10m bgl. Yields of the dug wells and dugcum- bored wells range between 100 and 450 m^3/day.

Aquifer Parameters

The aquifer parameters data available with CGWB from the exploration carried out in different parts of the district was taken into the consideration. In Alluvial areas the depth of the tube wells range from 36 to 169m with piezometric levels varying between 2 to 10m. The specific capacity of the wells ranges from 19 to 1658 m^3/day/m. The transmissivity values obtained at these tube wells ranges from 104 (Kosad) to 2965m^2/day (Vihan).

Behaviour of Water Levels

The depth to water level ranges from 2.48m (Palsana) to 17.75m (Mahuva 2) during May2012. Spatially, the depth to water level in the district ranges between 2m to 20m during May 2012. The major area of the district is covered by the water level of 5m to 10m bgl . There are seven isolated pockets in the district which shows the water level ranging between 2 to 5m. Similarly, there are four isolated pockets in the district which shows the water level ranging between 10 to 20m. There is no area where the water level of less than 2m and more than 20 meter.

The depth to water level ranges from 0.1.39m (Palsana) to 16.55m (Mahuva2) during November2012. Spatially, the depth to water level in the district ranges between less than 2m to nearly 20m during November 2012. In the major part of the district the water level ranges between 2 m and 5 m bgl. The water level ranging between 5m and 10 m bgl is found in considerably in larger part of the district. There are three isolated and pockets in the district where the water level of less than 2 m is also observed. Similarly water level ranging between 10 to 20m is seen in three isolated pockets in the district.

14.4.3 Water Quality

Water quality is the physical, chemical and biological characteristics of water. It is most frequently used with reference to a set of standards against which compliance can be assessed. The most common standards used to assess water quality are related to drinking water, safety of human contact, and for health of ecosystems. An understanding of the various factors influencing water quality is thus very important as human health is largely dependent on the quality of water available for our use. Water sampling Sites have been shown in **Fig. 14.1.**

Groundwater quality is quite good. However, total dissolved solids are a little higher than the desirable limits but within permissible limits. All other parameters are well within the desirable limits.

Table 14.4 Ground Water Quality at Project Site

Corridor- 1 Sample	Ganga Yamuna Aptt	Anuradha Society	Lambe Hanuman Temmple	Rajdarshan Apptt.	*Limits*
Date of Sampling	**11-03-2017**	**11-03-2017**	**11-03-2017**	**11-03-2017**	
Colour, Hazen	Colourless	Colourless	Colourless	Colourless	5 (15) Max
Odour	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable
Taste	Agreeable	Agreeable	Agreeable	Agreeable	Agreeable
Turbidity, NTU	3.5	3.2	2.8	3.1	1 (5) Max
pH	7.89	7.85	7.81	7.77	6.5-8.5 Max
Total Hardness as Caco3, Mg/l	258	261	256	271	200 (600) Max
Chloride as Cl, Mg/l	141	143	154	183	250 (1000) Max
Total Iron as Fe, Mg/l	0.08	0.09	0.08	0.09	0.3 Max
Total Dissolved Solids, Mg/l	1283	1379	1413	1354	500 (2000) Max
Sulphates as So4, Mg/l	176	91	88	93	200 (400) Max
Nitrates as No3, Mg/l	65.1	37.1	34.3	35.1	45 Max
Fluorides as F, Mg/l	3.19	2.56	2.47	2.55	1.0 (1.5) Max
Lead as Pb, Mg/l	BDL	BDL	BDL	BDL	0.01 Max
Copper as Cu,Mg/l	BDL	BDL	BDL	BDL	0.05 (1.5) Max
Manganese as Mn,Mg/l	BDL	BDL	BDL	BDL	0.1 (0.3) Max
Phenolic Compound as C6H5OH,Mg/l	BDL	BDL	BDL	BDL	0.001 (0.002) Max
Mercury as Hg,Mg/l	BDL	BDL	BDL	BDL	0.001 Max
Cadmium as Cd,Mg/l	BDL	BDL	BDL	BDL	0.01 Max
Selenium as Se, Mg/l	BDL	BDL	BDL	BDL	0.01 Max
Arsenic as As,Mg/l	BDL	BDL	BDL	BDL	0.05 Max
Cyanide as Cn,Mg/l	BDL	BDL	BDL	BDL	0.05 Max
Zinc as Zn, Mg/l	1.23	1.05	1.03	1.06	5 (15) Max
Detergent as MBAS, Mg/l	BDL	BDL	BDL	BDL	0.2 (1.0) Max
Chromium as Cr+6 ,Mg/l	BDL	BDL	BDL	BDL	0.05 Max
Total Alkalinity as Caco3,Mg/l	188.3	173.2	171.1	192.3	200 (600) Max
Aluminum as Al,Mg/l	BDL	BDL	BDL	BDL	0.03(2) Max
Boron as B, Mg/l	BDL	BDL	BDL	BDL	0.5(1) Max
Bacteriological Analysis					
Coliform,MPN/100Ml	Nil	Nil	Nil	Nil	10 Max
E-Coli/Ml	Negative	Negative	Negative	Negative	Negative

Corridor- 2 Sample	Manglam Apptt.	Sun Apptt.	Nr. Eichr Showroom	*Limits*
Date of Sampling	**11-03-2017**	**11-03-2017**	**11-03-2017**	
Colour, Hazen	Colourless	Colourless	Colourless	5 (15) Max
Odour	Unobjectionable	Unobjectionable	Unobjectionable	Unobjectionable
Taste	Agreeable	Agreeable	Agreeable	Agreeable
Turbidity, NTU	2.5	3.1	2.4	1 (5) Max
pH	7.59	7.65	7.61	6.5-8.5 Max
Total Hardness as Caco3, Mg/l	278	268	296	200 (600) Max
Chloride as Cl, Mg/l	151	146	163	250 (1000) Max
Total Iron as Fe, Mg/l	0.06	0.08	0.09	0.3 Max
Total Dissolved Solids, Mg/l	1295	1402	1427	500 (2000) Max
Sulphates as So4, Mg/l	177	99	96	200 (400) Max
Nitrates as No3, Mg/l	65.1	37.1	34.3	45 Max
Fluorides as F, Mg/l	2.15	2.43	2.41	1.0 (1.5) Max
Lead as Pb, Mg/l	BDL	BDL	BDL	0.01 Max
Copper as Cu,Mg/l	BDL	BDL	BDL	0.05 (1.5) Max
Manganese as Mn,Mg/l	BDL	BDL	BDL	0.1 (0.3) Max
Phenolic Compound as C6H5OH,Mg/l	BDL	BDL	BDL	0.001 (0.002) Max
Mercury as Hg,Mg/l	BDL	BDL	BDL	0.001 Max
Cadmium as Cd,Mg/l	BDL	BDL	BDL	0.01 Max
Selenium as Se, Mg/l	BDL	BDL	BDL	0.01 Max
Arsenic as As,Mg/l	BDL	BDL	BDL	0.05 Max
Cyanide as Cn,Mg/l	BDL	BDL	BDL	0.05 Max
Zinc as Zn, Mg/l	1.15	1.21	1.09	5 (15) Max
Detergent as MBAS, Mg/l	BDL	BDL	BDL	0.2 (1.0) Max
Chromium as Cr+6 ,Mg/l	BDL	BDL	BDL	0.05 Max
Total Alkalinity as Caco3,Mg/l	178.3	1751	169.7	200 (600) Max
Aluminum as Al,Mg/l	BDL	BDL	BDL	0.03(2) Max
Boron as B, Mg/l	BDL	BDL	BDL	0.5(1) Max
Bacteriological Analysis				
Coliform,MPN/100Ml	Nil	Nil	Nil	10 Max
E-Coli/Ml	Negative	Negative	Negative	Negative

14.5 METEOROLOGY

9.5.1 General

Surat has a tropical savanna climate (Köppen: Aw), moderated strongly by the Sea to the Gulf of Cambay. The summer begins in early March and lasts till June. April and May are the hottest months, the average maximum temperature being 37 °C. Monsoon begins in late June and the city receives about 1,200 millimetres of rain by the end of September, with the average maximum being 32 °C during those months. October and November see the retreat of the monsoon and a return of high temperatures till late November. Winter starts in December and ends in late

14.5.2 Temperature

The temperature data for Surat has been taken. The month-wise minimum & maximum temperatures have been given in **Table 14.5.**

Table 14.5 Normal Temperature at Surat

Month	Maximum Temperature ºC	Mean Daily Maximum Temperature, ºC	Minimum Temperature ºC	Mean Daily Minimum Temperature, ºC
January	38.3	30.8	4.4	15.2
February	41.7	32.3	5.6	16.7
March	44.0	35.4	8.9	20.7
April	45.6	36.	15.0	24.0
May	45.6	35.8	19.4	26.8
June	45.6	34.0	20.2	27.0
July	38.9	31.2	19.9	25.9
August	37.2	30.8	21.0	25.5
September	41.1	32.3	20.6	25.4
October	41.4	35.1	14.4	23.3
November	39.4	34.1	10.6	19.6
December	38.9	31.9	6.7	16.5
Annual	45.6	33.4	4.4	22.2

Source: India Meteorological Department, Govt. of India.

14.5.3 Rainfall

The detail of rainfall at the Surat (Santacruz) is given in **Table** 14**.6.**

Table 14.6 Month-wise Rainfall at Surat in mm

S. No.	Month	Rainfall	Peak Rainfall
1	January	1.5	
2	February	0.3	
3	March	04	
4	April	0.2	
5	May	3.9	
6	June	245.2	1188.9
7	July	466.3	
8	August	283.8	
9	September	151.8	
10	October	41.8	
11	November	7.1	
12	December	0.6	
Annual	Annual	1202.9	

Source: India Meteorological Department, Govt. of India.

14.5.4 Air Environment

The atmospheric concentrations of air pollutants were monitored at 39 locations near the proposed alignment during the month of January and March 2017. Locations of air monitoring station are shown in Figure 14.1. Air Monitoring was carried out for PM_{10}, NOx, SO_2 and CO. Results of the air quality monitoring are presented in **Table 14.7**.

Table 14.7 Ambient Air Quality Results µg/m3

S.No	Station Name	Date	Sampling	PM_{10}	$PM_{2.5}$	SO2	NOx	CO
	Regulatory Standards			**100**	**60.0**	**80**	**80**	**2000**
CORRIDOR-1SARTHANA TO DREAM CITY) OF SURAT METRO								
1	Sarthana	9 to11 Jan	Neelkanta Society	81	28.5	21.2	27.6	1160
				85	27.8	23.5	28.2	1205
2	Nature Park	9 to11 Jan	Krishna Hotel	83	28.2	23.4	26.1	1130
				87	27.1	24.2	29.2	1089
3	Varcha Chopati Garden	9 to11 Jan	Ganga Yamuna Aptt	85	29.0	20.4	24.7	998
				88	28.3	22.1	28.7	1127
4	Shri Swaminarayan Mandir Kalakunj	11 to13 Jan	Mamta Park-2	89	28.8	25.2	31.6	985
				86	28.3	23.7	29.4	910
5	Kapodara	11 to13 Jan	Balaji Kiriyana	74	25.2	23.5	33.6	1200
				87	26.2	26.1	33.5	1165

6	Labheshwar Chowk	11 to13 Jan	Star Computers	116	36.4	24.2	36.2	1160
				91	35.6	21.5	33.5	1205
7	Central Warehouse	13 to15 Jan	SBI	121	39.8	26.3	34.7	1240
				129	41.2	27.5	37.2	1190
8	Surat Railway Station	13 to15 Jan	Nr Temple	125	42.0	28.5	33.2	1150
				112	42.2	24.6	35.1	1320
9	Maskati Hospital	13 to15 Jan	Sheetal Fashions	117	39.6	27.6	37.4	1270
				123	39.2	28.4	36.1	1040
10	Chowk Bazar	16 to 18 Jan	Church	107	36.5	20.6	27.4	1070
				113	35.9	22.3	30.4	1130
11	Kadasrsha ni nal	16 to 18 Jan	Guard Post	119	43.5	18.5	29.2	1279
				108	42.8	19.2	26.4	1275
12	Majura Gate	16 to 18 Jan	DayalJi Mandap	89	32.0	21.4	30.6	1070
				93	33.3	23.7	33.2	1180
13	Roopali Canal	18 to20 Jan	Rajdarshan Aptt	77	30.1	17.4	26.1	1160
				81	31.1	19.3	28.6	1210
14	Althan Tenament	18 to20 Jan	ICICI Bank	79	29.8	17.4	23.9	860
				71	28.7	16.3	21.7	910
	Sarvodaya Vidyalaya	18 to20 Jan	Citizen Hardware	86	30.1	19.2	27.4	1090
				95	29.1	20.3	26.1	1180
15	AlthanGam	20 to22 Jan	Ridhi-Sidhi Aptt.	95	30.1	26.2	37.4	1290
				98	32.9	29.4	34.2	1270
16	Apcha Nagar	20 to22 Jan	Nr Transformer	89	30.1	24.1	35.2	1240
				93	30.3	26.7	37.6	1370
17	Surat Women ITI	20 to22 Jan	Safal residency	92	32.0	26.4	37.3	1250
				89	29.0	22.7	34.5	1190
18	Bhimrad	22 to24 Jan	Nr. Police Post	91	31.0	29.3	41.6	1690
				88	29.0	27.4	38.5	1670
19	Convention Centre	22 to24 Jan	Nr Market	93	31.0	23.7	34.5	1370
				91	32.0	26.7	37.2	1530
20	Dream City	22 to24 Jan	Nr. OH Tank	91	318.0	28.4	33.1	1360
				88	29.8	26.3	37.3	1270
	Corridor-2	**Beshan to Saroli**						
1	Bheshan	13 to 15 march	Nr STP	87	29.2	25.4	38.6	1470
				85	28.1	29.7	42.1	1430
2	Botanical Garden	13 to 15 march	Nr Temple	90	29.2	31.2	42.1	1590
				89	28.9	29.6	44.2	1660

3	Ugat Vaarigruh	13 to 15 march	Jyoti Marble	99	31.1	28.5	37.2	1190
				96	33.1	24.6	33.2	1150
4	Palanpur Road	15 to 17 march	Mangalam Aptt	94	31.1	27.6	35.1	1270
				92	31.5	20.6	37.4	1040
5	L.P.Savani school	15 to 17 march	Drishti Complex	88	29.8	22.3	36.1	1070
				84	28.9	26.4	27.4	1130
6	Performing Art Cetre	15 to 17 march	Pravag Aptt.	92	30.3	22.7	28.6	1279
				93	33.2	29.3	29.2	1070
7	AdajanGam	17 to 19 march	White Orchid	89	29.1	19.2	26.4	1180
				85	28.1	21.4	30.6	1160
8	Acquarium	17 to 19 march	Sun Aptt	92	32.1	23.7	33.2	1560
				89	29.9	17.4	26.1	1420
9	Badri Narayan Temple	17 to 19 march	Temple	88	28.8	19.3	28.6	1090
				85	27.9	17.4	21.7	1180
10	Athwa Chaupati	19 to 21 march	Hotel Krishna	92	28.9	16.3	27.4	1290
				89	29.1	19.2	26.1	1270
11	Majura Gate	19 to 21 march	Lovely Aquarium	99	34.1	26.4	37.4	1240
				97	32.8	26.2	34.2	1370
12	Udhana Darwaja	19 to 21 march	SB Centre	102	35.7	29.4	37.6	1250
				99	35.2	24.1	37.3	1190
13	Kamela Darwaja	21 to 23 march	Supr tex Tower	98	33.2	26.7	34.5	1690
				97	32.1	23.7	41.6	1670
14	Anjana Farm	21 to 23 march	Kakadiya Fabs	93	30.9	22.7	38.5	1370
				94	33.2	29.3	34.5	1560
15	Modal Town	21 to 23 march	Swasti k Plaza	91	29.8	27.4	28.6	1070
				88	28.2	23.7	29.2	1180
16	Magub Gam	23 to 25 march	Dharmendra Complex	89	29.0	17.4	30.6	1090
				87	29.9	16.3	33.2	1560
17	Bharat Cancer Hospital	23 to 25 march	Pandit Cargo	86	30.1	17.4	26.1	1420
				89	30.2	19.3	26.4	1160
18	Saroli	23 to 25 march	Eicher Showroom	92	31.0	23.7	30.6	1070
				93	33.1	27.4	33.2	1180

RSPM= Respirable Suspended Particulate Matter.

The results show that the concentration of RSPM (PM_{10}) is higher at all the locations whereas all other parameters are within permissible limits.

14.5.5 Noise Environment

Noise is responsible for adverse impacts on physical and mental health of the people. The other impacts are:

Physiological effects,
Hearing impairment,
Communication interference, and
Sleep disruption

Noise level survey was conducted along the alignment with an objective to establish the baseline noise levels and assess the impacts of total noise expected due to the proposed metro. Noise levels were measured at 39 locations in January and March 2017 for 24 hours. The locations of Noise level monitoring have been shown in **Fig. 14.1**. The noise levels so obtained are summarized in **Table 14.8.**

Table 14.8 Noise Levels

S.No	Station Name	Date		Lmax	Lmin	Leq	L_{10}	L_{50}	L_{90}
	CORRIDOR-1 (SARTHANA TO DREAM CITY) OF SURAT METRO								
1	Sarthana	9 to10 Jan	Day	80.8	585	73.2	76.5	64.7	61.4
			Night	73.3	45.5	59.4	67.9	53.2	50.8
2	Nature Park	9 to10 Jan	Day	84.4	58.3	69.1	74.5	63.9	61.1
			Night	73.9	46.6	57.5	61.2	56.2	53.9
3	Varcha Chopati Garden	9 to10 Jan	Day	86.8	58.6	70.5	73.2	69.1	61.4
			Night	78.1	54.6	66.9	75.4	64.2	59.3
4	Shri SwaminarayanMandir Kalakunj	10 to11 Jan	Day	92.2	56	71.8	75.3	66.3	62.5
			Night	84.5	51.9	62.3	69.2	60.4	55.2
5	Kapodara	10 to11 Jan	Day	85.7	67.3	77.3	80.9	74	70.2
			Night	79.4	51.2	63.2	71.9	59.3	54
6	Labheshwar Chowk	10 to11 Jan	Day	84	59.3	73.1	78.7	65.4	61.9
			Night	80.1	46.8	61.9	75.6	54.8	51.1
7	Central Warehouse	11 to12 Jan	Day	86.9	53.5	71.4	67.8	66.5	61.7
			Night	79.5	48.4	63.2	69.6	61.3	56.1
8	Surat Railway Station	11 to12 Jan	Day	85.7	57.8	70.5	79.7	66.5	61.6
			Night	71.6	46.2	55.9	67.2	54.2	51.7
9	Maskati Hospital	11 to12 Jan	Day	81.6	56.9	74.1	78.9	67.6	62.7
			Night	77.5	51.8	69.9	73.2	65.2	59.7

10	Chowk Bazar	12 to 13 Jan	Day	89.7	53.5	69.2	75.6	62.9	59
			Night	78.9	46.4	60.1	73.5	57.7	52.1
11	Kadarsha Ni Nal	12 to 13 Jan	Day	86.3	58.1	75.5	81.8	68.7	63.6
			Night	780	43.9	57.4	65.9	51.3	49.1
12	Majura Gate	12 to 13 Jan	Day	85.6	56.3	73.7	82.6	64.9	60.8
			Night	73.5	44.6	52	63.3	49.8	46.1
13	Roopali Canal	13 to14 Jan	Day	92.2	51	71.4	86.2	67.4	59.6
			Night	85.2	45.4	56.7	61.7	53	47.9
14	Althan Tenament	13 to14 Jan	Day	86.7	63.1	78.9	83.4	76.4	66.9
			Night	79.7	46.5	61.7	66.3	59.1	53.9
	Sarodaya Vidiyalaya	13 to14 Jan	Day	83.5	52.1	62.4	76.3	59.8	58.8
			Night	78.7	48.9	59.6	63.7	55.7	51.7
15	Althan Gam	14 to15 Jan	Day	85.3	54.1	70.5	73.1	67.5	62.6
			Night	81.9	48.9	62	68.8	60.5	57.1
16	Apcha Nagar	14 to15 Jan	Day	81.3	52.1	76.3	71.5	68.2	57.1
			Night	78.7	47.3	66.8	70.2	60.5	57.1
17	Surat Woman ITI	14 to15 Jan	Day	88.3	69	79.6	83.2	77.7	73
			Night	79.9	51	62.7	73.9	59.5	54.2
18	Bhimrad	15 to16 Jan	Day	88.2	69.2	78.6	81.8	72.9	69
			Night	75.2	50.8	62.3	72.8	58.9	52.7
19	Convention Centre	15 to16 Jan	Day	88	58.9	71.4	85.2	66.3	63.6
			Night	79.9	43.7	57.2	65.4	52.7	49.2
20	Dream City	15 to16 Jan	Day	81.6	56.3	73.7	76.9	69.4	65
			Night	70.1	45.3	59.2	63.5	55.5	51.2
Corridor-2 Beshan to Saroli									
1	Bheshan	13 to 14 march	Day	83.7	53.2	77.7	71.6	69.9	61.9
			Night	78.3	43.9	58.7	64.5	56.9	51.3
2	Botanical Garden	13 to 14 march	Day	89.4	56.1	78.5	84.1	67.2	60.7
			Night	83.8	46.5	59.9	66.7	54.1	51.4
3	Ugat Vaarigruh	13 to 14 march	Day	88.3	69	79.6	83.2	77.7	73
			Night	79.9	51	62.7	73.9	59.5	54.2
4	Palanpur Road	14 to 15 march	Day	85.6	56.3	73.7	82.6	64.9	60.8
			Night	81.9	48.9	62	68.8	60.5	57.1
5	L.P.Savani school	14 to 15 march	Day	83.7	53.2	77.7	71.6	69.9	61.9

			Night	73.5	44.6	52	63.3	49.8	46.1
6	Performing Art School	14 to 15 march	Day	79.7	46.5	61.7	66.3	59.1	53.9
			Night	83.5	52.1	62.4	76.3	59.8	58.8
7	Adajan Gam	15 to 16 march	Day	78.7	48.9	59.6	63.7	55.7	51.7
			Night	85.3	54.1	70.5	73.1	67.5	62.6
8	Aquarium	15 to 16 march	Day	81.9	48.9	62	68.8	60.5	57.1
			Night	81.3	52.1	76.3	72.2	68.2	57.1
9	Badri Narayan Temple	15 to 16 march	Day	78.7	47.3	66.8	70.2	60.5	57.1
			Night	88	58.9	71.4	85.2	66.3	63.6
10	Athwa Chaupati	16 to 17 march	Day	88.2	67.9	78.6	81.8	72.9	69
			Night	75.2	50.8	62.3	72.8	58.9	52.7
11	Majura Gate	16 to 17 march	Day	85.6	56.3	73.7	82.6	64.9	60.8
			Night	78.9	46.4	60.1	73.5	57.7	52.1
12	Udhana Darwaja	16 to 17 march	Day	86.3	58.1	75.5	81.8	68.7	63.6
			Night	78	43.9	57.4	65.9	51.3	49.1
13	Kamela Darwaja	17 to 18 march	Day	86.3	58.1	75.5	81.8	68.7	63.6
			Night	73.5	44.6	52	63.3	49.8	46.1
14	Anjana Farm	17 to 18 march	Day	92.1	51	71.4	86.4	67.4	59.6
			Night	85.2	45.4	56.7	61.7	53	47.9
15	Modal Town	17 to 18 march	Day	86.9	63.1	78.9	83.2	76.4	66.9
			Night	79.7	46.5	61.7	66.3	59.1	53.9
16	Magub Gam	18 to 19 march	Day	89.7	53.5	69.2	75.6	62.9	59.2
			Night	76.9	54.0	64.5	72.1	63.1	59.9
17	Bharat Cancer Hospital	18 to 19 march	Day	88.2	67.9	78.6	81.8	72.9	69
			Night	75.2	50.8	62.3	72.8	58.9	52.7
18	Saroli	18 to 19 march	Day	85.6	56.3	73.7	82.6	64.9	60.8
			Night	78.9	46.4	60.1	73.5	57.7	52.1

Allowable Noise Levels dB (A):

Category of Area/Zone	Day Time	Night Time	
Industrial Area	75	70	EPA-1986, Noise pollution (Regulation Control), Rule-2000, PCLS/02/1992, IVth Edition .
Commercial Area	65	55	
Residential Area	55	45	
Silence Area	50	40	

Day Time (6.00 Am-10.00 Pm); Night Time (10.00 Pm-6.00Am)

The observed noise level is higher than the permissible limits at all locations which may be due to heavy traffic movement and other activities on the roads.

14.5.6 Trees

Tree survey has been carried out along the proposed alignment. Tree with Girth at Breast Height (GBH) 30 cm have been counted. The alignment does not pass through any forest area. A total of 405 trees are located along the alignment and station area. Beshan Depot area in front portion of STP area is having 76 trees. Kajod Depot area may be having more trees. Thus, there is likelihood of felling of 481 trees. No endangered species of trees have been noticed during field survey. Trees have been found of indigenous and common species like Pipal, Mango, Khajur, Neem, Coconut, Palm, Babool, Ber, Gulmohar and Tadi etc.

14.6 SOCIO- ECONOMIC CONDITIONS

According to the 2011 India census, the population of Surat is 4,462,002. Surat has an average literacy rate of 89%, higher than the national average of 79.5%: male literacy is 93%, and female literacy is 84%. Males constitute 53% of the population and females 47%. Gujarati, Sindhi, Hindi, Marwari, Marathi, Tamil, Telugu, and Odia are the main languages spoken in Surat. In Surat, 13% of the population is under 6 years of age. Surat also has sizeable Parsi and Jewish populations. Dawoodi Bohra also made Surat (1787AD onward) its Dawat office in the era of 42nd Dai Syedna Yusuf Najmuddin. It continued up to the period of 51st Dai Taher Saifuddin (1915-1965AD), when it was shifted to Surat. The Seven Da'i al-Mutlaq of Dawoodi Bohras are laid to rest in the Rauzas at Devdi Mubarak, Surat.

Surat is known for its relaxed lifestyle and its cuisine. An old saying in Gujarati, **"Surat nu Jaman ane Kashi nu Maran**", means "Eat in Surat and Die in Kashi" to have the best for one's soul. Popular dishes include Locho (a favourties of the local, steamed snack made of gram flour and lenties, steamed to be eaten immediately. Locho is only available in Surat. Other are perennial favourites such as "ghari" (a type of sweet), petis, undhiyu, rasaawala khaman, Surati Khaman (spicy), Sarasiya Khaja, Khavsa, Aloopuri, Sosyo(soda) and surati Chinese. Unlike cuisines in other parts of Gujarat, Surti cuisine is quite spicy. Roadside kiosks, called "laaris" or "rekdis", are quite popular in Surat. Ponk is a roasted cereal that is available only in this part of the world.

The major Hindu festivals are celebrated in Surat. Navratri, Diwali and Ganesh Chaturthi are celebrated with great enthusiasm. The kite-flying festival of Uttarayan, which falls on Makar Sankranti – 14 or 15 January, or 'Vaasi Uttarayan', celebrated with great enthusiasm by flying kites and shouting 'Kai Po Che' (I have cut) – is very popular in the city. It is also well known for the celebration of Chandi Padvo which usually occurs in October and is a holiday unique to Surat. This day comes after one of the two biggest full moon days of the Hindu calendar year, Sharad Purnima , when residents of Surat have 'Ghari' and 'Bhusu' (Surti snacks).

Surat ranked 8th in India with a GDP of $40 billion in fiscal year 2011–2012 ($14 billion in 2010). Surat GDP in 2020 will be around $57 billion estimated by The City Mayors Foundation, an international think tank on urban affairs. The per capita GDP was $8,000 in 2010. Surat is known for diamonds, textiles, and for diamond-studded gold jewellery manufacturing. Surat registered a GDP growth of 11.5% for the seven fiscal years 2001–2008, the fastest growing GDP in India. The city accounts for:

90% of the world's total rough diamond cutting and polishing;
99.99% of the nation's total rough diamond cutting and polishing;
90% of the nation's total diamond exports;
40% of the nation's total man made fabric production;
28% of the nation's total man made fibre production;
18% of the nation's total man made fibre export; and
12% of the nation's total fabric production.

Surat is a major hub of diamond cutting and polishing. Gujarati diamond cutters, emigrating from East Africa, established the industry in 1901 and, by the 1970s, Surat-based diamond cutters began exporting stones to the US for the first time. Surat is the largest diamond hub in the entire world. It is to believed that 9 out of 10 diamonds in Manhattan, New York, is from Surat. Major diamond polishing factories are in Varracha and Kataragam area. Most of the diamond business is dominated by people from Saurastra.

Since it is known for producing textiles, including silk, Surat is known as the textile hub of the nation or the Silk City of India. It is very famous for its cotton mills and Surat Zari Craft. Surat is the biggest centre of MMF (man-made fibre) in India. It has a total of 381 dyeing and printing mills and 41,100 powerloom units. There are over a hundred thousand units and mills in total. The overall annual turnover is around 5 billion rupees. There are over 800 cloth wholesalers in Surat. It is the largest manufacturer of clothes in India,

and Surti dress material can be found in any state of India. Surat produces 9 million metres of fabric annually, which accounts for 60% of the total polyester cloth production in India. Now the city is focusing on increasing the exports of its textile.

Along with Textile and Diamond, there are other sectors in city like Petroleum, Oil,Gas, Shipping, Cement, Metallurgy, Paper, Chemical, Fertil iser etc. Hazira INA, Magadalla, Ichchhapore, Olpad near Surat city are having giant industries like ONGC, GAIL, ABG Shipyard, Narmada Cement, Ambuja Cement, Ultratech Cement, NTPC-KGPP Power Plant, ESSAR group of Industries (Steel/ Power/ Chemical/ Oil/ Shipping), L& T, Reliance Industries. There are many SME Domestic IT Companies present in Surat. MNC IT companies like IBM, TCS, HCL, Wipro have satellite or virtual branches in Surat. On 14 February 2014 Govt of Gujarat DST had handover STPI Surat at Bhestan-Jiav Road, Bhestan Near Udhana-Sachin BRTS Route.

14.7 SOCIO-ECONOMIC SURVEY

A socio-economic survey was undertaken for the proposed corridor to assess the socio-economic conditions of project-affected families/people and to examine the impacts of the proposed metro alignment on their conditions. There can be two types of impacts on the PAPs. One is the displacement of residential house and another is displacement of commercial establishments

It has been found during socio-economic survey that any residential structure is not affected by the metro route on both the alignments. Land is mainly required for viaduct, construction of stations and allied services, construction of Depot including laying of stabling lines, workshops, washing lines, administrative buildings and Water, Sewage and Effluent treatment systems in addition to storage facilities. Additionally, land is also required for RSS.

The survey has been undertaken on the corridors using structured questionnaire. It has been found that a total of 18 commercial structures are getting affected by the construction of proposed project. The commercial establishments have also employed 24 employees who are working for a period of 2 to 5 years.

14.8 ARCHAEOLOGICAL SITES

Surat city is having a few historical and archaeological structures. The details of such structures is given in Table 14.9.

Table 14.9 List of Archaeological Monuments in Surat

S. No.	Corridor	Code	Monument	Distance from Centre Line	Location
1	I	N-GJ 177	Dargah Khwaja Dana Sahab Rouza	400 M	Between Gandhi Bag and SMC Shopping Complex Stations
2	I	N-GJ 178	Old English Tomb	1450 M	Between SMC Hospital and Gandhi Bag Stations
3	I	N-GJ 179	Tomb of Khwaja Safar Sulemani	400 M	Between Gandhi Bag and SMC Shopping Complex Stations
4	I	N-GJ 180	Old Dutch and Armenian Tomb and Cemetry	1100 M	Between Gandhi Bag and SMC Shopping Complex Stations
5	I	N-GJ 181	Ancient Site at Kamrej	6500 M	Sarthana Station

There are many heritage sites in Surat which are very much far off from the corridor alignment. Their architectural features include a variety of European influences such as Dutch, English, Portuguese and traditional Indian features. The proposed alignment of Surat Metro corridor does not pass through or near any of the Archaeological monuments or heritage sites.

14.9 ENVIRONMENTAL IMPACTS ASSESSMENT

14.9.1 Environmental Impacts

This section identifies and appraises the negative impacts on various aspects of the environment likely to result from the proposed development. It is pertinent to mention that the negative environmental impacts listed below are based on the assumption that no negative impact mitigation measure or benefit enhancements are adopted. The negative environmental impacts are generally observed on the following parameters:

Land Environment
Water Environment

Air Environment
Noise Environment
Biological Environment
Socio-Economic Environment

The impacts on the above environmental components have been further assessed during various phases of project cycle namely project location, project design, construction and operation.

14.9.2 Impacts due to Project Location

During this phase, those impacts, which are likely to take place due to the layout of the project, have been assessed. These impacts are:

- Project Affected People (PAPs)
- Change of Land use;
- Loss of trees/forest;
- Utility/Drainage Problems,
- Socio-economic impacts;
- Impact on Historical and Cultural Monuments;

14.9.2.1 Project Affected People (PAPs)

There will be acquisition of private land and property in this project hence there are many PAPs as a result of the project activity. Detailed socio-economic assessment has been made for PAPs in Social Impact Assessment.

14.9.2.2 Change of Land Use

The required land (permanent & temporary) for the construction of the proposed alignment is both government as well as private land which shall be allotted by Surat Municipal Corporation. Private land will be acquired as per the provisions of The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act 2013 (Act 30 of 2013).

14.9.2.3 Loss of Forests/ Trees

The proposed metro lines are in urban/ city area and will not pass through any forests. Hence no loss to forest is anticipated due to the project. However, trees do exist in patches in the corridor selected for the project.

There are about 481 trees which are likely to be felled during construction. Trees are assets in purification of urban air, which by utilizing CO_2 from atmosphere, release oxygen into the air. However, with removal of these trees, the process for CO_2 conversion will get effected and the losses are reported below:

i)	Total number of Trees	:	481
ii)	Decrease in CO_2 absorption @ 21.8 Kg/ year tree for 8 years	:	83886.40 kg
iii)	Oxygen production @ 49 kg/ year tree for 8 years	:	188552 kg

The average consumption of oxygen for a person is about 182 kg/ year. It means these trees will meet the requirement of about 1036 people round the year. Trees help carbon sequestration acting as a carbon sink. By removing the carbon and storing it as cellulose, trees release oxygen back into the air.

14.9.2.4 Utility/ Drainage Problems

Metro lines are mostly planned to run through the urban area. The alignment will cross many properties, drains/ nalas, large number of sub-surface, surface and utility services, viz. sewer, water mains, storm water drains, telephone cables, overhead electrical transmission lines, electric pipes, roads, traffic signals etc. These utilities/ services are essential and have to be maintained in working order during different stages of construction by temporary/permanent diversions or by supporting in position. Since these affect construction and project implementation time schedule/ costs for which necessary planning/ action needs to be initiated in advance.

14.9.2.5 Socio-Economic Impact on PAPs

The metro alignment runs between Beshan and Saroli and Sarthana and Dream City Land is mainly required for viaduct, construction of stations and allied services, construction of Depot including laying of stabling lines, workshops, washing lines, administrative buildings and Water, Sewage and Effluent treatment systems in addition to storage facilities. Additionally, land is also required for RSS. There is no residential unit affected by the land acquisition for the proposed metro corridors.

It has been observed that the alignment is cutting across the Textile market building No. 451 just opposite Millenium Textile Market. It is anticipated that whole building will not be acquired and only corner shops directly influenced by he corridor alignment would be affected ad acquired. The corner of the building may be cut right from basement to the third floor.

A total of 18 commercial establishments are getting affected due to the project due to viaduct and stations. The commercial establishments are having 24 employees. Among the shops being affected 9 are located at SMC Shopping Complex, One at SMC Hospital Station on Corridor-1 between Sarthana- Dream City. Simultaneously there are 8 shops affected due to the sharp curve by the side of road overbridge at Textile Market 451 near Millenium Textile Market in basement and three floors between proposed Kamela Darwaja Station and Anjana Farm stations on Beshan- Saroli corridor of Surat Metro.

Shops at SMC Shopping Complex

Affected shops at Textile Market 451between Kamela Darwaja and Anjana Farm

14.9.2.6 Impact on Archaeological Sites

There is no historical monument having any archeological value in the close vicinity of the proposed alignment. Thus, on this aspect there would be no impact.

14.9.2.7 Impact on Sensitive Receptors

There are many sensitive receptors along the alignment like hospitals, schools and religious places but care has been taken to keep safe distance between the alignment and sensitive receptors. However, during operation stage care would be taken to provide noise barriers of suitable design between hospitals and the alignment to minimize the impact.

14.9.3 Impacts due to Project Design

Considered impacts, due to project designs are:

- Lighting,
- Risk Due to Earthquake.

14.9.3.1 Lighting

The platforms, concourse, staircase and escalator areas for the elevated stations will have adequate and uniform fluorescent lighting to provide pleasant and cheerful environment. It is proposed to adopt the norms prevailing in Metro for illumination. It is pertinent to note that care has been taken at design stage itself to avoid too much illuminating the stations. Maximum illumination level proposed is 200Lux which provides normal lighting.

14.9.3.2 Risk Due to Earthquake

The project area lies in Zone III of Bureau of Indian Standards (BIS) Seismic Zoning Map (**Fig. 14.2**). Seismic factor proposed by India Meteorological Department (IMD) for the purpose of design of Civil Engineering structures shall be incorporated suitably while designing the structures.

14.9.4 Impacts due to Project Construction

Although environmental hazards related to construction works are mostly of temporary nature. Appropriate measures should be included in the work plan and budgeted for. The most likely negative impacts related to the construction works are:

- Top Soil erosion, pollution and health risk at construction site,
- Traffic diversion and risk to existing building,
- Excavated soil disposal problems,
- Dust Generation,
- Increased water demand,
- Impact due to Supply of Construction Material,
- Disposal of Construction and Demolition Waste,
- Impacts due to batching plant and casting yard,

- Noise Pollution,

14.9.4.1 Soil Erosion, Pollution and Health Risk at Construction Site

Every care will be taken to avoid damage to the top soil. It shall be preserved and utilized. Problems could arise from dumping of construction spoils (Concrete, bricks) waste materials (from contractor camps) etc. causing surface and ground water pollution. However, it is proposed to have mix concrete directly from batching plant for use at site. Health risks include disease hazards due to lack of sanitation facilities in labour camps (water supply and human waste disposal) and insect vector disease hazards of local workers and disease hazards to the local population. Mitigation measures should include proper water supply, sanitation, drainage, health care and human waste disposal facilities. In addition to these, efforts need to be made to avoid water spills, adopt disease control measures and employment of local labour. Problems could arise due to difference in customs of workers from immigrant workers and local residents. These risks could be reduced by providing adequate facilities in worker's camps, raising awareness amongst workers and by employment of preferably local labour.

14.9.4.2 Traffic Diversions and Risk to Existing Buildings

During construction period, complete/ partial traffic diversions on road will be required, as most of the construction activities are on the central verge of road. Traffic would get affected on the roads. Rather than completely blocking the roads it will be advisable to make the narrow portion of roads as one way to allow for operation of traffic together with construction activities. Advance traffic updates/ information on communication systems will be an advantage to users of affected roads. The rail corridor does not pose any serious risk to existing buildings since there is safe distance between buildings and proposed corridor except at a few places where shops are affected due to the turning of alignment. Moreover, at many places facilities for station would affect open spaces and a few buildings which may be avoided by suitably adjusting the station layouts. Special care shall be taken for safety of the structures during construction.

14.9.4.3 Problems of Excavated Soil and Bentonite Disposal

The proposed alignment is elevated and thus the excavation would be limited to piers and their piling. The soil would be used for refilling at station site. If there would be some residual soil, it would be utilized by SMC for

internal use for refilling Depot sites and, if surplus, it would be disposed off at designated locations as per Surat Authority directions. Some Bentonite muck would also be generated in the project. Disposal of Bentonite would be at designated land fill site.

14.9.4.4 Air Pollution and Dust Generation

Transportation of earth and establishment of the material will involve use of heavy machinery like compactors, rollers, water tankers, and dumpers. This activity is machinery intensive resulting in dust generation. Simultaneously there would be fugitive gas emissions due to vehicular and machinery exhausts during their working during construction. However, this activity will be only short-term. Protective measures shall be undertaken during construction phase. Movement of trucks and other heavy equipment at construction site would generate dust during construction phase.

14.9.4.5 Water Pollution

Source of Water contamination will be from the washings and the surplus water from curing the structures which shall be diverted and passed through desilting chambers before letting it go outside the working site. Muck shall not be allowed to confluence with any water course. Controlled water should be used for curing.

14.9.4.6 Increased Water Demand

The water demand will increase during construction phase for meeting out drinking and domestic water requirement of workers. Sufficient water for construction purpose would be made available by SMC as it is responsible for water supply in Surat. Water requirement for construction of Metro will be met through the public supply. It is suggested to use treated STP water for the purpose of Construction. Proper care shall be taken while drawing water from public facilities to avoid any negative impact on the residents living in the vicinity of the project whose water demand is, in any case, met by SMC supplied water.

14.9.4.7 Impact due to Supply of Construction Material

Metro construction is a material intensive activity. Huge quantity of different construction materials will be required for construction of metro corridor. These shall be sourced from the nearest source. Quarry operations are

independently regulated activities and outside the purview of the project proponent. It is nonetheless, appropriate to consider the environmental implications in selection of quarry sources since poorly run operations create dust problems, contribute noise pollution, ignore safety of their employees, or cause the loss of natural resources. Although quarry operation is out of purview of the metro construction but, the construction material shall be sourced only from legalized and approved quarries.

14.9.4.8 Generation of Construction and Demolition Waste

Construction and demolition (C&D) debris is defined as that part of the solid waste stream that results from land clearing and excavation, and the construction, demolition, remodeling and repair of structures, roads and utilities. C&D waste includes concrete, stones and dirt generated during excavation (sometimes collectively referred to as "fill material" or rubble). C& D Waste may be generated from Pile caps, residual cement bags, residual steel scrap, excess construction material stacked at site etc. It is a waste stream that is separate and distinct from residential and commercial waste, commonly called municipal solid waste (MSW).

About 10-15% of the construction material such as waste material from contractor camps is left behind by the contractor as construction waste/ spoils. Dumping of construction waste/spoil in haphazard manner may cause surface and ground water pollution near the construction sites. The C& D waste would be handled and disposed-off to C&D waste processing facility or for back filling of low lying areas, leaving no significant impact on environment.

14.9.4.9 Impacts due to Casting Yard and Batching Plant

During construction phase, there would be establishment and operation of Batching Plant and Casting Yard which would be located in an area designated and allotted by SMC away from habitation. There would be requirement to get NOC (Consent to establish) and Consent to operate under water and air Acts from Gujarat Pollution Control Board at the time of establishing the facilities. Simultaneously, there would be requirement to get the authorization for storage and handling of hazardous chemicals to store and handle used oils and other such materials. The Application forms for seeking Consent to establish, Consent to Operate and Authorization for storage of Hazardous chemicals are available from the office of Gujarat Pollution Control Board at Surat.

There would be significant movement of men, material and machinery in batching plant and casting yard. It is expected that both batching and casting yard would be located at same complex. Huge quantity of Cement, aggregates and other construction materials would be used in batching plant and casting yard. There would be generation of dust, noise, flue gases and other contaminants from the working of heavy machinery for handling and transporting the construction materials. The mitigation measures have been elaborated in EMP.

14.9.4.10 Noise Pollution

The major sources of noise pollution during construction are movement of vehicles for transportation of construction material to the construction site and the noise generating activity at the construction site itself. The Metro construction is equipment intensive.

14.9.4.11 Vibrations During Construction

There may be vibration during piling operations due to working of heavy construction machinery and the movement of heavy transport vehicles, loading and unloading of materials etc. This would be a short-term activity and effort will be made to avoid piling operations during night between 11.00 pm to 5.00 am in the vicinity of residential areas.

14.9.4.12 Loss of Historical and Cultural Monuments

No historical/ cultural monuments will be lost because of the proposed development.

14.9.5 Impacts due to Project Operation

Along with many positive impacts, the project may cause the following negative impacts during operation of the project due to the increase in the number of passengers and trains at the stations:

- Noise pollution,
- Water supply and sanitation at Stations,
- Station refuse disposal and sanitation,
- Pedestrianization and visual issues
-

14.9.5.1 Noise Pollution

During the operation phase the main source of noise will be from running of metro trains. Noise radiated from train operations and track structures generally constitute the major noise sources. Airborne noise is radiated from elevated structures. The noise level at 2 m distance from the rail alignment is about 73 dB(A) as per the experience in operating metro system. The noise level reduces with distance logarithmically. At places, the alignment is likely to be passing close to the buildings which may affect the residents. At such places noise barriers would be used to minimize the noise impact in the vicinity of the alignment.

14.9.5.2 Water Supply and Sanitation at Stations

Public facilities such as water supply, sanitation and wash rooms are very much needed at the stations. The water requirement for stations would be for drinking, toilets, cleaning and also for other purpose like AC. Water Demand as per existing Metro corridors is calculated and presented in **Table 14.10**. It is assumed that there would be similar water requirements in Surat Metro as well. Raw water should be treated and brought to national drinking water standards, before used for consumption. In addition, water will be required for contractor's camps during construction. The water requirement for the stations will be met through the public water supply system or purpose built tube wells after taking necessary approvals from CGWA. However, as an environmental conservation measure, rainwater harvesting structure will also be constructed at stations and along the via-duct.

Table 14.10 Water Requirement at Stations

S. No.	Particular	Water Demand for each station KLD
1	At Stations for Drinking Purpose	6
2	At Elevated stations for AC, cleaning, chiller and other purposes	17
Total		**23**

Thus, there would be total water requirement of 897 KLD in 39 stations. However, arrangement of water will have to be made at each station separately.

14.9.5.3 Station Refuse

The collection and removal of refuse from stations in a sanitary manner is of great importance for effective vector control, nuisance abatement, aesthetic improvement and fire protection. The refuse from station includes:

- Garbage
- Rubbish and
- Floor Sweepings.

As per the available data from Delhi Metro Phase I and II and other operational metros, the solid waste generation is about 0.8 – 1.2 cum/day at elevated stations. At elevated stations, the solid waste generation is more due to airborne dust. At underground stations the waste generation would be lesser. However, it is estimated that a maximum of 31.2 to 46.8 cum of solid waste will be generated from the thirty nine stations of these corridor of Surat metro. The maintenance of adequate sanitary facilities for temporarily storing refuse on the premises is considered a responsibility of the project authorities. The storage containers for this purpose need to be designed. However, it is suggested that the capacity of these containers should not exceed 50 litres and these should be equipped with side handles to facilitate handling. To avoid odour and the accumulation of fly-supporting materials, garbage containers should be washed at frequent intervals.

14.9.5.4 Visual Impacts

The introduction of MRTS implies a change in streets through which it will operate. An architecturally well designed elevated section can be pleasing to the eyes of beholders. Recent MRTS projects have attempted to incorporate this objective in their designs. Since a low profile would cause the least intrusion, the basic elevated section has been optimised at this stage itself.

14.9.5.5 Vibrations

This corridor is elevated throughout the alignment. As per the experience from working metros particularly Delhi Metro it is found that the problem of ground vibration is generally felt in case of Underground sections. In elevated corridors there has been no complaint of vibration in the vicinity of alignments. Therefore, the vibration impact is not considered significant for most of the corridor section. However, preventive measures to reduce the vibration at

source would be applied in the rail design itself for elevated as well as underground sections.

In corridor-1 between Sarthana and Dream city there are six underground stations. Underground section would be app. 7.8 Km. Here the structures and buildings above the corridor or in the close vicinity of the alignment may feel the vibration impacts. The effects of ground-borne vibration include perceptible movement of the building floors, rattling of windows, shaking of items on shelves or hanging on walls, and rumbling sounds. In extreme cases, such vibration can damage buildings and other structures. In addition, the sound reradiated from vibrating room surfaces, referred to as ground-borne noise, may be audible in the form of a low-frequency rumbling sound.

The train wheels rolling on the rails create vibration energy transmitted through the track support system into the track bed or track structure. The amount of energy that is transmitted into the track structure depends strongly on factors such as how smooth the wheels and rails are and the resonance frequencies of the vehicle suspension system and the track support system. The vibration of the track or guideway structure excites the adjacent ground, creating vibration waves that propagate through the various soil and rock strata to the foundations of nearby buildings. The vibration propagates from the foundation throughout the remainder of the building structure. The maximum vibration amplitudes of floors and walls of a building often occur at the resonance frequencies of those building elements.

14.9.6 Impacts due to Depot

One Depot is proposed at Beshan near STP and another depot is proposed near village Khajod on Corridor-1. The depot will have following facilities:

Washing Lines,
Operation and Maintenance Lines,
Workshop, and
Offices.

These facilities could generate water and noise issues. The depot area may have to be filled up. Problems anticipated at depot sites are:

Water supply,
Oil Pollution,
Cutting of trees

Sanitation,
Effluent Pollution,
Noise Pollution,
Loss of livelihood,
Impact due to filling of area, and
Surface drainage.

Water Supply

Water supply will be required for different purposes in the depot. The water requirement for drinking will be 500 litre per day and 1,00,000 litre per day for other requirements (Departments and Contractors office). The water after conventional treatment can be processed through Reverse Osmosis (RO) technology for specific use such as final washing of equipment/ trains.

Oil Pollution

Oil spillage during change of lubricants, cleaning and repair processes, in the maintenance Depot cum workshop for maintenance of rolling stock, is very common. The spilled oil should be trapped in oil and grease trap. The collected oil would be disposed off to authorised collectors, so as to avoid any underground/ surface water contamination.

Noise Pollution

The main source of noise from depot is the operation of workshop. The roughness of the contact surfaces of rail and wheels and train speed are the factors which influence the magnitude of rail - wheel noise. The vibration of concrete structures also radiates noise. Due to less activity, no impact on the ambient noise is anticipated.

Solid Waste

At per available data, it is estimated that about 2 Ton per month of solid waste will be generated from the Depot site which will be taken by the cleaning contractor weekly and disposed to the Municipal waste disposal sites.

Sludge of the order of 250 kg/year is expected to be generated from the ETP/STP that will be stored in leak proof containers and disposed off as per State Pollution Control Board site.

According to experience and observation at operational DMRC depots, Oil and grease of the order of 2652 litres/ year will be produced in the Depot which will be disposed off through approved re-cyclers.

About 2.5 ton/month of iron turning of the PWL for the wheel profiling is likely to be generated from the Depot.

14.10 POSITIVE ENVIRONMENTAL IMPACTS OF THE PROJECT

Based on project particulars and existing environmental conditions, potential impacts that are likely to result from the proposed Surat metro corridors development have been identified and wherever possible these have been quantified. This section deals with the positive impacts of the project. The introduction of the corridor will also yield benefits from non-tangible parameters such as savings due to equivalent reduction in road construction and maintenance, vehicle operating costs, less atmospheric air pollution and socio-economic benefits of travel time, better accessibility, better comfort and quality of life. However, all benefits cannot be evaluated in financial terms due to non-availability of universally accepted norms. The parameters such as economic growth, improvement in quality of life, reduction in public health problems due to reduction in pollution, etc have not been quantified.

Various positive impacts have been listed under the following headings:

Employment Opportunities;
Enhancement of Economy;
Mobility, Safety and reduced accidents;
Traffic Congestion Reduction;
Reduced Fuel Consumption;
Reduced Air Pollution;
Reduction in Number of Buses/ Auto rickshaws, and

Employment Opportunities

The project is likely to be completed in a period of about 4 years. During this period manpower will be needed to take part in various activities. About 3500 persons are likely to work during peak period of activity. In operation phase of the project about 35 persons per kilo meter length of the corridor, ie (approx. 1400 persons) will be employed for operation and maintenance of the proposed system in shifts. Thus, the project would provide substantial direct employment. Besides, more people would be indirectly employed in allied activities and trades.

Enhancement of Economy

The proposed transport facility of SMC will facilitate sub-urban population to move quickly. With the development of metro corridors in Surat, it is likely that

more people will be involved in trade, commerce and allied services. SMC will, however, make it convenient for more people to move in the present suburban areas. This will reduce population pressure on transport facilities in the urban area.

Mobility Safety and Reduced Accidents

The metro network increases the mobility of people at faster rate. The proposed corridor will provide more people connectivity to other parts of the city. Metro journey is safe and result in reduced accidents on roads.

Traffic Congestion Reduction

To meet the forecast transport demand in the year 2026 and 2036, it is estimated that the number of buses will have to be more. During this period personalised vehicles may also grow. Together, they will compound the existing problems of congestion and delay. The proposed development will reduce journey time and hence congestion and delay. The substitution of 533890 persons in 2026 and 974310 persons in 2036 on this metro corridor of Surat metro for an average trip length of 10 Km may reduce about 133472 Petrol Car Equivalent (PCE) in 2026 and 243577 PCE units in 2036 assuming a switchover of 4 Person per PCE. The Asian Development Bank's "Transport Emissions Model" for the National Environment Commission has been used to predict/calculate the fuel consumption as well as the emissions of the harmful pollutants into the environment.

Reduced Fuel Consumption

On implementation of the project, it is estimated that both petrol and diesel consumption will get reduced. The saving will be due to two factors namely Reduction in vehicles and decongestion on roads. On the basis of assumption of 133472 vehicles for 10 Km (Average trip length for the corridor in 2026) at the speed of 30 KM/hr, the daily reduction in fuel consumption would be app 95447 litre Petrol and 18795 litre Diesel. Similarly in 2036 there is assumption of modal shift of 974310 persons equivalent to 243577 PCE for a length of about 10 Km. There will be reduction of 174185 litre Petrol and 34300 litre Diesel I 2036.

Reduced Air Pollution

Based on available data and assumptions, an attempt has been made to model the air quality scenario for future using Asian Development Bank's "Transport Emissions Model". On the basis of above referred assumptions, daily reduction in pollutants would be as given below:

Parameter		2026	2036
CO	:	8343.628 Kg	15226.534 Kg
CO_2	:	261662.766 Kg	477516.121 Kg
NOx	:	1064.407 Kg	1942.467 Kg
VOC	:	1014.961 Kg	1852.269 Kg
Particulates	:	21.245 Kg	38.771 Kg
SO_2	:	32.527 Kg	59.360 Kg

Carbon Credits

Due to savings in fuel and reduction in air pollution etc. carbon credit would be generated during operation of the metro rail similar to the experience with Delhi Metro Rail Corporation Ltd. However, at this stage calculation of carbon credits is not feasible which would be worked out after the system become operational.

Improvement of Quality of Life

Development of Metro rail in the city would lead to overall improvement of quality of life of local populace by virtue of availability of better transport facility at competitive rates, better road safety, reduced pollution, improved general health etc.

14.11 CHECKLIST OF IMPACTS

The impact evaluation determines whether a project development alternative is in compliance with existing standards and regulations. It uses acceptable procedures and attempts to develop a numeric value for total environmental impact. A transformation of the review of multiple environmental objectives into a single value or a ranking or projects is the final step in impact assessment. There are about hundred methods for carrying out impact assessment, which can be grouped into the following categories:

Ad-hoc method,
Checklist,
Matrix,
Network,
Overlays,
Environmental Index and
Cost Benefit analysis.

Each of the methods is subjective in nature and none of these is applicable in every case. Of the 7 methods listed above, checklist has been used and

presented. Checklist is a list of environmental parameters or impact indicators which encourages the environmentalist to consider and identify the potential impacts. A typical checklist identifying anticipated environmental impacts is shown in **Table 14.11**.

Table 14.11 Checklist of Impacts

S. No.	Parameter	Negative Impact	No Impact	Positive Impact
A.	**Impacts due to Project Location**			
i.	Displacement of People	*		
ii.	Change of Land use and Ecology	*		
iii.	Loss of Cultural and Religious Structures		*	
iv.	Socio-economic Impacts	*		
v.	Loss of Trees	*		
vi.	Drainage & Utilities Problems	*		
B.	**Impact due to Project Design**			
i.	Platforms - Inlets and Outlets		*	
ii.	Ventilation and Lighting		*	
iii.	Station Refuse	*		
iv.	Risk due to Earthquakes		*	
C.	**Impact due to Project Construction**			
i.	Top Soil Erosion, Pollution and Health risk	*		
ii.	Traffic Diversions and	*		
iii.	Risk to Existing Buildings	*		
iv.	Problems of Soil Disposal and Seepage Risk	*		
v.	Dust Generation	*		
vi.	Increased Water Demand	*		
vii.	Supply of Construction Material	*		
viii.	Construction and Demolition Waste	*		
ix.	Batching Plant and Casting Yard	*		
x.	Noise	*		
D.	**Impact due to Project Operation**			
i.	Oil Pollution	*		
ii.	Noise	*		
iii.	Water supply and sanitation	*		
iv.	Vibrations	*		
v.	Pedestrian Issues		*	
vi.	Visual Impacts		*	
vii.	Station Illumination		*	
viii.	Employment Opportunities			*
ix.	Enhancement of Economy			*
x.	Mobility			*
xi.	Safety			*
xii.	Traffic Congestion Reduction			*
xiii.	Less fuel Consumption			*

S. No.	Parameter	Negative Impact	No Impact	Positive Impact
xiv.	Less Air Pollution			*
xv.	Carbon dioxide Reduction			*
xvi.	Reduction in Buses			*
xvii.	Reduction in Infrastructure			*

14.12 PUBLIC CONSULTATION AND DISCLOSURE

Public consultation and participation is a continuous two way process, involving, promoting of public understanding of the processes and mechanisms through which developmental problems and needs are investigated and solved. The public consultation, as an integral part of environmental and social assessment process throughout the project preparation stage not only minimizes the risks and unwanted political propaganda against the project but also abridges the gap between the community and the project formulators, which leads to timely completion of the project and making the project people friendly.

Public consultations with the people of different sections of the society along the project alignment, shopkeepers, and influential persons of the project area will be made. Attention shall be given to potential vulnerable people like, squatters, encroachers, schedule caste, and other backward section (OBC) of society shall be consulted to make them aware and identify adverse impacts of the project.

A. Consultation with Stakeholders

As required for Category A projects, preliminary consultations were conducted at the early stage of EIA preparation, mostly involving local communities. Successive consultations shall be conducted by the SMC after the finalization of this report that includes representatives of local communities and entities

tasked with the regulation of the road development and environmental protection.

B. Compliance with Regulatory and Funding Agency Requirement

As per Indian Environmental Regulations, public hearing is not required, as railway projects do not attract EIA Notification 2006, amended 2009. Meaningful consultations will be undertaken. All the five principles of information dissemination, information solicitation, integration, co ordination and engagement into dialogue will be incorporated in the consultation process.

14.13 ENVIRONMENTAL MANAGEMENT PLAN

14.13.1 Management Plans

The Surat Metro Project will provide employment opportunity, quick mobility service and safety, traffic congestion reduction, less fuel consumption and air pollution on one hand and problems of muck disposal, traffic diversion, utility dislocation etc. on the other hand.

Protection, preservation and conservation of environment have always been a primary consideration in Indian ethos, culture and traditions. Management of Environment by provision of necessary safeguards in planning of the project itself can lead to reduction of adverse impacts due to a project. This chapter, therefore, spells out the set of measures to be taken during project construction and operation to mitigate or bring down the adverse environmental impacts to acceptable levels based on the proposed Environmental Management Plan (EMP).

The most reliable way to ensure that the plan will be integrated into the overall project planning and implementation is to establish the plan as a component of the project. This will ensure that it receives funding and supervision along with the other investment components. For optimal integration of EMP into the project, there should be investment links for:

Funding,
Management and training, and
Monitoring.

The purpose of the first link is to ensure that proposed actions are adequately financed. The second link helps in embedding training, technical assistance,

staffing and other institutional strengthening items in the mitigation measures to implement the overall management plan. The third link provides a critical path for implementation and enables sponsors and the funding agency to evaluate the success of mitigation measures as part of project supervision, and as a means to improve future projects. This chapter has been divided into three sections:

Mitigation measures,
Disaster management, and
Emergency measures.

14.13.2 Mitigation Measures

The main aim of mitigation measures is to protect and enhance the existing environment of the project. Mitigation measures have to be adopted during construction at all the construction sites including Batching Plant and Casting Yards on all the aspects. The mitigation measures to be adopted have been described under following heads:

Compensatory Afforestation,
Construction Material Management,
Labour Camp,
Energy Management
Hazardous Waste Management
Environmental Sanitation,
Utility Plan,
Air Pollution Control Measures,
Noise Control Measures,
Vibration Control Measures,
Traffic Diversion/Management,
Soil Erosion Control,
Water Supply, Sanitation and Solid Waste management,
Rain water harvesting
Management Plans for Depot
Training and Extension

14.13.2 Compensatory Afforestation

The objective of the afforestation program should be to develop natural areas in which ecological functions could be maintained on a sustainable basis. According to the results of the present study, it is found that about 481 trees are likely to be lost due to the project. Three saplings are to be planted for

felling a single tree. Hence 1443 trees need to be planted. Plantation program will be finalized in consultation with SMC and project proponent would provide the funds for compensatory afforestation as per government policy.

14.13.3 Construction Material Management – Storage and Procurement

The major construction material to be used for construction of the proposed corridor are coarse aggregates, cement, coarse sand, reinforcement steel, structural steel, water supply, drainage and sanitary fittings etc. The material will be loaded and unloaded by engaging labour at both the locations by the contractor.

The duties of the contractor will include monitoring all aspects of construction activities, commencing with the storing, loading of construction materials and equipment in order to maintain the quality. During the construction period, the construction material storage site is to be regularly inspected for the presence of uncontrolled construction waste. Close liaison with the SMC Officer and the head of the construction crew will be required to address any environmental issues and to set up procedures for mitigating impacts. The scheduling of material procurement and transport shall be linked with construction schedule of the project. The Contractor shall be responsible for management of such construction material during entire construction period of the project. Sufficient quantity of materials should be available before starting each activity. The contractor should test all the materials in the Government labs or Government approved labs in order to ensure the quality of materials before construction. This is also the responsibility of the contractor, which would be clearly mentioned in the contractor's agreement. Care shall be taken to avoid spillage of material during construction. Procurement of material would be from environment friendly source. The materials shall be procured from nearest available source and shall be transported in coverd trucks. All the material would be stored in a manner to avoid multiple handling for use in construction activities.

14.13.4 Labour Camp

The Contractor during the progress of work will provide, erect and maintain the necessary (temporary) living accommodation and ancillary facilities for labour to standards and scales approved by the SMC. All temporary accommodation must be constructed and maintained in such a fashion that uncontaminated water is available for drinking, cooking and washing. Safe

drinking water should be provided to the dwellers of the construction camps. Adequate washing and bathing places shall be provided, and kept in clean and drained condition. Construction camps are the responsibility of the concerned contractors and these shall not be allowed in the construction areas but sited away. Adequate health care is to be provided for the work force.

Sanitation Facilities: Construction sites and camps shall be provided sanitary latrines and urinals. Mobile STP/ septic tanks should be provided for the flow of used water outside the camp. Drains and ditches should be treated with bleaching powder on a regular basis. The sewage system for the camp must be properly designed, built and operated so that no health hazard occurs and no pollution to the air, ground or adjacent watercourses takes place. Garbage bins must be provided in the camp and regularly emptied and the garbage disposed off in a hygienic manner

Shelter at Workplace: At every workplace, shelter shall be provided free of cost, separately for use of men and women labourers. Sheds shall be maintained in proper hygienic conditions.

First aid facilities: At every workplace, a readily available first-aid unit including an adequate supply of sterilized dressing materials and appliances shall be provided. Suitable transport shall be provided to facilitate taking injured and ill persons to the nearest hospital.

Day Crèche Facilities: At every construction site, provision of a day crèche shall be worked out so as to enable women to leave behind their children. At construction sites where 25 or more women are ordinarily employed, at least a hut shall be provided for use of children under the age of 6 years belonging to such women. Huts shall be provided with suitable and sufficient openings for light and ventilation. Size of crèches shall vary according to the number of women workers employed.

14.13.5 Energy Management

The contractor shall use and maintain equipment so as to conserve energy and shall be able to produce demonstrable evidence of the same upon SMC request. Measures to conserve energy include but not limited to the following:

Use of energy efficient motors and pumps,
Use of energy efficient lighting, which uses energy efficient luminaries,

Adequate and uniform illumination level at construction sites suitable for the task,
Proper size and length of cables and wires to match the rating of equipment, and
Use of energy efficient air conditioner.

The contractor shall design site offices maximum daylight and minimum heat gain. The rooms shall be well insulated to enhance the efficiency of air conditioners and the use of solar films on windows may be explored.

14.13.6 Hazardous Waste Management

The contractor shall identify the nature and quantity of hazardous waste generated as a result of his activities and shall file a 'Request for Authorization' with Gujarat Pollution Control Board along with a map showing the location of storage area. Outside the storage area, the contractor shall place a 'display board', which will display quantity and nature of hazardous waste, on date. Hazardous Waste needs to be stored in a secure place. It shall be the responsibility of the contractor to ensure that hazardous wastes are stored, based on the composition, in a manner suitable for handling, storage and transport. The labeling and packaging is required to be easily visible and be able to withstand physical conditions and climatic factors. The contractor shall approach only Authorized Recyclers for disposal of Hazardous Waste, under intimation to the SMC.

14.13.7 Environmental Sanitation

Environmental sanitation also referred to as Housekeeping, is the act of keeping the working environment cleared of all unnecessary waste, thereby providing a first-line of defense against accidents and injuries. Contractor shall understand and accept that improper environmental sanitation is the primary hazard in any construction site and ensure that a high degree of environmental sanitation is always maintained. Environmental sanitation is the responsibility of all site personnel, and line management commitment shall be demonstrated by the continued efforts of supervising staff towards this activity.

General environmental sanitation shall be carried out by the contractor and at all times at Work Site, Construction Depot, Batching Plant, Labour Camp, Stores, Offices and toilets/urinals. The contractor shall employ a special group of environmental sanitation personnel to carry out following activities:

Full height fence, barriers, barricades etc. shall be erected around the site in order to prevent the surrounding area from excavated soil, rubbish etc, which may cause inconvenience to and endanger the public. The barricade especially those exposed to public shall be aesthetically maintained by regular cleaning and painting as directed by the Employer. These shall be maintained in one line and level.

The structure dimension of the barricade, material and composition, its colour scheme, SMC logo and other details.

All stairways, passageways and gangways shall be maintained without any blockages or obstructions. All emergency exits passageways, exits fire doors, break-glass alarm points, fire-fighting equipment, first aid stations, and other emergency stations shall be kept clean, unobstructed and in good working order.

All surplus earth and debris are removed/disposed off from the working areas to officially designated dumpsites. Trucks carrying sand, earth and any pulverized materials etc. in order to avoid dust or odour impact shall be covered while moving.

No parking of trucks/trolleys, cranes and trailers etc. shall be allowed on roads, which may obstruct the traffic movement.

Roads shall be kept clear and materials like: pipes, steel, sand boulders, concrete, chips and brick etc. shall not be allowed on the roads to obstruct free movement of road traffic.

Water logging or bentonite spillage on roads shall not be allowed.

Proper and safe stacking of material are of paramount importance at yards, stores and such locations where material would be unloaded for future use. The storage area shall be well laid out with easy access and material stored / stacked in an orderly and safe manner.

Flammable chemicals / compressed gas cylinders shall be safely stored.

Unused/surplus cables, steel items and steel scrap lying scattered at different places within the working areas shall be removed to identified locations.

All wooden scrap, empty wooden cable drums and other combustible packing materials, shall be removed from work place to identified location(s).

Empty cement bags and other packaging material shall be properly stacked and removed.

14.13.8 Utility Plan

The proposed Metro alignment runs along major arterial roads of the city, which serve Institutional, Commercial and Residential areas. A number of

sub-surface, surface and overhead utility services, viz. sewers, water mains, storm water drains, telephone cables, electrical transmission lines, electric poles, traffic signals etc. exists along the proposed alignment. These utility services are essential and have to be maintained in working order during different stages of construction by temporary / permanent diversions or by supporting in position. As such, these may affect construction and project implementation time schedule /costs, for which necessary planning / action needs to be initiated in advance. Prior to the actual execution of work at site, detailed investigation of all utilities and location will be undertaken well in advance by making trench pit to avoid damage to any utility. While planning for diversion of underground utility services e.g. sewer lines, water pipe lines, cables etc., during construction of Metro alignment, the following guidelines could be adopted:

Utility services shall be kept operational during the entire construction period and after completion of project. All proposals should therefore, ensure their uninterrupted functioning.

The elevated viaduct does not pose any serious difficulty in negotiating the underground utility services, especially those running across the alignment. In such situation, the spanning arrangement of the viaduct may be suitably adjusted to ensure that no foundation need be constructed at the location, where utility is crossing the proposed Metro alignment. In case of utility services running along the alignment either below or at very close distance, the layout of piles in the foundations shall be suitably modified such that the utility service is either encased within the foundation piles or remains clear of them.

14.13.8.1 Air Pollution Control Measures

During the construction period, the impact on air quality will be mainly due to increase in PM_{10} along haul roads and emission from vehicles and construction machinery. Though the estimation of air quality during construction shows some impact on ambient air quality, nevertheless certain mitigation measures which shall be adopted to reduce the air pollution are presented below:

The Contractor shall take all necessary precautions to minimise fugitive dust emissions from operations involving excavation, grading, and clearing of land and disposal of waste. He shall not allow emissions of fugitive dust from any transport, handling, construction or storage activity to remain

visible in atmosphere beyond the property line of emission source for any prolonged period of time without notification to the Employer.

The Contractor shall use construction equipment to minimize or control of air pollution. He shall maintain evidence of such design and equipment and make these available for inspection by Employer.

Contractor's transport vehicles and other equipment shall conform to emission standards fixed by Statutory Agencies of Government of India or the State Government from time to time. The Contractor shall carry out periodical checks and undertake remedial measures including replacement, if required, so as to operate within permissible norms.

The Contractor shall cover loads of dust generating materials like debris and soil being transported from construction sites. All trucks carrying loose material should be covered and loaded with sufficient free - board to avoid spills through the tailboard or sideboards.

The temporary dumping areas shall be maintained by the Contractor at all times until the excavate is re-utilized for backfilling or as directed by Employer. Dust control activities shall continue even during any work stoppage.

The Contractor shall place material in a manner that will minimize dust production. Material shall be minimized each day and wetted, to minimize dust production. During dry weather, dust control methods must be used daily especially on windy, dry days to prevent any dust from blowing across the site perimeter.

The Contractor shall water down construction sites as required to suppress dust, during handling of excavation soil or debris or during demolition. The Contractor will make water sprinklers, water supply and water delivering equipment available at any time that it is required for dust control use. Dust screens will be used, as feasible when additional dust control measures are needed especially where the work is near sensitive receptors.

The Contractor shall provide a wash pit or a wheel washing and/or vehicle cleaning facility at the exits from work sites such as construction depots and batching plants. At such facility, high-pressure water jets will be directed at the wheels of vehicles to remove all spoil and dirt.

14.13.10 Construction and Demolition Waste

Waste prevention, reuse and recycling can not only save money, but also generate broad environmental benefits, including the conservation of natural resources. Reuse and waste prevention reduce the air and water pollution associated with materials manufacturing and transportation. This saves

energy and reduces attendant greenhouse gas production. The recycling of many materials requires less energy than production from virgin stock, and can also reduce transportation requirements and associated impacts.

Opportunities for reducing C&D waste focus on three approaches, typically expressed as **Reduce-Reuse-Recycle**.

The source of C & D waste are pile caps, excess RMC and demolition material. An effort shall be made to recover embedded energy and to recycle the maximum quantity of C & D Waste to manufacture tiles, curb stones, paver block etc. The contractor shall store C&D waste separately at the site and sent to recycling facility periodically.

There shall be no disposal of any waste along storm water drains, canals and/ or any other water body or depression. Rather C & D waste shall be collected and sent to any authorized waste recycling facility.

14.13.11 Noise Control Measures

There will be an increase in noise level in nearby ambient air due to construction and operation of the Metro corridors. During construction, the exposure of workers to high noise levels especially near the machinery need to be minimized. This could be achieved by:

Job rotation,
Automation,
Construction of permanent and temporary noise barriers,
Use electric instead of diesel powered equipment,
Use hydraulic tools instead of pneumatic tools,
Acoustic enclosures should be provided for individual noise generating construction equipment like DG sets,
Scheduling and staggering truck loading, unloading and hauling operation,
Schedule and stagger work to avoid simultaneous activities which generate high noise levels,
Anti drumming floor and noise absorption material,
Low speed compressor, blower and air conditioner,
Mounting of under frame equipment on anti-vibration pad,
Smooth and gradual control of door,
Provision of sound absorbing material in the supply duct and return grill of air conditioner,

Sealing design to reduce the aspiration of noise through the gap in the sliding doors and piping holes, and
Sound proof compartments control rooms etc.

Special acoustic enclosures should be provided for individual noise generating equipment, wherever possible. Workers in sections where periodic adjustment of equipment/ machinery is necessary, should be provided with sound proof control rooms so that exposure to higher noise level is reduced. During construction, there may be high noise levels due to pile driving, use of compressors and drilling machinery. Effective measures should be taken during the construction phase to reduce the noise from various sources. The noise from air compressor can be reduced by fitting exhaust and intake mufflers.

The pile driving operation can produce noise levels up to 100 dB (A) at a distance of 25-m from site. Suitable noise barriers can reduce the noise levels to 70 dB (A) at a distance of 15m from the piles. A safety precaution as stipulated in IS: 5121 (1969) '*Safety Code for Piling and other Deep Foundation'* need to be adopted.

Noise level from loading and unloading of construction materials can be reduced by usage of various types of cranes and placing materials on sand or sandy bag beds.

14.13.12 Vibration Control

Mitigation can minimize the adverse effects of project ground-borne vibration on sensitive land uses. Vibrations arise due to rail - wheel interaction during operations. Vibrations could be reduced by improving track geometry, providing elastic fastenings, minimizing surface irregularities of wheel and rail, and separation of rail seat assembly from the concrete plinth with insertion of resilient and shock absorbing pad.

Adequate wheel and rail maintenance in controlling levels of ground-borne vibration is very important. Problems with rough wheels or rails can increase vibration levels by as much as 20 dB, negating the effects of even the most effective vibration control measures. It is rare that practical vibration control measures will provide more than 15- to 20-decibel attenuation. When ground-borne vibration problems are associated with existing rails and rolling stock, often the best control measure is to implement new or improved maintenance procedures. Grinding rough or corrugated rail and implementing wheel truing

to restore the wheel surface and contour may reduce vibration more than completely replacing the existing track system with floating slabs. Assuming that the track and vehicles are in good condition, the options to further reduce ground-borne vibration fit into one of seven categories: (1) maintenance procedures, (2) location and design of special track work, (3) vehicle modifications, (4) changes in the track support system, (5) building modifications, (6) adjustments to the vibration transmission path, and (7) operational changes.

Maintenance

Effective maintenance programs are essential for keeping ground-borne vibration levels under control. When the wheel and rail surfaces are allowed to degrade, the vibration levels can increase by as much as 20 dB compared with a new or well-maintained system. Maintenance procedures that are particularly effective at avoiding increases in ground-borne vibration include the following:

Rail grinding on a regular basis, particularly for rail that develops corrugations. Rail condition monitoring systems are available to optimize track conditions.

Wheel truing to re-contour the wheel, provide a smooth running surface, and remove wheel flats. The most dramatic vibration reduction results from removing wheel flats. However, significant improvements also can be observed simply from smoothing the running surface. Wheel condition monitoring systems are available to optimize wheel conditions.

Reconditioning vehicles, particularly when components such as suspension system, brakes, and wheels will be improved and slip-slide detectors will be installed.

Installing wheel condition monitoring systems to identify those vehicles most in need of wheel truing.

Location and Design of Special Track Work

Most vibration impact from a new train system is caused by wheel impacts at the special track work for turnouts and crossovers. Careful review of crossover and turnout locations during the preliminary engineering stage is an important step in minimizing potential for vibration impact. When feasible, the most effective vibration control measure is to relocate the special track work to a less vibration-sensitive area. Another approach is to install movable point or spring frogs that eliminate the gaps that occur when standard rail bound frogs are used. These special frogs significantly reduce vibration levels near

crossovers, and they are often specified because of their longer lifespan under repetitive high-speed conditions.

Vehicle Suspension

The ideal rail vehicle, with respect to minimizing ground-borne vibration, should have a low unsprung weight, a soft primary suspension, a minimum of metal-to-metal contact between moving parts of the truck, and smooth wheels that are perfectly round. A thorough dynamic analysis, including the expected track parameters, should be part of the specifications for the proposed high-speed train set.

Special Track Support Systems

When the vibration assessment indicates that vibration levels will be excessive, it is usually the track support system that is modified to reduce the vibration levels. Floating slabs, resiliency supported ties; high-resilience fasteners, ballast mats, and tire-derived aggregate (shredded tires) all have been used to reduce the levels of ground-borne vibration. To be effective, these measures must be optimized for the frequency spectrum of the vibration. While designing of the structures such as viaducts and pillars following points shall be taken into consideration:

A heavy rail section of 60-kg/m, 90 UTS, supported at every 60-cm. has been
proposed in order to prevent the development of surface irregularities on the rail.

The rail used shall be the one which is continuously welded which shall lead to reduction of noise/vibration especially on account of irregular track geometry and at curves.

Elastic fastening system is proposed to be used which shall reduce the vibration generated from rail- wheel interaction.

Monitoring requirements for vibrations at regular intervals throughout the construction period.

Pre-construction structural integrity inspections of historic and sensitive structures in project activity.

The ballast-less track is supported on two layers of rubber pads to reduce track noise and ground vibrations. The concept of a “low-noise” electric locomotive must be adopted at a very early stage of planning and must be followed up with detailed work throughout the project execution and operation. In addition, baffle walls as parapets will be constructed up to the rail level so as to reduce sound levels.

In addition, we have proposed to provide skirting of coach shell covering the wheel which will screen any noise coming from the rail wheel interaction as of propagating beyond the viaduct. In sensitive areas, track can be suitably designed so as to avoid propagation of noise to adjacent

structures.
The lower vibration can be achieved by providing bolster less type bogies having secondary air spring.

All these measures would be part of project cost.

14.13.13 Traffic Diversion/ Management

During construction, traffic is likely to be affected. Hence Traffic Diversion Plans are required in order to look for options and remedial measures so as to mitigate any traffic congestion situations arising out due to acquisition of road space during Metro construction of the corridor. Any reduction of road space during Metro construction results in constrained traffic flow. In order to retain satisfactory levels of traffic flow during the construction period; traffic management and engineering measures need to be taken. They can be road widening exercises, traffic segregation, one-way movements, traffic diversions on influence area roads etc. Maintenance of diverted roads in good working condition to avoid slow down and congestion shall be a prerequisite during construction period.

Various construction technologies are in place to ensure that traffic impedance is done at the minimum. They are:

The requirement would be mainly along the central verge/ side of the road.
As regards to the alignment cutting across a major traffic corridor, 'Box Girder Construction Technology' would be applied to prevent traffic hold-ups or diversions of any kind.

Only temporary diversion plans will be required during construction of the proposed Metro corridor. At the onset, all encroachments from road ROW will have to be removed. These encroachments vary from 'on-street' parking to informal activities.

Keeping in view the future traffic growth and reduction of carriageway due to Metro construction, implementation of traffic management/diversion plans shall become inevitable for ensuring smooth traffic movement and similar traffic diversion plans shall be formulated and followed during the execution stage.

Traffic Management Guidelines: The basic objective of the following guidelines is to lay down procedures to be adopted by contractor to ensure

the safe and efficient movement of traffic and also to ensure the safety of workmen at construction sites.

All construction workers should be provided with high visibility jackets with reflective tapes as most of viaduct and station works are on the right-of-way. The conspicuity of workmen at all times shall be increased so as to protect from speeding vehicular traffic.

Warn the road user clearly and sufficiently in advance.

Provide safe and clearly marked lanes for guiding road users.

Provide safe and clearly marked buffer and work zones

Provide adequate measures that control driver behavior through construction zones.

The primary traffic control devices used in work zones shall include signs, delineators, barricades, cones, pylons, pavement markings and flashing lights.

14.13.14 Soil Erosion Control

Prior to the start of the relevant construction, the Contractor shall submit to the SMC for approval, his schedules for carrying out temporary and permanent erosion/sedimentation control works are as applicable for the items of clearing and grubbing, roadway and drainage excavation, embankment/sub-grade construction, bridges and/ or other structures across water courses, pavement courses and shoulders. He shall also submit for approval his proposed method of erosion/sedimentation control on service road and his plan for disposal of waste materials. Work shall not be started until the erosion/sedimentation control schedules and methods of operations for the applicable construction have been approved by the project authority.

The surface area of erodible earth material exposed by clearing and grubbing, excavation shall be limited to the extent practicable. The Contractor may be directed to provide immediate control measures to prevent soil erosion and sedimentation that will adversely affect construction operations, damage adjacent properties, or cause contamination of nearby streams or other watercourses. Such work may involve the construction of temporary berms, dikes, sediment basins, slope drains and use of temporary mulches, fabrics, mats, seeding, or other control devices or methods as necessary to control erosion and sedimentation. Top soil shall be preserved by the contractor and stacked separately at designated place and utilize it to cover the refilled area and to support vegetation.

The Contractor shall be required to incorporate all permanent erosion and sedimentation control features into the project at the earliest practicable time

as outlined in his accepted schedule to minimize the need for temporary erosion and sedimentation control measures.

Temporary erosion/sedimentation and pollution control measures will be used to control the phenomenon of erosion, sedimentation and pollution that may develop during normal construction practices, but may neither be foreseen during design stage or associated with permanent control features on the Project. Under no conditions shall a large surface area of credible earth material be exposed at one time by clearing and grubbing or excavation without prior approval of the project authority.

The SMC may limit the area of excavation, borrow and embankment operations in progress, commensurate with the Contractor's capability and progress in keeping the finish grading, mulching, seeding and other such permanent erosion, sedimentation and pollution control measures, in accordance with the accepted schedule.

Temporary erosion is sometimes caused due to the Contractor's negligence, carelessness or failure to install permanent controls. Sedimentation and pollution control measures then become necessary as a part of the work as scheduled or ordered by the project authority, and these shall be carried out at the Contractor's own expense. Temporary erosion, sedimentation and pollution control work required, which is not attributed to the Contractor's negligence, carelessness or failure to install permanent controls, will be performed as ordered by the project authority.

14.13.15 Water supply, sanitation and solid waste management

During Construction

The public health facilities, such as water supply, sanitation and toilets are much needed at the stations. Water should be treated before use up to national drinking water standards. The collection and safe disposal of human wastes are among the most important problems of environmental health. The water carried sewerage solves the excreta disposal problems. The mobile STP/ septic tanks should be adopted for sewage disposal. The water for domestic consumption shall be sourced from public water supply or alternatively designated borewells may be installed with due permission from statutory authority prior to installation of borewell.

For Construction activity, there is a restriction to utilize groundwater all over the nation as per order of National Green Tribunal (NGT). Thus, construction

water shall be sourced from Surat Municipal Corporation which is responsible for sewage disposal in Surat area. Alternatively, contractor shall arrange tie up for surface water supply or tanker water supply for construction activity. Best option is to use treated STP water for construction activity.

Solid waste shall be stacked at designated place and when sufficient quantity accumulates it shall be disposed-off through covered trucks to land fill site designated and authorized by SMC.

During Operations

Practically, public facilities at stations have to be operated by regular staff or may be designated to any NGO working in the area in the field of sanitation as per policy of SMC.

Requirement of drinking water supply at an elevated station is about 6 KL/day. The water consumption for an elevated station to meet the requirements of its activities is 17 KLD. This shall be provided from SMC.

Solid waste will be generated at station is about 0.8 – 1.2 m^3/Day. The maintenance of adequate sanitary facilities for temporarily storing refuse on the premises is considered a responsibility of the project authority. The storage containers for this purpose need to be designed. However, it is suggested that the capacity of these containers should not exceed 50 litres and these should be equipped with side handles to facilitate handling. To avoid odour and the accumulation of fly-supporting materials, garbage containers should be washed at frequent intervals. This should be collected and transported to local municipal bins for onward disposal to disposal site by municipality. During operation, as mitigation measures rainwater harvesting will be carried out at stations and along the viaduct.

14.13.16 Rain Water Harvesting

To conserve and augment the storage of groundwater, it is suggested to construct rainwater harvesting structures of suitable capacity along the alignment and at stations. The stations shall be provided with the facility of rainwater harvesting and artificial recharge. The total length of the proposed alignment is about 40 km and there would be 39 stations. The estimated cost of rain water harvesting for elevated corridor is about Rs. 11 lakhs per km (Rs. 440 Lakh) and Rs. 3.5 lakhs per station (Rs. 136.50 Lakhs. The total cost of rainwater harvesting would be Rs. 576.50 Lakh.

14.13.17 Tree Protection

There is requirement of felling 481 trees during construction of Metro corridors in Surat. An attempt shall be made to minimize the tree felling. As remediation of tree felling it is suggested to plant 3 trees for each tree felled. Thus 1443 trees would be planted. Moreover, SMC would chalk out the plantation program in close coordination with Tree Authority SMC by making the payment for plantation work including after care for three years. An attempt would be made to minimize the felling of trees to the bare minimum while working and undertaking construction work. The left out trees shall be protected by providing metal or brick tree guard around the tree at a distance of one metre surrounding the tree. Scope of transplantation of trees would also be explored with discussion with the Tree Authority MCGM. A provision of 28.86 Lakh has been made @ Rs. 2000/- per tree to be planted and maintained for a period of three years.

14.13.18 Management Plans for Depot

The management plans for depot site includes:

Water Supply,
Oil Pollution Control,
Sewage/Effluent Pollution Control,
Surface Drainage,
Green belt development,
Rain water harvesting, and
Recycling of treated waste water.
Solid Waste Disposal

Water supply: About 300KLD of water will be required for operation and functioning of depot. This could be either taken from water supply Authority or through boring tube well into the ground after taking permission from Central Ground Water Authority. The ground water will need treatment depending upon its use. Domestic and some of the industrial application, a reverse Osmosis (RO) plant of 8 liter/ minute capacity will be appropriate. The water treatment plant flow chart is given in **Figure 14.4**. The estimated cost of water supply plant is about 120.50 Lakh. The cost for the plant in two depots will be Rs. 241 Lakhs.

Oil Pollution Control: The oil tends to form scum in sedimentation chambers, clog fine screens, interfere with filtration and reduce the efficiency

of treatment plants. Hence oil and grease removal tank has to be installed at initial stage of effluent treatments. Such tanks usually employ compressed air to coagulate the oil and grease and cause it to rise promptly to the surface. Compressed air may be applied through porous plates located in bottom of the tank. The tank may be designed for a detention period of 5 to 15 minutes.

Sewage/Effluent Pollution Control: About 80 KLD of sewage is likely to be generated at depot. The sewage could be treated up to the level so that it could be used for horticulture purpose in the campus and can also be discharged into the stream a process flow chart is presented in **Figure 14.5.** The estimated cost of one sewage treatment plant is about Rs 78.00 Lakh. The cost for the plant in two depots will be Rs. 156 Lakhs.

Expectedly about 63 KLD of effluent would be generated at Depot. The effluent will have oil, grease and, detergent as main pollutants. This has to be treated as per requirement of regulatory pollution control agency of the state (MSPCB). Process flow chart of effluent treatment plant is shown in **Figure 14.6.** The estimated cost of effluent treatment plant is about Rs 88.50 Lakh. The cost for the plant in two depots will be Rs. 177 Lakhs.

Surface Drainage: The depot area should have proper drainage. The Storm water of the depot will be collected through the drain. Rain water harvesting structures at different locations in the drains and for surplus storm water, the drainage system is to be connected to nearby disposal site. The drainage costs have been included in project cost.

Green belt development: The greenbelt development / plantation in the depot area not only functions as landscape features resulting in harmonizing and amalgamating the physical structures of proposed buildings with surrounding environment but also acts as pollution sink / noise barrier. In addition to augmenting present vegetation, it will also check soil erosion, make the ecosystem more diversified and functionally more stable, make the climate more conducive and restore balance. It is recommended to have a provision of Rs 80.00 Lakh in the cost estimate for the green belt development for 2 depots.

Rain water harvesting: To conserve and augment the storage of groundwater, it has been proposed to construct roof top rainwater harvesting structure of suitable capacity in the constructed depot site. A provision of Rs 70.00 Lakh for 2 depots has been kept in the cost estimate.

Recycling of treated waste water: The Waste Water to be generated at depots shall be treated by ETP & STP in each Depot. The treated waste water shall be recycled for horticulture work of the depot. About 64 KLD of treated waste water will be used for horticulture. The estimated cost of recycling of treated waste water is about Rs. 82 Lakh in the 2 depots.

Solid Waste Disposal

Effort shall be made to minimize the generation of solid wastes in the workshop by using preventive methods and recycling. Solid Wastes generated in the depot from the workshop in the form of filings and packaging materials etc. shall be stacked and stored separately and disposed-of periodically.

The costs of environment management measures have been included in the project cost as construction and civil costs of Depot.

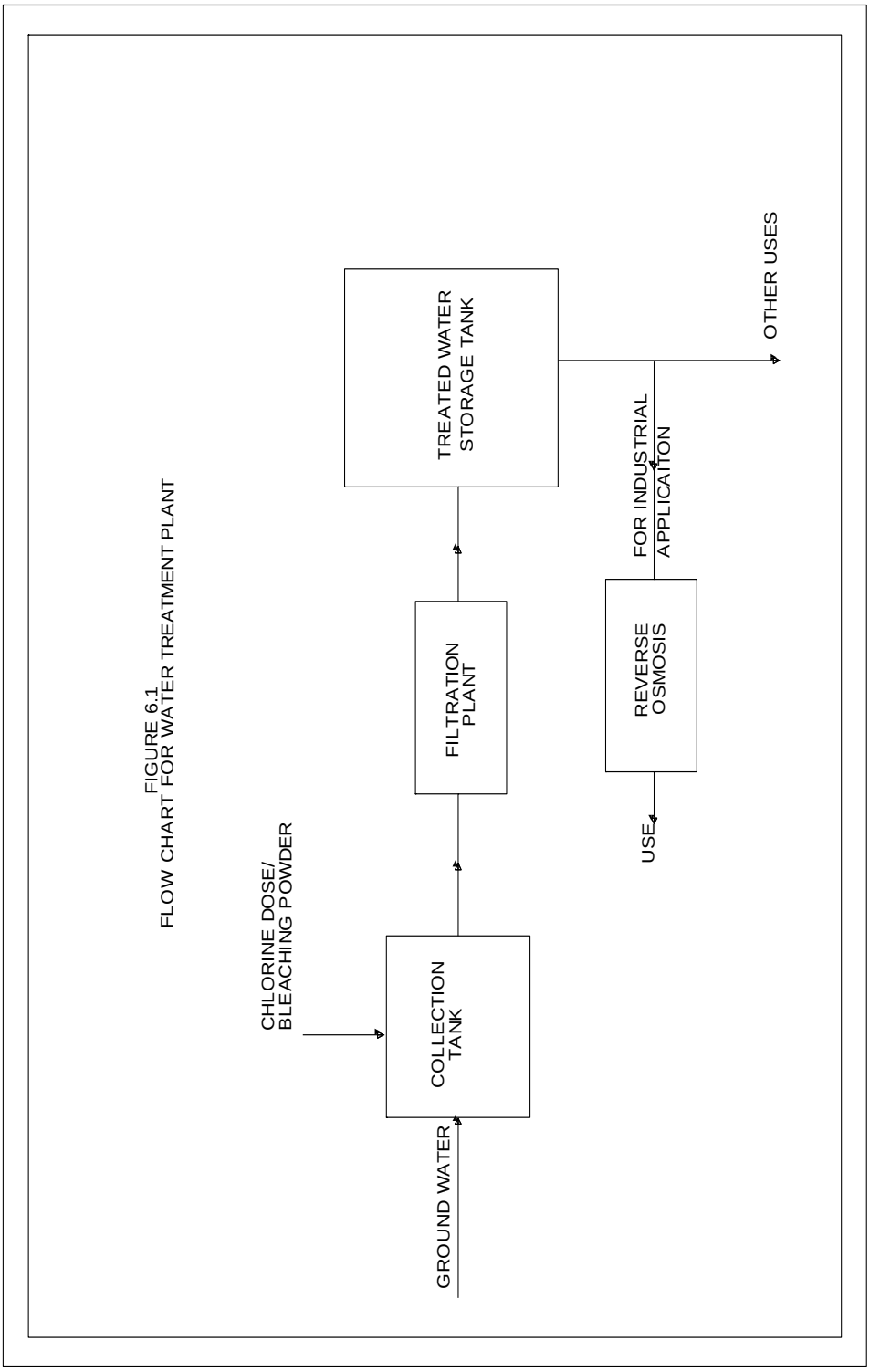


Fig. 14.4 Flow Chart for Water Treatment Plant

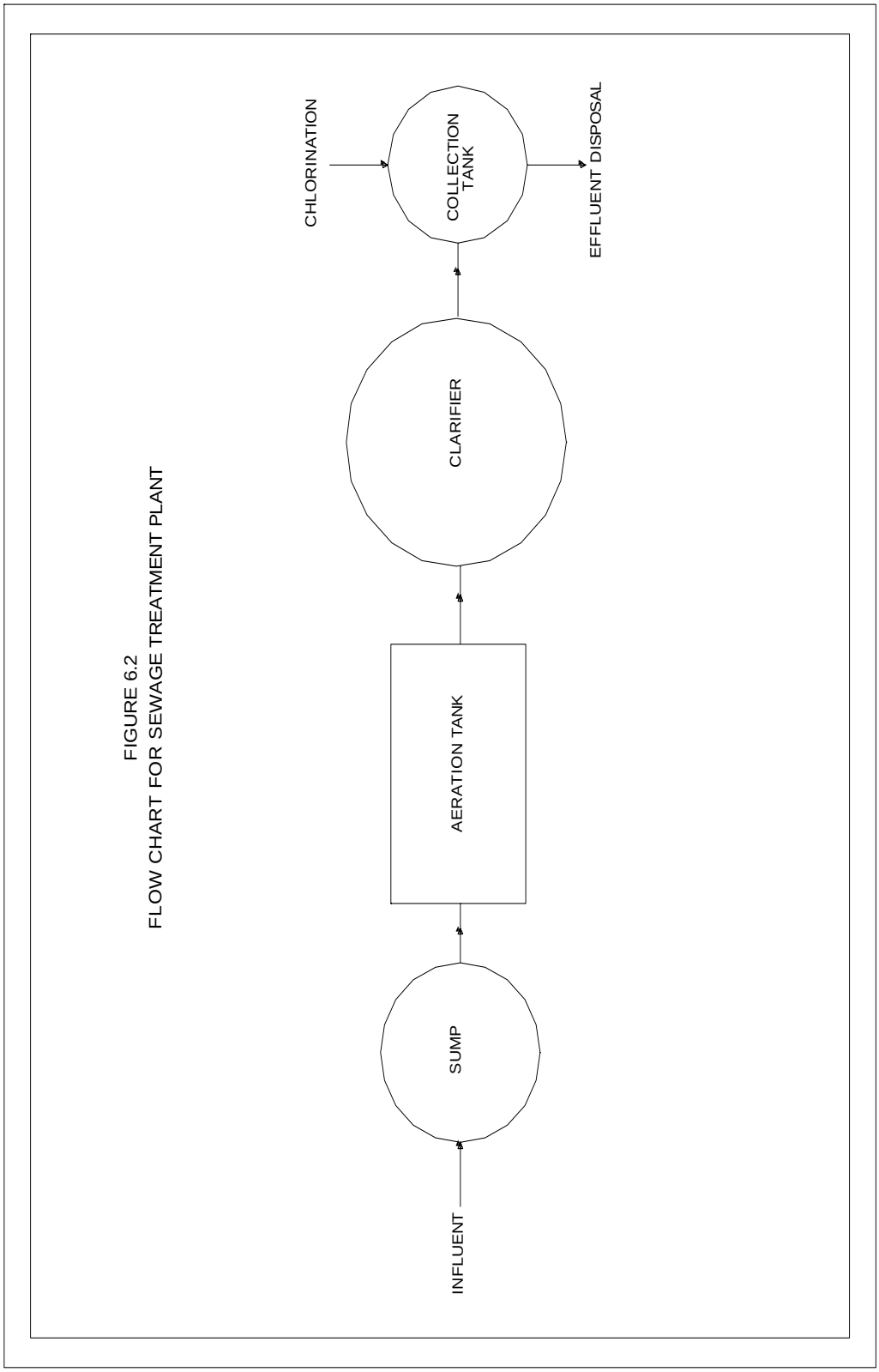


Fig. 14.5 Flow Chart for Sewage Treatment Plant

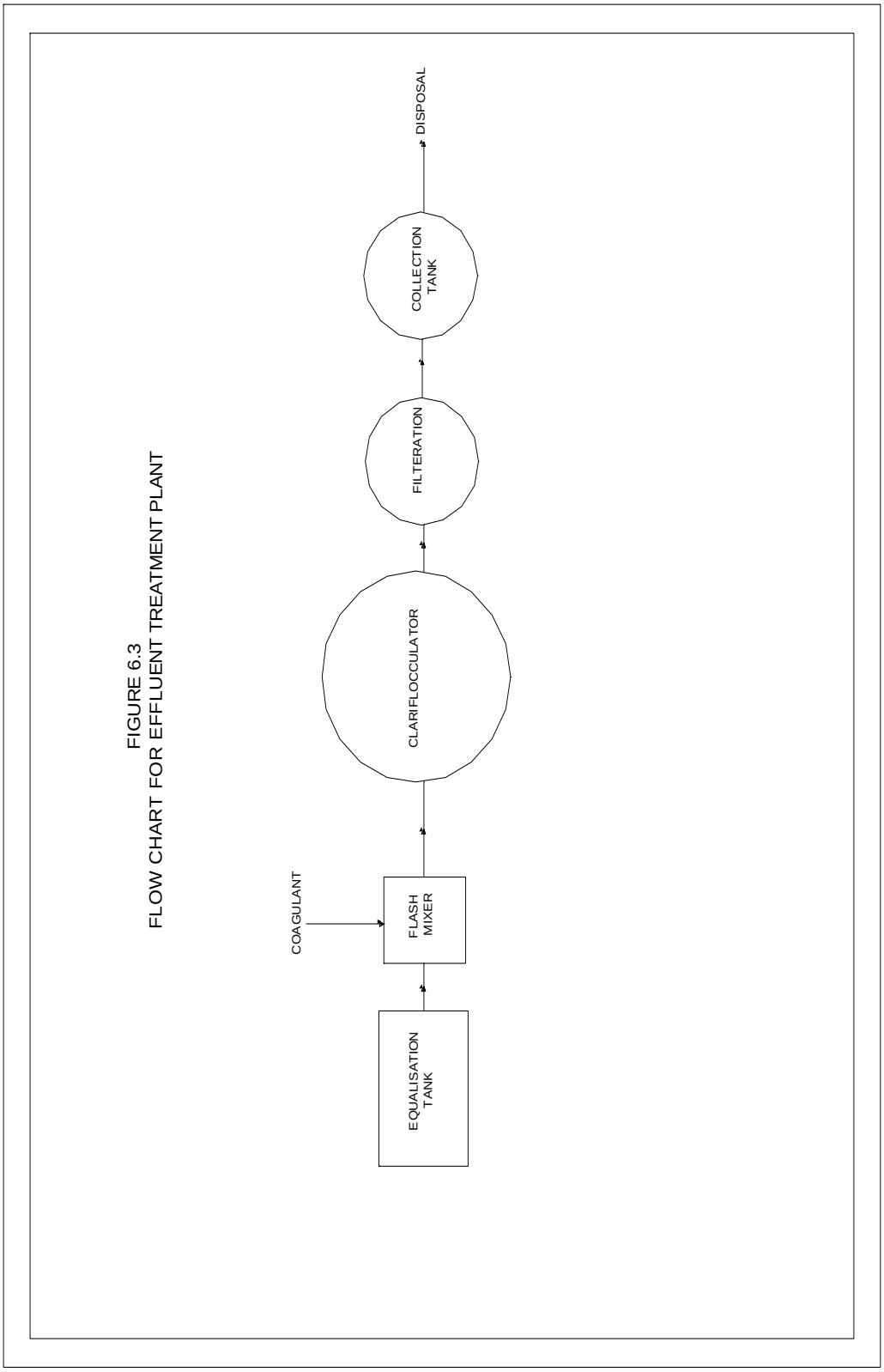


Fig. 14.6 Flow Chart for Effluent Treatment Plant

14.13.19 Disaster Management

Disaster is an unexpected event due to sudden failure of the system, external threats, internal disturbances, earthquakes, fire and accidents. The first step is to identify the causes which develop/ pose unexpected danger to the structural integrity of Metro overhead rail. The potential causes are excessive load, cracks, failure and malfunctioning of sensing instruments, accident, etc. This need to be looked into with care.

Preventive Action

Once the likelihood of a disaster is suspected, action has to be initiated to prevent a failure. Engineers responsible for preventive action should identify sources of repair equipments, materials, labour and expertise for use during emergency.

Reporting Procedures

The level at which a situation will be termed a disaster shall be specified. This shall include the stage at which the surveillance requirements should be increased both in frequency and details.

The Engineer-in-Chief should notify the officer for the following information:

Exit points for the public,
Safety areas in the tunnel/overhead rail, and
Nearest medical facility

Communication System

An efficient communication system is absolutely essential for the success of any disaster management plan. This has to be worked out in consultation with local authorities. More often, the entire communication system gets disrupted when a disaster occurs. The damage areas need to be clearly identified and provided with temporary and fool proof communication system.

Emergency Action Committee

To ensure coordinates action, an Emergency Action Committee should be constituted. The civic administrator may be the Chairman of this Committee. The committee may comprise of:

Station Manager concerned,
Police Officer of the area,
Surat Transport Corporation Representative,

Home Guard representative,
Fire Brigade representative,
Health Department representative,
Department of Information and Publicity, and
Non-Governmental Organization of the area

Emergency Action Committee will prepare the evacuation plan and procedures for implementation based on local needs and facilities available. The plan should include:

Demarcation of the areas to be evacuated with priorities,
Safe route to be used, adequacy of transport for evacuation, and traffic control,
Safe area and shelters,
Security of property left behind in the evacuated areas,
Functions and responsibilities of various members of evacuation teams, and
Setting up of joint control room

All personnel involved in the Emergency Action Plan should be thoroughly familiar with all the elements of the plan and their responsibilities. They should be trained through drills for the Emergency Action Plan. The staff at the site should be trained for problem detection, evaluation and emergency remedial measures. Individual responsibility to handle the segments in emergency plan must be allotted.

Success of an emergency plan depends on public participation, their response to warning notifications and timely action. Public has to be educated on the hazards and key role in disaster mitigation by helping in the planned evacuation and rescue operations.

It is essential to communicate by whom and how a declared emergency will be terminated. There should be proper notification to the public on de-alert signals regarding termination of the emergency. The notification should be clear so that the evacuees know precisely what to do when re-entering or approaching the affected areas.

14.13.20 Emergency Measures

The emergency measures are adopted to avoid any failure in the system such as lights, fire, means of escape etc. The aim of Emergency Action

Plan is to identify areas, population and structures likely to be affected due to a catastrophic event of accident. The action plan should also include preventive action, notification, warning procedures and co-ordination among various relief authorities. These are discussed in following sections.

Emergency Lighting

The emergency lights operated on battery power should be provided at each station. The battery system should supply power to at least 25% of the lights at the station, platforms, viaduct for a period of 2 hours.

Fire Protection

The building materials should be of appropriate fire resistance standard. The fire resistance period should be at least 2 hours for surface or over head structures. Wood shall not be used for any purpose, excluding artificial wood products, which are flame resistant. The materials which have zero surface burning characteristics need to be used. The electrical systems shall be provided with automatic circuit breakers activated by the rise of current as well as activated by over current. The design of a station will include provision for the following:

Fire prevention measures,
Fire control measures,
Fire detection systems,
Means of escape,
Access for fireman, and
Means of fire fighting

A. Fire Prevention and Safety Measures

Fire prevention measures will be designed and implemented to minimize the risk of outbreak of fire by appropriate choice, location and installation of various materials and equipment. In stations planning, potential sources of fire can be reduced by:

i. Fire Prevention

Use of non-combustible or smoke retardant materials where possible,
Rolling stock is provided with fire retarding materials, low smoke zero halogen type electric cable is also provide,
Provision of layout which permits ease of maintenance for equipment and cleaning of the station premises,

Provision of special storage spaces for combustible materials such as paint and oil,
Prohibition of smoking in fire prone areas,
Provision of cigarette and litter bins, and
Good housekeeping.

ii. Safety

Following provisions will be required from fire safety point of view:

Automatic sprinkler/detection system to be provided if floor area exceeds 750 sq.m

One wet riser-cum-down comer per 1000 sqm floor area with static underground storage tank, overhead tanks and pumps of suitable capacity with hydrants, first-aid reel, etc.

Portable fire non-aqueous extinguishers of Carbon Dioxide, chemical dry powder etc. at suitable places.

Automatic smokes venting facilities.

Two separate means of exit shall be provided, if more than 10 persons are working and the area exceeds 1400 sq.m.

Fire resisting doors shall be provided at appropriate places along the escape routes to prevent spread of fire and smoke.

The travel distance for fire escape shall not exceed 20 m where escape is available in more than one direction; the distance could be upto 40 m.

B. Fire Alarm and Detection System

A complete fire detection system with equipment complying with the requirements of Surat Fire Services shall be provided through out each station and ancillary buildings including entrance passageways, subways and adits etc. to give visual and audible indication of alarm conditions actuated by the operation of break glass contact or fire sensors e.g. detector heads, linear heat detecting cables etc. The system shall be operated from 24 V DC Power sources.

Manually operated call points shall be provided at every hydrant and nose reel points, station head wall, tail wall and other locations. Alarm bells shall be installed in each plant room complex at both platform and concourse level and shall be clearly audible at all points in the room/area.

Beam detector or heat detector shall be installed at roof level, ceiling and floor cavity, whilst linear detecting cables shall be installed in under platform cable ducts and cable shafts. Smoke probe units shall be installed in

rooms/compartments. When an alarm point is operated, the fire pump shall start to operate automatically. A station fire control and indicating panel shall be provided an installed in the station controllers room, for the control, indication and monitoring of the whole detection and fire fighting systems. While designing the fire fighting system, Surat Fire Services shall be taken into account for linking with the same.

C. Fire Control Measures

Control of the spread of fire and smoke will be achieved by partition of fire risk areas, planning for smoke extraction, and arrangement for smoke containment. Partition is aimed at limiting the extent of a fire. The openings must be capable of being sealed in the event of fire. With the exception of station public areas, a fire compartment will not exceed 1500 m^2. Partition of the public areas in stations is not practicable for operational reasons. The fire resistance period of this separated area should be about 3 hours.

D. Access for Fireman

A secondary access to the station, not used by passengers for evacuation, shall be available to fireman should the need arise. The entry point shall be easily accessible from the road. Access shall be available to all levels of the station. The minimum width of the stairs is 1.0 m and maximum height should not exceed 25 cm.

E. Emergency Door

The rolling stock is provided with emergency doors at both ends of the cab to ensure directed evacuation of passengers in case of any emergency including fire in the train.

14.14 SUMMARY OF ENVIRONMENTAL MANAGEMENT PLAN (EMP)

The environmental impacts stemming out of the proposed project can be mitigated with simple set of measures, dealing with careful planning and designing of the metro alignment and structures. Adequate provision of environmental clauses in work contracts and efficient contract management will eliminate or reduce significantly all possible problems. A common problem encountered during implementation of environmental management plans of such projects is lack of environmental awareness among engineers and managers concerned with day to day construction activities, which can be solved through regular environmental training programs. A set of preliminary EMP is presented in **Table 14.12,** which defines actions to be undertaken during the design stage, pre-construction, construction and operation stage of

the project. The effectiveness of environmental considerations will, however, depend on appropriate inclusion of these in the work contracts.

The major concern during the construction stage is that the contractors, due to lack of enforcement, would not practice good environmental sanitation (housekeeping) may intend to get unauthorized use of the easily available natural resources and other available infrastructure like roads and water resources. This would result in degradation of ambient air quality, water resources and land environment around the construction sites and workers camp. Improper management of earthwork and bridge construction activities would disrupt the natural drainage and increase soil erosion. Improper management may result in spillage of explosives into the hands of unsocial elements. Finally, the implementation of the mitigation actions requires that the project implementation unit would record an end-of-construction mitigation checklist, before releasing the final payment of any work contract.

Additionally, project authority should prepare and establish Environmental and Health Policy and Procedures as per earlier Phases and that should become an integral part of contract document.

Operational phase mitigation would involve good environmental sanitation (housekeeping) practice at metro establishments including effective solid waste collection and disposal, wastewater disposal, upbringing of plantations and green area. Protection of earth slopes in landslide prone area would be a very important task. During the operation period, the metro operating unit will be required to confirm receipt of the construction period mitigation report through the SMC and prepare a follow on timetable of actions.

TABLE 14.12 ENVIRONMENTAL MANAGEMENT ACTION PLAN (EMP)

Environmental Impact	Mitigation Measures Taken or To Be Taken	Time Frame	Implementing Organization	Responsible Organization
DESIGN PHASE				
Metro Alignment	The proposed corridor alignment was selected to minimise the land disturbance to avoid environmentally sensitive areas.	During Design	DPR and design consultant	SMC
Cultural Heritage	Avoided by adjustment of alignment.	During Design	DPR and design consultant	SMC
Flood	Bridges shall be well designed	During Design	DPR and design consultant	SMC
Inadequate design	Make sure that design provides for safety of structures against worst	DPR and detailed design	DPR and design	SMC

Environmental Impact	Mitigation Measures Taken or To Be Taken	Time Frame	Implementing Organization	Responsible Organization
provision for safety against seismological hazard	combination of forces in the probability of an earthquake likely to occur in seismic zone-III.	stage	consultant	
PRE –CONSTRUCTION STAGE				
Water requirement	The requirement of water for construction purpose etc shall be planned and shall be arranged from available and authorized sources in order to avoid digging of Tube wells.	Pre construction stage	Contractor	SMC/ EMP implementing agency
Disposal of final treated effluent from treatment plant	Options for final disposal shall be studied and the suitable disposal route shall be decided carefully to minimize the impact on receiving bodies. As far as possible zero discharge rules may be adopted.	During design stage / and pre construction of treatment plant	Contractor	SMC/ EMP implementing agency
Batching Plant and Casting Yard	These facilities to be located away from habitation. Consent to Establish and Consent to Operate to be taken from MPCB and to comply with all stipulations.	During Pre-construction Stage	Contractor	SMC/EMP implementing agency
CONSTRUCTION PHASE				
Environmental Management and Monitoring	This will include institutional requirements, training, environmental management and monitoring	During and after construction	Contractor	SMC/EMP implementing agency
Dust	Water should be sprayed during construction phase, wherever it is required to avoid dust. Vehicles delivering materials should be covered to reduce spills and dust blowing off the load.	During construction	Contractor	SMC/EMP implementing agency
Air Pollution	Vehicles and machinery are to be regularly maintained so that emissions conform to National and State AAQ Standards. No vehicle without valid PUC certificate would be allowed at Construction Sites.	Beginning with and continuing throughout construction period	Contractor	SMC/EMP implementing agency
Equipment Selection maintenance and operation	Construction plants and equipment will meet acceptable standards for emissions and will be maintained and operated in a manner that ensures that relevant air, noise, and discharge regulations are met.	During construction	Contractor	SMC/EMP implementing agency
Noise	Noise standard at processing sites, will be strictly enforced as per GOI noise standards. Workers in vicinity of strong noise will wear earplugs and their working time should be limited as a safety measure. At construction sites within 150m of sensitive receptors construction will be stopped from 22:00 to 06:00.	Beginning and through construction	Contractor	SMC/EMP implementing agency

Environmental Impact	Mitigation Measures Taken or To Be Taken	Time Frame	Implementing Organization	Responsible Organization
	Machinery to be provided noise barriers (Stone walls and plantation) for silence zones including schools and hospitals.			
Vibration	The vibration level limits at work sites adjacent to the alignment shall conform to the permitted values of peak velocity as given in Environmental Manual	Beginning and through construction	Contractor	SMC/EMP implementing agency
WATER				
Contamination from Wastes	All justifiable measures will be taken to prevent the wastewater produced in construction from entering directly into any rivers, drainage and irrigation system	Throughout construction period	Contractor	SMC/EMP implementing agency
Wastage of water	Measures shall be taken to avoid misuse of water. Construction agency shall be instructed accordingly to follow strict procedures while using the water for construction and drinking purpose.	Beginning with and continuing throughout construction	Contractor	SMC/EMP implementing agency
Sewerage disposal during construction at Service Centres	A minimum distance of any sewage or toilet facility from water sources should be 200 meters.	Throughout construction period	Contractor	SMC/EMP implementing agency
Sanitation and Waste Disposal in Construction Camps	Sufficient measures will be taken in the construction camps, i.e. provision of garbage tank and sanitation facilities. Waste in septic tanks will be cleared periodically. Drinking water will meet Indian National Standards. Garbage will be collected in a tank and disposed off daily. Special attention shall be paid to the sanitary condition of camps. Camps will be located at a minimum distance of 200 m from water sources.	Before and during building of construction camps	Contractor	SMC/ EMP implementing agency
SOIL				
Quarrying	Quarrying will be carried out at approved and licensed quarries only. All environmental mitigation measures shall be enforced at Quarry site also.	During construction	Contractor	SMC/ EMP implementing agency
FLORA AND FAUNA				
Loss of trees and Avenue Plantation	Areas of tree plantation cleared will be replaced according to Compensatory Afforestation Policy under the Forest Conservation Act. Ten trees will be planted against every tree felled as per norms.	During and after completion of construction activities	MCGM	MCGM

Environmental Impact	Mitigation Measures Taken or To Be Taken	Time Frame	Implementing Organization	Responsible Organization
SOCIAL				
Loss of Access	Temporary access should be built at the interchange and other roads.	During construction	Contractor	SMC/ Traffic department
Traffic jams and congestion	If there are traffic jams during construction, measures should be taken to relieve the congestion with the co-ordination of transportation and traffic police department	During construction	Contractor	SMC/ Traffic department
Safety with vehicles, people and livestock and signage	Safety education and fines. Allow for adequate traffic flow around construction areas Provide adequate signage, barriers and flag persons for safety precautions.	During construction	Contractor	SMC/ Traffic department
	Communicate to the public through radio, TV & newspaper announcements regarding the scope and timeframe of projects, as well as certain construction activities causing disruptions or access restrictions			
Increase in disease Water-borne Insect-borne Communicable diseases	Make certain that there is good drainage at all construction areas, to avoid creation of stagnant water bodies. Provide adequate sanitation and waste disposal at construction camps. Provide adequate health care for workers and locate camps away from vulnerable groups, if any	During construction At start-up Throughout construction	Contractor	SMC/ EMP implementing agency
Location of camps depots and storage areas	Location of camps depots and storage areas shall be as per the contract specifications.	Throughout construction	Contractor	SMC/ EMP implementing agency
OPERATION PHASE				
Noise and Vibration	Suitable measures should be considered where warranted. The public shall be educated about the regulations of noise and vibration pollution and its implications.	After completion of construction	SMC/EMP implementing agency	SMC/ EMP implementing agency
WATER				
Maintenance of Storm Water Drainage System	The urban drainage systems will be periodically checked and cleared so as to ensure adequate storm water flow.	Beginning and end of monsoon	SMC/EMP implementing agency	SMC/ EMP implementing agency

14.15 ENVIRONMENTAL MONITORING PLAN

14.15.1 Pre-construction Phase

The environmental monitoring programme is a vital process of any Environmental Management Plan (EMP) of development project for review of indicators and for taking immediate preventive action. This helps in signalling the potential problems resulting from the proposed project activities and will allow for prompt implementation of corrective measures. Historically, environmental monitoring has been integral part of works of SMC towards better environmental management of air, noise, vibration, water quality etc. both during construction and in operation. Generation of dust and noise are two main issues during any large construction activity. Degradation of water quality is another. The parameters are monitored in pre- construction, construction and operation phase and are based on the need to evaluate the deviation of environmental conditions from baseline environmental conditions due to construction and operation of the Metro. The environmental monitoring will be required during both construction and operational phases. The following parameters are proposed to be monitored:

Water Quality,
Air Quality,
Noise and Vibration,
Environmental Sanitation and Waste Disposal
Ecological Monitoring and Afforestation,
Workers Health and Safety

Environmental monitoring during pre-construction phase is important to know the baseline data and to predict the adverse impacts during construction and operations phases. Pre-construction phase monitoring has been done for the proposed project for air, noise, water, soil quality and ecology.

14.15.2 Construction Phase

During construction stage environmental monitoring will be carried out for air quality, noise levels and water quality. Keeping a broad view of the sensitive receptors and also the past experience of Metro projects, an estimate of locations has been made and are summarized in **Table 14.13.** The number could be modified based on need when the construction actually commences.

Water Quality

Since water contamination leads to various water related diseases, the project authorities shall establish a procedure for water quality surveillance and ensure safe water for the consumers. The water quality parameters are to be monitored during the entire period of project construction. Monitoring should be carried out by NABL certified laboratory. Water quality should be analyzed following the procedures given in standard methods. Parameters for monitoring will be as per BIS: 10500. The monitoring points could be ground and surface water.

Air Quality

Air quality should be monitored at the locations of baseline monitoring. The parameter recommended is Particulate Matter (PM_{10}). The contractor will be responsible for carrying out air monitoring during the entire construction phase under the supervision of project authority.

Noise and Vibration

The noise levels will be monitored at construction sites for entire phase of construction by the site contractor and under the supervision of project authority.

Workers Health and Safety

Monitoring of health risk issues that might arise throughout the project life time will be done. Epidemiological studies at construction sites and workers camp will be performed to monitor the potential spread of diseases. Regular inspection and medical checkups shall be carried out to worker's health and safety monitoring. Any reoccurring incidents such as irritations, rashes, respiratory problems etc shall be recorded and appropriate mitigation measures shall be taken. Contractor will be the responsible person to take care health and safety of workers during the entire period of the construction and project proponent is responsible to review/audit the health and safety measures/plans. The monitoring Schedule for Water Air, noise and ecology are presented in **Table 14.13.**

TABLE 14.13 CONSTRUCTION STAGE MONITORING SCHEDULE

Item	Parameter	Frequency and Duration	Locations
Air	PM_{10}	2×24 hours Twice a month During entire civil construction stage or even later, if directed by SMC	39 locations
Water	Groundwater quality (IS 10500:1991)	Once in 6months During entire civil construction stage or even later, if directed by SMC	8 locations
Noise	Noise Level	24hours Once a week	39 locations

	(Leq and Lmax)	During entire civil construction stage or even later, if directed by SMC	
Ecology	Felled and planted trees	Once a year till all trees that were to be planted by Gujarat Government on behalf of project authority, are planted	All the trees felled and newly planted

14.15.3 Operation Phase

Even though the environmental hazards during the operation phase of the project are minimal, the environmental monitoring will be carried out for air, noise, water, waste water, solid waste and ecology during operation phase of the project. The parameters monitored during operation will be PM_{10} for air, heavy metals for solid waste, pH, TSS, BOD, COD, oil and grease for waste water. However, water quality parameters that will be monitored will be as per BIS 10500. The monitoring schedule is presented in **Table 14.14**. The monitoring program shall be conducted by an external agency certified by NABL under the supervision of SMC. Project proponent (SMC) is responsible for successful environmental monitoring of the proposed project during operation phase.

TABLE14.14 OPERATION STAGE MONITORING SCHEDULE

Item	Parameter	Frequency and Duration	Locations
Air	PM_{10}	2×24hours Once a month For 3years	10 location
Water	Surface, Groundwater quality (IS 10500:1991)	Once a year For 3years	1+1 location
Noise	Noise Level (Leq)	24hours Once a year For 3years	39 locations (Sensitive Receptors)

The results of Air quality, water quality, waste-water will be submitted to management quarterly during construction phase and half yearly during operation phase.

14.16 COST ESTIMATES

14.16.1Summary of Costs

All costs involved in Environmental mitigation and management and monitoring has to be put on the account of Surat Metro Project corridors. A summary of these is presented in Table 14.15.

Table 14.15 Environmental Costs

S. No.	ITEM	COST Rs. lakh
1.	Rain Water Harvesting at stations and along alignment	576.50
2.	Air, Noise, vibration, Water, Waste Water, Solid waste, during construction and operation	50.00
3.	Ecological monitoring	10.00
4.	Tree Plantation 1443 trees @ Rs.2000/- per tree	28.86
5.	Water Treatment Plant	241.00
6.	Sewage Treatment Plant	156.00
7.	Effluent Treatment Plant	177.00
8.	Green Belt at Depot	80.00
9.	Rain water harvesting at Depot	70.00
10.	Recycling of treated waste water	82.00
	Total	**1471.36**

The compensation for loss of land, fire control, information systems and contractor's obligations has been incorporated in project costs.

The Environmental management plan should be implemented in phases so that optimum benefit could be achieved and should be synchronized with the construction schedules.

Table 14.16 Details of Cost of Resettlement and Rehabilitation (Rs. Lakh)

S. No.	Description	Entitlement	Unit	Quantity	Rate	Amount
1	Acquisition of Permanent land Private Land	Total Replacement Cost of land *	m^2			
2	Solatium	100% as per Act 30 of 2013.	m^2			
3	Acquisition of temporary private land	6% of total land cost per year for 3 years	m^2	-	-	
4	Acquisition of structures					
4.1	Residential PAPs	Area equivalent to affected area 20.91 m2 free of cost	Per unit	-	-	-
4.2	Commercial PAPs**	Area equivalent to affected area 20.91 m2 free of cost	Per unit	18		
5.	Subsistence Allowance ***	For a period of one year @Rs.3000/ month	Family	18	0.36	6.48
6.	Shifting Allowance ***	A lump sum shifting allowance of Rs. 50,000/-	No.	18	0.50	9.00
7.	One time resettlement allowance ***	For All the affected families excluding employees in shops	Per Family	18	0.50	9.00
9.	Independent Evaluation		LS		5.00	5.00
10.	Miscellaneous		LS		5.00	5.00
	Cost of R & R					**34.48**

* Land for the project to be provided by SMC

** refers the similar accommodation would be provided by SMC in the vicinity so additional cost is not given.

14.17 CONCLUSION

The proposed Metro line is proved to have significant positive effects to the development of Surat City. Benefits to the economy, traffic congestion reduction, quick and safety transport, employment opportunities, fuel consumption reduction, and air quality improvement are the obvious positive effects from this Metro line. Besides, the potential adverse environmental impacts on air quality (during construction phase), water environment, noise, solid waste, ecology, population resettlement are also taken into consideration. Based on these detailed potential adverse environmental impacts, appropriate mitigation measures have been developed for consideration. The EIA concluded that project impacts from both construction and operation will be minimal when mitigated through the use of prevailing current practices and appropriate technologies. With the implementation of the EMP and the monitoring plan, the Project is not expected to have significant adverse environmental impacts.

CHAPTER -15

SECURITY MEASURES FOR A METRO SYSTEM

15.1 INTRODUCTION

Metro is emerging as the most favoured mode of urban transportation system. The inherent characteristics of metro system make it an ideal target for terrorists and miscreants. Metro systems are typically open and dynamic systems which carry thousands of commuters. Moreover the high cost of infrastructure, its economic impotence, being the life line of city high news value, fear & panic and man casual ties poses greater threat to its security. Security is a relatively new challenge in the context of public transport. It addresses problems caused intentionally. Security differs from safety which addresses problems caused accidentally. Security problems or threats are caused by people whose actions aim to undermine or disturb the public transport system and/or to harm passengers or staff. These threats range from daily operational security problems such as disorder, vandalism and assault to the terrorist threat.

15.2 NECESSSITY OF SECURITY

It is well known that public transportation is increasingly important for urban areas to prosper in the face of challenges such as reducing congestion and pollution. Therefore, security plays an important role in helping public transport system to become the mode of choice. Therefore, excellence in security is a prerequisite for Metro system for increasing its market share. Metro railway administration must ensure that security model must keep pace rapid expansion of the metro and changing security scenario.

15.3 THREE PILLARS OF SECURITY

Security means protection of physical. Human and intellectual assets either from criminal interference, removal of destruction by terrorists or criminals or incidental to technological failures or natural hazardous events. There are three important pillars of security as mentioned under:

(i) The human factor; (ii) Procedures; and (iii)Technology

Staff engaging with the passengers create a sense of re-assurance which can not fully be achieved by technology. For human factor to be more effective staff has to be qualified, trained, well equipped and motivated. They should bs trained, drilled and tested. The security risk assessment is the first step for understanding the needs and prioritizing resources. The organization of security should be clear and consistent. Security incidents, especially major ones, often happen without warning. Emergency and contingency plans must be developed communicated and drilled in advance.

There are number of technologies which can be used to enhance security e.g. surveillance systems. The objectives of the security systems are to differ i.e., making planning or execution of on attack too difficult, detect the planned evidence before it occurs deny the access after in plan of attack has been made and to mitigate i.e. lessen the impact severity as the attack by appropriate digits.

15.4 PHASES OF SECURITY

There are three phases of security as under:

(i) Prevention

These are the measures which can prevent a security incidence from taking place. These can be identified by conducting a risk assessment and gathering intelligence. Prevention begins with the daily operational security -problems. Uncared for dirty, damaged property is a breeding ground for more serious crime.

(ii) Preparedness

Plans must be prepared to respond to incidents, mitigate the impact. Train staff accordingly and carry out exercises. The results of the risk assessment give a basis for such plans.

(iii) Recovery

Transport system must have laid down procedures/instructions for the quick recovery of normal service after an incident. Recovery is important for the financial health of the operation, but it also sends a clear message to public, it reassures passengers and gives them confidence to continue using the system. Communication is key to the quick restoration after such incidents. Restoration should ^also include an evaluation process for the lessons learnt.

15.5 RESPONSIBILITIES AND PARTNERSHIPS

Security is a sovereign function and hence is the responsibility of the state. Security in public requires clear governance. Responsibility should be clearly defined. In Surat City, it is the responsibility of Govt. of Gujarat (GoG) to ensure secured travelling to the public and Surat Metro.

15.6 PROPOSED PROVISIONS FOR SECURITY SYSTEM

1. CCTV coverage of all metro stations. With a provision of monitoring in the Station Security Room as well as at a Centralized Security Control Room with video wall, computer with access to internet TV with data connection, printer and telephone connection (Land Line and EPBX) for proper functioning, cluster viewing for stations. Cost of this is included in Telecom estimate.

2. Minimum one Baggage Scanners on all entry points (1 per AFC array). Additional requirement of baggage scanners at heavily crowed stations i.e at interchange may also be required. Cost of one baggage scanner is Rs. 15.0 Lacs approximately, on 2013 prices.

3. Multi-zone Door Frame Metal Detector (DFMD) minimum three per entry (2 per AFC array). The number can increase in view of the footfall at over crowded stations.

4. Hand held Metal Detector (HHMD) as per requirement of security agency, minimum two per entry, which varies from station to station with at least 1.5 per DFMD installed at the station.

5. Bomb Detection Equipments with modified vehicle as per requirement of security agency. One BDS team per 25 - 30 station will be required at par with present criteria of DMRC. Cost 1.25 crores including vehicle.

6. Bomb Blanket at least one per station and Depots shall be required.

7. Wireless Sets (Static and Hand Held) as per requirement of security agency.

8. Dragon light at least one per station and vital installation.

9. Mobile phones, land lines and EPBX phone connections for senior security officers and control room etc.

10. Dog Squads (Sniffer Dog), at least one dog for 4 metro stations which is at par with current arrangement of Delhi Metro. Dog Kennels alongwith provision for dog handlers and MI room will also be provided by metro train depot administration including land at suitable places line wise.

11. Bullet proof Morcha one per security check point (i.e. AFC array) and entry gate of metro train depot administration metro station.

12. Bullet proof jackets and helmets for QRTs and riot control equipments including space at nominated stations. One QRT Team looks after 5-6 metro stations as per present arrangement. One QRT consist of 5 personnel and perform duty in three shifts.

13. Furniture to security agency for each security room, and checking point at every entry point at stations. Scale is one office table with three chairs for security room and office of GO and one steel top table with two chairs for checking point.

14. Ladies frisking booth - 1 per security check point (AFC Arrey)
    Wooden Ramp - 1 per DFMD for security check points.

15. Wall mounted/ pedestal fan at security check point, ladies frisking booth and bullet proof morcha, as per requirement.

16. Physical barriers for anti scaling at Ramp area, low height of via duct by providing iron grill of appropriate height & design/concertina wire.

17. Adequate number of ropes. Queue managers, cordoning tapes, dragon search lights for contingency.

18. Iron grill at station entrance staircases, proper segregation of paid and unpaid by providing appropriate design grills etc.

19. Proper design of emergency staircase and Fireman entry to prevent unauthorized entry.

CHAPTER - 16

DISASTER MANAGEMENT MEASURES

16.1 INTRODUCTION

"Disaster is a crisis that results in massive damage to life and property, uproots the physical and psychological fabric of the affected communities and outstrips the capacity of the local community to cope with the situation." Disasters are those situations which cause acute distress to passengers, employees and outsiders and may even be caused by external factors. As per the disaster management act, 2005 *"disaster" means a catastrophe, mishap, calamity or grave occurrence in any area, arising from natural or manmade causes, or by accident or negligence which results in substantial loss of life or human suffering or damage to, and destruction of, property, or damage to, or degradation of, environment, and is of such a nature or magnitude as to be beyond the coping capacity of the community of the affected area".* As per World Health Organisation (WHO):

"Any occurrence that causes damage, economic disruption, loss of human life and deterioration of health and services on a scale sufficient to warrant an extra ordinary response from outside the affected community or area."

A disaster is a tragic event, be it natural or manmade, which brings sudden and immense agony to humanity and disrupts normal life. It causes large scale human suffering due to loss of life, loss of livelihood, damages to property and persons and also brings untold hardships. It may also cause destruction to infrastructure, buildings, communication channels essential services, etc.

16.2 NEED FOR DISASTER MANAGEMENT MEASURES

The effect of any disaster spread over in operational area of Surat Metro is likely to be substantial as Surat Metro is expected to deal with thousands of passengers daily in underground tunnels, viaducts and stations. Disaster brings about sudden and immense misery to humanity and disrupts normal human life in its established social and economic patterns. It has the potential to cause large

scale human suffering due to loss of life, loss of livelihood, damage to property, injury and hardship. It may also cause destruction or damage to infrastructure, buildings and communication channels of Metro. Therefore there is an urgent need to provide for an efficient disaster management plan.

16.3 OBJECTIVES

The main objectives of this Disaster Management Measures are as follows:

Save life and alleviate suffering.

Provide help to stranded passengers and arrange their prompt evacuation.

Instill a sense of security amongst all concerned by providing accurate information.

Protect Metro Rail property.

Expedite restoration of train operation.

Lay down the actions required to be taken by staff in the event of a disaster in Surat Metro in order to ensure handling of crisis situation in coordinated manner.

To ensure that all officials who are responsible to deal with the situation are thoroughly conversant with their duties and responsibilities in advance. It is important that these officials and workers are adequately trained in anticipation to avoid any kind of confusion and chaos at the time of the actual situation and to enable them to discharge their responsibilities with alertness and promptness.

16.4 LIST OF SERIOUS INCIDENTS REQUIRING USE OF PROVISIONS OF THE DISASTER MANAGEMENT MEASURES

Metro specific disasters can be classified into two broad categories e.g.: Man-made and Natural.

a. Man Made Disaster

1. Terrorist attack
2. Bomb threat/ Bomb blast
3. Hostage
4. Release of Chemical or biological gas in trains, stations or tunnels
5. Fire in metro buildings, underground/ elevated infrastructures, power stations, train depots etc.

6. Train accident and train collision/derailment of a passenger carrying train
7. Sabotage
8. Stampede

b. Natural Disaster

1. Earthquakes
2. Floods

16.5 PROVISIONS UNDER DISASTER MANAGEMENT ACT, 2005

A. The National Disaster Management Authority (NDMA)

Establishment of National Disaster Management Authority:-

(1) With effect from such date as the Central Government may, by notification in the Official Gazette appoint in this behalf, there shall be established for the purposes of this Act *(The Disaster Management Act, 2005)*, an authority to be known as the National Disaster Management Authority.

(2) The National Authority shall consist of the Chairperson and such number of other members, not exceeding nine, as may be prescribed by the Central Government and, unless the rules otherwise provide, the National Authority shall consist of the following:-

(a) The Prime Minister of India, who shall be the Chairperson of the National Authority, ex officio;
(b) Other members, not exceeding nine, to be nominated by the Chairperson of the National Authority.

(3) The Chairperson of the National Authority may designate one of the members nominated under clause (b) of sub-section (2) to be the Vice-Chairperson of the National Authority.

(4) The term of office and conditions of service of members of the National Authority shall be such as may be prescribed.

B. State Disaster Management Authority:

Establishment of State Disaster Management Authority:-

(1) Every State Government shall, as soon as may be after the issue of the notification under sub-section (1) of section 3, by notification in the Official Gazette, establish a State Disaster Management Authority for the State with such name as may be specified in the notification of the State Government.

(2) A State Authority shall consist of the Chairperson and such number of other members, not exceeding nine, as may be prescribed by the State Government and, unless the rules otherwise provide, the State Authority shall consist of the following members, namely:-

(a) The Chief Minister of the State, who shall be Chairperson, ex officio.

(b) Other members, not exceeding eight, to be nominated by the Chairperson of the State Authority;

(c) The Chairperson of the State Executive Committee, ex officio.

(3) The Chairperson of the State Authority may designate one of the members nominated under clause (b) of sub-section (2) to be the Vice-Chairperson of the State Authority.

(4) The Chairperson of the State Executive Committee shall be the Chief Executive Officer of the State Authority, ex officio: Provided that in the case of a Union territory having Legislative Assembly, except the Union territory of Delhi, the Chief Minister shall be the Chairperson of the Authority established under this section and in case of other Union territories, the Lieutenant Governor or the Administrator shall be the Chairperson of that Authority: Provided further that the Lieutenant Governor of the Union territory of Delhi shall be the Chairperson and the Chief Minister thereof shall be the Vice-Chairperson of the State Authority.

(5) The term of office and conditions of service of members of the State Authority shall be such as may be prescribed.

C. Command & Control at the National, State & District Level

The mechanism to deal with natural as well as manmade crisis already exists and that it has a four tier structure as stated below:-

(1) National Crisis Management Committee (NCMC) under the chairmanship of Cabinet Secretary

(2) Crisis Management Group (CMG) under the chairmanship of Union Home Secretary.

(3) State Level Committee under the chairmanship of Chief Secretary.

(4) District Level Committee under the Chairmanship of District Magistrate.

All agencies of the Government at the National, State and district levels will function in accordance with the guidelines and directions given by these committees.

D. Plans by Different Authorities at District Level and their Implementation

Every office of the Government of India and of the State Government at the district level and the local authorities shall, subject to the supervision of the District Authority:-

(a) Prepare a disaster management plan setting out the following, namely:-

(i) Provisions for prevention and mitigation measures as provided for in the District Plan and as is assigned to the department or agency concerned;

(ii) Provisions for taking measures relating to capacity-building and preparedness as laid down in the District Plan;

(iii) The response plans and procedures, in the event of, any threatening disaster situation or disaster;

(b) Coordinate the preparation and the implementation of its plan with those of the other organizations at the district level including local authority, communities and other stakeholders;

(c) Regularly review and update the plan; and

(d) Submit a copy of its disaster management plan, and of any amendment thereto, to the District Authority.

16.6 PROVISIONS AT METRO STATIONS/OTHER INSTALLATIONS

To prevent emergency situations and to handle effectively in case 'one arises' there needs to be following provisions for an effective system which can timely detect the threats and help suppress the same.

(A) FIRE DETECTION AND SUPPRESSION SYSTEM
(B) SMOKE MANAGEMENT
(C) ENVIRONMENTAL CONTROL SYSTEM (ECS)
(D) TUNNEL VENTILATION SYSTEM
(E) TRACK-WAY EXHAUST SYSTEM (TES)
(F) STATION POWER SUPPLY SYSTEM
(G) DG Sets & UPS
(H) LIGHTING SYSTEM
(I) STATION AREA LIGHTS
(J) TUNNEL LIGHTING
(K) TUNNEL LIGHTING CONTROL FROM BMS
(L) SEEPAGE SYSTEM
(M) WATER SUPPLY AND DRAINAGE SYSTEM
(N) SEWAGE SYSTEM
(O) ANY OTHER SYSTEM DEEMED NECESSARY

The above list is suggestive not exhaustive actual provisioning has to be done based on site conditions and other external and internal factors.

16.7 PREPAREDNESS FOR DISASTER MANAGEMENT

Being a technological complex system worked by new set of staff, with a learning curve to improve and stabilize with time, intensive mock drills for the staff concerned is very essential to train them to become fully conversant with the action required to be taken while handling emergencies.

They also need to be trained in appropriate communication skills while addressing passengers during incident management to assure them about their well being seeking their cooperation.

Since learning can only be perfected by 'doing' the following Mock Drills are considered essential:

a. Fire Drill
b. Rescue of a disabled train
c. Detrainment of passengers between stations
d. Passenger evacuation from station
e. Drill for use of rescue & relief train
f. Hot line telephone communication with state disaster management authority.

CHAPTER– 17

MULTI MODAL TRAFFIC INTEGRATION AT METRO STATIONS

17.1 BACKGROUND

The proposed metro in Surat has two corridors totaling 40.35 kms, consisting of 38 stations. Its two corridors Sarthana to Dream City and Bhesan to Saroli traverses across the city in the east-west and south west-north east direction respectively. Along its path, the corridors cover many important origins and destinations. However, its reach will not be enough to reach all origins and destinations in the city. Fortunately, Surat has planned for a multi-modal system, where the proposed metro will be complemented by the BRTS and Surat city bus services currently under implementation. The need is now to ensure that people have safe, comfortable and secure access to the above modes as well as seamless integration facilities.

17.2 PUBLIC TRANSPORT SYSTEM IN SURAT

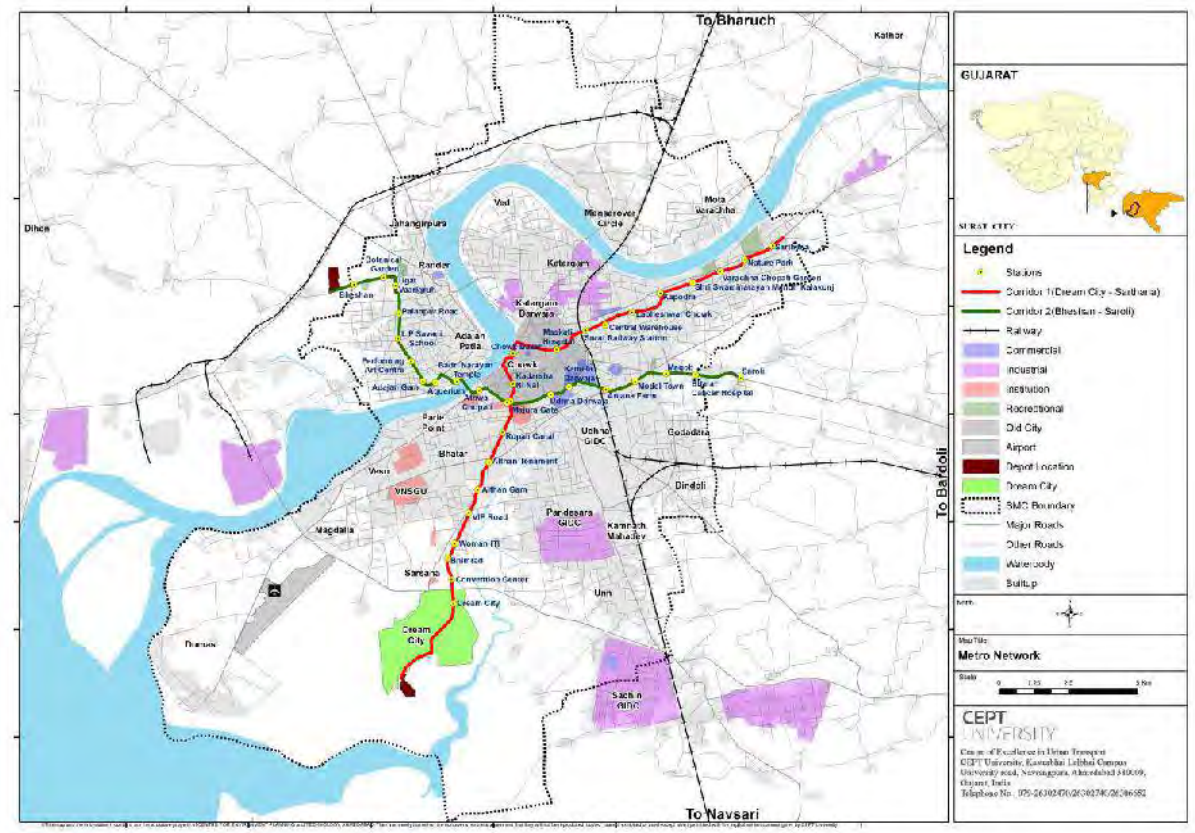
In terms of public transport services in the city, Surat has had a skeletal PT system for over a decade, leading to proliferation of auto-rickshaws operating as a shared mode on all major road corridors. The private bus services operating since 2007 have had poor service levels resulting in low ridership of around 50,000 passengers per day.

Realising the need for a quality public transport system, SMC is now trying to put in place multimodal PT system comprising of city bus services, BRTS and metro. The city introduced a Bus Rapid Transit System in the year 2014. Today, a total of 99 kms BRTS network is operational with a ridership of 67000 passengers per day. The average speed of BRTS buses is 24 kmph which is quite high as compared to existing city bus services and auto rickshaws.

The city is also in the process of instituting a new city bus services with a target of adding around 575 buses till the end of 2018. It started the new city bus services on 25 November 2016 and is currently operating 154 buses over 20 routes. Service operations are planned on 29 more routes which would be added in due course.

While planning of city services, care has been taken to take into consideration BRTS network and services. The bus services have thus been planned as a feeder and complementary service to the BRTS. Overlaps between bus routes and BRTS have been kept to a minimum for avoiding any inter-modal competition.

17.3 PLANNING FOR INTEGRATION OF PUBLIC TRANSPORT MODES



Map17.1: Proposed Metro corridor

The map above shows the metro corridors proposed in Surat. Along with the BRTS and city services, it is expected that the mode share of public transit in Surat will increase from the present 1% to 33% by 2046. However, there is a need to ensure that these three systems are integrated in such a manner that the commuters get benefit of single trip through fare integration as well enjoys a seamless physical transition from one mode to another. Along with this, there is a need to ensure that last mile connectivity through pedestrian and bicycle facilities are enhanced.

17.3.1 Last Mile Connectivity

Last mile connectivity actually impacts on overall ridership and usage of any mass transit system. It is essential to provide or improve last mile connectivity

and link them for ease of access for users. Even if street infrastructure that comprises of last mile connectivity falls outside Metro or BRT's jurisdiction and control, they remain critical components of an effective public transportation. The easier it is to access the Metro or BRT, the more likely people are to use it. It has been seen that access beyond a distance of 400-500m is not comfortable enough to attract people onto the public transport system. The unavailability of this type of service is one of the main constraints to the use of public transport in urban areas.

Last Mile Connectivity refers to the provision of travel service from home or workplace to the nearest public transportation mode. A trip is considered as the entire journey between origin and destination. Commuters may utilize and combine different modes of transport for the entire trip. Metro or BRTS may cater to a majority of this kind of trip, but commuters always need to complete the access and egress part on their own.

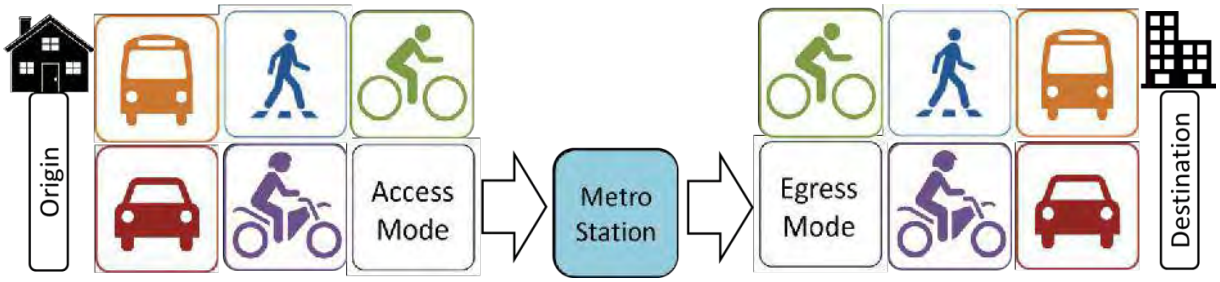


Figure 17.1: Components of trip

17.3.2 Physical Integration

Integration of NMT, para transit, feeder services and other public transport modes with metro stations is possible by planning and designing seamless interchange stations which facilitates physical integration of different access or egress modes.

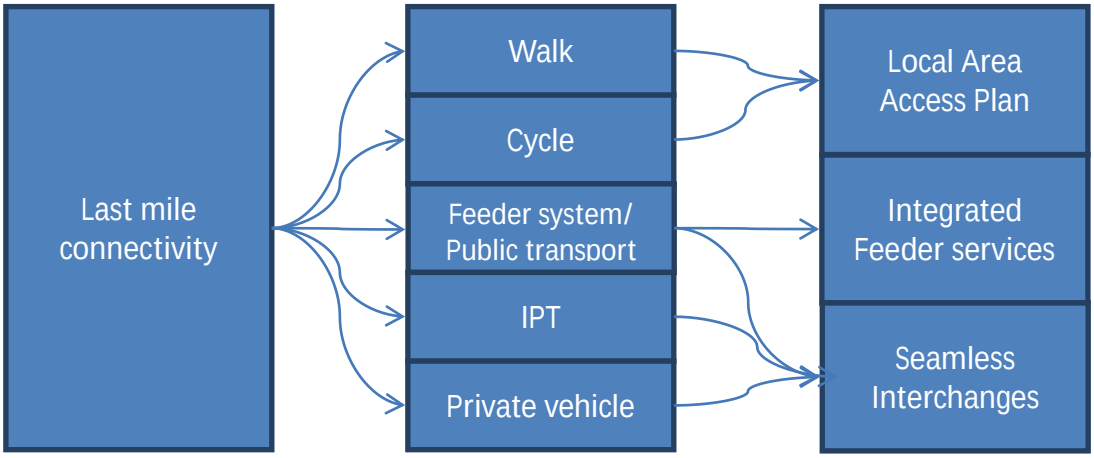


Figure17.2: Different aspects of Last Mile Connectivity

In order to develop an integrated public transport system, where one mode acts as a complementary mode to another mode rather than its competitor, a trunk and feeder system of transportation network is planned to be developed. Interchange points are required to be seamless and barrier- free movement of passengers. Thus, for the city of Surat, an effort has been made to thus identify locations and determine various typologies of interchanges that need to be developed for a seamless, barrier- free transportation experience.

17.3.3 Interchange stations

Inter change stations permit riders from particular mode of transport to move and board same or different mode in order to reach their desired destination. Interchanges can be of several types, depending on available modes of transport. Transfer may happen between the same modes or among different available public transport. Interchanges bring public transport together. For example metro station can have interchanges with metro on particular corridor connecting metro station of different corridor or, Metro stations may have interchange with BRTS, city services and even rail. Interchange helps to integrate various means of transport system under a common node.

Inter changes increases flexibility of movement and gives commuter option to avail multiple public transports at a common station. BRTS is the high capacity public transport system while metro and rail are considered to have even higher capacity than BRTS.

17.3.4 Level of Interchange

Interchanges can be of different levels, depending on the types of interchange. The four modes of transport mentioned above are under four different authorities. Surat Metro is a Special Purpose Vehicle (SPV) under Metro link Express for Gandhinagar and Ahmedabad (MEGA), BRTS and city bus services are under Surat Municipal Corporation and Railways comes under Indian Railways. Design aspect of interchange needs prior planning for seamless mobility and transfer from one mode to other. The multiple authorities or institutions involved needs to take necessary decisions for efficient functioning.

Level 1: Level I interchange refers to the first type of interchange, which is regional-city level interchange. In this case there are interchange facilities provided for transfer between regional transport nodes like railway station and high capacity transit mode like metro/BRTS. In case if Surat, railway station is one such node which is being developed as a multi-modal hub, where transfer facilities between regional rail, regional buses, metro, BRT and city buses would be developed. Sarthana is also a level 1 node providing interchanging opportunity between metro, BRTS and regional buses (GSRTC).

Level 2: Level 2 interchange in Surat refers to metro-metro interchange.

Level 3: Level 3 interchanges are those where metro has to be integrated with the BRTS system.

Level 4: Level 4 interchanges are those where metro has to be integrated with the open system of city bus services. In this level of interchange, city bus services will function more as a feeder to the Metro corridor.

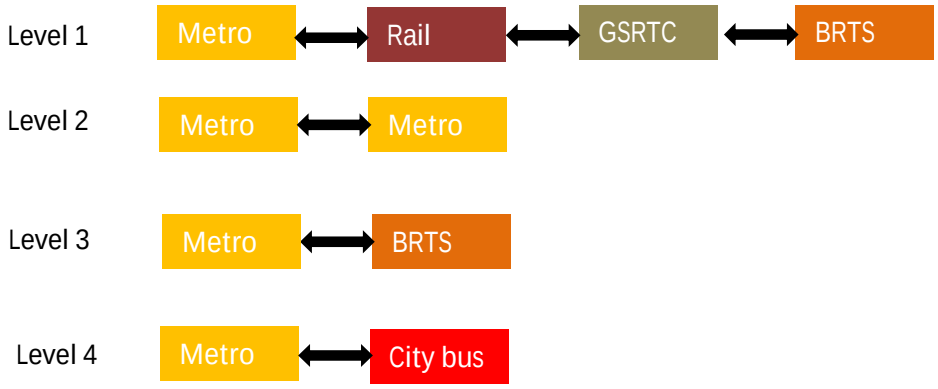


Figure 17. 1 Different levels of interchanges in Surat

Details of interchange and map showing the level of interchanges on metro corridor are shown below:

Table 17. 1 Levels of interchange for metro stations

Interchange Level	Interchange Station	Metro Corridor	Rail	Regional Bus	Metro	BRTS	Bus
Level 1 (Regional Node-Metro)	Railway station	Dream City to Sarthana					
	Sarthana	Dream City to Sarthana					
Level 2 (Metro-Metro)	Majura Gate	Dream City to Sarthana&Bhesan-Saroli					
Level 3 (Metro-BRT)	VIP Road	Dream City to Sarthana					
	Rupali Canal						
	Kapodra						
	Aquarium	Bhesan to Saroli					
	UdhanaDarwaja						
	Model Town						
	Magob						

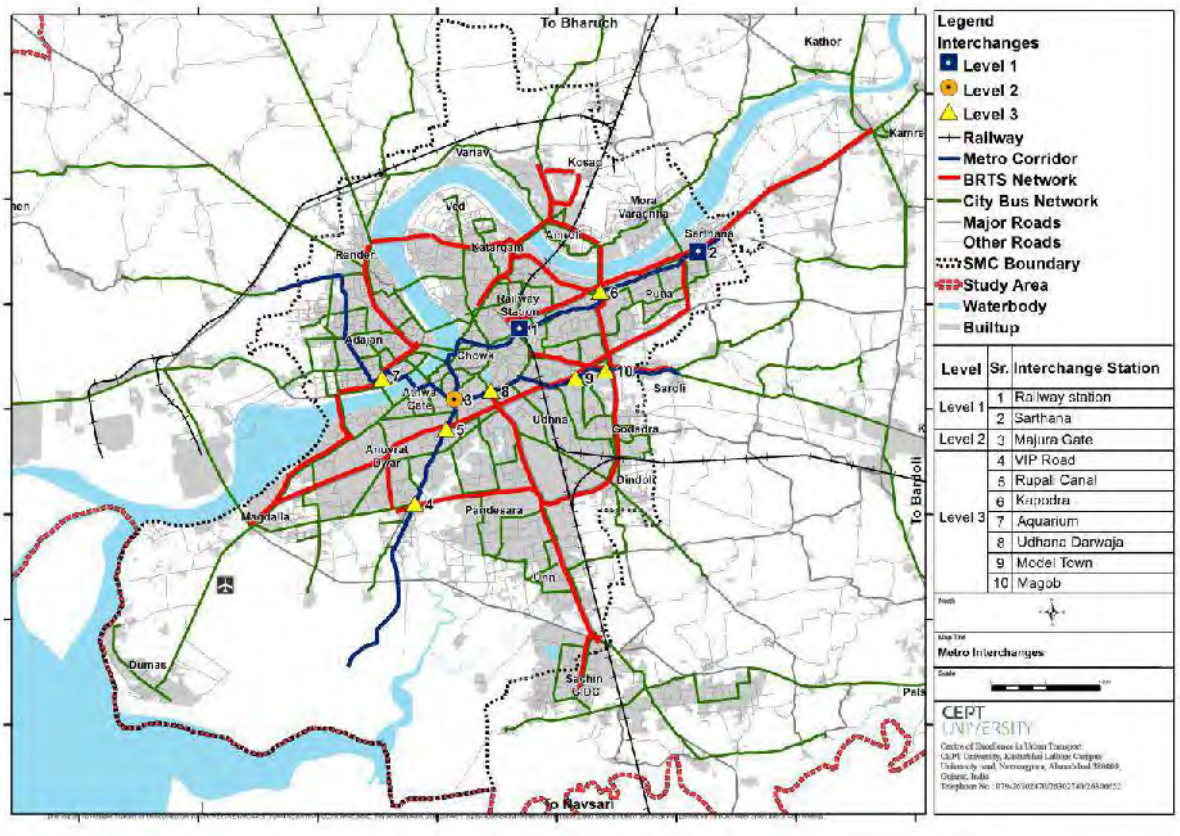


Figure17. 2 Different levels of metro interchanges in Surat

17.3.5 Fare integration

Fare integration involves ensuring that the public transport user pays for the entire journey once rather than paying separately for different journey legs. This helps in reducing travel time from origin to destination. The single fare system attracts passengers to use buses and MRTS because of the comfort and also saving of time in buying separate tickets for travel at bus/BRT Stops and MRTS stations. Such a single fare system can be implemented either through pre-paid travel cards or smart cards which help in reducing and eliminating the financial penalty imposed by inter change enabling users to select the most appropriate route through the network. In Surat, an integrated fare structure for the multi-modal PT system of metro-BRTS and city bus services is also being planned.

17.4 WAY FORWARD

In view of above deliberations in back ground, along with planning for Metro Rail System in any city, there is a need for providing a transportation system which is seamlessly integrated across all modes and provides first mile as well as last mile connectivity. It is also necessary that various public transportation modes including Inter-mediate Public Transport (IPT) and feeder buses etc. work together in order to facilitate increase in ridership to the Metro system and provide ease of using the Metro system by the public at large.

The basic level of connectivity for any mass transit system would be linking local areas to the nearest mass transit station. The concept of “Local Area Access Plan (LAAP)” can be proposed which will help identify the road network along the Metro corridor that should be taken up for pedestrian and NMT improvement. It is important to look at each metro station and the surrounding area in detail. The land use for a distance of 500m from the corridor should be surveyed to understand nature of existing development. All prominent activities such as schools, colleges, hospitals, malls etc. should be mapped to ensure that the LAAP caters to all of them. At the same time, for each station, feeder services in the form of BRTS and bus service can be identified.

CHAPTER 18

TRANSIT ORIENTED DEVELOPMENT (TOD) PLAN AND VALUE CAPTURE FINANCE (VCF)

18.1 BACKGROUND

` India is urbanizing at a rapid pace with urban population rising much faster than its total population. Level of urbanisation has increased from 17.29% in 1951 to 31.6 % in 2011. India is competing with the fastest growing countries in the world. The urban population in India, which is nearly 377 million is poised to grow to 600 million by 2030. The urban population of India contributes 65% of country's Gross Domestic Product (GDP), which is expected to grow to 75% in the next 15 years. With India witnessing a high economic growth, Indian cities are growing at a rate faster than other cities in the world.

Urbanization has led to horizontal growth of the cities thus creating problems of urban sprawl. This has resulted in increase of trip lengths and higher usage of private vehicles, problems of pollution and increased demand of infrastructure. To address these issues, many cities have strengthened their public transport by developing mass rapid transit systems (MRTS) such as metro rails and Bus Rapid Transit Systems (BRTS). It is however, important to efficiently use these systems by integrating the land use with the transport infrastructure to make the cities livable, healthy and smart.

New Metro Rail Policy 2017 emphasizes that "Transit Oriented Development (TOD)" with proposed intermodal integration, universal accessibility, adequate walkways and pathways for Non-Motorized Transport (NMT), stations for public bike sharing, commensurate parking lots for cycles and personal vehicles, as well as adequate arrangement for receiving and dispatch of feeder buses at all metro stations. The commitment by the State Government to adhere the guidelines issued by the central government w.r.t. TOD and adoption of VCF framework should be an integral part of the project proposal. The commitment should inter alia include commitment of transfer of the financial benefits accruing in the influence zone of the metro alignment on account of the TOD policies and VCF framework directly to the Special Purpose Vehicle (SPV)/agency

implementing the metro rail project. The project report should specify the proposed quantum of such benefits being transferred to the project. This requirement would form a mandatory part of all metro rail project proposals.

Commercial/Property development at stations and on other urban land has been used as a key instrument for maximizing revenues in metro rail/railway system in cities around the world. Notable examples are Hong Kong and Tokyo Metro rail implementing agencies should endeavour to maximize revenue through commercial development at stations and on land allocated for this purpose.

The McKinsey report has estimated that around Rs. 325,000 crore of urban infrastructure investments are required annually. The High Powered Expert Committee Report 2011 projects urban infrastructure requirement at 0.75%, which will increase to 1.5% of the GDP by 2032 (Rs. 97,500 crore to Rs. 195,000 crore annually). Presently, national Urban Missions are investing about Rs. 32,500 crore annually leading to an investment gap of nearly Rs. 65,000 crore.

Ordinarily, own sources of revenue in ULBs can be classified into three categories,

(a)Taxes levied by the municipality, (b) user charges levied for provision of civic services, and (c) fees and fines levied for performance of regulatory and other statutory functions. Octroi, which was one of the main sources of own income of the ULBs has been abolished, resulting in a serious dent on ULBs" resources. On the other hand, property tax, which is at present the main source of own resources is underused and has issues related to its narrow tax base, exemptions, etc. Furthermore, the State Governments are increasingly fixing the rate for services being provided by ULBs, even though these functions are mandated to be performed by ULBs under the 74th Constitutional Amendment. Overall, this has led to increased dependency on State Governments and reduction in efforts made by ULBs to mobilize resources.

Land is the most fundamental asset that is owned and managed by the States/ULBs and is a resource to generate revenues. Traditionally, States/ULBs have relied on direct sale of lands to raise funds, which is a less efficient form of resource mobilization, as compared to value capture. It is not that States/ULBs have not used Value Capture methods to raise resources. In fact, States/ ULBs are using different Value Capture methods, especially in urban areas, such as impact fee, betterment charges, etc. For example, the Mumbai Metropolitan Region Development Authority (MMRDA) and City and Industrial Development Corporation Limited (CIDCO) of Maharashtra have used different Value Capture

methods to finance infrastructure development in the urbanizing areas. Similarly, Haryana and Gujarat have successfully used land pooling schemes, where owners agree to exchange their lands for infrastructure services.

While States/ULBs have been developing and using some of the Value Capture Finance (VCF) methods, the Central Government Ministries/Departments have not yet systematically used VCF methods as a revenue generation tool. One reason is that land is a State subject and VCF Policies have to be made by the concerned State Governments. A promising way is to link the location and construction of the projects by the Central Government Ministries and their agencies with the existing VCF Policy of the generated within the area of influence of the projects. Alternatively, the State VCF Policy could be revised whenever new projects are being planned in order to capture full value being generated due to proposed investment in projects.

There is an increasing focus on creation of infrastructure by Ministries/Departments of Government of India and their agencies. For example, the Ministry of Ports is constructing a series of projects as part of the Sagarmala program. Moreover, the Delhi-Mumbai Industrial Corridor (DMIC) is being developed by the Department of Industrial Policy and Promotion (DIPP) and the Metro Rail projects by the Ministry of Urban Development (MoUD). All these projects have an area of influence in which they lead to increase in value of lands and buildings, creating opportunities for using value capture methods to mop up additional resources.

18.2 OBJECTIVES OF TOD

TOD integrates land use and transport planning to develop compact growth centers within the influence zone of 500-800 m on either side of the transit stations i.e. areas within walking distance, to achieve the following objectives:

To promote the use of public transport by developing high density zones in the influence area, which would increase the share of transit and walk trips made by the residents/ workers to meet the daily needs and also result in reduction in pollution and congestion in the influence area.

To provide all the basic needs of work/ job, shopping, public amenities, entertainment in the influence zone with mixed land-use development which would reduce the need for travel.

To establish a dense road network within the development area for safe and easy movement and connectivity of NMT and pedestrians between various uses as well as to transit stations.

To achieve reduction in the private vehicle ownership, traffic and associated parking demand.

To develop inclusive habitat in the influence area so that the people dependent on public transport can live in the livable communities within the walkable distance of transit stations.

To integrate the Economically Weaker Sections (EWS) and affordable housing in the influence zone by allocating a prescribed proportion of built-up area for them in the total housing supply.

To provide all kinds of recreational/entertainment/ open spaces, required for a good quality of life in the influence area.

To ensure development of safe society with special attention to safety of women, children, senior citizen and differently abled by making necessary amendments to the building bye laws.

To prevent urban sprawl by accommodating the growing population in a compact area with access to the transit corridor, which would also consolidate investments and bring down the infrastructure cost for development.

To reduce carbon footprints by shifting towards environmentally friendly travel options for the line haul as well as for access and egress trips.

18.3 PRINCIPLES OF TOD

TOD focuses on compact mixed use development around transit corridor such as metro rail, BRTS etc. International examples have demonstrated that though transit system facilitates transit oriented development, improving accessibility and creating walk able communities is equally important. Based on the objectives of National Urban Transport Policy, this TOD policy defines 12 Guiding Principles and 9 Supportive tools, as shown in Figure 18.1 and 18.2, for realizing the objectives of TOD.

Figure 18.1: TOD Principles

18.4 APPROACH FOR TOD IMPLEMENTATION

Influence Zone

The area in the immediate vicinity of the transit station, i.e. within a walking distance, having high density compact development with mixed land use to support all basic needs of the residents is called the influence zone of a transit station/ corridor.

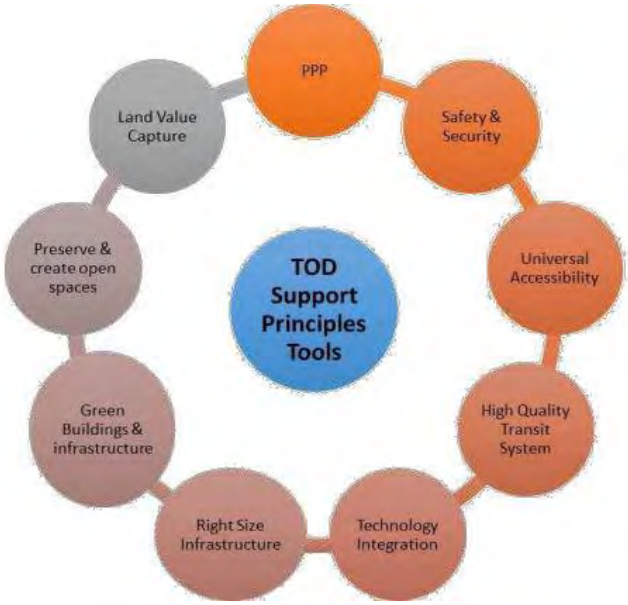


Figure 18.2: TOD Support Principles Tools

Figure 18.3: TOD Influence Zone

Influence zone is either established at a transit stations or along the transit corridors. It is generally up to a radius of nearly 500-800m of the transit station. Where the distance between the transit stations is less than 1 km and there is overlap in the influence area, it can be identified as a delineated zone (around 500m) on either side of the transit corridor within 10 - 12 minutes walking distance.

The area of influence, where the TOD is planned for implementation, should be demarcated and notified through master plan and local area plans before implementation. If in any case the TOD is to be implemented in a phased manner, the influence area of the TOD can also be notified in phases. The principles for delineating the influence area should be clearly indicated so that there is no speculation or confusion regarding the influence zone.

High Density Compact Development

TOD promotes densification in the influence area by providing higher Floor Area Ratio (FAR)/ Floor Space Index (FSI) and higher population & job density as compared to the area around and beyond the influence areas. To ensure

sustainable development, the minimum FAR should be 300 - 500%, and can be higher, depending on the city size. This will promote higher concentration of people within the walking distances of transit station, thereby increasing the ridership of the public transport and resulting in increased fare revenue, pollution and congestion reduction.

It is not necessary to keep the density and FAR norms consistent for the influence areas across the city. It can vary depending on the infrastructure available, land use zoning, transit capacity etc.

City willfollow green building norms, adopt renewal sources of energy such as solar and waste to energy options, adopt rain water harvesting and ground water recharge techniques, which would encourage water conservation, utilization of clean energy and promote sustainable waste management so as to make them self-sustaining through efficient use of resources and infrastructure.

Mixed Use Development

Mixed land use should be stipulated for development/ redevelopment in the TOD zone as it would reduce the need for travel by providing most of the activities such as shopping, entertainment and public amenities such as schools, parks, playgrounds, hospitals etc. within the walking distance of the residents. It would also improve the accessibility of the transit facilities and at the same time link origins and destinations, i.e. residences with work places or activity nodes. This would ensure better utilization of transit fleet by distributing loads in both directions, rather than creating unidirectional peak hour flows.

A blend of land-uses help in the optimization of physical infrastructure and resources, as all components like roads, parking, water, sewerage etc., remain functional at all times of the day.

The TOD benefits cannot be realized with the kind of developments that encourage the use of personalized vehicles. It is therefore imperative to restrict developments such as low-density housing, low-rise development, warehouses, petrol pumps/CNG stations, cremation ground and surface/Multilevel parking etc. in the influence area.

Mix of uses within the TOD can be achieved either by horizontal mixing i.e. separate activities in separate plots/ buildings or vertical mixing i.e. combining different activities within the same building.

To ascertain mixed use development along with the required street network and open spaces, the minimum plot area in the influence zone should be defined. The developer may, however, be permitted to undertake construction in a phased manner. In case, the individual landowners want to collaborate for development as per TOD norms, necessary provisions may be made to facilitate it. The landowner(s) may also be permitted to collaborate with developers in case they lack the required experience and institutional & financial capacity to undertake such development as per TOD norms. However, care needs to be taken that the amalgamated plots are redesigned to allow finer network of streets and dispersion of open spaces.

The mix of uses to be proposed shall be decided as per the local conditions and the trends in real estate market, however, the minimum percentage of built up area for housing, commercial and other amenities should be fixed. The use of balance built up area may depend on the prevailing market conditions and demand of the city.

(i) Value Capture Methods

A comparative study on land based financing tools being used in India and the world shows a large number of diverse VCF tools being used. The main types of VCF methods are given below.

Land value tax – is considered the most ideal Value Capture tool which apart from capturing any value increment, helps stabilize property prices, discourage speculative investments and is considered to be most efficient among all Value Capture methods. Maharashtra and Tamil Nadu, through State laws, have expanded the scope of this mechanism to cover urban land also. Globally, land value tax is widely used in Denmark, Australia, and New Zealand.

Fees for changing land use (agricultural to non-agricultural) – land revenue codes provide for procedures to obtain permission for conversion of land use from agricultural to non-agricultural use.

Betterment levy – one-time upfront charge on the land value gain caused by public infrastructure investment. This occurs in two forms – revenue source for improvement schemes and for specific projects. In India, the Mumbai Metropolitan Regional Development Authority (MMRDA) Act, 1974 provides for levying betterment charges for specific projects. The Hyderabad Municipal Corporation Act, 1955 originally provided for the levy of betterment charges to meet the costs of internal infrastructure and services in the case of development

projects. In the late nineties, the Government of Andhra Pradesh amended the Act to enhance the scope of such levy to include external betterment. Under this concept, the municipal authority is empowered to collect external betterment charges at the time of according approval to layouts or sub-divisions of plot or issue of building permit for the laying of trunk water lines, development of freeways/major roads, percent of the land value gain attributable to public investment.

Development charges (Impact fees) are area-based and link the development charge to the market value of land by carrying out periodic revisions. This is the most widely used land based fiscal tool in States. States like Andhra Pradesh, Gujarat, Maharashtra, Tamil Nadu and Madhya Pradesh levy Impact Fee and collect it upfront while granting development permissions. Impact fee is widely used to fund infrastructure in the United States. The Government of Andhra Pradesh in the late nineties also permitted Hyderabad Municipal Corporation to levy Impact Fees to mitigate the impacts of construction of commercial buildings, which lead to increase in traffic and necessitate decongestion measures. This is meant to address citywide problems emanating from high-density commercial development and is expected to be utilized for the Capital Improvement and Decongestion Plan. This includes works such as road widening, link roads, slip roads, parallel roads, junction improvements including traffic signals, flyovers, rail over-bridges, rail under-bridges, modern lighting on major roads, development of major storm water drains, riverfront and parks and for Geographic Information System (GIS) applications.

Transfer of Development Rights (TDRs) – used for trading development rights. Maharashtra, Karnataka and Gujarat have enabling laws for using TDRs for developing open spaces, promoting affordable housing, etc. In New York City, TDRs are given for preservation of the property owners for loss in revenue on their properties.

Premium on relaxation of rules or additional FSI/FAR – widely used in States such as Maharashtra, Karnataka, Gujarat, Tamil Nadu, etc. to allow for additional development rights beyond the permissible limits in the State Town Planning Laws and Regulations. Sale of additional Floor Area Ratio (FAR) is an important Value Capture tool in Brazil and France. The French Land-use Policy restricts the landowner's building right to a low baseline FAR and additional FAR has to be purchased.

Vacant Land Tax (VLT) – is applicable on those landowners who have not yet initiated construction on their lands. In Andhra Pradesh, the Greater Hyderabad Municipal Corporation (GHMC) imposes a tax of 0.5% of the registration value of the land if not used exclusively for agriculture purpose or is vacant without a building.

Tax Increment Financing (TIF) – is one of the most popular Value Capture tools in many developed countries, especially the United States. In TIF, the incremental revenues from future increases in property tax or a surcharge on the existing property tax rate is ring-fenced for a defined period to finance some new investment in the designated area. Tax Increment Financing tools are especially useful to finance new investments in existing habitations. Some of the Smart City Proposals have planned for TIF in their area-based developments (ABD).

Land Acquisition and Development – acquiring and developing land could be adopted as a useful Value Capture method to mobilize resources. In Hyderabad, impact fees are levied on all new developments within a one-kilometer wide growth corridor on both sides of the Outer Ring Road (ORR). Another innovative „Road widening Scheme" is being implemented in Hyderabad in which the Municipal Corporation gives additional FAR and relaxes zoning for property owners who give land free of cost for road widening.

Land pooling System (LPS) – is a form of land procurement where all land parcels in an area are pooled, converted into a layout, infrastructure developed, and a share of the land, in proportion to original ownership, returned as reconstituted parcels. In India, States such as Gujarat and Haryana have used land assembly programs where the owners agree to exchange their barren lands for infrastructure-serviced smaller plots. Gujarat has used these tools to guide the development of Ahmedabad city and its surrounding infrastructure. The State of Andhra Pradesh has used LPS to get land for Amravati, its new Capital City. Such LPS are also a common feature in countries like Japan and Germany.

(ii) Types of Value Capture

Tools like betterment levy, development charge, etc. have been extensively used across States whereas some tools like TDRs and VLT have been used less frequently. Value capture methods can be used in an area or can be specific to a project. Area-based value capture attempts to capture the basic appreciation of the value of the area as a result of infrastructure development,

while project-based value captures the appreciation of land and building values in the area of influence of the project. The area of influence determines the geographic extent of immediate positive impact of project investments. Table 18.1 gives the different VCF methods that can be applied to the two types of interventions.

Area-based application of Value Capture is best suited for urban areas. The area could be a locality, city or a larger planning area. On the other hand, project-based value capture can be used for projects being implemented by Ministries/Departments/Agencies of the Government of India. Some examples are given below.

- Ministry of Railways for high-speed rail projects and expansion of railway network through SPVs.
- Ministry of Road Transport and Highways for the phased implementation of the Indian National Expressway Network.
- Department of Industrial Policy and Promotion for setting up of Special Economic Zones (SEZs) and industrial corridors such as the Delhi Mumbai Industrial Corridor (DMIC).
- Ministry of Power for setting up power generation plants.
- Ministry of Shipping for projects requiring significant land resources such as cargo terminals, constructions of ferry and cruise terminals, and establishment of free trade zones.

Table 18.1. Value Capture Methods and Scale of Intervention

S. No.	Value Capture Method	Frequency of Incidence	Scale of Intervention
1.	Land value tax	Annual rates based on gain in land value uniformly	Area-based
2.	Fees for changing land use	One-time at the time of giving permission for change of land use	Area/Project-based
3.	Betterment levy	One-time while applying for project development rights	Area/Project-based

4.	Development charges (Impact fees)	One-time	Area-based
5.	Transfer of Development Rights	Transaction-based	Area/Project-based
6.	Premium on relaxation of rules or additional FSI	One-time	Area (Roads, railways)/Project(Metro)
7.	Vacant land tax	Recurring	Area-based
8.	Tax increment financing	Recurring and for a fixed period	Area-based
9.	Land Acquisition and Development	One-time upfront before project initiation	Area/Project-based
10.	Land Pooling System	One-time upfront before project initiation	Area/Project-based

(iii) Application of VCF Methods to Area

In order to capitalize on the full range of VCF tools to mobilize additional resources, the States/ULBs will have to use the Scope-Coverage-Optimization Model of Value Capture. Scope refers to the type of existing and new tools which can be used in the State/ULBs. Coverage is replication of an existing tool to all parts of the State; and Optimization is related to use of scientific methods to assess, levy and collect taxes from a range of VCF tools. Below, are given the steps for States/ULBs to assess the opportunities for using VCF tools to generate additional resources.

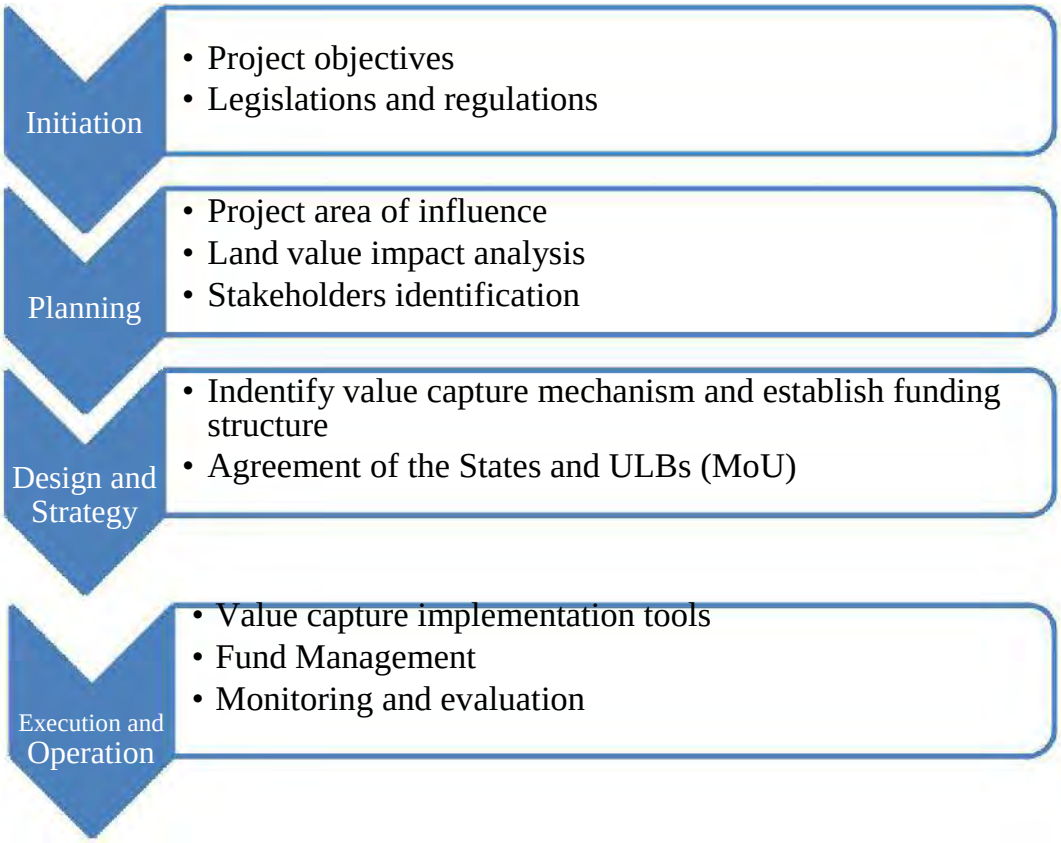


Figure 18.4: Steps Required for Project-based VCF

Scope - Different types of Value Capture tools being used in other States and countries reviewed and decided on the type which could be used in the Ahmedabad Area. Optimization - VCF methods based on an examination of the rates will be levied by State.

Coverage – Presently VCF tools are applied to small parts of the corridor and can easily be extended to other Areas. These will be identified and scaled-up.

After studying the scope, optimization possibilities and coverage of the Value Capture methods, the State will examine if existing Acts, rules, regulations and bye-laws have to be amended.

Finally, a mechanism for sharing of their venues through value capture between the States/ULBs and other entities will have to be designed and implemented.

(iv) Value Capture Financing (VCF) for TOD

Value Capture is based on the principle that private land and buildings benefit from public investments in infrastructure and policy decisions of the Government. Part of the increment in value of land and building should be captured to fund projects being set up for the public by the Central/ State government and the ULBs. The additional value is generated by actions other than the land owner"s direct investment. Land value capture is distinct from the user charges or fees that agencies collect for providing services.

The investment in transit system as well as increase in FAR and provision for mixed use development would result in increase in value of land within the influence zone. Land Value Capture can be used as a mechanism to finance the required upgradation of infrastructure and amenities within the influence zone and expansion of the public transport system.

Value capture financing is an opportunity for the private sector because the projects are planned for the overall development, thus increasing the value and are also backed by the government.

In TOD influence zones, land value capture can be done through enhanced or additional land value tax or one time betterment levy, development charges or impact fee, transfer of development rights (TDRs), or other such mechanisms which have been adopted in various states across the country and abroad.

The resources generated through various mechanism should be credited into a TOD fund created for funding the infrastructure upgradation/ maintenance, enhancement of viability of transit systems, development and maintenance of transit corridor and public transport etc. within and beyond the influence zone. The fund should be in the form of an escrow account, from which financing is provided to various agencies for the identified activities and the balance can be used by the ULBs for other development purposes such as public transport expansion.

The real-estate market can be erratic and therefore caution should be observed while estimating the revenue from land value capture. In case the revenues are lower than expectations, ULBs should prepare alternate financing plans to circumvent problems in implementing TOD.

(v) Statutory Framework

TOD policy should be notified as part of the Master Plan/Development plan of the city. The policy document should clearly outline the importance of the high capacity transit networks in the city’s development.

The vision of the Master Plan/ Development Plan should be resonated by all the stakeholders, especially those involved in infrastructure development and preparation of development plans. The building bye-laws and development control regulations would need to be amended to incorporate the changes required for implementing TOD.

The influence zone of the TOD should be clearly notified by the concerned authority to ensure that the infrastructure created in the influence zone is provided in a planned manner, the ULBs and the concerned authorities/agencies should prepare a comprehensive plan integrating all the utilities, physical infrastructure and essential facilities such as roads, sewers, drainage, electric lines, green spaces, police post, fire post, electric sub-stations, etc. The plan would be useful to assess the carrying capacity of the existing infrastructure and the upgradation needed to meet the increased demand once TOD is implemented.

18.5 TOD and Surat Metro

New Metro Policy-2017 inter-alia states that commitment by the State Government to adhere to the guidelines issued by the Central Government with regard to Transit Oriented Development(TOD) and adoption of Value Capture Finance(VCF) frame work should be an integral part of the project proposal. Further the Policy asks that the State Government commitment should inter alia include commitment of transfer of the financial benefits accruing in the influence zone of the metro alignment(s) on account with TOD policy and VCF framework directly to Special Purpose Vehicle(SPV) agency implementing the metro rail project.

As land along the proposed Metro Rail corridor is either held by the private parties or various governmental agencies, Surat Metro shall not be directly participating in the transit oriented development except at and around stations, depot and above parking to the extent permissible.

The project report is required to specify the proposed quantum of such benefits being transferred to the project. This requirement is supposed to form a mandatory part of all metro rail project proposals.

The corridor-wise total route length, underground length and elevated length for Surat Metro Rail Project(Phase-I) is shown in table 18.2 below :-

Table 18.2 – Route Lengths of Surat Metro Corridors

S.No.	Corridors	Underground (kms)	Elevated/At Grade (kms)	Total (kms)
01	Sarthana-Dream City	6.47 kms	15.14 kms	21.61 kms
02	Bhesan-Saroli	-	18.74 kms	18.74 kms
	Total	**6.47 kms**	**33.88Kms**	**40.35Kms**

18.6 Contribution of GoG towards Surat Metro Rail Project

Surat Metro Rail Project has been proposed to be developed in SPV mode with 50:50 equity contributions from Centre and State. In addition, as explained separately certain portion of Surat Metro has also been proposed to be undertaken on PPP mode.

In Surat City, most of the land proposed for the project belongs to Surat Municipal Corporation (SMC) or other Govt. agencies and some part belongs to private entities.

18.7 Potential of TOD And VCF In Surat Metro Corridors

In this connection it is clarified that the total cost of Surat Metro Rail Project (Phase-I) works out to be Rs. 9338 crores plus applicable taxes totalling to Rs 10,661 crores (inclusive of all taxes).

Once the contribution from the private sector for the proposed PPP portion amounting to Rs 396 crores (without taxes & duties) is excluded, the total cost of the project stands at Rs.8942 crore plus applicable taxes.

Thus against this total project cost of Rs. 9338 crore (plus applicable taxes), direct contributions from Transit Oriented Development (TOD) is expected to be Rs. 396 crores and as this amount is expected to directly flow to Surat Metro Project through the mechanism decided by Central Govt. and Government of Gujarat (GoG), it is only the balance amount of Rs. 8942 crores (plus applicable taxes) which is proposed to be funded by Central and State Governments by equity and other sources including Debt.

The Revenue of Surat Metro mainly consists of fare box collection, other property developments, advertisements, parkings, transit oriented developments(TOD), Value Capture Finance(VCF) etc.

Approximate revenue expected to be generated due to TOD along the Surat metro corridors of Phase-I, shall be amounting to Rs. 1343 crores and details of calculations forwarded by Surat Municipal Corporation (SMC) through their letter No.CE SP Cell/out/1199 dated 22-12-2017 have been depicted in Annexure-I. Further, it has been stated that 60% of total revenue from sale of FSI could be transferred towards MRTS project and considered for DPR purpose too. The prevailing rate of FSI chargeable is 40% of the Jantri rate. Regarding VCF, an amount equivalent to Rs. 2738 crores worked out from 5% cess on property tax and stamp duty charges has been considered and included in Chapter-20 (Financing Options, Fare Structure and Financial Viability)- Table 20.8.2. Prevailing FSI as per Surat Urban Development Authority(SUDA) is indicated in table 18.3 as under:-

Table 18.3 : Permissible FSI in Surat City (as per SUDA)

SN	Zone	Code	Permissible FSI		
			Base(Free)	Chargeable at 40% of Jantri Rate	Maximum
01	Gamtal	GM	2.0	Nil	2.0
02	Natural Growth of Village	GME	1.2	Nil	1.2
03	Residential/Commercial	R1/C2	1.8	0.9	2.7
				1.8	3.6(200 m on both sides of 36m wide road)
				2.2	4.0(for area falling within 200m on both sides of 45m wide road)
04	Industrial	IZ1 & IZ5	1.0	0.8 for commercial use	1.8
05	Agriculture	AG1	As per table 7.7.16 of CGDC regulations-2017		
06	Recreational Zone	REZ	0.25	Nil	0.25
07	Educational Purpose	KZ	1.2	Nil	1.2
08	Residential Zone	SPDS	0.6	3.4	4.0

Further, it has been assumed that Government of Gujarat (GoG) will either exempt the local taxes or reimburse the same and provide land worth Rs. 1234 crore on completion either free of cost or it shall provide interest free Sub-ordinate Debt in lieu thereof.

SUMMARY (SURAT METRO)									
SR. NO.	T.P.SCHEME NAME	PLOTS FALLING UNDER 200mt BUFFER FROM METRO LINE (NUMERS OF PLOTS)		PAID FSI AREA IN sq.mt		AMOUNT OF PREMIUME FOR PAID F.S.I			REMARKS
(1)	(2)	(3)		(4)		(5)			(6)
		"A" SOUTH-WEST ZONE (ATHWA)							
		INDUSTRIAL	OTHER THAN INDUSTRIAL (RESIDENTIAL, COMMERCIAL etc.)	INDUSTRIAL	OTHER THAN INDUSTRIAL (RESIDENTIAL,COMMERCIAL)	INDUSTRIAL	OTHER THAN INDUSTRIAL (RESIDENTIAL,COMMERCIAL)	TOTAL	
		3(a)	3(b)	4(a)	4(b)	5(a)	5(b)	5(c)	
1	T.P.S.NO. 6 (MAJURA-KHATODRA)	0	2	0	9825	0	106,112,160	106112160	
2	T.P.S.NO.9 (MAJURA)	0	1	0	5606	0	66146080	66146080	
3	T.P.S.NO.27 (BHATAR-MAJURA)	0	4	0	10386	0	83442040	83442040	
4	T.P.S.NO.28 (ALTHAN-BHATAR)	0	23	0	238203	0	142388860	1442388860	
5	T.P.S.NO.37 (ALTHAN-BHATAR)	0	13	0	109494	0	529244760	529244760	
6	T.P.S.NO.42 (BHIMRAD)	0	12	0	148975	0	633103680	633103680	
7	T.P.S.NO.43 (BHIMRAD)	0	8	0	121035	0	512787000	512787000	
							TOTAL	Rs. 337 Cr.	
		"B" WEST ZONE (RANDER)							
1	T.P.S.NO.9 (PALANPOR-BHESAN)	0	6	0	100439	0	321404160	321404160	
2	T.P.S.NO.10 (ADAJAN)	0	1	0	4730	0	24123000	24123000	
3	T.P.S.NO.13 (ADAJAN)	0	6	0	30831	0	178818640	178818640	
4	T.P.S.NO.29 (RANDER)	0	6	0	46774	0	169469080	169469080	
5	T.P.S.NO.31 (ADAJAN)	0	23	0	126511	0	1037404280	1037404280	
6	T.P.S.NO.32 (ADAJAN)	0	10	0	145446	0	1177400400	1177400400	
7	T.P.S.NO.42 (JAHAGIRABAD)	0	10	0	52481	0	201841200	201841200	
8	T.P.S.NO.43 (JAHAGIRABAD)	0	13	0	306214	0	1265708312	1265708312	
							TOTAL	Rs. 437 Cr.	
		"C" SOUTH EAST ZONE (LIMBAYAT)							
1	T.P.S.NO.7 (ANJNA)	0	1	0	12907	0	82607360	82607360	
2	T.P.S.NO.33 (DUMBHAL)	0	8	0	114169	0	409332000	409332000	
3	T.P.S.NO.53 (MAGOB-DUMBHAL)	0	5	0	54481	0	746386960	746386960	
4	T.P.S.NO.64 (DUMBHAL-MAGOB)	0	20	0	160123	0	1251435680	1251435680	
							TOTAL	Rs. 248Cr.	
		"D" EAST ZONE (VARACHHA)							
1	T.P.S.NO.16 (KAPADRA)	0	2	0	31616	0	237655220	237655220	
2	T.P.S.NO.20 (NANA VARACHHA - KAPADRA)	0	1	0	18616	0	107975120	107975120	
3	T.P.S.NO.21 (SARTHANA-SIMADA)	0	1	0	11614	0	38325540	38325540	
4	T.P.S.NO.22 (SARTHANA - VALAK)	0	1	0	24845	0	86956100	86956100	
5	T.P.S.NO.38 (NANA VARACHHA)	0	5	0	49031	0	313314980	313314980	
							TOTAL	Rs. 78Cr.	
		"E" CENTRAL ZONE							
1	T.P.S.NO. 2(NANPURA)	0	1	0	96369	0	2431577562	2431577562	
							TOTAL	Rs. 243Cr.	
							GRAND TOTAL A+B+C+D+E	Rs.1343 Cr.	

CHAPTER - 19

COST ESTIMATES

19.1 INTRODUCTION

Project Cost estimates for the Surat Metro Rail Project (Phase-I) network has been prepared covering civil, electrical, signalling and telecommunication works, rolling stock, environmental protection, rehabilitation, considering 750V DC third rail traction at December' 2018 price level, both for Capital and Operation & Maintenance costs.

While preparing cost estimates, various items have generally been grouped under three major heads on the basis of:-

(i) Route km. Length of alignment

(ii) No. of units of that item and

(iii) Item being an independent entity.

All items related with alignment, whether in underground or elevated or at-grade construction, permanent way , signaling and telecommunication, have been estimated on rate per route km/km basis. Route km. cost for underground alignment construction, excludes station lengths. Station lengths (190m) have to be done by cut and cover in general and by tunneling under compelling exceptional circumstances. The rates adopted for underground stations include cost of civil structures and architectural finishes. Similarly, cost of elevated and at grade stations includes civil work for station structures, architectural finishes, platform roofing, etc. Provisions for electrical and mechanical works, air conditioning, lifts, escalators, etc. have been worked out separately. These rates do not include cost of permanent way., power supply, signaling and telecommunication, automatic fare collection (AFC) installations, for which separate provisions have been made in the cost estimates. Similarly, for other items like Rolling stock, Traction & Power, VAC, etc. costs have been summed up separately. In remaining items, viz. land, utility diversions, rehabilitation, etc. the costs have been assessed on the basis of each item taken as an independent entity.

Initially cost estimate for Surat Metro (Phase-I) had been formulated adopting the Unit rates of various components of Ahmedabad Metro project. Escalation charges had been applied @5% per year to bring these costs to Current price level, wherever necessary in terms of MoUD's letter No.F.No.K-14011/58/2013-MRTS-1 (Vol.I) dated 27.10.2016.Taxes & Duties such as Custom Duty, CGST,SGST and IDST etc wherever applicable, had been worked out on the basis of prevailing rates and included in the cost estimates separately.

Now, on the request of MEGA Cost estimate has been revised, adopting the cost components in line with Ahmedabad Metro project (Ph-II) and Delhi Metro Phase-IV in accordance of MoUHA letter no. K-14011/5/2017-UT-II dated 23.01.2019 forwarded by MEGA vide their letter No. GMRCL/DPR Revision/JAN-19 dated 25-01-2019.

In accordance of MoHUA letter No. K-14011/2018-MRTS-III dated 15.02.2019 forwarded by MEGA through their e-mail dated 16.02.2019 contingencies have been removed on land component and cost estimate has been revised accordingly.

The overall Capital Cost for the Surat Metro rail network under Phase-I at December' 2018 price level works out to **Rs.10,661 Crores** including applicable Taxes & Duties of **1322.78 Crores** as tabulated hereunder.

Table 19.1 –Corridor-wise Details of Capital Cost

Sr. No.	Name of the corridor	Capital Cost (Rs. Crore)	Taxes & Duties (Rs. Crore)	Total (Rs. Crore)
1.	Sarthana – Dream City	5810	849.30	6659.30
2.	Bhesan–Saroli	3528	473.48	4001.48
	Total	**9338**	**1322.78**	**10660.78 =10,661**

Details and methodology of arriving at these costs are discussed in paras hereinafter.

19.1.1 PPP Model for Unbundled Components of Surat Metro

As communicated by Surat Municipal Corporation (SMC) vide their letter no. CE Spl Cell/Out/1199 Dated 22-12-2017, 04 nos stations as mentioned below

will be considered on PPP model through a developer to be engaged by metro SPV but with VGF/Land support to the PPP agencies from SMC:-

Corridor-1(Sarthana to Dream City)

(i) Rupali Canal
(ii) Althan tenement

Corridor-2(Bhesan to Saroli)

(i) Aquarium(Star Bazar)
(ii) Athwa Chaupati

However, the stations proposed above on PPP model may be changed depending upon financial feasibility.

In addition to above, it has also been conveyed that two nos elevated metro stations (including one for future requirement) inside Dream City area shall be implemented by Metro SPV with the VGF/Land support to the PPP agencies from DREAM Ltd.

For both the corridors of Surat Metro Project(Phase-I) three unbundled components namely (1)Lifts,(2) Escalators and (3) AFC gates are proposed under PPP model, in addition to civil cost of 05 nos elevated stations mentioned above. Details of cost under PPP model is given hereunder:-

Table 19.1(a) : Cost of proposed Components to be considered on PPP model

(In crores)

SN	Description	Estimated Cost*
(A)	**Corridor-1: Sarthana to Dream City**	
01	Station-Rupali Canal (civil cost)	26.00
02	Station-Althan Tenament (civil cost)	26.00
03	Station-Dream City (civil cost)	26.00
04	Lifts & Escalators (20 X 3.50)	70.00
05	Automatic Fare Collection(AFC) Gates (20 X 3.50)	70.00
	Sub-Total (a)	**218.00**
(B)	**Corridor-2: Bhesan to Saroli**	
01	Station-Aqurium (civil cost)	26.00
02	Station- Athwa Chaupati (civil cost)	26.00
03	Lifts & Escalators (18 X 3.50)	63.00

04	Automatic Fare Collection(AFC) Gates (18 X 3.50)	63.00
	Sub-Total(b)	**178.00**
	Grand Total (a)+(b)	**396.00**

*plus charges on account of applicable taxes, contingencies, GC etc

19.1.2 REQUIREMENT OF NEW METRO POLICY

Metro Rail Policy 2017 requires Metro Rail Administrations and States to have a greater focus on reducing the cost of construction and operations through indigenization and fostering competition along with standardization in various areas. This is in addition to the standardization being attempted for various components of the metro rail system at the level of Central Government.

19.1.2.1 INTRODUCTION OF METRO RAIL SYSTEMS AS MRTS IN INDIA

Before the advent of Delhi Metro Rail Project, the knowhow of modern state of the art, Metro Rail System was conspicuous by its absence. In the process of successfully constructing and operating more than 200 KM of Metro Rail in Delhi and NCR region (Phase I, II, and part phase III) DMRC has ushered a new era in standardizing construction cost, optimizing operations and maintenance cost, standardization of technology and components, indigenization and inducing competition. Suffices to say that emergence of large scale metro rail construction industry, systems suppliers and components manufacture growth is directly related to the success of upcoming metro systems in the country.

Success of DMRC has also resulted in many cities undertaking planning and construction of metro rail. In many such cities varying length of Metro Rail systems has also become operational. As in most such cases, the Detailed Project Report has been made by DMRC, there has been desired momentum in standardization of various technical aspects in varied components of Metro Rail systems across the country. In the instant proposal of Surat city also, DMRC has been engaged to prepare Detailed Project Report (DPR) for metro rail project.

19.1.2.2 FOCUS ON INDIGENIZATION

Before commencement of metro system, the country did not have where withal of constructing modern state of the art metro rail projects. Right from the world go, in Phase I DMRC has deliberately focused on large scale indigenization and that has had cascading effect on development of metro rail in other cities of the country itself. The indigenization effort of DMRC is explained in following paragraphs: -

a) Civil Construction:

Prior to emergence of metro construction, Indian EPC contractor's/ construction companies were not equipped to handle the difficult metro rail construction on large scale. Civil part of the Metro Rail comprises between 40-50% of the total project cost. In DMRC's. Phase-I project, bulk of the construction was done by the consortium of contractors led by the international majors and the Indian construction companies largely managed to get only a lesser pie as subcontractors to the international firms.

Subsequently, in further phases, the contract conditions were tweaked by DMRC to bring in domestic contracts to the fore and the bulk of the civil construction was done by the domestic firms and it has led to intense competition. It is hoped that today Indian firms in the area of Metro Rail construction are next to none in the world.

b) Metro Car Manufacturing:

Metro Car comprises 20-22 % of the total cost of Metro Rail construction cost. A modern metro car technologically requires advanced technology, manufacturing platform and vibrant components industry. Quite naturally before the advent of metro system in the year 1998, the country did not have any metro car manufacturing unit even though a small metro rail system at Kolkata was operational for quite some time.

Emergence and fast expansion of Delhi Metro Rail Project gave the fillip and provided the scaling up opportunity for indigenization of Metro Car production. Though the metro cars were imported initially but the contract conditions were such that eventually the manufacturer had to set up subsequently manufacturing facility in the country.

It is important to note that focus on indigenization of metro system resulted in setting up of first Metro Car Manufacturing unit by BEML in association with ROTEM at Bangalore. Delhi Metro's initiatives to indigenize manufacturing are also in consonance with the 'Make in India' initiative of the central government.

So far the trigger provided by DRC has resulted in setting up of three Metro Car Manufacturing centers in the country –one by BEML (Bangalore), one by Bombardier Transportation (at Savli Gujarat) and the latest by Alstom at Sri City near Chennai.

Besides factories for metro rail manufacturing so far eighteen sub systems also have been indigenized and it has resulted in a vibrant base of anciliiaries and trained human resources.

While Metro Rail contract conditions mandate that 75% of the metro cars have to be manufactured in India, sustained effort of indigenization has ensured that the 90% of the metro cars are being manufactured in India. It is worthwhile to note that the Indian Metro Car Units have started looking at markets outside India also for “Made in India” metro cars and first lot have already been exported to Australia.

Such a dedicated focus on increasing indigenization has a direct impact on optimizing and reducing the cost of metro cars.

c) Traction

Traction comprises approximately 10 Percent of the metro rail construction cost. Over past two decades metro organizations have made substantial headway in indigenization in this area and Indian contractors either on their own or in association with international firms have succeeded in the field.

d) Other Areas

Efforts of many metro organizations like Delhi Metro, Bangalore Metro, Chennai Metro etc. for indigenization in other areas like permanent way (track), E&M (particularly Lifts and Escalators) has induced competition, brought in indigenization and have also contributed to reduction of construction cost.

19.1.2.3 STANDARDIZATION

From the word “go” standardization in different areas of Metro Rail construction has been the prime focus of metro organizations. Even though initially DMRC was forced to adopt broad gauge it has since been able to standardize the gauge to Standard Gauge not only in Delhi Metro but has also been instrumental in adoption of the same for other metro rail projects too. This has provided common platform and opened new vista for cost reduction through standardization.

Similarly, DMRC has focused on standardization through optimization of design of civil works including the sub and super structure. Now large scale adoption of U-girder technology in the latest phases has been instrumental

in both reductions in cost as well as simultaneously increasing the speed of construction.

Standardization has also been introduced in station (both elevated and underground) construction for optimization of cost and aesthetic.

One area which has defied standardization and indigenization is signaling. It is primarily due to the fact that signaling after merger of most companies globally has become a oligopoly and signaling system of one OEM does not get synchronized with that of another manufacturer.

19.1.2.4 CONSTRUCTION COST REDUCTION

Cost of construction comprising broadly material component and labour component increases mainly with the lapse of time and results in substantial increase per unit increase in construction of the metro rail systems if one reckons just one element of annual inflation at the rate of 5%.

There has been second dampener as well with regard to the cost of both government and private land required for construction of Metro Rail. The cost of land in last one decade has gone up significantly and post implementation of new Land Acquisition Act.

If the unit cost of construction of Metro Rail projects is normalized for the above two factors (increase in cost due to elapse of time and impact of increasing cost of Land) the construction cost is reasonably well controlled both for elevated and underground development.

19.1.2.5 O & M COST REDUCTION

The last but not the least, requirement of Metro Rail Policy, 2017 is a sustained effort of Optimization and reduction of operations and maintenance cost.

There is a dire need to remain focused on optimizing and sustainably reducing the Operations and Maintenance cost in order to make the project economically viable. Towards the end of construction of phase II, DMRC undertook a major study to optimize and reduce the O & M cost and was able to implement most of the recommendations of the study.

As a starter DMRC was able to reduce O & M man power strength initially from 48 persons per kilometer to 40 and subsequently to 36 persons per kilometer at par with the same level of SMRT Singapore and efforts are being made to bring it further down, the same philosophy should be adopted by other metros in the country.

Apart from above many other systemic changes to asset management and maintenance to optimize the cost including introducing condition based maintenance should be adopted in place of time based maintenance.

Metro organizations should also remain focused on reducing the per unit energy consumption more particularly in the era of increasing per unit cost of energy and target to be upgraded to the highest benchmark viz. COMET etc.

19.1.2.6 CONCLUSION

The new Metro Rail Policy and Appraisal Frame work, 2017 have brought to fore the importance of sustained cost reduction in construction and operations & maintenance, have given thumps up to systemic indigenization and "Make in India" have deliberately provided for more and more standardization with sufficient leeway for innovation.

In this connection all metro organizations should continue to work proactively on their own, in association with other institutions and more particularly in active collaboration with the central government for further systemic improvement and cost reduction.

19.2 CIVIL ENGINEERING WORKS

19.2.1 Land

Land requirements have been kept to the barest minimum and worked out on area basis. Acquisition of private land has been minimised as far as possible. For underground and elevated alignment, no land is proposed to be acquired permanently, except small areas for locating entry/ exit structures, traffic integration etc. Elevated alignment is proposed to be located on the road verge and wherever, this is out side the road alignment, minimum land area about 20m wide is proposed for acquisition.

Cost of Govt. land is based on the rate presently being charged by the concerned authorities of Govt. of Gujarat (GoG)..

Private land for MRTS project shall be acquired by Surat Municipal Corporation (SMC) and compensation shall be paid as per Land Acquisition Act 1894. The average rate of private land has been adopted as per the prevailing rates in Surat City as intimated by SMC vide their e-mail dtd 27-03-2017.

Provision for cost of land required for resettlement and rehabilitation has been made in the cost estimates.

Required land shall have to be acquired in line with Resolution no JMN/1016/842A1 dated 17-05-2017 of Revenue department, Government of Gujarat for Ahmedabad Metro Rail project. In this resolution, it is stated that land required for depots, stations and associated works shall be allotted at jantri (circle) rate to MEGA. It is stated in the same resolution that land required for parking and resettlement/rehabilitation and such works shall be allotted at Rs 1 token rate to MEGA.

As decided in HPC meeting on 03-07-2017 at Gandhinagar, resolution no JMN/1016/842A1 dated 17-05-2017 passed for Ahmedabad Metro project shall be applicable for required land for Surat Metro also including land owned by SUDA and DREAM city.

In addition to the lands required permanently, some areas of land (mainly Govt.) are proposed to be taken over temporarily for construction depots. Ground rent charges @ 5% per year for a period of 4 years have been provided for in project cost estimates.

Details of the lands with their costs have been shown in corridor cost estimate.

19.2.2 Formation and Alignment

(i) **Underground section:** In the underground section work is proposed to be done by Tunnel Boring Machines, or Cut and Cover method, depending upon the site conditions. Rates adopted for cut and cover section, as well as for work to be done by T.B.M. are based on the rates adopted in Ahmedabad Metro Rail project (Ph-II) and Delhi Metro Phase-IV projects duly updated to December 2018 price level. Cost of mid section ventilation shaft wherever needed, has also been included.

(ii) **Elevated section:** A good portion of alignment is proposed with elevated viaduct and the rates adopted are based on the rates considered in Ahmedabad Metro Rail project (Ph-II) and Delhi Metro Phase-IV projects duly updated to December 2018 price level.

19.2.3 Stations

(i) **Underground Stations:** In the underground alignment, station lengths have to be done by cut and cover TBM or NATM. Rate proposed for stations (length 190 m) includes Cost of station structures, platforms, architectural finishes, etc, and provisions for electrical and mechanical works, V.A.C., Lifts and Escalators etc., have been made separately. Provisions for O.H.E., P. way, Signaling and Telecommunication, Automatic fare collection installations etc, have also been summed up in the cost estimates. Rates are based on the rates considered in Ahmedabad Metro Rail project (Ph-II) and Delhi Metro Phase-IV projects duly updated to the price level of December 2018, wherever required.

(ii) **Elevated Stations:** Rates adopted for elevated stations cover works of station structures, platforms, architectural finishes, covering, etc. Provisions for Electrical and Mechanical works have been made separately. Also provisions for Lifts and Escalators, Viaduct, P way, O.H.E., Signalling & Telecommunication works, Automatic fare collection installations etc, have been summed up in the cost estimates.

19.2.4 Permanent way

For underground and elevated alignment ballast-less track and for depot, ballasted track is proposed. Rates adopted are based on the rates considered in Ahmedabad Metro Rail project (Ph -II) and Delhi Metro Phase-IV projects duly updated to December 2018 price level.

19.3 DEPOT

Maintenance Depot have been planned as given hereunder:-

Table 19.2 - Maintenance Depot

S. No.	Corridor	Proposed Maintenance Depot
1.	Sarthana – Dream City	Dream City
2.	Bhesan–Saroli	Bhesan

19.4 UTILITY DIVERSIONS, ENVIRONMENTAL PROTECTION, MISCELLANEOUS OTHER WORKS

Provisions have been made to cover the cost of utility diversions, miscellaneous road works involved, road diversions, road signages etc. and environmental protection works on route km basis, based on the experience gained from the works done by DMRC in earlier similar projects.

19.5 REHABILITATION AND RESETTLEMENT

Provisions have been made on fair assessment basis, to cover cost of relocation of Jhuggies, shops, residential Houses on private land etc.

Provisions for barracks and security equipment for security personnels and Staff Quarters for O&M Wing have been made in the cost estimates on the basis of average cost involved per km length on the pattern of earlier projects duly updated to current price level.

19.6 TRACTION AND POWER SUPPLY

Provisions have been made to cover the cost of traction arrangements., Auxiliary sub stations, Receiving substations, service connection charges, SCADA and miscellaneous items, on route km basis separately for underground alignment, elevated and at-grade section as the requirements are different and costs are more for underground section.

Provisions towards cost of lifts, escalators for underground, elevated and at-grade stations have been made in the cost estimates. Rates provided are based on the rates considered in Ahmedabad Metro Rail project (Ph-II) and Delhi Metro Phase-IV projects duly updated to December 2018 price level. Provision for mid section shaft has been made separately.

19.7 SIGNALLING AND TELECOMMUNICATION WORKS

Rates adopted are based on the completion cost of similar works/projects done by DMRC for Delhi Metro in past. These rates include escalation during manufacturing and supply of equipment and their installation at site. Cost of Platform Screen Doors (PSD) for few stations has also been added in the respective corridors.

19.8 AUTOMATIC FARE COLLECTION

Adopted rates are based on the rates considered in Ahmedabad Metro Rail project (Ph-II) and Delhi Metro Phase-IV projects duly updated to December 2018 price level.

19.9 ROLLING STOCK

Cost per coach has been considered as INR 8.00 at December' 2018 level on the basis of cost adopted in Ahmedabad Metro Rail project (Ph-II) and Delhi Metro Phase-IV projects.

19.10 SECURITY

A lump sum provision for providing security infrastructure in the station premises has been made on running kilometre basis. Adopted rates are as per the completed projects by DMRC in past suitably escalated to current price level.

19.11 MULTIMODAL TRAFFIC INTEGRATION

A lump sum provision of Rs. 3.00 Crore per station has been made to have seamless integration of metro stations with other modes of transport. It is envisaged that in case this money is not sufficient for this purpose the deficient part of money will borne by the Urban Local Body (ULB) in whose area station is located.

19.12 GENERAL CHARGES AND CONTINGENCES

Provision @ 5% has been made towards general charges on all items, except cost of land, which also includes the charges towards Detailed Design Charges (DDC), etc. Provision for contingencies @ 3 % has been made on all items (except cost of land) including general charges.

19.13 Integration with Surat Railway Station MMTH project

In a series of meetings taken place at Surat in the office of The Commissioner, Surat Municipal Corporation(SMC), it had been intimated that Surat Railway station MMTH project shall be integrated with underground metro station, city bus/BRTS, interstate buses(GSRTC) and Indian Railways. Further, in view of Surat Railway station MMTH master plan, it has been decided to provide an underground **tunnel road** for vehicular traffic in the portion of MMTH along Lambe Hanuman road, which

shall be above the level of underground metro corridor and UG metro station with a sufficient cushion of 2 to 3 mtr from ground level. The estimated cost of proposed tunnel road shall be Rs 109.13 crores as intimated by Surat Municipal Corporation (SMC) vide their e-mail dated 2nd May' 2018.The cost of construction of this underground road shall be borne by SMC/GoG As such, the cost has not been added in the cost of Surat Metro Rail Project(Phase-I) for working out of financial indices such as FIRR,EIRR etc. Executing agency for tunnel road will be decided by Metro SPV in consultation with SMC.

19.14 CAPITAL COST ESTIMATES

The overall capital cost of all the MRTS Corridors under Phase-I of Surat Metro estimated at December' 2018 price level based on above consideration works out to Rs. **9338 Crores** without Taxes/Duties applicable Taxes and Duties have been worked out as Rs **1322.78** Crores in Table 19.3:-

Table 19.3

CORRIDOR-WISE ESTIMATED COST (Phase-I)
(At DECEMBER, 2018 Price Level)

Rs. In Crores

S.No.	Corridor	Estimated Amount	Taxes & Duties
01	Sarthana – Dream City	5810	849.30
02	Bhesan – Saroli	3528	473.48
	Total:	9338	1322.78
	Grand Total:	**Rs 10660.78 crores =Rs 10,661 Crores**	

19.14.1 Corridor-I :Sarthana – Dream City

The overall Capital Cost for this corridor estimated at December 2018 price level, based on the above considerations works out to **Rs 5810 Crores** without Taxes & Duties. Applicable Taxes & Duties have been worked out as **Rs.849.30 Crores.**

Table 19.4

Surat Metro corridor (Phase-I)						
Sarthana-Dream City (corridor-I)						
Capital Cost Estimate						
December 2018 level						
Total length = 21.61 km						
Total UG (including UG Ramp 0.465 km) = 6.47 km						
Elv (including elevated ramp) = 15.14 km						
Total Station = 20 nos, UG = 6, Elv = 14						
S. No.	Item	Unit	Rate	Qty.	Amount (Rs. in Cr.)	Remarks
				Without taxes		
1.0	**Land**					
1.1	Permanent					
a	Government	ha	15.73	32.45	510.44	
b	Private	ha	19.15	0.62	11.87	
1.2	Temporary					
a	Government	ha	3.15	18.00	56.70	
1.3	Property development(Government)	ha	15.73	3.76	59.14	
1.4	R & R incl. Hutments etc.	R.Km.	4.00	22.600	**90.40**	
	Subtotal (1)				728.56	
2.0	Alignment and Formation					
2.1	Underground section by Tunnel Boring Machine (TBM) (190m Each)	R.Km.	125.00	4.87	608.750	
2.2	Ramp Portion (Underground)	R.Km.	125.00	0.47	58.750	
2.3	Elevated section (viaduct) including in station length	R.Km.	37.00	15.14	560.18	
2.4	Depot entry connection	R.Km.	37.00	1.00	37.00	
	Subtotal (2)				1264.68	
3.0	Station Buildings					
3.1	Underground Station (Civil including finishes)	Each	125.00	6.00	750.00	
3.2	Underground Station (E&M including ECS,TVS, lift & escalator)	Each	50.00	6.00	300.00	
3.3	Extra for construction of two level station	Each	31.25	2.00	62.50	Labheshwar Chowk and Maskati Hospital stations to be constructed as two level stations

3.4	Elevated station (Civil including finishes) excluding viaduct in station portion	Each	26.00	14	364.00	
3.5	Elevated Station (E&M including lift & escalator)	Each	8.00	14	112.00	
3.6	Metro Bhawan & OCC bldg.					
a	Civil works	LS			90.00	
b	E&M works etc	LS			25.00	
3.7	Mid Shaft					
a	Mid Shaft (cost of Civil works)	Each	20.00	2.00	40.00	
b	Mid Shaft (cost of electrical works)	Each	7.43	2.00	14.86	
	Subtotal (3)				1758.36	
4.0	Depot					
4.1	Civil	Per Depot			105.00	
4.2	E&M & M&P	Per Depot			65.00	
	Subtotal (4)				170.00	
5.0	Permanent Way					
5.1	Ballastless track	R.Km.	6.60	22.60	149.16	
5.2	Ballasted track for Depot	T.Km.	3.90	10.00	39.00	
5.3	Addl. Concrete in UG (for derailment guard)	R.Km.	0.80	6.470	5.18	
5.4	Mass Spring System (MSS for 30% of UG length) to reduce Noise and Vibration effect	R.Km.	4.80	1.941	9.32	
	Subtotal (5)				202.65	
6.0	Traction & power supply					
6.1	750V DC Third Rail					
a	UG Section	R.Km.	12.50	6.470	80.88	
b	Elevated section	R.Km.	11.00	15.140	166.54	
c	RSS	Each	25.00	1.00	25.00	
d	Depot	T.Km.	3.00	10.000	30.00	
e	132 kV cabling from Grid to RSS	R.Km.	6.93	2.885	19.99	
	Subtotal (6)				322.41	
7.0	Signalling and Telecom.					
7.1	**Signalling**					
a	Main line including OCC	R.Km.	4.40	22.600	99.44	
b	Depot including DCC	T.Km.	3.20	10.00	32.00	
c	On Board Equipment	Per Train	1.70	15.00	25.50	
7.2	**Telecommunication**					
a	Station	Per Station	4.50	20.00	90.00	
b	Depot	Per Depot	3.50	1.00	3.50	
	Subtotal (7)				250.44	
8.0	Automatic Fare Collection(AFC) system	Per station	3.50	20.00	70.00	

	Subtotal (8)				70.00	
9.0	Platform Screen Doors(PSD)	Per station	3.00	20.00	60.00	
	Subtotal (9)				60.00	
10.0	Shifting of Miscellaneous Utilities					
10.1	Civil and E&M	R.Km.	6.00	22.600	135.60	
	Subtotal (10)				135.60	
11.0	Rolling Stock	Per coach	8.00	45	360.00	
	Subtotal (11)				360.00	
12.0	Security	Per station	0.37	20	7.40	
	Subtotal (12)				7.40	
13.0	**Staff quarter for O & M**					
a	Civil works	R.Km.	1.37	21.610	29.62	
b	EM works etc	R.Km.	0.32	21.610	6.97	
	Sub Total (13)				36.59	
14.0	**Multimodal Integration and Last mile connectivity**	Per station	3.00	20	60.00	
	Sub Total (14)				60.00	
15.0	Total of all items except Land				4698.13	
16.0	General Charges incl. Design charges @ 5 % on all items except land				234.91	
17.0	Total of all items including G. Charges except land				4933.04	
18.0	Continegencies @ 3 %				147.99	
19.0	Gross Total				5081.03	
	Cost without land			=	5081	
	Cost with land			=	5810	

Table 19.5

Details of Taxes and Duties

SARTHANA-DREAM CITY (CORRIDOR-I)

Basic Customs duty =	5.5000	%
CGST Customs Duty=	9.495	%
SGST Customs Duty=	9.495	%
Total Customs Duty=	**24.4900**	%
General IGST=	12	%
General CGST =	6	%
General SGST =	6	%

S. No.	Description	Total cost without Taxes & duties (Cr.)	Import Component	Taxes and duties								Total Taxes & Duties (Cr.)
				IMPORTED MATERIALS				INDEGENIOUS MATERIALS & SERVICES				
				CUSTOMS DUTY				GST				
				Basic Customs Duty (Cr.)	IGST (CGST portion) (Cr.)	IGST (SGST portion) (Cr.)	Total Customs Duty (Cr.)	GST RATE	CGST (Cr.)	SGST (Cr.)	Total GST (CGST & SGST) (Cr.)	
1	**Alignment & Formation**											
	Underground	667.50	0.23	8.44	14.58	14.58	37.60	12%	30.84	30.84	61.68	**99.28**
	Elevated, at grade & entry to Depot	597.18	0.30	9.85	17.01	17.01	43.87	18%	37.62	37.62	75.24	**119.12**
2	**Station Buildings**											
	Underground station - Civil works	790.00	0.18	7.82	13.50	13.50	34.82	12%	38.87	38.87	77.74	**112.56**
	Underground station - E&M works	314.86	0.20	3.46	5.98	5.98	15.42	18%	22.67	22.67	45.34	**60.76**
	Elevated station - Civil works	364.00	0.00	0.00	0.00	0.00	0.00	12%	21.84	21.84	43.68	**43.68**
	Elevated station - E&M works	112.00	0.00	0.00	0.00	0.00	0.00	18%	10.08	10.08	20.16	**20.16**
	Metro bhawan &OCC bldg. - Civil wor	90.00	0.00	0.00	0.00	0.00	0.00	12%	5.40	5.40	10.80	**10.80**
	Metro Bhawan &OCC bldg. - E&M work	25.00	0.00	0.00	0.00	0.00	0.00	18%	2.25	2.25	4.50	**4.50**
3	**Depot**											
	Civil works	105.00	0.00	0.00	0.00	0.00	0.00	12%	6.30	6.30	12.60	**12.60**
	EM works	65.00	0.10	0.36	0.62	0.62	1.59	18%	5.27	5.27	10.53	**12.12**
4	**P-Way**	202.65	0.90	10.03	17.32	17.32	44.67	12%	1.22	1.22	2.43	**47.10**
5	**Traction & power supply**	322.41	0.32	5.67	9.80	9.80	25.27	12%	13.15	13.15	26.31	**51.57**
6	**Signalling and Telecom.**	250.44	0.90	12.40	21.40	21.40	55.20	12%	1.50	1.50	3.01	**58.20**
7	**AFC**	70.00	0.80	3.08	5.32	5.32	13.71	18%	1.26	1.26	2.52	**16.23**
8	**PSD**	60.00	0.90	2.97	5.13	5.13	13.22	12%	0.36	0.36	0.72	**13.94**
7	**R & R hutments**	0.00	0.00	0.00	0.00	0.00	0.00	12%	0.00	0.00	0.00	**0.00**
8	**Misc.**											
	Civil works	181.87	0.00	0.00	0.00	0.00	0.00	12%	10.91	10.91	21.82	**21.82**
	EM works	57.72	0.00	0.00	0.00	0.00	0.00	18%	5.19	5.19	10.39	**10.39**
9	**Rolling stock**	360.00	0.88	17.42	30.08	30.08	77.58	18%	3.89	3.89	7.78	**85.36**
10	**Rent on Temporary Land**	56.70	0.00	0.00	0.00	0.00	0.00	12%	3.40	3.40	6.80	**6.80**
11	**General Consultancy charges**	234.91	0.00	0.00	0.00	0.00	0.00	18%	21.14	21.14	42.28	**42.28**
	Total	**4927.24**		**81.52**	**140.73**	**140.73**	**362.97**		**243.17**	**243.17**	**486.33**	**849.30**
	Total taxes & Duties									**SAY**		**849.30**
Rate of Taxes & Duties on Total cost without taxes & duties												**17.24%**
Total Central GST & Basic Customs duty												465.41
Total State GST												383.89
Total Taxes & Duties												**849.30**

19.14.2 Corridor-I :Bhesan – Saroli

The overall Capital Cost for this corridor estimated at December 2018 price level, based on the above considerations works out to **Rs 3528 Crores** without Taxes & Duties. Applicable Taxes & Duties, have been worked out as **Rs. 473.48 Crores.**

Table 19.6

Surat Metro corridor (Phase-I)					
BHESAN- SAROLI (corridor-II)					
Capital Cost Estimate					
					December 2018 level
Total length (Entirely Elevated)= 18.74 km					
Total Station = 18 nos (Elv)					
S. No.	Item	Unit	Rate	Qty.	Amount (Rs. in Cr.)
				Without taxes	
1.0	Land				
1.1	Permanent				
a	Government	ha	15.73	20.20	317.75
b	Private	ha	19.15	1.04	19.92
1.2	Temporary				
a	Government	ha	3.15	18.00	56.70
1.3	Property development(Government)	ha	15.73	2.01	31.62
1.4	R & R incl. Hutments etc.	R.Km.	4.00	19.74	78.96
	Subtotal (1)				504.94
2.0	Alignment and Formation				
2.1	Elevated section (viaduct) including station length	R.Km.	37.00	18.74	693.38
2.2	Depot entry connection	R.Km.	37.00	1.00	37.00
2.3	Extra for Important Bridge (at tapi River)	LS			60.00
	Subtotal (2)				790.38
3.0	Station Buildings				
3.1	Elevated station (Civil including finishes) excluding viaduct in station portion	Each	26.00	18.00	468.00
3.2	Elevated Station (E&M including lift & escalator)	Each	8.00	18.00	144.00
3.3	OCC bldg.				

a	Civil works	LS			10.00
b	E&M works etc	LS			11.03
	Subtotal (3)				633.03
4.0	Depot				
4.1	Civil	Per Depot			105.00
4.2	E&M & M&P	Per Depot			65.00
	Subtotal (4)				170.00
5.0	Permanent Way				
5.1	Ballastless track	R.Km.	6.60	18.74	123.68
5.2	Ballasted track for Depot	T.Km.	3.90	10.00	39.00
	Subtotal (5)				162.68
6.0	Traction & power supply				
6.1	750V DC Third Rail				
a	Elevated section	R.Km.	11.00	18.74	206.14
b	RSS	Each	25.00	1.00	25.00
c	Depot	T.Km.	3.00	10.00	30.00
d	132 kV cabling from Grid to RSS	R.Km.	6.93	2.00	13.86
	Subtotal (6)				275.00
7.0	Signalling and Telecom.				
7.1	**Signalling**				
a	Main line including OCC	R.Km.	4.40	19.74	86.86
b	Depot including DCC	T.Km.	3.20	10.00	32.00
c	On Board Equipment	Per Train	1.70	9.00	15.30
7.2	**Telecommunication**				
a	Station	Per Station	4.50	18.00	81.00
b	Depot	Per Depot	3.50	1.00	3.50
	Subtotal (7)				218.66
8.0	Automatic Fare Collection(AFC) system	Per station	3.50	18.00	63.00
	Subtotal (8)				63.00
9.0	Platform Screen Doors(PSD)	Per station	3.00	18.00	54.00
	Subtotal (9)				54.00
10.0	Shifting of Miscellaneous Utilities				
10.1	Civil and E&M	R.Km.	6.00	19.74	118.44
	Subtotal (10)				118.44
11.0	Rolling Stock	Per coach	8.00	27.00	216.00
	Subtotal (11)				216.00
12.0	Security	Per station	0.37	18.00	6.66
	Subtotal (12)				6.66
13.0	**Staff quarter for O & M**				

a	Civil works	R.Km.	1.37	19.74	27.06
b	EM works etc	R.Km.	0.32	18.74	6.05
	Sub Total (13)				33.10
14	**Multimodal Integration and Last mile connectivity**	Per station	3.00	18.00	54.00
	Sub Total (14)				54.00
15.0	Total of all items except Land				2794.95
16.0	General Charges incl. Design charges @ 5 % on all items except land				139.75
17.0	Total of all items including G. Charges except land				2934.70
18.0	Continegencies @ 3 %				88.04
19.0	Gross Total				3022.74
	Cost without land			=	3023
	Cost with land			=	3528

Table 19.7

Details of Taxes and Duties

BHESAN- SAROLI (CORRIDOR-II)

Basic Customs duty =	5.5000	%	
CGST Customs Duty=	9.495	%	
SGST Customs Duty=	9.495	%	
Total Customs Duty=	**24.4900**	%	
General IGST=	12	%	
General CGST =	6	%	
General SGST =	6	%	

S. No.	Description	Total cost without Taxes & duties (Cr.)	Import	Taxes and duties								Total Taxes & Duties (Cr.)
				IMPORTED MATERIALS				INDEGENIOUS MATERIALS & SERVICES				
				CUSTOMS DUTY				GST				
				Basic Customs Duty (Cr.)	IGST (CGST portion) (Cr.)	IGST (SGST portion) (Cr.)	Total Customs Duty (Cr.)	GST RATE	CGST (Cr.)	SGST (Cr.)	Total GST (CGST & SGST) (Cr.)	
1	**Alignment & Formation**											
	Underground	0.00	23%	0.00	0.00	0.00	0.00	12%	0.00	0.00	0.00	**0.00**
	Elevated, at grade & entry to Depot	730.38	30%					18%	46.01	46.01	92.03	**92.03**
2	**Station Buildings**											
	Underground station - Civil works	0.00	18%	0.00	0.00	0.00	0.00	12%	0.00	0.00	0.00	**0.00**
	Underground station - E&M works	0.00	20%	0.00	0.00	0.00	0.00	18%	0.00	0.00	0.00	**0.00**
	Elevated station - Civil works	468.00	0%					12%	28.08	28.08	56.16	**56.16**
	Elevated station - E&M works	144.00	0%	0.00	0.00	0.00	0.00	18%	12.96	12.96	25.92	**25.92**
	OCC bldg. - Civil works	10.00	0%					12%	0.60	0.60	1.20	**1.20**
	OCC bldg. - E&M works	11.03	0%	0.00	0.00	0.00	0.00	18%	0.99	0.99	1.99	**1.99**
3	**Depot**											
	Civil works	105.00	0%	0.00	0.00	0.00	0.00	12%	6.30	6.30	12.60	**12.60**
	EM works	65.00	10%	0.36	0.62	0.62	1.59	18%	5.27	5.27	10.53	**12.12**
4	**P-Way**	162.68	90%	8.05	13.90	13.90	35.86	12%	0.98	0.98	1.95	**37.81**
								0%				
5	**Traction & power supply**	275.00	32%	4.84	8.36	8.36	21.55	12%	11.22	11.22	22.44	**43.99**
6	**Signalling and Telecom.**	218.66	90%	10.82	18.69	18.69	48.19	12%	1.31	1.31	2.62	**50.82**
7	**AFC**	63.00	80%	2.77	4.79	4.79	12.34	18%	1.13	1.13	2.27	**14.61**
8	**PSD**	54.00	90%	2.67	4.61	4.61	11.90	12%	0.32	0.32	0.65	**12.55**
7	**R & R hutments**	0.00	0%	0.00	0.00	0.00		12%	0.00	0.00	0.00	**0.00**
8	**Misc.**											
	Civil works	161.38	0%	0.00	0.00	0.00	0.00	12%	9.68	9.68	19.37	**19.37**
	EM works	50.82	0%	0.00	0.00	0.00	0.00	18%	4.57	4.57	9.15	**9.15**
				0.00	0.00	0.00						
9	**Rolling stock**	216.00	88%	10.45	18.05	18.05	46.55	18%	2.33	2.33	4.67	**51.22**
10	**Rent on Temporary Land**	56.70	-	0.00	0.00	0.00		12%	3.40	3.40	6.80	**6.80**
11	**General Consultancy charges**	139.75	-	0.00	0.00	0.00		18%	12.58	12.58	25.15	**25.15**
	Total	**2931.40**		**39.97**	**69.01**	**69.01**	**177.99**		**147.75**	**147.75**	**295.49**	**473.48**
	Total taxes & Duties									**SAY**		**473.48**
	Rate of Taxes & Duties on Total cost without taxes & duties											**16.15%**
	Total Central GST & Basic Customs duty											256.73
	Total State GST											216.76
	Total Taxes & Duties											**473.48**

CHAPTER- 20

FINANCING OPTIONS, FARE STRUCTURE AND FINANCIAL VIABILITY

20.1 Introduction

The Surat Metro Rail Project consisting of two corridors (from Sarthna to Dream City and Bhesan to Saroli) is proposed to be constructed with an estimated cost of Rs.10661.00 Crore with land cost and all taxes. The route length of the metro system and estimated cost at December-2018 price level without and with all taxes are placed in table 20.1 as under:

Table: 20.1 Cost Details

Corridor no.	Name of Corridor	Distance (KMs)	Estimated cost with Land cost and without all taxes (Rs/Crore)	Estimated cost with land cost and all taxes (Rs/Crore)
1	Sarthana to Dream City	21.61	5810	6659.30
2	Bhesan to Saroli	18.74	3528	4001.48
Total		40.350	9338	10660.78 =10661

The estimated cost at December-2018 price level includes an amount of Rs.14.06 Crore plus applicable taxes as one-time charges towards security, i.e., cost of weapons, barricades and hand held and door detector machine etc. However, the recurring cost towards salary and allowances of security personal have not taken in to account in the FIRR calculation since providing required security at metro stations shall be the responsibility of state police department.

20.2 Costs

20.2.1 Investment Cost

20.2.1.1 For the purpose of calculating the Financial Internal Rate of Return (FIRR), the completion cost with all taxes except land and contingencies has been calculated by taking escalation factor @5.00% per annum. It has been assumed that the Government of Gujarat will provide the land worth Rs.1234 crore either free of cost or it shall provide Interest Free Subordinate Debt in lieu thereof. With the implementation of GST from 1st July 2017, the Excise Duty, Central Sales Tax/State VAT and Service Tax have been subsumed into Central Goods and Service Tax (CGST), State Goods and Service Tax (SGST), Integrated Goods and Service Tax (IGST). The effective CD rate under project import scheme, post GST, works out to 24.49% {Basic CD @ 5%, IGST (CGST & SGST) @ 18% and cess} on the imported portions, CGST and SGST @ 6% each and IGST @ 12% on indigenously manufactured items. The Interest Free Subordinate Debt

is repayable in 5 equal instalments after repayment of Multilateral/Overseas Development Assistance Loan.

It is assumed that the construction work will start on 01.04.2019 and is expected to be completed on 31.03.2024 with Revenue Opening Date (ROD) as 01.04.2024 for both the corridors. The total completion costs duly escalated and shown in the table 20.2 have been taken as the initial investment as below:-

Table: 20.2 Year –wise Investment

Figures in Rs. Crore

Financial Year	Estimated Cost including cost of land and all taxes & duties at December 2018 Price Level	Completion Cost including cost of land cost and all taxes & duties
2019-20	1,722	1,757
2020-21	2,439	2,595
2021-22	2,797	3,106
2022-23	2,382	2,751
2023-24	1,320	1,629
Total	10,660	11,838

20.2.2 Additional Investment

Total investment provided in the FIRR calculation towards requirement of additional rolling stock duly escalated @5% PA is placed in table 20.3 as under: -

Table: 20.3 Additional Investment towards Rolling Stock

(Rs/Crore)

Financial Year	No. of Cars	Amount
2026-27	39	799.00
2036-37	75	2504.00
2046-47	48	2610.00
TOTAL	**162**	**5913.00**

20.2.3 Operation & Maintenance (O&M) Costs

20.2.3.1 The Operation & Maintenance costs can be divided into three major parts:

(i) Staff costs

(ii) Maintenance cost which include expenditure towards upkeep and maintenance of the system and consumables

(iii) Energy costs, and

(iv) Service cost of Public Private Partnership (PPP) component

The requirement of staff has been assumed @ 30 persons per kilometre. The escalation factor used for staff costs is 9% per annum to provide for both escalation and growth in salaries.

The cost of other expenses is based on the actual O & M unit cost for the Delhi Metro Phase-II project. The average rate of electricity being paid in Surat City is Rs. 6 per unit. As per Metro Rail Policy 2017, it is mandatory now to involve Public Private Partnership (PPP) in some form for implementation, operation & maintenance, fare collection or any other unbundled activities of the proposed metro rail project. It has been proposed that activities involving a) civil construction works of 5 nos. of stations i.e. Rupali Canal, Althan Tenament of corridor 1, Aquarium (Star Bazar) and Athwa Chaupati of corridor 2 on SMC/SUDA land and Dream city stations on Dream City Land, b) Lifts & Escalator and c) AFC for all the stations may be given on PPP basis in this project. The cost of above unbundled activities may be funded by engaging concessionaire in line with Kochi Metro, Nagpur Metro and Noida Metro for a period of 10 years. The expected return to the concessionaire will be in between 12% to 16% from all sources. For calculation of FIRR, the cost of servicing PPP activities over a period of 10 years has been considered taking into account principal repayment and Interest @ 12% per year on PPP cost of proposed activities of Rs.452 Crore on completion. The O&M cost (excluding staff cost and Service cost of PPP component) has been obtained by providing an escalation of 5.00% per annum. The O&M costs have been tabulated in Table 20.4 as below for the both corridors:

Table: 20.4 Operations and Maintenance Costs (Both Corridors)

Rs. In Crore

Year			Staff	Maintenance Expenses	Energy	Serving Cost of PPP Component	Total
2024	-	2025	104.43	70.23	90.19	125.84	390.68
2025	-	2026	113.82	73.74	94.70	118.98	401.24
2026	-	2027	124.07	77.43	99.43	112.11	413.04
2027	-	2028	135.23	81.30	104.41	105.25	426.19
2028	-	2029	147.40	85.37	127.42	98.38	458.56
2029	-	2030	160.67	89.63	133.79	91.52	475.61
2030	-	2031	175.13	94.11	140.48	84.66	494.38
2031	-	2032	190.89	98.82	147.50	77.79	515.00
2032	-	2033	208.07	103.76	154.87	70.93	537.64
2033	-	2034	226.80	108.95	162.62	64.06	562.43
2034	-	2035	247.21	114.40	170.75	0.00	532.36
2035	-	2036	269.46	120.12	179.29	0.00	568.86
2036	-	2037	293.71	126.12	188.25	0.00	608.09
2037	-	2038	320.15	132.43	197.66	0.00	650.24
2038	-	2039	348.96	139.05	244.54	0.00	732.55
2039	-	2040	380.37	146.00	256.77	0.00	783.14
2040	-	2041	414.60	153.30	269.61	0.00	837.51

Year			Staff	Maintenance Expenses	Energy	Serving Cost of PPP Component	Total
2041	-	2042	451.91	160.97	283.09	0.00	895.97
2042	-	2043	492.59	169.02	297.24	0.00	958.85
2043	-	2044	536.92	177.47	312.11	0.00	1026.49
2044	-	2045	585.24	186.34	327.71	0.00	1099.29
2045	-	2046	637.91	195.66	344.10	0.00	1177.67
2046	-	2047	695.32	205.44	361.30	0.00	1262.07
2047	-	2048	757.90	215.71	379.37	0.00	1352.98
2048	-	2049	826.11	226.50	448.88	0.00	1501.50
Total			**8844.87**	**3351.87**	**5516.08**	**949.52**	**18662.34**

20.2.4 Depreciation

Although depreciation does not enter the FIRR calculation (not being a cash outflow) unless a specific depreciation reserve fund has been provided, in the present calculation, depreciation calculations are placed for purpose of record.

20.2.5 Replacement Cost

The replacement costs are provided for meeting the cost on account of replacement of equipment due to wear and tear. With the nature of equipments proposed to be provided, it is expected that only 50% of the Signalling and Telecom and 25% of electrical works would require replacement after 20 years.

Table: 20.4.1 Replacement Cost

Year	Cost	Esc Cost
2044-45	431.78	1,535.27
2045-46	431.78	1,612.04
Total		3,147.31

20.3 Revenues

The Revenue of Surat Metro mainly consists of fare box collection and other incomes from property development, advertisement, parking etc.

20.3.1 Fare box

The Fare box collection is the product of projected ridership per day and applicable fare structure based on trip distribution at different distance zones.

20.3.2 Traffic

20.3.2.1

a. The projected ridership as per CEPT studies conducted by Surat Municipal Corporation which is further moderated to 80% for FIRR calculation is as indicated in table 20.5 as below: -

Table: 20.5 Projected Ridership

Financial Year	Trips per day (lakhs) as per CEPT studies	Moderate (80%) Trips per day(lakhs)
2021-22	3.09	2.47
2024-25	3.54	2.83
2027-28	5.33	4.26
2037-38	9.74	7.79
2047-48	12.23	9.78

b. The growth rate for moderate traffic is assumed @14.65% Per Annum upto 2027-28, @ 6.21% Per Annum upto 2037-38, @2.30% Per Annum upto 2047-48, and thereafter @ 1% Per Annum.

20.3.2.2 Trip Distribution

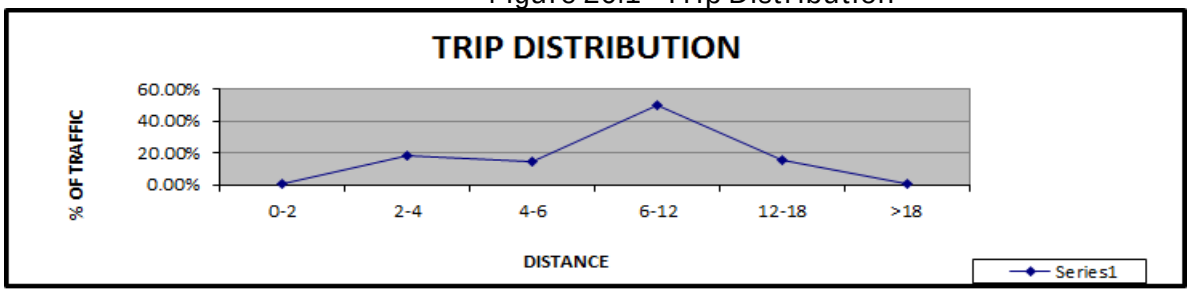
The trip distribution has been provided by Surat Municipal Corporation based on the CEPT study conducted, which is placed in Table 20.6 below: -

Table: 20.6 Trip Distribution (2024-25)

Distance in km	Percent distribution
0-2	0.90%
2-4	18.80%
4-6	14.40%
6-12	49.70%
12-18	15.40%
>18	0.80%
Total	100.00%

The graphic presentation of the same is placed below in Figure 20.1:-

Figure 20.1 –Trip Distribution



20.3.3 Fare Structure

The fare structure for the FY 2024-25 as per the proposed fare slabs is shown in the table 20.7 below: -

Table: 20.7 Fare Structure in 2024-25

Distance in kms.	Fare (Rs)
0-2	10
2-4	20
4-6	30
6-12	40
12-18	50
>18	60

Considering the increase in the Consumer Price Index (CPI) and input costs of operation since then, the fare structure has been escalated by using @14.00% once in every two years.

20.3.4 Non Fare Box Revenue

20.3.4.1 Income from property Business and Advertisement – Other revenues from Property Business and advertisement have been assumed @ 10% of the fare box revenues till year 2028-29 i.e. 5 years of operation and @ 20% thereafter during operations. Apart from development of property on metro stations and depot it is possible to raise resources through leasing of parking rights at stations, advertisement on trains and tickets, advertisements within stations and parking lots, advertisements on viaducts, columns and other metro structures, co-branding rights to corporate, naming rights, film shootings and special events on metro premises.

20.3.4.2 Income from PD Land – SPV/BOT operator will engage a developer/Concessionaire for generating rental income. It is assumed that about 4.70 Hectare. i.e., 14,10,000 square feet area will be available for property development with a FAR of 3. The developer will bring equity to the extent of Rs. 80.00 crore and the balance amount towards construction shall be raised by SPV as 12% Market Debt. The current rental revenue in Surat City is Rs. 40 per Sq. ft. The estimated development cost will be Rs.310.00 crore. It is assumed that the rental revenue will

accrue to the developer from the FY 2025-26 which has been escalated @5% every year. Out of the estimated rental income, apart from meeting maintenance expenditure, the developer will repay the loan and interest. After meeting these obligations and retaining 14% return on his equity with an escalation @5% every year, the residual rental earnings will accrue to SPV, which has been taken into account in the FIRR calculations. The details of PD income accrue to SPV is tabulated as under; -

Table 20.8.1 Estimated generation of Rental Income from PD

Rs. in Crore

Year	Construction cost	Rental Income	Maintenance Expenditure	Loan	IDC	Loan repayment	Bal Loan Amount	Interest on Loan @12%	Return @14% to the developer	Residual rental income to SPV
2020 - 2021	56			40	2		42		-16	
2021 - 2022	59			43	5		90		-16	
2022 - 2023	62			46	6		142		-16	
2023 - 2024	65			49	7		198		-16	
2024 - 2025	68			52	8		258		-16	
2025 - 2026		17	2			26	232	31	11	-53
2026 - 2027		24	2			26	206	28	12	-44
2027 - 2028		33	3			26	180	25	13	-34
2028 - 2029		53	5			26	154	22	14	-15
2029 - 2030		88	9			26	128	18	15	20
2030 - 2031		93	9			26	102	15	16	27
2031 - 2032		97	10			26	76	12	17	32
2032 - 2033		102	10			26	50	9	18	39
2033 - 2034		107	11			26	24	6	19	45
2034 - 2035		113	11			24	0	3	20	55
2035 - 2036		118	12						21	85
2036 - 2037		124	12						22	90
2037 - 2038		130	13						23	94
2038 - 2039		137	14						24	99
2039 - 2040		144	14						25	105
2040 - 2041		151	15						26	110
2041 - 2042		158	16						27	115
2042 - 2043		166	17						28	121
2043 - 2044		175	17						29	129
2044 - 2045		183	18						30	135
2045 - 2046		193	19						32	142
2046 - 2047		202	20						34	148
2047 - 2048		212	21						36	155
2048 - 2049		223	22						38	163
Total	310	3043	302	230	28	258		169	470	1763

20.3.4.3 Income from cess on Property tax and Stamp duty – It is proposed to levy a cess @ 5% on property tax and on stamp duty or any other levies (sale/purchase of property). A modest 3% annual increase is assumed in order to estimate the revenue over 25 years during the operation of the Metro. The average revenue collected by the Surat Municipal Corporation on account of property tax and stamp duty (sale/purchase of property) during the last 4 years is Rs.604 crore and Rs.703 crore respectively. Accordingly, the starting year i.e. 2024-25, the revenue from cess on property tax and on stamp duty worked out is Rs. 80 crore. However, considering the first year of implementation the expected revenue for the year 2024-25 is Rs. 40 crore i.e. 50% of

Rs. 80 crore. The above revenue is further enhanced by 3% annually. The year wise projected revenue from additional 5% cess on property tax and on stamp duty is tabulated in Table 20.8.2 below: -

Table 20.8.2 Revenue from 5% cess on property tax and on stamp duty

Rs. in Crore

Year			Income from cess on Property Tax	Income from cess on Stamp Duty	Total
2024	-	2025	19	21	40
2025	-	2026	38	43	81
2026	-	2027	39	44	83
2027	-	2028	40	45	85
2028	-	2029	41	46	87
2029	-	2030	42	47	89
2030	-	2031	43	48	91
2031	-	2032	44	49	93
2032	-	2033	45	50	95
2033	-	2034	46	52	98
2034	-	2035	47	54	101
2035	-	2036	48	56	104
2036	-	2037	49	58	107
2037	-	2038	50	60	110
2038	-	2039	52	62	114
2039	-	2040	54	64	118
2040	-	2041	56	66	122
2041	-	2042	58	68	126
2042	-	2043	60	70	130
2043	-	2044	62	72	134
2044	-	2045	64	74	138
2045	-	2046	66	76	142
2046	-	2047	68	78	146
2047	-	2048	70	80	150
2048	-	2049	72	82	154
Total			1273	1465	2738

20.3.4.4 Income from TOD & VCF – As per details provided by SMC, the total revenue to be generated from the sale of additional FSI along the metro corridors shall be Rs. 1343 crore out of which 60% for 15 years i.e. Rs. 54 crore per year shall be considered for metro project. The annual growth rate is considered at @ 3% per year. Accordingly, the income from TOD & VCF for the Year 2024-25 and 2025-26 is Rs. 31 crore (being the first year of implementation the revenue for the year 2024-25 has been assumed at 50% of expected revenue from TOD & VCF) and Rs. 65 crore respectively which is further escalated @ 3 % per year and shall be made available to metro authorities.

20.3.5 Revenue Maximization Plan for Surat Metro

Surat Metro may further enhance the non-fare box revenue base along with dogged focus to increase the fare box revenue. The multipronged approach is as under:-

20.3.5.1 Earnings from Stations (Advertisements, Kiosks, Parking, ATMs and other revenues) –

At a conservative level DPR has assumed the earnings from these sources @ 10% to 20% of the fare box revenue from operations. However, the effort of Surat Metro shall be to maximize these earnings.

(i) Advertisements:

a) ***Inside Station Advertisement****:* - All section/Lines of Metro may be awarded for advertisement purposes. Scope of the inside station advertisement may be display static advertisement on fixed advertisement boards/panels as well as through digital displays, putting up promotional canopies etc. Area of advertisement space may be fixed based on type of station i.e. elevated or underground station, size of station, footfall of station etc.

b) ***Outdoor Advertisement:*** - Commercial Advertisement may be allowed by putting Advertisement boards on Metro Civil Structure/Ancillary Buildings/Via Duct/FOBs/Metro Pillars. Scope of the Outdoor advertisement may be to put up static advertisement display at the Metro Pillars/FOBs/Metro Viaduct/Ancillary Building/Lift Lobby etc. Area of advertisement space may be decided based on type, size of station and footfall of station etc. and as per policy of Urban Local Body.

c) ***Advertisement on Trains:*** - MEGA may allow Commercial Advertisement on Metro Rail Trains; inside on prefabricated advertisement spaces and outside by wrapping of trains. Bidder may wrap 10% of available cars subject to maximum of 20% trains available in the line as well as may do advertisement on pre-designated spaces inside the Metro Trains.

d) ***Co-Branding:-*** Co-Branding by Suffixing/Prefixing of Brand Name with the Metro Station name may be allowed for selected metro stations. Co-branding rights may be awarded to the eligible/competent parties through tendering process. The Scope under this type of advertisement may be to Suffix the Brand name with the Metro Station name whereas for Stations. Colouring of the station in brand colour as well as inside and outside Advertisement rights for the station may also be permitted. The contract may be awarded generally for tenure of 10 years which may be further extendable by 5 years on mutually agreeable terms and conditions. The bidders quote the lump sum rate/annum for the total scope offered at the station.

e) ***Advertisement on Smart Card/Tokens****:* - One time commercial advertisement on Metro smart cards/tokens may be permitted. The rights of advertise may be awarded tendering process.

(i) Shops:

a) ***Built Up shops:*** - Built up shops on the selected Metro Stations may be available on license basis. These types of Shops may have an inbuilt structure and located within the Metro Stations. The Shops awarded to the eligible/competent parties through tendering process. The shops which fail to attract any bid through tendering process may be made available through Walk-in-basis Scheme (First come first serve basis) wherein the bidders may take the shops at reserve price.

b) **Bare spaces:** - Unutilized bare spaces may also be licensed. The bare spaces may be awarded to the eligible/competent parties through tendering process with a scope to do all permitted/legal commercial activities by developing the bare spaces into a commercial space at their own cost. The tenure for the spaces may be decided based on the area which is to be developed by the bidder.

(ii) Kiosk

a) *Kiosks:* - Leasing of small bare spaces to operate small *kiosks* for commuters facilitations may be allowed on the metro stations.

b) ***Tom spaces:*** -Unused TOM spaces may be leased to operate eateries etc. through tendering process.

c) **Health Monitors/BMI Machines:** - Small bare spaces may be allotted to install Health Monitor/BMI Machines as commuter facilitation.

d) **Parcel Machines:** - Small bare spaces may be allotted to install smart parcel delivery system as commuter facilitation.

e) **Water Vending Kiosk:** - Small bare spaces may be allotted to install Water Vending Kiosks as commuter facilitation.

f) **ATMs -** Bare spaces may be leased to Banks to install and operate their *ATMs/e-lobby* at Metro Stations as commuter facility.

vi) Telecoms:

a) ***Telecom Towers****:* - The spaces in metro corridor premise may be leased for installation of Telecom Towers to enhance/provide the mobile connectivity. Commuters may get better mobile connectivity at the Metro Stations and its vicinity and MEGA may earns revenue also.

b) **Small Cells: -** Leasing of spaces on DMRC pillars to install Small Telecom equipment to enhance the mobile connectivity may also be considered. The spaces may be allotted to the different telecom operators through tendering

mode for commuters to get better mobile connectivity at the vicinity of DMRC viaduct as well as for MEGA to earn revenue also.

20.4 Financial Internal Rate of Return (FIRR)

20.4.1 The Financial Internal Rate of Return (FIRR) obtained for 30 years life cycle business model including construction period without additional PD and TOD & VCF Income is **6.90%** and **with** additional PD and TOD & VCF Income is **7.38%.** The FIRR with all taxes & duties including land cost without additional PD and TOD & VCF Income and with additional PD and TOD & VCF Income is produced in Table 20.9.1 & 20.9.2 below:

Table: 20.9.1 FIRR with Central Taxes without additional PD and TOD & VCF Income

Figs in cr. (Rs.)

Year	Outflow						Inflow			Net Cash Flow
	Completion Cost	Additional Cost	Running Expenses	Replacement costs	Total Costs	Fare Box Revenue	PD &ADVT	Income from cess on Property Tax & Stamp Duty	Total Revenue	IRR
2019 - 2020	1757				1757				0	-1757
2020 - 2021	2595				2595				0	-2595
2021 - 2022	3106				3106				0	-3106
2022 - 2023	2751				2751				0	-2751
2023 - 2024	1629				1629				0	-1629
2024 - 2025	0	0	391		391	348	35	40	423	32
2025 - 2026	0	0	401		401	400	40	81	521	120
2026 - 2027	0	0	413		413	525	53	83	661	248
2027 - 2028	0	0	426		426	602	60	85	747	321
2028 - 2029	0	799	459		1258	726	73	87	886	-372
2029 - 2030	0	0	476		476	770	154	89	1013	537
2030 - 2031	0	0	494		494	931	186	91	1208	714
2031 - 2032	0	0	515		515	988	198	93	1279	764
2032 - 2033	0	0	538		538	1191	238	95	1524	986
2033 - 2034	0	0	562		562	1265	253	98	1616	1054
2034 - 2035	0	0	532		532	1530	306	101	1937	1405
2035 - 2036	0	0	569		569	1625	325	104	2054	1485
2036 - 2037	0	0	608		608	1967	393	107	2467	1859
2037 - 2038	0	0	650		650	2090	418	110	2618	1968
2038 - 2039	0	2504	733		3237	2432	486	114	3032	-205
2039 - 2040	0	0	783		783	2488	498	118	3104	2321
2040 - 2041	0	0	838		838	2901	580	122	3603	2765
2041 - 2042	0	0	896		896	2968	594	126	3688	2792
2042 - 2043	0	0	959		959	3466	693	130	4289	3330
2043 - 2044	0	0	1026		1026	3546	709	134	4389	3363
2044 - 2045	0	0	1099	1535	2635	4134	827	138	5099	2464
2045 - 2046	0	0	1178	1612	2790	4229	846	142	5217	2427
2046 - 2047	0	0	1262	0	1262	4937	987	146	6070	4808
2047 - 2048	0	0	1353	0	1353	5050	1010	150	6210	4857
2048 - 2049	0	2610	1502	0	4112	5820	1164	154	7138	3027
Total	11838	5913	18663	3147	39561	56929	11126	2738	70793	6.90%

Table: 20.9.1 FIRR with Central Taxes with additional PD and TOD & VCF Income

Figs in cr. (Rs.)

Year	Outflow							Inflow			Net Cash Flow
	Completion Cost	Additional Cost	Running Expenses	Replacement costs	Total Costs	Fare Box Revenue	PD &ADVT	Income from cess on Property Tax & Stamp Duty	Income from TOD & VCF	Total Revenue	IRR
2019 - 2020	1757				1757					0	-1757
2020 - 2021	2595				2595					0	-2595
2021 - 2022	3106				3106					0	-3106
2022 - 2023	2751				2751					0	-2751
2023 - 2024	1629				1629					0	-1629
2024 - 2025	0	0	391		391	348	35	40	31	454	63
2025 - 2026	0	0	401		401	400	-13	81	65	533	132
2026 - 2027	0	0	413		413	525	9	83	67	684	271
2027 - 2028	0	0	426		426	602	26	85	69	782	356
2028 - 2029	0	799	459		1258	726	58	87	71	942	-316
2029 - 2030	0	0	476		476	770	174	89	73	1106	630
2030 - 2031	0	0	494		494	931	213	91	75	1310	816
2031 - 2032	0	0	515		515	988	230	93	77	1388	873
2032 - 2033	0	0	538		538	1191	277	95	79	1642	1104
2033 - 2034	0	0	562		562	1265	298	98	81	1742	1180
2034 - 2035	0	0	532		532	1530	361	101	83	2075	1543
2035 - 2036	0	0	569		569	1625	410	104	85	2224	1655
2036 - 2037	0	0	608		608	1967	483	107	88	2645	2037
2037 - 2038	0	0	650		650	2090	512	110	91	2803	2153
2038 - 2039	0	2504	733		3237	2432	585	114	94	3225	-12
2039 - 2040	0	0	783		783	2488	603	118	49	3258	2474
2040 - 2041	0	0	838		838	2901	690	122	0	3713	2875
2041 - 2042	0	0	896		896	2968	709	126	0	3803	2907
2042 - 2043	0	0	959		959	3466	814	130	0	4410	3451
2043 - 2044	0	0	1026		1026	3546	838	134	0	4518	3492
2044 - 2045	0	0	1099	1535	2635	4134	962	138	0	5234	2599
2045 - 2046	0	0	1178	1612	2790	4229	988	142	0	5359	2569
2046 - 2047	0	0	1262	0	1262	4937	1135	146	0	6218	4956
2047 - 2048	0	0	1353	0	1353	5050	1165	150	0	6365	5012
2048 - 2049	0	2610	1502	0	4112	5820	1327	154	0	7301	3190
Total	11838	5913	18663	3147	39561	56929	12889	2738	1178	73734	7.38%

The various sensitivities with regard to increase /decrease in capital costs, O&M costs and fare-box revenues are placed in Table 20.10 below : -

Table: 20.10 FIRR SENSITIVITY
(With Additional PD and TOD & VCF Income)

Capital Cost with Central Taxes but without land cost			
20% increase in capital cost	10% increase in capital cost	10% decrease in capital cost	20% decrease in capital cost
6.31%	6.82%	8.02%	8.75%
REVENUE			
20% decrease in Fare Box revenue	10% decrease in Fare Box revenue	10% increase in Fare Box revenue	20% increase in Fare Box revenue
5.29%	6.41%	8.25%	9.03%
O&M COSTS			

10% increase in O&M cost	10% decrease in O&M cost
7.08%	7.68%

These sensitivities have been carried out independently for each factor.

20.5 Financing Options

Objectives of Funding: - The objective of funding metro systems is not necessarily enabling the availability of funds for construction but coupled with the objective of financial closure are other concerns, which are of no less importance: -

Ensuring low project cost

Ensuring debt funds at low rates of interest

Creating self sustainable system in the long run by

- Low infrastructure maintenance costs
- Longer life span
- Setting fares which minimise dependence on subsidies

Recovering returns from both direct and indirect beneficiaries

Rail based mass transit systems are characterised by heavy capital investments coupled with long gestation period leading to low financial rates of return although the economic benefits to the society are immense. Such systems generate externalities, which do not get captured in monetary terms and, therefore, do not flow back to the system. However, experience all over the world reveals that both construction and operations of metro are highly subsidised. Government involvement in the funding of metro systems is a foregone conclusion. Singapore had a 100% capital contribution from the government, Hong Kong 78% for the first three lines and 66% for the later 2 lines. The Phase-I, Phase-II as well as Phase-III of Delhi MRTS project, Chennai and Bengaluru metros are also funded with a mixture of equity and debt (ODA) by GOI & concerned State Governments.

20.5.1 ALTERNATIVE MODELS OF FINANCING

The financing option shall depend upon selection of the dedicated agency created to implement the project. The prominent models are: -

(i) Special Purpose Vehicle under the State Government Control (Delhi Metro Rail Corporation (DMRC) /Bangalore Metro Rail Corporation (BMRC)/Chennai Metro Rail Corporation/Mumbai Metro Rail Corporation

(ii) Design-Build-Finance-Operate-Transfer (DBFOT), and

(iii) Public Private Partnership (PPP)

20.5.1.1 SPV Model: - The corridor is a standalone one and therefore forming a separate SPV may be in the name of Surat Metro Rail Corporation may be desirable. MOUD vide letter no. F.No. K-14011/03/2017-UT-V-Part(1) dated 6th July 2017 has proposed for sharing of overall taxes post Goods and Service Tax (GST) regime in the ratio of 1:2. The funding pattern under this model (SPV) is placed in table 20.11 as under: -

Table: 20.11 Funding pattern under SPV model

Sl No	Sources	Amount(Rs./Crore)	% Contribution
1	Equity by GOI	1,487	15.70%
2	Equity by GOG	1,487	15.70%
3	SD for Central Taxes by GOG	408	4.30%
4	SD for Central Taxes by GOI	408	4.30%
5	Loan from Multilateral Bilateral Agency	5,435	57.35%
6	Upfront VCF and Proceeds from Monetization	250	2.65%
7	Total Cost Excluding Land R&R and PPP	9475	100.00%
8	SD for Land including R&R by GOG	1,234	
9	SD for State Taxes by GOG	677	
10	Sub Total	11,386	
11	PPP Component	452	
12	Sub Total	11,838	
13	IDC for Loan to be borne by GOG	182	
14	Total Completion Cost	12,020	

Surat Municipal Corporation (SMC) vide email dated 02.05.2018 stated that metro station at Surat Railway Station shall be integrated with Multi Model Traffic Hub (MMTH) project and one underground tunnel road shall also be provided above the metro station for vehicular traffic along Lambe Hanuman road. The estimated cost of Rs. 109.13 crore will be borne by SMC for above Multi Model Traffic Hub integration.

20.5.1.2 ODA Loan from JICA/ADB/AFD: JICA had provided loan assistance to Phase-I, Phase-II and ongoing Phase-III of Delhi MRTS projects to the extent of 60%, 54.47% & 51.32% respectively and also provided loan assistance to Chennai Metro, Mumbai Metro and MEGA. The Department of Economic Affairs, MOF, GOI vide letter no. F.1/9/2014-Japan.I dated 12th July 2018 has issued new set of guidelines on Operational Rules of the Japan-India Special ODA (Official Development Assistance)

for Metro and Railway Projects in India. As per the discussions made with JICA officials, JICA may extend only the modified step loan as per above said guidelines for the new projects in India at an interest rate of 0.20% per annum. The tenure of the loan is 40 years with 12 years moratorium period. JICA may fund the project to the extent of 85% of the cost of the project excluding the cost of the land, cost of Rehabilitation and Resettlement and taxes and duties. In case JICA not agreeing to provide loan for the Surat Metro project or provide less loan assistance than the projected, the difference shall have to be sourced either from Asian Development bank (ADB) or Agency Franchise de Development (AFD) or European Investment Bank (EIB) or Asian Infrastructure Investment Bank (AIIB) by GOI or from Domestic Financial Institutions against guarantee from GOI.

20.5.1.3 Loan from Domestic Financial Institutions: : Domestic Financial Institutions may have to be taped in the case of any shortfall in securing the targeted loan amount from bilateral/multilaterals institutions. In such a situation Government of India may have to provide sovereign guarantee as is being done to obtain the loan from JICA for Delhi Metro, ADB for Jaipur Metro, AFD for Kochi Metro and EIB for Lucknow Metro. The new SPV shall raise the required loan from IIFCL and or such other Domestic Financial Institutions. In order to keep the debt servicing sustainable so as to keep the Debt-Service Coverage Ratio (DSCR) above 1.2 at all points of time, interest subsidy between the domestic borrowing rate of interest and Multilateral/Overseas Development Loan rate of interest shall be provided to SPV.

20.5.1.4 Private Participation under SPV: As already stated in para 20.2.3, the private participation either for complete provisioning of metro rail or for some unbundled components will form an essential requirement for all metro rail project proposals seeking central financial assistance as per new Metro Policy 2017 issued by Ministry of Housing and Urban Affairs (MH&UA). The cost of unbundled activities of Rs.452 crore involving civil construction of 5 stations, AFC and Lifts & Escalators of all the corridors may be funded by engaging concessionaire in line with Kochi Metro, Nagpur Metro and Noida Metro for a period of 10 years.

20.5.1.5 DBFOT Model: - In this model, the private firm will be responsible for financing, designing, building, operating and maintaining of the entire project. The contribution of Government of Gujarat will be limited to cost of land only. Such a project become eligible for Viability Gap Funding (VGF) upto 20% from the Central Government provided the state government also contribute same or more amount towards the project. The metro being a social sector project not much private parties are available to bid for such a project. Besides quite expectedly the private operator may demand assured rate of return in the range of 16% to 18% or a comfort of guaranteed ridership.

The funding pattern assumed under this model excluding the cost of land with and without additional PD and TOD & VCF Income is placed in table 20.12.1 & 20.12.2 tabulated as under: -

Table: 20.12.1 Funding pattern under BOT –Combined (16% EIRR)
(Without Additional PD and TOD&VCF Income)

Particulars	With Taxes & Duties	
	Amount (Rs/Crore)	% Of contribution
VGF by GOI	2121	20.00%
VGF by GOG	5049	43.52%
Equity by Concessionaire	1145	11.57%
Concessionaire's debt @12% PA	2290	24.91%
Sub-Total	10605	100.00%
Land Free by GOG	1233	
Sub-Total	11838	
IDC	990	
Total	12738	

Table: 20.12.2 Funding pattern under BOT –Combined (16% EIRR)
(With Additional PD and TOD&VCF Income)

Particulars	With Taxes & Duties	
	Amount (Rs/Crore)	% Of contribution
VGF by GOI	2121	20.00%
VGF by GOG	4615	43.52%
Equity by Concessionaire	1227	11.57%
Concessionaire's debt @12% PA	2641	24.91%
Sub-Total	10605	100.00%
Land Free by GOG	1234	
Sub-Total	11838	
IDC	997	
Total	12835	

20.5.1.6 PPP Model: - In this model, Government funds the fixed infrastructure such as land and basic civil structures, and a private investor funds all the systems such as rolling stock, signalling, power supply, traction, track, fare collection, E&M works etc including station architectural finishes. An example of this is Delhi Metro Airport line. Under this arrangement the government's investment will be about 57% of the cost of the Project and the PPP operator funds the remaining 43%. Under this model the concessionaire, operates and maintains the system to the required and agreed service and safety levels. All the revenues will accrue to the Operator and at the end of the concession period the project is handed over to the owner. Ridership risks are taken by the operator or shared by the operator and owner. The PPP operator pays a specified amount every year to the Govt. out of his revenues. It could be that he may

need a viability gap funding (VGF) even. The VGF (positive or negative) will be known only after competitive bidding.

20.6. RECOMMENDATIONS

The FIRR of Surat metro rail project without & with additional PD and TOD & VCF income is **6.90% & 7.38%** and is financially viable. However, FIRR is not the only criteria to take up the metro project.

As per Metro Rail Policy 2017, issued by the Ministry of Housing and Urban Affairs, (MOH&UA), GOI, apart from financial viability, the economic and social viability of the project is also required to be assessed. The Economic Internal Rate of Return (EIRR) for any metro rail project proposal should be 14% and above for consideration of its approval. Accordingly, the metro corridors as discussed above are recommended for implementation provided the required EIRR works out to 14% or above. The total fund contribution of GOI & GOG under various alternatives is tabulated in table 20.13.

Table: 20.13

Rs. In crore

Particulars	SPV Model	BOT Model without Additional PD Income	BOT Model with Additional PD Income
GOI	1895	2121	2121
GOG	3806	6282	5849
Total	5701	8403	7970

Considering the difference, it is recommended to implement the project under SPV model as per the funding pattern given in Table 20.11.

The detailed cash flow statements under various alternatives are enclosed as per detail given below:-

Option	Table No.
SPV Model with JICA Loan	20.14
SPV Model with JICA Loan with Additional PD and TOD&VCF Income	20.15
BOT Model	20.16
BOT with Additional PD and TOD&VCF Income Model	20.17

The funding pattern assumed under SPV model is depicted in the pie chart i.e., Figure 20.2 as under: -

Figure 20.2
Funding pattern under SPV Model

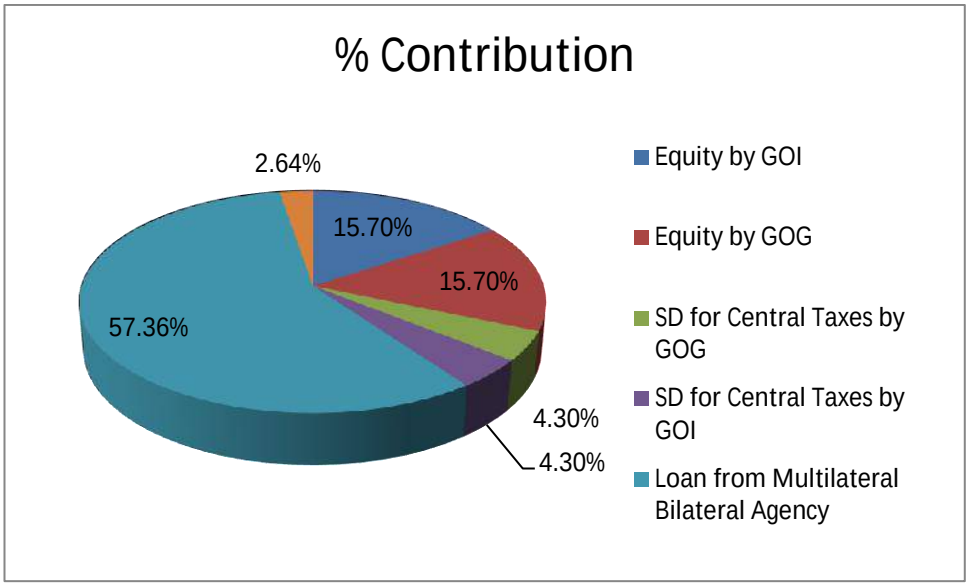


Table 20.14

Metro Corridor (1: Sarthana to Dream city and 2: Bhesan to Saroli)																								
CAPITAL COST-FIXED			11239													MB		0%	0.00%	0.00%			
CAPITAL COST - CURRENT			11838													Multilateral Loan		1.40%	100.00%	1.40%			
DOMESTIC FUNDING - BASE CASE																Front end Fee (one time)		0.20%		1.40%			
																Tenure of JICA Loan		12+28 Years					Rs/Crore
Year	Completion Cost	Additional Capital	Running Expenses	Depreciation	Replacement Cost	Total Cost	Fare box Revenue	PD & Advertisement	Income from cess on Property Tax & Stamp Duty	Total Revenue	Net Cash Flow for IRR	Equity from GOI & GOG+PPP component	Availability of cash	Cumulative cash	Cum. Loan	Loan	Repayment of Loan	IDC	Cumulative loan incl. IDC	Interest	Profit before Tax	Cash Balance	Cumulative Cash
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
2019 - 2020	1757					1757				0	-1757	1757	0	0	0	0	0	0	0				
2020 - 2021	2595					2595				0	-2595	1400	-1195	-1194	1194	1194	0	8	1202				
2021 - 2022	3106					3106				0	-3106	1000	-2106	-3301	3301	2106	0	42	3351				
2022 - 2023	2751					2751				0	-2751	988	-1763	-5064	5064	1763	0	59	5173				
2023 - 2024	1629					1629				0	-1629	1258	-371	-5435	5435	371	0	73	5617				
2024 - 2025	0	0	391	361		391	348	35	40	423	32	0	0	-5435	5435	0	0		5617	79	-407	-46	-46
2025 - 2026	0	0	401	361		401	400	40	81	521	120	0	0	-5435	5435	0	0		5617	79	-320	41	-5
2026 - 2027	0	0	413	361		413	525	53	83	661	248	0	0	-5435	5435	0	0		5617	79	-192	169	164
2027 - 2028	0	0	426	361		426	602	60	85	747	321		0	0	0	0	0		5617	79	-119	242	406
2028 - 2029	0	799	459	385		1258	726	73	87	886	-372		0	0	0	0	0		5617	79	-36	-450	-44
2029 - 2030	0	0	476	385		476	770	154	89	1013	537		0	0	0	0	0		5617	79	74	459	415
2030 - 2031	0	0	494	385		494	931	186	91	1208	714		0	0	0	0	0		5617	79	250	635	1050
2031 - 2032	0	0	515	385		515	988	198	93	1279	764		0	0	0	0	201		5417	79	300	485	1535
2032 - 2033	0	0	538	385		538	1191	238	95	1524	986		0	0	0	0	201		5216	76	526	710	2244
2033 - 2034	0	0	562	385		562	1265	253	98	1616	1054		0	0	0	0	201		5016	73	596	780	3024
2034 - 2035	0	0	532	385		532	1530	306	101	1937	1405		0	0	0	0	201		4815	70	949	1134	4158
2035 - 2036	0	0	569	385		569	1625	325	104	2054	1485		0	0	0	0	201		4614	67	1033	1217	5375
2036 - 2037	0	0	608	385		608	1967	393	107	2467	1859		0	0	0	0	201		4414	65	1409	1594	6969
2037 - 2038	0	0	650	385		650	2090	418	110	2618	1968		0	0	0	0	201		4213	62	1521	1705	8674
2038 - 2039	0	2504	733	460		3237	2432	486	114	3032	-205		0	0	0	0	201		4012	59	1780	-464	8210
2039 - 2040	0	0	783	460		783	2488	498	118	3104	2321		0	0	0	0	201		3812	56	1805	2064	10274
2040 - 2041	0	0	838	460		838	2901	580	122	3603	2765		0	0	0	0	201		3611	53	2252	2512	12786
2041 - 2042	0	0	896	460		896	2968	594	126	3688	2792		0	0	0	0	201		3411	51	2281	2541	15327
2042 - 2043	0	0	959	460		959	3466	693	130	4289	3330		0	0	0	0	201		3210	48	2822	3082	18408
2043 - 2044	0	0	1026	460		1026	3546	709	134	4389	3363		0	0	0	0	201		3009	45	2858	3117	21525
2044 - 2045	0	0	1099	506	1535	2635	4134	827	138	5099	2464		0	0	0	0	201		2809	42	3452	2222	23747
2045 - 2046	0	0	1178	554	1612	2790	4229	846	142	5217	2427		0	0	0	0	201		2608	39	3446	2187	25934
2046 - 2047	0	0	1262	554	0	1262	4937	987	146	6070	4808		0	0	0	0	201		2407	37	4217	4571	30505
2047 - 2048	0	0	1353	554	0	1353	5050	1010	150	6210	4857		0	0	0	0	201		2207	34	4269	4623	35128
2048 - 2049	0	2610	1502	632	0	4112	5820	1164	154	7138	3027		0	0	0	0	201		2006	31	4974	2795	37923
	11838.43	5913	18662	10854	3147	39561	56929	11126	2738	70793	6.90%	6403				5435	3611	182		1536	39740	37923	
											31232												

Sl No	Sources	Amount	% Contribution
1	Equity by GOI	1,487.69	15.70%
2	Equity by GOG	1,487.69	15.70%
3	SD for Central Taxes by GOG	407.36	4.30%
4	SD for Central Taxes by GOI	407.36	4.30%
5	Loan from Multilateral Bilateral Agency	5,435.16	57.36%
6	Upfront VCF and Proceeds from Monetisation	250.00	2.64%
7	Total Cost Excluding Land R&R and PPP	9475.27	100.00%
8	SD for Land including R&R by GOG	1,233.51	
9	SD for State Taxes by GOG	677.65	
10	Sub Total	11,386	
11	PPP Component	452.00	
12	Sub Total	11,838	
13	IDC for Loan to be borne by GOG	182	
14	Total Completion Cost	12,020	

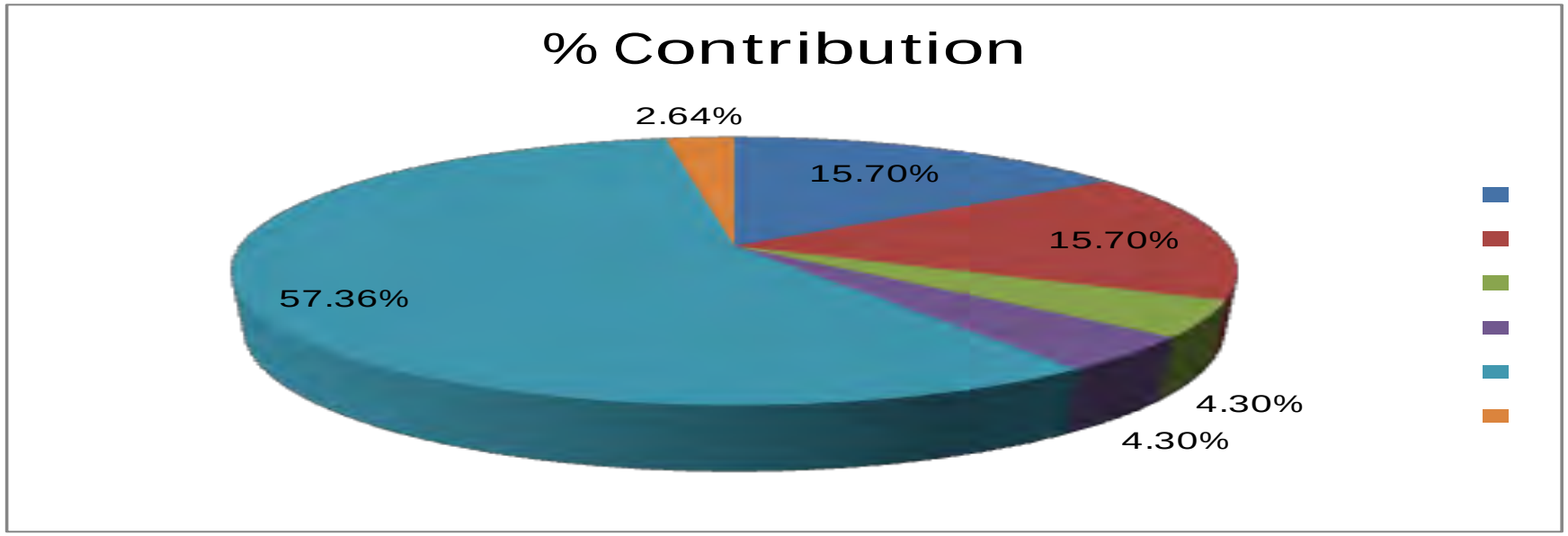
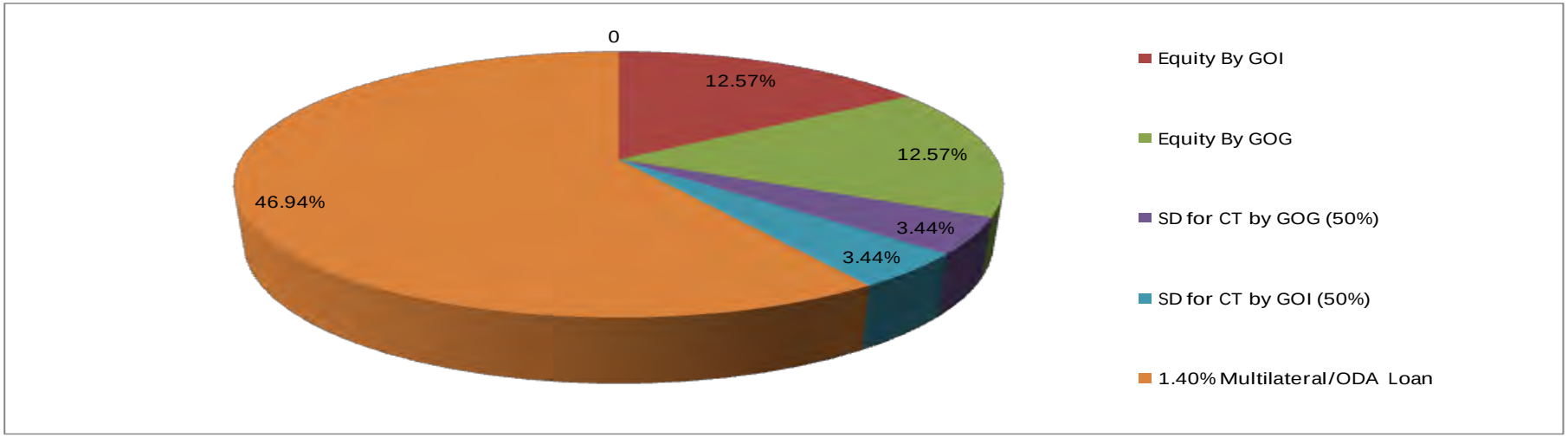


Table 20.15

	Metro Corridor (1: Sarthana to Dream city and 2: Bhesan to Saroli)																								
	CAPITAL COST-FIXED			11239														MB		0%	0.00%	0.00%			
	CAPITAL COST - CURRENT			11838														Multilateral Loan		1.40%	100.00%	1.40%			
	DOMESTIC FUNDING - BASE CASE																	Front end Fee (one time)		0.20%		1.40%			
																		Tenure of JICA Loan		12+28 Years				Rs/Crore	
	Year	Completion Cost	Additional Capital	Running Expenses	Depreciation	Replacement Cost	Total Cost	Fare box Revenue	PD & Advertisement	Income from cess on Property Tax & Stamp Duty	Income from TOD & VCF	Total Revenue	Net Cash Flow for IRR	Equity from GOI & GOG+PPP component	Availability of cash	Cumulative cash	Cum. Loan	Loan	Repayment of Loan	IDC	Cumulative loan incl. IDC	Interest	Profit before Tax	Cash Balance	Cumulative Cash
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
1	2019 - 2020	1756.83					1757					0	-1757	1430	-327	-327	327	327	0	2	329				
2	2020 - 2021	2594.57					2595					0	-2595	1927	-668	-994	994	668	0	9	1005				
3	2021 - 2022	3106.36					3106					0	-3106	917	-2189	-3184	3184	2189	0	40	3235				
4	2022 - 2023	2751.49					2751					0	-2751	905	-1846	-5030	5030	1846	0	57	5138				
5	2023 - 2024	1629.18					1629					0	-1629	1103	-526	-5556	5556	526	0	74	5738				
6	2024 - 2025	0.00	0	391	361		391	348	35	40	31	454	63	0	0	-5556	5556	0	0		5738	80	-378	-17	-17
7	2025 - 2026	0	0	401	361		401	400	-13	81	65	533	132	0	0	-5556	5556	0	0		5738	80	-310	51	34
8	2026 - 2027	0	0	413	361		413	525	9	83	67	684	271	0	0	-5556	5556	0	0		5738	80	-170	191	225
9	2027 - 2028	0	0	426	361		426	602	26	85	69	782	356		0	0	0	0	0		5738	80	-86	275	501
10	2028 - 2029	0	799	459	385		1258	726	58	87	71	942	-316		0	0	0	0	0		5738	80	18	-396	105
11	2029 - 2030	0	0	476	385		476	770	174	89	73	1106	630		0	0	0	0	0		5738	80	165	550	655
12	2030 - 2031	0	0	494	385		494	931	213	91	75	1310	816		0	0	0	0	0		5738	80	350	735	1390
13	2031 - 2032	0	0	515	385		515	988	230	93	77	1388	873		0	0	0	0	205		5533	80	408	588	1978
14	2032 - 2033	0	0	538	385		538	1191	277	95	79	1642	1104		0	0	0	0	205		5328	77	642	822	2800
15	2033 - 2034	0	0	562	385		562	1265	298	98	81	1742	1180		0	0	0	0	205		5124	75	720	900	3700
16	2034 - 2035	0	0	532	385		532	1530	361	101	83	2075	1543		0	0	0	0	205		4919	72	1086	1266	4966
17	2035 - 2036	0	0	569	385		569	1625	410	104	85	2224	1655		0	0	0	0	205		4714	69	1201	1381	6347
18	2036 - 2037	0	0	608	385		608	1967	483	107	88	2645	2037		0	0	0	0	205		4509	66	1586	1766	8113
19	2037 - 2038	0	0	650	385		650	2090	512	110	91	2803	2153		0	0	0	0	205		4304	63	1705	1885	9998
20	2038 - 2039	0	2504	733	460		3237	2432	585	114	94	3225	-12		0	0	0	0	205		4099	60	1972	-277	9721
21	2039 - 2040	0	0	783	460		783	2488	603	118	49	3258	2474		0	0	0	0	205		3894	57	1957	2212	11933
22	2040 - 2041	0	0	838	460		838	2901	690	122	0	3713	2875		0	0	0	0	205		3689	55	2361	2616	14549
23	2041 - 2042	0	0	896	460		896	2968	709	126	0	3803	2907		0	0	0	0	205		3484	52	2395	2650	17199
24	2042 - 2043	0	0	959	460		959	3466	814	130	0	4410	3451		0	0	0	0	205		3279	49	2942	3197	20397
25	2043 - 2044	0	0	1026	460		1026	3546	838	134	0	4518	3492		0	0	0	0	205		3074	46	2986	3241	23637
26	2044 - 2045	0	0	1099	506	1535	2635	4134	962	138	0	5234	2599		0	0	0	0	205		2869	43	3586	2351	25989
27	2045 - 2046	0	0	1178	554	1612	2790	4229	988	142	0	5359	2569		0	0	0	0	205		2664	40	3587	2324	28313
28	2046 - 2047	0	0	1262	554	0	1262	4937	1135	146	0	6218	4956		0	0	0	0	205		2459	37	4365	4714	33027
29	2047 - 2048	0	0	1353	554	0	1353	5050	1165	150	0	6365	5012		0	0	0	0	205		2254	34	4424	4773	37799
30	2048 - 2049	0	2610	1502	632	0	4112	5820	1327	154	0	7301	3190		0	0	0	0	205		2049	32	5136	2953	40752
		11838.43	5913	18662	10854	3147	39561	56929	12889	2738	1178	73734	7.38%	6282				5556	3689	182		1569	42648	40752	
													34172												

	Funding Pattern:-SPV Model			
	Equity By GOI	1487.66	12.57%	
	Equity By GOG	1487.66	12.57%	
	SD for CT by GOG (50%)	407.39	3.44%	
	SD for CT by GOI (50%)	407.39	3.44%	
	1.40% Multilateral/ODA Loan	5556.37	46.94%	
	Upfont VCF and Proceeds from Monetisation	250.00		
	Sub Total	9475.25		
	SD for Land by GOG	1233.50		
	SD for State Taxes by GOG	677.68		
	PPP	452.00		
	Total	11838.43	78.96%	
	IDC	182		
	Total	12020.43		



Metro Corridor (1: Sarthana to Dream city and 2: Bhesan to Saroli) — Table 20.16

	CAPITAL COST-FIXED					11239														MB		12%	100.00%	12.00%					
	CAPITAL COST - CURRENT					3870																0.00%	100.00%	0.00%					
	DOMESTIC FUNDING - BASE CASE																							12.00%					
																												Rs/Crore	
	Year			Completion Cost	VGF	Concessionaire Cost	Additional Capital	Running Expenses	Depreciation	Replacement Cost	Total Cost	Fare box Revenue	PD & Advertisement	Income from cess on Property Tax & Stamp Duty	Income from TOD & VCF	Total Revenue	Net Cash Flow for IRR	Concessioner Equity	Availability of cash	Cumulative cash	Cum. Loan	Loan	Repayment of Loan	IDC	Cumulative loan incl. IDC	Interest	Profit before Tax	Cash Balance	Cumulative Cash	Return on Equity (EIRR) Pre-Tax
	1					2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
1	2019	-	2020	1756.83	797.00	960					960					0	-960	245	-715	-715	715	715	0	43	758					-245
2	2020	-	2021	2594.57	1594.00	1001					1001					0	-1001	245	-756	-1470	1470	756	0	131	1644					-245
3	2021	-	2022	3106.36	1992.50	1114					1114					0	-1114	245	-869	-2339	2339	869	0	229	2742					-245
4	2022	-	2023	2751.49	2391.00	360					360					0	-360	245	-115	-2455	2455	115	0	288	3146					-245
5	2023	-	2024	1629.18	1195.50	434					434					0	-434	247	-187	-2641	2641	187	0	306	3638					-247
6	2024	-	2025			0	0	391	146		391	348	35	40	31	454	63	0	0	-2641	2641	0	0		3638	437	-519	-373	-373	-373
7	2025	-	2026			0	0	401	146		401	400	-13	81	65	533	132		0	-2641	2641	0	0		3638	437	-451	-305	-678	-305
8	2026	-	2027			0	0	413	146		413	525	9	83	67	684	271		0	-2641	2641	0	0		3638	437	-312	-166	-844	-166
9	2027	-	2028			0	0	426	146		426	602	26	85	69	782	356		0	0	0	0	0		3638	437	-227	-81	-925	-81
10	2028	-	2029			0	799	459	170		1258	726	58	87	71	942	-316		0	0	0	0	0		3638	437	-123	-752	-1677	-752
11	2029	-	2030			0	0	476	170		476	770	174	89	73	1106	630		0	0	0	0	121		3517	437	24	72	-1604	72
12	2030	-	2031			0	0	494	170		494	931	213	91	75	1310	816		0	0	0	0	121		3396	422	224	272	-1332	272
13	2031	-	2032			0	0	515	170		515	988	230	93	77	1388	873		0	0	0	0	121		3275	408	295	344	-988	344
14	2032	-	2033			0	0	538	170		538	1191	277	95	79	1642	1104		0	0	0	0	121		3153	393	541	590	-398	590
15	2033	-	2034			0	0	562	170		562	1265	298	98	81	1742	1180		0	0	0	0	121		3032	378	631	680	282	680
16	2034	-	2035			0	0	532	170		532	1530	361	101	83	2075	1543		0	0	0	0	121		2911	364	1009	1058	1340	1058
17	2035	-	2036			0	0	569	170		569	1625	410	104	85	2224	1655		0	0	0	0	121		2789	349	1136	1185	2524	1185
18	2036	-	2037			0	0	608	170		608	1967	483	107	88	2645	2037		0	0	0	0	121		2668	335	1532	1581	4105	1581
19	2037	-	2038			0	0	650	170		650	2090	512	110	91	2803	2153		0	0	0	0	121		2547	320	1663	1711	5817	1711
20	2038	-	2039			0	2504	733	245		3237	2432	585	114	94	3225	-12		0	0	0	0	121		2426	306	1942	-438	5378	-438
21	2039	-	2040			0	0	783	245		783	2488	603	118	49	3258	2474		0	0	0	0	121		2304	291	1938	2062	7440	2062
22	2040	-	2041			0	0	838	245		838	2901	690	122	0	3713	2875		0	0	0	0	121		2183	277	2354	2478	9918	2478
23	2041	-	2042			0	0	896	245		896	2968	709	126	0	3803	2907		0	0	0	0	121		2062	262	2400	2524	12442	2524
24	2042	-	2043			0	0	959	245		959	3466	814	130	0	4410	3451		0	0	0	0	121		1940	247	2959	3082	15524	3082
25	2043	-	2044			0	0	1026	245		1026	3546	838	134	0	4518	3492		0	0	0	0	121		1819	233	3014	3137	18661	3137
26	2044	-	2045			0	0	1099	309	2147	3246	4134	962	138	0	5234	1988		0	0	0	0	121		1698	218	3607	1648	20309	1648
27	2045	-	2046			0	0	1178	377	2254	3432	4229	988	142	0	5359	1927		0	0	0	0	121		1577	204	3601	1602	21912	1602
28	2046	-	2047			0	0	1262	377	0	1262	4937	1135	146	0	6218	4956		0	0	0	0	121		1455	189	4390	4645	26557	4645
29	2047	-	2048			0	0	1353	377	0	1353	5050	1165	150	0	6365	5012		0	0	0	0	121		1334	175	4460	4716	31273	4716
30	2048	-	2049			0	2610	1502	455	0	4112	5820	1327	154	0	7301	3190		0	0	0	0	121		1213	160	5184	2908	34181	2908
				11838	7970	3868	5913	18662	5649	4401	32845	56929	12889	2738	1178	73734	14.30%	1227				2641	2426	997		8150	41272	34181		16.00%
																	40889												

Funding Pattern:-SPV Model					
VGF By GOI			2121	20.00%	
VGF By GOG			4615	43.52%	
Equity by Concessionaire			1227	11.57%	
Concessionaire's debt @12% PA			2641	24.91%	
Total			10605	100.00%	
SD for Land by GOG			1234		
Sub Total			11838		11838.43
IDC			997		12835.43
Total			12835		24673.86
Total			13832.43		36512

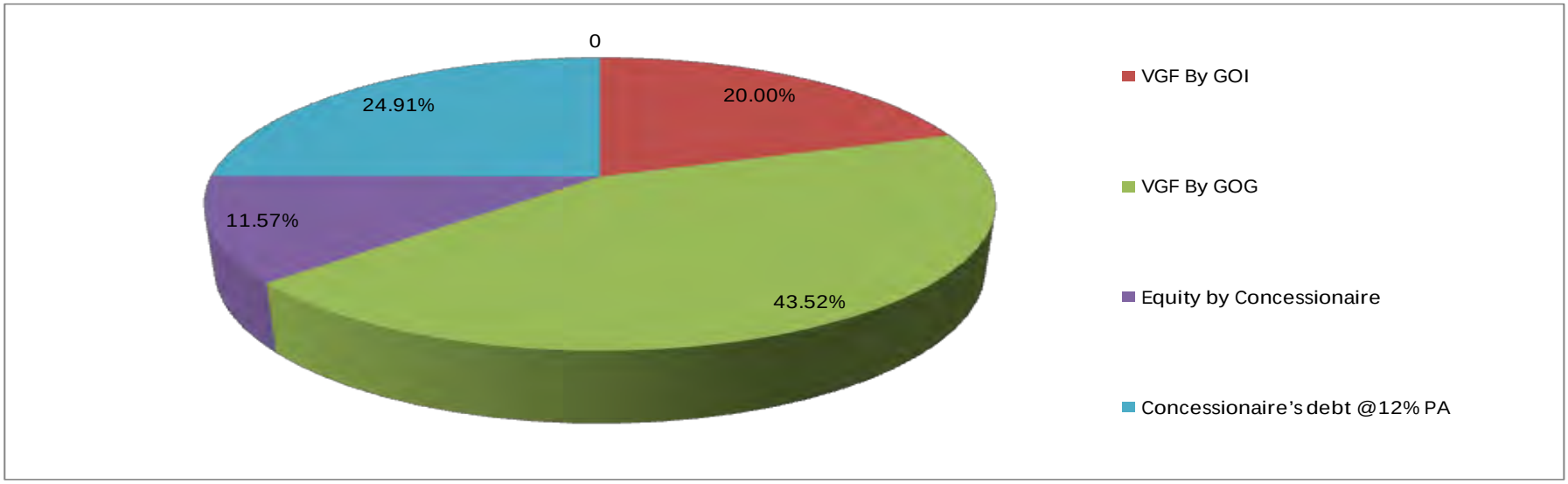
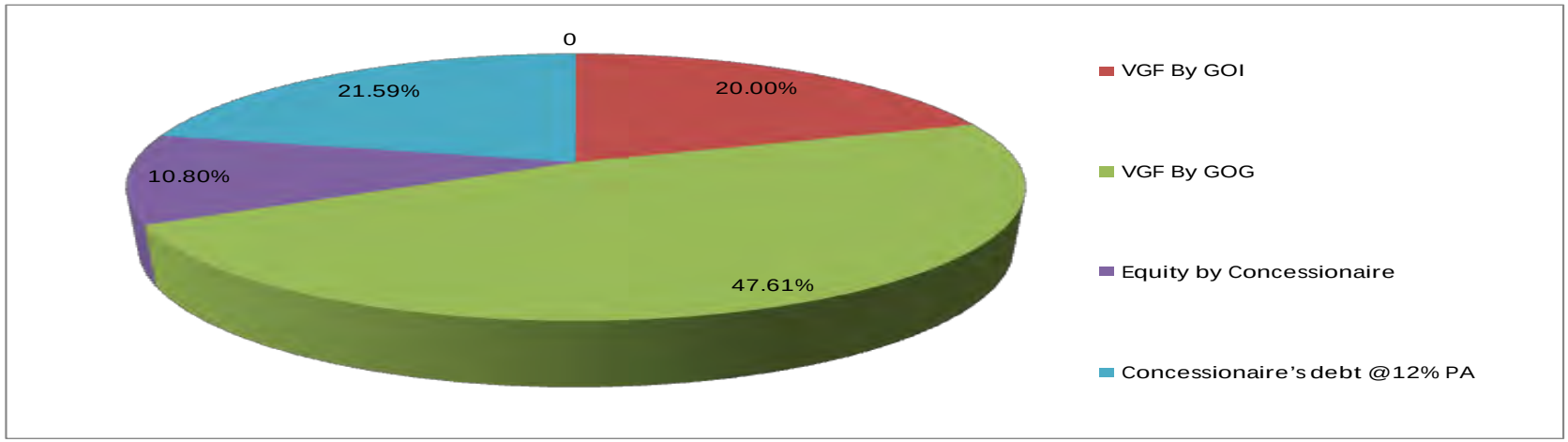


Table 20.17

	Metro Corridor (1: Sarthana to Dream city and 2: Bhesan to Saroli)																										
	CAPITAL COST-FIXED					11239													MB		12%	100.00%	12.00%				
	CAPITAL COST - CURRENT					3437															0.00%	100.00%	0.00%				
	DOMESTIC FUNDING - BASE CASE																						12.00%				
																										Rs/Crore	
	Year	Completion Cost	VGF	Concessionare Cost	Additional Capital	Running Expenses	Depreciation	Replacement Cost	Total Cost	Fare box Revenue	PD & Advertisement	Income from cess on Property Tax & Stamp Duty	Total Revenue	Net Cash Flow for IRR	Concessioner Equity	Availability of cash	Cumulative cash	Cum. Loan	Loan	Repayment of Loan	IDC	Cumulative loan incl. IDC	Interest	Profit before Tax	Cash Balance	Cumulative Cash	Return on Equity (EIRR) Pre-Tax
	1			2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
1	2019 - 2020	1756.83	840.33	916					916				0	-916	229	-687	-687	687	687	0	41	728					-229
2	2020 - 2021	2594.57	1680.67	914					914				0	-914	229	-685	-1372	1372	685	0	124	1537					-229
3	2021 - 2022	3106.36	2100.84	1006					1006				0	-1006	229	-777	-2149	2149	777	0	211	2525					-229
4	2022 - 2023	2751.49	2521.00	230					230				0	-230	229	-1	-2150	2150	1	0	258	2784					-229
5	2023 - 2024	1629.18	1260.50	369					369				0	-369	229	-140	-2290	2290	140	0	266	3190					-229
6	2024 - 2025			0	0	391	130		391	348	35	40	423	32	0	0	-2290	2290	0	0		3190	383	-480	-350	-350	-350
7	2025 - 2026			0	0	401	130		401	400	40	81	521	120		0	-2290	2290	0	0		3190	383	-393	-263	-614	-263
8	2026 - 2027			0	0	413	130		413	525	53	83	661	248		0	-2290	2290	0	0		3190	383	-265	-135	-748	-135
9	2027 - 2028			0	0	426	130		426	602	60	85	747	321		0	0	0	0	0		3190	383	-192	-62	-810	-62
10	2028 - 2029			0	799	459	154		1258	726	73	87	886	-372		0	0	0	0	0		3190	383	-109	-754	-1565	-754
11	2029 - 2030			0	0	476	154		476	770	154	89	1013	537		0	0	0	0	106		3084	383	1	48	-1516	48
12	2030 - 2031			0	0	494	154		494	931	186	91	1208	714		0	0	0	0	106		2977	370	190	237	-1279	237
13	2031 - 2032			0	0	515	154		515	988	198	93	1279	764		0	0	0	0	106		2871	357	253	300	-979	300
14	2032 - 2033			0	0	538	154		538	1191	238	95	1524	986		0	0	0	0	106		2765	345	488	535	-443	535
15	2033 - 2034			0	0	562	154		562	1265	253	98	1616	1054		0	0	0	0	106		2658	332	568	615	172	615
16	2034 - 2035			0	0	532	154		532	1530	306	101	1937	1405		0	0	0	0	106		2552	319	932	979	1151	979
17	2035 - 2036			0	0	569	154		569	1625	325	104	2054	1485		0	0	0	0	106		2446	306	1025	1073	2224	1073
18	2036 - 2037			0	0	608	154		608	1967	393	107	2467	1859		0	0	0	0	106		2339	293	1411	1459	3683	1459
19	2037 - 2038			0	0	650	154		650	2090	418	110	2618	1968		0	0	0	0	106		2233	281	1533	1581	5264	1581
20	2038 - 2039			0	2504	733	229		3237	2432	486	114	3032	-205		0	0	0	0	106		2127	268	1802	-579	4685	-579
21	2039 - 2040			0	0	783	229		783	2488	498	118	3104	2321		0	0	0	0	106		2020	255	1837	1959	6644	1959
22	2040 - 2041			0	0	838	229		838	2901	580	122	3603	2765		0	0	0	0	106		1914	242	2294	2417	9061	2417
23	2041 - 2042			0	0	896	229		896	2968	594	126	3688	2792		0	0	0	0	106		1808	230	2333	2456	11517	2456
24	2042 - 2043			0	0	959	229		959	3466	693	130	4289	3330		0	0	0	0	106		1701	217	2884	3007	14524	3007
25	2043 - 2044			0	0	1026	229		1026	3546	709	134	4389	3363		0	0	0	0	106		1595	204	2929	3052	17576	3052
26	2044 - 2045			0	0	1099	293	2147	3246	4134	827	138	5099	1853		0	0	0	0	106		1489	191	3515	1555	19131	1555
27	2045 - 2046			0	0	1178	361	2254	3432	4229	846	142	5217	1785		0	0	0	0	106		1382	179	3500	1500	20631	1500
28	2046 - 2047			0	0	1262	361	0	1262	4937	987	146	6070	4808		0	0	0	0	106		1276	166	4281	4536	25167	4536
29	2047 - 2048			0	0	1353	361	0	1353	5050	1010	150	6210	4857		0	0	0	0	106		1170	153	4343	4598	29764	4598
30	2048 - 2049			0	2610	1502	439	0	4112	5820	1164	154	7138	3027		0	0	0	0	106		1063	140	5057	2780	32544	2780
		11838	8403	3435	5913	18662	5249	4401	32411	56929	11126	2738	70793	14.34%	1145				2290	2127	900		7146	39736	32544		16.00%
														38382													

			`	
VGF By GOI			2121	20.00%
VGF By GOG			5049	47.61%
Equity by Concessionaire			1145	10.80%
Concessionaire's debt @12% PA			2290	21.59%
Total			10605	100.00%
SD for Land by GOG			1233	
Sub Total			11838	
IDC			900	
Total			12738	



Chapter-21

ECONOMIC APPRAISAL

21.1 INTRODUCTION

Economic benefits are social and environmental benefits which are quantified and then converted into money cost and discounted against the cost of construction and maintenance for deriving Economic Internal Rate of Return (EIRR). When actual revenue earned from fare collection, advertisement and property development are discounted against construction and maintenance cost, interest (to be paid) and depreciation cost, Financial Internal rate of Return (FIRR) is obtained. Therefore, EIRR is viewed from socio-economic angle while FIRR is an indicator of financial profitability and viability of any project

21.1.1 Economic appraisal of a project starts from quantification of measurable economic benefits in economic money values, which are basically the savings of resource cost due to introduction of the metro line. Economic savings are derived from the difference of the cost of the same benefit components under 'with' and 'without' metro line. Total net savings/or benefit is obtained by subtracting the economic cost of the project (incurred for construction (Capital) and maintenance (recurring) costs for the metro line) from the benefits out of the project in each year. Then the benefit value which would be negative during initial years becomes positive as years pass. Internal rate of return and benefit cost ratio are derived from the stream.

21.1.2 The sources from where economic savings occur are identified first. Although there are many kinds of primary, secondary and tertiary benefits, only the quantifiable components can be taken to measure the benefits. These components are quantified by linking with the number of passengers shifted and the passenger-km saved by the trips which are shifted from road / rail based modes to metro. It may be observed that first four benefit components given in **Table 21.1** are direct benefits due to shifting of trips to metro, but other benefit components are due to decongestion effect on the road.

Benefit components were first estimated applying market values then were converted into respective Economic values by using separate economic factors which are also given in **table 21.1**.

Table 21.1:Benefit Components due to Metro

Benefit Components
Annual Time Cost Saved by Metro Passengers
Annual Fuel Cost Saved by Metro Passengers
Annual Vehicle Operating Cost Saved by Metro Passengers
Emission Saving Cost
Accident Cost
Annual Time Cost Saved by Road Passengers
Annual Fuel Cost Saved by Road Passengers
Annual Infra Structure Maintenance Cost

21.2 VALUES ADOPTED FOR SOME IMPORTANT VARIABLES

Benefit components are converted (by applying appropriate unit cost) to money values (Rs.). Derivation procedures of some of the values used for economic analysis are shown in **Table 21.2**.

Table21.2: Values adopted for some important variables

	Values	Important variables
1	Rs.1.76/min (2017)	Time Cost derived from passenger's monthly income level (House hold Survey).
2	Market Rate (2017)	Fuel Cost (value of Petrol, Diesel and CNG).
3	Table 21.3	Vehicle Operating Cost (Derived from Life Cycle Cost of different passenger vehicles per km)
4	Table 21.4 (CPCB)	Emission (gm/km as per CPCB and UK Norms) Emission Saving Cost (adopted for Indian conditions in Rs/ton).
5	Table 21.5 (Accident Rate & Cost)	Accident Rate (No of fatal and all accidents per one Cr.KM). Accident costs are derived at current rate from published papers.
6	64.72%	Vehicle/ Passenger ratio (derived from traffic volume count and modal split within study area as reported in chapter2)
7	Road User Cost Study Model (CRRI-2010)	Fuel Consumption of vehicles at a given speed is derived
8	Rs.1.0 / vehicle km	Infra Structure Maintenance Cost is derived from published values on annual expenditure on roads and traffic and annual vehicle km
9	21.27 min	Weighted Average of all mode travel time saved for average trip length km journey after Shifting (Derived)
10	23.01 kmph	Operating Speed of all vehicles (weighted average)

Table21.3:
Vehicle Operating Cost (Other than fuel cost)in Rs.

Per Veh km Cost	2 Wh (MC)	2 Wh (SC)	4 Wh (Large)	4 Wh (Small)	Cab/ Sedan	RTV /Magic	3 Wh (Auto)	*BRT Bus*	Medium Bus	Mini Bus
Maintenance	0.14	0.12	3.19	2.06	6.16	2.62	1.47	*17.05*	7.12	3.74
Capital	2.88	2.40	19.23	12.02	24.03	9.61	5.77	*60.08*	24.03	14.42
Total	3.03	2.52	22.41	14.08	30.20	12.23	7.24	*77.14*	31.15	18.16

Table21.4:
Vehicle Emission factors 2021 (CPCB) and Cost in Rs.

Pollutants	2 Wh (MC)	2 Wh (SC)	4 Wh (Large)	4 Wh (Small)	Cab/ Sedan	RTV/ Magic	3 Wh (Auto)	BRT Bus	Medium Bus	Mini Bus
CO	1.400	1.400	0.580	1.390	1.500	2.450	1.240	3.900	2.479	3.720
HC	0.700	1.320	0.050	0.150	0.200	0.750	0.165	0.200	0.826	0.160
NOX	0.300	0.080	0.450	0.120	0.150	0.120	0.579	7.500	7.200	6.530
PM	0.050	0.050	0.050	0.020	0.020	0.080	0.165	0.240	0.579	0.240
SO2	0.001	0.002	0.003	0.003	0.004	0.006	0.010	0.025	0.017	0.025
ALL	2.45	2.85	1.13	1.68	1.87	3.41	2.16	11.87	11.10	10.68
CO2	28.58	24.99	156.55	139.51	154.00	177.00	160.00	800.00	787	360.00
Cost of all other emission Rs200/kg					Cost of CO2 emission Rs20/kg					

Table21.5: Accident Rate $ and Cost in Rs.(Year1999 Rate)

ACCIDENT	RATE	COST
FATAL	0.337	535849
INJURY	1.279	141351
MINOR	0.759	35338

$On the basis of a research paper

Traffic parameter values used for economic analysis are given inTable21.6.

Table21.6: Traffic parameter values Surat METRO

TRAFFICINPUT&	2024	2026	2031	2036	2041	2046
Trips/day	423757	533890	721231	974310	1091636	1223090
Line Length(km)	39.01	39.01	39.01	39.01	39.01	39.01
Average Trip length (km)	8.317	8.317	8.43	8.549	8.61	8.675
Passenger km	3524386	5605845	7572927	10230255	11462176	12842445
Passenger km/km	90346	143688	194108	262220	293796	329175

(As per Chapter 2 of DPR)

21.3 METRO CONSTRUCTION COST

21.3.1 Total actual cost of metro construction (**Capital Cost**) is derived for the year of estimation (2016) after considering cost of all major components such as Relocation and Rehabilitation (RR), Civil construction for underground and elevated portions, Stations and Depots, Track laying, Signaling and Telecommunication, Power traction line, Rolling stock, Manpower etc. On this

cost, 5% escalation factor per year is applied to obtain the **Completion Cost** as this expenditure is spread over next 5 years.

Recurring Cost includes energy cost, maintenance cost, and operation cost. Economic analysis period is taken from 2019-20 to 2048-49 out of which 5 years (2019-2024) are construction years and operation will start in 2024-2025. Additional capital expenditure may be incurred in the years 2028-29 (Rs.799Cr.), 2038-39 (Rs.2504Cr.), & in 2048-49 (Rs.2610Cr.) for purchase of more rolling stock. Major replacement cost (Rs.1535Cr.) and (Rs.1612 Cr.) is contemplated in 2044-2045 and 2045-46. This cost stream is generated with all taxes. Detail is shown in **Table 21.7.**

Table21.7 Estimated Completion and Recurring Cost

Year	Year	Completion Cost	Additional Capital	Running Expenses	Replacement Cost	Total Cost
Start	**Ending**	2	3	4	6	7
2019	2020	1757				1757
2020	2021	2595				2595
2021	2022	3106				3106
2022	2023	2751				2751
2023	2024	1629				1629
2024	2025	0	0	391		391
2025	2026	0	0	401		401
2026	2027	0	0	413		413
2027	2028	0	0	426		426
2028	2029	0	799	459		1258
2029	2030	0	0	476		476
2030	2031	0	0	494		494
2031	2032	0	0	515		515
2032	2033	0	0	538		538
2033	2034	0	0	562		562
2034	2035	0	0	532		532
2035	2036	0	0	569		569
2036	2037	0	0	608		608
2037	2038	0	0	650		650
2038	2039	0	2504	733		3237
2039	2040	0	0	783		783
2040	2041	0	0	838		838
2041	2042	0	0	896		896
2042	2043	0	0	959		959
2043	2044	0	0	1026		1026
2044	2045	0	0	1099	1535	2634
2045	2046	0	0	1178	1612	2790
2046	2047	0	0	1262	0	1262
2047	2048	0	0	1353	0	1353
2048	2049	0	2610	1502	0	4112

21.4 ECONOMIC BENEFITS

Benefits in terms of money value are **estimated directly from traffic figure for the horizon years** (2024, 2026, 2031 and 2041) and values for other years are interpolated on the basis projected traffic. Market values are used for calculating costs and then appropriate economic factors (see table21.1) are applied. For each year values of each benefit components are obtained and thus benefit stream is estimated. Accrued Benefit Components are shown in **Table21.8** and in **Figure 21.1.**

Table 21.8 Accrued Benefit Values

Benefit Components	Accrued Benefit Values between 2024-2048 in Cr.Rs.	Percent
TIME COST	106044	56.1%
VOC+FUEL COST	64101	33.9%
OTHER	18753	9.9%
	188898	100.0%

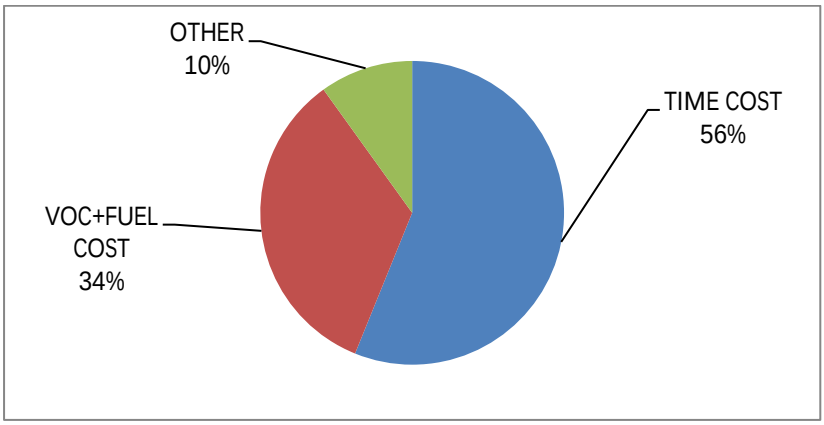


Figure 21.1 Percent of Accrued Benefits

It is seen that total cost of time saving by shifted passengers and road users is about 56%. **Figure21.1** also shows that benefits are also coming from VOC cost including fuel cost (34%), by shifted metro passengers and relieved road passengers. Environmental benefit from emission reduction, accident reduction and road maintenance cost (together) is10%.

Benefit stream is given in **Table 21.9**

Table 21.9 Year wise Economic Cost Values (Rs. Cr.) of Benefit Components (Stream)

YEAR		Daily Passenger Boarding on Metro	Annual Time Cost Saved by Metro Passengers in Cr. Rs.	Annual Fuel Cost Saved by Metro Passengers in Cr. Rs.	Annual Vehicle Operating Cost Saved saved by Metro Passengers in Cr. Rs.	Annual Time Cost Saved by Road Passengers in Cr. Rs.	Annual Fuel Cost Saved by Road Passengers in Cr. Rs.	Emission Saving Cost in Cr. Rs.	Accident Cost in Cr. Rs.	Annual Infra Structure Maintenance Cost	Total Savings
2024	2025	423757	818	211	355	52	15.4	37	2.21	84	1574
2025	2026	475646	964	238	418	61	18.1	44	2.59	99	1846
2026	2027	533890	1136	270	493	72	21.3	53	3.03	117	2165
2027	2028	566991	1267	289	551	80	23.9	60	3.36	131	2407
2028	2029	602145	1413	311	616	90	26.7	69	3.72	146	2676
2029	2030	639478	1576	335	691	101	30.1	79	4.12	164	2980
2030	2031	679125	1757	361	775	113	34.0	91	4.56	184	3319
2031	2032	721231	1959	389	868	127	38.3	104	5.04	206	3696
2032	2033	751752	2144	411	954	139	42.3	117	5.45	226	4040
2033	2034	783565	2347	435	1049	153	46.7	131	5.90	249	4416
2034	2035	816724	2568	460	1152	168	51.6	147	6.37	273	4827
2035	2036	851287	2811	486	1266	185	57.0	166	6.87	300	5278
2036	2037	887312	3076	514	1391	203	62.9	186	7.41	330	5771
2037	2038	924861	3367	560	1574	230	69.5	221	8.22	373	6402
2038	2039	964000	3685	593	1733	253	76.8	249	8.87	411	7009
2039	2040	1004794	4033	629	1907	278	84.8	281	9.57	452	7675
2040	2041	1047315	4487	677	2134	312	95.0	322	10.55	506	8542
2041	2042	1091636	4991	729	2388	349	106.5	369	11.64	566	9509
2042	2043	1116745	5553	784	2672	390	119.4	422	12.84	633	10587
2043	2044	1142431	6178	843	2990	437	133.8	484	14.16	708	11788
2044	2045	1168708	6874	907	3345	488	150.0	555	15.62	793	13128
2045	2046	1195590	7647	976	3743	546	168.1	635	17.22	887	14621
2046	2047	1223090	8508	1051	4188	611	188.5	728	18.99	992	16286
2047	2048	1251222	9465	1130	4687	684	211.3	834	20.95	1110	18143
2048	2049	1280002	10530	1216	5244	766	236.8	956	23.10	1243	20214

In this area, personalized modes (car and two wheelers) are dominant which have made vehicle by passenger ratio very high (64.72%).Average modal split (within motorized vehicle class) obtained from the traffic volume count and home interview survey shows that 65.61 % passenger (85.06% vehicular) trips are made by private modes and 32.8% passenger (14.90 % vehicular) are made by IPT. Trips carried by Bus is about 1.59% (0.05%vehicular). Expected average shift of passengers from road modes to metro is about 19.31% as may be seen in **table21.10**.

Table21.10:
Average modal split in study area

Source: Household Survey, CoE-UT,CEPT University	2Wh (MC)	2Wh (SC)	4Wh (Large)	4Wh (Small)	Cab /Sedan	3Wh (Auto)	Cluster /BRT Bus	Medium Bus
Share Passenger	41.44%	20.72%	1.73%	1.73%	15.54%	17.27%	0.62%	0.97%
Share Vehicle	58.21%	24.63%	0.89%	1.33%	6.00%	8.89%	0.02%	0.03%
Mode Split of Shifted Passengers	8.29%	4.14%	0.26%	0.26%	2.33%	3.45%	0.09%	0.48%

21.5 ECONOMICPERFORMANCEINDICATORS

After generating the cost and benefit stream table, economic performance indicators are derived and are presented in **table 21.11**. Project period is 2019-2048, With reference to completion cost of capital with tax, **EIRR** is found to be **17.20 %** and **B/C ratio** as **4.77** and with 12% discount, EIRR is **4.65%** and B/C ratio is **1.63**. NPV without discount is Rs **149337** Cr. And with 12% discount rate, NPV is Rs. **8287 Cr**. which shows that the project is economically viable.

Table21.11
Economic Indicator Values (on 2048-49)

	WITHOUT DISCOUNT	WITH DISCOUNT (12%)
Cumulative cost (Cr.)	39561	13127.87
Cumulative benefit(Cr.)	188898	21414.43
Benefit Cost Ratio	4.77	1.63
NPV(Cr.)	149337	8287
EIRR	17.20%	4.65%

21.6 SENSITIVITYANALYSIS

Sensitivity test on EIRR and B/C ratio values were carried out and the output is given in the **table 21.12**. 2047-48 is taken for the year of comparison. It is seen that even in the worst case (20% less traffic and 20 higher than estimated cost) EIRR value remained above 14.02% and after discount B/C ratio is more than one.

Table21.12
Sensitivity of EIRR

SENSITIVITY		WITHOUT DISCOUNT			WITH DISCOUNT		
TRAFFIC	COST	EIRR	B/C	COST	EIRR	B/C	COST
0%	0.00%	17.20%	4.77	39561	4.65%	1.63	13127.87
-10%	0.00%	16.60%	4.52	39561	4.11%	1.55	13127.87
-20%	0.00%	15.97%	4.27	39561	3.55%	1.46	13127.87
0%	10%	16.15%	4.34	43517	3.70%	1.48	14440.66
0%	20%	15.20%	3.98	47473	2.86%	1.36	15753.44
-10%	10%	15.56%	4.11	43517	3.18%	1.40	14440.66
-20%	20%	14.02%	3.56	47473	1.81%	1.22	15753.44

21.7 Quantified Benefits.

Environmental Benefits in monitory values are shown in previous tables. These benefits are estimated (in terms of quantity) first and then converted into money value. Quantity estimates are shown in **table 21.13 & 21.14.**

It is seen that in 2024-25, Metro passenger time saving will be 5.51 Cr hours, CO2 gas emission saving will be 39.87 thousand ton. Other emissions are toxic gases which will also be reduced 2.3 thousand ton due to less emission from the vehicles. 28 fatal road accidents will be reduced. Vehicle km saved will be 23 thousand km.

Table21.13
Environmental Benefits Quantified

Tons/Year	2024	2025	2026	2027	2028
CO	1278	1434	1610	1714	1826
HC	746	837	939	1000	1065
NOX	187	210	236	251	267
PM	54	61	68	72	77
SO2	3.45	3.87	4.34	4.63	4.93
CO2	39869	44751	50231	53492	56965
Total Emission Saved	42137	47296	53088	56535	60205
All Other Accident Saved	171	192	215	229	244
Fatal Accident Saved	28	32	36	38	40

Table21.14
Travel Benefits Quantified

Quantified Benefits in Horizon Years	2024	2025	2026	2027	2028
Annual Time Saved by Metro Passengers in Cr. Hr.	5.51	6.18	6.94	7.37	7.82
Annual Fuel Saved by Metro Passengers in thousand Tons.	36.75	41.63	47.17	50.72	54.55
Daily vehicles reduced (off the road)	49482	55542	62343	66390	70701
CO2 reduced in thousand tons	39.87	44.75	50.23	53.49	56.96
Other gases reduced in thousand tons	2.268	2.546	2.857	3.043	3.240
Reduced No of Fatal Accidents in Year	28.34	31.81	35.70	38.02	40.49
Reduced No of Other Accidents in year	142.36	159.79	179.36	191.01	203.41
Annual Vehicle km Reduced in Thousand Km.	23.04	25.86	29.03	30.91	32.92

CHAPTER - 22

IMPLEMENTATION PLAN

22.1 WAY FORWARD FOR IMPLEMENTING SURAT METRO PROJECT (Phase-I)

On receipt of the Detailed Project Report, following action will be required for implementing the Surat Metro (Phase-I):-

Approval to the Detailed Project Report to be taken from Gujarat State Government (Cabinet approval).

The DPR to be forwarded to the Ministry of Urban Development(GOI),Planning Commission and Finance Ministry with the request for approving the Metro project and for financial participation.

Signing of an MOU between Gujarat State Government and Government of India giving all details of the Joint Venture bringing out the financial involvement of each party, liability for the loans raised, the administrative control in the SPV, policy in regard to fare structure, operational subsidy, if any, etc.

A dedicated Special Purpose Vehicle (SPV) shall need to be set up, as Metro Company headquartered in other city namely Ahmedabad/ Gandhi Nagar may not be in a position to effectively implement Surat Metro Project and its subsequent Operation & Maintenance. An organization, depending upon the methodology of implementations, needs to be developed. In case the project is executed by appointing General Consultants, sufficient number of Engineers in SPV organization will have to be taken on deputation or from market. In case the work is intended to be done on deposit terms through an organization like DMRC, SPV's organization may be kept lean and thin.

The Metro Railways (Amendment) Act-2009 can readily be made use of for implementation of Surat Metro by declaring Surat City as Metropolitan Area in terms of clause- c of section 243 P of Constitution.

Request to GOI for a notification for making the Metro Railways (Amendment) Act 2009 applicable to Surat Metro.

In view of new Metro Rail Policy 2017, the State Government should formulate the funding plan for executing this project and get the same approved by the Government of India. The loan portion of the funding will have to be tied up by State Government in consultation with the Government of India.

The Government should freeze all developments along the corridors suggested. For any constructions near the proposed alignment a system of No Objection Certificate should be introduced as per provisions of Metro Railway Act so that infructuous expenditure at a later stage is avoided.

The Metro Railways (Amendment) Act-2009 can readily be made use of for implementation of Surat Metro by declaring Surat City as Metropolitan Area.

In view of new metro policy Unified Metropolitan Transport Authority (UMTA) should be formed within one year of Submission of Proposal of any metro project to GOI.

22.2 IMPLEMENTATION ON DELHI METRO/CHENNAI METRO MODEL

Special Purpose Vehicle (SPV) shall have to take action for appointment of General Consultants for project management including preparation of tender documents. Till the General Consultants are in position, SPV should appoint an interim Consultant for all preliminary and enabling jobs such as land acquisition, detailed design of civil structures, utility diversions, etc.

The proposed date of commissioning of the both corridor with suggested dates of important milestones is given in Table 22.1

Table 22.1 -Implementation Schedule through DMRC model Phase I

S. No.	Item of Work	Completion Period
1	Submission of Final DPR to State Govt.	D
2	Approval of DPR by State Government	D + 0.5momth
3	Submission of DPR for Approval of Ministry of Urban Development (MoUD).	D + 1month
4	Appoint interim Consultant for preliminary works	D + 3months
5	Approval of Central Government	D+4months

6	Arrangement of Funding Source/Agency	D +6months
7	Appoint General Consultant	D +12months
8	Tendering, Execution of works and Procurement of equipments, coaches and installations	D+57months
9	Testing and Commissioning	D +60moths
10	Revenue Operation	D +60months

22.3 ORGANISATIONAL SET-UP OF SPECIAL PURPOSE VEHICLE (SPV)

The SPV organization should be very lean but effective. It should consist of a non-executive Chairman, a Managing Director with full Executive Powers (in Schedule 'A') and three Functional Directors (in Schedule 'B') including Director (Finance). All the three Functional Directors will be full members of the Management Board. The Directors will be assisted by Heads of Departments in each of the major disciplines and they in turn will have Deputy Heads of Departments. The organization should be basically officer-oriented with only Personal Assistants and Technical Assistants attached to senior officers by eliminating unproductive layers of staff such as Peons, Clerks, etc. We strongly recommend that the total organizational strength is limited to 70 to 80 eliminating too many tiers to enable faster decision-making.

It is necessary for the SPV officers to get exposed to the Metro technology and Metro culture through study tours of some of the selected foreign Metros and Delhi/Calcutta Metros.

Implementing a metro project in a congested metropolis is indeed a challenge. In sheer size, magnitude and technical complexity there are no parallels to metro projects. Further, these projects are to be carried out in difficult urban environment without dislocating city life, while at the same time preserving the environment. The project involves integration of a number of complex technical systems some of these technologies used in these systems are totally new to the country each one of which is a major project by itself. Interfacing various system contracts is a difficult and highly skilled exercise. Side by side, timely and adequate funds have to be assured for implementation and lands, without encumbrances, have to be taken possession of in time. Clearances from the local authorities have to be taken which includes permission to cut trees, diversion of utilities, management of road traffic, etc., all of which will call for an efficient and competent project implementing agency.

Metro projects cannot be executed the way Government agencies execute projects in this country. Timely completion is very important to safeguard the financial viability. Competent and skilled technical personal to man such an organization are difficult to mobilize. In fact such experienced persons are not readily available in the country. Being a rail based project, for most of the systems such as rolling stock, signaling, telecommunication, traction power supply, etc., persons with railway background would be necessary. As systems & construction technology used in metro are much more advanced and sophisticated than the one used in Railways as these have to suit dense urban areas, Metro experience will enable faster & smoother execution and thus is desirable &therefore should be preferred.

Since newly set up SPV may not have the required expertise and experienced manpower to check and monitor the General Consultants it may be necessary to engage Prime Consultants from the very start of GC's assignment who will do this job on behalf of Surat Metro.

Delhi Metro Rail Corporation can also be considered straightaway for being appointed as General Consultant to Surat Metro which will reduce the construction time by 4 to 6 months.

22.4 CONTRACTS

22.4.1 Civil Works

It is proposed to carry out the civil works through following construction contracts-

(a) Viaduct Construction-It is suggested that each contract can be limited to about 5 to 6 kms in length including stations. Corridor wise number of contracts is expected to be as follows:

Table 22.2 -Construction Strategy (Elevated Portion)

Corridor	Length of Elevated Section (km)	Elevated Stations (Nos.)	Proposed Contract (i/c stations)
Sarthana to Dream City	15.14	14	3 Nos
Bhesan to Saroli	18.74	18	4 Nos.

(b) Underground Section: Following contracts are suggested for underground section including stations:

Table 22.3 - Construction Strategy (Underground Portion)

Corridor	Underground Section		Proposed Contracts
	Length (km)	Station (Nos.)	Underground section including stations
Sarthana to Dream City	6.47 Kms	6	2 Nos

Architectural finishes, fire fighting arrangements and general electrification will form part of civil contracts.

22.4.2 System Contracts

Design, construct and installation for Traction and Power Supply.

Design, construct and installation of Signal and Telecommunication works.

Design, construct and installation of lifts.

Design, construct and installation of escalators.

Design, construct and commissioning of Automatic Fare Collection System.

Design and supply of rolling stock.

Installation of track in Depot and on main line.

Design and installation of Signages.

22.4.3 Depot Contracts

The contracts are required for Civil and E&M works at Dream City Depot and Bhesan Depot. Each depot will have one package for civil works.

The number of contracts for supply of Depot Equipment may be decided as and when the work is in progress.

22.5 HIGH POWER COMMITTEE (HPC)

During the implementation of the project several problems with regard to acquisition of land, diversion of utilities, shifting of structures falling on the project alignment, rehabilitation of project affected persons, etc. are likely to arise. For expeditious resolution of these problems, an institutional mechanism needs to be

set up at the State Government level. Towards this end, it is recommended that a High Power Committee under the chairmanship of Chief Secretary, Gujarat should be set up. Other members of this Committee should be Secretaries of the concerned Departments of the State Government and Heads of civic bodies who will be connected in one way or the other with the implementation of the project. Commissioner of Surat Municipal Corporation (SMC) and Chief Executive Officer of Surat Urban Development Authority (SUDA)should also be the member of this committee. This Committee should meet once a month and sort out all problems brought before it by the SPV. For Delhi Metro also such a High Power Committee was set up and it proved very useful in smooth implementation of the Delhi Metro rail project.

22.6 LEGAL COVER FOR SURAT METRO

Construction of Surat Metro should commence soon. Thus there is immediate need to have a legislation to provide legal cover to the construction stage of Surat Metro.

Implementation of proposed Surat Metro can now be done under “The Metro Railways (Amendment) Act 2009”. The copies of the Gazette notification and the amendment are enclosed with this chapter.

22.7 CONCESSIONS FROM GOVERNMENT

Metro rail projects need very heavy investment. Loans have invariably to be taken to fund a part of the capital cost of the projects. These projects yield low financial internal rate of return. With reasonable fare level, servicing of these loans often pose problems. To make the project financially viable, therefore, the fares need to be substantially increased to socially un-acceptable levels. This results in the ridership coming down significantly, as it is sensitive to increases in the fare level. Thus the very objective of constructing the metro rail system to provide an affordable mode of mass travel for public is defeated. It, therefore, becomes necessary to keep the initial capital cost of a metro project as low as possible so that the fare level of the metro system can be kept at reasonable level.

As in the case of Delhi Metro, the State Government should exempt/reimburse the Gujarat Value Added Tax (VAT),GST etc to Surat Metro. It should also exempt the following: -

Tax on electricity required for operation and maintenance of the metro system.

Municipal Taxes.

As per the earlier policy 50% of the Central Taxes were to be paid by GOI as subordinate Debt and balance 50% were paid by the concerned State Government. However, the post- GST regime, GOI intends to pay only 1/3rd of GST as subordinate debt and balance 2/3rd needs to be paid by the concerned state Government. Gujarat state Government may pursue the central Government to extend the same benefit to Surat Metro also.

22.8 NEED FOR DEDICATED FUND FOR METRO PROJECTS

We also strongly recommend that the State Government start building up funds for the project through dedicated levies as has been done by other State Governments notably Karnataka.

To enable the State Governments to provide their share of equity in the Special Purpose Vehicles set up for such projects, it would be necessary to constitute a Special Metro Fund at the State Government level. The State Government should resort to imposition of dedicated levies for raising resources for these Funds. Areas where such dedicated levies are possible are given below:

Cess on the tax levies for registration of road vehicles.

A Green Surcharge on fuel (petrol, diesel).

The above two levies would also assist to discourage the use of personalized motorized vehicles and encourage the use of public transport, which would not only reduce the pollution level in the city but also reduce traffic congestion on the road.

A onetime Green Tax (Rs. 5000 to Rs. 10000 for four wheelers and Rs. 2000 for two wheelers) on existing vehicles registered in the City.

Receipts from traffic challans to be channeled to this Fund.

A 1 % turnover Tax on all shops, restaurants and hotels on a monthly basis.

A 5 % surcharge on Property Tax within the Corporation limits. In the financial analysis,5% cess has only been assumed

Metro Tax @ 2% on pay rolls of all establishments having more than 100 employees. Such cess is in existence in a number of Western countries for raising resources for metro rail. The employers' benefit a good deal by good Metro System.

Surcharge @ 10% on luxury tax on the earning of all 3-Star and above Hotels. Chinese cities have adopted this scheme to raise the funds for metro projects.

Densification of Corridor by way of selling of Floor Area Ratio (FAR) along the proposed metro corridors.

Revenue generated from change of land use along the metro corridors can be shared with metro SPV.

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रजिस्ट्री सं० डी० एल०-33004/99

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भारत का राजपत्र
The Gazette of India

असाधारण

EXTRAORDINARY

भाग II—खण्ड 3—उप-खण्ड (ii)

PART II—Section 3—Sub-section (ii)

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शहरी विकास मंत्रालय

(मैट्रो रेल प्रकोष्ठ)

अधिसूचना

नई दिल्ली, 7 सितम्बर, 2009

का.आ. 2279(अ).—केन्द्रीय सरकार, मैट्रो रेल (संशोधन) अधिनियम, 2009 (2009 का 34) की धारा 1 की उप-धारा (2) द्वारा प्रदत्त शक्तियों का प्रयोग करते हुए, 7 सितम्बर, 2009 को उस तारीख के रूप में नियत करती है, जिसको उक्त अधिनियम के उपबंध प्रवृत्त होंगे ।

[फा.सं. के-14011/40/2003-एमआरटीएस/मैट्रो]

विमल कुजूर, अवर सचिव

MINISTRY OF URBAN DEVELOPMENT

(Metro Rail Cell)

NOTIFICATION

New Delhi, the 7th September, 2009

S.O. 2279(E).—In exercise of the powers conferred by sub-section (2) of Section 1 of the Metro Railways (Amendment) Act, 2009 (34 of 2009) the Central Government hereby appoints the Seventh September, 2009 as the date on which the provisions of the said Act, shall come into force.

[F. No.K-14011/40/2003-MRTS/Metro]

BIMAL KUJUR, Under Secy.

3269 GI/2009

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इस भाग में भिन्न पृष्ठ संख्या दी जाती है जिससे कि यह अलग संकलन के रूप में रखा जा सके।
Separate paging is given to this Part in order that it may be filed as a separate compilation.

MINISTRY OF LAW AND JUSTICE
(Legislative Department)

New Delhi, the 27th August, 2009/Bhadra 5, 1931 (Saka)

The following Act of Parliament received the assent of the President on the 26th August, 2009, and is hereby published for general information:—

THE METRO RAILWAYS (AMENDMENT) ACT, 2009

No. 34 OF 2009

[*26th August, 2009.*]

An Act further to amend the Metro Railways (Construction of Works) Act, 1978 and to amend the Delhi Metro Railway (Operation and Maintenance) Act, 2002.

BE it enacted by Parliament in the Sixtieth Year of the Republic of India as follows:—

CHAPTER I
PRELIMINARY

Short title and commencement

1. (*1*) This Act may be called the Metro Railways (Amendment) Act, 2009.

(*2*) It shall come into force on such date as the Central Government may, by notification in the Official Gazette, appoint.

CHAPTER II

AMENDMENT TO THE METRO RAILWAYS (CONSTRUCTION OF WORKS) ACT, 1978

Amendment of section 1.

2. In the Metro Railways (Construction of Works) Act, 1978 (hereafter in this Chapter referred to as the Metro Railways Act), in section 1, in sub-section (*3*), for the portion beginning with the words "such other metropolitan city" and ending with the words "to that city accordingly", the following shall be substituted, namely:— 33 of 1978.

"the National Capital Region, such other metropolitan city and metropolitan area, after consultation with the State Government, and with effect from such date as may be specified in that notification and thereupon the provisions of this Act shall apply to the National Capital Region, such metropolitan city or metropolitan area accordingly:".

Substitution of words "metropolitan city" by words "metropolitan city, metropolitan area and National Capital Region".

3. In the Metro Railways Act, for the words "metropolitan city" occurring in clause (*h*) of sub-section (*1*) of section 2, clause (*c*) of sub-section (*1*) of section 4 and clause (*a*) of sub-section (*1*) of section 32, the words "metropolitan city, metropolitan area and the National Capital Region" shall be substituted.

Amendment of section 2.

4. In section 2 of the Metro Railways Act, in sub-section (*1*),—

(*i*) after clause (*h*), the following clause shall be inserted, namely:—

'(*ha*) "metropolitan area" shall have the meaning assigned to it in clause (*c*) of article 243P of the Constitution;';

(*ii*) after clause (*o*), the following clause shall be inserted, namely:—

'(*oa*) "National Capital Region" means the National Capital Region as defined in clause (*f*) of section 2 of the National Capital Region Planning Board Act, 1985;'. 2 of 1985

CHAPTER III

AMENDMENT TO THE DELHI METRO RAILWAY (OPERATION AND MAINTENANCE) ACT, 2002

Substitution of references to "metropolitan city of Delhi" by references to "National Capital Region and any other metropolitan area".

5. Throughout the Delhi Metro Railway (Operation and Maintenance) Act, 2002 (hereafter in this Chapter referred to as the Delhi Metro Railway Act), for the words "metropolitan city of Delhi" wherever they occur, the words "the National Capital Region, metropolitan city and metropolitan area" shall be substituted. 60 of 2002.

Amendment of section 1.

6. In section 1 of the Delhi Metro Railway Act, for sub-sections (*1*) and (*2*), the following sub-sections shall be substituted, namely:—

"(*1*) This Act may be called the Metro Railways (Operation and Maintenance) Act, 2002.

(*2*) It extends in the first instance to the National Capital Region and the Central Government may, by notification, after consultation with the State Government, extend this Act to such other metropolitan area and metropolitan city, except the metropolitan

city of Calcutta, and with effect from such date as may be specified in that notification, and thereupon the provisions of this Act shall apply to that metropolitan area or metropolitan city accordingly.".

Amendment of section 2.

7. In section 2 of the Delhi Metro Railway Act, in sub-section (*1*),—

(*i*) for clause (*a*), the following clauses shall be substituted, namely:—

'(*a*) "Central Government", in relation to technical planning and safety of metro railways, means the Ministry of the Government of India dealing with Railways;

(*aa*) "Claims Commissioner" means a Claims Commissioner appointed under section 48;';

(*ii*) for clause (*h*), the following clauses shall be substituted, namely:—

'(*h*) "metropolitan area" shall have the meaning assigned to it in clause (*c*) of article 243P of the Constitution;

(*ha*) "metropolitan city" means the metropolitan city of Bombay, Calcutta, Delhi or Madras;';

(*iii*) after clause (*k*), the following clause shall be inserted, namely:—

'(*ka*) "National Capital Region" means the National Capital Region as defined in clause (*f*) of section 2 of the National Capital Region Planning Board Act, 1985;'.

2 of 1985.

Amendment of section 6.

8. In section 6 of the Delhi Metro Railway Act, in sub-section (*2*), after clause (*b*), the following clauses shall be inserted, namely:—

"(*ba*) develop any metro railway land for commercial use;

(*bb*) provide for carriage of passengers by integrated transport services or any other mode of transport;".

Amendment of section 7.

9. Section 7 of the Delhi Metro Railway Act shall be renumbered as sub-section (*1*) thereof and after sub-section (*1*) as so renumbered, the following sub-section shall be inserted, namely:—

"(*2*) The Commissioner shall function under the administrative control of the Chief Commissioner of Railway Safety appointed under section 5 of the Railways Act, 1989.".

24 of 1989.

Substitution of new section for section 12.

10. For section 12 of the Delhi Metro Railway Act, the following section shall be substituted, namely:—

Annual report.

"12. The Chief Commissioner of Railway Safety shall, for each financial year, prepare in such form, and within such time, as may be prescribed, an annual report giving a full account of the activities of the Commissioners during the financial year immediately preceding the financial year in which such report is prepared and forward copies thereof to the Central Government.".

Amendment of section 13.

11. In section 13 of the Delhi Metro Railway Act, for the word "Commissioner", the words "Chief Commissioner of Railway Safety" shall be substituted.

Amendment of section 23.

12. In section 23 of the Delhi Metro Railway Act, in sub-section (*1*), for the words "Hindi and English", the words "Hindi, English and official language of the State in which such station is located" shall be substituted.

Amendment of section 26.

13. In section 26 of the Delhi Metro Railway Act, in sub-section (*1*), the words "a small" shall be omitted.

Amendment of section 34.

14. In section 34 of the Delhi Metro Railway Act, for sub-section (*4*), the following sub-section shall be substituted, namely:—

"(4) The Central Government and the State Government shall nominate one member each to the Fare Fixation Committee:

Provided that a person who is or has been an Additional Secretary to the Government of India or holds or has held an equivalent post in the Central Government or the State Government shall be qualified to be nominated as a member.".

Amendment of section 38.

15. In section 38 of the Delhi Metro Railway Act, in sub-section (2), for the words "Government of the National Capital Territory of Delhi", the words "State Government" shall be substituted.

Amendment of section 85.

16. In section 85 of the Delhi Metro Railway Act,—

(i) in sub-section (1), for the words "Government of the National Capital Territory of Delhi", the words "State Government" shall be substituted;

(ii) in sub-section (2), for the words "Government of the National Capital Territory of Delhi in the Delhi Gazette", the words "State Government" shall be substituted.

T.K. VISWANATHAN,
Secretary to the Govt. of India.

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CHAPTER – 23

CONCLUSIONS AND RECOMMENDATIONS

23.1 Surat has witnessed enormous growth during the last few years and is also known as a "Economic Power Hub" of Gujarat. The economy of Surat City is based mainly on two industries i.e. the textile industries of man made fibers/fabrics and the diamond cutting and polishing industry. Surat is one of the most dynamic cities of India having faster growth rate due to immigration from various parts of Gujarat and other States of India too. Rapid urbanization in the recent past has put the city's travel infrastructure to stress. Being thickly populated area, Surat's traffic needs cannot be met by only road-based system.

Surat's high population growth rate, coupled with high economic growth rate has resulted in an ever increasing demand for transport creating excessive pressure on the existent transport system. With high growth in transport demand over the years, congestion on roads, has been increasing due to phenomenal rise in private transport. Absence of an efficient full-fledged public transport system coupled with rapid growth in the use of personalized vehicles has led to high consumption of fossil fuels and increase in environmental pollution. The existing network of public transport systems including dedicated BRTS needs to be strengthened further in order to cope up with the rising demand of transport system especially when Surat is being developed as a '**Smart City'**. For this purpose provision of rail-based Metro system in the city has been considered.

Studies have brought out that a Medium Capacity Metro with carrying capacity of about 30,000 to 45,000 PHPDT will be adequate to meet not only the traffic needs for the present but for the future 30 to 40 years also. A Medium Metro System consisting of two Corridors namely (i) Sarthana to Dream City(21.61Kms) and (ii) Bhesan to Saroli (18.74 Kms.) at an estimated cost of **Rs. 6659.3 Crores and Rs. 4001.48 Crores** respectively (with all taxes & duties) to be made operational as recommended in 'Implementation Plan' chapter.

23.2 A detailed Environmental Impact Assessment Study has been carried out for the project. As a part of this Study, comprehensive environmental baseline data was collected, and both positive and negative impacts of the project were assessed in detail. The project has many positive environmental impacts like reduction in

traffic congestion, saving in travel time, reduction in air and noise pollution, lesser fuel consumption, lesser road accidents etc, with a few negative impacts (especially during implementation phase of the project) for which Environmental Management Plan has been suggested.

23.3 After examining the various options for execution of Surat Metro Project**,** it has been recommended that the project should be got executed through a SPV on DMRC funding pattern.

23.4 The fare structure has been estimated based on Delhi Metro fares decided by Fare Fixation Committee in 2009. Subsequently, for the purpose of assessing returns from the project, the fares have been revised every second year with an escalation of 12% every two years.

23.5 As in the case of Delhi Metro, the State Government should exempt/ reimburse the Gujarat Value Added Tax (VAT) to Surat Metro. It should also exempt the following:

Tax on electricity required for operation and maintenance of the metro system.

Municipal Taxes.

23.6 As per the earlier policy 50% of the Central Taxes were to be paid by GOI as subordinate Debt and balance 50% were paid by the concerned State Government. However, the post- GST regime, GOI intends to pay only 1/3rd of GST as subordinate debt and balance 2/3rd needs to be paid by the concerned state Government. Gujarat state Government may persue the central Government to extend the same benefit to Surat Metro also.

23.7 While the Financial Internal Rate of Return (FIRR) for the project has been assessed as **6.90%** (with all taxes& Duties) without additional PD,TOD & VCF income and **7.38%** with additional PD,TOD&VCF income(with all taxes& duties), and the Economic Internal Rate of Return (EIRR) works out to be **17.20%**

23.8 To avoid delays in processing the clearance for the Project, Government of Gujarat should approve project immediately on receipt of the DPR and forward the DPR to the Secretary, Ministry of Urban Development, Government of India, advising the GOI of the State Government's intention to take up the Project on DMRC pattern requesting for the latter's "in principle" clearance to go ahead with the Project.

23.9 Meanwhile the State Government should freeze all future developments along the proposed route of Surat Metro to avoid in-fructuous expenditure.

23.10 CONCLUSIONS AND RECOMMENDATIONS

The project is essentially a social necessity as it aims to improve the overall health of the city by way of reducing congestion on roads coupled with to reduced consumption of fossil fuels thereby reducing environmental pollution. This is evident from the fact the **EIRR** of the project is of the order of **17.20%** indicating that the project is very well justified in terms of social benefits that will accrue to the Society/Surat City. Therefore, implementation of this project is inescapable. Since the Gujarat Government is providing requisite land for PD and being social sector project, it is advisable to take up the job on DMRC model. Accordingly, the corridors are recommended for implementation.

As a way forward the metro network needs to be expanded in due course as the city population is already more than 50 lakhs and further city is growing in rapid manner. Therefore, the further study may be taken up when the execution of Phase-I is going on so that the DPR prepared and approved can be obtained before completion of Phase-I and the same project team can continue to work for Phase-II.